

ALGORITHMS, DATA STRUCTURES, AND APPLICATIONS

Computational Geometry

Chaman Singh Verma

July 2026

Preface

Computational geometry is the study of algorithms for solving geometric problems. It sits at the intersection of mathematics, computer science, and engineering, providing the theoretical foundations and practical tools for manipulating geometric objects in two and three dimensions.

This book presents a comprehensive collection of algorithms drawn from the **CompGeom** library. Each chapter provides a self-contained treatment of a family of related algorithms, including formal definitions, theoretical analysis, pseudocode implementations, complexity bounds, practical applications, and edge-case considerations.

The material is organized into twelve parts, progressing from foundations and mathematical preliminaries through fundamental planar algorithms, spatial data structures, proximity and tessellations, arrangements and duality, geometric optimization, curves and surfaces, motion and robotics, shape analysis, advanced algorithmic techniques, application domains, and finally modern research-level topics.

How to Use This Book

Each algorithm entry follows a consistent structure:

1. **Overview** — a high-level description of the problem and its significance.
2. **Definitions** — precise mathematical terminology.
3. **Theory** — the mathematical foundations and proofs.
4. **Pseudocode** — a readable implementation sketch.
5. **Parameter Selection** — practical tuning advice.
6. **Complexity** — time and space bounds.
7. **Applications** — real-world use cases.
8. **Edge Cases** — testing strategies and degenerate inputs.
9. **References** — original papers and textbooks.

Readers may approach the book topically or use it as a reference for individual algorithms. Cross-references between chapters are provided where algorithms build upon one another.

Introduction

Euclidean Geometry

Euclidean geometry studies points, lines, angles, distances, and shapes in a flat space governed by Euclid’s parallel postulate. In $\mathbb{R}^2$ and $\mathbb{R}^3$, the distance between points is measured by the Euclidean norm,

$$\|x - y\|_2 = \sqrt{\sum_i (x_i - y_i)^2},$$

and straight lines are the shortest paths between their points. Most classical computational-geometry algorithms in this book are Euclidean: orientation tests, convex hulls, Delaunay triangulations, Voronoi diagrams, line and segment intersections, polygon operations, and mesh algorithms all use this flat-space model either directly or as their local approximation.

Non-Euclidean Geometries

Non-Euclidean geometry changes one or more assumptions of Euclidean geometry, especially the behavior of parallel lines and the definition of distance. The two principal examples are:

- **Spherical or elliptic geometry:** points lie on a curved surface such as a sphere. “Straight” paths are great-circle arcs, and initially parallel geodesics can meet.
- **Hyperbolic geometry:** the space has negative curvature. A point and a line can have multiple distinct parallels through that point, and triangle angles sum to less than π .

More generally, differential geometry describes a curved space using a manifold, a metric tensor, and geodesics. This viewpoint is important for surface parameterization, geodesic distance, curvature, discrete exterior calculus, and spectral geometry. Algorithms designed for Euclidean input cannot automatically be transferred to curved spaces: Euclidean dot products, cross products, circumcenters, and straight-line interpolation may need to be replaced by metric-aware operations, local tangent-plane constructions, or geodesic computations. Several later chapters introduce these extensions.

Prerequisites

A basic understanding of data structures, algorithms, and linear algebra is assumed. Familiarity with discrete mathematics and introductory geometry will be helpful but is not strictly required, as key concepts are introduced inline.

Notation

Symbol	Meaning
$\mathbb{R}^d$	Euclidean space of dimension d
$\mathbb{R}^2, \mathbb{R}^3$	The 2D and 3D Euclidean planes
n	Number of input points or elements
k	Number of output features (intersections, layers, etc.)
$P = \{p_1, \dots, p_n\}$	A set of points
$CH(P)$	Convex hull of P
$DT(P)$	Delaunay triangulation of P
$VD(P)$	Voronoi diagram of P
$\mathcal{O}(f(n))$	Big-O complexity notation
$\Omega(f(n))$	Big-Omega complexity notation
$\Theta(f(n))$	Big-Theta complexity notation
$\alpha(n)$	Inverse Ackermann function
Δ	Laplace–Beltrami operator (or cotangent Laplacian)
d, δ	Exterior derivative and codifferential
$\star$	Hodge star operator
$\oplus, \ominus$	Minkowski sum and difference
$\langle \cdot, \cdot \rangle$	Inner product
$ \cdot $	Euclidean norm (or cardinality, context-dependent)
∂K	Boundary of a simplicial complex K
$\chi(K)$	Euler characteristic of K
β_p	p -th Betti number

Table 1: Principal notation used throughout the book.

Contents

Preface	iii
Introduction	v
Notation	vii
I Foundations	1
1 Introduction to Computational Geometry	3
1.1 What Is Computational Geometry?	3
1.2 Geometry as an Algorithmic Problem	3
1.3 Historical Development	4
1.4 Geometric Objects and Representations	4
1.5 Points, Lines, Segments, Rays, and Polygons	5
1.6 Combinatorial versus Numerical Geometry	5
1.7 The Real-RAM Model	6
1.8 Computational Models and Complexity	6
1.9 Degeneracies and General Position	7
1.10 Robustness and Numerical Precision	7
1.11 Symbolic Perturbation	8
1.12 Exact Geometric Computation	8
1.13 Major Families of Geometric Algorithms	8
1.14 Applications of Computational Geometry	9
1.15 Software Libraries and Experimental Geometry	10
1.16 Chapter Problems	10
1.17 Computational Projects	10
2 Mathematical Foundations	13
2.1 Vectors and Affine Spaces	13
2.2 Inner Products and Euclidean Distance	14
2.3 Lines and Hyperplanes	14
2.4 Affine Combinations	15
2.5 Convex Combinations	15
2.6 Orientation and Determinants	15
2.7 Signed Area and Signed Volume	16
2.8 Barycentric Coordinates	16
2.9 Half-Spaces	17
2.10 Hyperplanes and Separating Surfaces	17
2.11 Norms and Metrics	18
2.12 Transformations	18

2.13	Homogeneous Coordinates	19
2.14	Duality	19
2.15	Topological Preliminaries	20
2.16	Exercises	20
3	Geometric Predicates and Robust Computation	23
3.1	Orientation Predicate	23
3.2	Three-dimensional Orientation	23
3.3	In-circle Predicate	24
3.4	Robust Evaluation	24
3.5	Floating-Point Filters	24
3.6	Exact Arithmetic	25
3.7	Rational Arithmetic in Computational Geometry	25
3.8	Simulation of Simplicity	26
3.9	Robustness Failure Examples	26
II	Fundamental Planar Algorithms	27
4	Convex Hulls	29
4.1	Graham Scan	29
4.2	Monotone Chain	32
4.3	Chan's Algorithm	34
4.4	QuickHull	35
4.5	Convex Hull Peeling	37
5	Line-Segment Intersection	41
5.1	The Intersection Problem	41
5.2	Pairwise Intersection Testing	41
5.3	Sweep-Line Paradigm	43
5.4	Events and Event Queues	43
5.5	Sweep-Line Status Structures	44
5.6	Bentley–Ottmann Algorithm	44
5.7	Correctness	47
5.8	Complexity Analysis	47
5.9	Degenerate Intersections	48
5.10	Reporting versus Detecting Intersections	48
5.11	Red–Blue Segment Intersection	49
5.12	Contour of the Union of Rectangles	51
5.13	Applications	53
5.14	Exercises	53
5.15	Project: Interactive Sweep-Line Visualizer	54
6	Polygon Geometry	55
6.1	Polygon Boolean Operations	55
6.2	Line Arrangements	58
6.3	Minkowski Sums	61
6.4	Industrial Applications of Minkowski Sums	63
6.5	Polygon Symmetry	65
6.6	Polygon Matching	67
6.7	Line and Polygon Clipping	70
6.8	Polygon Partitioning	72

6.9	Convex Decomposition	72
6.10	Convex Partitioning	73
7	Polygon Triangulation	77
7.1	Ear Clipping	77
7.2	Monotone Decomposition	80
7.3	Incremental Delaunay Triangulation	82
7.4	Dynamic Delaunay Triangulation	85
7.5	Delaunay Flips	87
7.6	Constrained Delaunay Triangulation	90
7.7	Non-Obtuse Triangulation	91
III	Geometric Searching and Spatial Data Structures	93
8	Point Location	95
8.1	The Planar Point-Location Problem	95
8.2	Subdivision Representations	95
8.3	Slab Decomposition	96
8.4	Trapezoidal Maps	97
8.5	Randomized Incremental Construction	99
8.6	Search DAGs	100
8.7	Kirkpatrick's Hierarchy	101
8.8	Point Location in Triangulations	102
8.9	Dynamic Point Location	104
8.10	Practical Implementations	105
8.11	Exercises	105
9	Orthogonal Range Searching	107
9.1	Range Queries	107
9.2	One-Dimensional Range Searching	107
9.3	k-d Trees	108
9.4	Range Trees	110
9.5	Multilevel Data Structures	111
9.6	Fractional Cascading	111
9.7	Priority Search Trees	112
9.8	Reporting versus Counting	113
9.9	Higher-Dimensional Range Searching	114
9.10	Dynamic Range Searching	114
9.11	Curse of Dimensionality	114
9.12	Simplex Range Searching	115
9.13	Exercises	116
9.14	Project: Spatial Query Engine	117
10	Spatial Partitioning	119
10.1	K-Dimensional Trees	119
10.2	Quadtrees	123
10.3	R-Trees	127
10.4	Interval Trees	130
10.5	Segment Trees	132
10.6	Uniform Grids	135
10.7	Fixed-Radius Neighbor Search	137

10.8 Binary Space Partitioning Trees	139
10.9 Industrial Applications of BSP Trees	141
10.10 Space-Filling Curves	142
10.11 Hilbert Curve	142
10.12 Morton Curve	145
10.13 Peano Curve	148
10.14 Zigzag Curve	150
10.15 Spiral Walk	153
IV Proximity and Tessellations	157
11 Voronoi Diagrams	159
11.1 Voronoi Diagram	159
11.2 Voronoi Diagrams of Line Segments	160
11.3 Farthest-Point Voronoi Diagrams	162
11.4 Kinetic Voronoi Diagrams	163
11.5 Industrial Applications of Voronoi Diagrams	164
12 Delaunay Triangulations	165
12.1 Voronoi–Delaunay Duality	165
12.2 Empty-Circumcircle Property	167
12.3 Local Delaunay Criterion	169
12.4 Edge Flipping	171
12.5 Incremental Construction	172
12.6 Randomized Incremental Algorithm	174
12.7 Divide-and-Conquer Construction	176
12.8 Lifting to a Paraboloid	177
12.9 Relationship to Convex Hulls	179
12.10 Constrained Delaunay Triangulation	180
12.11 Quality Properties	182
12.12 Delaunay Refinement	184
12.13 Higher-Dimensional Delaunay Complexes	186
12.14 Intrinsic Delaunay Triangulation	188
12.15 Applications to Mesh Generation	190
12.16 Exercises	192
12.17 Project: Delaunay Triangulation from Scratch	192
13 Proximity Problems	193
13.1 Closest Pair of Points	193
13.2 Proximity Queries	196
13.3 Industrial Applications of Nearest Neighbor Search	198
13.4 Range Search	199
13.5 Point in Polygon (Winding Number)	202
13.6 Largest Empty Circle	204
13.7 Minimum Bounding Boxes	206
13.8 Maximum Overlap	208
13.9 Segment Intersection (Bentley-Ottmann)	211
13.10 Point Location via Trapezoidal Maps	214
13.11 Point Location in Triangulations (Walking)	219

V	Arrangements and Duality	223
14	Arrangements of Lines and Curves	225
14.1	Davenport–Schinzel Sequences	225
14.2	Lower Envelopes	228
14.3	Levels in Arrangements and Half-Plane Discrepancy	231
14.4	Union Volume Estimation	233
15	Geometric Duality	237
15.1	Point-Line Duality	237
15.2	Incidence Preservation	238
15.3	Order Reversal	239
15.4	Dual Arrangements	240
15.5	Convex Hulls through Duality	241
15.6	Half-Plane Intersection	244
15.7	Linear Programming through Duality	247
15.8	Applications to Optimization	251
15.9	Exercises	253
VI	Convexity and Geometric Optimization	255
16	Convex Polytopes	257
16.1	Convex Polytopes	257
16.2	V-Representations and H-Representations	257
16.3	Faces and Face Lattices	258
16.4	Supporting Hyperplanes	259
16.5	Separating Hyperplane Theorems	260
16.6	Carathéodory’s Theorem	260
16.7	Helly’s Theorem	261
16.8	Radon’s Theorem	261
16.9	Euler–Poincaré Relations	262
16.10	Polarity	263
16.11	Computing Higher-Dimensional Hulls	263
16.12	Exercises	268
17	Linear Programming in Geometry	271
17.1	Geometric Linear Programs	271
17.2	Half-Plane Intersection	271
17.3	Low-Dimensional Linear Programming	271
17.4	Duality and Applications	272
18	Intersection and Distance of Convex Objects	273
18.1	Convex Polygon Intersection	273
18.2	Separating Axis Theorem	274
18.3	Distance between Convex Sets	276
18.4	Support Functions	276
18.5	Minkowski Difference	277
18.6	Gilbert–Johnson–Keerthi Algorithm	277
18.7	Penetration Depth	279
18.8	Collision Detection	280
18.9	Continuous Collision Detection	281

18.10	Applications in Robotics and Simulation	282
18.11	Exercises	283
18.12	Enclosure and Convex Distance	283
18.13	Welzl's Algorithm	283
18.14	Convex Polygon Distance	286
VII	Curves, Surfaces, and Three-Dimensional Geometry	291
19	Computational Geometry in Three Dimensions	293
19.1	Polyhedral Representation	293
19.2	Three-Dimensional Predicates	293
19.3	Incremental Three-Dimensional Hulls	293
19.4	Degeneracies and Validation	294
19.5	Applications	294
20	Curves and Surfaces	295
20.1	Surface Modeling and Mesh Generation	295
20.2	Bézier Surfaces	295
20.3	NURBS Surfaces	298
20.4	Mesh Voxelization	300
20.5	Winding-Filtered Tetrahedral Meshing	302
20.6	Mesh Refinement	303
20.7	Approximate Convex Decomposition	307
20.8	Triangle to Quad Conversion	309
20.9	Voxel Grid Point Simplification	311
20.10	Surface Reconstruction	313
20.11	α -Shapes	318
20.12	Surface Parameterization and Geometry	322
20.13	BFF Parameterization	322
20.14	Harmonic Mapping	323
20.15	Functional Maps	326
20.16	Diffusion Distance	328
20.17	Mean Curvature Flow	329
20.18	Angle-Bounded Remeshing	332
20.19	Affine Heat Flow	333
20.20	Geodesic Distance	335
20.21	Medial Axis Transform	337
20.22	Straight Skeleton	340
20.23	Mesh Cut Loops	342
20.24	Mesh Tunnel Loops	344
20.25	Iterative Closest Point (ICP) Registration	346
21	Half-Edge Data Structure	353
21.1	Half-Edge Representation	353
21.2	Vertices, Half-Edges, and Faces	354
21.3	Twin, Next, and Prev Operators	355
21.4	Boundary and Manifold Handling	355
21.5	Adjacency and Neighborhood Queries	357
21.6	Traversal Operations	359
21.7	Comparison with Other Representations	361
21.8	Incidence and Storage Complexity	361

21.9 Construction from Polygon Soup	363
21.10 Applications in Mesh Processing	365
21.11 Exercises	367
22 Mesh Generation	369
22.1 Computational Meshes	369
22.2 Simplicial Complexes	369
22.3 Triangular and Tetrahedral Meshes	370
22.4 Mesh Quality Measures	370
22.5 Delaunay Mesh Generation	371
22.6 Constrained Delaunay Meshing	371
22.7 Delaunay Refinement	372
22.8 Ruppert's Algorithm	373
22.9 Chew's Algorithm	375
22.10 Advancing-Front Methods	376
22.11 Adaptive Meshing	377
22.12 Mesh Optimization	377
22.13 Finite-Element Applications	378
22.14 Exercises	378
22.15 Project: Adaptive Mesh Generator	378
23 Voxel Geometry and OpenVDB	381
23.1 Voxel Grids	381
23.2 Signed Distance Fields	383
23.3 Sparse Voxel Representations	384
23.4 OpenVDB: Hierarchical Sparse Volumes	385
23.5 Voxel-to-Mesh Conversion	387
23.6 Applications	388
VIII Motion, Visibility, and Robotics	389
24 Visibility	391
24.1 Visibility Problems	391
24.2 Visibility inside Polygons	392
24.3 Visibility Polygons	393
24.4 Computing Visibility from a Point	393
24.5 Guarding Polygons	396
24.6 Visibility Graphs	396
24.7 Shortest Paths among Obstacles	399
24.8 Geodesic Paths	400
24.9 Applications	401
24.10 Exercises	402
24.11 Distance Transforms and Shortest Paths	402
24.12 Fast Marching Method	402
24.13 Distance Map	406
24.14 Dijkstra's Algorithm	408
24.15 K-Geodesics	411
24.16 Fréchet Distance	414

25 Art Gallery Problem	417
25.1 Art Gallery Guard Placement	417
25.2 Polygon Kernel and Star-Shaped Polygons	420
26 Motion Planning	423
26.1 Configuration Space	423
26.2 Configuration-Space Obstacles	423
26.3 Exact Planning Methods	424
26.4 Roadmaps and Sampling-Based Planning	424
26.5 Planning Constraints and Validation	424
26.6 Applications	425
 IX Shape Analysis and Geometric Data	 427
27 Sampling	429
27.1 Poisson Disk Sampling	429
27.2 Halton Points	432
27.3 Uniform Random Point Generation	435
27.4 Random Line Segment Generation	438
27.5 Random Polygon Generation	440
27.6 Random Walk	442
27.7 Self-Avoiding Walk	445
28 Shape Representation and Reconstruction	449
28.1 Shape Representation	449
28.2 Implicit and Parametric Representations	449
28.3 Reconstruction from Point Clouds	450
28.4 Surface Reconstruction	452
28.5 Signed Distance Functions	456
28.6 Level Sets and Front Propagation	457
28.7 Poisson Surface Reconstruction	457
28.8 Applications	460
28.9 Exercises	461
29 Geometric Matching	463
29.1 Comparing Geometric Objects	463
29.2 Hausdorff Distance	464
29.3 Fréchet Distance	465
29.4 Discrete Fréchet Distance	465
29.5 Shape Matching	466
29.6 Point-Set Registration	467
29.7 Rigid Transformations	468
29.8 Iterative Closest Point	469
29.9 Geometric Hashing	471
29.10 Applications in Computer Vision	474
29.11 Exercises	475
30 Computational Topology	477
30.1 Mesh Analysis and Topology	477
30.2 Mesh Connected Components	477
30.3 Mesh Boundary Detection	480

30.4 Mesh Orientability	482
30.5 Manifold Verification	485
30.6 Euler Characteristic	488
30.7 Node Degree and Valence	491
30.8 Mesh Coloring	493
30.9 Mesh Reordering	496
30.10 Mesh Rigidity	498
30.11 Mesh Decimation via Quadric Error Metrics	501
30.12 Spectral Geometry and Shape Analysis	506
30.13 Shape Spectra	506
30.14 Shape Space Spectra	509
30.15 Odeco Frame Fields	510
30.16 Discrete Exterior Calculus	512
30.17 Hodge Star Computation	512
30.18 Exterior Derivative	515
30.19 Hodge Decomposition	518
30.20 Discrete Morse Theory	521
X Advanced Algorithmic Techniques	525
31 Randomized Geometric Algorithms	527
31.1 Why Randomization Helps	527
31.2 Randomized Incremental Construction	528
31.3 Backward Analysis	529
31.4 Conflict Graphs	530
31.5 Clarkson–Shor Technique	532
31.6 Random Sampling	534
31.7 ε -Nets	535
31.8 Applications to Convex Hulls	536
31.9 Applications to Delaunay Triangulations	538
31.10 Applications to Linear Programming	540
31.11 Exercises	541
32 Approximation Algorithms in Geometry	543
32.1 Why Approximation?	543
32.2 Geometric Approximation	544
32.3 ε -Approximations	544
32.4 Coresets	546
32.5 Approximate Nearest Neighbors	547
32.6 Locality-Sensitive Hashing	549
32.7 Well-Separated Pair Decomposition	551
32.8 Approximate Distance Queries	553
32.9 Geometric Clustering	553
32.10 k -Center and k -Median Problems	554
32.11 Minimum Enclosing Shapes	556
32.12 High-Dimensional Problems	557
32.13 Exercises	557
32.14 Project: Approximate Nearest-Neighbor Search	558

33 Packing Algorithms	559
33.1 Circle Packing	559
33.2 Rectangle Packing	562
33.3 Hopper Optimization	565
34 Dynamic and Kinetic Geometry	569
34.1 Dynamic Geometric Problems	569
34.2 Dynamic Point Sets	569
34.3 Dynamic Convex Hulls	570
34.4 Dynamic Proximity Structures	572
34.5 Moving Geometric Objects	573
34.6 Kinetic Data Structures	573
34.7 Certificates and Events	574
34.8 Kinetic Sorting	575
34.9 Kinetic Convex Hulls	577
34.10 Kinetic Closest Pairs	579
34.11 Applications to Tracking and Simulation	581
34.12 Exercises	582
XI Applications	583
35 Computational Geometry for Computer Graphics	585
35.1 Geometric Modeling	585
35.2 Spatial Acceleration Structures	585
35.3 Bounding Volume Hierarchies	585
35.4 Ray–Object Intersection	585
35.5 Hidden-Surface Problems	585
35.6 Clipping	585
35.7 Mesh Processing	585
35.8 Collision Detection	585
35.9 Real-Time Geometry	585
35.10 GPU-Based Geometric Computation	585
35.11 Case Study: Geometry Pipeline	585
35.12 Project	585
36 Computational Geometry for GIS	587
36.1 Geographic Data	587
36.2 Spatial Indexing	587
36.3 Map Overlay	587
36.4 Point-in-Region Queries	587
36.5 Proximity Analysis	587
36.6 Voronoi-Based Service Regions	587
36.7 Terrain Representation	587
36.8 Triangulated Irregular Networks	587
36.9 Digital Elevation Models	587
36.10 Map Generalization	587
36.11 Large-Scale Spatial Data	587
36.12 Case Study: Geographic Facility Location	587
36.13 Project	587

37 Computational Geometry for Robotics	589
37.1 Robot Geometry	589
37.2 Collision Detection	589
37.3 Configuration Spaces	589
37.4 Path Planning	589
37.5 Sensor Geometry	589
37.6 Localization	589
37.7 Mapping	589
37.8 SLAM and Geometric Structures	589
37.9 Manipulation Planning	589
37.10 Autonomous Navigation	589
37.11 Case Study: Mobile-Robot Planning	589
37.12 Project	589
38 Computational Geometry for Scientific Computing	591
38.1 Geometry in Numerical Simulation	591
38.2 Mesh Generation	591
38.3 Finite-Element Geometry	591
38.4 Domain Discretization	591
38.5 Adaptive Refinement	591
38.6 Particle Methods	591
38.7 Nearest-Neighbor Computations	591
38.8 Spatial Decomposition	591
38.9 Parallel Geometric Algorithms	591
38.10 Large-Scale Scientific Geometry	591
38.11 Case Study: PDE Mesh Construction	591
38.12 Project	591
38.13 Stochastic and Meshless PDE Methods	591
38.14 Stochastic PDE Solvers	591
38.15 Wave Kernel Signature	593
XII Modern and Research-Level Topics	597
39 High-Dimensional Computational Geometry	599
39.1 Geometry beyond Three Dimensions	599
39.2 High-Dimensional Convex Hulls	600
39.3 Voronoi and Delaunay Structures	602
39.4 Nearest-Neighbor Search	605
39.5 Dimensionality Reduction	607
39.6 Johnson–Lindenstrauss Lemma	610
39.7 Random Projections	612
39.8 Concentration of Measure	614
39.9 Curse of Dimensionality	616
39.10 Applications to Machine Learning	618
39.11 Exercises	619
40 Geometry of Data and Machine Learning	621
40.1 Geometric View of Data	621
40.2 Metric Spaces	622
40.3 Nearest-Neighbor Classification	623
40.4 Clustering Geometry	625

40.5	Convex Geometry in Machine Learning	629
40.6	Kernel Geometry	630
40.7	Manifold Learning	631
40.8	Geometric Embeddings	632
40.9	Graph-Based Geometric Learning	632
40.10	Optimal Transport: A Geometric Introduction	633
40.11	Computational Geometry for AI	634
40.12	Research Directions	637
40.13	Project	637
41	Research Frontiers	639
41.1	Robust and Certified Geometry	639
41.2	Massive Geometric Data Sets	640
41.3	Streaming Geometry	640
41.4	External-Memory Geometry	642
41.5	Geometric Algorithms for Autonomous Systems	644
41.6	Geometry in Machine Learning	645
41.7	Topological Data Analysis	645
41.8	Shape and Surface Reconstruction	646
41.9	Geometric Deep Learning	646
41.10	Differentiable Geometry	647
41.11	Geometry Processing	647
41.12	Open Problems in Computational Geometry	648
41.13	How to Read Computational Geometry Research	649
41.14	Designing Computational Experiments	649
41.15	From Graduate Study to Research	650
41.16	Computational Algebra and Similarity	651
41.17	Buchberger’s Algorithm	651
41.18	Resultant-Based Elimination	654
A	Mathematical Reference	657
A.1	Linear Algebra	657
A.2	Vector Calculus	657
A.3	Determinants	657
A.4	Affine Geometry	657
A.5	Convexity	657
A.6	Probability	657
A.7	Graph Theory	657
A.8	Asymptotic Analysis	657
B	Algorithms and Data Structures	659
B.1	Sorting	659
B.2	Binary Search	659
B.3	Balanced Search Trees	659
B.4	Priority Queues	659
B.5	Hashing	659
B.6	Graph Algorithms	659
B.7	Union–Find	659
B.8	Randomized Algorithms	659

C	Geometry Formula Reference	661
C.1	Vector Operations	661
C.2	Orientation Tests	661
C.3	Distance Formulas	661
C.4	Intersection Formulas	661
C.5	Circle Predicates	661
C.6	Areas and Volumes	661
C.7	Coordinate Transformations	661
C.8	Barycentric Coordinates	661
D	Implementation Guide	663
D.1	Designing a Geometry Kernel	663
D.2	Choosing Numeric Types	663
D.3	Floating-Point Error	663
D.4	Exact Predicates	663
D.5	Testing Geometric Software	663
D.6	Generating Adversarial Test Cases	663
D.7	Benchmarking	663
D.8	Visualization and Debugging	663
E	Computational Geometry Software	665
E.1	CGAL	665
E.2	GEOSE	665
E.3	Boost.Geometry	665
E.4	Qhull	665
E.5	Triangle	665
E.6	TetGen	665
E.7	SciPy Spatial Algorithms	665
E.8	Python Geometry Ecosystem	665
E.9	Choosing a Library	665
F	Graduate Projects	667
F.1	Convex Hull Laboratory	667
F.2	Sweep-Line Intersection Engine	667
F.3	Voronoi/Delaunay Package	667
F.4	Spatial Database	667
F.5	Mesh Generator	667
F.6	Collision-Detection System	667
F.7	Robot Path Planner	667
F.8	Point-Cloud Reconstruction	667
F.9	GIS Spatial Analysis System	667
F.10	Persistent-Homology Explorer	667
F.11	GPU Geometry Engine	667
F.12	Research Replication Project	667
G	Selected Solutions and Hints	669

List of Algorithms

1	Two-dimensional orientation test	23
2	Graham Scan for Convex Hull	31
3	Monotone Chain (Andrew’s Algorithm)	33
4	Chan’s Convex Hull Algorithm	35
5	QuickHull Algorithm	36
6	Convex Hull Peeling (Onion Peeling)	38
7	Segment Intersection Test	42
8	Bentley–Ottmann Segment Intersection	46
9	Red–Blue Segment Intersection by Sweep	50
10	Union Contour of Axis-Parallel Rectangles	52
11	Convex Decomposition via Triangle Merging	73
12	Minimum Convex Partition via Dynamic Programming	74
13	Ear Clipping Triangulation	79
14	Monotone Polygon Decomposition	81
15	Incremental Delaunay Triangulation	83
16	Dynamic Point Insertion	86
17	Dynamic Point Deletion	86
18	Delaunay Edge Flip Algorithm	89
19	Constrained Delaunay Triangulation	91
20	Non-Obtuse Triangulation	92
21	Point Location by Slab Decomposition	96
22	Build the Trapezoidal Map of a Segment Set	98
23	Query by History DAG	100
24	Kirkpatrick Point Location	102
25	Visibility Walk in a Triangulation	103
26	One-Dimensional Range Query	108
27	Orthogonal Range Query in a k -d Tree	109
28	2D Orthogonal Range Query	110
29	Simplex Range Counting with a Partition Tree	116
30	Fixed-Radius Neighbor Search	138
31	Hilbert Curve: Coordinates to Index	144
32	Morton Code: Coordinates to Index	146
33	Peano Curve: Coordinates to Index	149
34	Snake Scan Traversal	151
35	Diagonal Zigzag Scan (JPEG)	152
36	Square Spiral Walk	154
37	Voronoi Diagram via Delaunay Dual	160
38	Segment Voronoi Diagram by Plane Sweep	161
39	Smallest Enclosing Circle via Farthest-Point Voronoi	163
40	Kinetic Voronoi Maintenance	163
41	Voronoi Cells from a Delaunay Triangulation	166

42	Adaptive In-Circle Predicate	168
43	Local Delaunay Test	170
44	Lawson’s Flip Algorithm	171
45	Bowyer–Watson Incremental Insertion	173
46	Randomized Incremental Delaunay	175
47	Divide-and-Conquer Delaunay (Guibas–Stolfi)	177
48	Delaunay via Lifting	178
49	All Diagrams from One Convex Hull	180
50	CDT by Constraint Recovery	181
51	Quality Report	183
52	Ruppert Delaunay Refinement	185
53	Bowyer–Watson in Higher Dimensions	187
54	Intrinsic Delaunay Flip Algorithm	189
55	Divide-and-Conquer Closest Pair	195
56	BVH Distance Query	197
57	Orthogonal Range Search (Tree-based)	200
58	Point-in-Polygon via Winding Number	203
59	Largest Empty Circle	205
60	Minimum Bounding Box via Rotating Calipers	207
61	Maximum Overlap (1D Sweep)	210
62	Bentley–Ottmann Segment Intersection	213
63	Trapezoidal Map Query	217
64	Trapezoidal Map Construction (Incremental)	217
65	Locate in a Triangulation by Visibility Walk	220
66	Compute the k-Level of a Line Arrangement	232
67	Convex Hull via Dual Envelopes	243
68	Half-Plane Intersection via the Dual Hull	246
69	Seidel’s Randomized Incremental 2D LP	249
70	Randomized Incremental 3D Convex Hull	266
71	Gift Wrapping in 3D	266
72	Fourier–Motzkin Elimination of One Variable	267
73	Half-Plane Intersection	272
74	Intersection of Two Convex Polygons	274
75	Separating Axis Test for Two Convex Polygons	275
76	GJK Distance / Intersection Test	278
77	Expanding Polytope Algorithm (Penetration Depth)	280
78	Conservative Advancement (Time of Impact)	282
79	Incremental Three-Dimensional Convex Hull	294
80	Evaluate Bézier Surface	297
81	Evaluate NURBS Surface	299
82	Voxelize Mesh	301
83	Winding-Filtered Tet Meshing	303
84	Refine Mesh	305
85	CoACD	308
86	Greedy Triangle to Quad	310
87	Voxel Grid Simplify	312
88	Ball Pivoting Algorithm Reconstruction	316
89	α -Shape Computation (2D)	320
90	BFF Parameterization	323
91	Transfer Topology	325
92	Compute Functional Map	327

93	Compute Diffusion Distance	329
94	Mean Curvature Flow	331
95	Angle-Bounded Remesh	332
96	Affine Polygon Smoothing	334
97	Compute Geodesic via Heat Method	336
98	Medial Axis	338
99	Straight Skeleton	341
100	Find Cut Graph	343
101	Find Tunnel Loops	345
102	Iterative Closest Point (ICP) Registration	349
103	Enumerate Faces Around a Vertex	357
104	Walk a Face Boundary	359
105	Walk Around a Vertex	360
106	Weld Polygon-Soup Vertices	364
107	Build Half-Edge Structure from a Polygon Soup	364
108	Delaunay Refinement (Ruppert–Chew style)	373
109	Rotational Sweep for the Visibility Polygon	395
110	Rotational Sweep for the Visibility Graph	398
111	Shortest Path among Polygonal Obstacles	399
112	Shortest Path in a Triangulated Polygon (Funnel Algorithm)	400
113	Fast Marching Method	404
114	Meijster’s Distance Transform	407
115	Dijkstra’s Shortest Path Algorithm	410
116	K-Geodesics Clustering	412
117	Discrete Fréchet Distance	415
118	Probabilistic Roadmap Planner	424
119	Poisson Disk Sampling	431
120	Halton Sequence Generator	434
121	Uniform Random Points in Basic Shapes	437
122	Random Line Segment Generation	439
123	Random Polygon Generation	441
124	2D Random Walk	444
125	Random Walk on a Mesh	444
126	Generate Self-Avoiding Walk (Recursive)	447
127	Crust (Amenta–Bern)	453
128	Cocone (Amenta–Choi–Dey–Leekha)	454
129	Ball-Pivoting Algorithm (Bernardini et al.)	454
130	Poisson Surface Reconstruction	459
131	Hungarian Algorithm for the Assignment Problem	468
132	Iterative Closest Point	470
133	Geometric Hashing (Recognition)	473
134	Find Mesh Components	479
135	Detect Mesh Boundaries	481
136	Orient Mesh	484
137	Is Manifold	487
138	Calculate Euler Characteristic	490
139	Calculate Vertex Degrees	492
140	Get Vertex Degree	492
141	Color Mesh	495
142	RCM Reorder	497
143	Analyze Mesh Rigidity	500

144	Mesh Decimation via QEM	504
145	Shape-DNA Computation	508
146	Shape Space SpectralNet Usage	510
147	Odeco Frame Field Design	512
148	Compute Hodge Star Operators	514
149	Compute Exterior Derivatives	517
150	Hodge Decomposition of a 1-Form	520
151	Greedy Construction of Discrete Gradient Field	523
152	Generic Randomized Incremental Construction	529
153	Conflict-Graph Maintenance	531
154	Clarkson–Shor Sampling for Convex Hulls	533
155	Randomized ε -Net Construction	536
156	Randomized Incremental Convex Hull	537
157	Randomized Incremental Delaunay (Walk Variant)	539
158	Seidel’s Randomized Linear Programming	541
159	Random Sampling for ε -Nets	546
160	c -Approximate Nearest Neighbor via Box Decomposition	548
161	Locality-Sensitive Hashing: Index and Query	550
162	Well-Separated Pair Decomposition from a Split Tree	552
163	Farthest-Point k -Center (Gonzalez)	555
164	Lloyd’s Algorithm (k -Means Heuristic)	555
165	Core-Set Based $(1 + \varepsilon)$ -Approximate MEB	557
166	Dynamic Convex Hull Maintenance	571
167	Kinetic Sorted List	576
168	Kinetic Convex Hull	578
169	Kinetic Closest Pair via Kinetic Delaunay	580
170	Stochastic PDE Solver Usage	593
171	Wave Kernel Signature	595
172	Randomized Incremental Convex Hull in $\mathbb{R}^d$	601
173	Delaunay Complex in $\mathbb{R}^d$ by Lifting	604
174	Locality-Sensitive Hashing for (r, c) -Near Neighbor	606
175	Johnson–Lindenstrauss Embedding	609
176	Random Projection	613
177	k -Nearest-Neighbor Classification with a kd -Tree	624
178	k -Means via Lloyd Iteration	627
179	DBSCAN	627
180	Perceptron on a Linearly Separable Set	630
181	Learned Range Search	635
182	Streaming k -center (greedy, radius r)	641
183	External merge sort (skeleton)	643

List of Figures

4.1	Graham Scan: the anchor point p_1 (lowest y) sorts all other points by polar angle. The hull vertices (red) are traced in counter-clockwise order.	30
4.2	Convex hull peeling (onion peeling): successive convex hulls are removed, producing nested layers.	38
6.1	Polygon boolean operations: Union $A \cup B$ (center) and Intersection $A \cap B$ (right) of two overlapping rectangles.	56
6.2	Minkowski sum of a triangle A and a unit square B . The result is a pentagon with at most $n + m$ vertices.	62
7.1	Ear clipping: vertex v_3 is an “ear” since triangle (v_2, v_3, v_4) is entirely inside the polygon and contains no other vertices.	78
7.2	Delaunay property: (left) edge AC is illegal because D lies inside the circumcircle of ABC . (right) Flipping to edge BD satisfies the empty circumcircle property.	88
10.1	KD-tree: (left) recursive axis-aligned partitioning of 2D space; (right) the corresponding binary tree structure.	122
10.2	Quadtree: recursive 2D space partitioning. Quadrants with multiple points are subdivided; leaf quadrants contain at most one point.	124
10.3	Hilbert curve construction: each order recursively places four rotated copies in a U-shaped pattern.	143
10.4	Morton curve: the Z-shaped traversal is obtained by interleaving the binary representations of x and y	146
11.1	Voronoi diagram (colored cells) and its Delaunay triangulation dual (dashed gray edges). Each Voronoi vertex is the circumcenter of a Delaunay triangle.	160
13.1	Closest pair: divide-and-conquer splits points along a vertical line. The closest pair may cross the dividing strip.	194
13.2	Bentley–Ottmann sweep line: the vertical line moves left-to-right. The <i>status</i> maintains the vertical order of intersecting segments.	212
14.1	Lower envelope of three functions: the thick black curve is the pointwise minimum $\min\{f_1, f_2, f_3\}$	229
18.1	Welzl’s algorithm: the smallest enclosing circle is defined by at most 3 support points (red).	284
20.1	Bézier surface: a tensor-product surface defined by a control net of points.	296
20.2	1-to-4 triangle refinement: each edge is split at its midpoint, producing four congruent sub-triangles.	305

20.3 Geodesic vs. Euclidean distance on a curved surface. The geodesic (blue) follows the terrain.	336
24.1 Fast Marching Method: the wavefront propagates from the source. Points are categorized as Accepted, Trial, or Far.	404
25.1 Art gallery problem: a comb polygon with $n = 12$ vertices requires $\lfloor 12/3 \rfloor = 4$ guards in the worst case.	418
27.1 Poisson disk sampling: points are placed with minimum distance r between any pair.	430
27.2 Random walk on a grid: at each step, move to a uniformly random neighboring vertex.	443
30.1 A triangle mesh with vertices v_i , edges, and faces f_j . The Euler characteristic $\chi = V - E + F$ is a topological invariant.	478
30.2 (Left) Manifold mesh: every edge is shared by exactly two faces. (Right) Non-manifold: an edge shared by three faces violates the manifold property.	486
30.3 First three eigenfunctions of the Laplacian on a circle. The eigenvalues λ_k form the shape spectrum.	508
30.4 Discrete exterior derivative: d^0 maps vertex values to edge differences; d^1 maps edge values to face circulations.	516
30.5 Hodge decomposition: any 1-form ω splits uniquely into exact, co-exact, and harmonic components.	519
33.1 Circle packing: placing non-overlapping disks of varying radii inside a bounding region.	560
38.1 Walk on Spheres: from each interior point, jump to a random point on the largest inscribed sphere.	592
41.1 Buchberger's algorithm: S-polynomials are formed from pairs of generators and reduced.	652

Part I

Foundations

Chapter 1

Introduction to Computational Geometry

1.1 What Is Computational Geometry?

Computational geometry is the study of algorithms and data structures for problems that are stated in geometric terms: computing the convex hull of a point set, finding the intersection of two polygons, building a triangulation of a terrain, locating a point in a planar subdivision, or planning a motion for a robot that must avoid obstacles. It differs from classical geometry in that its objects of study are not shapes and their properties in the abstract, but the *computational tasks* associated with them, and its answers are not just existence statements but efficient algorithms with provable running times.

The field's objects are as diverse as its applications: points, line segments, polygons, polyhedra, arrangements of lines and planes, Voronoi diagrams, Delaunay triangulations, meshes, and curves and surfaces. Its methods combine discrete mathematics (the combinatorial structure of a geometric object), continuous mathematics (the geometry itself), and algorithm design (the data structures and complexity bounds). This combination is what makes the subject both useful and subtle: a geometric object has a continuous description with infinitely many points, yet its interesting features are captured by a finite combinatorial structure.

This book presents the subject through its algorithms, each developed with definitions, theory, pseudocode, complexity analysis, and edge cases. The chapters are largely self-contained and cross-reference one another where algorithms build on shared foundations. The mathematical vocabulary is collected in Chapter 2; this chapter describes the field itself.

1.2 Geometry as an Algorithmic Problem

The core intellectual step that created the field is the observation that geometric questions have *algorithmic content*: for most geometric objects, the natural computational questions are decision and construction problems whose answers are finite combinatorial structures, and the efficiency of computing those answers depends on clever algorithms rather than on more arithmetic. The convex hull of n points, for example, is a finite polygon whose complexity is at most n in the plane; the algorithmic question is how to compute it from the input points in the best possible time, which turns out to be $\Theta(n \log n)$ for comparison-based algorithms (Chapter 4).

Formulating geometry as algorithms requires three ingredients:

1. a **model of computation** that abstracts away floating-point arithmetic, so that complexity can be discussed rigorously (Section 1.7);

2. a **combinatorial encoding** of the output, so that the “result” of an algorithm is a well-defined object (e.g., a polygon, a planar subdivision, or a triangulation);
3. a **general-position assumption** that rules out degenerate inputs for the purposes of algorithm design, with methods to handle degeneracies when they occur (Section 1.9).

These three ingredients are developed in the next sections and recur throughout the book. They are what distinguish computational geometry from numerical geometry and from computational topology, whose relationships to the subject are described in Section 1.6 and in the chapters on discrete geometry and topology.

1.3 Historical Development

The origins of computational geometry lie in the early 1970s. Several questions were asked at roughly the same time: what is the complexity of computing the convex hull of n points? the closest pair among n points? the intersection of two polygons? Work by Shamos, who coined the name *computational geometry*, and by Preparata established the first algorithms and lower bounds, and the textbook of Preparata and Shamos [PS85] codified the field. Around the same time, Dobkin, Kirkpatrick, and others developed the divide-and-conquer and randomized-incremental techniques that remain central today.

Two developments in the late 1980s transformed the field. First, the discovery of efficient randomized incremental algorithms — Clarkson and Shor, and independently Mulmuley, for convex hulls and arrangements, and Guibas, Knuth, and Sharir for Delaunay triangulations [GKS92] — provided algorithms whose expected running times are optimal while being simple to implement. Second, the treatment of *robustness* became a research subject in its own right, culminating in the exact and adaptive predicate techniques of Section 1.12.

In the 1990s the field expanded in breadth: kinetic data structures, arrangements of surfaces, mesh generation, and the interplay with topology and with computational biology took center stage, and efficient libraries such as CGAL and the algorithms of this book’s CompGeom package brought the theory into practice (Section 1.15). Today computational geometry underlies computer graphics, geographic information systems, robotics, scientific computing, and data analysis, and its randomized and output-sensitive methods have influenced algorithm design well beyond geometry. Historical accounts appear in the bibliographies of [PS85] and [dBCvKO08].

1.4 Geometric Objects and Representations

Geometric objects admit several distinct representations, and choosing the right one is often the difference between an easy and an impossible algorithm. The principal representation families are:

Set-theoretic A shape is a subset of space, described either implicitly by membership tests and inequalities (half-spaces, implicit surfaces) or explicitly by its boundary (a polygon is a sequence of edges; a polyhedron is a collection of faces).

Combinatorial A shape is described by incidence data: vertices, edges, and faces and their adjacency, as in a planar subdivision (Chapter 8), a half-edge structure (Chapter 21), or a simplicial complex (Section 2.15).

Numerical A shape is described by coordinates and parametric equations, as with spline patches and NURBS in surface modeling (Chapter 20).

Point-cloud A shape is represented only by a sample of boundary points, from which connectivity must be reconstructed (Chapter 28).

The same physical object can be represented in several ways, and conversion between representations is itself a computational-geometry problem (rasterization, triangulation, polygonization, meshing). Many of the robustness and correctness issues in the field trace back to representation: a combinatorial structure asserts that two edges meet exactly at a point, while a numerical representation can only approximate that point. This tension between the exact combinatorial ideal and the approximate numerical reality is the subject of Sections 1.6 through 1.12.

1.5 Points, Lines, Segments, Rays, and Polygons

The primitive objects of planar computational geometry are:

- a **point** $p = (x, y)$;
- a **line**, given by a point and a direction, or by an equation $ax + by + c = 0$;
- a **segment** $[p, q]$, the set of convex combinations of its two endpoints (Section 2.3);
- a **ray** $p + tv$, $t \geq 0$;
- a **polygon**, a cyclic sequence of segments whose endpoints are the vertices; a polygon is **simple** if its boundary does not self-intersect.

In $\mathbb{R}^3$ the primitives are points, lines, planes, segments, and **polyhedra** (bounded unions of planar faces), treated in Chapters 19 and 16. These objects and their algebra — orientation tests (Section 2.6), distances, and intersections — are the vocabulary in which every algorithm in this book is written. Many algorithms reduce arbitrary input to these primitives: polygons are decomposed into triangles (Chapter 7), polyhedra into tetrahedra (Chapter 22), and surfaces into point clouds and meshes.

A recurring theme is the difference between an *exact* predicate (does this point lie to the left of that line?) and an *approximate* measurement (how far is it?). Predicates are the discrete decisions that determine combinatorial structure, and they must be answered consistently; measurements can be approximate. This theme is developed in Section 1.10.

1.6 Combinatorial versus Numerical Geometry

Computational geometry sits between two neighboring disciplines. On one side is *numerical* geometry: linear algebra, numerical analysis, and scientific computing, in which coordinates are real numbers, arithmetic is approximate, and results are approximations. On the other side is *combinatorial* and *discrete* geometry: the study of finite configurations of points and their convex hulls, arrangements, and polytopes, in which the interesting structure is the incidence pattern rather than the coordinates.

The combinatorial view is the source of the field's theoretical power. A Delaunay triangulation, for example, is defined by a predicate (the in-circle test, Section 3.3) and is a purely combinatorial object once the predicate values are fixed. Algorithms that compute it can be analyzed on the discrete level, with running times expressed in terms of n , the number of input points, and the output complexity k . The real-RAM model of Section 1.7 formalizes this by treating arithmetic on real numbers as exact and free.

The numerical view is the source of the field's practical difficulty. Floating-point arithmetic is neither exact nor consistent: the same algebraic expression evaluated in a different order can produce a different sign, and a predicate computed approximately can disagree with the exact result on which a combinatorial data structure was built. The resulting inconsistencies produce the classic failures of naive implementations (crashes in point location, invalid triangulations, inconsistent meshes) and motivate the exact-computation and robustness methods

of Sections 1.10–1.12. A practical geometry library — such as the CompGeom package that accompanies this book — must therefore combine combinatorial algorithms with numerically sound predicates.

1.7 The Real-RAM Model

The **real-RAM** (random-access machine) model treats a geometric algorithm as a program over real numbers: each memory cell holds a real number, and each basic operation (addition, multiplication, comparison, square root, accessing an array element) takes constant time and is exact. This model was introduced precisely to analyze geometric algorithms without the distraction of floating-point roundoff, and it is the model assumed throughout this book when discussing running times.

The real-RAM model has a well-known cost: because it can compute on real numbers exactly, it is more powerful than the integer/binary RAM, and some problems (e.g., deciding whether a real number is 0) become free rather than hard. But for the combinatorial questions of computational geometry, the model matches intuition: an algorithm that runs in $O(n \log n)$ comparison operations in the real-RAM also runs in $O(n \log n)$ on a binary machine for rational inputs, and lower bounds proved in the model (e.g., the $\Omega(n \log n)$ bound for convex hulls) carry over to comparison-based models on binary machines. For a rigorous account of the model and its variants see [dBCvKO08] and the references in [PS85].

In this book, pseudocode is written in the real-RAM spirit: it manipulates points and predicates as if they were exact primitives, and the question of how to evaluate a predicate exactly in finite arithmetic is delegated to the robustness machinery of Sections 1.10–1.12 and the implementation notes in the appendices.

1.8 Computational Models and Complexity

Within the real-RAM framework, algorithms are measured by their running time as a function of the input size n and, where relevant, the output size k . The standard notation is used throughout: $O(f(n))$, $\Omega(f(n))$, and $\Theta(f(n))$ for asymptotic upper, lower, and tight bounds, as defined in the notation table of the front matter and reviewed in Appendix A.

Beyond plain worst-case analysis, several refinements recur in the book:

Output-sensitive bounds express the cost in terms of the number k of items actually produced, so that a convex hull of k vertices from n points can be computed in $O(n \log k)$ rather than $O(n \log n)$ (Chapter 4) and a set of k segment intersections in $O((n + k) \log n)$ (Chapter 5).

Randomized bounds replace worst-case over inputs with expectation over the algorithm's own random choices, giving the clean $O(n \log n)$ expected bounds of the randomized incremental algorithms (Chapter 31).

Expected-case bounds average over a distribution of inputs, used for search structures such as the history DAG of Chapter 8.

Approximation relaxes the output requirement and measures the approximation factor, as in the ε -kernels and well-separated pair decompositions of Chapter 32.

A recurring structural fact is that many geometric problems in $\mathbb{R}^2$ are governed by the same combinatorial objects — planar graphs, arrangements, Voronoi diagrams — so their complexity bounds are tied together; the planar graphs and their Euler-characteristic bound (Section 2.15) explain why so many planar algorithms run in $O(n \log n)$.

1.9 Degeneracies and General Position

An input is **degenerate** when it violates the assumptions that an algorithm’s design (and analysis) silently makes. Typical planar degeneracies include three collinear points, four cocircular points, two coincident points, and vertical lines in contexts where a comparison by x -coordinate is used. Most algorithms in the classical theory are stated for inputs in **general position**: no three points collinear, no four cocircular, no two points sharing an x -coordinate, no point lying on a critical line. The general-position assumption is not a detail: it is what makes the algorithm’s description (e.g., “every vertex of a Delaunay triangulation has degree bounded by the number of cocircular points”) correct and its analysis clean.

Degeneracies matter in practice because they do occur — coordinates come from measurements and from snapping, and collinearity and cocircularity are common rather than rare. The standard strategies are:

- **Symbolic perturbation**: perturb the input infinitesimally so that all predicates have a definite sign, without changing the answer of the intended problem (Section 1.11).
- **Exact predicates**: evaluate the degeneracy test exactly and resolve ties by a rule, so the algorithm makes consistent choices even on degenerate input (Section 1.12).
- **Explicit handling**: design the algorithm to process degenerate configurations directly, as when convex-hull algorithms remove collinear points and segment-intersection algorithms handle overlaps (Chapters 4 and 5).

Each chapter of this book states its general-position assumptions and describes how its edge cases (collinear, coincident, cocircular, vertical, parallel) are treated; the testing sections document the corresponding test inputs.

1.10 Robustness and Numerical Precision

When an algorithm designed under general position and exact real arithmetic is implemented with double-precision floats, two kinds of failure occur. The first is *sign error*: a predicate whose exact value is a tiny positive number is evaluated as negative, flipping a decision that changes the combinatorial output — an invalid triangulation, a point located on the wrong side of an edge, a polygon whose boundary self-intersects. The second is *inconsistency*: different evaluations of equivalent predicates disagree, so the data structure built by the algorithm is not consistent with any exact computation, and downstream operations (e.g., point location) fail on it.

These failures are not cosmetic. In geometric modeling they produce non-watertight meshes; in CAD and simulation they produce invalid intersections; in collision detection they produce missed collisions. The chapter on geometric predicates (Chapter 3) develops the two standard remedies:

Adaptive predicates compute the predicate’s value first in floating point, then refine only when the result is provably uncertain; for inputs in general position the fast path succeeds almost always, so the cost is nearly that of a single floating-point evaluation [She97].

Exact computation evaluates the predicate exactly with arbitrary precision, guaranteeing the same sign as the exact arithmetic in all cases (Section 1.12).

Robustness has an engineering dimension as well: predicates must be evaluated identically wherever they appear (no duplicated, subtly different expressions), coordinate transformations must not change predicate signs, and tolerances (epsilon comparisons) should be used only where a geometric tolerance is genuinely intended, never as a substitute for exact predicate evaluation. These rules are followed by the implementations referenced in this book and are discussed in Appendix D.

1.11 Symbolic Perturbation

Symbolic perturbation resolves degeneracies without changing the input’s coordinates: instead of perturbing the points numerically, the algorithm evaluates predicates as if the points had been perturbed by an infinitesimal amount in a fixed direction, chosen so that every predicate has a definite value while all decisions that matter are unchanged. The classic scheme perturbs each point by ε raised to a distinguished power so that the perturbed configuration is in general position and the ordering induced by the perturbation is a total order that can be evaluated symbolically [Ede85].

Symbolic perturbation turns a degenerate input into a non-degenerate one at the predicate level, which lets an algorithm designed under general position run on arbitrary input without modification. Its main limitation is conceptual: some problems (for example, computing the arrangement of overlapping segments, or intersecting polygons that share edges) have legitimate answers that depend on how the degeneracy is resolved, and a blind perturbation can produce a different answer than the caller intended. For this reason the book’s chapters prefer explicit, documented handling of degenerate cases for the core data structures (triangulations, planar subdivisions, convex hulls), and mention symbolic perturbation as a technique where it is most natural, e.g. for point location and for the randomized incremental algorithms of Chapter 31.

1.12 Exact Geometric Computation

The **exact computation paradigm** guarantees that every predicate is evaluated with the same sign as the exact mathematical value, so the combinatorial output of an algorithm is exactly the output it would produce in the real-RAM model. This is achievable in principle for the predicates used in this book because they are determinants and rational combinations of the input coordinates, which can be evaluated exactly with multiprecision integer or rational arithmetic; in particular the orientation and in-circle predicates of Chapter 3 are signs of determinants, and their exact values can be computed by exact integer arithmetic on scaled coordinates.

Purely exact computation is often too slow to be the default, and the practical standard is the *adaptive* hybrid: evaluate the predicate in floating point, certify the sign with error bounds when the value is not too close to zero, and only fall back to exact arithmetic when the sign is uncertain. Shewchuk’s adaptive predicates [She97] show that for random (hence non-degenerate) inputs the fallback is exercised so rarely that the adaptive test is nearly as fast as the floating-point one. The geometry libraries discussed in Section 1.15 and the implementations accompanying this book follow this design.

The exact-computation approach has a strong correctness payoff: algorithms whose inputs are exact (integer or rational) coordinates, and whose outputs are combinatorial (triangulations, hulls, subdivisions), can be made fully deterministic and reproducible, which is a requirement for geometry used in scientific computing, manufacturing, and computer-aided design. When input coordinates are already the result of approximation (measurement, sensor data), exact predicates still provide the consistency guarantees that keep downstream combinatorial structures valid.

1.13 Major Families of Geometric Algorithms

The algorithms of this book fall into a few broad families that organize the parts of the book:

Fundamental predicates and constructions — orientation, in-circle, and other predicates (Chapter 3), convex hulls (Chapter 4), segment intersection (Chapter 5), and the whole polygon-operations suite (Chapter 6).

Spatial structures and searching — point location (Chapter 8), range searching (Chapter 9), spatial data structures (Chapter 10), and space-filling curves (Chapter 10.10).

Proximity and tessellations — Voronoi diagrams (Chapter 11), Delaunay triangulations (Chapter 12), and nearest-neighbor problems (Chapter 13).

Arrangements, duality, and optimization — line arrangements and envelopes (Chapter 14), point-line duality (Chapter 15), linear programming (Chapter 17), and convex intersection (Chapter 18).

Meshes, curves, and surfaces — triangulations (Chapter 7), polyhedral geometry (Chapter 19), curves and surfaces (Chapter 20), mesh generation (Chapter 22), and mesh analysis (Chapter 30.1).

Motion, visibility, and planning — visibility (Chapter 24), shortest paths (Chapter 24.11), and motion planning (Chapter 26).

Sampling, reconstruction, and matching — sampling and Poisson-disk methods (Chapter 27), surface reconstruction (Chapter 28), and shape matching (Chapter 29).

Topology and high dimension — computational topology (Chapter 30), randomized and approximation algorithms (Chapters 31 and 32), and high-dimensional and data-driven geometry (Chapters 39 and 40).

Each chapter follows the book’s template: overview, definitions, theory, pseudocode, parameter selection, complexity, applications, testing, and references, as described in the front matter.

1.14 Applications of Computational Geometry

Computational geometry is a service field: most of its algorithms exist because some application needs them. The application chapters of this book cover four domains in depth:

Computer graphics (Chapter 35) — rendering (hidden-surface removal, rasterization, ray tracing with spatial hierarchies), modeling (boolean operations, offsetting), and animation (collision detection, motion);

Geographic information systems (Chapter 36) — terrain modeling and TINs, map overlay, point-in-polygon queries, site selection, and spatial search;

Robotics (Chapter 37) — configuration spaces, collision detection, path planning, and manipulation;

Scientific computing (Chapter 38) — finite-element meshing, computational fluid dynamics, molecular and material modeling, and protein geometry.

Beyond these, geometric algorithms appear in computer-aided design and manufacturing (Chapter 33 covers packing and enclosure problems), in data analysis through nearest-neighbor and clustering methods (Chapter 13), in visualization, in geometric optics, and in many other settings. Each application chapter closes with the concrete product-level context in which the algorithms are deployed, so that the reader can see both the mathematics and the engineering that surrounds it.

1.15 Software Libraries and Experimental Geometry

Computational geometry is unusual among theoretical computer-science subjects in that its algorithms are routinely implemented, distributed, and used in production systems. The principal general-purpose libraries are CGAL (the Computational Geometry Algorithms Library), which provides industrial-grade implementations of most algorithms in this book with exact and adaptive predicates; the LEDA and more recent C++ libraries; and the Python ecosystem, including the CompGeom package that accompanies this book, whose API mirrors the algorithms described here. In addition, specialized libraries cover specific areas: Triangle and TetGen for meshing (Chapter 22), CGAL and the “delabella” and other Delaunay codes for triangulations (Chapter 12), and numerical packages such as scipy for lower-level operations (Appendix E).

Two practical lessons recur across these libraries and shape the book’s implementation advice. First, the robustness machinery of Sections 1.10–1.12 is not optional: production libraries invest heavily in exact predicates because their users depend on combinatorial validity. Second, performance is dominated by predicate evaluation and memory layout, so engineering details (predicate caching, branch-free evaluation, index-based data structures) matter as much as asymptotic bounds. The appendices of this book (Appendix D and Appendix E) collect the implementation-level guidance and library references, and experimental evaluation — comparing algorithms on real data, measuring predicate call counts and cache behavior — is a theme of the chapter testing sections.

1.16 Chapter Problems

1. Give three geometric problems and, for each, identify the combinatorial object that encodes its output.
2. Explain why the real-RAM model is convenient for the analysis of geometric algorithms, and state one setting in which the model gives a misleading picture.
3. Describe a degenerate input for (a) a convex-hull algorithm, (b) a Delaunay-triangulation algorithm, and (c) a point-location algorithm.
4. Give an example of a sign error that a floating-point evaluation of the orientation predicate can produce, and explain how adaptive evaluation avoids it.
5. Distinguish symbolic perturbation from exact predicates, and give a situation where the two lead to different outputs on the same input.
6. List the four object-representation families of Section 1.4 and, for each, name one algorithm in this book that operates on that representation.
7. Explain why an output-sensitive bound is preferable to a worst-case bound for reporting all intersections of many segments.
8. Identify, for each of the four application chapters, the two or three geometric problems that are most central to that domain.

1.17 Computational Projects

1. Implement the orientation predicate in floating point and with exact integer arithmetic, and compare their behavior on a set of nearly collinear inputs.
2. Build a small polygon library: read a simple polygon, compute its signed area (Section 2.7), and test point-in-polygon membership.

3. Implement a brute-force closest-pair algorithm and a divide-and-conquer version, and measure the crossover size on random point sets (compare with Chapter 13).
4. Generate a set of degenerate inputs (collinear, cocircular, coincident points) and record how a naive Delaunay implementation fails on them; then repeat with an exact in-circle predicate.
5. Use a geometric library of your choice to compute the convex hull and Delaunay triangulation of a public point dataset, and visualize both.
6. Construct a terrain (2.5-D) model from a point cloud and compare the natural-neighbor and nearest-neighbor interpolants on a regular grid (Chapters 11 and 27).

Chapter 2

Mathematical Foundations

This chapter collects the mathematical vocabulary used throughout the book: affine and vector spaces, inner products and metrics, lines and hyperplanes, convexity, orientation, barycentric coordinates, duality, and the topological notions needed to reason about meshes and complexes. It is not a full course in any of these subjects; rather, it fixes notation and states the facts that the algorithmic chapters rely on. Readers familiar with linear algebra and point-set topology can skip directly to the chapter bibliographic notes. References such as Matoušek’s *Lectures on Discrete Geometry* [Mat99] and de Berg et al.’s *Computational Geometry: Algorithms and Applications* [dBCvKO08] provide more complete treatments.

2.1 Vectors and Affine Spaces

Computational geometry is mostly carried out in $\mathbb{R}^d$, but it is useful to keep the distinction between *points* and *vectors*. A **vector** is an element of a vector space: it has a magnitude and a direction and can be added, scaled, and compared. A **point** is a location in space; it cannot be added to another point in a coordinate-free way. The two are related through the translation operation: the vector v carries a point p to the point $p + v$.

Definition 2.1 (Vector Space). A **vector space** over the field $\mathbb{R}$ is a set V together with a zero element $\mathbf{0}$, a vector addition $u + v$, and a scalar multiplication λv satisfying the standard axioms: associativity and commutativity of addition, distributivity, existence of additive inverses, and compatibility of scalar multiplication with multiplication in $\mathbb{R}$. A **linear combination** of vectors $v_1, \dots, v_k$ is $\lambda_1 v_1 + \dots + \lambda_k v_k$ with $\lambda_i \in \mathbb{R}$.

Definition 2.2 (Affine Space). An **affine space** over $\mathbb{R}$ is a set A of *points* together with a vector space V of *translations* and a mapping $(p, v) \mapsto p + v$ such that $p + \mathbf{0} = p$, $(p + u) + v = p + (u + v)$, and for every pair of points p, q there is a unique vector v with $q = p + v$, written $v = q - p$.

The distinction matters for a practical reason: every algorithm in this book that computes with coordinates implicitly chooses an origin. Results that are independent of that choice are called **affine-invariant**; convexity, collinearity, ratios of collinear lengths, and barycentric coordinates are affine-invariant, while absolute positions and lengths are not. Choosing a basis identifies the affine space $\mathbb{R}^d$ with d -tuples of reals, which is how all the pseudocode in this book manipulates geometry.

The **affine hull** of a set $S \subseteq A$ is the smallest affine subspace containing S . The affine hull of three non-collinear points is a plane, of four non-coplanar points is $\mathbb{R}^3$, and more generally of $d + 1$ affinely independent points in $\mathbb{R}^d$ is all of $\mathbb{R}^d$. This notion is developed further in Section 2.4. Point sets whose affine hull is low-dimensional (for example, collinear points in the plane) are called **degenerate** for algorithms that assume full dimension; the topic is treated in Section 1.9.

Example 2.3. In $\mathbb{R}^2$, the affine hull of two distinct points is the entire line through them, and the affine hull of any three non-collinear points is the whole plane. Algorithms that build, say, planar triangulations require the affine hull to be the plane; the Delaunay algorithm of Chapter 12 therefore assumes general position, perturbing degenerate inputs if necessary.

2.2 Inner Products and Euclidean Distance

The standard Euclidean structure on $\mathbb{R}^d$ comes from the **dot product** (inner product)

$$\langle u, v \rangle = \sum_{i=1}^d u_i v_i,$$

which induces the Euclidean norm $\|v\| = \sqrt{\langle v, v \rangle}$ and the Euclidean distance $\text{dist}(p, q) = \|p - q\|$.

Theorem 2.4 (Cauchy–Schwarz Inequality). *For all vectors $u, v \in \mathbb{R}^d$, $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$, with equality if and only if u and v are scalar multiples of one another.*

Cauchy–Schwarz implies the triangle inequality $\|u + v\| \leq \|u\| + \|v\|$, which is exactly the requirement that the Euclidean norm define a metric (Section 2.11). It also gives the cosine rule

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

for the angle θ between nonzero vectors, so the dot product lets us test orthogonality: $u \perp v$ iff $\langle u, v \rangle = 0$.

The dot product is the central computational primitive of the geometry algorithms in this book. Predicates such as orientation and in-circle tests (Section 3.1) are determinants built from dot products; projections onto lines and planes are dot products followed by scaling; and the closest-point queries of Chapter 24.11 all reduce to evaluating inner products. Because floating-point arithmetic can make a nearly zero inner product change sign, the robust evaluation methods of Section 3.4 are essential when a dot product determines a topological decision.

2.3 Lines and Hyperplanes

A **line** through a point p with direction $v \neq \mathbf{0}$ has the parametric form $\ell(t) = p + tv$, $t \in \mathbb{R}$. The line through two distinct points p and q is $\ell(t) = (1 - t)p + tq$. A **segment** restricts the parameter: $[p, q] = \{(1 - t)p + tq : 0 \leq t \leq 1\}$.

A **hyperplane** in $\mathbb{R}^d$ is the set of points x satisfying a nontrivial linear equation

$$H = \{x \in \mathbb{R}^d : \langle n, x \rangle = c\},$$

where $n \neq \mathbf{0}$ is a **normal vector** and c is a scalar. In $\mathbb{R}^2$ a hyperplane is a line; in $\mathbb{R}^3$ it is a plane. The normal n is unique up to nonzero scaling, and the unit normal $n/\|n\|$ is the direction perpendicular to the hyperplane.

The signed distance from a point x to the hyperplane H is

$$\delta(x) = \frac{\langle n, x \rangle - c}{\|n\|},$$

which is positive on one side of H and negative on the other. This signed distance is the basic building block for many predicates and proximity queries: the distance from a point to a line in $\mathbb{R}^2$ is $|\delta(x)|$, and the distance between a point and a plane in $\mathbb{R}^3$ is the same expression. Hyperplanes divide space into two **half-spaces** (Section 2.9), and every convex polyhedron can be written as the intersection of finitely many half-spaces, one per facet.

The point-to-line and point-to-plane distance formulas are used throughout Chapter 24.11; the sign of δ is the raw material of the orientation predicates of Section 3.1.

2.4 Affine Combinations

Definition 2.5 (Affine Combination). A point $x = \sum_{i=1}^k \lambda_i p_i$ is an **affine combination** of the points $p_1, \dots, p_k$ if $\sum_{i=1}^k \lambda_i = 1$.

The condition $\sum \lambda_i = 1$ is exactly what makes the combination independent of the choice of origin: translating all p_i by v moves x by $(\sum \lambda_i)v = v$. Affine combinations are closed under taking further affine combinations, so the set of all affine combinations of a point set is its affine hull (Section 2.1).

A set of points $p_1, \dots, p_k$ is **affinely independent** if no point is an affine combination of the others; equivalently, the vectors $p_2 - p_1, \dots, p_k - p_1$ are linearly independent. In $\mathbb{R}^d$ at most $d + 1$ points are affinely independent. An **affine frame** is an affinely independent set of $d + 1$ points $p_0, \dots, p_d$ in $\mathbb{R}^d$; every point has a unique representation as an affine combination of the frame, whose coefficients are the **barycentric coordinates** of Section 2.8.

Affine independence is the coordinate-level meaning of “general position” in Section 1.9: planar algorithms typically assume that no three points are collinear and no four points are concyclic, which is equivalent to all triples (resp. quadruples) being affinely independent (resp. affinely and spherically independent). The collinearity test itself is the orientation determinant of Section 2.6.

2.5 Convex Combinations

Definition 2.6 (Convex Combination and Convex Hull). An affine combination $x = \sum_{i=1}^k \lambda_i p_i$ is a **convex combination** if in addition $\lambda_i \geq 0$ for all i . A set C is **convex** if it contains every convex combination of its points. The **convex hull** $\text{conv}(S)$ of a set S is the set of all convex combinations of points of S ; it is the smallest convex set containing S .

Convex sets are closed under intersection and under Minkowski sums (Section 2.10 treats one of their most important structural properties). The convex hull of a finite point set is a convex polytope whose vertices form a subset of the input points; computing it is Chapter 4. The **vertices** of $\text{conv}(S)$ are exactly those points of S that are not convex combinations of the others.

Theorem 2.7 (Carathéodory). *In $\mathbb{R}^d$, every point of the convex hull of a set S is a convex combination of at most $d + 1$ points of S .*

Carathéodory’s theorem justifies the “small certificates” that appear in many algorithms: to check whether a point lies in the convex hull it suffices to search for a small subset witnessing the inclusion, which is the basis of the separation-based membership tests in Chapter 4 and Section 2.9. The proof is the standard one: write the point with the minimal number k of terms, and if $k > d + 1$ the difference vectors are linearly dependent, so one can perturb the coefficients to zero one of them while preserving convexity.

2.6 Orientation and Determinants

The **orientation** of an ordered point triple (p, q, r) in $\mathbb{R}^2$ records whether r lies to the left of, on, or to the right of the directed line from p to q . It is the sign of the determinant

$$\text{orient}(p, q, r) = \det \begin{pmatrix} 1 & 1 & 1 \\ p_x & q_x & r_x \\ p_y & q_y & r_y \end{pmatrix} = (q_x - p_x)(r_y - p_y) - (q_y - p_y)(r_x - p_x),$$

which equals $2 \cdot (\text{signed area})$ of the triangle $\triangle pqr$ (Section 2.7). The value is positive when the turning from $p \rightarrow q$ to $p \rightarrow r$ is counterclockwise, negative when clockwise, and zero exactly when the three points are collinear.

In $\mathbb{R}^3$, the orientation of an ordered quadruple (p, q, r, s) is the sign of the 4×4 determinant with rows $(1, p_x, p_y, p_z)$, etc., and records whether s lies above or below the plane through p, q, r .

Orientation is the fundamental **predicate** of computational geometry: it decides convexity of hulls, the validity of triangulations, point-in-polygon tests, segment intersection, and the legality of edge flips in the Delaunay construction (Chapter 12). Because the sign of a nearly zero determinant is unstable in floating point, production implementations use the adaptive exact evaluation described in Section 3.4. The chapter on geometric predicates (Section 3.1) develops these tests in detail.

Remark 2.8. Orientation tests can be interpreted as the sign of a $d + 1$ -dimensional determinant built from homogeneous coordinates, which unifies the 2D and 3D cases: see Section 2.13 for the coordinate transform and Section 2.6 references for the predicate literature.

2.7 Signed Area and Signed Volume

The **signed area** of an ordered polygon $p_1, \dots, p_n$ in the plane is given by the shoelace formula

$$A = \frac{1}{2} \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i),$$

with indices taken modulo n . The area is positive if the boundary is oriented counterclockwise and negative if clockwise; for a simple polygon the absolute value is the usual area. The shoelace formula underlies the area computations in Chapter 6 and the planarity/orientation fixes used in polygon repair.

In $\mathbb{R}^3$, the **signed volume** of a tetrahedron with ordered vertices p, q, r, s is $\frac{1}{6}$ times the scalar triple product

$$V = \frac{1}{6} \langle (q - p) \times (r - p), s - p \rangle,$$

which is $\frac{1}{6}$ of the orientation determinant of Section 2.6. The signed volume of a polyhedron is obtained by summing the signed volumes of tetrahedra formed with a fixed apex; this is the divergence-theorem (vector-calculus) way to compute volumes of arbitrary closed meshes, used in Chapter 19 for volume, center of mass, and moments.

Signed quantities are more than a convenience: they determine sidedness and orientation robustly and allow “signed” triangles to cancel outside a region, which is exactly how triangulated surface areas and enclosed volumes are computed without explicit inside/outside tests. The sign convention must be fixed consistently; the book uses counterclockwise (positive) polygons in the plane and positively oriented (right-handed) tetrahedra in space.

2.8 Barycentric Coordinates

Let $p_0, \dots, p_d$ be an affine frame of $\mathbb{R}^d$ (Section 2.4). Every point x has unique coefficients $(\lambda_0, \dots, \lambda_d)$ with $\sum_i \lambda_i = 1$ and $x = \sum_i \lambda_i p_i$, called the **barycentric coordinates** of x with respect to the frame.

For a triangle $\triangle ABC$ in the plane, the barycentric coordinates are the (signed) area ratios

$$\lambda_A = \frac{A(\triangle xBC)}{A(\triangle ABC)}, \quad \lambda_B = \frac{A(\triangle xCA)}{A(\triangle ABC)}, \quad \lambda_C = \frac{A(\triangle xAB)}{A(\triangle ABC)},$$

where $A(\cdot)$ is the signed area of Section 2.7. The point x lies inside the triangle iff all three coordinates are nonnegative; on the boundary iff one is zero; and outside iff some coordinate

is negative. This yields a robust point-in-triangle test built from three orientation evaluations, and it gives the linear interpolation formula used for color, texture, and data interpolation on meshes.

In a tetrahedron $p_0p_1p_2p_3$, the barycentric coordinates are the ratios of the four signed volumes

$$\lambda_i = \frac{V(\triangle xp_jp_kp_l)}{V(\triangle p_0p_1p_2p_3)},$$

where $\{i, j, k, l\}$ is a permutation of $\{0, 1, 2, 3\}$. Barycentric coordinates are the foundation of the linear finite-element basis on simplices used throughout Chapter 30.1 and Chapter 27; generalized (non-affine) coordinates for arbitrary polygons are used in the mesh and parameterization literature.

2.9 Half-Spaces

For a hyperplane $H = \{x : \langle n, x \rangle = c\}$, the two associated closed **half-spaces** are

$$H^+ = \{x : \langle n, x \rangle \geq c\}, \quad H^- = \{x : \langle n, x \rangle \leq c\},$$

and the open half-spaces are defined by strict inequalities. Every half-space is convex, and the intersection of any family of half-spaces is convex. The convex hull of Section 2.5 is related to half-spaces by a separation result: a convex set is the intersection of all half-spaces that contain it.

Theorem 2.9 (Hyperplane Separation). *Two disjoint nonempty convex sets A and B in $\mathbb{R}^d$ with one of them compact are strictly separated: there is a hyperplane $H = \{x : \langle n, x \rangle = c\}$ with $\langle n, a \rangle < c < \langle n, b \rangle$ for all $a \in A$, $b \in B$.*

The half-space representation is used pervasively:

- a convex polygon in the plane is the intersection of the half-spaces of its directed edges (each edge is traversed so that the polygon lies to the left);
- a convex polyhedron in $\mathbb{R}^3$ is the intersection of the half-spaces of its facets, one per face;
- the feasible region of a linear program is an intersection of half-spaces (Chapter 17), and the linear programming algorithms solve exactly the question of whether such an intersection is empty;
- point-in-convex-polygon and polygon-clipping algorithms test membership by checking the side of each edge half-space.

Because half-space intersection is how convex objects are stored, the boolean operations on convex solids of Chapter 18 reduce to the algebra of half-spaces and their boundary facets.

2.10 Hyperplanes and Separating Surfaces

Theorem 2.9 (Hyperplane Separation) is the one-dimensional separation that computational geometry uses constantly. In $\mathbb{R}^2$, two disjoint compact convex sets are separated by a line that can be moved until it is **supporting** both sets simultaneously, touching each in a point or a segment. Such a line is a **common supporting line** and exists exactly when the two sets are disjoint.

For convex polygons and convex polyhedra, separation can be decided algorithmically: two convex polygons in the plane are disjoint iff there is a line parallel to one of their edges that separates them, and two convex polyhedra in $\mathbb{R}^3$ are disjoint iff there is a separating plane

parallel to one of their faces, or a plane perpendicular to a cross product of a pair of edges. This is the **separating axis theorem**, the basis of the collision and intersection tests in Chapter 18 and of the broad-phase narrowing in motion planning (Chapter 26).

The Minkowski sum $A \oplus B = \{a + b : a \in A, b \in B\}$ reformulates separation: A and B are disjoint iff $\mathbf{0} \notin A \oplus (-B)$. Convexity is preserved by Minkowski sums, so the distance between two convex sets is the distance from the origin to the Minkowski difference, which is what the GJK distance algorithm computes (Chapter 18). This single equivalence is the geometric engine behind most collision-detection code in computer graphics and robotics.

2.11 Norms and Metrics

A **norm** on $\mathbb{R}^d$ is a function $\|\cdot\| \rightarrow [0, \infty)$ that is positive definite ($\|x\| = 0$ iff $x = \mathbf{0}$), homogeneous ($\|\lambda x\| = |\lambda| \|x\|$), and satisfies the triangle inequality. The standard family is the ℓ_p family

$$\|x\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}, \quad \|x\|_\infty = \max_i |x_i|,$$

with the Euclidean norm $\|\cdot\|_2$ from Section 2.2 as the $p = 2$ member. A **metric** on a set is a distance function that is positive definite, symmetric, and satisfies the triangle inequality; every norm induces a metric via $d(x, y) = \|x - y\|$.

Distances other than Euclidean arise naturally in geometry algorithms:

- the ℓ_1 (*Manhattan*) and ℓ_∞ (*Chebyshev*) metrics describe grid and axis-aligned motion and appear in L-shaped/rectilinear path problems (Chapter 24.11) and in the geometry of space-filling curves (Chapter 10.10);
- the **geodesic distance** on a surface is the length of the shortest path constrained to the surface, a non-Euclidean metric central to Chapters 27 and 30.1;
- in high dimension, ℓ_1 distances are often more stable than ℓ_2 for nearest-neighbor search, a phenomenon studied in Chapter 39.

The choice of metric changes which structures are “natural”: the Voronoi diagram and Delaunay triangulation (Chapters 11 and 12) are metric-dependent, and replacing the Euclidean metric with an ℓ_p metric changes the combinatorial complexity of the diagram. Metric properties also drive the correctness of nearest-neighbor algorithms in Chapter 13, which assume at minimum the triangle inequality.

2.12 Transformations

An **isometry** (rigid motion) of $\mathbb{R}^d$ preserves distances and is a composition of translations, rotations, and reflections. A **similarity** additionally permits uniform scaling, preserving angles and ratios of lengths. An **affine transformation** $x \mapsto Mx + t$, with M an invertible $d \times d$ matrix and t a translation vector, preserves collinearity, parallelism, ratios along lines, and affine combinations, but not lengths or angles in general. A **projective transformation** is the most general transformation that maps lines to lines; it preserves incidence and cross-ratios but not parallelism, and it is naturally expressed in homogeneous coordinates (Section 2.13).

	lines	parallelism	distances	angles
rigid	yes	yes	yes	yes
similarity	yes	yes	scaled	yes
affine	yes	yes	no	no
projective	yes	no	no	no

The properties preserved by each class determine which algorithms are invariant under which changes of coordinates. Clipping, convex hulls, and barycentric interpolation are affine-invariant; Voronoi diagrams and Delaunay triangulations are not (uniform scaling is harmless, anisotropic scaling breaks them). Applications in graphics and computer vision rely on the full projective group (Chapter 35); mesh algorithms typically use isometries and similarities (Chapters 30.1).

Rigid transformations are built from an orthonormal basis: a rotation matrix R satisfies $R^\top R = I$ and $\det R = 1$. Computationally, applying a rigid transformation to a set of points preserves every computed invariant of distance-based algorithms, which is why point-cloud registration in Chapter 29 searches over rigid motions only.

2.13 Homogeneous Coordinates

A point $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ is represented in **homogeneous coordinates** by any nonzero vector $(\tilde{x}_1, \dots, \tilde{x}_d, w)$ with $x_i = \tilde{x}_i/w$; the canonical choice is $(x_1, \dots, x_d, 1)$. Two homogeneous vectors represent the same point iff they are scalar multiples of one another, so the projective space is the quotient of $\mathbb{R}^{d+1} \setminus \{0\}$ by scaling. Points with $w = 0$ are **points at infinity**, one per direction.

The payoff is that translations become linear: the affine map $x \mapsto Mx + t$ is the linear map

$$\begin{pmatrix} \tilde{x} \\ w \end{pmatrix} \mapsto \begin{pmatrix} M & t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x} \\ w \end{pmatrix}.$$

Composing affine transformations is then ordinary matrix multiplication, and the composition of projective transformations is linear throughout (Section 2.12). In computer graphics this is why model, view, and projection matrices are kept as 4×4 matrices and why the perspective divide $x_i = \tilde{x}_i/w$ is a single final division.

Homogeneous coordinates also unify the orientation determinant: the d -dimensional orientation test of Section 2.6 is the determinant of the $(d+1) \times (d+1)$ matrix whose rows are the homogeneous vectors of the points. This is the algebraic reason the same predicate code scales from 2D to 3D, and the reason the in-circle test of Section 3.3 has a determinant of the same form. Hyperplane duality (Section 2.14) is also most simply stated in homogeneous coordinates, where points and hyperplanes are dual vectors.

2.14 Duality

Duality is a correspondence that swaps points with hyperplanes (or lines in the plane) while preserving incidence and order. In the plane, the standard **dual transform** maps the point $p = (a, b)$ to the line

$$p^* : y = ax - b,$$

and maps the line $\ell : y = mx - c$ to the point $\ell^* = (m, c)$. The transform is an involution up to the sign convention, preserves above/below relations, and maps vertical lines to points at infinity (Section 2.13). Its defining property is incidence preservation: p lies on ℓ iff ℓ^* lies on p^* .

Because convex hulls, intersections, and arrangements are transformed into dual objects with the same combinatorial structure, duality converts problems into easier ones. The convex hull of n points in the plane is dual to the intersection of the n upper half-planes below their dual lines; envelopes and lower envelopes (Chapter 14) are the dual picture of convex hulls; and the Delaunay triangulation is dual to the arrangement of lifted tangent planes (Section 12.8). Chapter 15 develops the planar theory, including the half-plane intersection formulation and its use in the randomized incremental algorithms.

In $\mathbb{R}^d$, the polar transform maps the point $a = (a_1, \dots, a_d)$ to the hyperplane $\{x : \langle a, x \rangle = 1\}$, and $0 \in \text{conv}(S)$ iff the corresponding half-spaces have empty intersection; this is the polarity

that underlies the convex-hull algorithms for higher dimensions and the certificates used in Chapter 17. All duality transforms preserve the ordering of points along a line through the relation between convex hull and half-space intersection, so the complexity results for hulls transfer directly to envelopes and back.

2.15 Topological Preliminaries

Computational geometry works mostly with finite combinatorial objects, but their global structure is topological, and a small vocabulary goes a long way. A set in $\mathbb{R}^d$ is **open** if every point has a small ball around it contained in the set, and **closed** if its complement is open; **compact** sets are closed and bounded. A set is **connected** if it cannot be split into two disjoint nonempty open pieces, and **path-connected** if any two points are joined by a continuous curve. A **homeomorphism** is a continuous bijection with continuous inverse; two sets are homeomorphic if one exists. The **Euler characteristic** of a 2D polyhedral surface is $V - E + F$, and for an orientable closed surface of genus g it equals $2 - 2g$; the same invariant in higher dimension is the alternating sum of the numbers of faces.

A **simplicial complex** is a finite collection K of simplices such that every face of a simplex in K is in K and the intersection of any two simplices is a face of both. A triangulated surface or mesh is a simplicial complex in this sense, and all the mesh algorithms of Chapters 30.1, 30, and 21 operate on the combinatorial structure of such a complex. The **underlying space** $|K|$ is the topological space formed by the union of the simplices; two complexes that triangulate the same space are homeomorphic.

The **manifold** concept lets us describe when a mesh is locally simple: a 2D surface is a manifold if every point has a neighborhood homeomorphic to a disk (or to a half-disk on the boundary). Manifold verification, orientation fixing, and genus computation are the subjects of the mesh-analysis chapter; the half-edge and half-face data structures of Chapter 21 are designed exactly to represent manifold meshes, and the discrete exterior calculus and cohomology algorithms of Chapter 30 compute the Betti numbers $\beta_0, \beta_1, \dots$ of such complexes. These foundations are summarized in the notation table of the front matter and developed in the appendix (Appendix A).

2.16 Exercises

1. Show that the set of affine combinations of a point set S is the smallest affine subspace containing S , and that convex combinations give the smallest convex set.
2. Prove that three points p, q, r are collinear iff the orientation determinant of Section 2.6 vanishes.
3. Derive the shoelace formula by decomposing a polygon into triangles from a fixed vertex and canceling signed areas.
4. Compute the barycentric coordinates of the incenter of a triangle in terms of side lengths, and verify that the three coordinates are nonnegative.
5. Prove that a point lies in a convex polygon iff it is on the left side of every directed edge (for a counterclockwise boundary).
6. Show that the intersection of the half-spaces of a convex polygon equals the polygon.
7. For two axis-aligned boxes, give a condition for disjointness using the separating axis theorem of Section 2.10.

8. Prove that the ℓ_1 , ℓ_2 , and ℓ_∞ norms are equivalent in the sense that each bounds the others up to constant factors, and give the constants for $\mathbb{R}^2$.
9. Verify that the point–line duality $p = (a, b) \mapsto y = ax - b$ of Section 2.14 preserves the above/below relation.
10. Compute the Euler characteristic of a triangulated tetrahedron and of a closed cube surface, and confirm $V - E + F = 2$.
11. Show that the two homogeneous vectors $(x_1, x_2, 1)$ and $(\lambda x_1, \lambda x_2, \lambda)$ represent the same point.
12. Prove that a rigid transformation preserves the distances of a point set, and that an arbitrary affine map need not.

Chapter 3

Geometric Predicates and Robust Computation

Geometric predicates are exact-sign tests that determine the combinatorial relationship between geometric objects. They are the foundation of the point, line, triangle, quadrilateral, tetrahedral, and hexahedral algorithms that follow. A predicate should return a discrete result such as positive, zero, or negative, rather than a constructed geometric object.

3.1 Orientation Predicate

For points $A, B, C \in \mathbb{R}^2$, define

$$\text{orient2d}(A, B, C) = (B_x - A_x)(C_y - A_y) - (B_y - A_y)(C_x - A_x).$$

The sign is positive when C lies to the left of the directed line $A \rightarrow B$, negative when it lies to the right, and zero when the three points are collinear. This test is used by convex hull, intersection, point-location, and triangulation algorithms.

Algorithm 1 Two-dimensional orientation test

```
1: function ORIENT2D( $A, B, C$ )  
2:    $u \leftarrow B_x - A_x$   
3:    $v \leftarrow B_y - A_y$   
4:    $w \leftarrow C_x - A_x$   
5:    $z \leftarrow C_y - A_y$   
6:   return  $uz - vw$   
7: end function
```

3.2 Three-dimensional Orientation

For points $A, B, C, D \in \mathbb{R}^3$, the signed volume predicate is

$$\text{orient3d}(A, B, C, D) = \det \begin{pmatrix} B - A \\ C - A \\ D - A \end{pmatrix}.$$

Its sign identifies which side of the oriented plane ABC contains D . Its zero case detects coplanarity. Tetrahedralization, volume-mesh validation, and tetrahedron–hexahedron conversion use this predicate.

3.3 In-circle Predicate

Given a counter-clockwise triangle ABC and a point D , the in-circle predicate determines whether D lies inside, on, or outside the circumcircle of ABC . It is evaluated from the sign of

$$\text{incircle}(A, B, C, D) = \det \begin{pmatrix} A_x & A_y & A_x^2 + A_y^2 & 1 \\ B_x & B_y & B_x^2 + B_y^2 & 1 \\ C_x & C_y & C_x^2 + C_y^2 & 1 \\ D_x & D_y & D_x^2 + D_y^2 & 1 \end{pmatrix}.$$

The test is central to Delaunay triangulation and edge-flip algorithms.

3.4 Robust Evaluation

Floating-point roundoff can change the sign of a nearly zero determinant and therefore change an algorithm's topology. Implementations should evaluate predicates with adaptive precision or exact arithmetic when the rounded result is uncertain. Epsilon thresholds may be useful for application-level tolerances, but they do not by themselves provide a consistent combinatorial result. Symbolic perturbation is another way to define deterministic answers for coincident, collinear, coplanar, or co-circular inputs.

The predicates in this chapter have constant arithmetic cost for fixed dimension. Their practical cost is dominated by the precision required to certify the sign.

3.5 Floating-Point Filters

3.5.1 Overview

A **floating-point filter** evaluates a predicate in floating point first and certifies the sign using an *a priori* error bound; only when the rounded result is too close to zero does it fall back to a slower, exact evaluation. Because ill-conditioned inputs are rare in practice, filters give the robustness of exact arithmetic at nearly floating-point speed [She97].

3.5.2 Static and Dynamic Filters

A **static filter** derives a fixed error bound from the magnitudes of the input coordinates before the computation begins; it is cheap but pessimistic. A **dynamic filter** computes the bound during the evaluation, tightening it as intermediate results are refined, so the exact path is entered only when it is genuinely needed [BBP01].

3.5.3 Adaptive Predicates

Shewchuk's adaptive predicates evaluate the orientation and in-circle determinants as a sequence of increasingly accurate approximations built from **expansion arithmetic**, stopping as soon as the sign is certified [She97]. The arithmetic rests on error-free transformations such as the *two-sum* and *two-product* primitives of Knuth and Priest: for IEEE 754 inputs these recover the roundoff error exactly, so the exact sign is obtained with no explicit multiple-precision representation [Pri91, Gol91]. In the common nondegenerate case the predicates run at roughly the speed of plain floating point.

3.5.4 Complexity

Each error-free transformation costs a small constant number of floating operations. The adaptive stage recomputes the determinant with increasing precision only along the path where roundoff

threatens the sign; the exact stage has cost proportional to the number of bits that must be carried.

3.6 Exact Arithmetic

3.6.1 Overview

The **exact geometric computation** (EGC) paradigm treats robustness as a non-issue by computing predicates exactly, and only when a new geometric object is constructed does it degrade to a certified numerical value [Yap97]. Exactness is achieved with arbitrary-precision integers, rationals, or floating-point expansions.

3.6.2 Integer and Rational Kernels

When input coordinates are given as integers (or rationals), every predicate sign is an exact integer computation. Fortune and Van Wyk generate specialized, unrolled evaluation code from an expression compiler so that the cost tracks the actual bit-length of the operands rather than paying a general-purpose library’s overhead [FVW93, FVW96]. The bit length grows only by a constant factor for each predicate: a 2D orientation of τ -bit integers needs $O(\tau)$ bits, and an in-circle test needs $O(\tau)$ as well.

3.6.3 Expansions

An **expansion** represents an exact value as the exact sum of non-overlapping floating-point components. The two-sum and two-product identities split a single floating-point addition or product into the rounded result and its exact error, and these identities compose into exact sums and products of arbitrary length [Pri91, She97]. This is the representation used by robust implementations of the orientation and in-circle predicates.

3.6.4 Exact Geometric Computation

Yap’s EGC framework adds *precision-driven* evaluation: expressions are evaluated to successively higher precision until a requested bound is met. It underwrites libraries such as CORE that combine rational and algebraic number types, and it extends exact computation beyond predicates to constructed objects like intersection points [Yap97, HMY04].

3.7 Rational Arithmetic in Computational Geometry

3.7.1 Overview

Rational arithmetic represents every number as a ratio of arbitrary-precision integers, so all predicates and all constructed points remain exact. It is the natural arithmetic for algorithms whose input is integer or rational, and it avoids the roundoff of floating point at the price of normalizing fractions and managing large numerators.

3.7.2 Application to Delaunay Triangulation

Karasick, Lieber, and Nackman showed that a Delaunay triangulation driven by rational arithmetic eliminates the failures caused by floating-point in-circle tests, at a cost that is acceptable for engineering input [KLN91]. The in-circle predicate for rational inputs reduces to a fixed combination of integer operations whose size is bounded by the input bit-length.

3.7.3 Trade-offs

Rational arithmetic is exact and uniform, but numerators can grow large when intersection points are repeatedly constructed. Hybrid designs combine a floating-point filter with a rational or integer fallback so that typical queries never touch the expensive exact kernel [FVW93, KLN91].

3.8 Simulation of Simplicity

3.8.1 Overview

Simulation of Simplicity (SoS) is a general technique for coping with degenerate inputs: it simulates an infinitesimal perturbation of the data that eliminates all coincident, collinear, coplanar, and co-circular configurations, yet the perturbation is never actually computed [EM90]. The programmer writes the algorithm for general position only; the tie-breaking is hidden inside the predicate implementations.

3.8.2 Construction

Each coordinate x_i is conceptually replaced by $x_i(\epsilon)$, a polynomial in an indeterminate ϵ , so that the perturbed predicate $P(x_0(\epsilon), \dots, x_{n-1}(\epsilon))$ is a nonzero polynomial whose sign at $\epsilon \rightarrow 0^+$ is the lexicographically first nonzero term. Evaluating only that leading term yields a deterministic, consistent answer for every degenerate input. Yap proved consistency for the corresponding symbolic perturbation scheme and extended the technique to arbitrary predicates [Yap88, Yap90].

3.8.3 Cost and Limitations

For determinant predicates the perturbed test costs only a small constant factor over the unperturbed one. SoS cannot make an inconsistent *algorithm* robust, and it ties degeneracies to a fixed symbolic order rather than to any geometric meaning; exact arithmetic with explicit degeneracy handling remains the alternative when the tie-break matters [EM90, HMY04].

3.9 Robustness Failure Examples

Floating-point predicates do not merely produce slightly wrong answers; they can make an algorithm crash, loop forever, or output an inconsistent structure. Kettner, Mehlhorn, Pion, Schirra, and Yap collected a suite of small, reproducible examples — such as a convex-hull routine whose topology changes under rotation, and a segment intersection where roundoff contradicts the planar subdivision — that expose these failures and are now used as regression tests for libraries like CGAL [KMP⁺08]. These examples motivate the filter and exact-arithmetic machinery above: certifying the sign of a predicate is the only way to guarantee a consistent combinatorial result.

Part II

Fundamental Planar Algorithms

Chapter 4

Convex Hulls

4.1 Graham Scan

The Graham scan is an efficient method for computing the convex hull of a finite set of points in the Euclidean plane. It was published by Ronald Graham in 1972 [Gra72] and is a fundamental algorithm in computational geometry. The algorithm finds all vertices of the convex hull ordered along its boundary.

4.1.1 Definitions

Definition 4.1 (Convex Hull). The *convex hull* of a set $S \subseteq \mathbb{R}^2$ is the smallest convex set containing all the points in S . Equivalently, it is the intersection of all convex sets containing S , or the set of all convex combinations of points in S :

$$\text{conv}(S) = \left\{ \sum_{i=1}^k \lambda_i p_i \mid k \in \mathbb{N}, p_i \in S, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1 \right\}.$$

Definition 4.2 (Cross Product in $\mathbb{R}^2$). For three points $A, B, C \in \mathbb{R}^2$, the *cross product* is defined as

$$\text{cross}(A, B, C) = (B.x - A.x)(C.y - A.y) - (B.y - A.y)(C.x - A.x).$$

It determines the orientation of the ordered triple (A, B, C) :

- Positive: Left turn (counter-clockwise).
- Negative: Right turn (clockwise).
- Zero: Collinear.

Geometrically, $|\text{cross}(A, B, C)|$ equals twice the signed area of triangle $\triangle ABC$.

Definition 4.3 (Anchor Point). The *anchor point* of a point set is the point with the lowest y -coordinate (and lowest x -coordinate in case of ties). It is guaranteed to lie on the convex hull.

4.1.2 Theory

The algorithm operates by first identifying an extreme point (the anchor) and sorting all other points based on the polar angle they make with this anchor. It then performs a linear scan through the sorted points, maintaining a stack of points that potentially form the hull. For each new point, the algorithm “backtracks” by popping points from the stack that would create a non-convex (right) turn. This ensures that the final stack contains only the vertices of the convex hull in counter-clockwise order.

Theorem 4.4 (Correctness of the Graham Scan). *Let P be a set of $n \geq 3$ points in general position in $\mathbb{R}^2$, and let p_0 be the anchor point. After sorting the remaining points by polar angle around p_0 , the Graham scan produces the vertices of $\text{conv}(P)$ in counter-clockwise order.*

Proof. The key invariant is that the stack always stores a sequence of points forming a left-turn chain. When a new point p_i is examined, any point(s) on the stack that would create a right turn (i.e., $\text{cross}(s_{k-1}, s_k, p_i) \leq 0$) are removed. Since every vertex of the convex hull must appear as a left turn when traversed counter-clockwise, no hull vertex is ever popped. Conversely, every non-hull point is eventually popped because it must lie inside the convex polygon formed by the hull vertices, and hence some subsequent hull vertex will create a right turn with it. The remaining stack therefore contains exactly the hull vertices in order. $\square$

Theorem 4.5 (Complexity of the Graham Scan). *The Graham scan computes the convex hull of n points in $O(n \log n)$ time and $O(n)$ space.*

Proof. The sorting step dominates with cost $O(n \log n)$. In the scan phase, each point is pushed onto the stack exactly once. Each point is popped at most once (since once popped, it is never re-examined). Thus the total number of push and pop operations is $O(n)$, making the scan phase $O(n)$ amortized. Space is $O(n)$ for the sorted list and stack. $\square$

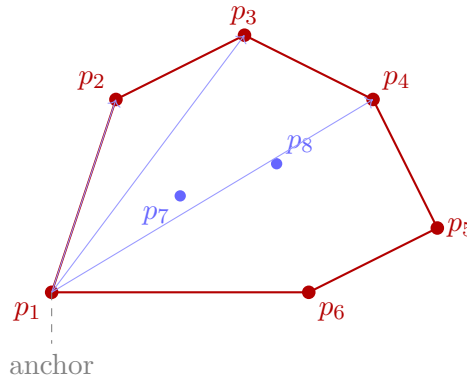


Figure 4.1: Graham Scan: the anchor point p_1 (lowest y) sorts all other points by polar angle. The hull vertices (red) are traced in counter-clockwise order.

4.1.3 Pseudocode

4.1.4 Parameters

- **Input:** A list of `Point2D` objects.
- **Sorting Criterion:** Polar angle relative to the anchor. In the implementation, `math.atan2` is often used, but cross-products can provide a more robust (and sometimes faster) comparison.

4.1.5 Complexity

- **Time Complexity:** $O(n \log n)$, where n is the number of points. This is dominated by the sorting step. The subsequent scan takes $O(n)$ time because each point is pushed and popped at most once.
- **Space Complexity:** $O(n)$ to store the sorted points and the stack.

Example 4.6 (Graham Scan Trace). Consider the point set $P = \{(0, 0), (1, 1), (2, 0), (1, -1), (0.5, 0.5)\}$.

Algorithm 2 Graham Scan for Convex Hull

```

1: function GRAHAMSCAN(points)
2:   if len(points) ≤ 2 then
3:     return points
4:   end if
5:   anchor ← min(points, key = λp : (p.y, p.x))
6:   sorted_points ← sorted(points, key = polar_angle_with_anchor)
7:   hull ← []
8:   for p in sorted_points do
9:     while len(hull) ≥ 2 and cross_product(hull[-2], hull[-1], p) ≤ 0 do
10:      hull.pop()
11:     end while
12:     hull.append(p)
13:   end for
14:   return hull
15: end function

```

1. The anchor point is (0,0) (lowest y , then lowest x).
2. Sorting by polar angle around (0,0) yields (approximately): (1, -1), (2, 0), (1, 1), (0.5, 0.5).
3. The scan proceeds: (0,0) and (1, -1) are placed on the stack. When (2,0) arrives, $\text{cross}((0,0), (1, -1), (2, 0)) = 1 \cdot 0 - (-1) \cdot 2 = 2 > 0$, so we keep the chain. When (1,1) arrives, we check $\text{cross}((1, -1), (2, 0), (1, 1)) = 1 \cdot 1 - (-1) \cdot (-1) = 2 > 0$: no pop. When (0.5, 0.5) arrives, $\text{cross}((2, 0), (1, 1), (0.5, 0.5)) = (-1)(0.5) - (1)(-1.5) = -0.5 + 1.5 = 1 > 0$: kept, but since (0.5, 0.5) lies in the interior of the hull formed by the other points, it will be interior to a final triangle and ultimately will not appear on the convex hull. (In practice, (0.5, 0.5) would be popped by a subsequent hull vertex check.)

The resulting hull is $\{(0, 0), (1, -1), (2, 0), (1, 1)\}$.

4.1.6 Usages

- Collision detection in physics engines.
- Shape analysis and pattern recognition.
- Geographic Information Systems (GIS) for defining boundary regions.
- Minimum bounding box calculations.

4.1.7 Testing and Edge Cases

- **Few Points:** 0, 1, or 2 points should return the input points.
- **Collinear Points:** Points lying on the same line. The algorithm should handle them by either including them or only keeping the endpoints (depending on the `cross_product <= 0` vs `< 0` logic).
- **Duplicate Points:** Multiple points at the same coordinates.
- **Degenerate Cases:** All points at the same location.
- **Vertical/Horizontal alignment:** Points forming a perfect grid.

4.1.8 References

- Graham, R. L. (1972). “An Efficient Algorithm for Determining the Convex Hull of a Finite Planar Set”. Information Processing Letters.
- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). “Introduction to Algorithms”. MIT Press.
- Wikipedia: Graham scan.

4.2 Monotone Chain

The Monotone Chain algorithm (also known as Andrew’s algorithm) is a method for computing the convex hull of a finite set of points in the plane. It was first described by A. M. Andrew in 1979 [And79]. It is generally faster than the Graham scan because it avoids the calculation of polar angles and trigonometric functions, relying solely on sorting and cross-product tests.

4.2.1 Definitions

Definition 4.7 (Upper and Lower Hull). Let P be a set of points sorted lexicographically. The *lower hull* is the sequence of vertices of $\text{conv}(P)$ from the leftmost to the rightmost point, traversed along the bottom boundary. The *upper hull* is the analogous sequence traversed along the top boundary. Together they partition the hull boundary into two monotone chains.

Definition 4.8 (Orientation Test). The *orientation test* for three points $A, B, C \in \mathbb{R}^2$ computes the sign of $\text{cross}(A, B, C)$. It determines whether the path $A \rightarrow B \rightarrow C$ makes a left turn, right turn, or is collinear.

4.2.2 Theory

The algorithm takes advantage of the fact that the convex hull can be split into upper and lower chains. By sorting the points lexicographically (first by x -coordinate, then by y -coordinate), we can build the lower hull by scanning the sorted points from left to right. A similar process (scanning either the reversed points or right-to-left) yields the upper hull. For each chain, the algorithm maintains a stack and removes the last point if it doesn’t form a “valid” turn relative to the previous points.

Theorem 4.9 (Correctness of Monotone Chain). *Let P be a set of $n \geq 2$ points in $\mathbb{R}^2$. After sorting P lexicographically, the Monotone Chain algorithm correctly computes $\text{conv}(P)$.*

Proof. The sorted order ensures the lower hull is built left-to-right with monotonically increasing x -coordinates. At each step, the invariant is that the stack maintains a chain with all cross products positive (left turns). If adding a point p would create a non-positive cross product with the last two stack points, the middle point cannot be a vertex of the convex hull (it is “inside” the angle formed by the other two). Removing it preserves the invariant. The upper hull is built symmetrically. The concatenation of lower and upper hulls (with endpoints removed to avoid duplication) gives the full convex hull boundary. $\square$

4.2.3 Pseudocode

4.2.4 Parameters

- **Input:** A list of `Point2D` objects.
- **Sorting Key:** Points are sorted primarily by their x -coordinate and secondarily by their y -coordinate.

Algorithm 3 Monotone Chain (Andrew's Algorithm)

```

1: function MONOTONECHAIN(points)
2:   if len(points) ≤ 2 then
3:     return points
4:   end if
5:   points.sort()
6:   lower ← []
7:   for p in points do
8:     while len(lower) ≥ 2 and cross_product(lower[-2], lower[-1], p) ≤ 0 do
9:       lower.pop()
10:    end while
11:    lower.append(p)
12:  end for
13:  upper ← []
14:  for p in reversed(points) do
15:    while len(upper) ≥ 2 and cross_product(upper[-2], upper[-1], p) ≤ 0 do
16:      upper.pop()
17:    end while
18:    upper.append(p)
19:  end for
20:  return lower[: -1] + upper[: -1]
21: end function

```

4.2.5 Complexity

- **Time Complexity:** $O(n \log n)$ due to sorting. The construction phase is strictly $O(n)$, as each point is added and removed at most once for each hull.
- **Space Complexity:** $O(n)$ to hold the sorted points and the resulting hull.

4.2.6 Usages

- Standard 2D convex hull calculation in many libraries (e.g., SciPy as a fallback).
- Calculating the area of a set of points.
- Simplifying complex polygons.
- GIS and map boundary creation.

4.2.7 Testing and Edge Cases

- **Vertical Lines:** Points with the same x -coordinate but different y -coordinates.
- **Collinear Points:** Handled by the ≥ 2 and $\text{cross_product} \leq 0$ conditions.
- **Identical Points:** Lexicographical sort handles duplicates naturally.
- **Small Sets:** Sets of 0, 1, or 2 points return early.
- **Closed Hulls:** Ensure the first point and last point of each chain match up correctly at the extremities.

4.2.8 References

- Andrew, A. M. (1979). “Another Efficient Algorithm for Convex Hulls in Two Dimensions”. Information Processing Letters.
- Preparata, F. P., & Shamos, M. I. (1985). “Computational Geometry: An Introduction”. Springer-Verlag.
- Wikipedia: Monotone chain.

4.3 Chan’s Algorithm

Chan’s algorithm is an **output-sensitive** algorithm for computing the convex hull of a set of n points in 2D or 3D. It combines the strengths of Graham scan (or Algorithm 3) and Jarvis march to achieve an optimal $O(n \log h)$ running time, where h is the number of vertices on the convex hull. It was introduced by Chan in 1996 [Cha96].

4.3.1 Definitions

Definition 4.10 (Output-Sensitive Algorithm). An algorithm is *output-sensitive* if its running time depends not only on the input size n but also on the output size (in this case, h , the number of hull vertices).

4.3.2 Theory

Chan’s algorithm partitions the point set into n/m groups of size m . It computes the convex hull of each group in $O(m \log m)$ time. Then, it uses a modified Jarvis march to find the global hull by performing binary searches (tangent queries) on the mini-hulls. By iteratively guessing $m = 2^{2^t}$, it reaches the optimal complexity.

Theorem 4.11 (Complexity of Chan’s Algorithm). *Chan’s algorithm computes the convex hull of n points in $O(n \log h)$ time and $O(n)$ space, where h is the number of vertices on the convex hull.*

Proof. The algorithm tries group sizes $m = 2, 4, 16, 256, \dots, 2^{2^t}$. For the correct guess where $m \geq h$, each group hull is computed in $O(m \log m)$ time, and the Jarvis march finds the hull in $O(m \log m)$ tangent queries (each taking $O(\log m)$ time on a group hull of size $\leq m$). The total work is $O((n/m) \cdot m \log m) = O(n \log m) = O(n \log h)$. All previous incorrect guesses $m' < h$ contribute at most $\sum_t O(n \log 2^{2^t}) = O(n)$ total work by a geometric series argument. $\square$

4.3.3 Pseudocode

4.3.4 Parameters

- **Group Size (m):** Controlled by the iteration parameter t .

4.3.5 Complexity

- **Time:** $O(n \log h)$, which is optimal.
- **Space:** $O(n)$.

4.3.6 Usages

Implemented in `compgeom.polygon.convex_hull.Chan`. Used for efficient hull calculation when the number of hull vertices is small.

Algorithm 4 Chan’s Convex Hull Algorithm

```

1: function CHANSCONVEXHULL( $P$ )
2:   for  $t = 1, 2, \dots$  do
3:      $m \leftarrow \min(2^{2^t}, n)$ 
4:     Partition  $P$  into groups of size  $m$ 
5:     Compute Monotone Chain hull for each group
6:      $\triangleright$  Try to find global hull using Jarvis march in  $m$  steps
7:     From current point, find next hull point by tangent query on each mini-hull
8:     if hull is closed then
9:       return hull
10:    end if
11:  end for
12: end function

```

4.3.7 Testing and Edge Cases

- **Testing:** Compare output with Graham Scan or SciPy’s Qhull. Verify all original points are inside the hull.
- **Edge Cases:** Collinear points, coincident points, small point sets ($n < 3$).

4.3.8 References

- Chan, T. M. (1996). “Optimal output-sensitive convex hull algorithms in two and three dimensions”. Discrete & Computational Geometry.
- Wikipedia: Chan’s algorithm.

4.4 QuickHull

QuickHull is a divide-and-conquer algorithm for computing the convex hull of a finite set of points. It is similar in design to the Quicksort sorting algorithm. It was independently published by several researchers, including Barber et al. (1996) [BDH96] and Eddy (1977) [Edd77], and is the basis of the widely used Qhull library.

4.4.1 Definitions

Definition 4.12 (Distance from a Line). The *perpendicular distance* from a point P to a line through points A and B is

$$d(P, \overline{AB}) = \frac{|\text{cross}(A, B, P)|}{\|B - A\|}.$$

Points farthest from a line segment are guaranteed to be vertices of the convex hull.

4.4.2 Theory

The algorithm begins by finding the leftmost and rightmost points, which are guaranteed to be on the hull. These points divide the point set into two subsets by a line. For each subset, the point farthest from the line is identified. This point, along with the two original points, forms a triangle. Points inside this triangle cannot be on the convex hull and are discarded. The process is then recursively applied to the two edges of the triangle that are not part of the original line.

Theorem 4.13 (QuickHull Average-Case Complexity). *The average-case time complexity of QuickHull is $O(n \log n)$ when points are distributed according to a smooth distribution (e.g., uniformly in a convex region).*

Proof. The argument mirrors that of Quicksort. On average, the farthest point partitions the remaining points into two roughly balanced subsets. If each recursive step discards a constant fraction of the remaining points, the depth of recursion is $O(\log n)$ and the total work is $O(n \log n)$. In the worst case (e.g., points on a circle), the partition is maximally unbalanced and the complexity degrades to $O(n^2)$. $\square$

4.4.3 Pseudocode

Algorithm 5 QuickHull Algorithm

```

1: function QUICKHULL(points)
2:   leftmost  $\leftarrow$  min(points)
3:   rightmost  $\leftarrow$  max(points)
4:   upper  $\leftarrow$  points above line (leftmost, rightmost)
5:   lower  $\leftarrow$  points below line (leftmost, rightmost)
6:   return [leftmost] + FindHull(leftmost, rightmost, upper) + [rightmost] +
      FindHull(rightmost, leftmost, lower)
7: end function
8: function FINDHULL(A, B, candidates)
9:   if candidates is empty then
10:    return []
11:  end if
12:  P  $\leftarrow$  arg maxp  $\in$  candidates dist_from_line(A, B, p)
13:  left_of_AP  $\leftarrow$  [p | p  $\in$  candidates  $\wedge$  is_left_of(A, P, p)]
14:  left_of_PB  $\leftarrow$  [p | p  $\in$  candidates  $\wedge$  is_left_of(P, B, p)]
15:  return FindHull(A, P, left_of_AP) + [P] + FindHull(P, B, left_of_PB)
16: end function

```

4.4.4 Parameters

- **Input:** A set of points (list of `Point2D`).
- **Recursive termination:** When the subset of candidate points for a segment is empty.

4.4.5 Complexity

- **Time Complexity:** Average case $O(n \log n)$. Worst case $O(n^2)$ when points are distributed such that the partition is highly unbalanced (e.g., points on a circle).
- **Space Complexity:** $O(n)$ to store the points and recursion stack.

4.4.6 Usages

- The primary algorithm in the `Qhull` library, used by SciPy, MATLAB, and R.
- General-purpose convex hull calculation in 2D and 3D.
- Collision detection and mesh simplification.

4.4.7 Testing and Edge Cases

- **Highly Symmetric Points:** Points on a circle can lead to the worst-case $O(n^2)$ performance.
- **Collinear Points:** Farthest point might not be unique; handle tie-breaking consistently.
- **Points inside the hull:** Ensure that the pruning (ignoring points inside triangles) is correct.
- **Numerical Precision:** Use an epsilon for distance and cross-product comparisons to avoid issues with floating-point arithmetic.

4.4.8 References

- Barber, C. B., Dobkin, D. P., & Huhdanpaa, H. (1996). “The Quickhull Algorithm for Convex Hulls”. ACM Transactions on Mathematical Software (TOMS).
- Eddy, W. F. (1977). “A New Convex Hull Algorithm for Planar Sets”. ACM Transactions on Mathematical Software (TOMS).
- Bykat, A. (1978). “Convex Hull of a Finite Set of Points in Two Dimensions”. Information Processing Letters.
- Wikipedia: Quickhull.

4.5 Convex Hull Peeling

Convex hull peeling, also known as “onion peeling”, is a recursive process of identifying and removing the convex hull of a point set. After the first hull is removed, the process is repeated on the remaining points until no points are left. This decomposition organizes a set of points into nested “layers,” similar to the layers of an onion. It was studied by Chazelle [Cha85], Overmars and van Leeuwen [OvL81], and is closely related to the concept of Tukey depth introduced by Tukey [Tuk75]. It is a fundamental technique for robust statistics, data depth analysis, and identifying outliers.

4.5.1 Definitions

Definition 4.14 (Convex Hull Layer). A *convex hull layer* is the set of points forming the boundary of the convex hull at a specific iteration of the peeling process.

Definition 4.15 (Convex Hull Depth). The *convex hull depth* (or layer number) of a point is the iteration number at which it is removed. Points on the outermost hull have depth 1. The innermost layer defines the **Tukey depth** of the deepest points.

4.5.2 Theory

The onion peeling of a point set P provides a natural way to measure the “centrality” of points.

1. Compute the convex hull $CH(P)$.
2. The points on the boundary of $CH(P)$ form the first layer L_1 .
3. Let $P' = P \setminus L_1$.
4. Repeat the process on P' to find $L_2, L_3, \dots, L_k$.

Theorem 4.16 (Layer Count). *For n points distributed uniformly at random in a unit square, the expected number of layers is $\Theta(n^{2/3})$.*

Proof. Each convex hull of a random point set in the unit square has $O(n^{1/3})$ vertices on average. Removing these vertices reduces the remaining set by this amount. The number of layers before the set is exhausted is thus $O(n/n^{1/3}) = O(n^{2/3})$. A matching lower bound follows from concentration arguments on the extreme point count. $\square$

In 2D, the total number of layers k is roughly $O(n^{2/3})$ for points distributed uniformly in a square. The structure of the layers can be used to define the Tukey Depth of points in a multivariate dataset.

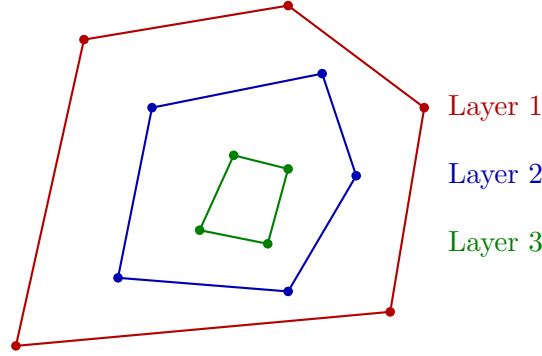


Figure 4.2: Convex hull peeling (onion peeling): successive convex hulls are removed, producing nested layers.

4.5.3 Pseudocode

Algorithm 6 Convex Hull Peeling (Onion Peeling)

```

1: function CONVEXHULLPEELING(points)
2:   layers  $\leftarrow []$ 
3:   current_points  $\leftarrow \text{set}(\text{points})$ 
4:   while current_points is not empty do
5:     if len(current_points) < 3 then
6:       layers.append(list(current_points))
7:       break
8:     end if
9:     hull_indices  $\leftarrow \text{ComputeConvexHull}(\text{list}(\text{current\_points}))$ 
10:    hull_points  $\leftarrow [\text{list}(\text{current\_points})[i] \mid i \in \text{hull\_indices}]$ 
11:    layers.append(hull_points)
12:    for p in hull_points do
13:      current_points.remove(p)
14:    end for
15:  end while
16:  return layers
17: end function

```

4.5.4 Parameters

- **Hull Algorithm:** Monotone Chain or Graham Scan are ideal for 2D. For 3D, QuickHull is standard.

- **Handling Collinearity:** Points that lie exactly on the edges of the convex hull can either be included in the current layer or treated as interior points for the next layer. Including them is more common in statistics.

4.5.5 Complexity

- **Naive Complexity:** $O(n^2)$ if a new $O(n \log n)$ hull is calculated for every point removed.
- **Efficient 2D Complexity:** $O(n \log n)$ using Chazelle’s algorithm [Cha85], which uses a more complex data structure to update the hull as points are removed.
- **Space Complexity:** $O(n)$ to store the point set and the layer indices.

4.5.6 Usages

- **Robust Statistics:** Identifying the “geometric median” of a point set by finding the deepest onion layer.
- **Outlier Detection:** Points in the first few layers are candidates for being outliers or “extreme” values.
- **Data Visualization:** Creating “bagplots” or “boxplots” for 2D data.
- **Pattern Recognition:** Describing the density and distribution of a point cloud.
- **Computational Geometry:** A preprocessing step for certain range-searching and visibility problems.

Example 4.17 (Onion Peeling by Hand). Consider a 3×3 grid of points $P = \{(i, j) : i, j \in \{0, 1, 2\}\}$ with 9 total points.

1. Layer 1 (outermost hull): The 8 boundary points $\{(0, 0), (1, 0), (2, 0), (2, 1), (2, 2), (1, 2), (0, 2), (0, 1)\}$.
2. After peeling, only $\{(1, 1)\}$ remains.
3. Layer 2: The single point $\{(1, 1)\}$.

The center point has hull depth 2, while all other points have depth 1.

4.5.7 Testing and Edge Cases

- **Perfectly Collinear Points:** All points should be removed in the first layer.
- **Points on a Circle:** All points should be removed in the first layer.
- **Grid Arrangement:** Verify that the peeling correctly identifies the nested squares or rectangles.
- **Small Sets:** Test with $n = 1, 2, 3$.
- **Symmetry:** For a symmetric distribution, the layers should maintain that symmetry.

4.5.8 References

- Chazelle, B. (1985). “On the convex layers of a planar set”. *IEEE Transactions on Information Theory*.
- Overmars, M. H., & Leeuwen, J. van. (1981). “Maintenance of configurations in the plane”. *Journal of Computer and System Sciences*.
- Tukey, J. W. (1975). “Mathematics and the picturing of data”. *Proceedings of the International Congress of Mathematicians*.
- Wikipedia: Convex layers.

Chapter 5

Line-Segment Intersection

5.1 The Intersection Problem

The **segment intersection problem** is the following: given n line segments in the plane, report all pairs that intersect, or decide whether any two intersect. Intersections can occur at interior points of both segments, at endpoints, or along overlapping portions of collinear segments. The problem is fundamental: map overlay in geographic information systems, design-rule checking in VLSI layout, window clipping and overdraw analysis in computer graphics, and the detection of crossings in geometric algorithms all reduce to it. Because two segments can intersect only if they are close in the vertical order, the problem is a showcase for the **sweep-line paradigm**, developed in Sections 5.3–5.6.

There are two versions of the problem with different complexity: *detection* (is the set pairwise disjoint?) and *reporting* (produce all k intersecting pairs). They are contrasted in Section 5.10. This chapter develops the classic Bentley–Ottmann algorithm [BO79], which reports all intersections among n segments with k crossings in $O((n + k) \log n)$ time, together with the degenerate-case handling and the specialized red–blue variant used for map overlay.

5.2 Pairwise Intersection Testing

5.2.1 1. Overview

The elementary operation on which every segment-intersection algorithm is built is the test of whether two segments s and t intersect. The test must be a *predicate* in the sense of Section 2.6: its output must be correct and consistent even on degenerate inputs, since an entire algorithm’s correctness rests on it.

5.2.2 2. Definitions

Definition 5.1 (Segment Intersection). Two segments $s = [p, q]$ and $t = [r, u]$ **intersect** if they share at least one point: $[p, q] \cap [r, u] \neq \emptyset$. The intersection may be a single point or, for collinear overlapping segments, a segment.

Definition 5.2 (Bounding Box). The **axis-aligned bounding box** of a segment $s = [p, q]$ is the rectangle $[\min(p_x, q_x), \max(p_x, q_x)] \times [\min(p_y, q_y), \max(p_y, q_y)]$.

5.2.3 3. Theory

Two segments intersect exactly when each crosses the line containing the other, or when they share an endpoint, or when they overlap collinearly. A clean formulation uses the orientation predicate *orient* of Section 2.6:

- s and t have intersecting interiors iff $\text{orient}(p, q, r)$ and $\text{orient}(p, q, u)$ have opposite signs and $\text{orient}(r, u, p)$ and $\text{orient}(r, u, q)$ have opposite signs;
- if any orientation is zero, the corresponding point lies on the line of the other segment, and a collinearity/overlap check decides.

The two-sides test alone is insufficient for endpoints and collinear overlaps, so the complete test combines orientation signs with a bounding-box check that filters quickly.

5.2.4 4. Pseudocode

Algorithm 7 Segment Intersection Test

Input: Two segments $s = [p, q]$ and $t = [r, u]$.

Output: True iff $s \cap t \neq \emptyset$.

```

1: function INTERSECTS( $s, t$ )
2:    $o_1 \leftarrow \text{orient}(p, q, r)$ ;  $o_2 \leftarrow \text{orient}(p, q, u)$ 
3:    $o_3 \leftarrow \text{orient}(r, u, p)$ ;  $o_4 \leftarrow \text{orient}(r, u, q)$ 
4:   if  $o_1 \neq o_2$  and  $o_3 \neq o_4$  then
5:     return TRUE ▷ proper crossing
6:   end if
7:   for  $\{o_1, o_2, o_3, o_4\}$  with  $o_i = 0$ : do
8:     if the witness point lies in both bounding boxes then
9:       return TRUE ▷ endpoint or collinear touch
10:    end if
11:  end for
12:  return FALSE
13: end function

```

5.2.5 5. Parameter Selection

- **Sign convention:** all four orientations must use the same convention (e.g., positive = counterclockwise); flipping one sign breaks the test.
- **Robustness:** the test is used as a predicate, so orientation values must be evaluated with the adaptive exact method of Section 3.4; an ε tolerance turns the test into a measurement and produces inconsistent combinatorial outputs.

5.2.6 6. Complexity

The test evaluates four orientation predicates and a constant number of comparisons, each $O(1)$, so pairwise testing of all $\binom{n}{2}$ pairs costs $O(n^2)$ time and $O(1)$ space. This is the baseline against which the output-sensitive algorithms of the rest of the chapter are measured.

5.2.7 7. Usages

- **Validity checks:** testing whether a polygon is simple, or a triangulation non-crossing, reduces to checking all edge pairs.
- **Brute-force baseline:** for small n ($n \lesssim 100$) the $O(n^2)$ all-pairs test is fastest and simplest, and it is the reference implementation used to validate sweep-line output in Section 5.6.

5.2.8 8. Testing Methods and Edge Cases

- **Shared endpoints:** two segments sharing exactly one endpoint must be reported as intersecting by the collinear/endpoint branch.
- **Collinear overlap:** segments on the same line overlapping in a segment, or merely touching at a point, must be distinguished.
- **Consistency:** the test must agree with itself under symmetric input order, so $\text{Intersects}(s, t)$ equals $\text{Intersects}(t, s)$ on all inputs.

5.2.9 9. References

The predicate formulation and the box-filtering strategy follow the standard treatment of the orientation test in Section 3.1 and the textbook treatments [PS85, BCKO08]. The robust evaluation of the orientations is due to Shewchuk [She97].

5.3 Sweep-Line Paradigm

The **sweep-line paradigm** processes a planar problem by moving an imaginary vertical line from left to right and maintaining the answer incrementally. At any position x , the sweep line $\ell_x = \{(x, y)\}$ meets a subset of the input objects; for segments, it intersects each segment in at most one point. The algorithm maintains two structures:

- the **sweep status**, the ordered list of objects intersected by ℓ_x , stored so that insertion, deletion, and neighbor queries cost $O(\log n)$ (Section 5.5);
- the **event queue**, the list of positions at which the status changes, maintained as a priority queue ordered by x -coordinate (Section 5.4).

The sweep advances from event to event. Between two consecutive events the status is static, so the algorithm is correct as long as every status change is registered as an event in advance or discovered during the sweep.

The paradigm appears throughout this book: it computes the contour of the union of rectangles (Section 5.12), the overlay of planar subdivisions (Section 5.11), and, in a rotating form, the convex hulls of Chapter 4 and the arrangement techniques of Chapter 14. Its strength is that it converts a global combinatorial question into a sequence of local updates; its weakness is that events are not always known in advance — the intersection of two segments is only discovered when the segments become adjacent in the status, which is the subtlety handled by the Bentley–Ottmann algorithm of Section 5.6.

5.4 Events and Event Queues

An **event** is a position of the sweep line at which the status changes. In the segment-intersection sweep the event types are:

left endpoint a segment enters the status at its left endpoint;

right endpoint a segment leaves the status at its right endpoint;

intersection two segments cross, and their vertical order swaps.

The event queue stores the events ordered by x -coordinate, and events with equal x must have a defined total order so that the algorithm is deterministic (Section 5.9). Endpoint events are known in advance and are inserted when the sweep starts. Intersection events are *discovered*:

when two segments become adjacent in the status, their intersection point is computed and inserted into the queue as a future event. If several pairs share the same intersection point, the event is inserted only once (the queue must be able to test membership, or each pair inserts its own event and duplicate processing is tolerated).

The queue must support, in $O(\log m)$ time with m events stored: insert a new event, extract the next event, and (for robustness against duplicate insertions) test membership. A balanced binary search tree with events ordered lexicographically by (x, y) provides all three operations. Because intersection events are inserted with x -coordinates beyond the current sweep position, the queue is a standard priority structure; the subtlety that the same intersection can be enqueued from several adjacent pairs is handled by the degenerate-case rules of Section 5.9.

5.5 Sweep-Line Status Structures

The **status structure** maintains the **vertical order** of the segments intersected by the sweep line: the sequence of segments sorted by the y -coordinate of their intersection with ℓ_x . It must support, in $O(\log n)$ time:

- insert a segment (at a left endpoint);
- delete a segment (at a right endpoint);
- return the segment immediately above and immediately below a given segment (or point).

A balanced binary search tree keyed by vertical order provides all of these operations. The comparison between two segments s and t is defined at the current sweep position x : $s \prec t$ iff s meets ℓ_x below t . The comparison is evaluated by computing the two intersection points with ℓ_x and comparing their y -coordinates; when x lies to the right of one segment's right endpoint, the segment is no longer in the status and is never compared.

The correctness of the whole approach rests on a locality lemma: two segments can intersect only if at some point during the sweep they are *adjacent* in the vertical order. Consequently, only adjacent pairs need to be tested for future intersections, and each adjacency change (insertion, deletion, or swap) triggers at most $O(1)$ new tests. This lemma is proved in Section 5.7.

5.6 Bentley–Ottmann Algorithm

5.6.1 1. Overview

The Bentley–Ottmann algorithm [B079] reports all k intersection points among n line segments in $O((n + k) \log n)$ time. It is the canonical output-sensitive algorithm for the planar segment-intersection problem and the template for every sweep-line algorithm in this book. The same technique is used for the red–blue intersection of Section 5.11 and for the union-of-rectangles contour of Section 5.12.

5.6.2 2. Definitions

Definition 5.3 (Upper and Lower Neighbor). At sweep position x , the **upper neighbor** of a segment s in the status is the segment directly above s in the vertical order, if any; the **lower neighbor** is the segment directly below, if any.

Definition 5.4 (Status Adjacency). Two segments are **adjacent** in the status at position x if no other segment lies between them in the vertical order at x .

5.6.3 3. Theory

Lemma 5.5 (Adjacent-Pair Lemma). *Let s and t intersect at a point p and assume no segment passes through p . Immediately before the sweep line reaches p , the segments s and t are adjacent in the vertical order; immediately after, their order is swapped.*

Proof. For a position x just to the left of p_x , the relative order of s and t is determined by their y -coordinates at x . If some segment u lay strictly between them for all x near p_x , then by continuity u would still be between them at $x = p_x$, contradicting that s and t meet at p . Hence just before the event no segment separates them, so they are adjacent. The swap is immediate because the two intersections with the sweep line cross at p . $\square$

The lemma justifies discovering intersection events only between adjacent segments: every intersection is found when its two segments first become adjacent, so testing only adjacent pairs (at most three new pairs per event) reports every intersection without missing any.

5.6.4 4. Pseudocode

5.6.5 5. Parameter Selection

- **General position:** the pseudocode assumes no vertical segments, no three segments through one point, and no coincident endpoints. Real inputs violate these assumptions; the ordering rules of Section 5.9 resolve the ties.
- **Ordering of events with equal x :** left endpoints must be processed before right endpoints at the same x , so that segments that touch at an endpoint are reported; the convention is fixed once and used throughout.
- **Right of the sweep:** an intersection is enqueued only if its x -coordinate is strictly to the right of the current event, so events are never inserted into the past; this prevents infinite loops.

5.6.6 6. Complexity

- **Time:** each of the $O(n + k)$ events performs $O(1)$ insertions, deletions, or neighbor queries, each costing $O(\log n)$; the total is $O((n + k) \log n)$.
- **Space:** the status holds $O(n)$ segments and the queue holds $O(n + k)$ events, so the space is $O(n + k)$.

The bound is optimal up to the $\log n$ factor for comparison-based algorithms when k is large, and $O((n + k) \log n)$ cannot be improved in the comparison model when $k = \Theta(n^2)$; the deterministic optimal algorithms of [Cha92, Bal95] reach $O(n \log n + k)$ but are substantially more complex and are rarely implemented.

5.6.7 7. Usages

- **Map overlay:** computing the overlay of two planar subdivisions (Section 5.11).
- **Design-rule checking:** detecting that VLSI wires intersect where they must not.
- **Graphics overdraw and clipping:** finding pairs of edges that cross within a window.
- **Verification:** sweep-line output is compared against the brute-force baseline of Section 5.2 in tests.

Algorithm 8 Bentley–Ottmann Segment Intersection

Input: A set S of n line segments in general position.**Output:** All intersection points of the segments in S .

```
1: function BENTLEYOTTMANN( $S$ )
2:    $Q \leftarrow$  empty priority queue ordered by  $(x, y)$ 
3:   for each segment  $s \in S$  do
4:      $Q.\text{insert}(\text{left}(s)); Q.\text{insert}(\text{right}(s))$ 
5:   end for
6:    $T \leftarrow$  empty balanced tree (vertical order)
7:   while  $Q$  is not empty do
8:      $e \leftarrow Q.\text{extract-min}()$ 
9:     if  $e$  is a left endpoint of  $s$  then
10:       $T.\text{insert}(s)$ ; let  $a, b$  be the neighbors of  $s$ 
11:      if  $a$  and  $s$  intersect to the right of the sweep then
12:         $Q.\text{insert}(\text{intersection}(a, s))$ 
13:      end if
14:      if  $s$  and  $b$  intersect to the right of the sweep then
15:         $Q.\text{insert}(\text{intersection}(s, b))$ 
16:      end if
17:    else if  $e$  is a right endpoint of  $s$  then
18:      let  $a, b$  be the neighbors of  $s$  in  $T$ 
19:       $T.\text{delete}(s)$ 
20:      if  $a$  and  $b$  intersect to the right of the sweep then
21:         $Q.\text{insert}(\text{intersection}(a, b))$ 
22:      end if
23:    else if  $e$  is an intersection of  $s$  and  $t$  then
24:      report  $e$ 
25:      swap  $s$  and  $t$  in  $T$ 
26:      if  $s$  and its new upper neighbor  $u$  intersect to the right then
27:         $Q.\text{insert}(\text{intersection}(s, u))$ 
28:      end if
29:      if  $t$  and its new lower neighbor  $v$  intersect to the right then
30:         $Q.\text{insert}(\text{intersection}(t, v))$ 
31:      end if
32:    end if
33:  end while
34: end function
```

5.6.8 8. Testing Methods and Edge Cases

- **Random sweeps:** generate random segment sets, compare the reported intersections against brute force, and verify both sets agree.
- **Vertical segments:** they break the “one point per segment” assumption; either rotate the input or handle them explicitly (Section 5.9).
- **Multiple crossings at one point:** several segments through a single point produce a “cluster” event; each pair must be reported exactly once.
- **No crossings:** verify that the algorithm terminates and reports nothing, i.e., no past events are ever enqueued.

5.6.9 9. References

The algorithm was introduced by Bentley and Ottmann [BO79]. Textbook treatments with correctness proofs appear in [PS85, BCKO08]. Optimal algorithms with reduced dependence on k were given by Chazelle [Cha92] and Balaban [Bal95].

5.7 Correctness

The Bentley–Ottmann algorithm is correct by induction on the sequence of events. The invariant is:

Immediately after processing the event at e , the status T reflects the vertical order of all segments at the sweep position $e_x + \varepsilon$, and every intersection to the left of the sweep line has been reported and every intersection immediately to the right involving a status-adjacent pair has been enqueued.

At the start of the sweep no intersections lie to the left, and the invariant holds vacuously. For the induction step one examines the three event types:

- a **left endpoint** inserts its segment into the correct vertical position (the tree key is the y -coordinate at the sweep line), and any future intersection with a neighbor is enqueued;
- a **right endpoint** deletes its segment; any intersection of the two neighbors that is uncovered by the deletion is enqueued;
- an **intersection event** is reported and the swap is performed; new adjacencies after the swap are tested and their intersections enqueued.

The Adjacent-Pair Lemma (Theorem 5.5) guarantees that no intersection is missed: every crossing pair is adjacent immediately before its crossing, at which time one of the three event handlers tests the pair and enqueues the event. Because events are processed in (x, y) -order and new events are only enqueued to the right of the sweep, no event is processed twice and the algorithm terminates. Thus the reported set is exactly the set of intersection points, with no duplicates.

5.8 Complexity Analysis

Let n be the number of segments and k the number of reported intersection points. The events processed are $2n$ endpoint events and at most k distinct intersection events. Each event performs $O(1)$ status operations at $O(\log n)$ each, and each status operation performs $O(1)$ queue insertions at $O(\log n)$ each, giving a total of $O((n + k) \log n)$ time. The queue can contain,

besides the $2n$ endpoint events, up to $O(n)$ pending intersection events per sweep position — the intersections of adjacent pairs — so the space is $O(n + k)$.

Two remarks about the bound. First, when k is small the algorithm is nearly linear in n ; when k is large (e.g., a grid-like configuration with $k = \Theta(n^2)$), the output size dominates. Second, the $\log n$ factor per reported intersection is not inherent to the problem: the optimal algorithms of [Cha92, Bal95] achieve $O(n \log n + k)$ by scheduling status operations in batches. The extra $\log n$ per event is the price of the simple “always enqueue new adjacencies” rule, and for most practical inputs the simple algorithm is preferred because of its dramatically lower constant factors and ease of robust implementation.

The decision version of the problem (does any pair intersect?) has a lower bound of $\Omega(n \log n)$ in the comparison model, by reduction from element uniqueness under a geometric encoding, so the sweep-based decision test is optimal.

5.9 Degenerate Intersections

Real inputs violate general position, and the Bentley–Ottmann algorithm must be extended to remain correct. The principal degenerate cases are:

Coincident endpoints When two segments share an endpoint, the events at that x must be processed in a fixed order: all left endpoints, then all intersection events at that point, then all right endpoints. This ordering ensures that segments meeting only at an endpoint are reported exactly once.

Three or more segments through one point The pairwise crossings all occur at the same event. A cluster is processed by rotating the segments around the point: the segments that pass through the point change vertical order all at once, and every pair must be reported. The standard method reorders the affected segments at the event and tests the new adjacencies, reporting every pair of segments through the point.

Vertical segments A vertical segment meets every sweep line in its x -range in a whole segment, not a point, so the “one intersection per segment” ordering breaks. The common fix is to treat vertical segments by a perturbation or by using (x, slope) orderings; alternatively the input is rotated slightly so no segment is vertical.

Collinear overlapping segments Two collinear segments that overlap have infinitely many common points; the “intersection point” event model does not apply. Overlaps are handled either by splitting the segments at their overlap endpoints and treating the pieces separately, or by a separate collinearity pass that reports overlaps as segments rather than points.

Duplicate segments Coincident segments are an extreme overlap; the split-based handling reduces them to the duplicate-endpoint case.

The unifying technique for all of these cases is **simulation of simplicity** (Section 1.11): a fixed symbolic perturbation makes every predicate definite, and the algorithm runs in general position on the perturbed input. The rules above are the explicit, predictable version of that idea, and they are what a production implementation should encode.

5.10 Reporting versus Detecting Intersections

The segment-intersection problem has two forms. **Detection** asks whether the input contains any intersecting pair and returns a yes/no answer; **reporting** asks for the set of all k intersecting pairs (or points). The distinction drives both the complexity and the algorithm design:

- reporting has output size k , so its cost must include k ; the Bentley–Ottmann bound $O((n + k) \log n)$ is the correct measure;
- detection can stop at the first intersection and needs no output structure, so its worst-case cost is at most $O(n \log n)$ — the cost of running the sweep until the first event of type intersection, or of running a simpler proximity-based test;
- **counting** (report k without listing the pairs) is an intermediate form studied in [BO79, Cha92].

The lower bound for detection is $\Omega(n \log n)$ in the comparison model, by the standard reduction from element uniqueness, and the sweep-line algorithm for detection is therefore asymptotically optimal. Reporting in time $O(n \log n + k)$ is also optimal, achieved by [Cha92, Bal95]; the Bentley–Ottmann algorithm is optimal in the sense that its $\log n$ factor per output element is unavoidable for the natural class of algorithms that update a sorted status at every event.

In practice, most applications need reporting (map overlay, layout checking), and the practical question is rarely asymptotic optimality but robustness on degenerate inputs and a small constant factor, which favors the classic algorithm.

5.11 Red–Blue Segment Intersection

5.11.1 1. Overview

The **red–blue segment intersection** problem [MS88] reports the intersections between two sets of segments, the red set R and the blue set B , under the promise that no two red segments intersect and no two blue segments intersect. The promise holds in the canonical application, computing the overlay of two planar subdivisions: the edges of one subdivision form the red set, the edges of the other the blue set, and each color class is non-self-intersecting. With $n = |R| + |B|$ and k red–blue intersections, the problem admits a sweep-line algorithm running in $O((n + k) \log n)$ time, where the monochromatic promise reduces the event handling: within each color no swapping or overlap events occur.

5.11.2 2. Definitions

Definition 5.6 (Red–Blue Set). A pair (R, B) of segment sets is a **red–blue set** if no two segments in R intersect and no two segments in B intersect. A **red–blue intersection** is an intersection point of a segment of R with a segment of B .

Definition 5.7 (Subdivision Overlay). The **overlay** of two planar subdivisions is the planar subdivision whose edges are the union of the two edge sets, split at every edge crossing. Computing the overlay is exactly reporting all red–blue intersections and then splitting the edges there.

5.11.3 3. Theory

The monochromatic promise simplifies the sweep in two ways. First, since segments of one color do not cross, their relative order never changes and they can be maintained without swap events. Second, an intersection event always involves a red and a blue segment, so the status can be maintained as two interleaved lists (red order and blue order), and a candidate pair is tested only when a red and a blue segment become adjacent in the combined order. The counting version of the problem is the basis of the **Klee–Mairson** measure problems and of the “symmetrized” segment-intersection counting of [MS88].

Algorithm 9 Red–Blue Segment Intersection by Sweep

Input: Red set R and blue set B forming a red–blue set.**Output:** All red–blue intersection points.

```
1: function REDBLUEINTERSECTIONS( $R, B$ )
2:    $Q \leftarrow$  priority queue of all endpoints, then discovered events
3:    $T \leftarrow$  empty balanced tree (vertical order over both colors)
4:   while  $Q$  is not empty do
5:      $e \leftarrow Q.\text{extract-min}()$ 
6:     if  $e$  is a left endpoint of segment  $s$  then
7:        $T.\text{insert}(s)$ ; let  $a, b$  be the neighbors of  $s$ 
8:       test  $a$  against  $s$  and  $s$  against  $b$  for a crossing to the right; enqueue it
9:     else if  $e$  is a right endpoint of segment  $s$  then
10:      let  $a, b$  be the neighbors of  $s$ ;  $T.\text{delete}(s)$ 
11:      if  $a, b$  have different colors and cross to the right then
12:        enqueue
13:      end if
14:     else if  $e$  is an intersection of  $s$  (red) and  $t$  (blue) then
15:       report  $e$ ; swap  $s$  and  $t$  in  $T$ ; test new adjacencies to the right
16:     end if
17:   end while
18: end function
```

5.11.4 4. Pseudocode**5.11.5 5. Parameter Selection**

- **Color symmetry:** the algorithm is symmetric in the two colors; only the monochromatic promise matters, not which color is “red”.
- **Overlay output:** for the map-overlay application the intersection points are used to split edges; the sweep should return the crossing pairs, not just the points, so the two edges can be cut.

5.11.6 6. Complexity

- **Time:** $O((n + k) \log n)$ with $n = |R| + |B|$ and k crossings.
- **Space:** $O(n + k)$.

The bound matches Bentley–Ottmann, but the promise typically makes the constant smaller and, more importantly, the monochromatic order never changes, which is used by the optimal algorithms [CGL85, MS88] to reach $O(n \log n + k)$.

5.11.7 7. Usages

- **Map overlay in GIS:** overlaying two maps (parcels and flood zones, roads and districts) is red–blue intersection followed by edge splitting.
- **Boolean operations on polygons:** union/intersection of two simple polygons first computes their overlay (Chapter 6).
- **VLSI and printed-circuit checking:** comparing two wiring layers for accidental contacts.

5.11.8 8. Testing Methods and Edge Cases

- **Red–blue violations:** a test must deliberately feed self-intersecting color classes and verify the promise is checked or the behavior documented.
- **Touching at vertices:** a red vertex lying on a blue segment is a red–blue intersection; the endpoint-ordering rule of Section 5.9 must report it exactly once.
- **Overlapping collinear edges:** shared boundary between two subdivisions produces collinear overlap; resolve by splitting as in Section 5.9.

5.11.9 9. References

The problem and its $O((n+k) \log n)$ sweep solution were introduced by Mairson and Stolfi [MS88]; its connection to planar overlay is developed in the textbook treatments [BCKO08, PS85].

5.12 Contour of the Union of Rectangles

5.12.1 1. Overview

The contour of the union of rectangles problem asks for the boundary of the region covered by a set of axis-parallel rectangles. It is a classic application of the sweep-line paradigm: a vertical line sweeps the plane while a status structure maintains the active set of rectangles, and the contour is assembled from the events at which the union boundary changes. Applications include VLSI mask checking, building footprints in GIS, and windowing in computer graphics [BCKO08].

5.12.2 2. Definitions

Definition 5.8 (Union Contour). The *contour* of a set of rectangles is the boundary of their union: the set of points on the union’s frontier that are not interior to any rectangle.

Definition 5.9 (Contour Break Point). A *break point* is a point on the contour where the boundary changes direction, occurring at rectangle corners or at intersections between rectangle edges.

5.12.3 3. Theory

At a sweep position x , the vertical line cuts the union of the rectangles into a set of disjoint intervals — the union of the y -intervals of the rectangles active at x . The boundary of the union consists of pieces of rectangle edges, so the contour is described exactly by the changes in the interval set. When the sweep passes a rectangle’s left edge, the y -interval of that rectangle is inserted; passing a right edge removes it. A vertical contour piece exists wherever the number of active rectangles changes, and horizontal contour pieces connect consecutive break points on the same rectangle edge. The problem therefore reduces to maintaining the union of intervals under insertions and deletions, and reporting the top/bottom boundary of that union at each event.

5.12.4 4. Pseudocode

5.12.5 5. Parameter Selection

- **Interval structure:** the active y -intervals are maintained in a balanced tree keyed by bottom coordinate; the union is maintained incrementally, merging intervals that overlap.
- **Edge ordering:** left and right edges at the same x are processed left edges first so that touching rectangles are handled consistently.

Algorithm 10 Union Contour of Axis-Parallel Rectangles

Input: n axis-parallel rectangles.**Output:** The contour (list of break points) of their union.

```
1: function UNIONCONTOUR( $R_1, \dots, R_n$ )
2:    $E \leftarrow$  all  $2n$  vertical edges, each tagged open/close and sorted by  $x$ 
3:    $C \leftarrow$  empty set of intervals (active  $y$ -intervals, kept sorted)
4:    $P \leftarrow$  empty list of break points
5:   for each vertical edge  $e$  in  $E$  by  $x$  do
6:     before:  $U \leftarrow$  union of intervals in  $C$ 
7:     if  $e$  is a left edge then
8:        $C \leftarrow C \cup \{\text{interval}(e)\}$ 
9:     else
10:       $C \leftarrow C \setminus \{\text{interval}(e)\}$ 
11:    end if
12:    after:  $U' \leftarrow$  union of intervals in  $C$ 
13:    report the changed portions of the top and bottom of  $U$  vs  $U'$  as contour edges
14:    append their break points to  $P$ 
15:  end for
16:  return  $P$ 
17: end function
```

- **Output assembly:** the contour is emitted as horizontal edges (from break points on the top/bottom of the union) plus vertical edges (where the active set changes), connected into closed loops afterwards.

5.12.6 6. Complexity

With n axis-parallel rectangles, the sweep processes $2n$ vertical edge events, each involving an $O(\log n)$ update of the interval status structure. The running time is $O(n \log n)$; the output has $O(k)$ segments, where k is the number of contour break points.

5.12.7 7. Usages

- **VLSI mask checking:** the contour of the union of mask features is exactly the shape that fabrication sees, and design-rule checks (minimum spacing, enclosure) run on the contour.
- **GIS building footprints:** merging overlapping parcel or building footprints requires their union contour.
- **Graphics windowing:** the visible region of overlapping windows is the union contour of their rectangles.

5.12.8 8. Testing Methods and Edge Cases

- **Overlapping rectangles:** nested, identical, and partially overlapping rectangles must all merge correctly with no duplicate break points.
- **Touching rectangles:** rectangles sharing an edge must produce a single contour without a spurious internal edge.
- **Verification:** compare the contour area (by the shoelace formula, Section 2.7) with an independent area computation by inclusion–exclusion.

5.12.9 9. References

The union-of-rectangles problem and its sweep solution are covered in [BCKO08, Ben77]; the connection to interval maintenance and to Klee’s measure problem is developed in [Ben77, BO79].

5.13 Applications

Segment intersection algorithms are the engine behind several major application areas:

Map overlay Combining two thematic maps in a geographic information system reduces to red–blue intersection and edge splitting (Section 5.11); the overlay result is a new planar subdivision, and the GIS application chapter develops the pipeline (Chapter 36).

Design-rule checking In VLSI layout, wires must not intersect where their netlists forbid contact; reporting all intersections of a wire layer is a segment-intersection query over axis-aligned and diagonal segments.

Computer graphics Window clipping, overdraw analysis, and boolean operations on polygons all begin with edge-intersection reporting; the graphics chapter uses these in modeling and rendering (Chapter 35).

CAD and CAM Verifying that a tool path does not self-intersect, and detecting collisions of planar outlines, are segment-intersection checks.

Robotics and motion planning In the plane, the validity of a path is checked against obstacle edges by intersection tests, and the configuration-space computations of Chapter 26 use them.

Validity of geometric structures Checking that a triangulation is non-crossing, that a polygon is simple, and that a planar subdivision is well-formed all reduce to segment-intersection tests over edge sets.

In each of these settings the practical requirements are the same: the underlying intersection predicate must be exact and consistent (Sections 5.2 and 1.12), degenerate inputs must be handled by documented rules (Section 5.9), and the sweep must be output-sensitive because k can be very small or very large depending on the data.

5.14 Exercises

1. Prove the Adjacent-Pair Lemma of Section 5.6 when several segments pass through the intersection point.
2. Show that testing all pairs is $\Omega(n^2)$ even when the output is empty, by giving an input with no intersections whose *bounding boxes* all pairwise overlap.
3. Explain why an intersection event must be enqueued only when it lies strictly to the right of the current sweep position, and give an input on which enqueueing past events loops forever.
4. Adapt the Bentley–Ottmann algorithm to vertical segments without rotation, describing the exact ordering predicate.
5. Compute the running time of Bentley–Ottmann for an input with $k = 0$ (no crossings) and for a “spiral” input where every segment crosses every other ($k = \binom{n}{2}$).

6. Design a test oracle for red–blue intersection that uses the pairwise predicate of Section 5.2 but exploits the monochromatic promise.
7. Show how the contour of the union of rectangles changes when two rectangles touch at a single corner, and verify that the sweep of Section 5.12 handles it.
8. Prove that the decision version of segment intersection has an $\Omega(n \log n)$ lower bound by reduction from element uniqueness.

5.15 Project: Interactive Sweep-Line Visualizer

Build an interactive tool that visualizes the sweep-line algorithm (Section 5.6):

1. **Input:** allow the user to place segments with the mouse, or load segments from a file.
2. **Sweep:** animate the vertical line moving left to right, drawing the status order as a ranked list beside the sweep line.
3. **Events:** highlight the current event; for an intersection event, flash the two segments and report the point.
4. **Status:** display the vertical order explicitly and update it with each event, so that insertions, deletions, and swaps are visible.
5. **Verification:** include a brute-force button that highlights every intersecting pair, and color any pair the sweep missed.
6. **Degenerate inputs:** add a library of degenerate examples (vertical segments, clusters, collinear overlaps) and confirm the implementation follows the rules of Section 5.9.
7. **Analysis:** report n , k , the number of events processed, and the number of predicate evaluations, and compare the sweep against brute force on random inputs.

The project can be implemented with the `compgeom` package’s segment predicates and with a small GUI layer; a terminal-based animation is sufficient if a full GUI is not desired.

Chapter 6

Polygon Geometry

6.1 Polygon Boolean Operations

6.1.1 1. Overview

Polygon Boolean operations are a set of geometric algorithms used to compute the result of applying set operations (union, intersection, difference, and symmetric difference) to two or more polygons [Vat92]. These operations are essential in computer graphics, CAD/CAM, and GIS for manipulating complex shapes and defining regions of interest.

6.1.2 2. Definitions

Definition 6.1 (Polygon Union). The *union* $A \cup B$ of two polygons A and B is the region covered by at least one of the input polygons.

Definition 6.2 (Polygon Intersection). The *intersection* $A \cap B$ of two polygons A and B is the region shared by both input polygons.

Definition 6.3 (Polygon Difference). The *difference* $A - B$ is the region inside polygon A but outside polygon B .

Definition 6.4 (Symmetric Difference). The *symmetric difference* $A \Delta B$ is the region inside either A or B , but not both. Formally, $A \Delta B = (A - B) \cup (B - A)$.

Definition 6.5 (Winding Rule). A *winding rule* (e.g., Even-Odd or Non-Zero winding number) is a rule used to determine the “inside” of a self-intersecting or hole-containing polygon. The Even-Odd rule considers a point inside if a ray from it crosses an odd number of edges; the Non-Zero rule considers the winding number (total signed crossings).

6.1.3 3. Theory

The most common approach to polygon Boolean operations is based on **edge clipping and subdivision** [GH98]. The algorithm typically involves:

1. **Intersection Detection:** Finding all points where edges of polygon A intersect edges of polygon B .
2. **Edge Subdivision:** Splitting edges at intersection points into smaller segments.
3. **Edge Classification:** For each segment, determining if it is “Inside,” “Outside,” or “Shared” relative to the other polygon.

4. **Loop Assembly:** Collecting the appropriate segments based on the desired operation and assembling them into new closed loops.

Theorem 6.6 (Polygon Boolean Complexity). *Let A and B be two simple polygons with n and m vertices respectively, producing k intersection points. The Boolean operations $A \cup B$, $A \cap B$, $A - B$, and $A \Delta B$ can all be computed in $O((n + m + k) \log(n + m))$ time using a sweep-line algorithm [Vat92].*

Proof. The algorithm proceeds in three phases. In phase 1, we find all k edge-edge intersections using a Bentley–Ottmann sweep line in $O((n + m + k) \log(n + m))$ time. In phase 2, we subdivide and classify each edge segment in $O(1)$ per segment (total $O(n + m + k)$). In phase 3, we assemble result polygons by traversing the doubly-linked edge structures in $O(n + m + k)$ time. The overall bound is dominated by the sweep. $\square$

Example 6.7 (Polygon Intersection Trace). Let A be the square $[0, 3] \times [0, 3]$ and B the square $[1, 4] \times [1, 4]$.

1. **Intersection detection:** The edges intersect at $(1, 3)$, $(3, 1)$, $(1, 1)$, and $(3, 3)$. Wait: A 's right edge $x = 3$ intersects B 's bottom edge $y = 1$ at $(3, 1)$; A 's top edge $y = 3$ intersects B 's left edge $x = 1$ at $(1, 3)$. There are $k = 2$ intersection points: $(3, 1)$ and $(1, 3)$.
2. **Edge subdivision:** A 's right edge is split into $[(3, 0), (3, 1)]$ and $[(3, 1), (3, 3)]$; A 's top edge is split into $[(0, 3), (1, 3)]$ and $[(1, 3), (3, 3)]$. Similarly for B 's edges.
3. **Classification:** For $A \cap B$, we keep only segments that lie inside both polygons: $[(3, 1), (3, 3)]$ (right edge of A , inside B), $[(1, 3), (3, 3)]$ (top edge of A , inside B), plus segments from B inside A .
4. **Assembly:** The resulting intersection is the square $[1, 3] \times [1, 3]$ with vertices $(1, 1)$, $(3, 1)$, $(3, 3)$, $(1, 3)$.

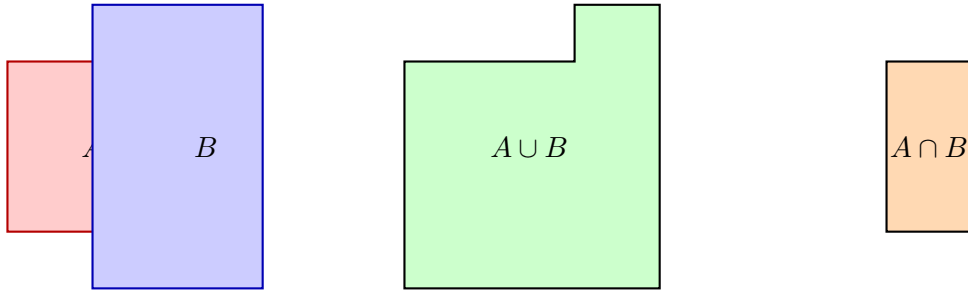


Figure 6.1: Polygon boolean operations: Union $A \cup B$ (center) and Intersection $A \cap B$ (right) of two overlapping rectangles.

6.1.4 4. Pseudo code

6.1.5 5. Parameters Selections

Definition 6.8 (Precision (Epsilon)). The *precision* ϵ is a floating-point tolerance used to determine if a vertex lies on an edge or if two edges are collinear. It is crucial for numerical robustness.

Definition 6.9 (Consistent Winding). *Consistent winding* requires that both input polygons use the same orientation convention (e.g., both counter-clockwise). Most algorithms assume counter-clockwise orientation for all inputs.

```

1: function POLYGONBOOLEAN(A, B, operation)      ▷ 1. Split edges at intersection points
2:   subdivided_A ← SplitEdges(A, B)
3:   subdivided_B ← SplitEdges(B, A)
                                                    ▷ 2. Classify segments
4:   classified_A ← Classify(subdivided_A, B)
5:   classified_B ← Classify(subdivided_B, A)
                                                    ▷ 3. Filter segments based on operation
6:   result_segments ← []
7:   if operation == "UNION" then
8:     result_segments += [s for s in classified_A if s.is_outside]
9:     result_segments += [s for s in classified_B if s.is_outside]
10:  else if operation == "INTERSECTION" then
11:    result_segments += [s for s in classified_A if s.is_inside]
12:    result_segments += [s for s in classified_B if s.is_inside]
13:  end if
                                                    ▷ 4. Assemble into polygons
14:  return AssemblePolygons(result_segments)
15: end function

```

6.1.6 6. Complexity

Theorem 6.10 (Polygon Boolean Complexity Summary). *For two simple polygons A (n vertices) and B (m vertices) with k intersection points:*

1. **Time:** $O((n + m + k) \log(n + m))$ using sweep-line intersection detection [Vat92].
2. **Space:** $O(n + m + k)$ for the subdivided edges and output polygon.

6.1.7 7. Usages

- **CSG (Constructive Solid Geometry):** Building complex 3D objects from simple primitives.
- **GIS Clipping:** Extracting spatial data within a specific boundary (e.g., finding all roads in a county).
- **PCB Design:** Calculating overlapping areas of copper layers or solder masks.
- **UI Layout:** Determining visible regions of windows and buttons.

6.1.8 8. Testing methods and Edge cases

- **Collinear Edges:** Edges that overlap exactly or partially.
- **Shared Vertices:** Polygons touching at a single point or along a common edge.
- **Holes:** Polygons containing internal voids or “islands.”
- **Self-Intersecting Inputs:** May require preprocessing into a set of simple polygons.
- **Degeneracies:** Zero-area polygons resulting from subtraction.

6.1.9 9. References

- Vatti, B. R. (1992). “A generic solution to polygon clipping”. Communications of the ACM.
- Greiner, G., & Hormann, K. (1998). “Efficient clipping of arbitrary polygons”. ACM TOG.
- Weiler, K., & Atherton, P. (1977). “Hidden surface removal using polygon area subdivision”. SIGGRAPH.
- [Wikipedia: Boolean operations on polygons](#)

6.2 Line Arrangements

6.2.1 1. Overview

A **line arrangement** is the subdivision of the 2D plane induced by a set of n lines. It is a fundamental structure in computational geometry that encodes all intersection points and the topological relationships between the lines (vertices, edges, and faces) [EOS86]. Arrangements provide a geometric framework for solving problems involving duality, range searching, and visibility.

6.2.2 2. Definitions

Definition 6.11 (Arrangement). Given a set L of n lines, the *arrangement* $\mathcal{A}(L)$ is the subdivision of the plane into vertices (intersection points), edges (maximal connected portions of lines between vertices), and faces (maximal connected regions of the plane not containing any line).

Definition 6.12 (Zone). The *zone* of a line ℓ in an arrangement $\mathcal{A}(L)$ is the set of faces in $\mathcal{A}(L)$ that are intersected by ℓ .

Definition 6.13 (DCEL). The *Doubly-Connected Edge List* (DCEL) is the standard data structure for representing arrangements. It stores for each edge: an origin vertex, a destination vertex, a pointer to the next edge around the face, and a pointer to the twin (opposite) edge.

6.2.3 3. Theory

For n lines in **general position** (no two lines parallel, no three lines concurrent):

- Number of vertices: $\binom{n}{2} = \frac{n(n-1)}{2}$.
- Number of edges: n^2 .
- Number of faces: $\frac{n^2+n+2}{2}$.

The standard way to build an arrangement is through **incremental construction**:

1. Start with a simple arrangement of two lines.
2. Add lines one by one. For each new line ℓ_i :
 - Find the face containing one end of ℓ_i .
 - Trace the path of ℓ_i through the arrangement using the **Zone Theorem**.
 - Split edges and faces as ℓ_i passes through them.

Theorem 6.14 (Zone Theorem). *In an arrangement of n lines, the zone of a single line ℓ has $O(n)$ complexity, i.e., it intersects $O(n)$ edges and faces [EOS86].*

Proof. Let ℓ be the new line. As ℓ traverses the arrangement from left to right, each time it crosses an edge of the arrangement, it enters a new face. Each existing line l_j intersects ℓ in at most one point, contributing at most 2 edges to the zone boundary (one above ℓ , one below). Since there are $n - 1$ other lines, the total zone complexity is at most $2(n - 1) + 1 = O(n)$. $\square$

Theorem 6.15 (Arrangement Construction Complexity). *An arrangement of n lines can be constructed in $O(n^2)$ time and stored in $O(n^2)$ space using the incremental algorithm based on the Zone Theorem [EOS86].*

Proof. We add n lines one at a time. For the i -th line, the Zone Theorem guarantees that its zone in the arrangement of the first $i - 1$ lines has $O(i)$ complexity. The cost of inserting line i is therefore $O(i)$. Summing over all n lines: $\sum_{i=1}^n O(i) = O(n^2)$. $\square$

Example 6.16 (Arrangement of 3 Lines). Consider three lines in general position: $\ell_1 : y = 0$, $\ell_2 : y = x$, $\ell_3 : y = -x + 2$.

- $\binom{3}{2} = 3$ vertices: $(0, 0)$, $(1, 0)$, $(1, 1)$.
- $3^2 = 9$ edges.
- $\frac{9+3+2}{2} = 7$ faces: 1 bounded triangle and 6 unbounded regions.

The bounded face is the triangle with vertices $(0, 0)$, $(1, 0)$, and $(1, 1)$.

6.2.4 4. Pseudo code

<pre> 1: function BUILDARRANGEMENT(lines) 2: arrangement $\leftarrow$ InitializeDCEL() 3: for line in lines do 4: face $\leftarrow$ LocateFace(arrangement, line.start_point) 5: current_p $\leftarrow$ line.start_point 6: while not IsFinished(current_p) do 7: exit_edge $\leftarrow$ FindExitEdge(face, line) 8: arrangement.SplitFace(face, line, exit_edge) 9: face $\leftarrow$ arrangement.GetAdjacentFace(exit_edge) 10: current_p $\leftarrow$ arrangement.GetIntersection(line, exit_edge) 11: end while 12: end for 13: return arrangement 14: end function </pre>	<pre> $\triangleright$ 1. Initialize with a large bounding box $\triangleright$ 2. Add lines incrementally $\triangleright$ 3. Locate start face $\triangleright$ 4. Trace the line through the arrangement (Zone Walk) $\triangleright$ Find which edge of the current face the line exits through $\triangleright$ Split the face and update the DCEL $\triangleright$ Move to the adjacent face </pre>
---	--

6.2.5 5. Parameters Selections

Definition 6.17 (DCEL Representation). The *Doubly-Connected Edge List (DCEL)* is the standard choice for representing arrangements as it allows for efficient traversal of vertices, edges, and faces in $O(1)$ per step.

Definition 6.18 (General Position Handling). If lines can be parallel or concurrent (degenerate), the construction algorithm must use robust geometric predicates and degeneracy handling. In practice, a symbolic perturbation or an epsilon-based approach is used.

6.2.6 6. Complexity

Theorem 6.19 (Arrangement Complexity Summary). *For n lines in general position:*

1. **Time:** $O(n^2)$ for construction using the incremental algorithm with the Zone Theorem.
2. **Space:** $O(n^2)$ to store all vertices, edges, and faces.
3. **Face Count:** $\frac{n^2+n+2}{2} = O(n^2)$.

6.2.7 7. Usages

- **Duality Problems:** Many problems involving points can be solved by mapping them to lines in a dual plane and analyzing their arrangement (e.g., finding the maximum number of collinear points) [CGL85].
- **Visibility:** Computing the visibility graph of a set of obstacles.
- **Motion Planning:** Analyzing the complexity of the configuration space for a robot.
- **Range Searching:** Identifying all points within a specific region.
- **Statistical Analysis:** Finding the Theil-Sen estimator or other robust regression lines.

6.2.8 8. Testing methods and Edge cases

- **Parallel Lines:** Verify that the algorithm correctly creates regions between lines that never intersect.
- **Concurrent Lines:** Ensure that three or more lines meeting at a single point are handled without creating zero-length edges.
- **Horizontal/Vertical Lines:** Test with lines aligned to the axes.
- **Small n :** Verify the formula for $n = 1, 2, 3$.
- **Bounding Box:** Ensure the arrangement handles lines that extend to infinity correctly by using a sufficiently large bounding volume or symbolic coordinates.

6.2.9 9. References

- Edelsbrunner, H., O'Rourke, J., & Seidel, R. (1986). "Constructing arrangements of lines and hyperplanes with applications". SIAM Journal on Computing.
- Chazelle, B., Guibas, L. J., & Lee, D. T. (1985). "The power of geometric duality". BIT Numerical Mathematics.
- Berg, M. de, Cheong, O., Kreveld, M. van, & Overmars, M. (2008). "Computational Geometry: Algorithms and Applications". Springer-Verlag.
- Wikipedia: [Arrangement of lines](#)

6.3 Minkowski Sums

6.3.1 1. Overview

The **Minkowski sum** of two sets of points A and B is the set of all possible sums of a point from A and a point from B [LP83]. In computational geometry, this operation is primarily used with polygons (2D) or polyhedra (3D). It is a fundamental tool for motion planning, collision detection, and calculating the offset of shapes.

6.3.2 2. Definitions

Definition 6.20 (Minkowski Sum). The *Minkowski sum* of two point sets $A, B \subseteq \mathbb{R}^2$ is defined as $A \oplus B = \{a + b \mid a \in A, b \in B\}$.

Definition 6.21 (Minkowski Difference). The *Minkowski difference* is $A \ominus B = \{a - b \mid a \in A, b \in B\} = A \oplus (-B)$. Two shapes A and B intersect if and only if $\mathbf{0} \in A \ominus B$.

Definition 6.22 (Convex Polygon Minkowski Sum). For two *convex polygons* P and Q with n and m vertices respectively, their Minkowski sum $P \oplus Q$ is also a convex polygon with at most $n + m$ vertices.

6.3.3 3. Theory

For two convex polygons P and Q with n and m vertices, the Minkowski sum is computed by:

1. Collect all the edges of P and Q .
2. Orient all edges counter-clockwise.
3. Sort all edges by their polar angle.
4. Chain the edges together starting from the sum of the bottom-left vertices of P and Q .

Theorem 6.23 (Convex Minkowski Sum Complexity). *The Minkowski sum of two convex polygons P (n vertices) and Q (m vertices) can be computed in $O(n + m)$ time and space [GS87].*

Proof. The edges of both polygons, when sorted by polar angle, form two sorted sequences (since both are convex). Merging two sorted sequences of total length $n + m$ takes $O(n + m)$ time. The resulting edge chain is traversed in $O(n + m)$ to form the sum polygon. $\square$

For **non-convex polygons**, the Minkowski sum is much more complex. The standard approach is to:

1. Decompose both non-convex polygons into convex pieces (e.g., using convex decomposition).
2. Compute the Minkowski sum for every pair of convex pieces ($P_i \oplus Q_j$).
3. Compute the union of all the resulting convex Minkowski sums.

Theorem 6.24 (Non-Convex Minkowski Sum Complexity). *For two simple (possibly non-convex) polygons P (n vertices) and Q (m vertices), the Minkowski sum can be computed in $O(n^2 m^2)$ worst-case time.*

Proof. A convex decomposition of P yields at most $O(n)$ convex pieces, and similarly $O(m)$ for Q . Computing each pairwise sum $P_i \oplus Q_j$ takes $O(|P_i| + |Q_j|)$ time by Theorem 6.23. The total work for $O(nm)$ pairwise sums is $O(n^2 m^2)$ in the worst case (accounting for the union step). $\square$

Example 6.25 (Minkowski Sum of Two Squares). Let $P = \{(0, 0), (1, 0), (1, 1), (0, 1)\}$ (unit square) and $Q = \{(0, 0), (2, 0), (2, 2), (0, 2)\}$ (2×2 square). Both are CCW.

1. Edges of P : $(1, 0), (0, 1), (-1, 0), (0, -1)$.
2. Edges of Q : $(2, 0), (0, 2), (-2, 0), (0, -2)$.
3. Sorted by angle: $(1, 0), (2, 0), (0, 1), (0, 2), (-1, 0), (-2, 0), (0, -1), (0, -2)$.
4. Starting from $(0, 0) + (0, 0) = (0, 0)$, chain the edges: $(0, 0) \rightarrow (1, 0) \rightarrow (3, 0) \rightarrow (3, 1) \rightarrow (3, 3) \rightarrow (2, 3) \rightarrow (0, 3) \rightarrow (0, 2) \rightarrow (0, 0)$.
5. Result: A 3×3 square (after merging collinear edges), with 4 vertices.

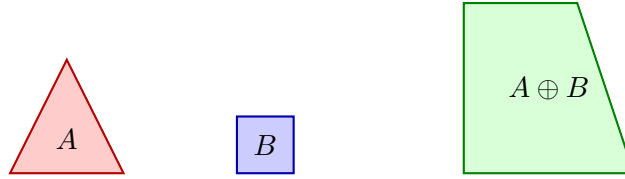


Figure 6.2: Minkowski sum of a triangle A and a unit square B . The result is a pentagon with at most $n + m$ vertices.

6.3.4 4. Pseudo code

```

1: function MINKOWSKI SUM CONVEX(P, Q)           ▷ P and Q are CCW convex polygons
2:   edges_P ← GetOrientedEdges(P)
3:   edges_Q ← GetOrientedEdges(Q)
4:   all_edges ← sorted(edges_P + edges_Q, key=lambda e: atan2(e.dy, e.dx))
                                                    ▷ Sort all edges by polar angle
                                                    ▷ Start at the sum of bottom-left vertices
5:   start_P ← min(P, key=lambda p: (p.y, p.x))
6:   start_Q ← min(Q, key=lambda p: (p.y, p.x))
7:   current ← start_P + start_Q
8:   result ← [current]
9:   for e in all_edges do
10:     current ← current + e.vector
11:     result.append(current)
12:   end for
13:   return result[:-1]                             ▷ Remove duplicate start/end
14: end function

```

6.3.5 5. Parameters Selections

Definition 6.26 (Input Orientation Requirement). Polygons **MUST** be oriented consistently (e.g., both counter-clockwise) before computing the Minkowski sum. Inconsistent orientation produces incorrect edge orderings.

Definition 6.27 (Convex Decomposition Method). The choice of convex decomposition (exact vs. approximate) for non-convex inputs significantly affects the performance and quality of the non-convex Minkowski sum.

6.3.6 6. Complexity

Theorem 6.28 (Minkowski Sum Complexity Summary). 1. **Convex Case:** $O(n+m)$ time and space for convex polygons P (n vertices) and Q (m vertices) [GS87].

2. **Non-Convex Case:** $O(n^2m^2)$ worst-case time for simple polygons.

6.3.7 7. Usages

- **Motion Planning (Configuration Space):** Expanding obstacles by the shape of the robot. If the robot's center is at x , it collides with obstacle O if $x \in O \oplus (-\text{Robot})$ [LP83].
- **Collision Detection:** The GJK (Gilbert-Johnson-Keerthi) algorithm uses Minkowski differences to check for intersections efficiently.
- **CAD Offsetting:** Creating a “buffer” or “padding” around a shape (mitered offsets).
- **Morphological Operations:** Dilation of 2D images or 3D volumes.

6.3.8 8. Testing methods and Edge cases

- **Points and Lines:** Minkowski sum of a polygon and a single point is just a translation. Sum of a polygon and a line segment is a “swept” version of the polygon.
- **Collinear Edges:** Edges from P and Q with the same angle should be merged into a single longer edge in the result.
- **Degenerate Shapes:** Polygons with zero area or zero vertices.
- **Non-Convex Union:** For non-convex sums, verify that internal “voids” are correctly handled or filled if necessary.
- **Origin Symmetry:** Minkowski sum of P and its reflection $-P$ is always centrally symmetric.

6.3.9 9. References

- Guibas, L. J., & Seidel, R. (1987). “Computing convolutions by reciprocal search”. Discrete & Computational Geometry.
- Lozano-Pérez, T. (1983). “Spatial Planning: A Configuration Space Approach”. IEEE Transactions on Computers.
- Fogel, E., & Halperin, D. (2006). “Exact Minkowski Sums of Convex Polygons”. CGAL Documentation.
- [Wikipedia: Minkowski addition](#)

6.4 Industrial Applications of Minkowski Sums

The Minkowski sum $A \oplus B = \{a + b \mid a \in A, b \in B\}$ (Section 6.3) is the algebraic engine behind an unusually broad set of industrial problems, because any operation that “grows” one shape by another, or that checks whether two shapes can touch, is a Minkowski computation. The following are the principal industrial use cases.

- **Robot Motion Planning (Configuration Space):** Growing every obstacle by the reflected robot shape $O \oplus (-\text{Robot})$ reduces the moving-robot collision problem to a point-robot problem: the robot at translation x collides iff $x \in O \oplus (-\text{Robot})$. This is the classical **C-space** construction of Lozano-Pérez, and it underlies the collision checking of commercial offline-programming and simulation suites for industrial manipulators (for example, ABB RobotStudio and FANUC Roboguide) (Section 26.2) [LP83].
- **Collision Detection and GJK:** The Gilbert–Johnson–Keerthi distance algorithm computes the distance between convex shapes from a simplex inside the Minkowski difference $P \ominus Q$; the origin lies in $P \ominus Q$ exactly when the shapes intersect. GJK powers the collision layer of commercial physics middleware (PhysX, Havok, Bullet) used in games, film, and robotics simulation (Section 18.6).
- **Assembly and Nesting (No-Fit Polygon):** The no-fit polygon (NFP) used in 2D irregular cutting and packing is the Minkowski sum of one piece with the reflection of the other (Section 33.3 and the definition 33.15); the NFP encodes all placements where the pieces touch, driving the nesting engines of commercial sheet-metal, textile, and packaging software (SigmaNEST, Lantek, Optimation).
- **Machine Toolpath Generation:** Offsetting a polygonal pocket by its tool radius via the Minkowski sum of the pocket with a disk yields the set of admissible tool-center positions, from which parallel or contour-parallel toolpaths are derived in commercial CAM systems (Mastercam, Siemens NX CAM, Autodesk Fusion 360) for CNC machining and laser/waterjet cutting.
- **Vehicle Swept-Path Analysis:** Computing the swept region of a vehicle turning along a path is a Minkowski sum of the chassis footprint with the path curve. It is the core of AutoTURN (Transoft Solutions), the industry-standard vehicle swept-path package used by over 90% of U.S. state departments of transportation for truck-turning clearance, bus terminals, and loading-dock design, and of the turning templates in the AASHTO Green Book.
- **Autonomous Driving (Occupancy Grids):** Inflating detected obstacles by the vehicle’s safety margin (a Minkowski sum with a disk or box) builds the expanded cost map over which planners find collision-free paths; the same growth bounds the car’s shape against lane geometry in commercial automated-driving stacks.
- **Image Morphology (Dilation) in Industrial Vision:** Grayscale and binary dilation is the Minkowski sum of the image set with a structuring element, and erosion is the Minkowski difference. These are the workhorses of defect detection, OCR, and inspection in commercial machine-vision systems (Cognex, Keyence).
- **Geometric Tolerancing (GD&T):** The maximum material condition and envelope tolerances in mechanical tolerancing are computed with Minkowski sums of the nominal profile with tolerance zones, bounding the worst-case assembled geometry in commercial tolerance-analysis packages.

6.4.1 Industrial-Scale Software

Minkowski-sum-based operations are implemented in CGAL (Minkowski_sum_2, NFP and offsetting), clipper and Clipper2 (polygon offsetting; Clipper is the slicing kernel of Ultimaker Cura), Boost.Polygon (polygon Boolean operations and offsetting), OMPL (C-space construction, the planning library of the robotics middleware MoveIt), and the bullet, PhysX, and Havok physics engines (GJK for collision). See also the motion-planning treatment in Section 26.2 [LP83, GS87].

6.5 Polygon Symmetry

6.5.1 1. Overview

Polygon symmetry detection is the problem of identifying whether a given polygon possesses symmetries such as rotational symmetry or reflectional (axial) symmetry. A polygon is symmetric if it looks identical after a specific geometric transformation. Detecting these symmetries is a classic problem in computational geometry with applications in computer vision, robotics, and geometric design [Ata85].

6.5.2 2. Definitions

Definition 6.29 (Reflectional Symmetry). A polygon has *reflectional (axial) symmetry* if there exists a line ℓ (the axis of symmetry) such that reflecting the polygon across ℓ maps it onto itself.

Definition 6.30 (Rotational Symmetry). A polygon has *rotational symmetry* of order $k > 1$ if rotating it by $360^\circ/k$ around its centroid leaves it unchanged.

Definition 6.31 (Symmetry Centroid). The *centroid* of a polygon is the average position of all the points (or vertices, for a regular polygon). For a polygon with rotational symmetry, the centroid must be the fixed point of all rotations.

6.5.3 3. Theory

The most efficient algorithm for detecting polygon symmetry is based on **string matching** [Ata85], similar to the polygon congruence algorithm.

1. Represent the polygon as a sequence of (angle, edge-length) pairs: $S = (a_1, l_1, a_2, l_2, \dots, a_n, l_n)$.
2. **Rotational Symmetry:** Find all cyclic shifts of S that are identical to S itself. This can be done using the **Knuth-Morris-Pratt (KMP)** algorithm by searching for S in $S + S$ (excluding the trivial match at position 0).
3. **Reflectional Symmetry:** Check if the sequence S is a cyclic shift of its own reverse sequence S^R . If it is, the axes of symmetry must pass through either a vertex or the midpoint of an edge.

Theorem 6.32 (Polygon Symmetry Detection Complexity). *All rotational and reflectional symmetries of a polygon with n vertices can be detected in $O(n)$ time using the string-matching approach [Hig86].*

Proof. The signature generation takes $O(n)$ time. KMP preprocessing of the pattern S takes $O(n)$, and searching S in $S + S$ takes $O(n)$, giving $O(n)$ for rotational detection. Similarly, KMP on the reversed signature takes $O(n)$ for reflectional detection. The total is $O(n)$. $\square$

Example 6.33 (Symmetry of a Regular Hexagon). A regular hexagon has:

- Rotational symmetry of order 6 ($k = 6$): rotations by $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$.
- 6 axes of reflectional symmetry: 3 passing through opposite vertices, 3 passing through midpoints of opposite edges.

The signature S of a regular hexagon is $(120^\circ, s, 120^\circ, s, \dots)$ for edge length s . All 6 cyclic shifts of S match S itself, confirming order 6. The reversed signature S^R matches S (cyclically), confirming 6 reflection axes.

```

1: function DETECTSYMMETRIES(polygon)
2:   sig ← GenerateSignature(polygon)
3:   n ← len(polygon)
4:    $\triangleright$  1. Detect Rotational Symmetry  $\triangleright$  Find all occurrences of sig in sig + sig
5:   rotations ← KMP_AllMatches(sig + sig[:-1], sig)
6:   order_k ← len(rotations)
7:    $\triangleright$  2. Detect Reflectional Symmetry
8:   sig_rev ← ReverseSignature(sig)
9:   reflection_matches ← KMP_AllMatches(sig + sig, sig_rev)
10:  axes ← []
11:  for match_pos in reflection_matches do  $\triangleright$  Calculate the axis line based on match position
12:    axis ← CalculateAxis(match_pos, polygon)
13:    axes.append(axis)
14:  end for
15:  return order_k, axes
16: end function
17: function GENERATESIGNATURE(polygon)  $\triangleright$  Returns (alpha_1, d_1, alpha_2, l_2, ...)  $\triangleright$ 
    where alpha is interior angle and d is edge length
18:   pass
19: end function

```

6.5.4 4. Pseudo code

6.5.5 5. Parameters Selections

Definition 6.34 (Symmetry Precision). Symmetry detection is highly sensitive to noise. An *epsilon tolerance* should be used when comparing angles and lengths to account for floating-point imprecision.

Definition 6.35 (Signature Choice). Using interior angles and edge lengths makes the detection invariant to translation, rotation, and scaling. The signature uniquely determines a polygon's shape up to congruence (or similarity if lengths are normalized).

6.5.6 6. Complexity

Theorem 6.36 (Polygon Symmetry Complexity Summary). *For a polygon with n vertices:*

1. *Signature Generation: $O(n)$.*
2. *Rotational Check: $O(n)$ using KMP.*
3. *Reflectional Check: $O(n)$ using KMP on the reversed signature.*
4. *Total Time: $O(n)$ [Ata85].*
5. *Space: $O(n)$ for the signature.*

6.5.7 7. Usages

- **Computer Vision:** Recognizing symmetric objects (like tools, symbols, or parts) in images regardless of their orientation.
- **Robotics:** Planning grasps for symmetric objects (where multiple orientations are equivalent).

- **Geometric Design (CAD):** Enforcing symmetry constraints during modeling.
- **Chemistry:** Identifying the point group symmetry of molecular structures represented as polygons or polyhedra.
- **Art and Architecture:** Analyzing and generating symmetric patterns and tilings.

6.5.8 8. Testing methods and Edge cases

- **Regular Polygons:** A regular n -gon should have rotational symmetry of order n and n axes of reflectional symmetry.
- **Isosceles Triangle:** Should have one axis of symmetry and no rotational symmetry ($k = 1$).
- **Rectangle:** Should have rotational symmetry of order 2 and two axes of symmetry.
- **Parallelogram:** Should have rotational symmetry of order 2 but no axes of symmetry (in general).
- **No Symmetry:** Test on random, scalene polygons to ensure they return order 1 and zero axes.
- **Precision Issues:** Test with nearly symmetric polygons to verify the epsilon handling.

6.5.9 9. References

- Atallah, M. J. (1985). “On symmetry detection”. IEEE Transactions on Computers.
- Highnam, P. T. (1986). “Optimal algorithms for finding the symmetries of a planar point set”. Information Processing Letters.
- Wolter, J. D., Woo, T. C., & Volz, R. A. (1985). “Optimal algorithms for detecting relevant symmetries in polygons”. IEEE Transactions on Robotics and Automation.
- Wikipedia: [Symmetry in geometry](#)

6.6 Polygon Matching

6.6.1 1. Overview

Polygon matching is the problem of determining whether two given polygons P and Q are identical under a set of transformations (congruence) or whether one is a scaled version of the other (similarity) [ACH⁺91]. This is a specialized form of shape matching that focuses on the exact geometric properties of the boundary. It is essential for pattern recognition, computer vision, and verifying geometric designs.

6.6.2 2. Definitions

Definition 6.37 (Polygon Congruence). Two polygons are *congruent* if they have the same shape and size. They can be mapped to each other via translation, rotation, and (optionally) reflection.

Definition 6.38 (Polygon Similarity). Two polygons are *similar* if they have the same shape but possibly different sizes. They can be mapped via congruence plus uniform scaling.

Definition 6.39 (Geometric Signature). A *geometric signature* is a sequence of values (like edge lengths and angles) that uniquely describes a polygon’s shape and is invariant under certain transformations.

6.6.3 3. Theory

The most robust way to check for polygon matching is to construct a **string representation** of the polygon and use string-matching algorithms [ACH⁺91].

1. For each vertex, calculate its **internal angle** and the **length of the edge** following it.
2. Form a sequence: $S = (a_1, l_1, a_2, l_2, \dots, a_n, l_n)$.
3. To handle rotation, treat the sequence as a circular string.
4. Two polygons are congruent if their sequences are cyclic shifts of each other (or one is a cyclic shift of the reversed other, for reflection).
5. Two polygons are similar if their angle sequences are identical and their edge length sequences are proportional.

The **Knuth-Morris-Pratt (KMP)** algorithm can be used to find cyclic shifts in linear time by searching for sequence S in $S + S$.

Theorem 6.40 (Polygon Congruence via String Matching). *Two simple polygons P and Q with n and m vertices respectively can be tested for congruence in $O(n + m)$ time, or $O(n)$ if $n = m$ [ACH⁺91].*

Proof. Generate the geometric signature S_P for P in $O(n)$ and S_Q for Q in $O(m)$. If $n \neq m$, reject. Otherwise, use KMP to search for S_Q in $S_P + S_P$ (for rotation) and in $S_P + S_P$ for S_Q^R (for reflection). Each KMP search takes $O(n)$ time. $\square$

Example 6.41 (Congruence Check). Consider two triangles:

- $P = \{(0, 0), (3, 0), (0, 4)\}$: Edges $(3, 0)$, $(-3, 4)$, $(0, -4)$. Angles: 90° at $(0, 0)$, $\arctan(4/3) \approx 53.1^\circ$ at $(3, 0)$, $\arctan(3/4) \approx 36.9^\circ$ at $(0, 4)$.
- $Q = \{(1, 1), (4, 1), (1, 5)\}$: Same edge lengths $(3, 5, 4)$ and same angles. Signature of Q is a cyclic shift of signature of P .

KMP finds the match, confirming congruence.

6.6.4 4. Pseudo code

6.6.5 5. Parameters Selections

Definition 6.42 (Matching Precision). Matching should use a small epsilon for floating-point comparisons of angles and lengths. The choice of epsilon affects both false positives and false negatives.

Definition 6.43 (Allowed Transformations). Decide if reflection (mirror symmetry) is allowed. If not, only check for cyclic shifts (rotation and translation). For similarity matching, divide all edge lengths by the total perimeter or the longest edge.

6.6.6 6. Complexity

Theorem 6.44 (Polygon Matching Complexity Summary). *For two polygons with n vertices each:*

1. **Signature Generation:** $O(n)$ each.
2. **Matching:** $O(n)$ using KMP for cyclic shifts [ACH⁺91].
3. **Total Time:** $O(n)$ for congruence; $O(n)$ for similarity (after normalization).
4. **Space:** $O(n)$ to store the signatures.

```
1: function ISCONGRUENT(P, Q)
2:   if len(P) != len(Q) then
3:     return False
4:   end if

5:   sigP ← GenerateSignature(P)
6:   sigQ ← GenerateSignature(Q)

7:   if IsCyclicShift(sigP, sigQ) then
8:     return True
9:   end if
10:  if IsCyclicShift(sigP, Reverse(sigQ)) then
11:    return True
12:  end if
13:  return False
14: end function

15: function GENERATESIGNATURE(polygon)
16:  sig ← []
17:  n ← len(polygon)
18:  for i in range(n) do
19:    p1 ← polygon[i-1]
20:    p2 ← polygon[i]
21:    p3 ← polygon[(i+1)%n]
22:    angle ← CalculateAngle(p1, p2, p3)
23:    length ← Distance(p2, p3)
24:    sig.append((angle, length))
25:  end for
26:  return sig
27: end function

28: function ISCYCLICSHIFT(S1, S2)
29:   return KMP_Search(S1 + S1, S2)
30: end function
```

▷ 1. Generate signatures

▷ 2. Handle reflection (optional)

▷ KMP search for S2 in S1 + S1

6.6.7 7. Usages

- **Computer Vision:** Identifying an object in an image by matching its contour to a template.
- **CAD/CAM:** Finding duplicate parts in a complex mechanical assembly.
- **Architecture:** Identifying repeating patterns or tiles in a floor plan.
- **Geographic Information Systems (GIS):** Matching building footprints across different map revisions.
- **Character Recognition:** Matching the strokes of a drawn letter to a font database.

6.6.8 8. Testing methods and Edge cases

- **Rotated Polygons:** Verify that a polygon matches its rotated version.
- **Mirrored Polygons:** Test if the algorithm correctly identifies reflections.
- **Regular Polygons:** A regular n -gon should have n different valid cyclic shifts.
- **Degenerate Polygons:** Test on triangles or polygons with collinear vertices.
- **Self-Intersecting Polygons:** Ensure the signature is robust for non-simple polygons.
- **Numerical Sensitivity:** Test with polygons having very small edges or angles.

6.6.9 9. References

- Arkin, E. M., Chew, L. P., Huttenlocher, D. P., Kedem, K., & Mitchell, J. S. (1991). “An efficiently computable metric for comparing polygonal shapes”. IEEE Transactions on Pattern Analysis and Machine Intelligence.
- Manber, U. (1989). “Introduction to Algorithms: A Creative Approach”. Addison-Wesley.
- Wikipedia: [Congruence \(geometry\)](#)

6.7 Line and Polygon Clipping

6.7.1 1. Overview

Clipping removes the portions of geometric objects lying outside a given window. Line clipping restricts a line segment to a convex window, and polygon clipping restricts a polygon to a convex window, yielding a new polygon. These operations are central to graphics viewport clipping and to Boolean operations on polygons [SH74, CB78].

6.7.2 2. Definitions

Definition 6.45 (Clip Window). A *clip window* is a (usually convex) region against which an object is clipped; the result is the intersection of the object with the window.

Definition 6.46 (Cyrus–Beck Clipping). The *Cyrus–Beck* algorithm clips a line segment to a convex polygon by computing, for each boundary edge, the interval of parameters $t \in [0, 1]$ for which the segment lies on the inner side, and intersecting these intervals [CB78].

Definition 6.47 (Sutherland–Hodgman Clipping). The *Sutherland–Hodgman* algorithm clips a polygon to a convex window by successively clipping the polygon against each boundary edge of the window, producing an output polygon at every stage [SH74].

6.7.3 3. Theory

Cyrus–Beck expresses the segment as $p(t) = p_0 + t(p_1 - p_0)$ and, for each window edge with inward normal n_i through point q_i , keeps parameters satisfying $n_i \cdot (p(t) - q_i) \geq 0$. Intersecting all such t -intervals yields the clipped segment. Sutherland–Hodgman clips the vertex list once per window edge, retaining vertices on the inside and inserting intersection points where an edge crosses the clipping line.

6.7.4 4. Pseudo code

```

function CLIPPOLYGON( $P$ , window  $W$ )
   $output \leftarrow P$ 
  for each boundary edge  $e$  of  $W$  do
     $input \leftarrow output$ ,  $output \leftarrow \emptyset$ 
    for each edge  $(s, p)$  of  $input$  do
      if  $p$  inside  $e$  then
        if  $s$  not inside  $e$  then
           $output \leftarrow$  intersection of  $(s, p)$  with  $e$ 
        end if
         $output \leftarrow p$ 
      else if  $s$  inside  $e$  then
         $output \leftarrow$  intersection of  $(s, p)$  with  $e$ 
      end if
    end for
  end for
  return  $output$ 
end function

```

6.7.5 5. Parameters Selections

- **Window convexity:** Sutherland–Hodgman assumes a convex window; for non-convex windows the polygon is first decomposed or the operation handled by general Boolean intersection.
- **Tolerance:** use a small epsilon when classifying vertices as inside or on the boundary.

6.7.6 6. Complexity

Cyrus–Beck clips a segment against an m -edge convex window in $O(m)$ time. Sutherland–Hodgman clips an n -vertex polygon against an m -edge window in $O(nm)$ time.

6.7.7 7. Usages

- **Graphics:** viewport and scissor clipping in the rendering pipeline.
- **CAD and GIS:** windowing queries and map clipping to regions of interest.
- **Boolean operations:** clipping is the primitive behind polygon intersection [Vat92].

6.7.8 8. Testing methods and Edge cases

- **Window fully inside:** the polygon is returned unchanged.
- **Polygon fully outside:** an empty polygon is returned.

- **Vertex on boundary:** verify the on-edge case is not counted twice.
- **Degenerate output:** clipping to a triangle or segment must not produce repeated vertices.

6.7.9 9. References

- Sutherland, I. E., & Hodgman, G. W. (1974). “Reentrant polygon clipping”. Communications of the ACM.
- Cyrus, M., & Beck, J. (1978). “Generalized two- and three-dimensional clipping”. Computers & Graphics.
- Wikipedia: [Sutherland–Hodgman algorithm](#)

6.8 Polygon Partitioning

6.9 Convex Decomposition

Convex decomposition is the process of partitioning a non-convex (concave) polygon into a set of disjoint convex polygons. This is a fundamental operation in computational geometry, as many algorithms work more efficiently on convex shapes.

6.9.1 Definitions

Definition 6.48 (Convex Polygon). A polygon P is *convex* if for every pair of points $a, b \in P$, the line segment $\overline{ab}$ lies entirely within P .

Definition 6.49 (Diagonal). A *diagonal* of a polygon is a line segment connecting two non-adjacent vertices that lies entirely within the interior of the polygon.

6.9.2 Theory

The implementation uses a **triangle merging** approach.

1. Triangulate the polygon (e.g., using ear clipping [Mei75]).
2. For each shared diagonal between two adjacent triangles/polygons, check if removing the diagonal results in a convex polygon.
3. If it does, merge the two parts.

Theorem 6.50 (Hertel–Mehlhorn Bound). *Any convex decomposition produced by the Hertel–Mehlhorn triangle-merging heuristic has at most $4k^*$ parts, where k^* is the optimal (minimum) number of convex parts.*

6.9.3 Pseudocode

6.9.4 Parameters

- **Triangulation Algorithm:** Ear clipping is used for simplicity.

6.9.5 Complexity

- **Time:** $O(n^2)$ for simple merging heuristics.
- **Space:** $O(n)$.

Algorithm 11 Convex Decomposition via Triangle Merging

```

1: function CONVEXDECOMPOSITION( $P$ )
2:    $T \leftarrow \text{Triangulate}(P)$ 
3:    $D \leftarrow$  list of all interior diagonals of  $T$ 
4:   for each  $d$  in  $D$  do
5:     Let  $P_1, P_2$  be the parts sharing diagonal  $d$ 
6:     if Merged( $P_1, P_2$ ) is convex at endpoints of  $d$  then
7:       Merge  $P_1$  and  $P_2$  into a single part
8:     end if
9:   end for
10:  return final parts
11: end function

```

6.9.6 Usages

Implemented in `compgeom.polygon.polygon.decomposer.convex_decompose_polygon`. Used in physics engines, motion planning, and collision detection.

6.9.7 Testing and Edge Cases

- **Testing:** Verify all resulting parts are convex. Check that the union of parts matches the original area.
- **Edge Cases:** Polygons with holes, self-intersecting polygons (unsupported), highly narrow “snaky” polygons.

6.9.8 References

- Hertel, S., & Mehlhorn, K. (1983). “Fast triangulation of simple polygons”. FCT.
- Wikipedia: Polygon decomposition.

6.10 Convex Partitioning

Convex partitioning is the problem of dividing a non-convex (concave) polygon into the **minimum possible number** of convex sub-polygons. While a simple decomposition (like triangulation) results in $n - 2$ convex parts, an optimal partition typically results in significantly fewer. This problem is solvable in polynomial time for simple polygons, making it a powerful tool for geometric simplification.

6.10.1 Definitions

Definition 6.51 (Simple Polygon). A polygon is *simple* if it has no self-intersections and no holes.

Definition 6.52 (Reflex Vertex). A vertex v of a polygon is *reflex* if the interior angle at v exceeds π (i.e., 180°). Reflex vertices are the source of non-convexity.

Definition 6.53 (Optimal Convex Partition). An *optimal convex partition* of a polygon P is a decomposition into the minimum number k^* of convex sub-polygons whose union equals P and whose interiors are pairwise disjoint.

6.10.2 Theory

The minimum number of convex parts is determined by how reflex vertices are “resolved” by diagonals. Each reflex vertex must be incident to at least one diagonal that splits its reflex angle.

The **Keil–Snoeyink algorithm** uses **dynamic programming** to find the optimal partition of a simple polygon in $O(n^3)$ time (or $O(r^2n)$ where r is the number of reflex vertices).

1. Define $DP[i][j]$ as the minimum number of convex parts in the sub-polygon formed by vertices $V_i, \dots, V_j$ and the diagonal V_iV_j .
2. The algorithm systematically builds up solutions for larger sub-polygons by combining solutions for smaller ones and checking if the resulting combination is convex.
3. The optimal solution handles cases where multiple reflex vertices are resolved by a single diagonal.

Theorem 6.54 (Lower Bound on Convex Parts). *Any convex partition of a simple polygon with r reflex vertices requires at least $\lceil r/2 \rceil + 1$ convex parts.*

Proof. Each reflex vertex must be “cut” by at least one diagonal to reduce its interior angle below π . A single diagonal can resolve at most two reflex vertices (one at each endpoint). Thus at least $\lceil r/2 \rceil$ diagonals are needed, producing at least $\lceil r/2 \rceil + 1$ parts. $\square$

6.10.3 Pseudocode

Algorithm 12 Minimum Convex Partition via Dynamic Programming

```

1: function MINIMUMCONVEXPARTITION(polygon)
2:    $n \leftarrow \text{len}(\text{polygon.vertices})$ 
3:    $\text{reflex} \leftarrow [v \mid v \in \text{polygon} \wedge \text{IsReflex}(v)]$ 
4:   if  $\text{reflex}$  is empty then
5:     return  $[\text{polygon}]$ 
6:   end if
7:    $dp \leftarrow \text{array}[n][n]$ 
8:   for  $\text{length} = 2$  to  $n - 1$  do
9:     for  $i = 0$  to  $n - \text{length} - 1$  do
10:       $j \leftarrow i + \text{length}$ 
11:      if  $\neg \text{IsDiagonal}(i, j, \text{polygon})$  then
12:         $dp[i][j] \leftarrow \infty$ 
13:        continue
14:      end if
15:       $dp[i][j] \leftarrow \min_{k \in (i+1, \dots, j)} (dp[i][k] + dp[k][j])$ 
16:      if  $\text{ResolvesReflex}(i, j)$  then
17:         $dp[i][j] \leftarrow dp[i][j] - 1$ 
18:      end if
19:    end for
20:  end for
21:  return  $\text{ExtractPartition}(dp, \text{polygon})$ 
22: end function

```

6.10.4 Parameters

- **Polygon Type:** This optimal algorithm is for **simple polygons**. For polygons with holes, the problem is NP-hard.

- **Goal:** This specifically minimizes the number of **parts**. Other algorithms might minimize the **total length** of diagonals (Minimum Weight Convex Partition).

6.10.5 Complexity

- **Time Complexity:** $O(n^3)$ for the general dynamic programming approach. More optimized versions can reach $O(r^2n)$.
- **Space Complexity:** $O(n^2)$ to store the DP table.

6.10.6 Usages

- **Path Planning:** Breaking a complex floor plan into the fewest convex “rooms” to simplify robot navigation.
- **Computer Vision:** Representing a complex silhouette as a small number of convex components for shape matching.
- **Physics Engines:** Reducing the number of primitives needed for collision detection while maintaining accuracy.
- **Pattern Recognition:** Identifying structural features in complex polygonal datasets.

Example 6.55 (Optimal Partition of an L-Shape). Consider an L-shaped polygon with vertices (in order):

$$(0, 0), (3, 0), (3, 1), (1, 1), (1, 3), (0, 3).$$

This polygon has $r = 1$ reflex vertex at $(1, 1)$. By the lower bound (Theorem 6.54), at least $\lceil 1/2 \rceil + 1 = 2$ convex parts are needed. A single diagonal from $(1, 1)$ to $(0, 0)$ or from $(1, 1)$ to $(3, 3)$ would resolve the reflex vertex, yielding exactly 2 convex parts. This is optimal.

6.10.7 Testing and Edge Cases

- **Convex Polygon:** Should return the original polygon (1 part).
- **Star Polygon:** An optimal partition might be smaller than the number of reflex vertices.
- **“C” Shape:** Should be split into 3 convex parts optimally.
- **“L” Shape:** Should be split into 2 parts.
- **Numerical Precision:** Correctness of `IsDiagonal` and `IsReflex` is critical.
- **Holes:** Verify the algorithm correctly identifies that holes make the optimal solution much harder.

6.10.8 References

- Keil, J. M., & Snoeyink, J. S. (2002). “On the time complexity of optimal convex polygon partitioning”. Computational Geometry.
- Hertel, S., & Mehlhorn, K. (1983). “Fast triangulation of simple polygons”. FCT.
- Chazelle, B., & Dobkin, D. P. (1985). “Optimal convex decompositions”. Computational Geometry.
- Wikipedia: Convex polygon.

Chapter 7

Polygon Triangulation

7.1 Ear Clipping

Ear clipping is a simple and intuitive algorithm for triangulating a simple polygon. It is based on the Two-Ears Theorem, which states that every simple polygon with at least four vertices has at least two “ears” that can be removed to form a smaller simple polygon. By repeatedly clipping these ears, the entire polygon is eventually decomposed into triangles. The result was proved by Meisters [Mei75].

7.1.1 Definitions

Definition 7.1 (Simple Polygon). A polygon is *simple* if it does not self-intersect and has no holes.

Definition 7.2 (Convex Vertex). A vertex V_i of a polygon is *convex* if the interior angle at V_i is less than π (180°).

Definition 7.3 (Ear). A vertex V_i of a simple polygon is an *ear* if the triangle formed by V_{i-1}, V_i, V_{i+1} lies entirely within the polygon and contains no other vertices of the polygon in its interior.

Definition 7.4 (Barycentric Coordinates). *Barycentric coordinates* provide a coordinate system relative to a triangle. For a point P and triangle $\triangle ABC$, the barycentric coordinates $(\lambda_1, \lambda_2, \lambda_3)$ satisfy $P = \lambda_1 A + \lambda_2 B + \lambda_3 C$ with $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and $\lambda_i \geq 0$ if and only if P lies inside $\triangle ABC$.

7.1.2 Theory

The algorithm proceeds by identifying a vertex that satisfies two conditions:

1. The vertex is **convex** (the turn from V_{i-1} to V_i to V_{i+1} is counter-clockwise for CCW polygons).
2. The triangle formed by V_{i-1}, V_i, V_{i+1} contains no other vertices of the polygon in its interior.

Once such an ear is found, the triangle (V_{i-1}, V_i, V_{i+1}) is recorded, and the vertex V_i is removed from the polygon’s vertex list. This process is repeated until only three vertices remain, which form the final triangle.

Theorem 7.5 (Two-Ears Theorem). *Every simple polygon with $n \geq 4$ vertices has at least two non-overlapping ears.*

Theorem 7.6 (Ear Clipping Correctness). *The ear clipping algorithm produces a triangulation of any simple polygon with n vertices, consisting of exactly $n - 2$ triangles.*

Proof. We proceed by induction on n . For $n = 3$, the polygon is already a triangle. For $n \geq 4$, by the Two-Ears Theorem [Mei75], the polygon has at least one ear. Removing this ear produces a simple polygon with $n - 1$ vertices. By the inductive hypothesis, the smaller polygon can be triangulated into $(n - 1) - 2 = n - 3$ triangles. Adding the ear triangle yields $n - 2$ triangles in total. $\square$

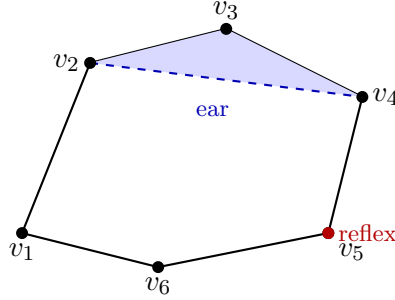


Figure 7.1: Ear clipping: vertex v_3 is an “ear” since triangle (v_2, v_3, v_4) is entirely inside the polygon and contains no other vertices.

7.1.3 Pseudocode

7.1.4 Parameters

- **Input:** A list of `Point2D` representing a simple polygon in counter-clockwise order.
- **Precision:** An epsilon should be used in cross-product and point-in-triangle tests to handle floating-point inaccuracies.

7.1.5 Complexity

- **Time Complexity:** $O(n^2)$ for a polygon with n vertices. In each step, we check $O(n)$ vertices for “ear-ness,” and each check takes $O(n)$ time to verify that no other points are inside.
- **Space Complexity:** $O(n)$ to store the triangles and the remaining vertex indices.

Example 7.7 (Ear Clipping Trace). Consider a quadrilateral with vertices (CCW): $V_0 = (0, 0)$, $V_1 = (2, 0)$, $V_2 = (2, 1)$, $V_3 = (0, 1)$. This is a convex rectangle, so every vertex is an ear.

1. Check V_0 : $\text{cross}(V_3, V_0, V_1) > 0$ and no other vertex is inside $\triangle V_3 V_0 V_1$. It is an ear.
2. Record triangle (V_3, V_0, V_1) and remove V_0 .
3. Remaining vertices: V_1, V_2, V_3 form a triangle.

Result: Two triangles (V_3, V_0, V_1) and (V_1, V_2, V_3) , which is the expected $n - 2 = 2$ triangles for a quadrilateral.

7.1.6 Usages

- Converting arbitrary polygons into triangles for rendering via OpenGL or DirectX.
- Preprocessing geometry for physics simulation and collision detection.
- Calculating the area or centroid of complex shapes.

Algorithm 13 Ear Clipping Triangulation

```
1: function TRIANGULATE(polygon)
2:   indices  $\leftarrow [0, 1, \dots, n - 1]$ 
3:   triangles  $\leftarrow []$ 
4:   while len(indices) > 3 do
5:     for i = 0 to len(indices) - 1 do
6:       u  $\leftarrow$  indices[i - 1]
7:       v  $\leftarrow$  indices[i]
8:       w  $\leftarrow$  indices[(i + 1) mod len(indices)]
9:       if IsEar(u, v, w, indices, polygon) then
10:        triangles.append((u, v, w))
11:        indices.pop(i)
12:        break
13:      end if
14:    end for
15:  end while
16:  triangles.append(tuple(indices))
17:  return triangles
18: end function
19: function ISEAR(u, v, w, current_indices, polygon)
20:  if cross_product(polygon[u], polygon[v], polygon[w])  $\leq 0$  then
21:    return False
22:  end if
23:  for index in current_indices do
24:    if index  $\in \{u, v, w\}$  then
25:      continue
26:    end if
27:    if PointInTriangle(polygon[index], polygon[u], polygon[v], polygon[w]) then
28:      return False
29:    end if
30:  end for
31:  return True
32: end function
```

7.1.7 Testing and Edge Cases

- **Concave Polygons:** Ensure the algorithm correctly skips reflex (non-convex) vertices.
- **Collinear Vertices:** Handled by strict inequality in the convexity check.
- **Self-intersecting Polygons:** Basic ear clipping may fail or produce incorrect results; the input must be simple.
- **Sliver Triangles:** Very thin polygons can lead to numerical stability issues.
- **Degenerate Polygons:** Polygons with 0, 1, or 2 vertices.

7.1.8 References

- Meisters, G. H. (1975). “Polygons have ears”. The American Mathematical Monthly.
- Eberly, D. (2002). “Triangulation by Ear Clipping”. Geometric Tools.
- Wikipedia: Polygon triangulation.

7.2 Monotone Decomposition

Monotone decomposition is the process of partitioning a simple polygon into a set of monotone polygons. A polygon is monotone with respect to a line L if every line orthogonal to L intersects the polygon in at most one connected component (a single line segment). This is typically a preprocessing step for more efficient polygon triangulation, as monotone polygons can be triangulated in linear time [PS85].

7.2.1 Definitions

Definition 7.8 (Monotone Polygon). A polygon is *monotone* with respect to a direction d if every line orthogonal to d intersects the polygon in at most one connected component (at most one line segment).

Definition 7.9 (Y-Monotone Polygon). A polygon is *y-monotone* if every horizontal line intersects the interior in at most one connected component. Equivalently, the polygon has no “cusp” vertices where both neighbors have y -coordinates on the same side.

Definition 7.10 (Cusp Vertex). A *cusp vertex* is a vertex where both adjacent vertices have y -coordinates either all above or all below it. Cusps violate y -monotonicity and are classified as:

- **Start/Split vertex:** interior is below, neighbors have higher y .
- **End/Merge vertex:** interior is above, neighbors have lower y .

7.2.2 Theory

The algorithm uses a **plane sweep** approach, moving a horizontal line from top to bottom. The key to making a polygon y -monotone is to eliminate all split and merge vertices by adding diagonals. A split vertex is connected to a vertex above it (the “helper” of the edge to its left), and a merge vertex is connected to a vertex below it when the sweep line reaches it. By the time the sweep is finished, all cusps that violated monotonicity are resolved.

Theorem 7.11 (Monotone Decomposition Correctness). *The sweep-line algorithm adds at most r diagonals to produce a decomposition of a simple polygon with r reflex vertices into at most $r + 1$ y -monotone sub-polygons.*

Proof. Each split or merge vertex requires exactly one diagonal to resolve the cusp. There are at most r such vertices (each reflex vertex is either a split, merge, or regular vertex; only split and merge vertices require diagonals). Adding one diagonal per such vertex resolves all monotonicity violations, yielding a partition into monotone pieces. $\square$

7.2.3 Pseudocode

Algorithm 14 Monotone Polygon Decomposition

```

1: function MONOTONEDECOMPOSITION(polygon)
2:   events  $\leftarrow$  sorted(polygon.vertices, key =  $\lambda v : (-v.y, v.x)$ )
3:   status  $\leftarrow$  BalancedBST()
4:   diagonals  $\leftarrow []$ 
5:   for v in events do
6:     if IsStartVertex(v) then
7:       HandleStartVertex(v, status)
8:     else if IsEndVertex(v) then
9:       HandleEndVertex(v, status, diagonals)
10:    else if IsSplitVertex(v) then
11:      HandleSplitVertex(v, status, diagonals)
12:    else if IsMergeVertex(v) then
13:      HandleMergeVertex(v, status, diagonals)
14:    else
15:      HandleRegularVertex(v, status, diagonals)
16:    end if
17:  end for
18:  return SplitPolygon(polygon, diagonals)
19: end function

```

7.2.4 Parameters

- **Sweep Direction:** Usually vertical (y -axis), but can be any arbitrary direction if the polygon is rotated.
- **Data Structures:** A balanced binary search tree (BST) for the status and a priority queue for events are standard.

7.2.5 Complexity

- **Time Complexity:** $O(n \log n)$ due to the sorting of vertices and the $O(\log n)$ operations on the balanced BST for each vertex.
- **Space Complexity:** $O(n)$ to store the polygon, the status tree, and the resulting diagonals.

7.2.6 Usages

- **Triangulation:** The first stage of $O(n \log n)$ polygon triangulation algorithms.
- **Trapezoidal Decomposition:** Often used in conjunction with monotone partitioning.
- **Path Planning:** Simplifying complex obstacles into monotone regions for easier navigation.

7.2.7 Testing and Edge Cases

- **Horizontal Edges:** Vertices with identical y -coordinates require consistent tie-breaking in the sort.
- **Complex Cusps:** Multiple split/merge vertices at the same height.
- **Strict Monotonicity:** Ensuring that horizontal lines intersecting a vertex or edge are handled correctly.
- **Non-Simple Polygons:** Monotone decomposition is generally defined for simple polygons; holes require special handling (treating them as merge/split events).

7.2.8 References

- Berg, M. de, Cheong, O., Kreveld, M. van, & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer-Verlag.
- Lee, D. T., & Preparata, F. P. (1977). “Location of a point in a planar subdivision and its applications”. SIAM Journal on Computing.
- Wikipedia: Monotone polygon.

7.3 Incremental Delaunay Triangulation

Incremental Delaunay triangulation is a fundamental algorithm for constructing a Delaunay triangulation of a set of points in the plane. The algorithm builds the triangulation by adding points one at a time and locally updating the mesh to maintain the Delaunay property. It was independently proposed by Bowyer [Bow81] and Watson [Wat81]. It is valued for its relative simplicity and efficiency, especially when combined with spatial sorting techniques.

7.3.1 Definitions

Definition 7.12 (Delaunay Property). A triangulation of a point set P is *Delaunay* if the circumcircle of every triangle contains no other point of P in its interior.

Definition 7.13 (In-Circle Test). The *in-circle test* determines whether a point D lies inside, on, or outside the circumcircle of triangle ABC . Given the determinant:

$$\text{InCircle}(A, B, C, D) = \begin{vmatrix} a_x - d_x & a_y - d_y & (a_x - d_x)^2 + (a_y - d_y)^2 \\ b_x - d_x & b_y - d_y & (b_x - d_x)^2 + (b_y - d_y)^2 \\ c_x - d_x & c_y - d_y & (c_x - d_x)^2 + (c_y - d_y)^2 \end{vmatrix}$$

the sign indicates: positive (inside), zero (on), negative (outside), assuming counter-clockwise orientation of ABC .

Definition 7.14 (Edge Flip). An *edge flip* replaces the common edge of two adjacent triangles in a triangulation with the other diagonal of their union quadrilateral.

7.3.2 Theory

The algorithm relies on the fact that any triangulation can be converted into a Delaunay triangulation through a series of edge flips. When a new point P is inserted into a triangle T , it is connected to the three vertices of T , creating three new triangles. The edges of these new triangles (which were the edges of T) may now be “illegal.” The algorithm recursively checks and flips these edges until the Delaunay property is restored globally.

Theorem 7.15 (Delaunay Triangulation Existence and Uniqueness). *For a set of n points in $\mathbb{R}^2$ in general position (no four points co-circular), the Delaunay triangulation exists and is unique.*

Theorem 7.16 (Incremental Delaunay Correctness). *The incremental insertion algorithm with edge legalization produces a valid Delaunay triangulation.*

Proof. The proof proceeds by induction on the number of inserted points. After inserting the first three points, the single triangle trivially satisfies the Delaunay property. When point p is inserted into triangle T , the three new triangles may violate the Delaunay property only along the edges shared with neighboring triangles. The LEGALIZEEDGE procedure checks the in-circle condition for each such edge. If violated, flipping the edge locally restores the Delaunay property for that edge. Since each flip can only create at most two new potentially illegal edges (among the new neighbors), the process terminates. The total number of flips is $O(n)$ in the expected case. $\square$

7.3.3 Pseudocode

Algorithm 15 Incremental Delaunay Triangulation

```

1: function INCREMENTALDELAUNAY(points)
2:    $ST \leftarrow \text{CreateSuperTriangle}(\text{points})$ 
3:    $\text{Triangulation} \leftarrow \{ST\}$ 
4:   for  $p$  in  $\text{points}$  do
5:      $T \leftarrow \text{FindTriangleContaining}(p, \text{Triangulation})$ 
6:      $\text{SplitTriangle}(T, p, \text{Triangulation})$ 
7:   end for
8:    $\text{RemoveSuperTriangleVertices}(\text{Triangulation})$ 
9:   return  $\text{Triangulation}$ 
10: end function
11: function SPLITTRIANGLE( $T, p, \text{Triangulation}$ )
12:    $(A, B, C) \leftarrow T.\text{vertices}$ 
13:    $\text{NewTris} \leftarrow [(p, A, B), (p, B, C), (p, C, A)]$ 
14:   Replace  $T$  with  $\text{NewTris}$  in  $\text{Triangulation}$ 
15:   for each  $\text{edge}$  in  $\{AB, BC, CA\}$  do
16:      $\text{LegalizeEdge}(p, \text{edge}, \text{Triangulation})$ 
17:   end for
18: end function
19: function LEGALIZEEDGE( $p, \text{edge}(U, V), \text{Triangulation}$ )
20:    $\text{Neighbor} \leftarrow \text{FindNeighbor}(\text{edge}, \text{Triangulation})$ 
21:   if  $\text{Neighbor}$  is None then
22:     return
23:   end if
24:    $W \leftarrow \text{Neighbor.vertex\_opposite\_to}(\text{edge})$ 
25:   if  $\text{InCircle}(p, U, V, W)$  then
26:      $\text{FlipEdge}(\text{edge}(U, V))$  to  $\text{edge}(p, W)$ 
27:      $\text{LegalizeEdge}(p, \text{edge}(U, W), \text{Triangulation})$ 
28:      $\text{LegalizeEdge}(p, \text{edge}(W, V), \text{Triangulation})$ 
29:   end if
30: end function

```

7.3.4 Parameters

- **Point Location:** Using a **Visibility Walk** or a **Point Grid** (spatial index) drastically speeds up finding the triangle containing the new point.
- **Insertion Order:** Sorting points along a **Hilbert Curve** or **Morton Code** ensures that subsequent points are geographically close, keeping the walk short.

7.3.5 Complexity

- **Time Complexity:** Average case $O(n \log n)$ with spatial sorting and efficient point location. Worst case $O(n^2)$ for highly structured point sets without randomization or sorting.
- **Space Complexity:** $O(n)$ to store the triangles and point coordinates.

7.3.6 Usages

- Generating high-quality meshes for Finite Element Method (FEM) analysis.
- Interpolation of scattered data (e.g., creating contour maps from elevation points).
- Pathfinding in robotics and video games.
- Dual representation of Voronoi diagrams.

7.3.7 Testing and Edge Cases

- **Co-circular Points:** Four or more points on the same circle. The algorithm will choose one valid triangulation, but the result may not be unique.
- **Collinear Points:** Points lying on a single line. Requires robust predicates or a small perturbation (epsilon).
- **Points on Edges:** If a point falls exactly on an existing edge, the edge should be split into four triangles instead of three.
- **Super-Triangle Selection:** Must be large enough to contain all points, but not so large that it causes numerical overflow.

7.3.8 References

- Bowyer, A. (1981). “Computing Dirichlet tessellations”. The Computer Journal.
- Watson, D. F. (1981). “Computing the n-dimensional Delaunay tessellation with application to Voronoi polytopes”. The Computer Journal.
- Guibas, L., & Stolfi, J. (1985). “Primitives for the manipulation of general subdivisions and the computation of Voronoi diagrams”. ACM Transactions on Graphics.
- Wikipedia: Delaunay triangulation.

7.4 Dynamic Delaunay Triangulation

Dynamic Delaunay triangulation is the process of maintaining a Delaunay triangulation as points are inserted into or deleted from the set. Unlike static construction, where the entire point set is known beforehand, dynamic algorithms must update the mesh locally to restore the Delaunay property while maintaining efficiency. This is critical for applications involving moving objects, real-time data streams, and interactive geometry editing.

7.4.1 Definitions

Definition 7.17 (Dynamic Triangulation). A *dynamic triangulation* maintains a valid triangulation under a sequence of point insertions and deletions, guaranteeing the triangulation invariant (here, the Delaunay property) after each update.

7.4.2 Theory

Insertion

Insertion typically uses the **Incremental Algorithm** (Algorithm 15):

1. **Locate**: Find triangle T containing new point P .
2. **Split**: Split T into three new triangles by connecting P to its vertices.
3. **Legalize**: Recursively check the edges of the new triangles and perform **Delaunay Flips** until all edges are legal (satisfy the empty circumcircle property).

Deletion

Deletion is more complex:

1. **Remove**: Remove the vertex V and all incident edges, leaving a star-shaped polygon (the “hole”).
2. **Re-triangulate**: Triangulate the hole. A simple way is to use a recursive “ear clipping” approach that specifically chooses diagonals satisfying the Delaunay property.
3. **Legalize**: Alternatively, perform any triangulation of the hole and then flip edges until it is Delaunay.

Theorem 7.18 (Expected Flip Complexity). *For n points inserted in random order, the total number of edge flips during incremental Delaunay triangulation is $O(n)$ in expectation.*

Proof. Guibas, Knuth, and Sharir [GS85] showed that the randomized analysis of edge flips leads to an amortized $O(1)$ flips per insertion. The key observation is that each flip can be “charged” to a pair of points whose circumcircle relation changes, and the total number of such chargeable events is bounded by $O(n)$ by a planar graph argument. $\square$

7.4.3 Pseudocode

Dynamic Insertion

Dynamic Deletion

7.4.4 Parameters

- **Point Location Strategy**: For high performance, use a **Stochastic Walk** or a **Directed Acyclic Graph (DAG)** (e.g., the Delaunay Tree) to achieve $O(\log n)$ location time.

Algorithm 16 Dynamic Point Insertion

```
1: function INSERTPOINT(triangulation,  $p$ )
2:    $T \leftarrow \text{FindTriangleContaining}(\text{triangulation}, p)$ 
3:    $\text{new\_triangles} \leftarrow \text{SplitTriangle}(T, p)$ 
4:    $\text{edges\_to\_check} \leftarrow \text{GetOuterEdges}(\text{new\_triangles})$ 
5:   while  $\text{edges\_to\_check}$  is not empty do
6:      $e \leftarrow \text{edges\_to\_check.pop}()$ 
7:     if  $\text{IsIllegal}(e, \text{triangulation})$  then
8:        $\text{new\_e} \leftarrow \text{Flip}(e, \text{triangulation})$ 
9:        $\text{edges\_to\_check.extend}(\text{GetSurroundingEdges}(\text{new\_e}))$ 
10:    end if
11:  end while
12: end function
```

Algorithm 17 Dynamic Point Deletion

```
1: function DELETEPOINT(triangulation,  $v$ )
2:    $\text{neighbors} \leftarrow \text{GetAdjacentVertices}(v, \text{triangulation})$ 
3:    $\text{RemoveVertex}(v, \text{triangulation})$ 
4:    $\text{FillHoleDelaunay}(\text{neighbors}, \text{triangulation})$ 
5: end function
```

- **Robustness:** Use robust predicates to prevent infinite loops during flipping caused by precision errors.

7.4.5 Complexity

- **Insertion:** $O(\log n)$ average for location, $O(1)$ expected for flipping (though $O(n)$ worst-case).
- **Deletion:** $O(\log n)$ average to find the vertex, $O(k)$ to re-triangulate where k is the degree of the vertex.
- **Space:** $O(n)$ to store the triangles and point location structures.

7.4.6 Usages

- **Terrain Modeling:** Adding new survey points to a digital elevation model.
- **Robotics:** Updating a map as a robot moves and discovers new obstacles.
- **Fluid Simulation:** Re-meshing around a moving boundary (e.g., a boat moving through water).
- **Mesh Refinement:** Splitting triangles to improve resolution in high-gradient areas.
- **Data Visualization:** Real-time plotting of scattered data points.

7.4.7 Testing and Edge Cases

- **Co-circular Points:** Ensure both insertion and deletion handle cases where four or more points lie on a circle (Delaunay triangulation is not unique).
- **Points on Boundary:** Handle points added to or removed from the convex hull correctly.

- **Degenerate Deletion:** Deleting a vertex that reduces the convex hull to a line or a single point.
- **Multiple Insertions at same position:** Handle duplicate points by ignoring them or updating metadata.
- **High-Valence Vertices:** Test deletion of vertices shared by many (e.g., 50+) triangles.

7.4.8 References

- Bowyer, A. (1981). “Computing Dirichlet tessellations”. The Computer Journal.
- Watson, D. F. (1981). “Computing the n-dimensional Delaunay tessellation with application to Voronoi polytopes”. The Computer Journal.
- Guibas, L., Knuth, D. E., & Sharir, M. (1992). “Randomized incremental construction of Delaunay and Voronoi diagrams”. Algorithmica.
- Devillers, O. (1999). “On handling transitions between different structures in geometric modeling”. Proceedings of the 11th Canadian Conference on Computational Geometry.
- Wikipedia: Delaunay triangulation.

7.5 Delaunay Flips

The edge flip algorithm is a local transformation technique used to convert any valid triangulation of a set of points into a **Delaunay triangulation**. It is based on the Lawson Flip Theorem [Law77], which states that any two triangulations of the same set of points are connected by a series of edge flips. This process iteratively improves the “quality” of the triangles by ensuring they satisfy the Delaunay (empty circumcircle) property.

7.5.1 Definitions

Definition 7.19 (Illegal Edge). An interior edge e shared by triangles $T_1 = (A, B, C)$ and $T_2 = (B, C, D)$ is *illegal* if D lies inside the circumcircle of $\triangle ABC$ (equivalently, if $\text{InCircle}(A, B, C, D) > 0$).

Definition 7.20 (Angle-Vector). The *angle-vector* of a triangulation is the sorted (non-decreasing) list of all interior angles. Delaunay flips lexicographically maximize this vector, yielding the triangulation that maximizes the minimum angle.

7.5.2 Theory

For any interior edge e shared by triangles $T_1 = (A, B, C)$ and $T_2 = (B, C, D)$, we check the **In-Circle** condition for vertex D relative to triangle ABC . If D is inside the circumcircle of ABC , then the edge BC is illegal.

Flipping an illegal edge BC produces a new edge AD and two new triangles ABD and ACD . It is a proven property that:

1. Flipping an illegal edge increases the minimum angle of the triangulation.
2. An illegal edge can only exist if the four vertices form a convex quadrilateral.
3. By repeatedly flipping illegal edges, the algorithm will eventually converge to the unique Delaunay triangulation (assuming no four points are co-circular).

Theorem 7.21 (Lawson’s Flip Theorem). *Any triangulation of a point set in general position can be transformed into the Delaunay triangulation by a finite sequence of edge flips. Furthermore, the number of flips is at most $O(n^2)$.*

Proof. Each flip strictly increases the angle-vector in lexicographic order. Since there are finitely many triangulations of n points (at most $O(27^n)$ by a result of Sharir and others), the process must terminate. The bound of $O(n^2)$ flips follows from an amortized analysis of the number of illegal edges created per flip. $\square$

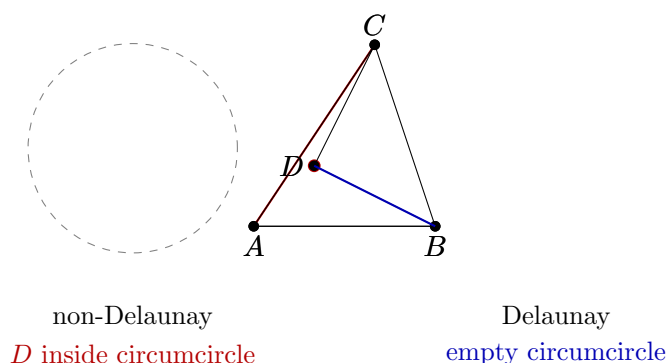


Figure 7.2: Delaunay property: (left) edge AC is illegal because D lies inside the circumcircle of ABC . (right) Flipping to edge BD satisfies the empty circumcircle property.

7.5.3 Pseudocode

7.5.4 Parameters

- **Robust Predicates:** The `InCircle` test must be implemented using robust geometric predicates (e.g., using `orient2d` and `incircle` from Shewchuk [She97]) to handle floating-point precision issues.
- **Data Structure:** A **Half-Edge** or **Quad-Edge** data structure is essential for efficiently finding neighboring faces and edges.

7.5.5 Complexity

- **Time Complexity:** In the worst case, $O(n^2)$ flips may be required for n points. However, starting from a “good” initial triangulation (like one from a sweep-line), it is often much faster.
- **Space Complexity:** $O(n)$ to store the triangulation and the queue of edges.

7.5.6 Usages

- **Mesh Refinement:** Improving the quality of an existing mesh for Finite Element analysis.
- **Incremental Construction:** Maintaining a Delaunay triangulation as points are added or moved.
- **Terrain Modeling:** Converting a set of elevation samples into a high-quality TIN (Triangulated Irregular Network).
- **Fluid Simulation:** Re-meshing to avoid thin, sliver triangles that cause numerical instability.

Algorithm 18 Delaunay Edge Flip Algorithm

```
1: function DELAUNAYFLIP(triangulation)
2:   illegal_edges  $\leftarrow$  Queue()
3:   for edge in triangulation.interior_edges do
4:     if IsIllegal(edge, triangulation) then
5:       illegal_edges.push(edge)
6:     end if
7:   end for
8:   while illegal_edges is not empty do
9:     edge  $\leftarrow$  illegal_edges.pop()
10:    if IsIllegal(edge, triangulation) then
11:      new_edge  $\leftarrow$  Flip(edge, triangulation)
12:      for neighbor in GetNeighboringEdges(new_edge) do
13:        if neighbor.is_interior then
14:          illegal_edges.push(neighbor)
15:        end if
16:      end for
17:    end if
18:  end while
19:  return triangulation
20: end function
21: function ISILLEGAL(edge(B, C), triangulation)
22:   T1  $\leftarrow$  triangulation.GetFace(edge)
23:   T2  $\leftarrow$  triangulation.GetOppositeFace(edge)
24:   A  $\leftarrow$  T1.vertex_opposite(edge)
25:   D  $\leftarrow$  T2.vertex_opposite(edge)
26:   return InCircle(A, B, C, D)
27: end function
```

7.5.7 Testing and Edge Cases

- **Co-circular Points:** Four points on a circle mean both diagonals are “legal.” The algorithm should terminate consistently.
- **Non-Convex Quadrilateral:** Ensure the flip logic only applies to convex quadrilaterals (though an illegal edge is guaranteed to be a diagonal of a convex quad).
- **Boundary Edges:** Edges on the convex hull cannot be flipped.
- **Sliver Triangles:** Verify that flips successfully remove extremely thin triangles.
- **Convergence:** Test on highly structured point sets (like a grid) to ensure the algorithm doesn’t loop infinitely due to precision errors.

7.5.8 References

- Lawson, C. L. (1977). “Software for C^1 surface interpolation”. Mathematical Software.
- Guibas, L., & Stolfi, J. (1985). “Primitives for the manipulation of general subdivisions and the computation of Voronoi diagrams”. ACM Transactions on Graphics.
- Shewchuk, J. R. (1997). “Adaptive Precision Floating-Point Arithmetic and Fast Robust Geometric Predicates”. Discrete & Computational Geometry.
- Wikipedia: Delaunay triangulation.

7.6 Constrained Delaunay Triangulation

Constrained Delaunay Triangulation (CDT) is a version of Delaunay triangulation that forces certain edges (constraints) into the triangulation. It is essential for representing domains with specific boundaries, holes, or internal features that must be preserved.

7.6.1 Overview

While a standard Delaunay triangulation maximizes the minimum angle and satisfies the “empty circumcircle” property for all points, it may not respect the boundaries of a non-convex polygon. CDT modifies the Delaunay criteria to ensure that all constraint edges are present in the final mesh, while maintaining the Delaunay property for all other edges as much as possible.

7.6.2 Implementation Details

The CDT is implemented using an iterative edge-flip approach:

1. **Initial Triangulation:** The domain (including holes) is initially triangulated using a standard polygon triangulation algorithm.
2. **Constraint Enforcement:** Bridge edges are added to connect holes to the outer boundary, creating a single degenerate polygon that is then triangulated.
3. **Edge Flipping:** All internal edges that are not part of the constraints are placed in a queue. Edges are flipped if they violate the Delaunay condition, provided the flip doesn’t cross a constraint and maintains the domain’s validity.

Algorithm 19 Constrained Delaunay Triangulation

```

1: function CONSTRAINEDDELAUNAYTRIANGULATION(outer, holes)
2:   triangulation  $\leftarrow$  TriangulateDomain(outer, holes)
3:   edge_queue  $\leftarrow$  GetInteriorNonConstraintEdges(triangulation)
4:   while edge_queue is not empty do
5:     e  $\leftarrow$  edge_queue.pop()
6:     if IsIllegal(e) and  $\neg$  CrossesConstraint(e) then
7:       Flip(e, triangulation)
8:       edge_queue.extend(GetAffectedEdges(e))
9:     end if
10:  end while
11:  return triangulation
12: end function

```

7.6.3 Pseudocode**7.6.4 Usage Example**

```

from compgeom.mesh import constrained_delaunay_triangulation
from compgeom import Point2D

outer = [Point2D(0,0), Point2D(10,0), Point2D(10,10), Point2D(0,10)]
holes = [[Point2D(4,4), Point2D(6,4), Point2D(6,6), Point2D(4,6)]]

triangles, constrained_edges = constrained_delaunay_triangulation(outer, holes)

```

7.6.5 References

- Chew, L. Paul. “Constrained Delaunay Triangulations.” *Algorithmica*, 1989.
- Sloan, S. W. “A Fast Algorithm for Generating Constrained Delaunay Triangulations.” *Computers & Structures*, 1993.

7.7 Non-Obtuse Triangulation

Non-Obtuse Triangulation is the process of subdividing a 2D domain into triangles such that no internal angle exceeds 90° .

7.7.1 Overview

Most standard triangulation algorithms (including Delaunay) may produce obtuse triangles. For certain numerical simulations (like finite element analysis or fluid dynamics), obtuse triangles can introduce errors or numerical instability. A non-obtuse triangulation ensures that all angles are acute or right angles.

7.7.2 Implementation Details

The `NonObtuseTriangulator` uses an iterative Steiner point insertion strategy:

1. **Initial Mesh:** Start with a Delaunay triangulation of the input point set.
2. **Detection:** Identify triangles with an angle $> 90^\circ$.

3. **Refinement:** For each obtuse triangle, insert a new vertex (Steiner point) at the midpoint of its longest edge.
4. **Re-triangulation:** Update the Delaunay triangulation with the new point.
5. **Iteration:** Repeat until no obtuse triangles remain or a maximum number of Steiner points is reached.

7.7.3 Pseudocode

Algorithm 20 Non-Obtuse Triangulation

```
1: function NONOBTUSETRIANGULATE(points, max_steiner)
2:   mesh  $\leftarrow$  DelaunayTriangulation(points)
3:   steiner_count  $\leftarrow$  0
4:   while hasObtuse(mesh) and steiner_count < max_steiner do
5:     T  $\leftarrow$  findObtuseTriangle(mesh)
6:     edge  $\leftarrow$  longestEdge(T)
7:     midpoint  $\leftarrow$  midpoint(edge)
8:     InsertPoint(mesh, midpoint)
9:     steiner_count  $\leftarrow$  steiner_count + 1
10:  end while
11:  return mesh
12: end function
```

7.7.4 Usage Example

```
from compgeom.mesh.surface.trimesh.non_obtuse_triangulation import NonObtuseTriangulator
from compgeom import Point2D
```

```
pts = [Point2D(0, 0), Point2D(10, 0), Point2D(5, 0.1)]
triangulator = NonObtuseTriangulator(pts)
mesh = triangulator.triangulate(max_steiner=100)
```

7.7.5 References

- Bern, M., et al. “Provably Good Mesh Generation.” Journal of Computer and System Sciences, 1994.
- CG:SHOP 2025 Challenge: “Minimum Non-Obtuse Triangulation.”

Part III

Geometric Searching and Spatial Data Structures

Chapter 8

Point Location

8.1 The Planar Point-Location Problem

The **point-location problem** is: given a planar subdivision (a partition of the plane into disjoint faces bounded by edges) and a query point, find the face containing the query point. Point location is the workhorse query of computational geometry: locating a click in a map, a street address in a census tract, a site in a Voronoi diagram, or a point in a triangulated terrain are all instances. Formally, the subdivision is a straight-line planar graph whose edges and vertices partition the plane into faces; a query returns the face, the edge, or the vertex that contains the point, with the convention that points on a boundary are resolved by a consistent rule.

The difficulty of point location is entirely a matter of the data structure built on the subdivision. A naive scan over faces takes $O(n)$ time per query; the goal is to answer each query in $O(\log n)$ time with $O(n)$ storage after $O(n \log n)$ (or $O(n)$) preprocessing. This chapter develops the classical solutions in increasing order of sophistication: slab decomposition (Section 8.3), trapezoidal maps with randomized incremental construction (Sections 8.4–8.6), Kirkpatrick’s hierarchy (Section 8.7), and the walking methods used for triangulations (Section 8.8). Dynamic and practical aspects follow in Sections 8.9–8.10.

The simplest special case is point location in a convex polygon, which is two binary searches (Section 8.1); the general case requires the full machinery of this chapter. Point location is dual in spirit to the range searching of Chapter 9: there the points were fixed and the ranges varied, here the subdivision is fixed and the query points vary.

8.2 Subdivision Representations

A planar subdivision must be stored in a representation that supports the two operations a point-location structure needs: enumerating the edges around a face, and testing adjacency between faces. The standard representation is the **half-edge (doubly connected edge list, DCEL)**: each undirected edge is stored as two directed *half-edges* with opposite directions, each owning its incident face and the previous/next half-edge around that face. The DCEL supports $O(1)$ queries of “which face is left of this half-edge?” and traversal around a face, which the incremental and hierarchical point-location algorithms of this chapter require.

Representations differ in what they are designed for:

- the **DCEL** is the reference representation for general planar subdivisions and the trapezoidal-map construction of Section 8.4;
- a **triangulation** stored as a triangle list or as an index-based neighbor structure supports the walking methods of Section 8.8;

- the **arrangement** representation of Chapter 14 stores subdivisions of the plane by line segments and supports the same location queries.

The choice of representation affects robustness: an inconsistent DCEL (cracked edges, duplicate vertices) breaks location queries, which is why the half-edge machinery of Chapter 21 and the mesh validity checks of Chapter 30.1 matter in practice.

8.3 Slab Decomposition

8.3.1 1. Overview

The first point-location structure, due to Dobkin and Lipton, decomposes the plane into vertical **slabs** by extending every vertex upward and downward to infinity. Within a slab, the subdivision edges crossing it appear as non-crossing vertical order; a query then reduces to two binary searches: one on the slab's x -coordinate and one on the vertical order of the edges inside the slab.

8.3.2 2. Definitions

Definition 8.1 (Slab). Given a planar subdivision with vertices sorted by x -coordinate, a **slab** is the open vertical strip between two consecutive vertex x -coordinates. The edges of the subdivision crossing a slab are pairwise non-crossing and therefore have a total vertical order.

8.3.3 3. Theory

Within a slab the answer is determined by the vertical order: a query point (x_q, y_q) falls between two consecutive edges (or below the lowest or above the highest), and the face is read off from the interval. Since the edge order differs from slab to slab, each slab stores its own sorted list. The two binary searches locate the slab ($O(\log n)$) and the interval ($O(\log n)$), giving $O(\log n)$ query time.

8.3.4 4. Pseudocode

Algorithm 21 Point Location by Slab Decomposition

Input: Subdivision with slabs $S_1, \dots, S_m$ (each with sorted edge order), query point q .

Output: The face of the subdivision containing q .

```

1: function SLABLOCATE( $q$ )
2:    $i \leftarrow \text{binarySearch}(\text{vertex } x\text{-coordinates}, q_x)$ 
3:    $j \leftarrow \text{binarySearch}(S_i.\text{edges sorted by } y\text{-order}, q_y)$ 
4:   return the face stored for interval  $j$  of slab  $S_i$ 
5: end function
```

8.3.5 5. Parameter Selection

- **Vertex order:** vertices with equal x must be ordered consistently so the slab boundaries are well defined; this is a degeneracy issue of the kind treated in Section 5.9.
- **Representation:** each slab stores, per interval, a pointer to the incident face; storing only the edges and reconstructing the face from the DCEL halves the storage.

8.3.6 6. Complexity

- **Storage:** $O(n^2)$: there are $O(n)$ slabs and each slab can hold $O(n)$ edges.
- **Preprocessing:** $O(n^2)$ by sorting each slab's edges.
- **Query:** $O(\log n)$.

The quadratic storage is the reason slab decomposition is of historical interest only; the trapezoidal-map structure of Section 8.4 achieves $O(n)$ space by merging the slab lists.

8.3.7 7. Usages

- **Small subdivisions:** for a few hundred edges the quadratic storage is acceptable and the structure is trivially robust.
- **Teaching:** the structure introduces the vertical-decomposition idea that the trapezoidal map refines.

8.3.8 8. Testing Methods and Edge Cases

- **Points on edges and vertices:** the query must consistently return the edge/vertex or the incident face; pick one convention and test it.
- **Vertical edges:** an edge with constant x lies on a slab boundary; it must be assigned to exactly one of the two adjacent slabs.

8.3.9 9. References

Slab decomposition is presented in the early literature and textbook treatments of point location [PS85, BCKO08]; its quadratic space is the motivation for the linear-space structures that follow.

8.4 Trapezoidal Maps

8.4.1 1. Overview

The **trapezoidal map** (vertical decomposition) refines a planar subdivision into *trapezoids*: from each vertex and each segment endpoint, a vertical extension is drawn upward and downward until it hits a segment of the subdivision. The resulting trapezoids have at most two vertical sides and two (possibly missing) non-vertical sides, and every face of the original subdivision is a union of trapezoids. The map has $O(n)$ trapezoids, and point location in it is the basis of the randomized incremental algorithm of Section 8.5.

8.4.2 2. Definitions

Definition 8.2 (Trapezoid). A **trapezoid** of a trapezoidal map is a cell whose top and bottom boundaries are (possibly degenerate) segments of the input and whose left and right boundaries are vertical extensions from vertices or segment endpoints.

Definition 8.3 (Vertical Decomposition). The **vertical decomposition** of a set of segments is the partition of the plane induced by drawing, from each vertex, the maximal vertical segment that stays inside the complement of the segment set.

8.4.3 3. Theory

The key property of the trapezoidal map is its size: with n segments in general position, the map has at most $6n + 3$ trapezoids. This linear bound follows because each segment endpoint creates at most two new trapezoids (its top and bottom extensions enter and leave existing trapezoids). Since each trapezoid is adjacent to its vertical-side neighbors, the map is a planar subdivision of the plane, and locating a point in the original subdivision is equivalent to locating it in the trapezoidal map.

8.4.4 4. Pseudocode

Algorithm 22 Build the Trapezoidal Map of a Segment Set

Input: n line segments in general position.

Output: The trapezoidal map $\mathcal{T}$ and a history DAG for queries.

```

1: function BUILDTRAPEZOIDALMAP( $S$ )
2:    $T \leftarrow$  trapezoid covering the bounding box of  $S$ 
3:   for segment  $s$  in a random order do
4:     locate the trapezoid(s) of  $T$  containing  $s$ 
5:     split the affected trapezoids by the vertical extensions of  $s$ 's endpoints
6:     cut the segment into the trapezoids it crosses, updating adjacency
7:     record the split in the history DAG
8:   end for
9:   return  $T$ 
10: end function

```

8.4.5 5. Parameter Selection

- **Insertion order:** the segments are inserted in random order; the expected number of trapezoids ever created is $O(n \log n)$ because a segment's endpoints fall into expected $O(\log n)$ trapezoids over the construction.
- **Bounding box:** a large box avoids unbounded trapezoids and simplifies the data structure.
- **General position:** if endpoints share x -coordinates or segments touch, the extensions are handled by the tie-breaking rules of Section 5.9.

8.4.6 6. Complexity

- **Storage:** $O(n)$.
- **Preprocessing:** expected $O(n \log n)$ time, expected $O(n)$ space, under a random insertion order [Sei91, BCKO08].
- **Query:** expected $O(\log n)$ via the history DAG of Section 8.6.

8.4.7 7. Usages

- **Map overlay:** computing the overlay of two subdivisions starts by trapezoidal decomposition of one of them (Section 5.11).
- **Computing arrangements:** the trapezoidal map is the standard structure for building arrangements of line segments (Chapter 14).
- **Voronoi and Delaunay queries:** locating a point in a Voronoi diagram to answer nearest-site queries.

8.4.8 8. Testing Methods and Edge Cases

- **Count invariant:** verify the number of trapezoids against the $6n + 3$ bound on random segments.
- **Endpoint ties:** two segments sharing an endpoint, and a vertex lying exactly above another, must produce the same map regardless of insertion order.
- **Consistency:** locate random points both in the trapezoidal map and by brute-force face scan, and require identical answers.

8.4.9 9. References

The trapezoidal map and its randomized incremental construction with a history DAG are due to Mulmuley and to Seidel [Sei91]; the textbook treatment is in [BCK008].

8.5 Randomized Incremental Construction

8.5.1 1. Overview

The trapezoidal map is built incrementally: segments are inserted one at a time, in a random order, and each insertion splits the trapezoids the segment crosses. The construction simultaneously maintains a **history DAG** (Section 8.6) that turns every construction step into a query step. The expected behavior is remarkable: although a single insertion can change many trapezoids, the expected total work over a random order is $O(n \log n)$, and the expected query time is $O(\log n)$.

8.5.2 2. Definitions

Definition 8.4 (Insertion Step). Inserting a segment s into the trapezoidal map splits the trapezoids that s crosses: the vertical extensions of s 's endpoints create up to four new trapezoids, and s itself bisects the crossed trapezoids.

8.5.3 3. Theory

The analysis has two parts. First, the expected number of trapezoids created by inserting n segments in random order is $O(n \log n)$: a given trapezoid survives until one of the two input segments bounding it is inserted, and a backward analysis shows each insertion destroys expected $O(1)$ trapezoids created so far. Second, the history DAG depth is expected $O(\log n)$: a query point's trajectory changes only when one of the trapezoids on its path is split, and backward analysis bounds the number of such splits by the expected $O(\log n)$.

8.5.4 4. Pseudocode

The insertion procedure is given in Algorithm 22; the query procedure uses the history DAG built alongside:

8.5.5 5. Parameter Selection

- **Randomization:** the random order is essential for the expected bounds; a deterministic bad order (e.g., left-to-right) makes the depth $\Theta(n)$.
- **Memory of the DAG:** the DAG has expected $O(n)$ nodes but can grow; production implementations periodically rebuild the structure when the number of nodes exceeds a constant times the map size.

Algorithm 23 Query by History DAG

Input: History DAG rooted at the initial bounding trapezoid, query point q .**Output:** The trapezoid (hence face) containing q .

```
1: function QUERYDAG( $q$ )
2:    $v \leftarrow \text{root}; t \leftarrow \text{trapezoid of } v$ 
3:   while  $v$  has children do
4:      $v \leftarrow \text{the child of } v \text{ containing } q$ 
5:      $t \leftarrow \text{trapezoid of } v$ 
6:   end while
7:   return  $t$ 
8: end function
```

8.5.6 6. Complexity

- **Expected time:** $O(n \log n)$ preprocessing, $O(\log n)$ per query.
- **Expected space:** $O(n)$ for the map plus $O(n)$ for the DAG.

All bounds hold with high probability under a uniformly random insertion order [Sei91].

8.5.7 7. Usages

- **Interactive graphics:** clicking to select a face of a map or mesh.
- **Voronoi nearest-site queries:** locating the query point in the Voronoi diagram returns its nearest site.
- **CSG and boolean geometry:** face classification in polygon boolean operations (Chapter 6).

8.5.8 8. Testing Methods and Edge Cases

- **Order invariance:** the final map must be the same for any insertion order; the DAG differs but queries must agree.
- **Random-point sweep:** query millions of random points and compare against a brute-force reference.
- **Boundary queries:** points exactly on an input segment must resolve to a consistent side by the fixed convention.

8.5.9 9. References

The randomized incremental construction is presented in [Sei91] and [BCKO08]; the general framework of randomized incremental algorithms is developed in Chapter 31.

8.6 Search DAGs

The **search DAG** (history DAG) is the data structure that turns the incremental construction of Section 8.5 into a query structure. Every time an insertion splits a trapezoid t into trapezoids $t_1, \dots, t_j$, the node for t is given children $t_1, \dots, t_j$ in the DAG. A query walks from the root: at each node, test which child contains the query point, descend, and repeat until reaching a leaf (a trapezoid of the final map). The leaf identifies the face.

The structure is a DAG and not a tree because a trapezoid created by an early insertion can be split again later, so several construction paths lead to the same trapezoid; conversely, the query is correct because the trapezoids containing the query point at every construction step form a nested chain, and each child test is an $O(1)$ containment check against a trapezoid. The expected depth is $O(\log n)$, which is the expected query time. The DAG has expected $O(n)$ nodes (one per trapezoid ever created), though the constant is larger than the map's.

The search-DAG idea is the special case for point location of the general **history-based** technique for randomized incremental algorithms: the history of the construction doubles as the query structure, which is why the Delaunay-based point location of Section 8.8 also uses a history DAG (the **Delaunay hierarchy** of [Dev02]). For implementations, the DAG's memory is the practical constraint, and rebuilding once the node count exceeds a threshold keeps both memory and cache behavior under control.

8.7 Kirkpatrick's Hierarchy

8.7.1 1. Overview

Kirkpatrick's algorithm [Kir83] achieves *worst-case* $O(\log n)$ point location with $O(n)$ space by building a hierarchy of progressively coarser triangulations: a sequence of triangulations $T_0 \supset T_1 \supset \dots \supset T_m$ where each level refines the next and each vertex of T_{i+1} keeps pointers to the constant number of vertices of T_i it covers. The construction requires the subdivision to be a triangulation with no vertices of degree larger than a constant, which is achieved by a preliminary refinement step.

8.7.2 2. Definitions

Definition 8.5 (Independent Set). An **independent set** of a planar graph is a set of vertices no two of which are adjacent. In a planar triangulation of n vertices, an independent set of size at least $n/18$ with bounded degree always exists and can be found in linear time.

Definition 8.6 (Kirkpatrick Hierarchy). A **Kirkpatrick hierarchy** is a sequence of triangulations $T_0, T_1, \dots, T_m$ where T_{i+1} is obtained from T_i by removing an independent set of bounded-degree vertices and retriangulating the holes, so that T_{i+1} has at most a constant fraction of the vertices of T_i .

8.7.3 3. Theory

Each level removes an independent set of bounded degree, so the remaining graph is still a triangulation (holes are retriangulated), and the size decreases by a constant factor per level, giving $m = O(\log n)$ levels. Retriangulating a hole around a removed vertex of degree at most d produces at most d new triangles, and each new triangle stores a pointer to the (constant number of) triangles it overlaps in the previous level. A query descends level by level: at level $i+1$, the located triangle points to its $O(1)$ overlapping triangles in level i , one of which contains the query point.

8.7.4 4. Pseudocode

8.7.5 5. Parameter Selection

- **Preprocessing:** refine the input subdivision into a triangulation with bounded degree by inserting a bounded number of Steiner points per edge and face; this adds $O(n)$ vertices.
- **Independent-set choice:** any maximal independent set of bounded degree works; greedily selecting low-degree vertices keeps the retriangulation cheap.

Algorithm 24 Kirkpatrick Point Location

Input: Triangulation T_0 and its Kirkpatrick hierarchy, query point q .**Output:** The triangle of T_0 containing q .

```
1: function KIRKPATRICKLOCATE( $q$ )
2:    $i \leftarrow m$ ;  $t \leftarrow$  the single triangle of  $T_m$ 
3:   while  $i > 0$  do
4:      $t' \leftarrow$  the triangle of  $T_{i-1}$  overlapping  $t$  that contains  $q$ 
5:      $t \leftarrow t'$ ;  $i \leftarrow i - 1$ 
6:   end while
7:   return  $t$ 
8: end function
```

8.7.6 6. Complexity

- **Storage:** $O(n)$.
- **Preprocessing:** $O(n)$ (with $O(n)$ Steiner points).
- **Query:** $O(\log n)$ worst case, with a small constant (about 12 pointer dereferences per level in the classic implementation).

8.7.7 7. Usages

- **Worst-case guarantees:** the only linear-space planar structure with a deterministic $O(\log n)$ query is the choice when query-time variability is unacceptable.
- **Static triangulations:** terrains and meshes that are built once and queried many times.

8.7.8 8. Testing Methods and Edge Cases

- **Degree bound:** verify the refinement step produced a triangulation whose maximum degree is bounded; the independent-set lemma fails otherwise.
- **Overlap pointers:** for every triangle of level $i + 1$, all its overlapping level- i triangles must cover it exactly once.
- **Consistency:** brute-force comparison on random queries, plus boundary queries on vertices and edges.

8.7.9 9. References

The hierarchy and its analysis are due to Kirkpatrick [Kir83]; textbook treatments appear in [BCKO08, PS85].

8.8 Point Location in Triangulations

8.8.1 1. Overview

A triangulation (of a point set, a terrain, or a mesh) is located by walking through the triangles: from a known starting triangle, cross the edge that the query point lies to the other side of, and repeat until the triangle containing the query point is found. Walking methods are the method of choice inside triangulation and mesh algorithms because they require almost no preprocessing (often none) and run in expected $O(\log n)$ time when the start is well chosen. Walking methods locate a query point by crossing mesh edges (2D) or faces (3D) from a starting simplex toward

the query point. The visibility walk and jump-and-walk are described in Section 13.11, which covers both triangles and tetrahedra and the expected $O(\log n)$ walk length with a spatial-index start.

8.8.2 2. Definitions

Definition 8.7 (Visibility Walk). In a **visibility walk**, from the current triangle t , test the three edges of t ; if the query point lies to the outside of one of them, cross that edge into the adjacent triangle and repeat.

Definition 8.8 (Jump-and-Walk). **Jump-and-walk** first finds a start triangle near the query point using a spatial index (a grid, a k -d tree, or a space-filling-curve index), then runs a visibility walk from it.

8.8.3 3. Theory

The visibility walk is correct if the triangulation is a valid triangulation of its domain and the walk never cycles; for a Delaunay triangulation, the walk moves monotonically closer to the query point and terminates. The number of steps is proportional to the graph distance in the triangulation, which in the worst case is $\Theta(\sqrt{n})$ for a triangular grid but in practice is small. The **Delaunay hierarchy** [Dev02] reduces the walk to expected $O(\log n)$ by building a hierarchy of randomly sampled triangulations: each level walks on a coarser triangulation and jumps into the next. Jump-and-walk [MSZ99] achieves expected $O(\sqrt{n})$ steps on random Delaunay triangulations without any hierarchy, and near-constant practical time when the grid start is well chosen.

8.8.4 4. Pseudocode

Algorithm 25 Visibility Walk in a Triangulation

Input: Triangulation with neighbor pointers, start triangle t_0 , query point q .

Output: The triangle containing q .

```

1: function VISIBILITYWALK( $t_0, q$ )
2:    $t \leftarrow t_0$ 
3:   while true do
4:     for each edge  $e$  of  $t$  do
5:       if  $q$  lies strictly outside the half-plane of  $e$  then
6:          $t \leftarrow$  neighbor of  $t$  across  $e$ ; break
7:       end if
8:     end for
9:     if no edge was crossed then
10:      return  $t$ 
11:    end if
12:  end while
13: end function

```

8.8.5 5. Parameter Selection

- **Start choice:** the start triangle determines the walk length; a grid or k -d-tree index (Chapter 10) gives a near-optimal start in expected $O(\log n)$ time.
- **Walk order:** checking edges in a fixed cyclic order makes the walk deterministic; when the query lies outside the domain, the walk should terminate by reaching the boundary.

- **Degeneracy:** if the query lies exactly on an edge, the convention “belongs to the higher-indexed face” must be fixed to keep the walk from cycling.

8.8.6 6. Complexity

- **Preprocessing:** none for the bare walk; $O(n \log n)$ for a spatial index.
- **Query:** worst-case $\Theta(\sqrt{n})$ steps on a grid; expected $O(\log n)$ with a Delaunay hierarchy; expected $O(\sqrt{n})$ for jump-and-walk on random triangulations.

8.8.7 7. Usages

- **Delaunay and Voronoi algorithms:** inserting a point into a Delaunay triangulation first locates it (Chapter 12); the history DAG of Section 8.6 is the standard accelerator there.
- **Finite elements:** locating integration points in a mesh (Chapter 30.1).
- **Terrain interpolation:** locating a query position in a TIN before interpolating elevation (Chapter 36).

8.8.8 8. Testing Methods and Edge Cases

- **Outside points:** for a bounded triangulation, verify the walk reports “outside” when the query is beyond the domain.
- **On-edge and on-vertex queries:** fix and test the convention so the walk terminates.
- **Consistency:** compare against the brute-force face scan on random and adversarial queries.

8.8.9 9. References

Walking methods are covered in [BCKO08] and in the specialized literature on triangulation point location: the Delaunay hierarchy [Dev02] and the jump-and-walk analysis [MSZ99].

8.9 Dynamic Point Location

When the subdivision itself changes (edges inserted, deleted, moved), the point-location structure must be updated. The main dynamic scenarios are:

Incremental construction The trapezoidal map of Section 8.5 is inherently incremental: each insertion updates the map and the history DAG in expected $O(\log n)$ time, so “point location while building” is free.

Deletions Removing segments from a trapezoidal map requires merging trapezoids and updating the DAG; the expected cost is $O(\log^2 n)$ with the standard techniques [Ove83].

Moving subdivisions When vertices move along known trajectories (kinetic subdivisions), the combinatorial structure changes only at discrete events; kinetic data structures maintain point-location structures across events (Chapter 34).

For triangulations, insertions and deletions are local flips (Section 8.8), and walking locates the affected triangles; the Delaunay hierarchy supports deletions with expected $O(\log n)$ updates. The general theory of transforming static search structures into dynamic ones is Overmars’ transform-and-conquer method [Ove83], which yields logarithmic amortized update costs for most planar point-location structures at the price of larger constants.

8.10 Practical Implementations

Production point-location code differs from the textbook algorithms in predictable ways. The robust-predicate discipline of Section 3.4 is mandatory: a query whose orientation tests are inconsistent can walk into a cycle or return the wrong face, and the failures are silent. Implementations therefore:

- **Use exact predicates:** every edge-side test in the walk, the DAG descent, and the hierarchy descent is an orientation predicate evaluated with adaptive exact arithmetic (Chapter 3).
- **Fix boundary conventions:** points on edges and vertices are resolved by a documented rule (e.g., “open on the right side”) applied uniformly by all code paths.
- **Index the start:** for triangulations, a grid or space-filling-curve index (Chapter 10.10) gives jump-and-walk a constant-time start on realistic data, beating the DAG in cache behavior despite the same asymptotics.
- **Use cache-friendly layouts:** traversing the DCEL or the triangle neighbors is pointer-chasing; layouts that store triangles in adjacency order (space-filling curves, breadth-first numbering) improve walk performance severalfold.
- **Build the structure once:** for a static mesh queried many times, a k -d-tree start plus visibility walk is usually the best engineering trade-off; for a subdivision that changes, the trapezoidal map with a history DAG remains the reference.

The practical guidance for the reader is to start with a visibility walk over a k -d-tree start (simplest to implement correctly with exact predicates), benchmark against the application’s query distribution, and move to a trapezoidal map or Delaunay hierarchy only when asymptotic behavior demands it. The spatial data structures of Chapter 10 and the space-filling curves of Chapter 10.10 provide the indexing primitives.

8.11 Exercises

1. Prove that a convex polygon can be point-located with two binary searches: first find the vertex sector by x , then test the relevant edge.
2. Show that the trapezoidal map of n segments has at most $6n + 3$ trapezoids.
3. Explain why the history DAG of Section 8.6 is a DAG and not a tree, and why queries still terminate at the leaf.
4. Prove the independent-set lemma used by Kirkpatrick’s hierarchy: any planar triangulation with n vertices has an independent set of size at least $n/18$ with bounded degree.
5. Give a query point and a triangulation on which the visibility walk of Section 8.8 takes $\Theta(\sqrt{n})$ steps, and explain why a Delaunay hierarchy avoids this.
6. Compare the expected query time, space, and preprocessing of slab decomposition, the trapezoidal map, Kirkpatrick’s hierarchy, and jump-and-walk for $n = 10^3, 10^6$.
7. Implement a brute-force locator and validate a trapezoidal-map implementation on random segments, verifying the $6n + 3$ count and the agreement of the DAG with the brute-force answers on 10^5 random queries.
8. Extend the validation to boundary queries (points on edges and vertices) and to degenerate inputs (vertical segments, shared endpoints), and document the boundary convention used.

Chapter 9

Orthogonal Range Searching

9.1 Range Queries

A **range query** asks for the points of a fixed set $P \subset \mathbb{R}^d$ that lie inside a query region Q : the output is either the *count* $|P \cap Q|$ or the *report* of all points in $P \cap Q$. The queries are answered by a data structure built once from P , and the structure is judged by its storage, its query time, and its construction and update costs. The problem is fundamental because it is the primitive behind database range selection, geographic windowing, multi-dimensional indexing, and the nearest-neighbor queries of Chapter 13.

The query region Q determines the difficulty of the problem:

Orthogonal ranges Q is an axis-aligned box (a rectangle in the plane, a box in $\mathbb{R}^d$), possibly with only some sides specified (a 3-sided rectangle, a half-plane strip). These queries support the elegant tree structures of this chapter.

Simplex ranges Q is a half-space or a simplex; these queries require the partition-tree and cutting techniques of Section 9.12.

General ranges Q is a disk, a polygon, or a set of several ranges; these are typically answered by reduction to the orthogonal or simplex case.

This chapter develops the classical orthogonal range-searching hierarchy: one-dimensional search (Section 9.2), k -d trees (Section 9.3), range trees with multilevel structures (Sections 9.4–9.5), fractional cascading (Section 9.6), priority search trees (Section 9.7), and the higher-dimensional and dynamic extensions (Sections 9.9–9.10). The statistical structure of points in high dimensions is treated in Section 9.11 and Chapter 39.

9.2 One-Dimensional Range Searching

9.2.1 Overview

In one dimension the points are a set of n real numbers and the query is an interval $[x_1, x_2]$: report (or count) the points in the interval. This is the simplest nontrivial range-searching problem, and the ideas it introduces — sorted order, binary search, and split nodes — are the basis of every structure in this chapter.

9.2.2 Definitions

Definition 9.1 (Range Counting and Reporting). For a set P of n reals and an interval $I = [x_1, x_2]$, the **counting query** asks for $|P \cap I|$ and the **reporting query** asks for the sorted list of points in $P \cap I$.

9.2.3 Theory

Sort the points as $p_1 < p_2 < \dots < p_n$. The points in the query interval form a contiguous block $p_{i+1}, \dots, p_j$, where i is the number of points $\leq x_1$ and j the number of points $< x_2$. Both indices are found by binary search on the sorted array. A balanced binary search tree keyed by value stores the same information with the additional ability to support insertion and deletion (Section 9.10).

9.2.4 Pseudocode

Algorithm 26 One-Dimensional Range Query

Input: Sorted array $A[1..n]$ of distinct points, query interval $[x_1, x_2]$.

Output: The points of A in $[x_1, x_2]$.

```

1: function RANGEQUERY( $A, x_1, x_2$ )
2:    $i \leftarrow \text{binarySearch}(A, x_1)$                                  $\triangleright$  first index with  $A[i] \geq x_1$ 
3:    $j \leftarrow \text{binarySearch}(A, x_2)$                                  $\triangleright$  last index with  $A[j] \leq x_2$ 
4:   return  $A[i..j]$ 
5: end function

```

9.2.5 Complexity

- **Time:** counting in $O(\log n)$; reporting in $O(\log n + k)$ with $k = |P \cap I|$ (the output itself needs k time).
- **Storage:** $O(n)$.

9.2.6 Usages

- **Database queries:** selecting records whose key lies in an interval.
- **Component of higher-dimensional structures:** every range tree node stores a one-dimensional structure of exactly this kind (Section 9.4).

9.2.7 References

The one-dimensional structure is elementary; its textbook treatment appears in [BCKO08] and the analysis of its use inside multilevel structures in [Ben80].

9.3 k-d Trees

9.3.1 Overview

The *k-d tree* partitions the point set recursively with axis-aligned splitting hyperplanes: at each internal node, the points are split by a hyperplane perpendicular to one coordinate axis into two subsets of equal cardinality, each stored in a child. The result is a balanced binary tree whose leaves are points. *k-d trees* support orthogonal range queries by pruning any subtree whose cell is disjoint from the query range. A brief outline appears here; the full data structure — construction, balancing, and query pruning — is developed in Section 10.1 (Chapter 10).

9.3.2 Definitions

Definition 9.2 (Split Plane and Cell). Each k -d tree node v stores a **split hyperplane** $x_d = c$ (perpendicular to the split axis d) and is associated with a **cell**, the axis-aligned box containing the points of its subtree. The two children own the two halves of the cell on either side of the split plane.

9.3.3 Theory

The query on an orthogonal range Q descends the tree: if a node's cell does not intersect Q , the whole subtree is pruned; if the cell is contained in Q , the whole subtree is reported; otherwise the search continues in both children. The number of nodes visited is bounded by $O(n^{1-1/d} + k)$ in the worst case for a box query in $\mathbb{R}^d$ with n points; the exponent reflects that pruning eliminates entire subtrees only in dimensions with “thick” splits. The k -d tree is the structure of choice when the dimension is small ($d \leq 3$) and the points are static or slowly changing, because its storage is exactly linear and its constants are tiny.

9.3.4 Pseudocode

Algorithm 27 Orthogonal Range Query in a k -d Tree

Input: k -d tree root with cells, orthogonal query box Q .

Output: All points of the tree inside Q .

```

1: function KDQUERY( $v, Q$ )
2:   if  $v$  is a leaf then
3:     return  $v$ .point if it lies in  $Q$ 
4:   end if
5:   if  $\text{cell}(v) \cap Q = \emptyset$  then
6:     return  $\emptyset$ 
7:   end if
8:   if  $\text{cell}(v) \subseteq Q$  then
9:     return all points of the subtree
10:  end if
11:  return KDQUERY(left( $v$ ),  $Q$ )  $\cup$  KDQUERY(right( $v$ ),  $Q$ )
12: end function

```

9.3.5 Complexity

- **Storage:** $O(n)$.
- **Construction:** $O(n \log n)$ by median finding at each node.
- **Query:** $O(n^{1-1/d} + k)$ worst case for orthogonal ranges; the expected behavior on typical data is much better (logarithmic plus output).

9.3.6 Usages

- **Nearest-neighbor search:** the branch-and-bound version of the k -d tree is the workhorse of Chapter 13.
- **Windowing:** reporting all points in a query rectangle, used in GIS and graphics.
- **Dynamic point sets:** reinsertion-based updates make the k -d tree usable online (Section 9.10).

9.3.7 References

The k -d tree was introduced by Bentley [Ben75]; the analysis of its range queries and the branch-and-bound nearest-neighbor search are due to Friedman, Bentley, and Finkel [FBF77]; textbook treatments appear in [BCKO08, PS85].

9.4 Range Trees

9.4.1 Overview

The **range tree** answers orthogonal box queries in the plane in $O(\log^2 n + k)$ time with $O(n \log n)$ storage, and in d dimensions in $O(\log^d n + k)$ time with $O(n \log^{d-1} n)$ storage. Unlike the k -d tree, its query time is near-logarithmic in n and independent of the answer count's position on the $n^{1-1/d}$ curve; it is the textbook example of a **multilevel** (or layered) data structure.

9.4.2 Definitions

Definition 9.3 (Range Tree). A **range tree** for a set P of points in $\mathbb{R}^2$ is a balanced binary search tree T on the x -coordinates of P . Each node v stores the subset P_v of points in its subtree, organized as a one-dimensional balanced tree (an **associate structure**) on their y -coordinates.

9.4.3 Theory

A box query $[x_1, x_2] \times [y_1, y_2]$ is answered in two stages. First, search the primary tree for the x -range and collect the **canonical subset**: the $O(\log n)$ nodes whose subtrees exactly cover the points with x -coordinates in $[x_1, x_2]$. Then, for each such node v , query v 's associate y -tree for the interval $[y_1, y_2]$ and report the output. Counting is the same, with each associate query returning a count. The two-stage structure is correct because the points with $x \in [x_1, x_2]$ are exactly the disjoint union of the canonical nodes' subsets, and each is filtered by the y -query.

9.4.4 Pseudocode

Algorithm 28 2D Orthogonal Range Query

Input: Range tree T on P , box $Q = [x_1, x_2] \times [y_1, y_2]$.

Output: All points of P in Q .

```

1: function RANGETREEQUERY( $T, Q$ )
2:   collect the  $O(\log n)$  canonical nodes  $v_1, \dots, v_m$  covering  $[x_1, x_2]$ 
3:   for each canonical node  $v_i$  do
4:     report the points of  $v_i$ 's  $y$ -tree with  $y \in [y_1, y_2]$ 
5:   end for
6: end function

```

9.4.5 Complexity

- **Storage:** $O(n \log n)$; each point appears in one associate tree per level, of which there are $\log n$.
- **Construction:** $O(n \log n)$ by building the associate trees bottom-up.
- **Query:** $O(\log^2 n + k)$ in 2D; the canonical search costs $O(\log n)$ and each of the $O(\log n)$ associate queries costs $O(\log n)$.

9.4.6 Usages

- **Orthogonal windowing:** the canonical-subset mechanism is the standard answer for exact-logarithmic queries on static data.
- **Dual problems:** counting points in a query box is the direct analogue of the database “box select”.
- **Reduction to 1D:** any problem solvable on a sorted set in $O(\log n)$ transfers to $O(\log^2 n)$ in the plane via the range tree, which is how the range tree is used as a black box inside other algorithms.

9.4.7 References

The range tree (as the “layered range tree” using multidimensional divide-and-conquer) is due to Bentley [Ben80]; the higher-dimensional analysis and the fractional-cascading speedup are developed in [BCKO08] and [Wil85].

9.5 Multilevel Data Structures

A **multilevel data structure** answers a query by filtering through a hierarchy of one-dimensional structures, one level per dimension. The range tree is the canonical example: the primary level filters on x , and each canonical node contains a secondary structure that filters on y . The construction recurses: a d -dimensional range tree has a primary tree on the first coordinate and, at each node, a $(d - 1)$ -dimensional range tree on the remaining coordinates, with the base case $d = 1$ being the balanced tree of Section 9.2.

The general design principle is: if a query on a set can be answered in $Q(n)$ time and the associate structures can be built in $S(n)$ storage per node, then the two-level structure answers in $O(\log n)$ (canonical search) times $Q(n)$ and stores $O(n)$ times the per-node storage. Applied recursively, this yields the d -dimensional bounds of Section 9.9. The same principle underlies many geometric data structures beyond range searching; it is sometimes called **multidimensional divide-and-conquer** [Ben80].

Two implementation notes matter in practice. First, the associate structures at a node contain the points of the subtree, so a point participates in one structure per level, giving the $\log n$ blow-up in storage. Second, when the query filter is a contiguous interval in the associated coordinate, the associate structure can be a sorted array and the query becomes two binary searches — this is the form used when fractional cascading (Section 9.6) is applied.

9.6 Fractional Cascading

9.6.1 Overview

A 2D range-tree query spends $O(\log n)$ time finding the canonical nodes and then performs $O(\log n)$ binary searches, one per canonical node, for a total of $O(\log^2 n)$. Each binary search looks for the same y -value in a related sorted list, and the repeated searches are wasteful: the list at a node and the list at its children are highly correlated. **Fractional cascading** removes the second $\log n$ factor by passing the search position from each list to the next, reducing the query time to $O(\log n + k)$.

9.6.2 Definitions

Definition 9.4 (Fractional Cascading). **Fractional cascading** is a technique for sequences of sorted lists $L_1, L_2, \dots$ where each list’s successor can be searched faster by *cascading*: every

element of L_{i+1} stores a pointer to the closest elements in L_i , so that a query value's position in L_i determines its position in L_{i+1} in $O(1)$ time rather than $O(\log n)$.

9.6.3 Theory

Each y -list in the range tree is replaced by a **cascading array**: for every element, a pair of pointers to the nearest elements of the same value range in the lists of the node's children. A query that has found its position in the root's y -list then walks down the tree, spending $O(1)$ per level to derive the position in the child's list. Searching all $O(\log n)$ canonical y -lists therefore costs $O(\log n)$ total instead of $O(\log^2 n)$. Because the lists are now linked, the storage remains $O(n \log n)$ but the constant grows by a small factor (each element keeps two extra pointers).

9.6.4 Complexity

- **Query:** $O(\log n + k)$ for 2D orthogonal boxes.
- **Storage:** $O(n \log n)$.
- **Construction:** $O(n \log n)$.

9.6.5 Usages

- **Optimal orthogonal searching:** together with careful canonical decomposition, fractional cascading gives the optimal $O(\log n + k)$ 2D structure of Willard [Wil85].
- **Any repeated-scan hierarchy:** the technique applies wherever the same value is searched in related sorted lists, which occurs in several other chapters (e.g., computing lower envelopes).

9.6.6 References

Fractional cascading was introduced by Chazelle and Guibas [CG86]; its application to range trees is the textbook treatment of [BCKO08] and the optimal-structure result of [Wil85].

9.7 Priority Search Trees

9.7.1 Overview

A **priority search tree** [McC85] answers **3-sided** orthogonal queries — ranges of the form $[x_1, x_2] \times (-\infty, y]$ — in $O(\log n + k)$ time with only $O(n)$ storage. The improvement over the range tree's $O(n \log n)$ storage comes from combining the x -order (tree structure) and the y -order (heap order) in a single binary tree, a “heap on y over a search tree on x ”.

9.7.2 Definitions

Definition 9.5 (Priority Search Tree). A **priority search tree** for a set P of n points is a binary tree where the root holds the point of minimal y , the remaining points are split by x -median into the left and right subtrees, and each subtree is itself a priority search tree. Every node stores the x -range (min and max x) of its subtree.

9.7.3 Theory

A 3-sided query $[x_1, x_2] \times (-\infty, y]$ is answered by a pruning search: a node is visited only if its x -range intersects the query interval. When a visited node's point has y -coordinate at most y , the point is reported (the heap order guarantees that if the root of a subtree fails the y -test, no point of that subtree passes it). The number of visited nodes is $O(\log n + k)$, giving the near-logarithmic bound with linear storage. An axis-aligned rectangle query can be decomposed into two 3-sided queries, so priority search trees also answer full orthogonal range queries, though with $O(k)$ reporting and $O(\log n)$ counting structure trade-offs different from the range tree.

9.7.4 Complexity

- **Storage:** $O(n)$.
- **Query (3-sided):** $O(\log n + k)$.
- **Insertion/deletion:** $O(\log n)$.

9.7.5 Usages

- **Dynamic 3-sided queries:** the heap order supports cheap insertions, making the structure attractive when points arrive online.
- **Component of box structures:** 3-sided queries are a standard primitive into which general orthogonal queries are decomposed.

9.7.6 References

The structure is due to McCreight [McC85]; its relation to the range tree and its use in dynamic structures are covered in [BCKO08] and [Ove83].

9.8 Reporting versus Counting

Range queries come in two flavors whose complexity differs in an important way. **Reporting** queries must list the k points in the range, so any structure costs at least $\Omega(k)$ per query, and a structure with $O(\log n)$ search time can report in $O(\log n + k)$. **Counting** queries return only the number $|P \cap Q|$, so their time is independent of k ; the challenge is that counting often requires storing more information per canonical subset (a subtree size, or a precomputed count) while spending as little time as possible inside the structure.

The two versions have different optimality landscapes:

- for counting, a range tree stores the size of each y -tree, so a box count costs $O(\log^d n)$ with $O(n \log^{d-1} n)$ storage;
- for reporting, the same tree costs $O(\log^d n + k)$, where k includes the output; priority search trees and k -d trees trade query time for output time differently;
- lower bounds differ as well: counting is easier than reporting in the sense that the output term is absent, and decision-tree arguments give $O(n \log n)$ -ish bounds for the static 2D case, while reporting has the trivial $\Omega(k)$ component.

In practice most applications need reporting, and the k term dominates; counting appears in statistics (counts of points in half-spaces for classification) and in the simplex setting of Section 9.12, where the near-linear-space lower bound $\Omega(n^{1-1/d})$ applies to counting and reporting alike.

9.9 Higher-Dimensional Range Searching

The range-tree construction generalizes directly to $\mathbb{R}^d$: a d -dimensional range tree has a primary tree on the first coordinate and a $(d-1)$ -dimensional range tree as each node's associate structure (Section 9.5). The query cost and storage follow by induction:

- **Range tree:** $O(\log^d n + k)$ query time and $O(n \log^{d-1} n)$ storage in d dimensions.
- **k -d tree:** $O(n^{1-1/d} + k)$ query time with $O(n)$ storage in d dimensions.
- **Priority search tree:** $O(n \log n)$ -free — it answers 3-sided queries in $\mathbb{R}^2$ in $O(\log n + k)$ with $O(n)$ storage; in higher dimensions its role is taken by more specialized structures.

The range tree's storage grows polynomially in the dimension (the $n \log^{d-1} n$ factor), so for d fixed it is the near-logarithmic structure of choice; for variable dimension the polynomial storage becomes prohibitive. The k -d tree, with its strictly linear storage, is the practical default when d is moderate (say $d \leq 10$), because the $n^{1-1/d}$ query factor is small until d approaches the data size. The extreme range — where every structure degenerates toward linear scans — is the curse of dimensionality of Section 9.11. Simplex range searching in higher dimensions, with its near-optimal $O(n^{1-1/d}(\log n)^{O(1)})$ bounds, is treated in Section 9.12.

9.10 Dynamic Range Searching

When points are inserted and deleted, a range-searching structure must maintain its invariants online. The classical framework is the transform-and-conquer method of Overmars [Ove83], which converts any static structure into a dynamic one at the cost of a logarithmic factor per operation, provided the static structure can be rebuilt and queried efficiently. The main strategies are:

Binary-search-tree dynamics A dynamic range tree keeps the primary tree balanced with rotations; each rotation changes only $O(1)$ associate structures locally, so insertions and deletions cost $O(\log^2 n)$ in 2D (an $O(\log n)$ factor for the tree update times an $O(\log n)$ factor for the associate-structure updates).

Rebuilding Overmars' method maintains several static structures of geometrically growing sizes (partial rebuilding): a new point is inserted into the smallest structure, and structures are merged when they exceed a size threshold. Querying all levels costs $O(\log n)$ times the static query cost.

k -d tree updates Insertions into a k -d tree are handled by attaching a point at the appropriate leaf; deletions mark points and periodically rebuild the subtree. The amortized update cost is $O(\log n)$ and queries remain $O(n^{1-1/d} + k)$.

The fundamental limitation is that updates interfere with fractional cascading and heap orders (Section 9.7), so the fully dynamic optimal 2D structure is considerably more complex than the static one; in practice, the field's recommendation is to use a dynamic range tree for moderate update rates and a k -d tree when updates are frequent and the dimension is low.

9.11 Curse of Dimensionality

The **curse of dimensionality** is the phenomenon that all data structures for orthogonal range searching deteriorate as the dimension d grows. The k -d tree's query factor $n^{1-1/d}$ approaches n as $d \rightarrow \infty$, so for large d the query degenerates to a linear scan; the range tree's storage

$n \log^{d-1} n$ becomes astronomical in d ; and, most fundamentally, the lower bounds of Section 9.12 show that any structure with near-linear storage must spend $\Omega(n^{1-1/d})$ time per query in the worst case. For $d = \log n$, the exponents meet: every near-linear-space structure answers queries in time at least $n^{1-1/\log n} = n^{1-o(1)}$, i.e., no better than scanning.

The cause is combinatorial: the number of distinct orthogonal ranges that a point set can define grows exponentially in d , so a structure that wants to answer all of them with few canonical subsets cannot exist. Two practical responses are standard. First, for moderate d ($d \leq 20$), indexing schemes such as the k -d tree and the space-filling-curve index of Chapter 10.10 still perform well on real data, whose intrinsic dimension is far below d ; the analysis in Chapter 39 explains when this holds. Second, when d is genuinely large, exact range searching is abandoned in favor of approximate structures — locality-sensitive hashing, approximate nearest neighbors, and sketches — whose guarantees are probabilistic rather than worst-case (Chapter 39).

9.12 Simplex Range Searching

9.12.1 Overview

Range searching asks for the set of points of a fixed point set P that lie inside a query region Q . When Q is a simplex (a half-plane, triangle, or half-space), the problem is called **simplex range searching**. It is one of the most fundamental query problems in computational geometry, with applications in databases, GIS, and computer graphics. The standard solutions use **partition trees**, **cuttings**, and ε -**nets** to answer counting queries in roughly $O(n^{1-1/d})$ time with linear storage, which is optimal in the worst case for near-linear space.

9.12.2 Definitions

Definition 9.6 (Simplex Range Counting). Given a set P of n points in $\mathbb{R}^d$ and a query simplex Δ , the *simplex range counting* problem asks for $|P \cap \Delta|$. The *reporting* variant asks for all points in $P \cap \Delta$.

Definition 9.7 (ε -Net). Let $(X, \mathcal{R})$ be a range space of points and subsets (ranges) with VC dimension d . A subset $N \subseteq X$ is an ε -*net* if every range that contains at least $\varepsilon|X|$ points also contains a point of N .

9.12.3 Theory

A random sample of size $O(\frac{d}{\varepsilon} \log \frac{1}{\varepsilon})$ is an ε -net with high probability [HW87]. Cuttings generalize this idea: a $(1/r)$ -*cutting* of an arrangement of n lines subdivides the plane into $O(r^2)$ cells, each crossed by at most n/r lines. Cuttings can be built deterministically in $O(nr)$ time [CF90], and hierarchical cuttings yield partition trees.

Theorem 9.8 (Simplex Range Searching Bounds). *For a set of n points in $\mathbb{R}^d$, a partition tree of linear size can answer a simplex counting query in $O(n^{1-1/d} (\log n)^{O(1)})$ time and a reporting query in $O(n^{1-1/d} (\log n)^{O(1)} + k)$ time, where k is the number of reported points [Mat92, CF90]. This is optimal for near-linear space: any data structure using $O(n \text{ polylog } n)$ space requires $\Omega(n^{1-1/d})$ query time in the worst case.*

9.12.4 Pseudocode

9.12.5 Complexity

- **Storage:** $O(n)$ for a partition tree; cutting-based structures use $O(n \text{ polylog } n)$.
- **Counting Query:** $O(n^{1-1/d} (\log n)^{O(1)})$ time.

Algorithm 29 Simplex Range Counting with a Partition Tree

```

1: function COUNTSIMPLEX(node,  $\Delta$ )
2:                                      $\triangleright$  node is a cell with an associated canonical subset
3:   if node.cell  $\cap \Delta = \emptyset$  then
4:     return 0
5:   end if
6:   if node.cell  $\subseteq \Delta$  then
7:     return node.count
8:   end if
9:   return  $\sum_{child} \text{COUNTSIMPLEX}(child, \Delta)$ 
10: end function

```

- **Reporting Query:** add $O(k)$ output time.
- **Construction:** $O(n \log n)$ with randomized incremental methods, or $O(nr)$ for a single cutting.

9.12.6 Usages

- **Database and GIS** polygon windowing queries.
- **Geometric statistics:** counting points in a query half-space for classification and density estimation.
- **Levels and duality:** by duality, half-plane range counting is equivalent to counting vertices of an arrangement below the query line, connecting to levels (Section 14.3).

9.12.7 Testing Methods and Edge Cases

- **Degenerate queries:** simplex boundaries through points, empty and full queries.
- **Consistency:** compare counting against brute force for small n ; compare reporting to counting by checking the output size.

9.12.8 References

- Haussler, D., & Welzl, E. (1987). “ ϵ -Nets and Simplex Range Queries”. Discrete & Computational Geometry.
- Matoušek, J. (1992). “Reporting Points in Halfspaces” Computational Geometry: Theory and Applications.
- Chazelle, B., & Friedman, J. (1990). “A Deterministic View of Random Sampling and Its Use in Geometry”. Combinatorica.
- Matoušek, J. (1999). “Geometric Discrepancy: An Illustrated Guide”. Springer.

9.13 Exercises

1. Show that a 1D range tree (Section 9.2) is just a sorted array, and that counting needs only the binary-search indices.
2. Prove that the k -d tree query visits $O(n^{1-1/d} + k)$ nodes in the worst case for a box query: argue that a query box whose side is unbounded prunes half the subtrees per level, while a thin box cuts many subtrees.

3. Verify the range-tree bounds: each point is stored in exactly one associate tree per level, so storage is $O(n \log n)$, and the query visits $O(\log n)$ canonical nodes.
4. Explain why fractional cascading reduces the query time by a factor of $\log n$ and why the storage stays $O(n \log n)$.
5. Design a priority search tree that answers the query $[x_1, x_2] \times (-\infty, y]$ and prove the visited-node bound $O(\log n + k)$.
6. Compare the asymptotic costs of range trees and k -d trees for orthogonal reporting in $d = 2, 3, 10$, and state which you would choose for each.
7. Argue from the lower bound of Section 9.12 that no near-linear-space structure can answer counting queries faster than $n^{1-1/d}$ in the worst case, and interpret the bound for $d = \log n$.
8. Implement a brute-force counter and verify a range tree on random point sets, checking that reporting agrees with brute force and that counting equals the reported cardinality.

9.14 Project: Spatial Query Engine

Build a spatial query engine over a point dataset:

1. **Index:** implement a 2D range tree (primary tree on x , associate sorted arrays on y) and a k -d tree.
2. **Queries:** support box reporting and box counting, and a point-in-polygon query by bounding-box filtering followed by an exact test.
3. **Verification:** compare both indexes against a brute-force scan on random and adversarial point sets, checking output equality and reporting counts.
4. **Measurement:** plot query time and storage against n and against the query box size; verify the predicted $O(\log^2 n + k)$ and $O(n^{1-1/2} + k)$ behaviors.
5. **Extras:** add the 3-sided query with a priority search tree, and a nearest-neighbor query reusing the k -d tree (Chapter 13).
6. **Visualization:** draw the points, the query boxes, and the reported points; optionally animate the recursive subdivision of the k -d tree.

Chapter 10

Spatial Partitioning

Geometric algorithms depend on data structures that organize points, lines, regions, and higher-dimensional objects for efficient construction and query. This chapter presents hierarchical and ordered structures for spatial searching: KD-trees, quadtrees, R-trees, interval trees, and segment trees. Together they support nearest-neighbor queries, range searching, collision detection, intersection reporting, and other geometric operations.

10.1 K-Dimensional Trees

10.1.1 1. Overview

A **KD-tree** is a space-partitioning data structure for organizing points in a k -dimensional space [Ben75]. It is a binary tree where every node is a k -dimensional point and every non-leaf node represents a splitting hyperplane. This structure is extremely efficient for range searches and nearest neighbor queries in low-to-medium dimensional spaces. A first treatment of k -d trees in the context of orthogonal range queries appears in Section 9.3; this section gives the full data structure, including construction, balancing, and query pruning.

10.1.2 2. Definitions

Definition 10.1 (Splitting Hyperplane). A *splitting hyperplane* is a $(k - 1)$ -dimensional surface that divides the point set into two subsets. It is always perpendicular to one of the coordinate axes.

Definition 10.2 (Depth). The *depth* d of a node is its level in the tree. The splitting axis is chosen as $\text{axis} = d \bmod k$.

Definition 10.3 (KD-tree Median). The *median* is the point chosen to split the set at each node, ensuring the tree is balanced. Choosing the exact median guarantees that both subtrees contain at most $\lfloor n/2 \rfloor$ points.

Definition 10.4 (Bounding Box). The *bounding box* of a node is the region of space represented by that node and all its descendants. At depth d , the bounding box is an axis-aligned rectangle restricted along axis $d \bmod k$ to at most one side of the splitting hyperplane.

10.1.3 3. Theory

A KD-tree is built by recursively partitioning the set of points. At each level, the algorithm chooses one of the k dimensions and splits the points into two groups: those whose coordinate in that dimension is less than the median and those whose coordinate is greater.

Theorem 10.5 (KD-tree Construction Complexity). *Building a KD-tree on n points in $\mathbb{R}^k$ takes $O(n \log n)$ time when using a linear-time median selection algorithm at each level, and $O(nk \log n)$ time when sorting at each level.*

Proof. At each level of recursion we partition at most n points. If we use an $O(n)$ median-finding algorithm (e.g., quickselect), the total work at level i is $O(n)$, and the tree has $O(\log n)$ levels (assuming balanced splits), giving $O(n \log n)$ total. With sorting, each level costs $O(n \log n)$ for a total of $O(n \log^2 n)$, but a pre-sorted approach or median-of-medians reduces this to $O(n \log n)$. $\square$

For **Nearest Neighbor Search**, the tree is traversed from the root down to the leaf containing the query point. The distance from the query to this leaf is our initial “best” distance. Then, the algorithm backtracks up the tree, checking if any other branches could contain a closer point. This is done by checking if the query point’s distance to a node’s splitting hyperplane is smaller than the current best distance.

Theorem 10.6 (KD-tree Nearest Neighbor Search). *For n points in $\mathbb{R}^k$ with fixed k , the expected nearest neighbor search time in a randomly built KD-tree is $O(\log n)$.*

Proof. The expected number of nodes visited during a nearest neighbor query is $O(2^k \log n)$ for uniformly distributed points [FBF77]. For fixed k , this simplifies to $O(\log n)$. The key insight is that the pruning criterion—skipping a subtree when the splitting hyperplane is farther than the current best—eliminates most branches. In the worst case (e.g., all points equidistant from the query), no pruning occurs and $O(n)$ nodes are visited. $\square$

Example 10.7 (KD-tree Construction in $\mathbb{R}^2$). Consider the point set $S = \{(2, 3), (5, 4), (9, 6), (4, 7), (8, 1), (7, 2)\}$.

1. **Root (depth 0, axis x):** Median x -coordinate is 5 (from sorted x : 2, 4, 5, 7, 8, 9). Node: (5, 4). Left subtree: $\{(2, 3), (4, 7)\}$; right subtree: $\{(9, 6), (8, 1), (7, 2)\}$.
2. **Left child (depth 1, axis y):** Median y -coordinate of $\{(2, 3), (4, 7)\}$ is 3 or 7. Node: (2, 3). Right child: (4, 7).
3. **Right child (depth 1, axis y):** Median y -coordinate of $\{(9, 6), (8, 1), (7, 2)\}$ is 2. Node: (7, 2). Left child: (8, 1); right child: (9, 6).

The resulting tree partitions $\mathbb{R}^2$ into rectangular regions, each containing exactly one point.

10.1.4 4. Pseudo code

10.1.5 5. Parameters Selections

Definition 10.8 (Dimension Cycling). *Dimension cycling* is the strategy for choosing the splitting axis at each level of the KD-tree. The standard choice is $\mathbf{d} \% \mathbf{k}$, but choosing the dimension with the largest variance (spread) can lead to more efficient searches.

Definition 10.9 (Leaf Size). The *leaf size* is the maximum number of points stored in a single leaf node. Storing a few points (e.g., 5–10) in each leaf instead of one reduces recursion depth and can improve cache performance.

10.1.6 6. Complexity

Theorem 10.10 (KD-tree Complexity Summary). *For n points in $\mathbb{R}^k$:*

1. **Construction:** $O(n \log n)$ time and $O(n)$ space.

```

1: function BUILDKDTree(points, depth=0)
2:   if not points then
3:     return None
4:   end if
5:    $k \leftarrow \text{len}(\text{points}[0])$ 
6:    $\text{axis} \leftarrow \text{depth} \bmod k$ 
7:   Sort points by axis
8:    $\text{mid} \leftarrow \text{len}(\text{points})/2$ 
9:    $\text{node} \leftarrow \text{Node}(\text{points}[\text{mid}])$ 
10:   $\text{node.left} \leftarrow \text{BuildKDTree}(\text{points}[:\text{mid}], \text{depth} + 1)$ 
11:   $\text{node.right} \leftarrow \text{BuildKDTree}(\text{points}[\text{mid}+1:], \text{depth} + 1)$ 
12:  return node
13: end function
14: function NEARESTNEIGHBOR(root, query, depth=0, best=None)
15:  if root is None then
16:    return best
17:  end if
18:   $k \leftarrow \text{len}(\text{query})$ 
19:   $\text{axis} \leftarrow \text{depth} \bmod k$ 
20:   $\text{dist} \leftarrow \text{EuclideanDistance}(\text{root.point}, \text{query})$ 
21:  if best is None or  $\text{dist} < \text{EuclideanDistance}(\text{best.point}, \text{query})$  then
22:     $\text{best} \leftarrow \text{root}$ 
23:  end if
24:   $\text{next\_branch} \leftarrow \begin{cases} \text{root.left} & \text{if } \text{query}[\text{axis}] < \text{root.point}[\text{axis}] \\ \text{root.right} & \text{otherwise} \end{cases}$ 
25:   $\text{other\_branch} \leftarrow \begin{cases} \text{root.right} & \text{if } \text{query}[\text{axis}] < \text{root.point}[\text{axis}] \\ \text{root.left} & \text{otherwise} \end{cases}$ 
26:   $\text{best} \leftarrow \text{NearestNeighbor}(\text{next\_branch}, \text{query}, \text{depth} + 1, \text{best})$ 
27:  if  $|\text{query}[\text{axis}] - \text{root.point}[\text{axis}]| < \text{EuclideanDistance}(\text{best.point}, \text{query})$  then
28:     $\text{best} \leftarrow \text{NearestNeighbor}(\text{other\_branch}, \text{query}, \text{depth} + 1, \text{best})$ 
29:  end if
30:  return best
31: end function

```

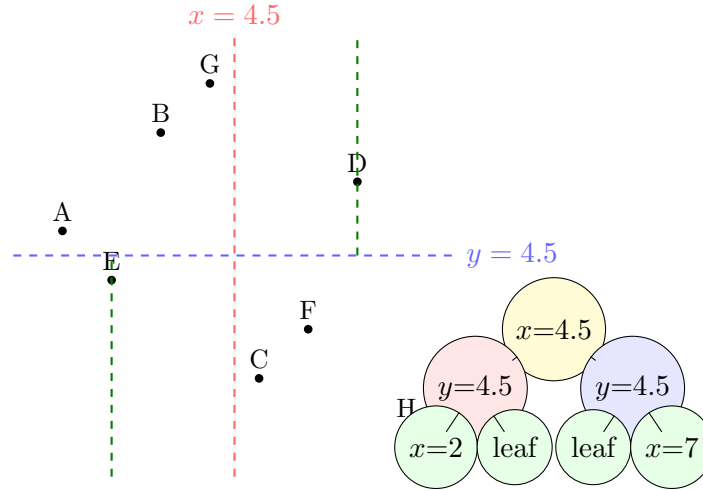


Figure 10.1: KD-tree: (left) recursive axis-aligned partitioning of 2D space; (right) the corresponding binary tree structure.

2. **Nearest Neighbor Search:** $O(\log n)$ expected time for fixed k ; $O(n)$ worst case.

3. **Range Search:** $O(n^{1-1/k} + t)$ time for reporting t points in a rectangular range, for $k \geq 2$ [FBF77].

Proof. Construction follows from the median-based partitioning argument (Theorem 10.5). The range search bound $O(n^{1-1/k} + t)$ is obtained by observing that at each level i , at most $O(2^{ki/k})$ nodes can straddle the query boundary, and summing over $O(\log n)$ levels yields the stated bound. The space is $O(n)$ since each point is stored in exactly one leaf. $\square$

Remark 10.11. The “curse of dimensionality” causes the range search bound to degrade to $O(n)$ when k is large (e.g., $k > 20$). In such regimes, alternative structures like locality-sensitive hashing or random-projection trees are preferred.

10.1.7 7. Usages

- **Computational Geometry:** Fast point location and range searching.
- **Machine Learning:** K-Nearest Neighbors (KNN) classification and clustering.
- **Computer Graphics:** Ray tracing (finding the first object a ray intersects) and photon mapping.
- **Data Compression:** Efficient representation of multidimensional datasets.
- **GIS:** Finding the closest point of interest (e.g., “find the nearest gas station”).

10.1.8 8. Testing methods and Edge cases

- **Duplicate Points:** Ensure the tree correctly handles multiple points at the same coordinates.
- **Points on Boundaries:** Correctly assign points that lie exactly on a splitting hyperplane.
- **Empty Tree:** Handle $n = 0$ cases gracefully.
- **Very High Dimensions:** The “Curse of Dimensionality” affects KD-trees ($k > 20$); performance should be monitored.

- **Skewed Data:** Test on points that are nearly collinear or form very narrow clusters.

10.1.9 9. References

- Bentley, J. L. (1975). “Multidimensional binary search trees used for associative searching”. Communications of the ACM.
- Friedman, J. H., Bentley, J. L., & Finkel, R. A. (1977). “An Algorithm for Finding Best Matches in Logarithmic Expected Time”. ACM Transactions on Mathematical Software.
- Samet, H. (2006). “Foundations of Multidimensional and Metric Data Structures”. Morgan Kaufmann.
- Wikipedia: [k-d tree](#)

10.2 Quadtrees

10.2.1 1. Overview

A **quadtree** is a tree data structure in which each internal node has exactly four children. It is most commonly used to partition a two-dimensional space by recursively subdividing it into four quadrants (North-West, North-East, South-West, South-East) [FB74]. Quadtrees are the 2D equivalent of Octrees (used in 3D) and are essential for efficient spatial querying, image compression, and collision detection.

10.2.2 2. Definitions

Definition 10.12 (Quadtree Root). The *root* of a quadtree is the top-level node representing the entire bounding area.

Definition 10.13 (Quadtree Leaf). A *leaf* is a node that contains actual data points or a uniform value and has no children.

Definition 10.14 (Quadrant). A *quadrant* is one of the four rectangular regions created by splitting a parent node’s region in half along both the x and y axes. The four quadrants are denoted NW, NE, SW, and SE.

Definition 10.15 (Quadtree Capacity). The *capacity* of a quadtree node is the maximum number of points a single node can hold before it must be split into four children.

10.2.3 3. Theory

A quadtree is built by starting with a bounding box and inserting points one by one. If a point is inserted into a leaf node that is already at its capacity, the node “subdivides” into four children, and all points in the node are redistributed to the correct child.

Theorem 10.16 (Quadtree Subdivision Property). *When a quadtree node with capacity c is subdivided, at most 4 nodes of depth $d + 1$ are created, each covering exactly one quarter of the parent’s area. After subdivision, each child contains at most c of the original points.*

Proof. By construction, the four quadrants partition the parent’s bounding box into four disjoint rectangular regions whose union equals the parent region. Each of the original c (or fewer) points in the parent falls into exactly one quadrant based on its coordinates, so no point is lost or duplicated. $\square$

For **Range Querying** (e.g., finding all points within a circle or rectangle), the quadtree allows us to quickly discard entire branches of the tree if their bounding box does not intersect the query area. This pruning results in $O(\log n)$ performance for localized searches.

Example 10.17 (Quadtree Insertion Trace). Consider a quadtree with capacity $c = 1$ and bounding box $[0, 8] \times [0, 8]$.

1. Insert $(2, 3)$: Root is empty, store directly.
2. Insert $(6, 7)$: Root is at capacity, subdivide into four quadrants centered at $(4, 4)$. $(2, 3)$ falls in SW, $(6, 7)$ falls in NE.
3. Insert $(5, 1)$: Root subdivided, $(5, 1)$ falls in SE quadrant (empty), stored there.
4. Insert $(1, 6)$: Falls in NW quadrant (empty), stored there.

All four children now each hold one point, and no further subdivision is needed until a fifth point lands in an occupied quadrant.

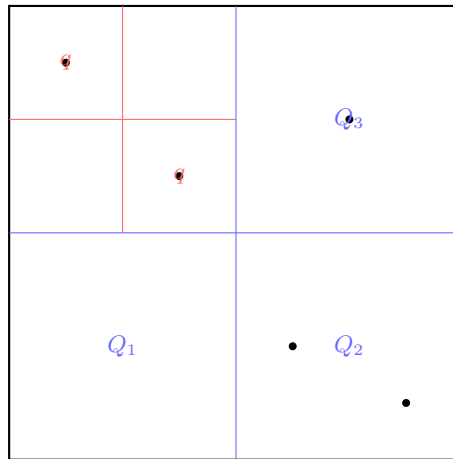


Figure 10.2: Quadtree: recursive 2D space partitioning. Quadrants with multiple points are subdivided; leaf quadrants contain at most one point.

10.2.4 4. Pseudo code

10.2.5 5. Parameters Selections

Definition 10.18 (Quadtree Node Capacity). The *node capacity* is typically set to 4–16 points. Higher capacity reduces the number of internal nodes but increases the linear search time within each node.

Definition 10.19 (Maximum Depth). The *maximum depth* is a limit on subdivision to prevent infinite recursion in the case of overlapping or very dense points.

10.2.6 6. Complexity

Theorem 10.20 (Quadtree Complexity). For n uniformly distributed points in a quadtree with capacity c :

1. **Construction:** $O(n \log n)$ on average; $O(n \cdot \text{depth})$ worst case for clustered data.
2. **Range Query:** $O(\log n + t)$ for localized queries reporting t points [Sam84].

```
1: function INSERT(node, point)
2:   if not node.boundary.contains(point) then
3:     return False
4:   end if
5:   if len(node.points) ≥ node.capacity and not node.is_divided then
6:     node.points.append(point)
7:     return True
8:   end if
9:   if not node.is_divided then
10:    node.subdivide()
11:  end if
12:  if node.nw.insert(point) then
13:    return True
14:  end if
15:  if node.ne.insert(point) then
16:    return True
17:  end if
18:  if node.sw.insert(point) then
19:    return True
20:  end if
21:  if node.se.insert(point) then
22:    return True
23:  end if
24: end function
25: function QUERY(node, range, result)
26:   if not node.boundary.intersects(range) then
27:     return
28:   end if
29:   for p in node.points do
30:     if range.contains(p) then
31:       result.append(p)
32:     end if
33:   end for
34:   if node.is_divided then
35:     Query(node.nw, range, result)
36:     Query(node.ne, range, result)
37:     Query(node.sw, range, result)
38:     Query(node.se, range, result)
39:   end if
40: end function
```

▷ Try inserting into children

▷ Check points in current node

▷ Check children

3. **Space:** $O(n)$ on average, though internal node overhead can grow with depth.

Remark 10.21. For highly non-uniform point distributions, quadtree depth can reach $O(n)$ in the worst case. Using a **point quadtree** variant (instead of region quadtree) or a loose quadtree can mitigate this.

10.2.7 7. Usages

- **Collision Detection:** Quickly finding pairs of objects that are close enough to collide.
- **Image Compression:** Representing regions of uniform color with a single large node (e.g., in the TGA or PNG-like formats).
- **Video Games:** View frustum culling (rendering only objects that are visible within the camera's FOV).
- **GIS:** Storing and querying map features like road networks or city locations.
- **Particle Systems:** Efficiently calculating local forces (e.g., gravity or repulsion) in many-body simulations.

10.2.8 8. Testing methods and Edge cases

- **Uniform Point Distribution:** Verify that the tree divides evenly into a balanced structure.
- **Points on Boundaries:** Ensure points exactly on the dividing lines are assigned to exactly one child.
- **High-Density Clusters:** Ensure the “capacity” and “max depth” parameters prevent excessive subdivision.
- **Empty Areas:** Verify that large regions with no points are represented by a single empty leaf node.
- **Resizing:** Test if the quadtree can handle a dynamic bounding box or if it requires a fixed initial area.

10.2.9 9. References

- Finkel, R. A., & Bentley, J. L. (1974). “Quad trees: A data structure for retrieval on composite keys”. *Acta Informatica*.
- Samet, H. (1984). “The Quadtree and Related Hierarchical Data Structures”. *ACM Computing Surveys*.
- Samet, H. (2006). “Foundations of Multidimensional and Metric Data Structures”. Morgan Kaufmann.
- Wikipedia: [Quadtree](#)

10.3 R-Trees

10.3.1 1. Overview

An **R-tree** is a tree data structure used for efficient spatial indexing of multidimensional information, particularly objects represented as Minimum Bounding Rectangles (MBRs) [Gut84]. Unlike KD-trees or quadtrees, which partition the space itself, R-trees group the objects into hierarchical, potentially overlapping regions. This makes R-trees exceptionally well-suited for disk-based databases and large-scale spatial datasets.

10.3.2 2. Definitions

Definition 10.22 (Minimum Bounding Rectangle). The *Minimum Bounding Rectangle (MBR)* of a node is the smallest rectangle (axis-aligned) that contains all the elements in that node's subtree.

Definition 10.23 (R-tree Leaf Node). A *leaf node* stores pointers to the actual geometric objects and their MBRs.

Definition 10.24 (R-tree Internal Node). An *internal node* stores a pointer to a child node and the MBR of that child.

Definition 10.25 (R-tree Balance Property). An R-tree is always *height-balanced*: all leaf nodes appear at the same depth. This is analogous to the B-tree balance property.

10.3.3 3. Theory

R-trees are built by grouping nearby objects and enclosing them in their MBR.

1. **Insertion:** To insert an object, we start at the root and recursively choose a child node whose MBR requires the least expansion to include the new object.
2. **Node Splitting:** When a node exceeds its capacity, it must be split into two. Since the choice of split affects search performance, heuristics are used to minimize the area and overlap of the two new MBRs (e.g., Quadratic Split or Linear Split).
3. **Search:** To find objects in a query region, we traverse all branches whose MBRs intersect the query. Because MBRs at the same level can overlap, multiple branches may need to be explored.

Theorem 10.26 (R-tree Search Complexity). *For n objects stored in an R-tree with node capacity M and height $h = O(\log_M n)$, a window query reports t objects in expected time $O(M^{h-1} + t)$ under uniform object distribution. For the R*-tree [BKSS90], the average number of disk accesses per query is significantly reduced due to minimized MBR overlap.*

Remark 10.27. The worst-case search time for an R-tree is $O(n)$ when all MBRs overlap heavily, effectively degrading to a linear scan. The R*-tree [BKSS90] mitigates this by using a more sophisticated split heuristic and forced reinsertion.

10.3.4 4. Pseudo code

Example 10.28 (R-tree Insertion). Consider inserting objects $A = [0, 1] \times [0, 1]$, $B = [3, 4] \times [0, 1]$, $C = [0, 1] \times [3, 4]$, $D = [3, 4] \times [3, 4]$, and $E = [1, 2] \times [1, 2]$ into an R-tree with capacity $M = 2$.

1. Insert A : root is a leaf with $[A]$.
2. Insert B : root becomes $[A, B]$. MBR is $[0, 4] \times [0, 1]$.

```
1: function SEARCH(node, query_rect)
2:   results  $\leftarrow$  []
3:   if node.is_leaf then
4:     for obj in node.objects do
5:       if Intersects(obj.mbr, query_rect) then
6:         results.append(obj)
7:       end if
8:     end for
9:   else
10:    for child in node.children do
11:      if Intersects(child.mbr, query_rect) then
12:        results.extend(Search(child, query_rect))
13:      end if
14:    end for
15:  end if
16:  return results
17: end function
18: function INSERT(node, obj)
19:   if node.is_leaf then
20:     node.add(obj)
21:     if node.count  $\geq$  MAX_CAPACITY then
22:       return Split(node)  $\triangleright$  Returns two nodes
23:     end if
24:   else  $\triangleright$  1. Choose subtree that requires least area expansion
25:     child  $\leftarrow$  ChooseBestSubtree(node, obj)
26:     split_result  $\leftarrow$  Insert(child, obj)
27:      $\triangleright$  2. Update node's MBR
28:     node.mbr  $\leftarrow$  Union(node.mbr, obj.mbr)
29:      $\triangleright$  3. Handle split if it occurred in child
30:     if split_result is not None then
31:       node.add_child(split_result)
32:       if node.child_count  $\geq$  MAX_CAPACITY then
33:         return Split(node)
34:       end if
35:     end if
36:   end if
37:   return None
38: end function
```

3. Insert C : root overflows. Split into two nodes: $\{A, C\}$ (MBR $[0, 1] \times [0, 4]$) and $\{B\}$ (MBR $[3, 4] \times [0, 1]$). A new root is created.
4. Insert D : descends to the B -node, making it $\{B, D\}$.
5. Insert E : the MBR of $\{A, C\}$ is $[0, 1] \times [0, 4]$ and the MBR of $\{B, D\}$ is $[3, 4] \times [0, 4]$. Object $E = [1, 2] \times [1, 2]$ overlaps neither MBR exactly, but whichever node requires least expansion is chosen.

10.3.5 5. Parameters Selections

Definition 10.29 (R-tree Split Algorithm). **Guttman’s Quadratic Split** [Gut84] is a common splitting heuristic. The **R*-tree** variant [BKSS90] uses a more complex split that also considers overlap and perimeter, leading to significantly better search performance.

Definition 10.30 (R-tree Min/Max Capacity). Typically, nodes are required to be at least 40% full to maintain balance and storage efficiency. The maximum capacity M depends on the disk page size.

10.3.6 6. Complexity

Theorem 10.31 (R-tree Complexity Summary). *For n objects stored in an R-tree with height h :*

1. **Insertion:** $O(\log_M n)$ expected. Worst case $O(n)$ if many splits propagate.
2. **Search:** $O(M^{h-1} + t) = O(n^{1-1/\alpha} + t)$ for localized queries [Gut84].
3. **Space:** $O(n)$ for storage, plus $O(n/M)$ internal nodes.

10.3.7 7. Usages

- **Spatial Databases:** The core indexing mechanism for PostGIS, Oracle Spatial, and MySQL Spatial.
- **GIS:** Indexing road networks, parcels, and satellite imagery.
- **Network Analysis:** Routing and nearest-neighbor searches in complex maps.
- **Computer Graphics:** Managing large scenes with many individual models.
- **Internet of Things (IoT):** Real-time tracking and querying of moving sensors or vehicles.

10.3.8 8. Testing methods and Edge cases

- **Highly Overlapping Objects:** Ensure the tree doesn’t degrade into a linked list when many objects occupy the same space.
- **Large Objects:** A single large rectangle can intersect many MBRs, causing search to slow down.
- **Empty Search:** Verify the algorithm correctly identifies when no objects are in the query area.
- **Bulk Loading:** For static datasets, use a “Sort-Tile-Recursive” (STR) bulk loader to build a nearly optimal tree.
- **Deletion:** R-trees handle deletions by merging nodes if they fall below minimum capacity.

10.3.9 9. References

- Guttman, A. (1984). “R-trees: A dynamic index structure for spatial searching”. SIGMOD.
- Beckmann, N., Kriegel, H. P., Schneider, R., & Seeger, B. (1990). “The R*-tree: An efficient and robust access method for points and rectangles”. SIGMOD.
- Samet, H. (2006). “Foundations of Multidimensional and Metric Data Structures”. Morgan Kaufmann.
- Wikipedia: [R-tree](#)

10.4 Interval Trees

10.4.1 1. Overview

An **interval tree** is a balanced binary search tree used to store segments (intervals) and allow for efficient range queries [Ede80]. Specifically, it is designed to answer the question: “Which of the stored intervals overlap with a given query interval $[x, y]$?” Interval trees are an essential tool in computational geometry for solving 1D segment intersection problems and are a key component of more complex spatial structures.

10.4.2 2. Definitions

Definition 10.32 (Interval). An *interval* is a pair of numbers $[start, end]$ with $start \leq end$.

Definition 10.33 (Interval Overlap). Two intervals $[a, b]$ and $[c, d]$ *overlap* if and only if $a \leq d$ and $c \leq b$.

Definition 10.34 (Interval Tree Node). Each node in an interval tree stores a single interval and the maximum value *max* found in its subtree. For any node N , $N.max = \max(N.interval.end, N.left.max, N.right.max)$.

10.4.3 3. Theory

The interval tree is built using the **start point** of each interval as the key in a standard binary search tree. Each node additionally maintains the maximum endpoint of all intervals in its subtree. This “max” property is key for efficient pruning during a query.

To search for an interval overlapping with $[q_{start}, q_{end}]$:

1. Check if the current node’s interval overlaps.
2. If the left child exists and its *max* is $\geq q_{start}$, we must search the left branch (there could be an overlap there).
3. Otherwise, we only search the right branch.

This pruning logic ensures that we don’t explore subtrees that cannot possibly contain an overlapping interval.

Theorem 10.35 (Interval Tree Query Complexity). *Given n intervals stored in an interval tree, all intervals overlapping a query interval $[q_{start}, q_{end}]$ can be reported in $O(k + \log n)$ time, where k is the number of reported intervals [Ede80].*

Proof. The search descends from the root to a leaf along at most $O(\log n)$ nodes. At each node, checking for overlap costs $O(1)$. The left subtree of a node is explored only if its *max* value exceeds q_{start} . By the max property, any interval in the left subtree with $end \geq q_{start}$ must have been accounted for, so we visit at most $O(k)$ additional nodes. The total cost is $O(\log n + k)$. $\square$

Example 10.36 (Interval Tree Overlap Query). Build an interval tree on $\{[15, 20], [10, 30], [17, 19], [5, 20], [12, 15]\}$ ordered by start point.

- Root: $[15, 20]$, $max = 40$.
- Query $[18, 22]$: Check root $[15, 20]$: overlaps ($15 \leq 22$ and $18 \leq 20$). Report it. Left child exists and $max = 30 \geq 18$, so recurse left. Right child: $[30, 40]$, $max = 40 \geq 18$, so recurse right. Continue pruning based on max values.

Reported intervals: $[15, 20]$, $[10, 30]$, $[17, 19]$, $[5, 20]$.

10.4.4 4. Pseudo code

```

1: function INSERT(node, interval)
2:   if node is None then
3:     return Node(interval, interval.end)
4:   end if
5:   if interval.start < node.interval.start then
6:     node.left  $\leftarrow$  Insert(node.left, interval)
7:   else
8:     node.right  $\leftarrow$  Insert(node.right, interval)
9:   end if
                                      $\triangleright$  Update the max endpoint in this subtree
10:  node.max  $\leftarrow$  max(node.max, interval.end)
11:  return node
12: end function
13: function FINDOVERLAPS(node, query, results)
14:   if node is None then
15:     return
16:   end if
                                      $\triangleright$  Check current node
17:   if Overlaps(node.interval, query) then
18:     results.append(node.interval)
19:   end if
                                      $\triangleright$  Choose branch
20:   if node.left is not None and node.left.max  $\geq$  query.start then
21:     FindOverlaps(node.left, query, results)
22:   end if
                                      $\triangleright$  Always check right if there's potentially an overlap
                                      $\triangleright$  We can skip if
                                     node.interval.start < query.end
23:   if node.right is not None and node.right.max  $\geq$  query.start then
24:     FindOverlaps(node.right, query, results)
25:   end if
26: end function

```

10.4.5 5. Parameters Selections

Definition 10.37 (Interval Tree Balancing). To maintain $O(\log n)$ performance, the underlying BST should be a self-balancing tree such as an **AVL Tree** or a **Red-Black Tree**.

Definition 10.38 (Interval Type). The interval tree can handle open, closed, or half-open intervals with consistent comparison logic by adjusting the overlap predicate.

10.4.6 6. Complexity

Theorem 10.39 (Interval Tree Complexity Summary). *For n intervals:*

1. **Construction:** $O(n \log n)$ to insert n intervals into a balanced tree.
2. **Query:** $O(k + \log n)$ where k is the number of overlapping intervals found [[Ede80](#)].
3. **Space:** $O(n)$ to store each interval exactly once.

10.4.7 7. Usages

- **Resource Allocation:** Finding all booked time slots that conflict with a new appointment.
- **VLSI Design:** Design rule checking (checking for overlapping wires on a chip).
- **Database Indexing:** Efficiently searching through time-series data or ranges.
- **Ray Tracing:** 1D interval trees are used to find all objects a ray passes through.
- **Video Games:** Managing “life spans” or “event timelines.”

10.4.8 8. Testing methods and Edge cases

- **Contained Intervals:** An interval $[5, 10]$ fully containing another $[6, 7]$.
- **Sharing Boundaries:** Intervals $[5, 10]$ and $[10, 15]$ overlapping only at 10.
- **Large and Small Max values:** Ensure the `node.max` correctly propagates from the leaves to the root.
- **Disconnected Intervals:** Searching for overlaps in a region with no intervals.
- **Identical Intervals:** Multiple nodes with the exact same $[start, end]$.

10.4.9 9. References

- Edelsbrunner, H. (1980). “Dynamic data structures for orthogonal intersection queries”. Technical Report.
- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). “Introduction to Algorithms”. MIT Press.
- Wikipedia: [Interval tree](#)

10.5 Segment Trees

10.5.1 1. Overview

A **segment tree** is a versatile tree data structure used for storing information about intervals or segments [[Ben77](#)]. It is primarily used for range queries where we need to find the sum, minimum, or maximum of a specified range of values in an array that can be updated. In computational geometry, segment trees are used to solve problems involving orthogonal segments and rectangles.

10.5.2 2. Definitions

Definition 10.40 (Segment Tree). A *segment tree* is a binary tree where each node represents an interval of an array. The root represents $[0, n - 1]$, and each internal node representing $[L, R]$ has children representing $[L, \lfloor (L + R)/2 \rfloor]$ and $[\lfloor (L + R)/2 \rfloor + 1, R]$.

Definition 10.41 (Segment Tree Leaf Node). A *leaf node* represents a single element of the array.

Definition 10.42 (Lazy Propagation). *Lazy propagation* is a technique to optimize range updates by storing pending updates at internal nodes and pushing them down only when those children are accessed, maintaining $O(\log n)$ performance per operation.

10.5.3 3. Theory

A segment tree for an array of n elements is built by recursively splitting the array in half. The root node represents the interval $[0, n - 1]$. Each internal node representing $[L, R]$ has children representing $[L, mid]$ and $[mid + 1, R]$.

To solve a **Range Sum Query** for $[qL, qR]$:

1. If the current node's interval is fully within $[qL, qR]$, return the node's sum.
2. If the current node's interval is completely outside $[qL, qR]$, return 0.
3. Otherwise, recurse to both children and return the sum of their results.

Theorem 10.43 (Segment Tree Query Complexity). *Any range query (sum, min, max, etc.) on a segment tree of size n can be answered in $O(\log n)$ time.*

Proof. Any query range $[qL, qR]$ can be decomposed into at most $2 \log_2 n$ disjoint node intervals. This follows from the observation that at each level of the tree, at most two nodes are “partially overlapping” (one on each side of the decomposition), while all others are either fully inside or fully outside. Thus we visit $O(\log n)$ nodes at each of the $O(\log n)$ levels. $\square$

For **Range Updates**, a naive approach would update all relevant leaves, taking $O(n)$ time. **Lazy Propagation** stores the update at the highest relevant internal node and pushes it down only when those children are accessed, maintaining $O(\log n)$ performance.

Theorem 10.44 (Lazy Propagation Complexity). *With lazy propagation, both point updates and range updates on a segment tree of size n take $O(\log n)$ time.*

Proof. A range update modifies at most $O(\log n)$ nodes (the same decomposition argument as in Theorem 10.43). Each node's lazy tag is pushed down at most once before its children are accessed, contributing $O(1)$ amortized work per level. Thus the total work per update is $O(\log n)$. $\square$

Example 10.45 (Segment Tree Range Query). Build a segment tree on $A = [2, 1, 5, 3, 4, 2]$.

- Root: interval $[0, 5]$, sum = 17.
- Left child $[0, 2]$: sum = 8; right child $[3, 5]$: sum = 9.
- Query $[1, 4]$: The range $[1, 4]$ partially overlaps $[0, 2]$ and $[3, 5]$. Recurse into $[1, 2]$ (sum = $1 + 5 = 6$) and $[3, 4]$ (sum = $3 + 4 = 7$). Result: $6 + 7 = 13$.

```
1: function BUILD(arr, node, start, end)
2:   if start == end then
3:     tree[node]  $\leftarrow$  arr[start]
4:     return
5:   end if
6:   mid  $\leftarrow$  (start + end) / 2
7:   Build(arr, 2*node, start, mid)
8:   Build(arr, 2*node+1, mid+1, end)
9:   tree[node]  $\leftarrow$  tree[2*node] + tree[2*node+1]
10: end function
11: function RANGEQUERY(node, start, end, qL, qR)
12:   if qR < start or end < qL then
13:     return 0 ▷ Completely outside
14:   end if
15:   if qL  $\leq$  start and end  $\leq$  qR then
16:     return tree[node] ▷ Completely inside
17:   end if
18:   mid  $\leftarrow$  (start + end) / 2
19:   p1  $\leftarrow$  RangeQuery(2*node, start, mid, qL, qR)
20:   p2  $\leftarrow$  RangeQuery(2*node+1, mid+1, end, qL, qR)
21:   return p1 + p2
22: end function
```

10.5.4 4. Pseudo code

10.5.5 5. Parameters Selections

Definition 10.46 (Segment Tree Operator). The segment tree can store *any associative operator*: Sum, Min, Max, GCD, matrix product, etc. The choice of operator determines the type of query the tree supports.

Definition 10.47 (Segment Tree Size). For an array of n elements, the segment tree requires $4n$ space in an array-based implementation (since a complete binary tree on n leaves has fewer than $4n$ nodes).

10.5.6 6. Complexity

Theorem 10.48 (Segment Tree Complexity Summary). *For an array of n elements stored in a segment tree:*

1. **Build**: $O(n)$ time and $O(n)$ space.
2. **Point Update**: $O(\log n)$ time.
3. **Range Query**: $O(\log n)$ time.
4. **Range Update (lazy)**: $O(\log n)$ time.

10.5.7 7. Usages

- **Range Minimum Query (RMQ)**: Finding the smallest value in a subarray.
- **Computational Geometry (Bentley-Ottmann)**: Used as the status structure for identifying intersections between vertical segments during a horizontal sweep.

- **Rectangle Union:** Calculating the area of the union of several rectangles (Klee’s measure problem).
- **Dynamic Connectivity:** Keeping track of connectivity in a graph that is undergoing edge insertions and deletions.
- **Statistics:** Calculating running averages or variances in real-time data streams.

10.5.8 8. Testing methods and Edge cases

- **Single Element Ranges:** Querying and updating a range where $qL == qR$.
- **Full Array Query:** Querying the entire range $[0, n - 1]$ should return the root node’s value.
- **Overlapping Updates:** Multiple range updates on overlapping segments should be correctly accumulated by lazy propagation.
- **Power of Two vs. Arbitrary n :** Ensure the tree correctly handles array sizes that are not powers of two.
- **Negative Values:** If using Min/Max, ensure negative numbers are handled correctly (e.g., initial values should be $-\infty$ or $+\infty$).

10.5.9 9. References

- Bentley, J. L. (1977). “Algorithms for Klee’s measure problem”. Technical Report.
- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). “Introduction to Algorithms”. MIT Press.
- Wikipedia: [Segment tree](#)

10.6 Uniform Grids

10.6.1 1. Overview

A **uniform grid** partitions the plane into congruent axis-aligned cells of fixed side length δ , storing each point in the cell that contains it. Queries inspect only the cells that can possibly contain the answer, giving a simple and very fast spatial index for uniformly distributed data [BCKO08]. Grids are also the workhorse of spatial sorting, spatial hashing, and image-based algorithms.

10.6.2 2. Definitions

Definition 10.49 (Uniform Grid). A *uniform grid* of side length δ maps each point p to the cell

$$\text{cell}(p) = \left(\left\lfloor \frac{p_x}{\delta} \right\rfloor, \left\lfloor \frac{p_y}{\delta} \right\rfloor \right),$$

and a cell index stores all points that map to it, typically in a hash table keyed by the cell coordinates.

Definition 10.50 (Query Neighborhood). A query point q examines the cells within distance r of q : for a nearest-neighbor query with search radius r , only the $(2\lceil r/\delta \rceil + 1)^2$ cells in the Chebyshev neighborhood of $\text{cell}(q)$ need inspection.

10.6.3 3. Theory

The choice of δ trades cell density against the number of cells examined per query. For n points distributed uniformly over an area A , taking $\delta \approx \sqrt{A/n}$ keeps a constant expected number of points per cell, so a range query of radius r inspects $O((r/\delta)^2)$ cells. The grid degenerates when the data is highly non-uniform; hierarchical grids that adapt cell size locally are one remedy [BCKO08].

10.6.4 4. Pseudo code

```
function BUILDGRID( $P, \delta$ )
   $T \leftarrow$  empty hash table
  for each  $p \in P$  do
     $T[\text{cell}(p)] \leftarrow T[\text{cell}(p)] \cup \{p\}$ 
  end for
  return  $T$ 
end function

function NEARESTCANDIDATECELLS( $q, r, \delta$ )
  for each cell  $c$  with  $\|c - \text{cell}(q)\|_\infty \leq \lceil r/\delta \rceil$  do
    emit  $T[c]$ 
  end for
end function
```

10.6.5 5. Parameters Selections

- **Cell side length δ :** choose $\delta \approx \sqrt{A/n}$ for uniform data; for queries, a common heuristic is δ equal to the average query radius.
- **Hash function:** use a multiplicative hash of the cell coordinates (e.g., “hashing the cell pair”) to spread adjacent cells across buckets.
- **Boundary points:** points exactly on a cell boundary must be assigned to exactly one cell, consistently.

10.6.6 6. Complexity

Build time and space are $O(n)$. A range query of radius r inspects $O((r/\delta)^2)$ cells, each holding an expected constant number of points for uniform data; worst-case behavior is poor for clustered or pathological inputs [BCKO08].

10.6.7 7. Usages

- **Collision detection:** pairs of objects that cannot collide are those in distant cells.
- **Nearest neighbor queries:** spatial sorting in Delaunay construction, and “point grid” acceleration in triangulation [BCKO08].
- **Image processing and games:** pixel neighborhoods and broad-phase culling in a game engine.

10.6.8 8. Testing methods and Edge cases

- **Empty cells:** queries near the boundary must tolerate cells with no stored points.

- **Non-uniform data:** verify query cost on strongly clustered inputs; consider an adaptive or hierarchical variant.
- **Large coordinates:** guard against overflow when computing $\lfloor p_x/\delta \rfloor$.

10.6.9 9. References

- de Berg, M., Cheong, O., van Kreveld, M., & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer.
- Wikipedia: [Spatial hashing](#)

10.7 Fixed-Radius Neighbor Search

10.7.1 1. Overview

Fixed-radius neighbor search (also called **ball query** or **range search with radius r**) reports, for a query point q , all stored points within a prescribed distance r . It is the natural tool for billion-point 2D/3D point clouds—sensor networks, particle simulations, and point-cloud processing—where the query distance is known in advance. The problem is a special case of range search (Section 13.4) with a ball as the range, but the fixed radius permits data structures, most notably the uniform grid, that answer queries in time proportional to the output size plus the boundary of the query ball [BCKO08].

10.7.2 2. Definitions

Definition 10.51 (Fixed-Radius Query). Given a set P of n points in $\mathbb{R}^d$ and a radius r , a *fixed-radius query* at q reports

$$N_r(q) = \{p \in P : \|p - q\|_2 \leq r\}.$$

The answer count $|N_r(q)|$ is the *ball count*.

Definition 10.52 (Grid Search Region). In a uniform grid of cell side δ , the cells that may contain points within distance r of q are those whose Chebyshev distance from $\text{cell}(q)$ is at most $\lceil r/\delta \rceil$; every point of $N_r(q)$ lies in this region because a point farther than r in one coordinate is automatically farther than r in Euclidean distance.

10.7.3 3. Theory

A uniform grid with cell side $\delta \geq r$ guarantees that only the $(2\lceil r/\delta \rceil + 1)^d$ cells of the Chebyshev neighborhood of $\text{cell}(q)$ need to be scanned. Setting $\delta = r$ minimizes the number of cells examined to 3^d while keeping every answer within the candidate region; each candidate point is then accepted or rejected by the exact Euclidean test $\|p - q\|_2 \leq r$.

Theorem 10.53 (Grid Query Cost). *For n points distributed uniformly over a region of volume V and a query radius r with cell side $\delta = r$, a fixed-radius query examines $O(1)$ cells in the worst case over the choice of query point, and each cell holds an expected $O(nr^d/V)$ points. Hence the expected query time is $O(nr^d/V + |N_r(q)|)$, which for dense clouds (large nr^d/V) reduces to output-size dominated work [BCKO08].*

For sparse clouds or non-uniform data, tree structures give better worst-case bounds: a k -d tree (Section 10.1) or octree (Section 10.2) reports $N_r(q)$ in $O(n^{1-1/d} + k)$ time with $O(n)$ space, where $k = |N_r(q)|$.

Algorithm 30 Fixed-Radius Neighbor Search

```
function BUILDFIXEDRADIUSINDEX(points, r)
     $\delta \leftarrow r$ 
     $T \leftarrow$  empty hash table keyed by cell coordinates
    for each  $p \in P$  do
         $T[\text{cell}(p)] \leftarrow T[\text{cell}(p)] \cup \{p\}$ 
    end for
    return  $T$ 
end function
function FIXEDRADIUSQUERY(q, r, T)
     $c \leftarrow \text{cell}(q)$ 
     $out \leftarrow \emptyset$ 
    for each cell  $c'$  with  $\|c' - c\|_\infty \leq 1$  do
        for each  $p \in T[c']$  do
            if  $\|p - q\|_2 \leq r$  then  $\triangleright$  exact Euclidean test
                 $out \leftarrow out \cup \{p\}$ 
            end if
        end for
    end for
    return  $out$ 
end function
```

10.7.4 4. Pseudo code**10.7.5 5. Parameter Selection**

- **Cell side δ :** setting $\delta = r$ keeps the query neighborhood to 3^d cells. Smaller cells reduce per-cell work but increase the number of cells scanned; larger cells invert the trade-off.
- **Hash function:** hash the cell coordinates to a dense bucket array so that adjacent cells spread evenly (as in Section 10.6).

10.7.6 6. Complexity

- **Space:** $O(n)$ cells plus $O(n)$ stored points.
- **Build time:** $O(n)$.
- **Query time:** expected $O(nr^d/V + |N_r(q)|)$ on a grid; $O(n^{1-1/d} + |N_r(q)|)$ worst case on a k-d tree or octree.

10.7.7 7. Usages

- **Sensor networks:** finding all sensors within communication range of a query point.
- **Particle simulations:** building interaction lists within the cutoff radius.
- **Point-cloud processing:** pairing points within a tolerance for surface reconstruction and registration.
- **Collision detection:** broad-phase culling of objects within a blast or detection radius.

10.7.8 8. Testing methods and Edge cases

- **Boundary cells:** points exactly at distance r may lie on a shared cell boundary; assign each point to exactly one cell and rely on the exact test $\|p - q\|_2 \leq r$ for inclusion.
- **Ball count:** compare against a brute-force scan for small n ; verify the reported set has exactly the brute-force size.
- **Clustered data:** test strongly non-uniform inputs, where the grid degenerates and the tree variant is preferable.
- **Outside domain:** queries whose ball extends past the domain bounds must skip empty hash buckets.
- **Large coordinates:** guard against overflow when computing $\lfloor p_x/r \rfloor$.

10.7.9 9. References

- de Berg, M., Cheong, O., van Kreveld, M., & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer.
- Wikipedia: [Nearest neighbor search](#)

10.8 Binary Space Partitioning Trees

10.8.1 1. Overview

A **binary space partitioning (BSP) tree** recursively partitions space by a hyperplane into two half-spaces, building a binary tree whose leaves correspond to convex regions of the partition [FKN80]. BSP trees were introduced for hidden-surface removal and are used for painter’s-algorithm sorting, collision detection, and image compression; the k-d tree of Section 10.1 is the special case where every splitting plane is axis-aligned.

10.8.2 2. Definitions

Definition 10.54 (BSP Tree). A *BSP tree* on a set of objects S is a binary tree where each internal node stores a hyperplane h ; the left subtree contains the objects of S entirely in the back half-space of h , the right subtree those in the front half-space, and objects crossing h are split and stored in both subtrees. Leaves are homogeneous convex regions.

Definition 10.55 (Autopartition). An *autopartition* uses, at every node, the supporting line of one of the input segments as the splitting line. Restricting splitting lines to input edges bounds both the number of pieces and the size of the tree.

10.8.3 3. Theory

For a set of n disjoint segments in the plane, a BSP tree is built by recursive subdivision: choose a splitting line h , split every segment that crosses h into two fragments, and recurse on each half-plane. The number of fragments can grow, but for disjoint line segments there is an autopartitioning scheme producing a tree of size $O(n^2)$ in the worst case and $O(n \log n)$ expected under random choices [BCKO08]. In $\mathbb{R}^3$, BSP trees of linear size exist for suitably restricted inputs, and the tree supports painter’s-algorithm depth sorting and point-location queries in $O(n)$ time.

10.8.4 4. Pseudo code

```
function BUILD BSP( $S$ )  
  if  $|S| \leq 1$  then  
    return leaf with  $S$   
  end if  
  choose a splitting line  $h$  from  $S$   
   $S^+ \leftarrow \emptyset$ ,  $S^- \leftarrow \emptyset$   
  for each segment  $s \in S$  do  
    if  $s$  lies in front of  $h$  then  
       $S^+ \leftarrow S^+ \cup \{s\}$   
    else if  $s$  lies in back of  $h$  then  
       $S^- \leftarrow S^- \cup \{s\}$   
    else  
      split  $s$  into  $s_1, s_2$  at  $h$ ; add to  $S^+$  and  $S^-$   
    end if  
  end for  
  return node( $h$ , BSP( $S^-$ ), BSP( $S^+$ ))  
end function
```

10.8.5 5. Parameters Selections

- **Splitting strategy:** random autopartitioning gives $O(n \log n)$ expected tree size and is easy to implement; greedy strategies that minimize splits improve performance on real data [NAT90].
- **Axis-aligned splits:** restricting to vertical or horizontal lines yields a k-d tree, trading generality for a simpler construction.
- **Robustness:** fragmenting segments requires exact predicates to classify points against the splitting line.

10.8.6 6. Complexity

For n disjoint segments in the plane, an autopartition can be built in $O(n \log n)$ expected time and yields $O(n \log n)$ expected fragments; worst-case size is $O(n^2)$. A point-location or visibility query descends one root-to-leaf path in $O(n)$ time [BCKO08, FKN80].

10.8.7 7. Usages

- **Hidden-surface removal:** the painter's algorithm draws the back-to-front order given by the tree [FKN80].
- **Collision detection:** convex leaf regions allow fast rejection tests in robotics and games.
- **Discrete BSP / image compression:** the Doom engine's rendering and classic BSP-based image coding [NAT90].

10.8.8 8. Testing methods and Edge cases

- **Coincident segments:** overlapping or identical input segments must be assigned consistently to one side.
- **Segments on the splitting line:** define a deterministic tie-break for segments lying exactly on h .

- **Non-uniform inputs:** verify tree size does not blow up to $O(n^2)$ on adversarial configurations.

10.8.9 9. References

- Fuchs, H., Kedem, Z. M., & Naylor, B. F. (1980). “On visible surface generation by a priori tree structures”. *Computer Graphics (SIGGRAPH)*.
- Naylor, B., Amanatides, J., & Thibault, W. (1990). “Merging BSP trees yields polyhedral set operations”. *Computer Graphics (SIGGRAPH)*.
- de Berg, M., Cheong, O., van Kreveld, M., & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer.
- Wikipedia: [Binary space partitioning](#)

10.9 Industrial Applications of BSP Trees

BSP trees were invented for the painter’s algorithm in graphics, but they became the structural foundation of the first-generation real-time 3D game engines, and the axis-aligned special case (the kd-tree) remains the ray-traversal kernel of commercial production renderers [FKN80, NAT90]. The following are the principal industrial use cases.

- **First-Person Shooter Engines (Licensed Middleware):** The Doom engine stored each level as a BSP tree of subsectors, giving back-to-front visibility without per-polygon sorting. The lineage continued in the Quake engines (id Tech 2 and id Tech 3), which were licensed to outside studios—most notably a heavily modified id Tech 3, which shipped as *Call of Duty* (2003), the first release of a franchise that has generated billions in revenue. The BSP leaves in these engines also serve the collision and line-of-sight queries of the game physics [NAT90].
- **Brush Worlds and CSG in Commercial Level Design:** Level geometry in the Source engine is assembled from convex BSP brushes; the editor performs BSP-based CSG (union, subtraction) to carve rooms and archways, and the resulting tree provides visibility culling, portal connectivity, and collision in a single structure. Source powered commercial titles such as *Counter-Strike: Source*, *Team Fortress 2*, and *Half-Life 2*.
- **Shadow Volumes in the Doom 3 Engine:** The BSP tree’s leaf depth ordering lets a stencil shadow volume terminate where it enters an occluded cell. Doom 3 (id Tech 4) popularized this combination, and the technique entered the real-time shadow pipelines of subsequent commercial engines.
- **Production Renderers (kd-Tree Ray Traversal):** The axis-aligned BSP, or kd-tree (Section 10.1), accelerates ray-vs-scene traversal by pruning half of space at each node, and has long served as the acceleration structure of commercial offline renderers such as V-Ray (Chaos), which is licensed by architecture, product-design, and visual-effects firms for photorealistic lighting, shadows, and global illumination.
- **Robotics and Automation Collision Checking:** Each BSP node’s half-space test rejects whole branches of the tree in $O(1)$ time, giving broad-phase collision rejection for robotic manipulators and automated guided vehicles operating against static polygonal cell and warehouse models.

10.9.1 Industrial-Scale Software

BSP-based structures are implemented in the **Doom** and **Quake**/ id Tech engines (level geometry, PVS, and collision), the **Source** engine (brush worlds), and the kd-tree ray-traversal cores of commercial renderers such as V-Ray [FKN80, NAT90].

10.10 Space-Filling Curves

10.11 Hilbert Curve

10.11.1 Overview

The Hilbert curve is a continuous fractal space-filling curve first described by David Hilbert in 1891 [Hil91]. It is a mapping between 1D and 2D (or higher-dimensional) space that preserves locality: points that are close together on the 1D curve are also close together in the 2D plane. This property makes the Hilbert curve an invaluable tool for multidimensional indexing, data compression, and spatial analysis [Sag94].

10.11.2 Definitions

Definition 10.56 (Space-Filling Curve). A continuous function $f: [0, 1] \rightarrow [0, 1]^2$ whose range contains the entire 2-dimensional unit square. Formally, f is *surjective*: for every point $(x, y) \in [0, 1]^2$, there exists $t \in [0, 1]$ such that $f(t) = (x, y)$ [Sag94].

Definition 10.57 (Order (n)). The number of recursive subdivisions. An order- n Hilbert curve covers a grid of $2^n \times 2^n$ cells and visits all 4^n cells exactly once. The total curve length at order n is $2^n - 1$ cell-to-cell transitions.

Definition 10.58 (Locality Preservation). The property where distance on the curve (Hilbert index) correlates strongly with Euclidean distance in space. Specifically, for points $p, q \in [0, 1]^2$ with Hilbert indices $h(p), h(q)$, the Hilbert distance $|h(p) - h(q)|$ is bounded by a polynomial function of the Euclidean distance $\|p - q\|$ [MJFS01].

Definition 10.59 (U-shape). The basic motif of the Hilbert curve, which is rotated and reflected to form higher orders. The four orientations of the U-shape determine how the sub-curves connect at each level of recursion.

10.11.3 Theory

The Hilbert curve is constructed recursively. For order $n = 1$, it is a simple U-shape connecting four cells. To create order $n + 1$:

1. The space is divided into four quadrants.
2. Four order- n curves are placed in the quadrants.
3. The bottom-left curve is rotated 90° clockwise.
4. The bottom-right curve is rotated 90° counter-clockwise and reflected.
5. The four curves are connected by three segments.

The transformation from 2D coordinates (x, y) to a 1D index d (and vice versa) involves bitwise operations and coordinate rotations. As the order $n \rightarrow \infty$, the curve fills the entire square.

Theorem 10.60 (Locality Bound of the Hilbert Curve). *For any two points $p, q \in [0, 1]^2$ with Hilbert indices $h(p)$ and $h(q)$, there exists a constant $c > 0$ such that:*

$$|h(p) - h(q)| \leq c \cdot \|p - q\|_1 \cdot \max\left(\frac{1}{\|p - q\|_1}, 1\right)^{\log_2 d},$$

where d is the dimension. In 2D, this simplifies to the bound that points within a Hilbert-index window of width w are contained in a square of side length $O(\sqrt{w})$ [MJFS01].

Proof. The proof follows from the self-similar structure of the Hilbert curve. At each level k of the recursion, the space is partitioned into 4^k cells, each of side length 2^{-k} . Two points in the same level- k cell have Hilbert indices differing by at most 4^k . Conversely, if their Hilbert indices differ by Δ , they must share a common ancestor cell at level $k = \lceil \log_4 \Delta \rceil$, which has side length $2^{-k} \leq 2\sqrt{\Delta}$. This gives the $O(\sqrt{w})$ spatial bound for a window of width w [MJFS01]. $\square$

Example 10.61 (Hilbert Curve of Order 2). An order-2 Hilbert curve visits all $4^2 = 16$ cells in a 4×4 grid. Starting from the bottom-left corner $(0, 0)$:

$$\begin{aligned} (0, 0) &\rightarrow (1, 0) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (0, 3) \rightarrow (1, 3) \rightarrow (1, 2) \rightarrow \\ &(2, 2) \rightarrow (2, 3) \rightarrow (3, 3) \rightarrow (3, 2) \rightarrow (3, 1) \rightarrow (2, 1) \rightarrow (2, 0) \rightarrow (3, 0) \end{aligned}$$

Points that are consecutive on the curve are always Manhattan neighbors (differing in exactly one coordinate by 1). The four quadrants each contain an order-1 U-shape, rotated and reflected as prescribed by the recursive construction [Sag94].

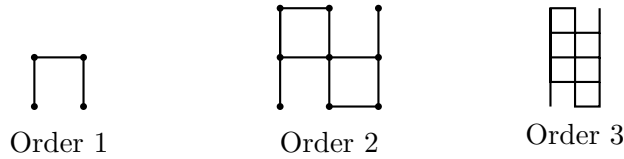


Figure 10.3: Hilbert curve construction: each order recursively places four rotated copies in a U-shaped pattern.

10.11.4 Pseudocode

Algorithm 31 converts 2D coordinates to a 1D Hilbert index using $O(n)$ bitwise operations. The inverse mapping `HilbertToXY` follows the same logic in reverse [Hil91, Sag94].

10.11.5 Parameter Selection

- **Order (n):** Typically chosen so that 2^n matches the resolution of the spatial grid.
- **Dimension:** While most common in 2D, the Hilbert curve can be extended to any number of dimensions k .

10.11.6 Complexity

- **Transformation:** $O(n)$ bitwise operations for an order- n curve. For a fixed grid size, this is essentially $O(1)$.
- **Space:** $O(1)$ auxiliary space to perform the coordinate mapping.

Algorithm 31 Hilbert Curve: Coordinates to Index

```
function XYTOHILBERT( $n, x, y$ )
   $d \leftarrow 0$ 
   $s \leftarrow 2^{n-1}$ 
  while  $s > 0$  do
     $rx \leftarrow (x \ \& \ s) > 0$ 
     $ry \leftarrow (y \ \& \ s) > 0$ 
     $d \leftarrow d + s \cdot s \cdot ((3 \cdot rx) \oplus ry)$  ▷ Rotate and flip
     $x, y \leftarrow \text{RotateHilbert}(s, x, y, rx, ry)$ 
     $s \leftarrow \lfloor s/2 \rfloor$ 
  end while return  $d$ 
end function

function ROTATEHILBERT( $n, x, y, rx, ry$ )
  if  $ry = 0$  then
    if  $rx = 1$  then
       $x \leftarrow n - 1 - x$ 
       $y \leftarrow n - 1 - y$ 
    end if return  $y, x$ 
  end if return  $x, y$ 
end function
```

10.11.7 Usages

- **Spatial Indexing:** Organizing data in databases (like MongoDB or GeoJSON) for faster range queries.
- **Image Dithering:** Ordering pixels for error diffusion to minimize visual artifacts.
- **Parallel Computing:** Mapping multidimensional tasks to linear processing units while maintaining local data relationships (e.g., in domain decomposition).
- **Data Compression:** Improving the performance of compression algorithms by clustering similar nearby values.
- **Cache Optimization:** Storing 2D arrays in Hilbert order to improve the L1/L2 cache hit rate for local access patterns.

10.11.8 Testing Methods and Edge Cases

- **Invertibility:** Verify that $\text{HilbertToXY}(\text{XYtoHilbert}(x, y)) == (x, y)$ for all points in a grid.
- **Adjacency:** Verify that points with sequential indices $d, d + 1$ are always Manhattan neighbors in the 2D grid.
- **Large Order:** Test the transformation with $n = 31$ to ensure bitwise operations don't overflow 64-bit integers.
- **All Quadrants:** Test points in each of the four main quadrants to ensure rotation and reflection logic is correct.

10.11.9 References

- Hilbert, D. (1891). “Über die stetige Abbildung einer Linie auf ein Flächenstück”. *Mathematische Annalen*.
- Sagan, H. (1994). “Space-Filling Curves”. Universitext.
- Moon, B., Jagadish, H. V., Faloutsos, C., & Saltz, J. H. (2001). “Analysis of the Clustering Properties of the Hilbert Space-Filling Curve”. *IEEE Transactions on Knowledge and Data Engineering*.
- Wikipedia: Hilbert curve

10.12 Morton Curve

10.12.1 Overview

The Morton curve (also known as Z-order) is a space-filling curve that maps multidimensional data to one dimension while maintaining some level of spatial locality. It is much simpler to implement than the Hilbert curve, as it is based on bit-interleaving of coordinates [Mor66]. The resulting path forms a “Z” shape at each scale, which is why it is commonly called Z-order. It is a fundamental technique for spatial indexing in computer graphics and databases [Sam84].

10.12.2 Definitions

Definition 10.62 (Bit-Interleaving). Creating a single number by alternating bits from two or more source numbers. For 2D coordinates, the i -th bit of the Morton code is formed by placing the i -th bit of y at position $2i + 1$ and the i -th bit of x at position $2i$.

Definition 10.63 (Z-Order Index). The 1D index (Morton code) resulting from bit-interleaving. For (x, y) with binary representations $x = (x_n \dots x_0)_2$ and $y = (y_n \dots y_0)_2$:

$$Z(x, y) = (y_n x_n y_{n-1} x_{n-1} \dots y_0 x_0)_2.$$

Definition 10.64 (Locality of Z-Order). The property where points close in space are also close on the 1D curve. Z-order preserves locality well, but not as consistently as the Hilbert curve—there are “jumps” at certain power-of-two boundaries where spatially distant cells receive consecutive indices [Sam84].

10.12.3 Theory

For 2D coordinates (x, y) , the Morton code is computed by taking the binary representation of x and y and interleaving them:

- $x = (x_n x_{n-1} \dots x_1 x_0)_2$
- $y = (y_n y_{n-1} \dots y_1 y_0)_2$
- $Z = (y_n x_n y_{n-1} x_{n-1} \dots y_0 x_0)_2$

This process is extremely fast because it can be implemented with simple bitwise shifts and masks. Geometrically, this recursively partitions the space into four quadrants, ordering them in a Z-pattern: $(0, 0) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (1, 1)$.

Theorem 10.65 (Morton Code Uniqueness). *The bit-interleaving map $Z: \mathbb{Z}_{\geq 0}^2 \rightarrow \mathbb{Z}_{\geq 0}$ is a bijection. Every pair of non-negative integers (x, y) maps to a unique Morton code $Z(x, y)$, and the inverse map (de-interleaving) uniquely recovers (x, y) from Z .*

Proof. The map separates the even and odd bit positions: the even-position bits of Z are exactly the bits of x , and the odd-position bits are exactly the bits of y . Since the bit extraction operations (shifting and masking) are deterministic and invertible, the map is a bijection. Concretely, $x = Z \& 0x55555555$ and $y = (Z \gg 1) \& 0x55555555$ recover the original coordinates. $\square$

Example 10.66 (Morton Code for (3, 5)). Convert $(x, y) = (3, 5)$ to a Morton code. In binary: $x = (011)_2$ and $y = (101)_2$. Interleaving:

$$Z = (y_2 x_2 y_1 x_1 y_0 x_0)_2 = (101111)_2 = 47_{10}.$$

Verifying the inverse: extract even bits of $47 = (101111)_2$: positions 0, 2, 4 give $110_2 = 3 = x$. Extract odd bits (positions 1, 3, 5): $101_2 = 5 = y$. The mapping is invertible as guaranteed by Theorem 10.65 [Mor66].

5	6	9	10
4	7	8	11
3	2	13	12
0	1	14	15

Morton (Z-order) curve: bit-interleaving

Figure 10.4: Morton curve: the Z-shaped traversal is obtained by interleaving the binary representations of x and y .

10.12.4 Pseudocode

Algorithm 32 Morton Code: Coordinates to Index

```

function XYTOMORTON( $x, y$ )
     $morton\_code \leftarrow 0$  ▷ Process bits one by one
    for  $i$  in range( $MAX\_BITS$ ) do ▷ Extract  $i$ -th bit from  $x$  and  $y$ 
         $bit\_x \leftarrow (x \gg i) \& 1$ 
         $bit\_y \leftarrow (y \gg i) \& 1$ 
        ▷ Interleave bits:  $y_i$  at position  $2i + 1$ ,  $x_i$  at position  $2i$ 
         $morton\_code \leftarrow morton\_code | (bit\_x \ll (2 \cdot i))$ 
         $morton\_code \leftarrow morton\_code | (bit\_y \ll (2 \cdot i + 1))$ 
    end for
    return  $morton\_code$ 
end function

▷ Faster version using bit manipulation magic (for 16-bit  $x, y$ )
function INTERLEAVEBITS( $x$ )
     $x \leftarrow (x | (x \ll 8)) \& 0x00FF00FF$ 
     $x \leftarrow (x | (x \ll 4)) \& 0x0F0F0F0F$ 
     $x \leftarrow (x | (x \ll 2)) \& 0x33333333$ 
     $x \leftarrow (x | (x \ll 1)) \& 0x55555555$  return  $x$ 
end function

```

Algorithm 32 shows both the straightforward bit-by-bit interleaving and the optimized “bit manipulation magic” version that performs the interleave in $O(1)$ constant time using only shifts and masks [Mor66].

10.12.5 Parameter Selection

- **Bit Depth:** The number of bits used for each coordinate (e.g., 16 bits for a $2^{16} \times 2^{16}$ grid).
- **Interleaving Order:** Conventionally y bits are placed in higher positions, but x -first is also used.

10.12.6 Complexity

- **Transformation:** $O(1)$ for fixed-size integers using bit manipulation (or $O(n)$ where n is bits).
- **Space:** $O(1)$ auxiliary space.

10.12.7 Usages

- **Quadtrees/Octrees:** Morton codes can be used to uniquely represent nodes in a linear quadtree/octree, making them extremely fast to navigate [Sam84].
- **Texture Mapping:** Storing textures in Morton order (often called “swizzling”) significantly improves GPU texture cache hit rates by ensuring local pixels are stored close together in memory.
- **Spatial Databases:** Efficient range queries and nearest neighbor searches.
- **Video Compression:** Partitioning macroblocks in a locality-preserving order.
- **GPGPU Optimization:** Ordering parallel threads to match memory access patterns.

10.12.8 Testing Methods and Edge Cases

- **Invertibility:** Ensure bit-interleaving can be reversed (de-interleaving) to recover the original (x, y) (Theorem 10.65).
- **Boundary Points:** Test the mapping at $(0, 0)$ and the maximum possible coordinate (e.g., $(2^{16} - 1, 2^{16} - 1)$).
- **Adjacency:** Note that Z-order has “jumps” where sequential points on the curve are geographically distant. Verify that these occur only at specific power-of-two boundaries.
- **High Dimensions:** Test that the code works correctly for 3D (x, y, z) coordinates.

10.12.9 References

- Morton, G. M. (1966). “A computer-oriented geodetic data base and a new technique in file sequencing”. IBM Technical Report.
- Samet, H. (1984). “The Quadtree and Related Hierarchical Data Structures”. ACM Computing Surveys.
- Wikipedia: Z-order curve

10.13 Peano Curve

10.13.1 Overview

The Peano curve is the first known example of a space-filling curve, discovered by Giuseppe Peano in 1890 [Pea90]. It is a continuous mapping from a 1D unit interval to a 2D unit square. Unlike the Hilbert curve, which is based on a 2×2 subdivision, the Peano curve is based on a 3×3 subdivision. It is an important foundational structure in mathematical analysis and fractal geometry [Sag94].

10.13.2 Definitions

Definition 10.67 (Space-Filling Curve (Peano)). A continuous function $f: [0, 1] \rightarrow [0, 1]^2$ whose image is all of $[0, 1]^2$. The Peano curve was the first such function to be constructed, answering a question posed by Cantor about whether a continuous bijection between $[0, 1]$ and $[0, 1]^2$ exists [Pea90].

Definition 10.68 (Order (n) of the Peano Curve). The number of recursive subdivisions. An order- n Peano curve covers a grid of $3^n \times 3^n$ cells, visiting all 9^n cells exactly once.

Definition 10.69 (Locality of the Peano Curve). The property of mapping nearby 1D points to nearby 2D points. The Peano curve preserves locality, but with a different structure than the Hilbert curve due to its ternary subdivision [Sag94].

10.13.3 Theory

The Peano curve is constructed recursively using a 3×3 grid [Sag94].

1. For order $n = 1$, the space is divided into 9 squares.
2. The curve visits these squares in an S-like pattern, ensuring that it enters and exits each square in a way that allows for continuous connection between neighbors.
3. Each of the 9 squares is then recursively replaced by a smaller 3×3 Peano curve. Some of these sub-curves must be reflected or rotated to maintain the overall continuity.

Mathematically, the Peano curve is defined by a base-3 representation of coordinates. Interleaving these base-3 digits (trit) provides the 1D index, though the digits must be flipped for certain sub-quadrants to preserve the curve's continuity.

Theorem 10.70 (Peano Curve is Continuous and Surjective). *The Peano curve $f: [0, 1] \rightarrow [0, 1]^2$ is a continuous surjection: for every $(x, y) \in [0, 1]^2$, there exists $t \in [0, 1]$ such that $f(t) = (x, y)$. However, f is not injective: there exist points in $[0, 1]^2$ with multiple pre-images.*

Proof. Continuity follows from the uniform convergence of the sequence of piecewise-linear approximations f_n . At order n , the curve visits all 9^n cells in the $3^n \times 3^n$ grid, and consecutive cells share an edge. The maximum distance between $f_n(t)$ and $f_{n+1}(t)$ is $O(3^{-n})$, so $\{f_n\}$ converges uniformly. The uniform limit of continuous functions is continuous. Surjectivity follows because every cell in the grid is visited at every order, so every point in $[0, 1]^2$ is a limit of visited cells. Non-injectivity arises because the curve must “double back” at certain junction points to maintain continuity [Pea90, Sag94]. $\square$

Example 10.71 (Peano Curve of Order 1). An order-1 Peano curve visits all 9 cells in a 3×3 grid. A standard traversal in S-pattern order (with appropriate reflections) visits:

$$(0, 0) \rightarrow (1, 0) \rightarrow (2, 0) \rightarrow (2, 1) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow$$

$$(0, 2) \rightarrow (1, 2) \rightarrow (2, 2)$$

This forms a backward-S shape. For order 2, each of these 9 cells is replaced by a 3×3 sub-grid with appropriate reflections to maintain continuity at the junctions.

10.13.4 Pseudocode

Algorithm 33 Peano Curve: Coordinates to Index

```

function PEANOINDEX(n, x, y)  ▷ Mapping coordinates (x, y) to a 1D index  ▷ Based on
base-3 representation
  idx ← 0
  for i in range(n − 1, −1, −1) do
    trit_x ←  $\lfloor x/3^i \rfloor \bmod 3$ 
    trit_y ←  $\lfloor y/3^i \rfloor \bmod 3$ 
    ▷ Handle reflections for even-numbered segments

    if x_is_reversed then
      trit_x ← 2 − trit_x
    end if
    if y_is_reversed then
      trit_y ← 2 − trit_y
    end if
    ▷ Segment index (0–8)

    segment ← trit_y · 3 + trit_x
    idx ← idx · 9 + segment
    ▷ Update reflection state for next level

    UpdateReflectionState(trit_x, trit_y)
  end for
  return idx
end function

```

10.13.5 Parameter Selection

- **Order (*n*):** Determines the resolution of the grid ($3^n \times 3^n$).
- **Grid Size:** Must be a power of 3.

10.13.6 Complexity

- **Transformation:** $O(n)$ base-3 digit operations.
- **Space:** $O(1)$ auxiliary space.

10.13.7 Usages

- **Theoretical Analysis:** A fundamental counter-example in topology and real analysis (showing a continuous mapping from 1D to 2D) [Pea90].
- **Image Processing:** Scanning images in a way that respects local structures, similar to Hilbert scans.
- **Data Organization:** Multidimensional indexing (though base-2 curves like Hilbert and Morton are much more common in digital hardware).
- **Fractal Design:** Used in generative art and architectural patterns.

10.13.8 Testing Methods and Edge Cases

- **Continuity:** Verify that for any index i , the Euclidean distance between `Peano(i)` and `Peano(i+1)` is exactly 1 unit in the grid.
- **Full Coverage:** Verify that every (x, y) coordinate in a $3^n \times 3^n$ grid is mapped to a unique index $0 \dots 9^n - 1$.
- **Boundary Cases:** Test coordinates $(0, 0)$ and the maximum coordinate $(3^n - 1, 3^n - 1)$.
- **Reflections:** Carefully test the “flipping” logic in the S-pattern to ensure the curve doesn’t have breaks.

10.13.9 References

- Peano, G. (1890). “Sur une courbe, qui remplit toute une aire plane”. *Mathematische Annalen*.
- Sagan, H. (1994). “Space-Filling Curves”. Universitext.
- Wikipedia: Peano curve

10.14 Zigzag Curve

10.14.1 Overview

The zigzag curve (or snake scan) is a simple space-filling curve that visits every cell in a 2D grid in a back-and-forth pattern. Unlike Hilbert or Morton curves, it does not have fractal properties and is not designed for recursive locality preservation. Instead, it is highly efficient for sequential data access in rectangular arrays and is a common technique in image and video compression [PM92].

10.14.2 Definitions

Definition 10.72 (Zigzag Scan). A path that traverses a grid row-by-row, alternating the direction of traversal for each row. Even-indexed rows are traversed left-to-right, and odd-indexed rows are traversed right-to-left.

Definition 10.73 (Locality of Zigzag Scan). Sequential points on a zigzag curve are geographically close within a row, but there are large jumps when moving between rows. The horizontal locality is perfect (adjacent indices map to adjacent cells), but vertical locality is poor compared to fractal curves.

Definition 10.74 (Raster Scan). A similar scan that always goes in the same direction (e.g., left-to-right). Zigzag is preferred in some cases because it maintains some horizontal locality at the row transitions, avoiding the large horizontal jumps that a raster scan has at each row boundary.

10.14.3 Theory

The zigzag scan is constructed by iterating through the grid row by row.

- For **even-indexed rows** $(0, 2, \dots)$, the curve moves from left to right.
- For **odd-indexed rows** $(1, 3, \dots)$, the curve moves from right to left.

In image compression (like JPEG), a more specific **diagonal zigzag scan** is used. This scan traverses an 8×8 block of frequency coefficients in order of increasing frequency. This ensures that the low-frequency coefficients (which carry most of the image information) are grouped together, making them easier to compress [PM92].

Theorem 10.75 (Zigzag Scan Covers All Cells). *For a $W \times H$ grid, the snake scan produces a path of length $W \cdot H$ that visits every cell (x, y) with $0 \leq x < W$ and $0 \leq y < H$ exactly once. Consecutive cells in the path are always Manhattan neighbors.*

Proof. The scan processes H rows, each containing W cells. For each row y , the cells are enumerated in order: if y is even, the cells $(0, y), (1, y), \dots, (W-1, y)$ are visited; if y is odd, the cells $(W-1, y), (W-2, y), \dots, (0, y)$ are visited. Within a row, consecutive cells differ by exactly 1 in the x -coordinate. Between rows, the last cell of row y and the first cell of row $y+1$ share the same x -coordinate (since row $y+1$ starts from the opposite side), so they are Manhattan neighbors. The total number of cells visited is $H \cdot W$, and since each (x, y) with $0 \leq x < W$, $0 \leq y < H$ appears in exactly one row, the coverage is complete and without repetition. $\square$

Example 10.76 (Snake Scan on a 4×3 Grid). A 4×3 grid traversed by snake scan:

$(0, 0) \rightarrow (1, 0) \rightarrow (2, 0) \rightarrow (3, 0) \rightarrow (3, 1) \rightarrow (2, 1) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (1, 2) \rightarrow (2, 2) \rightarrow (3, 2)$

Row 0 (even): left to right. Row 1 (odd): right to left. Row 2 (even): left to right. All 12 cells are visited exactly once, and consecutive cells are Manhattan neighbors as guaranteed by Theorem 10.75.

10.14.4 Pseudocode

Standard Snake Scan

Algorithm 34 Snake Scan Traversal

```

function SNAKESCAN(width, height)
     $path \leftarrow []$ 
    for  $y$  in range(height) do
        if  $y \bmod 2 = 0$  then ▷ Even row: left to right
            for  $x$  in range(width) do
                 $path.append((x, y))$ 
            end for
        else ▷ Odd row: right to left
            for  $x$  in range(width - 1, -1, -1) do
                 $path.append((x, y))$ 
            end for
        end if
    end for return  $path$ 
end function

```

Diagonal Zigzag (JPEG style)

The diagonal zigzag (Algorithm 35) is particularly important in JPEG compression [PM92], where it orders DCT coefficients so that low-frequency entries come first, enabling efficient run-length encoding.

Algorithm 35 Diagonal Zigzag Scan (JPEG)

```
function DIAGONALZIGZAG(size)                                ▷ Scan a square block diagonally
    path ← []
    for s in range(2 · size − 1) do
        if s mod 2 = 0 then                                    ▷ Up-right diagonal
            for y in range(size) do
                x ← s − y
                if 0 ≤ x < size then
                    path.append((x, y))
                end if
            end for
        else                                                    ▷ Down-left diagonal
            for x in range(size) do
                y ← s − x
                if 0 ≤ y < size then
                    path.append((x, y))
                end if
            end for
        end if
    end for return path
end function
```

10.14.5 Parameter Selection

- **Direction:** Can be row-major (standard snake) or diagonal (compression).
- **Block Size:** For JPEG, the block size is fixed at 8×8 .

10.14.6 Complexity

- **Transformation:** $O(1)$ to compute a coordinate from an index (and vice versa) using modular arithmetic.
- **Space:** $O(1)$ auxiliary space.

10.14.7 Usages

- **Image/Video Compression (JPEG, MPEG):** Ordering DCT coefficients to group low-frequency data, which improves the efficiency of run-length encoding (RLE) [PM92].
- **Display Technology:** Scanning pixels in CRT or LCD screens.
- **Robotics/3D Printing:** Planning the path for a nozzle or sensor to cover a flat area (rasterizing a plane).
- **Sensor Networks:** Ordering data collection from a grid of sensors.

10.14.8 Testing Methods and Edge Cases

- **Full Coverage:** Verify that every (x, y) in the grid appears exactly once in the scan.
- **Continuity:** Verify that consecutive points in the path are Manhattan neighbors.
- **Non-Square Grids:** Ensure the algorithm handles $W \neq H$ correctly.
- **Zero/One dimensions:** Test on $1 \times N$ or $N \times 1$ grids.

10.14.9 References

- Pennebaker, W. B., & Mitchell, J. L. (1992). “JPEG: Still Image Data Compression Standard”. Van Nostrand Reinhold.
- Wikipedia: Zigzag

10.15 Spiral Walk

10.15.1 Overview

A spiral walk is a space-filling traversal that starts at a center point and visits cells in a 2D grid in an expanding spiral pattern. Unlike Hilbert or Morton curves, which are recursive and locality-preserving, the spiral walk is intuitive and ensures that the distance from the starting point increases monotonically (on average). It is used for localized searching, image scanning, and data visualization.

10.15.2 Definitions

Definition 10.77 (Archimedean Spiral). A continuous curve where the distance from the origin is proportional to the angle: $r = a + b\theta$. The discrete square spiral is an approximation to the Archimedean spiral on an integer grid.

Definition 10.78 (Square Spiral). A discrete version of the spiral restricted to a square grid. It visits cells in layers, where layer k consists of the $8k$ cells forming the perimeter of a $(2k + 1) \times (2k + 1)$ square centered at the origin.

Definition 10.79 (Radius (r) of a Spiral Walk). The maximum Manhattan or Chebyshev distance from the starting point during the walk. After visiting all cells within radius r , the total number of cells visited is $(2r + 1)^2$.

10.15.3 Theory

The square spiral walk is typically performed by moving in layers [SUW64].

1. **Layer 0:** The starting cell $(0, 0)$.
2. **Layer 1:** The 8 neighbors surrounding the center.
3. **Layer k :** The $8k$ cells that form the perimeter of a $(2k + 1) \times (2k + 1)$ square.

The walk follows a repeating pattern of directions (e.g., Right, Up, Left, Down) with increasing segment lengths. For example:

- Right 1, Up 1
- Left 2, Down 2
- Right 3, Up 3
- Left 4, Down 4
- ...

Theorem 10.80 (Spiral Walk Coverage). After completing layer k (i.e., after $N = (2k + 1)^2$ steps), the spiral walk has visited every cell (x, y) with $|x| \leq k$ and $|y| \leq k$. The total number of steps to cover a square of radius r is $(2r + 1)^2$.

Proof. Layer k adds the cells on the perimeter of the $(2k+1) \times (2k+1)$ square that are not already covered by layers $0, \dots, k-1$. The number of cells on this perimeter is $(2k+1)^2 - (2(k-1)+1)^2 = (2k+1)^2 - (2k-1)^2 = 8k$. By induction, after completing layer k , the walk has visited all cells within the $(2k+1) \times (2k+1)$ square, totaling $\sum_{j=0}^k 8j + 1 = (2k+1)^2$ cells [SUW64]. $\square$

Example 10.81 (First 25 Steps of a Square Spiral). The first 25 steps of the square spiral (covering a 5×5 square):

$(0, 0) \rightarrow (1, 0) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow (-1, 1) \rightarrow (-1, 0) \rightarrow (-1, -1) \rightarrow (0, -1) \rightarrow (1, -1) \rightarrow$
 $(2, -1) \rightarrow (2, 0) \rightarrow (2, 1) \rightarrow (2, 2) \rightarrow (1, 2) \rightarrow (0, 2) \rightarrow (-1, 2) \rightarrow (-2, 2) \rightarrow (-2, 1) \rightarrow$
 $(-2, 0) \rightarrow (-2, -1) \rightarrow (-2, -2) \rightarrow (-1, -2) \rightarrow (0, -2) \rightarrow (1, -2) \rightarrow (2, -2)$

After $25 = (2 \cdot 2 + 1)^2$ steps, all cells in the 5×5 square are covered, as guaranteed by Theorem 10.80.

10.15.4 Pseudocode

Algorithm 36 Square Spiral Walk

```

function SPIRALWALK(num_steps)
     $x, y \leftarrow 0, 0$ 
     $path \leftarrow [(x, y)]$ 
    ▷ Directions: Right, Up, Left, Down

     $dx \leftarrow [1, 0, -1, 0]$ 
     $dy \leftarrow [0, 1, 0, -1]$ 
     $step\_limit \leftarrow 1$ 
     $direction \leftarrow 0$ 
    while  $\text{len}(path) < num\_steps$  do ▷ Move in current direction for 'step_limit' steps
        for  $_$  in  $\text{range}(2)$  do ▷ Two directions share the same limit
            for  $_$  in  $\text{range}(step\_limit)$  do
                if  $\text{len}(path) \geq num\_steps$  then
                    break
                end if
                 $x \leftarrow x + dx[direction]$ 
                 $y \leftarrow y + dy[direction]$ 
                 $path.append((x, y))$ 
            end for
            ▷ Change direction
             $direction \leftarrow (direction + 1) \bmod 4$ 
        end for
        ▷ Increase segment length after every two turns
         $step\_limit \leftarrow step\_limit + 1$ 
    end while
    return  $path$ 
end function

```

10.15.5 Parameter Selection

- **Starting Direction:** Can be any of the four cardinal directions.
- **Orientation:** Clockwise vs. Counter-clockwise.
- **Step Size:** Standard is 1 unit, but can be larger for sparse sampling.

10.15.6 Complexity

- **Time Complexity:** $O(n)$ where n is the number of steps.
- **Space Complexity:** $O(n)$ to store the path coordinates.

10.15.7 Usages

- **Search Algorithms:** Finding the nearest point of interest starting from a query location (e.g., “nearest gas station” in a local grid).
- **Computer Vision:** Scanning a local neighborhood around a keypoint or feature.
- **Procedural Generation:** Growing a city or a structure from a central “seed” location.
- **Data Visualization:** Arranging objects around a center based on their priority or timestamp (e.g., a “spiral timeline”).
- **Cryptography:** Permuting bits or pixels in a non-linear but deterministic way.

10.15.8 Testing Methods and Edge Cases

- **Full Coverage:** Verify that every (x, y) coordinate within a k -layer square is visited exactly once.
- **Monotonicity:** Ensure that for any two sequential points P_i, P_{i+1} , their coordinates change by exactly 1 unit.
- **Large Step Count:** Verify the coordinates don’t drift or skip cells for $n > 10^6$.
- **Start at Edge:** Handle cases where the spiral is constrained within a bounding box and must stop at the edges.

10.15.9 References

- Stein, M. L., Ulam, S. M., & Wells, M. B. (1964). “A Visual Display of Some Properties of the Distribution of Primes”. The American Mathematical Monthly. (The Ulam Spiral).
- Gardner, M. (1971). “The Extraordinary Properties of the Mathematical Constant Pi”. Scientific American.
- Wikipedia: Spiral

Part IV

Proximity and Tessellations

Chapter 11

Voronoi Diagrams

11.1 Voronoi Diagram

A Voronoi diagram partitions the plane into regions based on distance to a set of sites. The cell of a site contains the points closer to it than to any other site. The Voronoi diagram is the geometric dual of the Delaunay triangulation [Vor08, Aur91].

11.1.1 Definitions

Definition 11.1 (Voronoi Cell). For sites $S = \{s_1, \dots, s_n\} \subset \mathbb{R}^2$, the Voronoi cell of s_i is

$$V(s_i) = \{p \in \mathbb{R}^2 \mid \|p - s_i\| \leq \|p - s_j\| \text{ for all } j \neq i\}.$$

Definition 11.2 (Voronoi Vertex). A Voronoi vertex is a point where three or more cells meet. It is equidistant from its closest sites and corresponds to the circumcenter of a Delaunay triangle.

11.1.2 Delaunay Duality

The duality is given by the following correspondences:

1. A Delaunay vertex corresponds to a Voronoi cell.
2. A Delaunay triangle corresponds to a Voronoi vertex.
3. A Delaunay edge corresponds to a Voronoi edge on a perpendicular bisector.

Theorem 11.3 (Duality of Delaunay and Voronoi Diagrams). *For sites in general position, the Delaunay triangulation and Voronoi diagram are planar duals and both have linear complexity.*

11.1.3 Pseudocode

11.1.4 Complexity and Applications

Construction through a Delaunay triangulation takes $O(n \log n)$ time and $O(n)$ space. A bounding box is required to cap cells belonging to sites on the convex hull. Applications include facility location, robot path planning, territory modeling, and nearest-neighbor search.

11.1.5 Testing and Edge Cases

Sites on the convex hull produce unbounded cells, co-circular sites produce non-unique vertices, collinear sites create degeneracies, and duplicate sites must be merged or rejected.

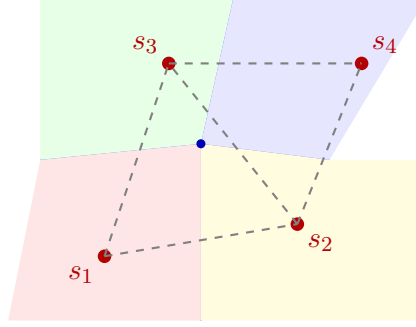


Figure 11.1: Voronoi diagram (colored cells) and its Delaunay triangulation dual (dashed gray edges). Each Voronoi vertex is the circumcenter of a Delaunay triangle.

Algorithm 37 Voronoi Diagram via Delaunay Dual

```

1: function COMPUTEVORONOI(sites, bounding_box)
2:    $DT \leftarrow \text{DelaunayTriangulation}(\text{sites})$ 
3:    $C \leftarrow \{\text{Circumcenter}(T) : T \in DT.\text{triangles}\}$ 
4:   for  $site$  in  $sites$  do
5:      $star \leftarrow \text{FindTrianglesSharingVertex}(site, DT)$ 
6:      $boundary \leftarrow \text{SortAround}(site, C[star])$ 
7:      $cell \leftarrow \text{ClipPolygon}(boundary, bounding\_box)$ 
8:     Store  $cell$ 
9:   end for
10:  return all stored cells
11: end function

```

11.2 Voronoi Diagrams of Line Segments

The generalized Voronoi diagram of a set of disjoint line segments $S = \{s_1, \dots, s_n\}$ partitions the plane according to the Euclidean distance to the closest point of each segment. Segment sites arise whenever features are modeled as one-dimensional objects: road networks, rivers, and walls in GIS, or the edges of polygonal obstacles in robot motion planning [LD81, OBSC00]. The medial axis of a polygonal domain is a subset of this diagram.

11.2.1 Definitions

Definition 11.4 (Distance to a Segment). The distance from a point p to a closed segment $s = \overline{ab}$ is

$$\text{dist}(p, s) = \min_{q \in s} \|p - q\|_2,$$

attained at the orthogonal projection of p onto the supporting line of s clamped to $[a, b]$; if the projection falls outside the segment, the minimum is attained at the nearer endpoint.

Definition 11.5 (Voronoi Cell of a Segment). The Voronoi cell of segment s_i is

$$V(s_i) = \{p \in \mathbb{R}^2 \mid \text{dist}(p, s_i) \leq \text{dist}(p, s_j) \text{ for all } j \neq i\}.$$

11.2.2 Theory

The bisector of two sites is the locus of points equidistant from them. For segment sites the bisector is composed of

- a **line** between two points (or between two parallel segment interiors),

- a **parabola** whose focus is a segment endpoint and whose directrix is the supporting line of the other segment, or
- an **angle bisector** between two non-parallel supporting lines.

Consequently the edges of a segment-site Voronoi diagram are straight segments or parabolic arcs, and its cells are star-shaped but not necessarily convex.

Theorem 11.6 (Complexity of the Segment Voronoi Diagram). *The Voronoi diagram of n disjoint line segments has $O(n)$ vertices, edges, and cells, and can be computed in $O(n \log n)$ time.*

Proof sketch. Each segment contributes a bounded number of features, and the planar structure of the diagram bounds the total complexity linearly. The $O(n \log n)$ construction time is achieved by Fortune’s sweep-line algorithm generalized to segment sites—the beach line then alternates between parabolic arcs (foci at segment endpoints) and line segments—and by divide-and-conquer constructions [For87, LD81]. $\square$

11.2.3 Pseudocode

Algorithm 38 Segment Voronoi Diagram by Plane Sweep

```

1: function SEGMENTVORONOI(segments)
2:    $Q \leftarrow$  site events (segment endpoints) and circle events
3:    $T \leftarrow$  balanced tree holding the beach line of lines and parabolas
4:   initialize output structure for each segment
5:   while  $Q$  is not empty do
6:      $e \leftarrow Q.\text{pop}()$ 
7:     if  $e$  is a site event then
8:       insert the new beach-line arc into  $T$ 
9:       detect and schedule new circle events
10:    else ▷ circle event: an arc disappears
11:      create a Voronoi vertex and its incident edges
12:      remove the arc and update circle events
13:    end if
14:  end while
15:  return edges and vertices, clipped to the bounding box
16: end function

```

11.2.4 Complexity and Applications

The diagram needs $O(n \log n)$ time and $O(n)$ space. Applications include

- the **medial axis** of a polygonal domain (a subset of the diagram),
- **nearest-obstacle** and clearance queries for a robot among polygonal obstacles, and
- **1-skeleton** neighborhood queries in road and river networks.

11.2.5 Testing and Edge Cases

Collinear overlapping segments, parallel segments (whose bisectors are parallel lines), shared endpoints, and segments touching the bounding box must be handled; predicates for parabolic bisectors need care near focus degeneracies.

11.3 Farthest-Point Voronoi Diagrams

The farthest-point Voronoi diagram assigns to each site the region of points for which that site is the *farthest* among all sites. It is the exact analogue of the ordinary Voronoi diagram with min replaced by max, and it is the key structure for computing the diameter and the smallest enclosing circle of a point set [SH75, OBSC00].

11.3.1 Definitions

Definition 11.7 (Farthest-Point Voronoi Cell). For sites $S = \{s_1, \dots, s_n\} \subset \mathbb{R}^2$, the farthest-point Voronoi cell of s_i is

$$V^*(s_i) = \{p \in \mathbb{R}^2 \mid \|p - s_i\| \geq \|p - s_j\| \text{ for all } j \neq i\}.$$

11.3.2 Theory

Theorem 11.8 (Structure of the Farthest-Point Voronoi Diagram). *A site has a non-empty farthest-point Voronoi cell if and only if it is a vertex of the convex hull of S . If the hull $\mathcal{H}(S)$ has m vertices, the farthest-point Voronoi diagram has m cells and $m - 2$ vertices, is a tree, and its edges lie on perpendicular bisectors of pairs of hull vertices.*

The diagram is the dual of the Delaunay triangulation of the hull vertices restricted to the upper side of the lifted paraboloid: a farthest-point Voronoi vertex is the center of a circle through three hull vertices that contains all of S . Both the nearest- and farthest-point diagrams can be obtained from the same sweep-line construction by duality [For87].

11.3.3 Applications: Diameter and Smallest Enclosing Circle

The principal use of the farthest-point Voronoi diagram is optimization over the point set:

- **Diameter:** the two points realizing the diameter of S are the endpoints of an edge of the farthest-point Delaunay diagram of the hull; the diameter is found in $O(n \log n)$ time by rotating calipers on the hull.
- **Smallest enclosing circle:** the center of the minimum-radius circle enclosing S is either a farthest-point Voronoi vertex (the center of a circle through three hull vertices that contains all points) or the midpoint of a diameter pair. Welzl's randomized algorithm [Wel91] solves the same problem directly in expected linear time.

11.3.4 Pseudocode

11.3.5 Complexity and Applications

The hull takes $O(n \log n)$ time; the farthest-point Voronoi diagram of the m hull vertices is a tree with m cells and takes $O(m)$ time and space once the hull is known. Scanning its vertices costs $O(m)$, for an overall $O(n \log n)$.

11.3.6 Testing and Edge Cases

Collinear point sets (the diagram degenerates to parallel slabs), co-circular hull vertices, and hulls with very few vertices (with two points every cell is a half-plane) are the main edge cases.

Algorithm 39 Smallest Enclosing Circle via Farthest-Point Voronoi

```

1: function SMALLESTENCLOSINGCIRCLE(points)
2:    $H \leftarrow \text{ConvexHull}(\text{points})$ 
3:    $FPV \leftarrow \text{FarthestPointVoronoi}(H)$ 
4:    $best \leftarrow$  circle through the two points of the diameter of  $H$ 
5:   for vertex  $v$  of  $FPV$  do
6:      $C \leftarrow$  circle centered at  $v$  through its three hull vertices
7:     if  $\text{radius}(C) < \text{radius}(best)$  and  $C$  encloses all points then
8:        $best \leftarrow C$ 
9:     end if
10:  end for
11:  return  $best$ 
12: end function

```

11.3.7 References

- Shamos, M. I. (1978). “Computational Geometry”. PhD thesis, Yale University.
- Fortune, S. (1987). “A sweepline algorithm for Voronoi diagrams.”
- Okabe, A., Boots, B., Sugihara, K., & Chiu, S. N. (2000). “Spatial Tessellations: Concepts and Applications of Voronoi Diagrams”. Wiley.

11.4 Kinetic Voronoi Diagrams

A kinetic Voronoi diagram maintains the diagram of moving sites. Its dual is maintained through a kinetic Delaunay triangulation. Certificates based on the in-circle predicate predict when an edge becomes invalid; the next event is then processed by an edge flip.

Algorithm 40 Kinetic Voronoi Maintenance

```

1: function MAINTAINKINETICVORONOI(points, trajectories)
2:    $DT \leftarrow \text{DelaunayTriangulation}(\text{points})$ 
3:    $Q \leftarrow \text{PriorityQueueOfCertificateFailures}(DT, \text{trajectories})$ 
4:   while  $Q$  is not empty do
5:      $event \leftarrow Q.\text{pop}()$ 
6:     Flip the invalid edge in  $DT$ 
7:     Update certificates for affected edges
8:   end while
9: end function

```

The event-driven structure processes $O(\log n)$ priority-queue work per event. Degenerate cases include zero-velocity sites, collisions, simultaneous co-circular events, and sites leaving the region of interest.

- Voronoi, G. (1908). “Nouvelles applications des paramètres continus.”
- Fortune, S. (1987). “A sweepline algorithm for Voronoi diagrams.”
- Aurenhammer, F. (1991). “Voronoi Diagrams: A Survey.”

11.5 Industrial Applications of Voronoi Diagrams

Voronoi diagrams appear throughout industry because the dual Delaunay triangulation (Section 12.1) and the nearest-site assignment they encode are the exact solution to many “assign each point to the closest resource” problems. The following are the principal industrial use cases.

- **Retail and Service-Area Analysis (GIS):** Retail chains, banks, and emergency services use the Voronoi diagram (“Thiessen polygons”) of existing sites to compute service territories and to find the location of a new facility that maximizes the underserved area—the site whose insertion grows the smallest uncovered Voronoi cell the most. Thiessen-polygon generation is a standard tool in commercial GIS—Esri ArcGIS (3D Analyst) and MapInfo—sold to thousands of planning departments [OBSC00].
- **Wireless and Cellular Network Planning:** The Voronoi cells of base stations or small cells are the coverage regions under nearest-site association; load balancing and cell planning are performed directly on these cells, and the Delaunay graph gives the interference and handover topology. Commercial RF-planning suites (Atoll, Planet) sold to carriers and network operators build on this decomposition [OBSC00].
- **Logistics and Field-Service Territory Design:** Voronoi territories of distribution centers or service technicians partition demand points so that every customer is served by the closest resource; commercial territory- and route-optimization products (for example, ORTEC and Descartes) ship Voronoi-based territory design, recomputed as fleet and demand change.
- **Reservoir Simulation Grids (PEBI/Voronoi):** The perpendicular-bisector (PEBI) simulation grids of the oil and gas industry are Voronoi diagrams whose cells honor wells, faults, and fracture networks. Commercial reservoir-modeling software such as Roxar RMS (Emerson) sells this gridding as part of its seismic-to-simulation workflow.
- **Exact Nearest-Site Queries in Geodata Stacks:** Databases answer “nearest X to each location” exactly by materializing Voronoi cells: PostGIS (ST_VoronoiPolygons, via GEOS) exposes this in the SQL stacks of commercial geodata platforms and location services.

11.5.1 Industrial-Scale Software

Production implementations of the Voronoi diagram and its Delaunay dual are available in CGAL (2D and 3D Voronoi), `scipy.spatial` (Voronoi, Delaunay), `Boost.Polygon`, `voro++` (3D parallel Voronoi for molecular and physics simulations), and PostGIS (ST_VoronoiPolygons for spatial databases) [Aur91, OBSC00].

Chapter 12

Delaunay Triangulations

12.1 Voronoi–Delaunay Duality

12.1.1 1. Overview

The Delaunay triangulation of a point set P is the straight-line dual of the Voronoi diagram (Section 11.2): two sites $p, q \in P$ are connected by a Delaunay edge exactly when their Voronoi cells share a boundary edge. The dual of a Voronoi diagram of points in general position is a planar straight-line graph whose bounded faces are triangles, so the dual is a triangulation of the convex hull of P . The two diagrams form a planar dual pair: Voronoi vertices (circumcenters) correspond to Delaunay triangles, Voronoi edges to Delaunay edges, and Voronoi faces to Delaunay vertices [dBCvKO08, OBSC00].

12.1.2 2. Definitions

Definition 12.1 (Voronoi Cell). For a site $p \in P$, the Voronoi cell $V(p)$ is the set of points of the plane at least as close to p as to any other site:

$$V(p) = \{x \in \mathbb{R}^2 : \|x - p\| \leq \|x - q\| \text{ for all } q \in P\}.$$

Definition 12.2 (Straight-Line Dual). The *Delaunay triangulation* $DT(P)$ is the graph with vertex set P and an edge pq for every pair of sites whose Voronoi cells $V(p)$ and $V(q)$ share a common edge of positive length.

Definition 12.3 (General Position). A planar point set is in *general position* if no four points are cocircular and no three are collinear. Under this assumption every Voronoi vertex has degree three and the dual graph is a genuine triangulation.

12.1.3 3. Theory

Theorem 12.4 (Duality). *Let P be a set of n points in general position in the plane. The straight-line dual of the Voronoi diagram $VD(P)$ is the Delaunay triangulation $DT(P)$, with the following correspondences:*

- *each Voronoi vertex, the circumcenter of three sites p, q, r , corresponds to the Delaunay triangle pqr , whose circumcircle passes through the three sites;*
- *each Voronoi edge lies on the perpendicular bisector of its dual Delaunay edge;*
- *each Voronoi face $V(p)$ corresponds to the Delaunay vertex p .*

Proof sketch. A point of the plane equidistant from exactly three sites p, q, r is their circumcenter; it is the unique point where $V(p)$, $V(q)$, and $V(r)$ meet, so it is a Voronoi vertex and its dual is the triangle pqr . A point equidistant from exactly two sites p, q lies on their perpendicular bisector, so the shared boundary of $V(p)$ and $V(q)$ is a straight segment and the dual edge pq is a straight segment. Distinct such segments meet only at Voronoi vertices, which are endpoints of the corresponding dual edges, so the straight-line embedding is planar. $\square$

The planar duality yields exact size bounds.

Corollary 12.5 (Size). *If h points of P lie on the boundary of the convex hull of P , then $DT(P)$ has exactly $2n - 2 - h$ triangles and $3n - 3 - h$ edges.*

Proof. The planar graph $DT(P)$ has n vertices, e edges, and $t + 1$ faces where t is the number of triangular faces. Euler's formula gives $n - e + (t + 1) = 2$. The unbounded face has h boundary edges and each triangle contributes three edge-sides, so $3t + h = 2e$. Eliminating e gives $t = 2n - 2 - h$ and $e = 3n - 3 - h$. $\square$

12.1.4 4. Pseudocode

Algorithm 41 Voronoi Cells from a Delaunay Triangulation

Input: A Delaunay triangulation $DT(P)$.

Output: The Voronoi diagram $VD(P)$.

```

1: function VORONOIFROMDELAUNAY( $DT(P)$ )
2:   for each triangle  $T = (p, q, r)$  do
3:      $c_T \leftarrow \text{circumcenter}(p, q, r)$ 
4:   end for
5:   for each Delaunay edge  $(p, q)$  do
6:     emit the segment through the circumcenters of the one or two triangles sharing  $(p, q)$ ,
       clipped to  $V(p) \cap V(q)$ 
7:   end for
8:   return the arrangement of circumcenters and connecting segments
9: end function

```

12.1.5 5. Parameter Selection

- **General position:** real data is rarely in general position; algorithms either break cocircular ties symbolically (simulation of simplicity) or deterministically (e.g., by a lexicographic rule on the point order).
- **Representation:** a half-edge (DCEL) structure stores both diagrams at once; each Delaunay edge keeps a pointer to its dual Voronoi edge and vice versa.

12.1.6 6. Complexity

- **Time:** $O(n \log n)$ to construct $DT(P)$ by any of the algorithms of Sections 12.6–12.7; the dual $VD(P)$ costs $O(n)$ more.
- **Space:** $O(n)$ for either diagram; both together remain $O(n)$ because both are planar graphs.

12.1.7 7. Usages

- **Meshing:** Delaunay refinement (Section 12.12) exploits the empty-circumcircle property inherited from the duality.
- **Nearest-site queries:** point location in $VD(P)$ answers “closest site” queries in $O(\log n)$.
- **Surface reconstruction:** the alpha complex of the Delaunay complex (Section 12.13) reconstructs surfaces from point clouds.

12.1.8 8. Testing Methods and Edge Cases

- **Cocircular points:** four cocircular sites give a dual quadrilateral; verify that the two triangulations obtained by the two diagonals are both accepted and reported consistently.
- **Collinear sites:** three collinear sites create a degenerate (zero-area) Voronoi vertex; the dual must not emit zero-area triangles.
- **Duplicate sites:** coincident sites have an empty Voronoi cell; the dual must ignore the duplicate.

12.1.9 9. References

- de Berg, M., Cheong, O., van Kreveld, M., Overmars, M. [dBCvKO08], Ch. 7 and 9.
- Okabe, A., Boots, B., Sugihara, K., Chiu, S. N. [OBSC00].
- Aurenhammer, F. [Aur91].
- Voronoi, G. [Vor08].

12.2 Empty-Circumcircle Property

12.2.1 1. Overview

The defining property of a Delaunay triangulation is that no vertex of the point set lies inside the circumcircle of any triangle. This *empty-circumcircle property* is the local certificate that makes the Delaunay triangulation easy to recognize, to verify, and to maintain under insertions and flips. It is equivalent to the Voronoi duality of Section 12.1: a triangle whose circumcircle is empty has a circumcenter that is a Voronoi vertex.

12.2.2 2. Definitions

Definition 12.6 (Empty Circumcircle). A triangle T of a triangulation of P satisfies the *empty-circumcircle property* if the open disk bounded by the circumcircle of T contains no point of P .

Definition 12.7 (In-Circle Test). Let a, b, c, d be points with a, b, c oriented counterclockwise. The *in-circle predicate* returns true if d lies strictly inside the circle through a, b, c :

$$\text{InCircle}(a, b, c, d) = \det \begin{pmatrix} a_x & a_y & a_x^2 + a_y^2 & 1 \\ b_x & b_y & b_x^2 + b_y^2 & 1 \\ c_x & c_y & c_x^2 + c_y^2 & 1 \\ d_x & d_y & d_x^2 + d_y^2 & 1 \end{pmatrix} > 0.$$

The determinant is zero exactly when the four points are cocircular.

Definition 12.8 (Delaunay Triangulation). A triangulation of P is a *Delaunay triangulation* if every triangle satisfies the empty-circumcircle property.

12.2.3 3. Theory

Theorem 12.9 (Empty Circle Characterizes Delaunay). *A triangulation of a point set P in general position is the Delaunay triangulation if and only if every triangle has an empty circumcircle.*

Proof sketch. ($\Leftarrow$) If every triangle of a triangulation $\mathcal{T}$ has an empty circumcircle, then each triangle's circumcenter is a Voronoi vertex of $VD(P)$: the open disk is empty, so the circumcenter is as close to the three triangle vertices as to any other site. The triangulation $\mathcal{T}$ is then the straight-line dual of $VD(P)$, hence the Delaunay triangulation. ($\Rightarrow$) If $\mathcal{T}$ is the Delaunay triangulation, its triangles are dual to Voronoi vertices, whose circumcircles contain no site in their interior. $\square$

The in-circle determinant is closely tied to the orientation test. Writing the four points as lifted points on the paraboloid $z = x^2 + y^2$, the determinant is (up to sign) the oriented volume of the tetrahedron a^+, b^+, c^+, d^+ , which explains both the sign convention and the numerical behavior (Section 12.8).

12.2.4 4. Pseudocode

Algorithm 42 Adaptive In-Circle Predicate

Input: points a, b, c oriented counterclockwise and query point d .

Output: true if d is inside the circle through a, b, c .

```

1: function INCIRCLE( $a, b, c, d$ )
2:    $\Delta \leftarrow$  determinant of the  $4 \times 4$  matrix of Definition 12.7, computed by floating-point arithmetic
3:   if  $|\Delta|$  safely larger than the error bound then
4:     return  $\Delta > 0$ 
5:   else
6:     recompute  $\Delta$  exactly using arbitrary-precision arithmetic
7:     return  $\Delta > 0$ 
8:   end if
9: end function

```

Adaptive evaluation (Section 12.12 uses the same technique) runs at full floating-point speed in the common case and only falls back to exact arithmetic near degenerate configurations [She97].

12.2.5 5. Parameter Selection

- **Filtering:** cheap floating-point filters with rigorous error bounds reject most inputs before exact arithmetic is invoked (Section 3.4).
- **Exact construction:** when circumcenters themselves are inserted (Delaunay refinement), exact predicates still help decide legality even though the new points are irrational.

12.2.6 6. Complexity

- **Time:** the predicate is $O(1)$; with adaptive filtering the expected cost is a small constant number of floating-point operations.
- **Space:** $O(1)$.

12.2.7 7. Usages

- **Legality tests:** the flip criterion of Section 12.4 calls `InCircle` once per edge.
- **Verification:** checking that a mesh produced by any method is Delaunay reduces to one predicate per interior edge.
- **Voronoi computation:** empty circles certify Voronoi vertices.

12.2.8 8. Testing Methods and Edge Cases

- **Cocircular queries:** verify that the predicate returns exactly zero and that callers treat the result as “legally Delaunay” (either diagonal is acceptable).
- **Counterclockwise orientation:** the sign convention assumes a, b, c counterclockwise; test both orientations.
- **Large coordinates:** exact fallback must be exercised on coordinates far from the origin, where floating-point filters are inconclusive.

12.2.9 9. References

- Guibas, L., Stolfi, J. [GS85].
- Shewchuk, J. R. [She97], adaptive robust predicates.
- de Berg, M., et al. [dBCvKO08], Ch. 9.

12.3 Local Delaunay Criterion

12.3.1 1. Overview

Checking every triangle of a triangulation against all other points would be expensive. The *local Delaunay criterion* reduces the global empty-circle condition to a test on each interior edge: an interior edge is locally Delaunay when the vertex of one incident triangle does not lie inside the circumcircle of the other. A triangulation in which every interior edge is locally Delaunay is globally Delaunay [dBCvKO08].

12.3.2 2. Definitions

Definition 12.10 (Locally Delaunay Edge). Let $e = (u, v)$ be an interior edge shared by triangles (u, v, w) and (u, v, z) . The edge e is *locally Delaunay* if z does not lie strictly inside the circumcircle of (u, v, w) ; by symmetry this is equivalent to w not lying inside the circumcircle of (u, v, z) .

Definition 12.11 (Legal and Illegal Edges). An interior edge that is not locally Delaunay is *illegal*; the other diagonal of the quadrilateral (u, w, v, z) is the unique candidate replacement that is legal in the Delaunay sense.

12.3.3 3. Theory

Theorem 12.12 (Local Implies Global). *A triangulation of a point set in general position is the Delaunay triangulation if and only if every interior edge is locally Delaunay.*

Proof sketch. The forward direction is immediate from the empty-circumcircle property. For the converse, suppose every interior edge is locally Delaunay and let T be any triangle. If T 's circumcircle contained a point p of P , consider the segment from p to a vertex of T ; walking across the triangulation from the face containing p to T crosses an interior edge whose flip strictly increases the minimum angle of the incident quadrilateral, contradicting the assumption that all edges are locally Delaunay (the standard “no edge can be illegal” exchange argument of Lawson [Law77]). $\square$

The theorem is the basis of all flip-based algorithms: it suffices to fix illegal edges one at a time.

12.3.4 4. Pseudocode

Algorithm 43 Local Delaunay Test

```

1: function ISLOCALLYDELAUNAY( $u, v, w, z$ )
2:   if  $z$  lies strictly inside the circle through  $u, v, w$  then
3:     return FALSE  $\triangleright e = (u, v)$  is illegal
4:   end if
5:   return TRUE
6: end function

```

12.3.5 5. Parameter Selection

- **Boundary edges:** edges on the convex hull have no second triangle and are always treated as locally Delaunay.
- **Cocircularity:** when the four vertices of a quadrilateral are cocircular, both diagonals are locally Delaunay; pick either consistently.

12.3.6 6. Complexity

- **Time:** $O(1)$ per edge using the in-circle predicate.
- **Overall:** a full verification pass over a triangulation costs $O(n)$ predicate calls.

12.3.7 7. Usages

- **Edge flipping** (Section 12.4) flips precisely the edges that fail this test.
- **Incremental insertion** re-checks only the edges of the affected neighborhood after each insertion.

12.3.8 8. Testing Methods and Edge Cases

- **Non-convex quadrilaterals:** after a flip the new edge may not be valid for non-convex quadrilaterals; verify the quadrilateral is convex before flipping.
- **Tie breaking:** cocircular quadrilaterals must be reported as legal to guarantee termination.

12.3.9 9. References

- Lawson, C. L. [Law77].
- de Berg, M., et al. [dBCvKO08], Sec. 9.3.

12.4 Edge Flipping

12.4.1 1. Overview

Given any triangulation of a point set, the Delaunay triangulation can be reached by repeatedly replacing illegal edges with their other diagonal. *Edge flipping* is both an algorithm in its own right (Lawson’s algorithm) and the local repair primitive inside incremental and divide-and-conquer constructions.

12.4.2 2. Definitions

Definition 12.13 (Flip). A *flip* replaces an interior edge (u, v) shared by triangles (u, v, w) and (u, v, z) with the other diagonal (w, z) , provided the quadrilateral (u, w, v, z) is convex.

12.4.3 3. Theory

Theorem 12.14 (Lawson’s Flip Algorithm). *Starting from an arbitrary triangulation of a point set in general position, repeatedly flipping illegal edges terminates with the Delaunay triangulation. The number of flips is at most $O(n^2)$.*

Proof sketch. Each flip of an illegal edge strictly increases the lexicographically sorted list of angles of the triangulation (the minimum angle never decreases, and strictly increases at every flip). Since there are finitely many triangulations of a fixed point set and the angle list is a monotone quantity that is bounded above, the process terminates. Termination can only occur when no illegal edge remains, which by Theorem 12.12 is the Delaunay triangulation. The $O(n^2)$ bound follows because the Delaunay triangulation has $O(n)$ edges and a “history” argument bounds the total number of successful flips by the number of edges ever created. $\square$

12.4.4 4. Pseudocode

Algorithm 44 Lawson’s Flip Algorithm

Input: a triangulation $\mathcal{T}$ of P .

Output: the Delaunay triangulation of P .

```

1: function FLIPTODELAUNAY( $\mathcal{T}$ )
2:    $Q \leftarrow$  all interior edges of  $\mathcal{T}$ 
3:   while  $Q$  is not empty do
4:      $e \leftarrow Q.\text{pop}()$ 
5:     if not IsLocallyDelaunay( $e$ ) then
6:        $e' \leftarrow$  other diagonal of the two triangles sharing  $e$ 
7:       flip  $e$  to  $e'$ 
8:       push the (up to four) newly created edges onto  $Q$ 
9:     end if
10:  end while
11:  return  $\mathcal{T}$ 
12: end function
```

12.4.5 5. Parameter Selection

- **Queue discipline:** processing edges in stack or queue order both terminate; a heap keyed by “amount of violation” reaches convergence faster in practice.
- **Convexity check:** only flip when the quadrilateral is convex; non-convex quadrilaterals are kept unchanged.

12.4.6 6. Complexity

- **Time:** $O(n^2)$ flips in the worst case, each $O(1)$; with a priority queue the expected cost on random inputs is near-linear.
- **Space:** $O(n)$ for the triangulation and the queue.

12.4.7 7. Usages

- **Post-processing:** any meshing algorithm can be followed by a flip pass to enforce the Delaunay property (e.g., before Delaunay refinement, Section 12.12).
- **Local repair:** incremental insertion (Section 12.5) legalizes only the edges of the modified neighborhood.

12.4.8 8. Testing Methods and Edge Cases

- **Termination:** verify the algorithm terminates on adversarially ordered inputs, including nearly cocircular configurations.
- **Cocircularity:** ties must be resolved without oscillation; prefer the lexicographically smaller diagonal.
- **Collinear input:** a triangulation of collinear points has no interior edges; the algorithm must do nothing.

12.4.9 9. References

- Lawson, C. L. [Law77].
- Guibas, L., Stolfi, J. [GS85].
- de Berg, M., et al. [dBCvKO08], Sec. 9.3.

12.5 Incremental Construction

12.5.1 1. Overview

Incremental construction builds the Delaunay triangulation by inserting the points of P one at a time, maintaining the Delaunay property after every insertion. Each step (i) locates the triangle containing the new point, (ii) splits it into three triangles, and (iii) legalizes the edges of the affected neighborhood by flipping. Two families of algorithms realize this scheme: the flip-based incremental algorithm of Section 12.4 and the cavity-based Bowyer–Watson algorithm [Bow81, Wat81].

12.5.2 2. Definitions

Definition 12.15 (Conflict). A new point p *conflicts* with a Delaunay triangle T if p lies inside the circumcircle of T . The set of conflicting triangles is the *conflict zone* of p ; it forms a connected set whose union is a simple polygon, the *cavity* of p .

12.5.3 3. Theory

Theorem 12.16 (Cavity Repair). *Let p be a point not yet in $DT(P)$. The union of the Delaunay triangles of $DT(P)$ whose circumcircle contains p is a star-shaped polygon with respect to p , and retriangulating its boundary with the spokes from p to the boundary vertices yields the Delaunay triangulation of $P \cup \{p\}$.*

Proof sketch. Every Delaunay triangle has an empty circumcircle before insertion. The conflict zone is connected: two conflicting triangles share an edge only if the cavity boundary stays simple, and the empty-circle condition guarantees that the zone is exactly the set of triangles whose circumcenter is hidden from p . The new triangles p -to-boundary-edge have empty circumcircles: if one contained a site q , then the circumcircle of the original conflicting triangle it replaced would contain q as well, contradicting Delaunay-ness. Thus the repair is Delaunay. $\square$

12.5.4 4. Pseudocode

Algorithm 45 Bowyer–Watson Incremental Insertion

Input: point set P ; a surrounding “big triangle” that encloses all of P .

Output: the Delaunay triangulation of P .

```

1: function INSERTDELAUNAY( $P$ )
2:    $\mathcal{T} \leftarrow$  single big triangle
3:   for each  $p \in P$  do
4:     cavity  $\leftarrow \{T \in \mathcal{T} : p \in \text{circumcircle}(T)\}$ 
5:     remove cavity triangles from  $\mathcal{T}$ 
6:      $\mathcal{B} \leftarrow$  boundary edges of the cavity
7:     for each edge  $(a, b) \in \mathcal{B}$  do
8:       add triangle  $(a, b, p)$  to  $\mathcal{T}$ 
9:     end for
10:  end for
11:  remove all triangles incident to a vertex of the big triangle
12:  return  $\mathcal{T}$ 
13: end function

```

The flip-based variant replaces the cavity step: split the located triangle into three, then legalize by flipping outward using Section 12.4’s repair pass.

12.5.5 5. Parameter Selection

- **Point location:** locating the triangle containing p can be done with a walk, a jump-and-walk heuristic, or a history DAG; the history DAG gives the best worst-case behavior (Section 13.11).
- **Big triangle:** its vertices must be far enough that no circumcircle of the final mesh meets them; remove its incident triangles at the end.

12.5.6 6. Complexity

- **Time:** $O(n \log n)$ expected for random insertion orders and $O(n^2)$ worst case for adversarial orders without a history structure.
- **Space:** $O(n)$.

12.5.7 7. Usages

- **Dynamic point sets:** insertions support kinetic meshing and adaptive refinement (Section 12.12).
- **3D:** Bowyer–Watson generalizes directly to d dimensions (Section 12.13) and is the workhorse of TetGen-style tetrahedral meshers.

12.5.8 8. Testing Methods and Edge Cases

- **Point on an edge:** locate both incident triangles and merge the cavity appropriately.
- **Point coincident with an existing vertex:** skip insertion (exact predicate) to avoid zero-area triangles.
- **Points outside the current hull:** the cavity is still connected; verify no spurious edges cross the convex hull.

12.5.9 9. References

- Bowyer, A. [Bow81].
- Watson, D. F. [Wat81].
- de Berg, M., et al. [dBCvKO08], Ch. 9.

12.6 Randomized Incremental Algorithm

12.6.1 1. Overview

Randomized incremental construction inserts the points in a uniformly random permutation. Randomization makes the expected size of the conflict zone small at every step, so the total expected cost drops to $O(n \log n)$, matching the divide-and-conquer bound with a much simpler implementation. This is the algorithm behind the classic randomized incremental Delaunay/Voronoi construction [GKS92].

12.6.2 2. Definitions

Definition 12.17 (Conflict Graph). The *conflict graph* links each inserted point to the triangles of the current triangulation whose circumcircle contains it. When a new point is inserted, its conflict set is read from the graph, and the graph is updated for the new triangles.

12.6.3 3. Theory

Theorem 12.18 (Randomized Incremental Delaunay). *Inserting the points of P in a uniformly random order, maintaining the conflict graph, computes the Delaunay triangulation in $O(n \log n)$ expected time and $O(n)$ expected space.*

Proof sketch (backward analysis). Consider the insertion that creates a particular triangle T of the final triangulation. Reverse the process: delete points one by one in reverse random order, and call T a “created” triangle at the moment its third vertex is deleted. Conditioning on T being present at that time, its three vertices are a uniformly random triple among the k remaining points that lie in $T \cup \{\text{interior}\}$, so the probability that a specific triangle is created at step k is $O(3/k)$. Summing over the $O(n)$ triangles of the final diagram gives expected total conflict-zone work $\sum_k O(3/k \cdot k) = O(n)$ for the conflicts plus $O(n \log n)$ for point location through the history DAG, yielding the claimed bound. $\square$

12.6.4 4. Pseudocode

Algorithm 46 Randomized Incremental Delaunay

Input: point set P .

Output: the Delaunay triangulation of P .

```

1: function RANDOMIZEDDELAUNAY( $P$ )
2:    $\sigma \leftarrow$  uniformly random permutation of  $P$ 
3:    $\mathcal{T} \leftarrow$  big triangle;  $D \leftarrow$  empty history DAG
4:   for  $i = 1, \dots, n$  do
5:      $p \leftarrow \sigma[i]$ 
6:      $T \leftarrow$  leaf of  $D$  whose region contains  $p$  (walk down  $D$ )
7:     cavity  $\leftarrow$  conflict zone of  $p$  in  $\mathcal{T}$ 
8:     retriangulate the cavity with  $p$  (Section 45)
9:     add the new triangles to  $D$  as children of their parent triangle
10:  end for
11:  return  $\mathcal{T}$  with big-triangle triangles removed
12: end function

```

12.6.5 5. Parameter Selection

- **Random seed:** a fixed seed makes runs reproducible; permuting only once amortizes the cost.
- **History DAG:** without it, expected time degrades to $O(n^2)$ for point location; the DAG adds $O(n)$ space.

12.6.6 6. Complexity

- **Time:** $O(n \log n)$ expected, $O(n^2)$ worst case.
- **Space:** $O(n)$ expected.

12.6.7 7. Usages

- **Reference implementation:** the simplicity of the randomized version makes it the algorithm of choice for correctness testing of other constructions.
- **Basis for higher dimensions:** the same backward-analysis argument extends to d dimensions (Section 12.13).

12.6.8 8. Testing Methods and Edge Cases

- **Adversarial orders:** compare the output against the divide-and-conquer result for the same input with many different permutations.
- **Duplicate and collinear points:** the conflict graph must not record degenerate conflicts; test inputs with repeated coordinates.

12.6.9 9. References

- Guibas, L., Knuth, D. E., Sharir, M. [GKS92].
- Mulmuley, K. [Mul90].

- Seidel, R. [Sei91].
- de Berg, M., et al. [dBCvKO08], Sec. 9.4.

12.7 Divide-and-Conquer Construction

12.7.1 1. Overview

The divide-and-conquer construction of Guibas and Stolfi sorts the points by x -coordinate, triangulates each half recursively, and merges the two triangulations in linear time. The merge proceeds along a growing “baseline” that is the unique edge connecting the two halves and respects the empty-circumcircle criterion [GS85].

12.7.2 2. Definitions

Definition 12.19 (Lower Tangent). Let $DT(P_L)$ and $DT(P_R)$ be the Delaunay triangulations of the left and right halves. The *lower tangent* (b_l, b_r) is the unique edge from a vertex b_l of the left hull to a vertex b_r of the right hull that is an edge of the convex hull of $P_L \cup P_R$.

12.7.3 3. Theory

Theorem 12.20 (Delaunay Merge). *Given the Delaunay triangulations of P_L and P_R (with P_L entirely left of P_R), the Delaunay triangulation of $P_L \cup P_R$ is obtained by repeatedly crossing edges between the two halves, always keeping the crossed edge locally Delaunay, and deleting the edges it destroys. The merge runs in $O(n)$ time.*

Proof sketch. The Delaunay triangulation of $P_L \cup P_R$ contains the lower tangent, which is a Delaunay edge (its two endpoints have an empty circle through them). Starting from the lower tangent, the next edge of the cross-hull is found by walking the convex hulls of the two halves and testing the in-circle predicate: a new edge is legal if the triangles on both sides satisfy Definition 12.10. Each candidate edge is considered a constant number of times as the baseline advances monotonically, so the total work is proportional to the number of cross edges, $O(n)$. $\square$

12.7.4 4. Pseudocode

12.7.5 5. Parameter Selection

- **Balanced split:** splitting at the median gives the $O(n \log n)$ recurrence $T(n) = 2T(n/2) + O(n)$.
- **Same- x coordinates:** tie-break by y (or perturb) so the halves are cleanly separable.

12.7.6 6. Complexity

- **Time:** $O(n \log n)$ worst case.
- **Space:** $O(n)$.

12.7.7 7. Usages

- **Deterministic construction:** unlike the randomized algorithm, gives worst-case guarantees without a random source.
- **Baseline for benchmarking:** its robust implementation by Guibas and Stolfi remains the reference for exactness in practice [GS85].

Algorithm 47 Divide-and-Conquer Delaunay (Guibas–Stolfi)**Input:** point set P sorted by x -coordinate.**Output:** the Delaunay triangulation of P .

```

1: function DELAUNAYDNC( $P$ )
2:   if  $|P| \leq 3$  then
3:     return trivial triangulation of  $P$ 
4:   end if
5:   split  $P$  into  $P_L$  and  $P_R$  by the median  $x$ -coordinate
6:    $\mathcal{T}_L \leftarrow \text{DelaunayDnC}(P_L)$ ;    $\mathcal{T}_R \leftarrow \text{DelaunayDnC}(P_R)$ 
7:   return Merge( $\mathcal{T}_L, \mathcal{T}_R$ )
8: end function
9: function MERGE( $\mathcal{T}_L, \mathcal{T}_R$ )
10:   $(b_l, b_r) \leftarrow$  lower tangent of  $\mathcal{T}_L \cup \mathcal{T}_R$ 
11:  repeat
12:     $c \leftarrow$  the endpoint (in  $P_L$  or  $P_R$ ) that, with  $(b_l, b_r)$ , satisfies the empty-circumcircle
    test
13:    add edge  $(b_l, c)$  or  $(c, b_r)$  as the next cross edge, crossing if it does not intersect
    existing edges
14:  until no further legal cross edge exists
15:  return the combined triangulation
16: end function

```

12.7.8 8. Testing Methods and Edge Cases

- **Merge correctness:** cross-check every cross edge against the in-circle predicate; assert no two cross edges cross.
- **Vertical splits:** points with equal x must not fragment the recursion.
- **Small sizes:** the $n \leq 3$ base case must handle collinear triples without creating zero-area triangles.

12.7.9 9. References

- Guibas, L., Stolfi, J. [GS85].
- de Berg, M., et al. [dBCvKO08], Sec. 9.1–9.2.

12.8 Lifting to a Paraboloid**12.8.1 1. Overview**

Lifting maps each planar site $p = (x, y)$ to the point $p^+ = (x, y, x^2 + y^2)$ on the paraboloid $z = x^2 + y^2$. A plane intersects the paraboloid in a circle projected onto the xy -plane, which converts the in-circle predicate into a half-space (orientation) test in $\mathbb{R}^3$ and reduces Delaunay construction to a 3D convex-hull problem [Bro79, dBCvKO08].

12.8.2 2. Definitions

Definition 12.21 (Lifted Point). The *lift* of $p = (x, y)$ is $p^+ = (x, y, x^2 + y^2)$. The lower hull of the lifted point set is the set of facets visible from below (i.e., with the plane normal pointing downward in z).

12.8.3 3. Theory

Theorem 12.22 (Lifting Theorem). *Let P be a set of points in general position, and let P^+ be their lifts to the paraboloid $z = x^2 + y^2$. The vertical projection onto the xy -plane of the lower convex hull of P^+ is exactly the Delaunay triangulation $DT(P)$.*

Proof sketch. A point $q = (x, y, x^2 + y^2)$ lies on the plane H through a^+, b^+, c^+ if and only if (x, y) lies on the circle through a, b, c ; it lies below H if and only if (x, y) is strictly inside that circle. Therefore the in-circle predicate of Definition 12.7 is the sign of the oriented volume of the tetrahedron $a^+b^+c^+d^+$. A triangle of the projection is a facet of the lower hull exactly when all other lifted points lie above its plane, i.e., when the corresponding circumcircle is empty. Hence lower-hull facets and Delaunay triangles coincide. $\square$

12.8.4 4. Pseudocode

Algorithm 48 Delaunay via Lifting

Input: point set P .

Output: the Delaunay triangulation of P .

```

1: function DELAUNAYBYLIFTING( $P$ )
2:    $P^+ \leftarrow \{(x, y, x^2 + y^2) : (x, y) \in P\}$ 
3:    $\mathcal{H} \leftarrow$  3D convex hull of  $P^+$ 
4:   for each facet of  $\mathcal{H}$  whose outward normal has a negative  $z$ -component do
5:     project the facet vertices back to the  $xy$ -plane and emit the triangle
6:   end for
7:   return the projected lower hull
8: end function
```

12.8.5 5. Parameter Selection

- **Which hull:** only the *lower* hull matters; computing the full hull is wasteful but harmless.
- **Exact arithmetic:** lifted coordinates are quadratic, so predicates grow in magnitude; adaptive filters (Section 12.2) keep the common case fast.

12.8.6 6. Complexity

- **Time:** $O(n \log n)$ for a 3D hull algorithm that is output sensitive; note that the lifted set lies on a strictly convex surface, so its hull has $O(n)$ facets, not the worst-case $O(n^2)$.
- **Space:** $O(n)$.

12.8.7 7. Usages

- **Conceptual tool:** the lifting view unifies Delaunay triangulation, Voronoi diagrams, and convex hulls (Section 12.9).
- **Reduction:** any 3D hull routine (e.g., Qhull) becomes a Delaunay mesher, which is precisely how several numerical packages compute triangulations.

12.8.8 8. Testing Methods and Edge Cases

- **Cocircularity:** cocircular points lift to coplanar points; verify the hull routine tolerates coplanar facets.
- **Scaling:** quadratic lifts amplify coordinate ranges; test with coordinates spanning several orders of magnitude.

12.8.9 9. References

- Brown, K. Q. [Bro79].
- Edelsbrunner, H. [Ede87].
- de Berg, M., et al. [dBCvKO08], Sec. 11.4.

12.9 Relationship to Convex Hulls

12.9.1 1. Overview

The lifting theorem of Section 12.8 is one of three equivalent views of the same object. The Delaunay triangulation is (i) the dual of the Voronoi diagram, (ii) the projection of the lower hull of the lifted points, and (iii) the projection of the convex hull of points on a sphere via stereographic projection. This section records the two hull relations used in practice and their polar counterpart for Voronoi diagrams.

12.9.2 2. Definitions

Definition 12.23 (Upper Envelope of Tangent Planes). For each site p , let H_p be the tangent plane to the paraboloid $z = x^2 + y^2$ at the lifted point p^+ . The *upper envelope* of the planes $\{H_p\}$ is the pointwise maximum of their z -values.

12.9.3 3. Theory

Theorem 12.24 (Voronoi Diagram as Upper Hull). *The vertical projection onto the xy -plane of the upper envelope of the tangent planes H_p is the Voronoi diagram $VD(P)$: a point x of the plane lies in the projection of H_p exactly when p is its nearest site.*

Proof sketch. The tangent plane H_p at p^+ is $z = 2p_x x + 2p_y y - (p_x^2 + p_y^2)$. The vertical gap between H_p and the paraboloid at the point above x is $|x|^2 - (2p \cdot x - |p|^2) = \|x - p\|^2$. Therefore H_p is above H_q at x if and only if $\|x - p\| \leq \|x - q\|$, i.e., if and only if p is the nearest site. Projecting the upper envelope gives precisely the Voronoi cells. $\square$

Theorem 12.25 (Delaunay and Farthest Delaunay). *The lower hull of the lifted points projects to the (nearest) Delaunay triangulation, while the upper hull projects to the farthest-point Delaunay triangulation, whose dual is the farthest-point Voronoi diagram.*

Proof sketch. By symmetry with Theorem 12.22, a facet of the *upper* hull corresponds to a circle that contains all points of P in its closed disk and passes through three hull vertices; its projection is a farthest Delaunay triangle. The dual cells are the regions where a given hull vertex is the *farthest* site, which is the farthest-point Voronoi cell. $\square$

12.9.4 4. Pseudocode

The practical consequence is that a single 3D convex-hull computation yields all four objects:

Algorithm 49 All Diagrams from One Convex Hull

```
1: function ALLDIAGRAMS( $P$ )
2:    $\mathcal{H} \leftarrow$  3D hull of lifted points  $P^+$ 
3:   lower facets  $\rightarrow$  project  $\rightarrow DT(P)$ 
4:   upper facets  $\rightarrow$  project  $\rightarrow$  farthest Delaunay
5:   dualize  $DT(P) \rightarrow VD(P)$  (Section 41)
6:   return ( $DT(P), VD(P)$ )
7: end function
```

12.9.5 5. Parameter Selection

- **Stereographic alternative:** projecting the points from the paraboloid onto a sphere gives a hull computation in $\mathbb{R}^3$ with numerically better conditioning than the quadratic lift.
- **Output sensitivity:** for convex input the hull is small; exploit output-sensitive hull algorithms.

12.9.6 6. Complexity

- **Time:** $O(n \log n)$ for the 3D hull (the lifted set has $O(n)$ facets).
- **Space:** $O(n)$.

12.9.7 7. Usages

- **Unified implementation:** one hull routine provides Delaunay, Voronoi, and farthest-point structures for testing and cross-validation.
- **Pedagogy:** the three-way equivalence (Voronoi, Delaunay, hull) is the central organizing idea of the field.

12.9.8 8. Testing Methods and Edge Cases

- **Consistency:** verify DT , VD , and the hull all agree on random and degenerate inputs.
- **Coplanar facets:** hulls with coplanar facets must be resolved by triangulation or symbolic perturbation.

12.9.9 9. References

- Brown, K. Q. [Bro79].
- Edelsbrunner, H. [Ede87].
- de Berg, M., et al. [dBCvKO08], Sec. 11.4.

12.10 Constrained Delaunay Triangulation

12.10.1 1. Overview

A *constrained Delaunay triangulation* (CDT) is a triangulation of a planar straight-line graph (PSLG) that contains every constraint segment as a union of edges, while remaining as close as possible to the Delaunay criterion in the triangles that do not cross a constraint. CDTs are the standard representation for meshing domains with internal boundaries, such as faults, wells, or material interfaces.

12.10.2 2. Definitions

Definition 12.26 (Visibility). A segment uv is *visible* from a point p if the open segment pu meets no constraint segment, and likewise for $p v$.

Definition 12.27 (Constrained Delaunay Edge). An edge e of a triangulation of a PSLG is *constrained Delaunay* if it is a constraint segment, or if there exists a circle through its two endpoints whose open disk contains no vertex visible to e .

Definition 12.28 (Constrained Delaunay Triangulation). A *CDT* of a PSLG is a triangulation containing every constraint segment in which every edge is constrained Delaunay.

12.10.3 3. Theory

Theorem 12.29 (Existence and Uniqueness of the CDT). *Every planar straight-line graph admits a constrained Delaunay triangulation, and it is unique: the diagonal of every quadrilateral in the CDT is chosen so that the constraints are respected and the remaining edges satisfy Definition 12.27.*

Proof sketch. Existence follows from the flip viewpoint: take any triangulation that contains the constraints, then flip any illegal edge that is not a constraint; each flip reduces a lexicographic angle potential, so termination yields a triangulation with no illegal unconstrained edges, which is a CDT by the analogue of Theorem 12.12 restricted to visible vertices. Uniqueness follows because the empty-circle test, restricted to visible points, decides every non-constrained diagonal independently of the rest of the triangulation [Che89, dBCvKO08]. $\square$

12.10.4 4. Pseudocode

Algorithm 50 CDT by Constraint Recovery

Input: a PSLG G (points plus constraint segments).

Output: the constrained Delaunay triangulation of G .

```

1: function CONSTRAINEDDELAUNAY( $G$ )
2:    $\mathcal{T} \leftarrow$  Delaunay triangulation of the vertices of  $G$ 
3:   for each constraint segment  $s$  of  $G$  do
4:     if  $s$  is not a union of edges of  $\mathcal{T}$  then
5:       locate the polygons crossed by  $s$ 
6:       retriangulate them so that  $s$  becomes a sequence of edges
7:       legalize the affected region with flip-based repair, keeping every constraint edge
       fixed
8:     end if
9:   end for
10:  return  $\mathcal{T}$ 
11: end function

```

12.10.5 5. Parameter Selection

- **Segment insertion order:** inserting constraints one at a time with local repair is simple; batch processing via constrained triangulation is faster for many segments.
- **Conforming variants:** if constraints may be *split* by new vertices (conforming Delaunay), Ruppert-style refinement (Section 12.12) can be used to enforce both the constraints and the quality bounds.

12.10.6 6. Complexity

- **Time:** $O(n^2)$ worst case for flip-based constraint recovery; expected $O(n \log n)$ for random inputs.
- **Space:** $O(n)$.

12.10.7 7. Usages

- **Finite-element meshing:** constraints model material interfaces, wells, and boundaries (see the reservoir and mesh applications in Section 12.15).
- **Terrain modeling:** breaklines and rivers are constraints in the TIN surfaces of GIS (Section 11.5).

12.10.8 8. Testing Methods and Edge Cases

- **Intersecting constraints:** the input PSLG must be planar; verify that segments only meet at endpoints, or insert intersection points first.
- **Overlapping constraints:** deduplicate coincident segments.
- **Constraints along the hull:** ensure hull edges remain part of the triangulation.

12.10.9 9. References

- Chew, L. P. [Che89].
- Shewchuk, J. R. [She98].
- de Berg, M., et al. [dBCvKO08], Sec. 9.5.

12.11 Quality Properties

12.11.1 1. Overview

Among all triangulations of a point set, the Delaunay triangulation is in a precise sense the “best”. It maximizes the minimum angle, which is exactly the criterion that guarantees well-shaped triangles for finite-element computation: skinny triangles produce ill-conditioned stiffness matrices and large interpolation errors.

12.11.2 2. Definitions

Definition 12.30 (Minimum Angle). The *minimum angle* of a triangulation is the smallest angle over all of its triangles. A triangulation T_1 is *angle-optimal* if no other triangulation of the same point set has a larger minimum angle.

Definition 12.31 (Aspect Ratio). The *aspect ratio* (or radius ratio) of a triangle is the ratio of its circumradius to its inradius; it is a scale-invariant measure of shape quality, equal to 1 for equilateral triangles and unbounded for skinny ones.

12.11.3 3. Theory

Theorem 12.32 (Maximin Angle Property). *Among all triangulations of a point set in general position, the Delaunay triangulation maximizes the minimum angle (and, more strongly, is lexicographically optimal in the sorted list of angles).*

Proof sketch. Flipping an illegal edge replaces two angles α, β (the angles opposite the illegal edge) with angles α', β' satisfying $\min(\alpha', \beta') > \min(\alpha, \beta)$: the smallest of the four angles of the quadrilateral strictly increases. Since Lawson’s algorithm (Section 12.4) reaches the Delaunay triangulation by such flips and each flip only improves the minimum angle, no triangulation can have a larger minimum angle. The lexicographic claim follows by the same monotonicity on the sorted angle sequence. $\square$

Corollary 12.33 (Angle Lower Bound). *Every triangle of the Delaunay triangulation of points in general position has minimum angle at least 30° when restricted to its circumcircle’s diameter bound; more relevantly, Delaunay refinement (Section 12.12) can push the bound to any prescribed value by inserting points.*

12.11.4 4. Pseudocode

Quality measures are cheap to evaluate and are used to drive refinement:

Algorithm 51 Quality Report

Input: a triangulation $\mathcal{T}$ and a target minimum angle $\theta_{\min}$.

Output: the list of triangles violating the quality bound.

```

1: function REPORTSKINNYTRIANGLES( $\mathcal{T}, \theta_{\min}$ )
2:   violators  $\leftarrow \square$ 
3:   for each triangle  $T \in \mathcal{T}$  do
4:      $\theta_T \leftarrow$  minimum angle of  $T$ 
5:     if  $\theta_T < \theta_{\min}$  then
6:       append  $T$  to violators
7:     end if
8:   end for
9:   return violators
10: end function

```

12.11.5 5. Parameter Selection

- **Quality metric:** minimum angle, circumradius-to-shortest-edge ratio, and radius ratio are equivalent up to constants for bounded shapes; pick the one matched to the solver’s error estimate.
- **Target angle:** angles above $\approx 30^\circ$ typically require more Steiner points; the mesh-size trade-off drives Section 12.12’s parameters.

12.11.6 6. Complexity

- **Time:** $O(n)$ to evaluate quality over all triangles; one constant-time formula per triangle.
- **Space:** $O(n)$ for the report.

12.11.7 7. Usages

- **Mesh quality control:** FEM and CFD pipelines validate meshes against minimum-angle or aspect-ratio thresholds before solving.
- **Refinement driver:** skinny-triangle reports feed the circumcenter-insertion loop (Section 12.12).

12.11.8 8. Testing Methods and Edge Cases

- **Degenerate triangles:** collinear vertices yield zero angles; the report must flag them without division-by-zero.
- **Extreme aspect ratios:** verify scale invariance under uniform scaling and rotation.

12.11.9 9. References

- Lawson, C. L. [Law77].
- Ruppert, J. [Rup95].
- Shewchuk, J. R. [She98].

12.12 Delaunay Refinement

12.12.1 1. Overview

Delaunay triangulations of a *point set* can still contain skinny triangles when the points are badly distributed. *Delaunay refinement* (Chew [Che89], Ruppert [Rup95]) inserts new points—circumcenters of skinny triangles—until every triangle satisfies a quality bound. Ruppert’s algorithm and its successors are the workhorses of 2D quality mesh generation (Section 12.15) and are the reason angle bounds such as “no angle below 20.7°” appear in every meshing manual.

12.12.2 2. Definitions

Definition 12.34 (Skinny Triangle). A triangle is *skinny* if its circumradius-to-shortest-edge ratio $\rho = R/s$ exceeds a threshold B ; equivalently its minimum angle is below $\arcsin(1/(2B))$.

Definition 12.35 (Encroached Segment). A segment of the domain boundary (or a constraint) is *encroached* by a vertex v if v lies inside the circle with the segment as diameter. An encroached segment must be split at its midpoint before any circumcenter inside its influence region is inserted.

12.12.3 3. Theory

Theorem 12.36 (Ruppert’s Algorithm). *Let a planar straight-line graph be input with no two segments forming an angle below a threshold ϕ (typically 90°). Ruppert’s refinement algorithm terminates and produces a mesh in which every triangle has circumradius-to-shortest-edge ratio at most $B \geq \sqrt{2}$, hence minimum angle at least $\arcsin(1/(2B))$ (about 20.7° for $B = \sqrt{2}$), and whose size is within a constant factor of the smallest possible mesh satisfying the bound.*

Proof sketch. Termination rests on a shortest-edge lemma: inserting the circumcenter of a skinny triangle (or the midpoint of an encroached segment) always creates edges of length at least L times the length of the offending short edge, for a fixed $L = \sin \theta$ with $B \geq \sqrt{2}$. Since every newly inserted point is a constant fraction away from all existing points, the smallest edge

length can decrease by at most a fixed factor, and after finitely many reductions the refinement stops. Optimality follows because every inserted circumcenter lies in a region that must contain a Steiner point in any mesh achieving the bound, so the count is within a constant factor of optimum [Rup95, She98]. $\square$

12.12.4 4. Pseudocode

Algorithm 52 Ruppert Delaunay Refinement

Input: PSLG G , ratio bound B .

Output: a quality Delaunay mesh of G .

```

1: function RUPPERTREFINE( $G, B$ )
2:    $\mathcal{T} \leftarrow$  conforming Delaunay triangulation of  $G$ 
3:    $Q \leftarrow$  queue of encroached segments and skinny triangles
4:   while  $Q$  is not empty do
5:      $x \leftarrow Q.\text{pop}()$  (prefer encroached segments)
6:     if  $x$  is an encroached segment then
7:       insert its midpoint; re-triangulate locally; enqueue new encroached segments and
       skinny triangles
8:     else  $\triangleright x$  is a skinny triangle
9:        $c \leftarrow$  circumcenter( $x$ )
10:      if some segment near  $c$  is encroached then
11:        split that segment at its midpoint instead
12:      else
13:        insert  $c$ ; re-triangulate locally; enqueue new skinny triangles
14:      end if
15:    end if
16:  end while
17:  return  $\mathcal{T}$ 
18: end function

```

12.12.5 5. Parameter Selection

- **Ratio bound B :** $B = \sqrt{2}$ gives the classic 20.7° guarantee; larger B (e.g., $B = 2$) yields fewer points but weaker angles; Chew's variant attains 30° .
- **Small input angles:** angles below 60° in the input PSLG force special handling (segment splitting at graded lengths) because circumcenter insertion near acute corners can create arbitrarily short edges.
- **Sizing fields:** refinement with a user sizing function (graded meshes) replaces the uniform bound B by a local target edge length.

12.12.6 6. Complexity

- **Time:** $O(n \log n)$ expected for the refinement loop, where n is the output size.
- **Space:** $O(n)$.
- **Output size:** within a constant factor of the optimal mesh achieving the same angle bound (Theorem 12.36).

12.12.7 7. Usages

- **Quality mesh generation:** the meshers behind Section 12.15 (Triangle, CGAL, TetGen in 2D analog form) implement Ruppert’s algorithm directly.
- **Conforming CDTs:** refinement yields a conforming Delaunay triangulation that contains the constraints and satisfies angle bounds, combining Section 12.10 with quality guarantees.

12.12.8 8. Testing Methods and Edge Cases

- **Acute input angles:** verify termination when the input contains 30° corners by checking the segment-splitting path is taken.
- **Encroachment chains:** an encroached midpoint can encroach on neighbors; the queue must drain without infinite loops (the $B \geq \sqrt{2}$ condition prevents cascades).
- **Verification:** after termination, assert every triangle has ratio $\leq B$ and every segment is unencroached.

12.12.9 9. References

- Chew, L. P. [Che89].
- Ruppert, J. [Rup95].
- Shewchuk, J. R. [She98].

12.13 Higher-Dimensional Delaunay Complexes

12.13.1 1. Overview

All of the theory extends to $\mathbb{R}^d$ with the empty-circumsphere property replacing the empty circle. The Delaunay complex of a point set in $\mathbb{R}^d$ is a simplicial complex of d -simplices, and its restriction to simplices whose circumradius is bounded—the *alpha complex*—is the standard structure for surface reconstruction from point clouds. Weighted (regular) triangulations generalize the construction to sites with radii, replacing circles by power regions.

12.13.2 2. Definitions

Definition 12.37 (Empty Circumsphere). A simplex T of a point set in $\mathbb{R}^d$ is *Delaunay* if the open ball bounded by the circumsphere of T contains no point of the set. The *Delaunay complex* is the collection of all such simplices; its maximal d -simplices tile the convex hull of the point set when the points are in general position.

Definition 12.38 (Alpha Complex). For a real $\alpha \geq 0$, the *alpha complex* consists of all Delaunay simplices whose circumradius is at most α (and, recursively, all faces of those simplices). Varying α from 0 to ∞ produces the *alpha shape* family interpolating from the empty set to the convex hull.

Definition 12.39 (Regular Triangulation). For weighted points (p_i, r_i) , the *power distance* of x to p_i is $\|x - p_i\|^2 - r_i^2$. A *regular triangulation* (weighted Delaunay triangulation) is defined by empty *power spheres*: no point of the set lies in the power region of any simplex. Its dual is the power (weighted Voronoi) diagram.

12.13.3 3. Theory

Theorem 12.40 (Delaunay Complex in Higher Dimensions). *Let P be a set of points in general position in $\mathbb{R}^d$. The Delaunay complex is a triangulation of the convex hull of P . Lifting $p \mapsto (p, \|p\|^2)$ to the paraboloid in $\mathbb{R}^{d+1}$ shows that the Delaunay complex is the projection of the lower hull of the lifted points, exactly as in two dimensions (Theorem 12.22).*

Proof sketch. The determinant form of the in-circle predicate generalizes verbatim to a $(d+2) \times (d+2)$ determinant whose sign decides whether a point lies inside a circumsphere. All two-dimensional arguments—cavity repair, flips, lifting, randomized incremental construction—carry through with the same proofs, replacing circles by spheres and triangles by d -simplices. $\square$

In three dimensions the situation differs in one essential way: the Delaunay tetrahedralization need not be unique, and no flip sequence suffices to reach it from an arbitrary tetrahedralization. Moreover the worst-case complexity grows.

Theorem 12.41 (Complexity in Fixed Dimension). *In $\mathbb{R}^d$ the Delaunay complex has $\Theta(n^{\lceil d/2 \rceil})$ simplices in the worst case; in $\mathbb{R}^3$ it has $O(n^2)$ tetrahedra and can be constructed by randomized incremental insertion in expected $O(n^2)$ time and $O(n \log n)$ time for well-distributed inputs.*

12.13.4 4. Pseudocode

Algorithm 53 Bowyer–Watson in Higher Dimensions

Input: point set P in $\mathbb{R}^d$, enclosing $(d+1)$ -simplex.

Output: the Delaunay complex of P .

```

1: function INSERTDELAUNAYD( $P$ )
2:    $\mathcal{K} \leftarrow$  single enclosing  $(d+1)$ -simplex
3:   for each  $p \in P$  do
4:     cavity  $\leftarrow \{T \in \mathcal{K} : p \in \text{circumsphere}(T)\}$ 
5:     remove the cavity; add the fan of simplices from  $p$  to each boundary facet of the
       cavity
6:   end for
7:   remove simplices incident to the enclosing simplex
8:   return  $\mathcal{K}$ 
9: end function

```

12.13.5 5. Parameter Selection

- **Dimension:** the code is generic in d but the practical sweet spot is $d \leq 4$; beyond that the $\Theta(n^{\lceil d/2 \rceil})$ size bound makes exact construction prohibitive and approximate methods (spatial hashing, approximate NNS) are used instead.
- **Weights:** regular triangulations encode anisotropic or radius-bearing sites (spheres, molecules); they are the default when inputs have sizes.

12.13.6 6. Complexity

- **Time:** expected $O(n \log n)$ in $\mathbb{R}^2$; $\Theta(n^{\lceil d/2 \rceil})$ worst case in $\mathbb{R}^d$.
- **Space:** the output itself dominates; in $\mathbb{R}^3$ up to $O(n^2)$ tetrahedra.

12.13.7 7. Usages

- **Surface reconstruction:** the alpha complex (at a suitable α) reconstructs surfaces from scanner output, as used in the reverse engineering pipelines of Section 12.15.
- **Tetrahedral meshing:** Delaunay tetrahedralizations of 3D point sets (with refinement for quality) are the foundation of TetGen and CGAL volume meshing.
- **Molecular modeling:** regular triangulations compute power/Voronoi decompositions of atom models for solvent-accessible surface areas.

12.13.8 8. Testing Methods and Edge Cases

- **Cospherical points:** five cospherical points make the tetrahedralization non-unique; verify consistent tie-breaking.
- **Slivers:** four nearly coplanar points produce a near-degenerate tetrahedron; report the smallest radius ratio to check mesh quality.
- **Degenerate lower dimensions:** points embedded in a lower-dimensional affine subspace (all points coplanar in $\mathbb{R}^3$) must be handled by projecting to the subspace first.

12.13.9 9. References

- Bowyer, A. [Bow81]; Watson, D. F. [Wat81].
- Edelsbrunner, H., Mücke, E. P. [EM94a], alpha shapes.
- Edelsbrunner, H. [Ede87].
- Okabe, A., et al. [OBSC00], Ch. 4.

12.14 Intrinsic Delaunay Triangulation

The Euclidean Delaunay triangulation is defined through empty circumcircles, a notion that depends on the ambient metric of $\mathbb{R}^2$. On a polyhedral surface $\mathcal{M}$ triangulated by triangles that do not lie in a common plane, this extrinsic test is meaningless: the two triangles sharing an edge lie in different tangent planes, so their circumcircles (as circles in $\mathbb{R}^3$) cannot be compared. An *intrinsic Delaunay triangulation* instead uses only the intrinsic metric of the surface, i.e., the edge lengths of the mesh, to decide legality. This section follows the flip-based construction of Bobenko and Springborn [BS07] and its practical implementation by Fisher et al. [FSBS07].

Definition 12.42 (Intrinsic Delaunay Edge). Let $e = (i, j)$ be an interior edge of a triangulated polyhedral surface, shared by triangles T_1 and T_2 , and let α_e and β_e be the angles opposite e in T_1 and T_2 , respectively. The edge e is *intrinsically Delaunay* if

$$\alpha_e + \beta_e \leq \pi.$$

A triangulation is *intrinsic Delaunay* if every interior edge is intrinsically Delaunay.

Because each triangle of the surface is intrinsically a Euclidean triangle, the angles α_e and β_e are computed from the three edge lengths of the respective triangles using the law of cosines. The criterion therefore depends only on the intrinsic metric, never on how the triangles are embedded in $\mathbb{R}^3$.

Definition 12.43 (Cotangent Weight). For an interior edge e , the *cotangent weight* is $w_e = \cot \alpha_e + \cot \beta_e$, where α_e, β_e are the angles opposite e . The identity

$$\cot \alpha_e + \cot \beta_e = \frac{\sin(\alpha_e + \beta_e)}{\sin \alpha_e \sin \beta_e}$$

shows that $w_e \geq 0$ if and only if $\alpha_e + \beta_e \leq \pi$. Thus a triangulation is intrinsic Delaunay if and only if all cotangent weights are non-negative.

The connection to the Laplace–Beltrami operator motivates the definition. The cotangent Laplacian of the mesh has off-diagonal entries $-\frac{1}{2}w_e$ and diagonal entries summing to zero; it is positive semi-definite exactly when the triangulation is intrinsic Delaunay. Negative weights, which can appear in an ordinary triangulation containing obtuse triangles, are responsible for spurious oscillations in discrete harmonic functions and for unstable heat propagation. Intrinsic Delaunay re-triangulation removes this defect at no cost in resolution.

12.14.1 Algorithm

The construction proceeds by flipping illegal edges exactly as in the planar case (Section 7.5), but with two differences:

1. The legality test uses the intrinsic angle sum $\alpha_e + \beta_e$ rather than the planar in-circle predicate.
2. After a flip, the new diagonal e' of the quadrilateral must be given an intrinsic length. The two triangles sharing e are, intrinsically, two Euclidean triangles glued along e ; unfolding them into a common plane yields a flat (possibly non-convex) quadrilateral, and the new edge length is the length of the other diagonal of this quadrilateral, computed from the four known edge lengths.

Algorithm 54 Intrinsic Delaunay Flip Algorithm

```

1: function INTRINSICDELAUNAY(triangulation)
2:   queue  $\leftarrow$  all interior edges of triangulation
3:   while queue is not empty do
4:      $e \leftarrow$  queue.pop()
5:      $(\alpha_e, \beta_e) \leftarrow$  angles opposite  $e$  in its two incident triangles
6:     if  $\alpha_e + \beta_e > \pi$  then
7:        $e' \leftarrow$  other diagonal of the quadrilateral formed by the two triangles
8:        $\ell(e') \leftarrow$  intrinsic length of  $e'$  from the unfolded quadrilateral
9:       flip  $e$  to  $e'$ 
10:      for  $f$  in the two newly created edges do
11:        push(queue,  $f$ )
12:      end for
13:    end if
14:  end while
15:  return triangulation
16: end function

```

Theorem 12.44 (Existence of Intrinsic Delaunay Triangulation). *Every triangulation of a polyhedral surface admits an intrinsic Delaunay re-triangulation of the same vertices, obtainable by a finite sequence of edge flips [BS07, FSBS07].*

Proof sketch. Each flip replaces an edge whose opposite angles sum to more than π by the other diagonal. An angle-sum decrease argument shows that every flip strictly decreases a global potential over the set of triangulations of the intrinsic metric; since the set of triangulations of a fixed vertex set is finite, the process terminates. When the intrinsic Delaunay condition determines every diagonal uniquely, the resulting triangulation is unique; otherwise ties are broken consistently, for example lexicographically over edges. $\square$

Remark 12.45. The intrinsic Delaunay triangulation of a mesh is the dual of the *geodesic Voronoi diagram* of its vertices on the surface, just as the planar Delaunay triangulation is the dual of the planar Voronoi diagram. This explains why the intrinsic construction is the “correct” generalization for surface processing.

12.14.2 Properties

- **Metric invariance:** The result depends only on edge lengths, so it is invariant under isometric deformations of the surface (bending without stretching).
- **Non-negative weights:** All cotangent weights are non-negative, making the cotangent Laplacian positive semi-definite and well-behaved for PDEs, harmonic maps, and the heat method.
- **Same vertex set:** The algorithm changes only connectivity, never vertex positions; it is a pure re-tiling of the intrinsic metric.
- **Compatibility:** The intrinsic flip criterion coincides with the planar one on flat regions of the mesh, so the algorithm reproduces the usual Delaunay triangulation on planar surfaces.

12.14.3 Complexity

- **Time:** Each flip is $O(1)$; the total number of flips is $O(n^2)$ in the worst case and $O(n)$ in typical cases for meshes with reasonable initial quality [FSBS07].
- **Space:** $O(n)$ for the mesh and the queue of candidate edges.

12.14.4 Applications

- **Discrete Laplace–Beltrami:** Guaranteeing positive semi-definiteness of the cotangent operator for spectral analysis and shape signature computation.
- **Heat methods:** Stabilizing the heat kernel and geodesic-distance approximation on meshes with obtuse triangles.
- **Harmonic and biharmonic fields:** Removing spurious extrema caused by negative cotangent weights in, e.g., parameterization and deformation [BS07].
- **Discrete exterior calculus:** Restoring the non-negativity of the Hodge-star edge weights discussed in the discrete exterior calculus chapter (Section 30.17).

12.15 Applications to Mesh Generation

Delaunay meshes are the workhorse of industrial mesh generation because the empty-circumcircle (empty-circumsphere in 3D) property and the associated refinement algorithms of Section 12.12 produce element-quality guarantees that are provably unobtainable with other triangulation schemes. The following are the principal industrial use cases.

- **Finite Element Analysis (FEM):** Delaunay meshes with the refinement guarantees of Section 12.12 bound the minimum angle of every triangle away from zero, which is necessary for the conditioning of the stiffness matrix and the convergence of FEM solvers in structural and thermal analysis. The commercial FEA market—**ANSYS**, **Abaqus** (Dassault Systèmes), **COMSOL**, and **Altair**—sells solvers whose mesh generators are exactly this refinement scheme [She98].
- **Computational Fluid Dynamics (CFD):** Commercial CFD meshers (**ANSYS** Fluent meshing, **SIMULIA**, **STAR-CCM+**, and **Pointwise**) generate quality tetrahedral meshes from CAD geometry; the empty-sphere property keeps high-aspect-ratio elements near the anisotropic inflation layers while preserving the angle bounds that stabilize pressure–velocity coupling on aircraft, turbine, and vehicle models.
- **Reservoir Simulation (Voronoi/PEBI Simulation Grids):** The perpendicular-bisector (PEBI) grids used for reservoir flow simulation are Voronoi diagrams, and their construction is driven by the dual Delaunay triangulation. Commercial reservoir-modeling suites such as **Roxar RMS** (Emerson) build unstructured simulation grids whose cells honor wells, faults, and fracture networks, on which multiphase flow and transport solvers run.
- **Medical Imaging and Surgical Planning:** Delaunay tetrahedralization of segmented CT and MRI volumes produces the volumetric meshes used in finite-element modeling of bone, soft tissue, and blood flow. Commercial planning platforms such as **Materialise Mimics** sell patient-specific modeling for surgical guides, implants, and biomechanical simulation.
- **Reverse Engineering and Metrology:** Surface reconstruction from laser-scan point clouds builds a Delaunay (or alpha-shape, Section 12.13) complex of the scanned part. Commercial scanning pipelines such as **PolyWorks** (InnovMetric) and **Geomagic** (3D Systems) ship this reconstruction and the resulting mesh to downstream CAD and inspection.
- **Geographic Information Systems (GIS):** Conforming Delaunay triangulations of terrain point clouds produce the triangulated irregular networks (TINs) that back digital elevation models, visibility analysis, and contouring in **Esri ArcGIS 3D Analyst** and comparable commercial GIS; the conforming refinement also generates the meshes for watershed and flood modeling.
- **Shape and Topology Optimization:** The conforming mesh produced by Delaunay refinement with sizing fields adapts element density to the stress field, accelerating the density-based optimization loop in commercial tools such as **Altair OptiStruct**, **SIMULIA Tosca**, and **nTopology** for automotive and aerospace lightweighting.

12.15.1 Industrial-Scale Software

The algorithms of this chapter are implemented at production scale in the open-source meshers **TetGen** (tetrahedralization and constrained Delaunay), **CGAL** (2D and 3D Delaunay and regular triangulations), **Triangle** (2D CDT and refinement, embedded under license in commercial GIS and numerical products), **Gmsh** (unstructured mesh generation), and **MMG** (remeshing and metric adaptation), as well as in commercial tools such as **ANSYS**, **COMSOL**, and **Pointwise** for CFD. Their shared core is the incremental insertion and flip algorithm of Sections 12.5 and 12.4, coupled with the quality refinement guarantees of Section 12.12.

12.16 Exercises

12.17 Project: Delaunay Triangulation from Scratch

Chapter 13

Proximity Problems

13.1 Closest Pair of Points

The closest pair of points problem is a fundamental challenge in computational geometry that involves finding the two points in a set of n points that have the minimum Euclidean distance between them. While a naive brute-force approach takes $O(n^2)$ time, more efficient algorithms exist, including a deterministic $O(n \log n)$ divide-and-conquer approach by Shamos and Hoey [SH75] and Bentley and Shamos [BS76], and a randomized $O(n)$ grid-based approach.

13.1.1 Definitions

Definition 13.1 (Closest Pair). Given a set P of n points in $\mathbb{R}^2$, the *closest pair* is the pair (p_i, p_j) with $i \neq j$ that minimizes the Euclidean distance $\|p_i - p_j\|$.

Definition 13.2 (Euclidean Distance). For two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$, the *Euclidean distance* is $d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Definition 13.3 (Divide and Conquer Strategy). A *divide-and-conquer* strategy recursively partitions the input into two halves, solves each half independently, and merges the results. For the closest pair problem, the merge step requires examining only a narrow strip of width $2d$ centered on the dividing line, where $d = \min(d_L, d_R)$.

13.1.2 Theory

Divide and Conquer Approach

The set of points is first sorted by x -coordinate and then split into two halves by a vertical line. The minimum distance is recursively found in the left half (d_L) and the right half (d_R). Let $d = \min(d_L, d_R)$. The global minimum could still be a pair of points where one is in the left half and one is in the right half. To check this, we only need to consider points within a vertical “strip” of width $2d$ centered on the dividing line.

Within this strip, for any point P , we only need to check points Q that are within distance d of P . If the strip points are sorted by y -coordinate, it can be proven that each point only needs to be compared with a constant number (at most 7) of succeeding points in the sorted list to find a pair closer than d .

Theorem 13.4 (Strip Lemma). *Let S be a set of points in a $2d \times d$ rectangle, sorted by y -coordinate. For each point $p \in S$, at most 7 other points in S can lie within distance d of p .*

Proof. Divide the $2d \times d$ rectangle into 8 squares of side $d/2$. By the pigeonhole principle, if two points lie in the same square, their distance is at most $\frac{d}{2}\sqrt{2} < d$. However, each square can

contain at most one point from a pair with mutual distance $\geq d$, giving at most 8 points total (including p). Thus p needs to compare with at most 7 others. $\square$

Theorem 13.5 (Closest Pair Correctness). *The divide-and-conquer closest pair algorithm correctly finds the closest pair of n points in $\mathbb{R}^2$.*

Proof. By induction on n . The base cases ($n \leq 3$) are solved by brute force. For the recursive case, the minimum distance is either entirely within the left half, entirely within the right half, or crosses the dividing line. The recursive calls find d_L and d_R , and $d = \min(d_L, d_R)$. By the Strip Lemma (Theorem 13.4), only points within the strip need to be checked for a closer cross-pair, and the constant-bounded comparisons ensure no cross-pair closer than d is missed. $\square$

Randomized Grid Approach

This approach uses a grid with cell size $d \times d$. Each cell in the grid contains at most one point (if d is the current minimum distance). When a new point is added, we check its own cell and the 8 neighboring cells. If a closer point is found, the minimum distance d is updated, and the grid is rebuilt with the new cell size. In the randomized version, the expected number of rebuilds is $O(\log n)$, leading to an overall expected time of $O(n)$.

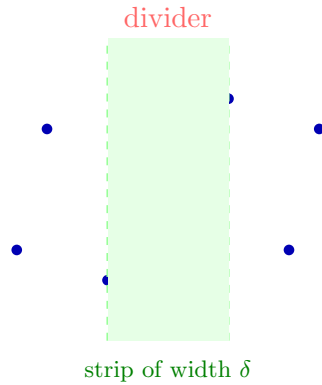


Figure 13.1: Closest pair: divide-and-conquer splits points along a vertical line. The closest pair may cross the dividing strip.

13.1.3 Pseudocode

13.1.4 Parameters

- **Algorithm Choice:** Use D&C for general-purpose deterministic results. Use Grid-based for massive streaming datasets where $O(n \log n)$ is too slow.
- **Base Case:** Brute force for $n < 4$ to avoid excessive recursion overhead.

13.1.5 Complexity

- **Divide and Conquer:** $O(n \log n)$ time, $O(n)$ space.
- **Grid-based:** $O(n)$ expected time, $O(n)$ space.

Example 13.6 (Closest Pair by Hand). Consider $P = \{(0, 0), (1, 3), (2, 2), (4, 1), (5, 4)\}$, sorted by x .

1. Split at $x = 2$: left $L = \{(0, 0), (1, 3), (2, 2)\}$, right $R = \{(4, 1), (5, 4)\}$.

Algorithm 55 Divide-and-Conquer Closest Pair

```

1: function CLOSESTPAIR(points_sorted_x, points_sorted_y)
2:   if len(points_sorted_x) ≤ 3 then
3:     return BruteForce(points_sorted_x)
4:   end if
5:   mid ← len(points_sorted_x) // 2
6:   left_x ← points_sorted_x[: mid]
7:   right_x ← points_sorted_x[mid :]
8:   (d_L, p1_L, p2_L) ← ClosestPair(left_x, points_sorted_y_left)
9:   (d_R, p1_R, p2_R) ← ClosestPair(right_x, points_sorted_y_right)
10:  d ← min(d_L, d_R)
11:  best_pair ← (p1_L, p2_L) if d_L < d_R else (p1_R, p2_R)
12:  strip ← [p | p ∈ points_sorted_y ∧ |p.x − points_sorted_x[mid].x| < d]
13:  for i = 0 to len(strip) − 1 do
14:    for j = i + 1 to min(i + 8, len(strip)) − 1 do
15:      if dist(strip[i], strip[j]) < d then
16:        d ← dist(strip[i], strip[j])
17:        best_pair ← (strip[i], strip[j])
18:      end if
19:    end for
20:  end for
21:  return (d, best_pair)
22: end function

```

2. Recurse on *L*: brute force gives $d_L = \|(1, 3) - (2, 2)\| = \sqrt{2}$.
3. Recurse on *R*: brute force gives $d_R = \|(4, 1) - (5, 4)\| = \sqrt{10}$.
4. $d = \sqrt{2}$. Strip: points with $|x - 2| < \sqrt{2}$, i.e., $x \in (0.586, 3.414)$: $\{(1, 3), (2, 2)\}$.
5. Check strip pairs: only $(1, 3)$ and $(2, 2)$ at distance $\sqrt{2}$: no improvement.
6. Result: closest pair is $(1, 3)$ and $(2, 2)$ with distance $\sqrt{2} \approx 1.414$.

13.1.6 Usages

- **Collision Detection:** Finding the two closest objects in a physics simulation.
- **Cluster Analysis:** Identifying tight groupings of data points.
- **Air Traffic Control:** Detecting potential collisions between aircraft.
- **Bioinformatics:** Comparing molecular structures or sequences.

13.1.7 Testing and Edge Cases

- **Identical Points:** Points with the exact same coordinates (distance 0).
- **Vertical/Horizontal alignment:** Points forming a perfect grid or line.
- **Large Coordinates:** Precision issues with squared distances $(x^2 + y^2)$ for very large x, y .
- **Sparse vs. Dense:** Performance on points scattered far apart vs. points clustered tightly together.
- **Verification:** Always compare with an $O(n^2)$ brute-force implementation for small inputs during testing.

13.1.8 References

- Shamos, M. I., & Hoey, D. (1975). “Closest-point problems”. IEEE FOCS.
- Khuller, S., & Matias, Y. (1995). “A simple randomized sieve algorithm for the closest-pair problem”. Information and Computation.
- Bentley, J. L., & Shamos, M. I. (1976). “Divide-and-conquer in multidimensional space”. ACM STOC.
- Wikipedia: Closest pair of points problem.

13.2 Proximity Queries

Proximity queries are a class of geometric problems focused on finding the spatial relationships between objects, specifically their distances and relative positions. These queries answer questions like “Which object is closest to this point?”, “Are these two objects intersecting?”, or “How far apart are these two shapes?” Proximity queries are the foundation of collision detection, spatial data mining, and robotics.

13.2.1 Definitions

Definition 13.7 (Nearest Neighbor). Given a set $P \subseteq \mathbb{R}^d$ and a query point q , the *nearest neighbor* of q in P is the point $p^* \in P$ minimizing $\|q - p^*\|$.

Definition 13.8 (K-Nearest Neighbors). The *k-nearest neighbors* of a query point q in a set P are the k points of P closest to q , ordered by distance.

13.2.2 Theory

Proximity queries are categorized by the type of geometric objects involved:

1. **Point-Point**: Handled using **KD-Trees** [Ben75] or **Quadtrees** for efficient $O(\log n)$ retrieval.
2. **Point-Mesh**: Finding the closest point on a surface. This involves searching a spatial index (like an **AABB Tree** or **BVH**) and calculating the distance to each triangle in the candidate leaves.
3. **Mesh-Mesh**: Calculating the distance between two 3D models. The **GJK (Gilbert-Johnson-Keerthi)** algorithm is the standard for convex shapes, while BVH hierarchies are used for general non-convex meshes.
4. **All-Pairs**: Finding all pairs of objects within a distance r . This is solved using **Spatial Hashing** or sweep-line algorithms.

The efficiency of these queries relies heavily on **Spatial Partitioning** (dividing the space) or **Bounding Volume Hierarchies** (grouping the objects).

Theorem 13.9 (KD-Tree Nearest Neighbor). A KD-tree built on n points in $\mathbb{R}^d$ supports nearest neighbor queries in $O(n^{1-1/d})$ worst-case time.

Algorithm 56 BVH Distance Query

```

1: function DISTANCE(nodeA, nodeB, current_min)
2:   if BoxDistance(nodeA.bbox, nodeB.bbox)  $\geq$  current_min then
3:     return current_min
4:   end if
5:   if nodeA.is_leaf and nodeB.is_leaf then
6:     for primA in nodeA.primitives do
7:       for primB in nodeB.primitives do
8:          $d \leftarrow$  PrimitiveDistance(primA, primB)
9:         current_min  $\leftarrow$  min(current_min, d)
10:      end for
11:    end for
12:    return current_min
13:  end if
14:  if nodeA.volume > nodeB.volume then
15:    for childA in nodeA.children do
16:      current_min  $\leftarrow$  Distance(childA, nodeB, current_min)
17:    end for
18:  else
19:    for childB in nodeB.children do
20:      current_min  $\leftarrow$  Distance(nodeA, childB, current_min)
21:    end for
22:  end if
23:  return current_min
24: end function

```

13.2.3 Pseudocode**13.2.4 Parameters**

- **Bounding Volume Type:** **AABB** (fast intersection), **OBB** (tighter fit), or **Bounding Spheres** (rotation invariant).
- **Spatial Index:** Use **KD-Trees** for static points and **R-Trees** or **BVHs** for moving objects.

13.2.5 Complexity

- **Point NN:** $O(\log n)$ average using KD-Trees. $O(n^{1-1/d})$ worst case in d dimensions [Ben75].
- **Closest Pair:** $O(n \log n)$ deterministic or $O(n)$ randomized.
- **Collision Detection:** $O(n \log n)$ for broad-phase, $O(1)$ for narrow-phase (convex primitives).

13.2.6 Usages

- **Physics Engines:** Real-time collision detection in video games (e.g., PhysX, Bullet).
- **Machine Learning:** Clustering (K-Means) and classification (KNN).
- **Robotics:** Obstacle avoidance and sensor fusion.
- **Molecular Modeling:** Detecting van der Waals overlaps between atoms.
- **GIS:** Finding the nearest points of interest (e.g., hospitals or fire stations).

13.2.7 Testing and Edge Cases

- **Identical Objects:** Distance should be zero.
- **Touching Objects:** Distance should be zero or very small.
- **Nested Objects:** Ensure the algorithm detects if one object is entirely inside another.
- **Large Aspect Ratios:** Test with very long and thin objects which can “leak” through poorly fitted bounding volumes.
- **Precision:** Verify distances for objects separated by many orders of magnitude.

13.2.8 References

- Gilbert, E. G., Johnson, D. W., & Keerthi, S. S. (1988). “A fast procedure for computing the distance between complex objects in three-dimensional space”. *IEEE Journal on Robotics and Automation*.
- Ericson, C. (2004). “Real-Time Collision Detection”. Morgan Kaufmann.
- Samet, H. (2006). “Foundations of Multidimensional and Metric Data Structures”. Elsevier.
- Wikipedia: Nearest neighbor search.

13.3 Industrial Applications of Nearest Neighbor Search

Nearest neighbor search (NNS)—finding the stored point closest to a query point, or its k -nearest generalization of Definition 13.8—is the most frequently used geometric primitive in industry because a surprising number of problems reduce to “find the most similar known item.” The data structures of Sections 10.1 and 10.7 (and, for high dimensions, locality-sensitive hashing) make it practical at billions of points [Ben75, Sam06]. The following are the principal industrial use cases.

- **Recommendation Systems:** Annoy was built at Spotify specifically for music recommendations—millions of tracks and users in a low-dimensional embedding space, queried for similar items—and the same nearest-neighbor formulation powers the recommendation engines of the largest web platforms.
- **Semantic Search and Vector Databases:** Embedding-based semantic search is served by approximate nearest-neighbor (ANNS) indexes (HNSW, FAISS, DiskANN) at the core of commercial vector databases (Pinecone, Weaviate, Qdrant, Milvus, Elasticsearch kNN), a market sold to enterprise search, retrieval-augmented generation, and chatbot deployments [MY20].
- **Reverse Image and Visual Search:** Content-based retrieval encodes images (or frames) into embedding vectors and returns the k most visually similar items by nearest-neighbor search in the embedding space, powering reverse image search, product visual search, and duplicate detection in retail and content platforms.
- **Geospatial Services:** Mapping, ride-hailing, and navigation apps answer “closest restaurant” or “nearest pickup point” through spatial nearest-neighbor queries over point-of-interest databases (Section 10.1), a query type materialized at billion-query scale by location platforms.

- **Autonomous Driving and Robotics:** Nearest-neighbor search over a precomputed point cloud aligns sensor scans (point-cloud registration, ICP), matches features for SLAM loop closure, and finds the nearest obstacle for collision checking and path planning in commercial autonomous-driving and robotics stacks.
- **Drug Discovery and Molecular Similarity:** Molecular similarity by fingerprints or descriptors is a nearest-neighbor query over compound databases, used to find the closest known active molecule to a candidate drug; structure- and ligand-based virtual screening is sold commercially by computational-chemistry vendors.
- **Fraud Detection and Anomaly Detection:** Transactions embedded into a feature space are flagged when their nearest neighbors are predominantly fraudulent or when the distance to all known-good neighbors exceeds a learned threshold—an anomaly-detection formulation of NNS shipped in commercial fraud and security analytics platforms.
- **Bioinformatics and Genomics:** DNA and protein sequences compared by edit distance use nearest-neighbor search over a string-space index to find the closest gene, detect homology, and cluster metagenomic samples, powering commercial sequencing-analysis pipelines.
- **Pattern Recognition and Classification:** The k -nearest neighbors classifier (k -NN) labels a query point by the majority class of its k nearest training points; it remains a production baseline in biometric identification, OCR, and spam filtering.

13.3.1 Industrial-Scale Software

Exact and approximate NNS is implemented in `scipy.spatial.cKDTree` (KD-tree), `FLANN` and `FAISS` (approximate, billion-scale), `HNSWlib` (graph-based ANNS), `Annoy` (random-projection forests), and `PostGIS` (spatial nearest-neighbor SQL), in addition to the KD-tree and fixed-radius structures of Sections 10.1 and 10.7 [MY20].

13.4 Range Search

Range searching is a fundamental task in computational geometry where the goal is to efficiently retrieve all geometric objects (typically points) that lie within a specified query region (the “range”). Query regions are usually simple shapes like axis-aligned rectangles (orthogonal range search), circles, or half-planes. Efficient range searching is the backbone of spatial databases, geographic information systems (GIS), and computer graphics. Foundational work on range search data structures was done by Bentley [Ben75] and Lueker [Lue78].

13.4.1 Definitions

Definition 13.10 (Range Search). Given a set P of n points in $\mathbb{R}^d$ and a query region R , the *range search* problem asks to either report all points $p \in P \cap R$ (range reporting) or count $|P \cap R|$ (range counting).

Definition 13.11 (Orthogonal Range Query). An *orthogonal range query* is a range search where the query region is an axis-aligned hyper-rectangle $R = [x_1, x_2] \times [y_1, y_2] \times \dots$.

13.4.2 Theory

The efficiency of range search depends on the dimensionality and the type of range.

1. **1D Range Search:** Solved using a balanced binary search tree in $O(\log n + k)$ time.

2. Orthogonal Range Search (d -dimensions):

- **KD-Trees:** Good for medium dimensions; $O(\sqrt{n} + k)$ query time in 2D.
- **Range Trees:** Faster query time $O(\log^d n + k)$ but higher space $O(n \log^{d-1} n)$.
- **Quadrees:** Effective for non-uniform data distributions.

3. **Circular/Spherical Range Search:** Often handled using KD-trees with a distance-based pruning condition or specialized structures like **Ball Trees**.

4. **Non-Orthogonal Range Search:** General shapes can be approximated by rectangles or handled using more complex structures like **Simplex Range Reporting**.

The fundamental tradeoff in range search is between **Query Time**, **Preprocessing Time**, and **Space Complexity**.

Theorem 13.12 (Range Tree Query Time). *A range tree on n points in $\mathbb{R}^d$ uses $O(n \log^{d-1} n)$ space and supports orthogonal range reporting queries in $O(\log^d n + k)$ time, where k is the number of reported points.*

13.4.3 Pseudocode

Algorithm 57 Orthogonal Range Search (Tree-based)

```

1: function RANGESEARCH(node, query_range, results)
2:   if node is None then
3:     return
4:   end if
5:   if  $\neg \text{Intersect}(\text{node.bbox}, \text{query\_range})$  then
6:     return
7:   end if
8:   if FullyContains(query_range, node.bbox) then
9:     results.extend(node.all_points)
10:    return
11:  end if
12:  if node.is_leaf then
13:    for  $p$  in node.points do
14:      if query_range.contains( $p$ ) then
15:        results.append( $p$ )
16:      end if
17:    end for
18:  else
19:    for child in node.children do
20:      RangeSearch(child, query_range, results)
21:    end for
22:  end if
23: end function

```

13.4.4 Parameters

- **Dimensionality:** For $d > 20$, most spatial data structures degrade to linear search (the “Curse of Dimensionality”).

- **Data Distribution:** Use **Quadtrees** or **R-Trees** for highly clustered data; use **KD-Trees** or **Range Trees** for more uniform data.
- **Memory:** If memory is tight, KD-trees are preferred ($O(n)$ space).

13.4.5 Complexity Summary

Structure	Space	Query (2D Reporting)
KD-Tree	$O(n)$	$O(\sqrt{n} + k)$
Range Tree	$O(n \log n)$	$O(\log^2 n + k)$
Quadtree	$O(\text{depth} \cdot n)$	$O(\text{depth} + k)$
R-Tree	$O(n)$	$O(\log n + k)$ (avg)

13.4.6 Usages

- **Spatial Databases:** Finding all customers within a specific zip code or radius.
- **Computer Graphics:** Frustum culling and identifying all objects hit by a blast radius.
- **GIS:** Overlaying different map layers (e.g., finding all houses in a flood zone).
- **Machine Learning:** Preprocessing for K-Nearest Neighbors (KNN).
- **Collision Detection:** Finding all pairs of objects that might be colliding.

Example 13.13 (1D Range Search). Given sorted points $P = \{1, 3, 5, 7, 9, 11, 13\}$ and query range $R = [4, 10]$, a binary search locates the first point ≥ 4 (which is 5) and the last point ≤ 10 (which is 9). All points in this contiguous subsequence $\{5, 7, 9\}$ are returned in $O(\log n + k)$ time, where $k = 3$.

13.4.7 Testing and Edge Cases

- **Empty Query:** Ensure the algorithm returns an empty list for a range containing no points.
- **Full Range Query:** Verify that a range covering the entire domain returns all n points.
- **Points on Boundaries:** Consistently handle points exactly on the edge of the query rectangle.
- **Overlapping Points:** Ensure duplicate points are correctly counted/reported.
- **Very Large Coordinates:** Precision check for queries far from the origin.

13.4.8 References

- Bentley, J. L. (1975). “Multidimensional binary search trees used for associative searching”. Communications of the ACM.
- Lueker, G. S. (1978). “A data structure for orthogonal range queries”. IEEE FOCS.
- Berg, M. de, Cheong, O., Kreveld, M. van, & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer-Verlag.
- Wikipedia: Range searching.

13.5 Point in Polygon (Winding Number)

The Winding Number algorithm is a robust method for determining whether a point lies inside a given polygon. It calculates the total number of times the polygon's boundary wraps around the test point. A winding number of zero indicates the point is outside the polygon, while a non-zero value indicates it is inside. This method is particularly superior to the Ray Casting algorithm when dealing with self-intersecting polygons. Analysis of the winding number method for arbitrary polygons is given by Sunday [Sun01] and Hörmann and Agathos [HA01].

13.5.1 Definitions

Definition 13.14 (Winding Number). The *winding number* $wn(P, C)$ of a point P with respect to a closed curve C is the total number of times C winds counter-clockwise around P . Formally, it equals $\frac{1}{2\pi} \oint_C d\theta$ where θ is the angle from P to a point on C .

Definition 13.15 (Is-Left Test). The *Is-Left test* determines whether a point P is to the left of the directed line from V_1 to V_2 :

$$\text{IsLeft}(V_1, V_2, P) = (V_2.x - V_1.x)(P.y - V_1.y) - (P.x - V_1.x)(V_2.y - V_1.y).$$

This is equivalent to the signed area of $\triangle V_1 V_2 P$ (up to a factor of 2).

13.5.2 Theory

The algorithm counts the net number of full revolutions the polygon makes around the point P . As we traverse the polygon's edges:

- Every time an edge crosses the positive x -ray of P going **upward**, we increment the winding number.
- Every time an edge crosses the positive x -ray of P going **downward**, we decrement the winding number.

Crucially, an edge is only counted if the point P is actually to the left of the upward edge (or to the right of the downward edge), ensuring that the crossing actually “wraps” around the point.

Theorem 13.16 (Winding Number Correctness). *A point P lies inside a simple polygon C if and only if $wn(P, C) \neq 0$. More precisely, the winding number counts the number of times C encircles P .*

Proof. The winding number is a topological invariant related to the degree of the map $\theta : C \rightarrow S^1$ defined by $\theta(q) = \frac{q-P}{\|q-P\|}$. By the Jordan Curve Theorem, for a simple closed curve, $wn(P, C) = \pm 1$ for interior points and 0 for exterior points. The sign depends on the orientation of C . The incremental counting algorithm correctly tracks the accumulated angle by detecting upward and downward crossings of the horizontal ray from P . $\square$

13.5.3 Pseudocode

13.5.4 Parameters

- **Input:** A test `Point2D` and a list of `Point2D` vertices representing the polygon.
- **Winding Rule:** Most applications use the **Non-Zero Winding Rule** (inside if $wn \neq 0$), though some use the **Odd-Even Rule** (wn is odd).

Algorithm 58 Point-in-Polygon via Winding Number

```

1: function POINTWINDINGNUMBER( $P$ , polygon)
2:    $wn \leftarrow 0$ 
3:   for each edge  $(V1, V2)$  in polygon do
4:     if  $V1.y \leq P.y$  then
5:       if  $V2.y > P.y$  then
6:         if  $\text{IsLeft}(V1, V2, P) > 0$  then
7:            $wn \leftarrow wn + 1$ 
8:         end if
9:       end if
10:    else
11:      if  $V2.y \leq P.y$  then
12:        if  $\text{IsLeft}(V1, V2, P) < 0$  then
13:           $wn \leftarrow wn - 1$ 
14:        end if
15:      end if
16:    end if
17:  end for
18:  return  $wn$ 
19: end function
20: function ISLEFT( $V1, V2, P$ )
21:  return  $(V2.x - V1.x) * (P.y - V1.y) - (P.x - V1.x) * (V2.y - V1.y)$ 
22: end function

```

13.5.5 Complexity

- **Time Complexity:** $O(n)$ where n is the number of vertices. Each edge is visited exactly once.
- **Space Complexity:** $O(1)$ auxiliary space, assuming the polygon vertices are already stored.

Example 13.17 (Winding Number for a Square). Consider the unit square with vertices $(0,0), (1,0), (1,1), (0,1)$ traversed counter-clockwise. Testing point $P = (0.5, 0.5)$:

1. Edge $(0,0) \rightarrow (1,0)$: $V1.y = 0 \leq 0.5$, $V2.y = 0 \not> 0.5$: no crossing.
2. Edge $(1,0) \rightarrow (1,1)$: $V1.y = 0 \leq 0.5$, $V2.y = 1 > 0.5$: upward crossing. $\text{IsLeft} = (1-1)(0.5-0) - (0.5-1)(1-0) = 0 + 0.5 = 0.5 > 0$: $wn \leftarrow 1$.
3. Edge $(1,1) \rightarrow (0,1)$: $V1.y = 1 \not\leq 0.5$: downward branch. $V2.y = 1 \not\leq 0.5$: no crossing.
4. Edge $(0,1) \rightarrow (0,0)$: $V1.y = 1 \not\leq 0.5$: downward branch. $V2.y = 0 \leq 0.5$: downward crossing. $\text{IsLeft} = (0-0)(0.5-1) - (0.5-0)(0-1) = 0 + 0.5 = 0.5 > 0$, but we need < 0 : not counted (this is correct since the edge goes down and P is to its right).

Final $wn = 1 \neq 0$: point is inside. For $P = (2, 0.5)$ (outside), no crossings contribute: $wn = 0$.

13.5.6 Usages

- **SVG/PostScript Rendering:** Determining which areas of a complex path should be filled.
- **GIS Analysis:** Checking if a GPS coordinate falls within a specific territorial boundary.
- **Video Games:** Hitbox detection for complex, non-convex characters or objects.

13.5.7 Testing and Edge Cases

- **Point on Edge/Vertex:** The algorithm's behavior on the boundary depends on the strictness of the inequalities ($<$ vs $\leq$).
- **Horizontal Edges:** Handled correctly because they fail both the $V1.y \leq P.y < V2.y$ and $V2.y \leq P.y < V1.y$ conditions.
- **Self-intersecting Polygons:** A “figure-eight” polygon will have regions with winding numbers of 1, -1 , and 0.
- **Degenerate Polygons:** Polygons with fewer than 3 vertices or zero area.

13.5.8 References

- Sunday, D. (2001). “Inclusion of a Point in a Polygon”. Journal of Graphics Tools.
- Hörmann, K., & Agathos, A. (2001). “The point in polygon problem for arbitrary polygons”. Computational Geometry.
- Wikipedia: Point in polygon.

13.6 Largest Empty Circle

The largest empty circle problem asks for the circle of maximum radius whose center is within the convex hull of a given set of points P , and whose interior contains no points from P . This problem is a fundamental challenge in computational geometry with direct applications in facility location and urban planning.

13.6.1 Definitions

Definition 13.18 (Largest Empty Circle). Given a set $P \subseteq \mathbb{R}^2$, the *largest empty circle* is the circle C^* with maximum radius r^* such that C^* contains no point of P in its interior and its center lies within $\text{conv}(P)$.

13.6.2 Theory

The largest empty circle must have its center at one of two types of locations:

1. **Voronoi Vertices:** A Voronoi vertex is the circumcenter of three points from P . If this vertex is inside the convex hull, it defines an empty circle passing through those three points.
2. **Intersection of Voronoi Edges and Convex Hull:** The center could be at the intersection of a Voronoi edge and an edge of the convex hull.

The algorithm for finding the largest empty circle is:

1. Compute the **Voronoi Diagram** (Algorithm 37) of the points P .
2. Iterate through all Voronoi vertices. For each vertex inside the convex hull, calculate the distance to its closest point (its radius).
3. Iterate through all intersections of Voronoi edges with the convex hull boundary. For each intersection, calculate the radius of the empty circle centered there.
4. The largest of all these radii defines the largest empty circle.

Theorem 13.19 (Optimality of Voronoi Vertices). *The center of the largest empty circle of a point set P in general position is either a Voronoi vertex of P or lies on the intersection of a Voronoi edge with the boundary of $\text{conv}(P)$.*

13.6.3 Pseudocode

Algorithm 59 Largest Empty Circle

```

1: function LARGESTEMPTYCIRCLE(points)
2:    $CH \leftarrow \text{ConvexHull}(\text{points})$ 
3:    $VD \leftarrow \text{Voronoi}(\text{points})$ 
4:    $\text{max\_radius} \leftarrow 0$ 
5:    $\text{best\_center} \leftarrow \text{None}$ 
6:   for  $v$  in  $VD.\text{vertices}$  do
7:     if  $\text{PointInPolygon}(v, CH)$  then
8:        $r \leftarrow \text{Distance}(v, VD.\text{GetClosestSite}(v))$ 
9:       if  $r > \text{max\_radius}$  then
10:         $\text{max\_radius} \leftarrow r$ 
11:         $\text{best\_center} \leftarrow v$ 
12:       end if
13:     end if
14:   end for
15:   for  $\text{edge}$  in  $VD.\text{edges}$  do
16:     for  $\text{ch\_edge}$  in  $CH.\text{edges}$  do
17:        $p \leftarrow \text{Intersect}(\text{edge}, \text{ch\_edge})$ 
18:       if  $p$  is not  $\text{None}$  then
19:          $r \leftarrow \text{Distance}(p, VD.\text{GetClosestSite}(p))$ 
20:         if  $r > \text{max\_radius}$  then
21:            $\text{max\_radius} \leftarrow r$ 
22:            $\text{best\_center} \leftarrow p$ 
23:         end if
24:       end if
25:     end for
26:   end for
27:   return  $\text{best\_center}, \text{max\_radius}$ 
28: end function

```

13.6.4 Parameters

- **Precision:** Finding the intersection of Voronoi edges and the convex hull requires careful handling of floating-point inaccuracies.
- **Region Constraint:** The problem can be generalized to find the largest empty circle within any bounding polygon (not just the convex hull).

13.6.5 Complexity

- **Time Complexity:** $O(n \log n)$ to build the Voronoi diagram and convex hull. Finding intersections takes $O(n \log n)$ or $O(n)$ depending on the implementation.
- **Space Complexity:** $O(n)$ to store the Voronoi and convex hull structures.

13.6.6 Usages

- **Facility Location:** Finding the location for a new facility (e.g., a toxic waste dump or a new shopping center) that is as far as possible from all existing points of interest.
- **Urban Planning:** Determining where to place a new park to maximize the distance from the nearest noise sources.
- **Robotics:** Path planning to stay as far from obstacles as possible (the “safest” path follows Voronoi edges).
- **Signal Coverage:** Identifying the largest “blind spot” in a cellular or Wi-Fi network.

13.6.7 Testing and Edge Cases

- **Square/Rectangle Alignment:** For four points in a square, the center should be at the centroid.
- **Points on a Circle:** All Voronoi vertices should coincide at the center of the circle.
- **Collinear Points:** The convex hull is a line segment, so the circle will have zero radius.
- **Sparse vs. Dense:** Ensure the algorithm handles large empty spaces correctly.
- **Large Coordinates:** Precision check for very large x, y .

13.6.8 References

- Shamos, M. I., & Hoey, D. (1975). “Closest-point problems”. IEEE FOCS.
- Preparata, F. P., & Shamos, M. I. (1985). “Computational Geometry: An Introduction”. Springer-Verlag.
- Wikipedia: Largest empty sphere.

13.7 Minimum Bounding Boxes

A minimum bounding box (MBB) is the smallest rectangle (in terms of area or perimeter) that contains all the points in a given set P . Unlike the axis-aligned bounding box (AABB), which is restricted to the coordinate axes, the minimum bounding box can be rotated to find the optimal fit. This problem is fundamental in object recognition, shape analysis, and spatial data compression.

13.7.1 Definitions

Definition 13.20 (Minimum Bounding Box). The *minimum bounding box* of a point set P is the rectangle of minimum area (or perimeter) containing all points of P , with no restriction on orientation.

Definition 13.21 (Rotating Calipers). The *rotating calipers* technique is an algorithmic paradigm that uses a set of lines (“calipers”) touching a convex polygon at extreme points and rotates them simultaneously to compute geometric properties in linear time [Tou83].

13.7.2 Theory

The minimum bounding box of a point set P is identical to the MBB of its convex hull $\text{conv}(P)$. Freeman and Shapira (1975) proved that one edge of the minimum-area bounding box **MUST** be collinear with an edge of the convex hull.

The **Rotating Calipers** algorithm uses this property:

1. Compute the convex hull $\text{conv}(P)$.
2. Place four “calipers” (lines) touching the convex hull at its extreme points $(x_{\min}, x_{\max}, y_{\min}, y_{\max})$.
3. Rotate the four lines simultaneously around the hull.
4. Each time a line becomes collinear with an edge of the hull, calculate the area of the rectangle formed by the four lines.
5. After a 90° rotation, the minimum area found is the MBB.

Theorem 13.22 (Rotating Calipers Correctness). *At least one edge of the minimum-area bounding rectangle is collinear with an edge of the convex hull.*

Proof. Let R^* be the minimum-area bounding rectangle, and let e be an edge of R^* . If no edge of $\text{conv}(P)$ is parallel to e , then we can rotate R^* slightly to decrease its area while still containing all points, contradicting optimality. Hence at least one edge of R^* must be parallel to an edge of $\text{conv}(P)$. $\square$

13.7.3 Pseudocode

Algorithm 60 Minimum Bounding Box via Rotating Calipers

```

1: function MINIMUMBOUNDINGBOX(points)
2:    $CH \leftarrow \text{ConvexHull}(\text{points})$ 
3:    $\text{min\_area} \leftarrow \infty$ 
4:    $\text{best\_rectangle} \leftarrow \text{None}$ 
5:   for  $i = 0$  to  $\text{len}(CH) - 1$  do
6:      $\text{edge} \leftarrow CH[i + 1] - CH[i]$ 
7:      $\text{angle} \leftarrow \text{atan2}(\text{edge.y}, \text{edge.x})$ 
8:      $\text{rotated\_hull} \leftarrow \text{Rotate}(CH, -\text{angle})$ 
9:      $\text{min\_x}, \text{max\_x}, \text{min\_y}, \text{max\_y} \leftarrow \text{GetBounds}(\text{rotated\_hull})$ 
10:     $\text{area} \leftarrow (\text{max\_x} - \text{min\_x}) * (\text{max\_y} - \text{min\_y})$ 
11:    if  $\text{area} < \text{min\_area}$  then
12:       $\text{min\_area} \leftarrow \text{area}$ 
13:       $\text{best\_rectangle} \leftarrow \text{RotateBack}(\text{Rectangle}(\text{min\_x}, \text{max\_x}, \text{min\_y}, \text{max\_y}), \text{angle})$ 
14:    end if
15:  end for
16:  return  $\text{best\_rectangle}$ 
17: end function

```

13.7.4 Parameters

- **Objective Function:** Usually **Area** is minimized, but **Perimeter** can also be an objective.
- **Convex Hull Algorithm:** Graham Scan (Algorithm 2) or Monotone Chain (Algorithm 3) are suitable.

13.7.5 Complexity

- **Time Complexity:** $O(n \log n)$ to build the convex hull, and $O(n)$ for the rotating calipers scan.
- **Space Complexity:** $O(n)$ to store the convex hull.

13.7.6 Usages

- **Object Recognition:** Normalizing the orientation of an object before comparing it to a database.
- **Packaging and Bin Packing:** Finding the most space-efficient way to fit an object into a shipping container.
- **Spatial Indexing:** Using tighter-fitting MBBs instead of AABBs in R-trees to reduce overlap and improve search performance.
- **Computer Vision:** Finding the orientation of a rectangular object (e.g., a credit card or a license plate).
- **Geographic Information Systems (GIS):** Describing the extent of spatial features.

13.7.7 Testing and Edge Cases

- **Points forming a Circle:** The MBB should be a square.
- **Already Axis-Aligned Rectangle:** The MBB should match the AABB.
- **Points on a Line:** The MBB will have zero area.
- **Rotated Rectangle:** Ensure the algorithm finds the correct orientation and area.
- **Precision:** Use robust arithmetic for cross-products and distance calculations.

13.7.8 References

- Freeman, H., & Shapira, R. (1975). “Determining the minimum-area encasing rectangle for an arbitrary closed curve”. *Communications of the ACM*.
- Toussaint, G. T. (1983). “Solving geometric problems with the rotating calipers”. *Proceedings of MELECON '83*.
- Wikipedia: Minimum bounding box.

13.8 Maximum Overlap

The maximum overlap problem asks for a point (or region) in space that is covered by the maximum number of geometric objects from a given set. For example, given a set of 1D intervals, finding the time when the most tasks are active simultaneously. In 2D, this often involves finding the location covered by the most rectangles or disks. It is a fundamental problem in scheduling, resource allocation, and signal processing.

13.8.1 Definitions

Definition 13.23 (Depth of a Point). The *depth* of a point p with respect to a set of objects $\mathcal{F}$ is the number of objects in $\mathcal{F}$ that contain p .

Definition 13.24 (Maximum Depth). The *maximum depth* is $\max_{p \in \mathbb{R}^d} \text{depth}(p, \mathcal{F})$, i.e., the maximum number of overlapping objects at any point.

13.8.2 Theory

1D Case (Intervals)

For n intervals $[s_i, e_i]$, the maximum overlap can be found using a **sweep-line** approach:

1. Treat all start and end points as “events.”
2. Sort events by coordinate.
3. Traverse events: increment a counter for every start point and decrement for every end point.
4. The maximum value reached by the counter is the maximum overlap.

2D Case (Rectangles)

For axis-aligned rectangles, the problem is solved by sweeping a vertical line across the plane:

1. As the sweep line hits the left edge of a rectangle, the corresponding vertical interval is added to a **Segment Tree**.
2. As it hits the right edge, the interval is removed.
3. The Segment Tree maintains the maximum “coverage” count across its entire range efficiently.

Theorem 13.25 (1D Maximum Overlap Sweep Correctness). *The sweep-line algorithm correctly computes the maximum overlap of n intervals in $O(n \log n)$ time.*

13.8.3 Pseudocode

13.8.4 Parameters

- **Boundary Inclusion:** Decide if intervals are open (a, b) or closed $[a, b]$. This affects how events at the same coordinate are handled during the sort.
- **Data Structure:** For 2D, a Segment Tree with lazy propagation is required to achieve $O(n \log n)$ time.

13.8.5 Complexity

- **1D Complexity:** $O(n \log n)$ due to sorting the $2n$ endpoints.
- **2D Complexity:** $O(n \log n)$ using a sweep-line and a Segment Tree.
- **Space Complexity:** $O(n)$ to store the events and the tree structure.

Example 13.26 (1D Sweep Trace). Consider intervals $[1, 4]$, $[2, 5]$, $[3, 6]$. Events sorted: $(1, +1)$, $(2, +1)$, $(3, +1)$, $(4, -1)$, $(5, -1)$, $(6, -1)$. The sweep yields counts: 1, 2, 3, 2, 1, 0. Maximum overlap is 3, achieved at any point in $[3, 4]$.

Algorithm 61 Maximum Overlap (1D Sweep)

```
1: function MAXIMUMOVERLAP1D(intervals)
2:   events  $\leftarrow []$ 
3:   for s, e in intervals do
4:     events.append((s, +1))
5:     events.append((e, -1))
6:   end for
7:   events.sort(key =  $\lambda x : (x[0], -x[1])$ )
8:   max_count  $\leftarrow 0$ 
9:   current_count  $\leftarrow 0$ 
10:  best_pos  $\leftarrow \text{None}$ 
11:  for pos, type in events do
12:    current_count  $\leftarrow \text{current\_count} + \text{type}$ 
13:    if current_count > max_count then
14:      max_count  $\leftarrow \text{current\_count}$ 
15:      best_pos  $\leftarrow \text{pos}$ 
16:    end if
17:  end for
18:  return max_count, best_pos
19: end function
```

13.8.6 Usages

- **Scheduling:** Finding the peak number of concurrent users or tasks.
- **Genomics:** Identifying “hotspots” where many DNA reads map to the same genomic location.
- **VLSI Design:** Finding regions with the highest density of circuit components to prevent overheating.
- **Finance:** Detecting when the most stock market orders are active in a specific price range.
- **Sensor Networks:** Identifying regions covered by the most sensors (redundancy analysis).

13.8.7 Testing and Edge Cases

- **Disjoint Intervals:** Maximum overlap should be 1.
- **Identical Intervals:** Maximum overlap should be n .
- **Nested Intervals:** Ensure the algorithm handles intervals fully contained within each other.
- **Zero-Length Intervals:** Points treated as intervals.
- **Large Coordinates:** Precision check for very distant intervals.
- **Touching Boundaries:** Verify behavior when intervals touch at a single point (overlap of 1 or 2 depending on inclusion rule).

13.8.8 References

- Bentley, J. L., & Wood, D. (1980). “Algorithms for spectral and overlapping rectangles”. SIAM Journal on Computing.

- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). “Introduction to Algorithms”. MIT Press.
- Wikipedia: Interval tree.

13.9 Segment Intersection (Bentley-Ottmann)

The segment intersection problem asks for all intersection points among a set of n line segments in the 2D plane. While a brute-force approach checks every pair of segments in $O(n^2)$ time, the Bentley-Ottmann algorithm [BO79] uses a **sweep-line paradigm** to achieve a complexity that is sensitive to the number of intersections k . It is a cornerstone algorithm in computational geometry for applications in CAD, GIS, and computer graphics.

13.9.1 Definitions

Definition 13.27 (Sweep Line). A *sweep line* is an imaginary vertical line that moves from left to right across the plane, processing events in order of x -coordinate.

Definition 13.28 (Sweep-Line Status). The *sweep-line status* is a data structure (typically a balanced BST) that maintains the vertical order of segments currently intersecting the sweep line, ordered by their y -coordinates at the sweep line’s current position.

Definition 13.29 (Output-Sensitive Algorithm). An algorithm is *output-sensitive* if its running time depends on both the input size and the output size. For Bentley-Ottmann, the time is $O((n + k) \log n)$ where k is the number of intersection points reported.

13.9.2 Theory

The Bentley-Ottmann algorithm is based on the observation that two segments can only intersect if they are **adjacent** in the vertical order at some position along the sweep line.

1. Initialize the event queue with all segment endpoints.
2. Pop the next event point P .
3. **If P is a left endpoint:** Insert the segment into the status structure. Check for intersections with its immediate neighbors (above and below).
4. **If P is a right endpoint:** Check if its neighbors (above and below) now intersect each other. Remove the segment from the status structure.
5. **If P is an intersection point:** Report the intersection. Swap the order of the two intersecting segments in the status structure. Check for new intersections with their new neighbors.

This process ensures that all k intersections are found by only checking pairs of segments that are “geographically” close.

Theorem 13.30 (Bentley-Ottmann Correctness). *The Bentley-Ottmann algorithm correctly reports all intersection points among n line segments. Each intersection is reported exactly once.*

Theorem 13.31 (Bentley-Ottmann Complexity). *The Bentley-Ottmann algorithm runs in $O((n + k) \log n)$ time and $O(n + k)$ space, where n is the number of segments and k is the number of intersections.*

Proof. There are $2n$ endpoint events and k intersection events. Each event requires $O(\log n)$ time for BST operations (insert, delete, swap) and $O(1)$ neighbor checks. Each intersection event is generated at most once (when two segments become adjacent), and each event is processed once. The total number of events is $O(n + k)$, each costing $O(\log n)$, giving $O((n + k) \log n)$ total time. Space is $O(n + k)$ for the event queue (which stores at most $O(n + k)$ events at any time) and $O(n)$ for the status structure. $\square$

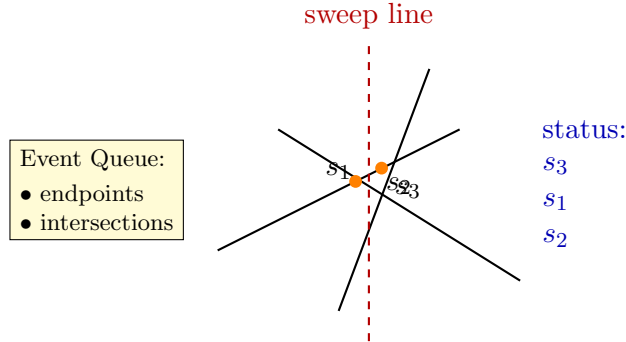


Figure 13.2: Bentley–Ottmann sweep line: the vertical line moves left-to-right. The *status* maintains the vertical order of intersecting segments.

13.9.3 Pseudocode

13.9.4 Parameters

- **Precision:** Highly sensitive to floating-point errors. Use robust geometric predicates for intersection calculation and segment ordering.
- **Degeneracies:** Multiple segments intersecting at the same point or vertical segments require careful handling.

13.9.5 Complexity

- **Time Complexity:** $O((n + k) \log n)$, where n is the number of segments and k is the number of intersections.
- **Space Complexity:** $O(n + k)$ to store the event queue and status structure.

Example 13.32 (Bentley-Ottmann Trace). Consider three segments: $s_1 = \{(0, 0), (3, 3)\}$, $s_2 = \{(0, 3), (3, 0)\}$, $s_3 = \{(1, 0), (1, 3)\}$.

1. Events sorted by x : $(0, \text{left}, s_1)$, $(0, \text{left}, s_2)$, $(1, \text{left}, s_3)$, $(3, \text{right}, s_1)$, $(3, \text{right}, s_2)$.
2. After processing $x = 0$: Status order (at $x = 0^+$) is s_2 (above), s_1 (below). Check intersection: they meet at $(1.5, 1.5)$: event enqueued.
3. At $x = 1$: s_3 inserted. Status: s_2, s_3, s_1 . Check s_3 with s_2 : they meet at $(1, 2)$: event enqueued. Check s_3 with s_1 : meet at $(1, 1)$: event enqueued.
4. At $x = 1$ (intersection $(1, 1)$): swap s_1, s_3 . New neighbors checked.
5. At $x = 1$ (intersection $(1, 2)$): swap s_2, s_3 .
6. At $x = 1.5$ (intersection $(1.5, 1.5)$): swap s_1, s_2 .
7. After $x = 3$: all segments removed. Three intersection points reported.

Algorithm 62 Bentley-Ottmann Segment Intersection

```

1: function BENTLEYOTTMANN(segments)
2:   events  $\leftarrow$  PriorityQueue()
3:   for s in segments do
4:     events.push(Event(s.left, LEFT_ENDPOINT, s))
5:     events.push(Event(s.right, RIGHT_ENDPOINT, s))
6:   end for
7:   status  $\leftarrow$  BalancedBST()
8:   intersections  $\leftarrow$  []
9:   while events is not empty do
10:    event  $\leftarrow$  events.pop()
11:    p  $\leftarrow$  event.point
12:    if event.type == LEFT_ENDPOINT then
13:      s  $\leftarrow$  event.segment
14:      status.insert(s, p.x)
15:      above, below  $\leftarrow$  status.neighbors(s)
16:      CheckIntersection(s, above, events)
17:      CheckIntersection(s, below, events)
18:    else if event.type == RIGHT_ENDPOINT then
19:      s  $\leftarrow$  event.segment
20:      above, below  $\leftarrow$  status.neighbors(s)
21:      status.remove(s)
22:      CheckIntersection(above, below, events)
23:    else if event.type == INTERSECTION then
24:      s1, s2  $\leftarrow$  event.segments
25:      intersections.append(p)
26:      status.swap(s1, s2)
27:      new_above  $\leftarrow$  status.above(s2)
28:      new_below  $\leftarrow$  status.below(s1)
29:      CheckIntersection(s2, new_above, events)
30:      CheckIntersection(s1, new_below, events)
31:    end if
32:  end while
33:  return intersections
34: end function

```

13.9.6 Usages

- **Map Overlay:** Combining two maps (e.g., roads and counties) to find where they cross.
- **Boolean Operations:** Preprocessing for polygon union/intersection.
- **CAD/CAM:** Detecting errors in mechanical drawings (e.g., overlapping paths).
- **VLSI Design:** Checking for short circuits in circuit board traces.
- **Path Planning:** Determining if a path intersects any obstacles.

13.9.7 Testing and Edge Cases

- **No Intersections:** The algorithm should only process $2n$ events and return empty.
- **All segments intersect at one point:** Verify that the event queue handles multiple events at the same location.
- **Vertical Segments:** Ensure the ordering logic correctly handles segments with the same x -coordinate.
- **Collinear Segments:** Overlapping segments on the same line.
- **Duplicate Intersections:** Ensure each intersection point is reported only once if multiple segments meet.

13.9.8 References

- Bentley, J. L., & Ottmann, T. A. (1979). “Algorithms for reporting and counting geometric intersections”. IEEE Transactions on Computers.
- Berg, M. de, Cheong, O., Kreveld, M. van, & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer-Verlag.
- Wikipedia: Bentley–Ottmann algorithm.

13.10 Point Location via Trapezoidal Maps

The point location problem asks, given a planar subdivision and a query point, which face of the subdivision contains that point. It is one of the most fundamental problems in computational geometry, appearing as a subroutine in polygon operations, mesh processing, GIS overlays, and visibility computations. A classical solution is the *trapezoidal map* combined with a directed acyclic graph (DAG) search structure, which supports $O(\log n)$ expected query time after $O(n \log n)$ expected preprocessing [Mul90, Sei91]. The randomized incremental construction avoids the degeneracy analysis required by deterministic approaches.

13.10.1 Definitions

Definition 13.33 (Trapezoid). A *trapezoid* T in a trapezoidal map is a region bounded by two vertical walls (left and right sides), a *top edge* e_{top} , and a *bottom edge* e_{bot} . The left and right boundaries are vertical segments from the bottom edge to the top edge. Each trapezoid is uniquely identified by its left x -coordinate, right x -coordinate, and the two edges that bound it from above and below.

Definition 13.34 (Trapezoidal Map). Given a set S of n non-intersecting line segments in $\mathbb{R}^2$, the *trapezoidal map* $\mathcal{T}(S)$ is the subdivision of the plane induced by extending vertical rays from every endpoint of every segment in S upward and downward until they hit a segment or a bounding box. The result is a partition of the plane into $O(n)$ trapezoids.

Definition 13.35 (DAG Search Structure). The *DAG search structure* $D(\mathcal{T})$ is a directed acyclic graph with leaf nodes representing the trapezoids of $\mathcal{T}$ and internal nodes representing *branching queries*. There are two types of internal nodes: *x-nodes* (containing a point p , branching left if the query has $x < p.x$) and *y-nodes* (containing a segment $e = (p_1, p_2)$, branching left if the query is above e and right otherwise). A root-to-leaf path encodes a sequence of decisions that identifies exactly one trapezoid.

Definition 13.36 (*x-Node*). An *x-node* stores a point $p = (p.x, p.y)$. For a query point q , the search proceeds to the left child if $q.x < p.x$ and to the right child otherwise. An *x-node* corresponds to an endpoint of a segment.

Definition 13.37 (*y-Node*). A *y-node* stores a segment $e = (p_1, p_2)$ with $p_1.x < p_2.x$. For a query point q , the search proceeds to the left child if q lies above e (i.e., the orientation of (p_1, p_2, q) is counter-clockwise) and to the right child otherwise. A *y-node* corresponds to a segment in S .

13.10.2 Theory

Construction

The trapezoidal map is built via a *randomized incremental algorithm*. Starting with a single bounding trapezoid that encompasses the entire input, the segments $s_1, s_2, \dots, s_n$ are inserted in a uniformly random order. When a new segment s_k is inserted, the existing trapezoids intersected by s_k are located by walking along the trapezoid adjacency graph. Each intersected trapezoid is split into at most four new trapezoids, and the DAG is updated accordingly. Because the insertion order is random, the expected depth of the DAG is $O(\log n)$.

The DAG is a *dag* (directed acyclic graph), not a tree, because multiple internal nodes can share the same child. This sharing is critical: when a segment is inserted and splits an existing trapezoid, the new *x-node* and *y-node* are wired into the DAG at the position of the old leaf, so other paths to that leaf are implicitly updated without duplicating the subtree.

Query Algorithm

To locate a query point q , we descend from the root of the DAG:

1. At an *x-node* with point p : go left if $q.x < p.x$, right otherwise.
2. At a *y-node* with segment $e = (p_1, p_2)$: go left if q is above e (counter-clockwise orientation), right otherwise.
3. At a leaf: return the trapezoid stored at that leaf.

The above-below test is performed using the orientation predicate (cross product):

$$\text{IsAbove}(q, (p_1, p_2)) \iff (p_2.x - p_1.x)(q.y - p_1.y) - (p_2.y - p_1.y)(q.x - p_1.x) > 0.$$

Theorem 13.38 (Trapezoidal Map Complexity). *Let S be a set of n non-intersecting line segments in $\mathbb{R}^2$. The randomized incremental trapezoidal map construction with a DAG search structure has expected preprocessing time $O(n \log n)$ and expected space $O(n)$. The expected query time to locate a point in the trapezoidal map is $O(\log n)$, where the expectation is over the random insertion order.*

Proof. We analyze the expected size of the search structure using backwards analysis. Consider the DAG after k segments have been inserted. For any trapezoid T that exists after all n segments are inserted, T was created at some step $k < n$ when the last segment that affects T was inserted. The probability that a particular segment among $\{s_1, \dots, s_k\}$ is the last to affect T is at most $2/k$ (since T is bounded by at most two segments, and any one of them could be the last inserted among the k). Therefore, the expected number of trapezoids created at step k is at most 2. Summing over $k = 1, \dots, n$, the total expected number of trapezoids (and hence DAG nodes) is $O(n)$.

The expected depth of a leaf in the DAG is bounded by the expected number of segments whose x -nodes or y -nodes lie on the root-to-leaf path. By the same backwards analysis, the expected depth of the deepest leaf is $O(\log n)$. Since each level of the DAG is traversed in $O(1)$ time, the expected query time is $O(\log n)$.

The expected total work across all n insertions is $O(n \log n)$ by an amortized argument: at step k , the new segment intersects $O(1)$ expected trapezoids, each of which is split in $O(1)$ time, and the DAG update is $O(1)$ per affected node. $\square$

Example 13.39 (Trapezoidal Map Construction). Consider two non-intersecting segments $s_1 = \{(1, 1), (3, 3)\}$ and $s_2 = \{(2, 0), (4, 2)\}$ inside a bounding box $[0, 5] \times [0, 5]$.

1. **Initialization:** The bounding trapezoid T_0 spans the entire box. The DAG root is a leaf containing T_0 .
2. **Insert s_1 :** The segment s_1 splits T_0 into trapezoids above and below it. A y -node for s_1 is created at the root. Its left child is a leaf for the trapezoid above s_1 ; its right child is a leaf for the trapezoid below s_1 .
3. **Insert s_2 :** The segment s_2 crosses through the lower trapezoid (the right child of the s_1 y -node). That trapezoid is split: new x -nodes for the endpoints $(2, 0)$ and $(4, 2)$ are inserted, and additional y -nodes for s_2 are created. The upper trapezoid (above s_1) is unaffected.
4. **Query for $q = (3, 2.5)$:** Starting at the root (y -node for s_1): Is $(3, 2.5)$ above s_1 ? Cross product: $(3-1)(2.5-1) - (3-1)(3-1) = 2 \cdot 1.5 - 2 \cdot 2 = -1 < 0$, so q is below s_1 : go right. Descend through the DAG nodes for the lower region until a leaf is reached, identifying the trapezoid containing q .

13.10.3 Pseudocode

The following pseudocode is adapted from the `TrapezoidalMap` class in the source implementation.

13.10.4 Parameters

- **Bounding Box:** A sufficiently large bounding box must be chosen to contain all segments and query points. The bounding box defines the initial trapezoid and the vertical extent of the map.
- **Precision:** The orientation predicate `IsAbove()` uses a cross-product computation. For inputs with nearly collinear segments, a robust predicate with an epsilon tolerance is recommended to avoid numerical instability.
- **Randomization:** The insertion order of segments should be uniformly random to guarantee the expected $O(\log n)$ query depth. A poor (adversarial) ordering can lead to $O(n)$ worst-case depth.

Algorithm 63 Trapezoidal Map Query

```

1: function LOCATE(root, q)
2:   curr  $\leftarrow$  root
3:   while curr is not a leaf do
4:     if curr is an x-node then
5:       p  $\leftarrow$  curr.value  $\triangleright$  a point
6:       if q.x < p.x then
7:         curr  $\leftarrow$  curr.left_child
8:       else
9:         curr  $\leftarrow$  curr.right_child
10:      end if
11:    else  $\triangleright$  y-node
12:      e  $\leftarrow$  curr.value  $\triangleright$  a segment (p1, p2)
13:      if IsAbove(q, e) then
14:        curr  $\leftarrow$  curr.left_child
15:      else
16:        curr  $\leftarrow$  curr.right_child
17:      end if
18:    end if
19:  end while
20:  return curr.value  $\triangleright$  the containing trapezoid
21: end function
22: function ISABOVE(q, e = (p1, p2))
23:  return (p2.x - p1.x) · (q.y - p1.y) - (p2.y - p1.y) · (q.x - p1.x) > 0
24: end function

```

Algorithm 64 Trapezoidal Map Construction (Incremental)

```

1: function TRAPEZOIDALMAP(bbox)
2:   p1, p2  $\leftarrow$  bbox  $\triangleright$  bounding box corners
3:   T0  $\leftarrow$  new Trapezoid(p1, p2, top-edge, bottom-edge)
4:   root  $\leftarrow$  new node containing T0
5:   T0.node  $\leftarrow$  root
6:   return root
7: end function
8: function INSERTSEGMENT(root, sk)
9:   Tstart  $\leftarrow$  Locate(root, sk)
10:  Walk from Tstart to the right along adjacent trapezoids intersected by sk
11:  for each intersected trapezoid T do
12:    Split T into at most four new trapezoids using sk
13:    Replace the leaf for T in the DAG with a new subtree of x-nodes and y-nodes
14:  end for
15: end function

```

- **Segment Validity:** The input segments must be non-intersecting (except possibly at shared endpoints). Intersecting segments must be split at their intersection points before building the trapezoidal map.

13.10.5 Complexity

- **Preprocessing Time:** $O(n \log n)$ expected, where n is the number of segments. Each insertion processes $O(1)$ expected trapezoids, and the walk to find intersected trapezoids is amortized $O(1)$ per segment.
- **Space:** $O(n)$ expected. The DAG stores $O(n)$ nodes (both internal and leaf), and each trapezoid stores $O(1)$ data plus adjacency pointers.
- **Query Time:** $O(\log n)$ expected. The DAG depth is $O(\log n)$ by backwards analysis, and each level is processed in $O(1)$ time.
- **Worst Case:** $O(n)$ query time and $O(n^2)$ space in the worst case (adversarial input ordering), though this is extremely unlikely with randomized construction.

13.10.6 Usages

- **Planar Subdivision Queries:** Determining which region of a Voronoi diagram, triangulation, or map overlay contains a given point.
- **Polygon Operations:** Point location is a subroutine in polygon clipping, union, and intersection algorithms (e.g., Weiler–Atherton).
- **Mesh Processing:** Finding which triangle or tetrahedron contains a query point in finite element analysis and computer graphics.
- **GIS and Cartography:** Identifying which administrative region, soil type, or land-use zone a GPS coordinate falls within.
- **Visibility and Ray Tracing:** Determining the first surface hit by a ray in a 2D scene represented as a planar subdivision.
- **Robotics:** Localization within a known map by matching sensor observations to map regions.

13.10.7 Testing and Edge Cases

- **Point on Segment Boundary:** The orientation predicate may return zero for points exactly on a segment. The implementation should define a consistent convention (e.g., treat collinear as “not above”).
- **Point on Vertical Wall:** Queries on the vertical boundary between two trapezoids should consistently return one of the two trapezoids.
- **Empty Map:** A map with no segments should have a single bounding trapezoid; all queries should return that trapezoid.
- **Single Segment:** The map should contain exactly two trapezoids (above and below the segment) inside the bounding box.
- **Degenerate Segments:** Horizontal or vertical segments should be handled without creating zero-width trapezoids.

- **Collinear Endpoints:** Multiple segments sharing an endpoint should produce consistent x -node ordering in the DAG.
- **Large Input:** Verify that the expected $O(\log n)$ query depth holds empirically for $n > 10,000$ segments by measuring average path length.

13.10.8 References

- Mulmuley, K. (1990). “A fast planar partition algorithm, I”. *Journal of Symbolic Computation*.
- Seidel, R. (1991). “A simple and fast incremental randomized algorithm for computing trapezoidal decompositions and for point location”. *Computational Geometry: Theory and Applications*.
- Berg, M. de, Cheong, O., Kreveld, M. van, & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer-Verlag.
- Guibas, L. J., & Sedgwick, R. (1983). “A dichromatic framework for balanced trees”. *IEEE FOCS*.
- Wikipedia: Point location.

13.11 Point Location in Triangulations (Walking)

13.11.1 Overview

Given a triangulation or tetrahedral mesh with billions of elements, the **walking** method locates the simplex containing a query point by moving across edges (2D) or faces (3D) from a starting simplex toward the query point. Unlike the trapezoidal map of Section 13.10 (see also the dedicated point location chapter, Section 8.8), which is restricted to planar subdivisions, walking works directly on a mesh in any dimension and needs no auxiliary search structure beyond the mesh adjacency itself. It is the method of choice inside incremental Delaunay construction (Section 12.5 and 12.4) and for FEM and particle codes that repeatedly query a fixed mesh.

13.11.2 Definitions

Definition 13.40 (Visibility Walk). A *visibility walk* from a starting simplex σ_0 toward query point q repeatedly crosses the facet of the current simplex that is *visible* from q : among the facets whose separating hyperplane leaves q in the opposite half-space, move across the first such facet found. The walk terminates at a simplex containing q .

Definition 13.41 (Jump-and-Walk). A *jump-and-walk* first selects a starting simplex σ_0 using a spatial index (a grid, k-d tree, or Delaunay tree) that returns a simplex close to q , then walks from σ_0 . The jump makes the walk length shorter in expectation than a walk from a fixed or random start.

Definition 13.42 (Straight Walk). A *straight walk* from σ_0 moves, at every step, across the facet intersected by the line segment from the current position to q . It tends to take fewer steps than a visibility walk but can fail on non-convex or non-Delaunay meshes; the visibility walk is robust on any triangulation.

13.11.3 Theory

Theorem 13.43 (Visibility Walk Terminates). *A visibility walk terminates at a simplex containing q on any triangulation of a convex domain. On a Delaunay triangulation it terminates on any simple domain [Dev02].*

Proof sketch. Each crossing strictly decreases the distance from the current simplex to q in the sense of the supporting lines of the crossed facets: the walk cannot cross the same facet twice, because the crossed hyperplanes separate q from the already visited simplices. Finiteness follows from the finite number of facets. The Delaunay case follows because every vertex sees at most one successive facet, which guarantees progress on non-convex domains as well [Dev02]. $\square$

Theorem 13.44 (Expected Walk Length). *On a Delaunay triangulation of n points, the expected number of steps of a walk starting at a random simplex is $O(\sqrt{n})$; with a jump-and-walk whose starting simplex is chosen near q by a grid or k -d tree, the expected number of steps is $O(\log n)$ for uniformly distributed points [Dev02].*

Proof sketch. The expected $O(\sqrt{n})$ bound follows from the fact that the walk crosses $O(\sqrt{n})$ cells when n points are distributed over a bounded region: the number of simplices crossed is bounded by the number of simplices intersected by a segment, which is $O(\sqrt{n})$ in expectation for the Delaunay triangulation of a uniform set. A spatial-index jump reduces the distance from the start to the query point, which reduces the expected crossing count to $O(\log n)$ [Dev02]. $\square$

13.11.4 Pseudocode

Algorithm 65 Locate in a Triangulation by Visibility Walk

```

1: function LOCATEWALK(mesh, q, start)
2:    $\sigma \leftarrow \text{start}$ 
3:   while true do
4:      $\text{facet} \leftarrow$  first facet of  $\sigma$  such that  $q$  and  $\sigma \setminus \text{facet}$  lie on opposite sides of  $\text{aff}(\text{facet})$ 
5:     if no such facet exists then  $\triangleright q$  is inside  $\sigma$ 
6:       return  $\sigma$ 
7:     end if
8:      $\sigma \leftarrow$  neighbor of  $\sigma$  across  $\text{facet}$   $\triangleright$  cross the visible facet
9:   end while
10: end function

11: function JUMPANDWALK(mesh, q, grid)
12:    $\text{cells} \leftarrow \text{NearestCandidateCells}(q, r, \delta)$   $\triangleright$  grid jump, Section 10.7
13:    $\sigma_0 \leftarrow$  a simplex intersecting a cell in  $\text{cells}$ , closest to  $q$ 
14:   return LOCATEWALK(mesh, q,  $\sigma_0$ )
15: end function

```

The 3D case is identical with *facets* replaced by the four triangular faces of a tetrahedron: cross the face whose plane separates q from the tetrahedron's opposite vertex. In 2D the facet is the single edge separating q from the opposite vertex.

13.11.5 Parameter Selection

- **Start simplex:** for Delaunay construction with spatially sorted (Hilbert or Morton) insertion, reuse the simplex found for the previous point; for queries, use a grid or k -d tree jump.

- **Robust predicates:** use adaptive orientation tests (`orient2d`, `orient3d`) to avoid looping near degeneracies [She97].
- **Tie-breaking:** when q lies exactly on a facet, pick one adjacent simplex consistently to avoid infinite loops.

13.11.6 Complexity

- **Time:** expected $O(\sqrt{n})$ steps from a random start; $O(\log n)$ with a grid or k-d tree jump on uniform data.
- **Space:** $O(1)$ beyond the mesh itself; the grid jump adds $O(n)$.
- **No preprocessing:** unlike the trapezoidal map, walking needs no auxiliary DAG; it uses only the mesh adjacency.

13.11.7 Usages

- **Incremental Delaunay construction:** locating the triangle containing each inserted point [Bow81, Wat81].
- **FEM post-processing:** reporting which element contains a query point or interpolation target.
- **Ray tracing and probing** of volumetric (tetrahedral) meshes.

13.11.8 Testing and Edge Cases

- **Termination:** compare against the trapezoidal map location (Section 13.10) for 2D on the same mesh; results must agree.
- **Boundary points:** queries exactly on an edge or vertex must return a consistent simplex.
- **Non-convex meshes:** use the visibility walk (not the straight walk) and verify termination on meshes with concavities.
- **Large input:** measure average walk length over n queries and confirm it matches the theoretical $O(\log n)$ or $O(\sqrt{n})$ growth.
- **Counterexample checks:** the straight walk can fail on non-Delaunay meshes; assert the visibility walk is always used there.

13.11.9 References

- Devillers, O. (2002). “The Delaunay hierarchy”. International Journal of Foundations of Computer Science.
- Bowyer, A. (1981). “Computing Dirichlet tessellations”. The Computer Journal.
- Watson, D. F. (1981). “Computing the n -dimensional Delaunay tessellation”. The Computer Journal.
- Shewchuk, J. R. (1997). “Adaptive precision floating-point arithmetic and fast robust geometric predicates”. Discrete & Computational Geometry.

Part V

Arrangements and Duality

Chapter 14

Arrangements of Lines and Curves

14.1 Davenport–Schinzel Sequences

14.1.1 Overview

A Davenport–Schinzel sequence is a sequence of symbols that satisfies specific constraints on the alternating sub-sequences it can contain. These sequences provide a powerful combinatorial framework for analyzing the complexity of the **lower envelope** of a set of functions. In computational geometry, they are used to bound the complexity of arrangements of curves, motion planning visibility, and dynamic geometric structures. The theory originated with Davenport and Schinzel [DS65] and was developed into a comprehensive geometric theory by Sharir and Agarwal [SA95].

14.1.2 Definitions

Definition 14.1 (Davenport–Schinzel Sequence). A sequence $U = (u_1, u_2, \dots, u_m)$ over an alphabet A of n symbols is a **Davenport–Schinzel sequence of order s** (denoted $DS(n, s)$) if:

1. No two adjacent symbols are the same: $u_i \neq u_{i+1}$ for all $1 \leq i < m$.
2. It does not contain any alternating sub-sequence of length $s + 2$ of the form $a, b, a, b, \dots$ (alternating $s + 1$ times between two distinct symbols $a, b \in A$).

Definition 14.2 ($\lambda_s(n)$). $\lambda_s(n)$ denotes the maximum possible length of a Davenport–Schinzel sequence of order s on n distinct symbols. These functions grow near-linearly for any fixed s :

$$\lambda_0(n) = 1, \quad \lambda_1(n) = n, \quad \lambda_2(n) = 2n - 1, \quad \lambda_3(n) = \Theta(n \alpha(n)),$$

where $\alpha(n)$ is the inverse Ackermann function.

Definition 14.3 (Inverse Ackermann Function). The **inverse Ackermann function** $\alpha(n)$ is the function such that $\alpha(n) \leq 4$ for all practical values of n (up to $n = 10^{80}$, the number of atoms in the observable universe). Formally, $\alpha(n)$ is the inverse of the Ackermann function $A(k)$, which grows faster than any primitive recursive function.

Definition 14.4 (Lower Envelope). The **lower envelope** of a collection of n functions $\{f_1, \dots, f_n\}$ defined on a common domain D is the function $L(x) = \min_{1 \leq i \leq n} f_i(x)$. The *complexity* of the lower envelope is the number of maximal continuous pieces that compose it.

14.1.3 Theory

The importance of DS sequences stems from their relationship to function intersections. If we have n continuous functions such that any two functions intersect at most s times, then the sequence of function indices that form the **lower envelope** is a Davenport–Schinzel sequence of order s .

For example:

- **Order** $s = 1$: Functions intersect at most once (e.g., lines). $\lambda_1(n) = n$.
- **Order** $s = 2$: Functions intersect at most twice (e.g., parabolas). $\lambda_2(n) = 2n - 1$.
- **Order** $s = 3$: Functions intersect at most three times. $\lambda_3(n) = \Theta(n\alpha(n))$, where $\alpha(n)$ is the extremely slow-growing inverse Ackermann function.

This theory allows us to prove that the complexity of the lower envelope of n line segments is $O(n\alpha(n))$, which is much smaller than the total number of intersections $O(n^2)$.

Theorem 14.5 (DS Sequence Length Bounds). *For $s \geq 1$ and $n \geq 1$:*

1. $\lambda_1(n) = n$.
2. $\lambda_2(n) = 2n - 1$.
3. For $s \geq 3$, $\lambda_s(n) = n \cdot \alpha(n)^{O(1)}$, where $\alpha(n)$ is the inverse Ackermann function. More precisely, $\lambda_3(n) = 2n\alpha(n) + O(n)$.

These bounds are tight.

Proof. The cases $s = 1$ and $s = 2$ are elementary. For $s \geq 3$, the upper bound was established by Sharir [SA95] using a charging argument on the “blocks” of the sequence and the properties of the Ackermann hierarchy. The matching lower bound is obtained by an explicit construction based on the composition sequences of the Ackermann function. See [SA95, Chapters 2–4] for the complete proof. $\square$

Observation 14.6. The growth rate of $\lambda_s(n)$ for increasing s reflects the increasing complexity of the lower envelope as the number of allowed intersections between function pairs grows. For $s = 1$ (lines), the envelope has $O(n)$ complexity. For $s = 3$ (line segments), the complexity jumps to $O(n\alpha(n))$ —a subtlety that arises from the non-monotonic behavior of segments.

14.1.4 Pseudocode

Computing the Lower Envelope (Recursive)

14.1.5 Parameter Selection

- **Order** (s): Depends on the type of geometric primitives (e.g., $s = 1$ for lines, $s = 2$ for circles, $s = 3$ for line segments).
- **Alphabet Size** (n): The number of geometric objects.

14.1.6 Complexity

- **Maximum Length:** $\lambda_s(n)$ is near-linear for any fixed s . For example, $\lambda_3(n) = O(n\alpha(n))$.
- **Construction Time:** The lower envelope can be computed in $O(\lambda_s(n) \log n)$ time using a divide-and-conquer approach.

```

1: function LOWERENVELOPE(functions, start, end)
2:   if |functions| = 1 then
3:     return functions[0]
4:   end if
5:   mid  $\leftarrow$  |functions|/2
6:   left_env  $\leftarrow$  LowerEnvelope(functions[: mid], start, end)
7:   right_env  $\leftarrow$  LowerEnvelope(functions[mid :], start, end)
8:   merged_env  $\leftarrow$  MergeEnvelopes(left_env, right_env)
9:   return merged_env
10: end function
11: function MERGEENVELOPES( $E1$ ,  $E2$ )
12:   intersections  $\leftarrow$  FindIntersections( $E1$ ,  $E2$ )
13:   return ResultingSequence( $E1$ ,  $E2$ , intersections)
14: end function

```

14.1.7 Usages

- **Arrangements:** Bounding the number of edges on a single cell or the lower envelope of an arrangement of curves.
- **Motion Planning:** Analyzing the complexity of the “free space” for a robot moving among obstacles.
- **Visibility:** Calculating the complexity of the region visible from a point in a dynamic environment.
- **Kinetic Data Structures:** Tracking the minimum value among a set of moving points.
- **Shortest Paths:** Computing the visibility graph of a set of segments.

14.1.8 Testing Methods and Edge Cases

- **Non-Intersecting Functions:** The envelope should simply follow the lowest function for the entire range.
- **Multiple Intersections:** Ensure the algorithm correctly captures all points where the lower envelope switches from one function to another.
- **Touching vs. Crossing:** Handle cases where functions are tangent to each other without crossing.
- **Vertical Segments:** Ensure the interval logic handles vertical discontinuities.
- **Complexity Check:** Verify that the number of segments in the resulting envelope is within the DS bound $\lambda_s(n)$.

14.1.9 References

- Davenport, H., & Schinzel, A. (1965). “A combinatorial problem connected with differential equations”. *American Journal of Mathematics*.
- Sharir, M., & Agarwal, P. K. (1995). “Davenport–Schinzel Sequences and Their Geometric Applications”. Cambridge University Press.
- Atallah, M. J. (1985). “Some dynamic computational geometry problems”. *Computers & Mathematics with Applications*.

- Wikipedia: [Davenport–Schinzel sequence](#)

14.2 Lower Envelopes

14.2.1 Overview

The lower envelope of a set of functions $f_1, f_2, \dots, f_n$ is the pointwise minimum of those functions: $L(x) = \min_i f_i(x)$. In computational geometry, the lower envelope is a fundamental structure used to solve problems related to visibility, optimization, and arrangements. For example, the lower envelope of a set of lines forms the boundary of their intersection (a convex polygon in the dual plane). The computational aspects of envelopes were studied extensively by Edelsbrunner [Ede87] and Hershberger [Her89].

14.2.2 Definitions

Definition 14.7 (Lower Envelope). The **lower envelope** of a family of functions $\{f_1, \dots, f_n\}$ on a domain $D \subseteq \mathbb{R}$ is the function $L : D \rightarrow \mathbb{R}$ defined by $L(x) = \min_{1 \leq i \leq n} f_i(x)$. The envelope is piecewise composed of segments of the individual functions.

Definition 14.8 (Upper Envelope). The **upper envelope** of a family of functions $\{f_1, \dots, f_n\}$ is $U(x) = \max_{1 \leq i \leq n} f_i(x)$. The upper envelope of a set of functions is the lower envelope of their negations.

Definition 14.9 (Minimization Diagram). A **minimization diagram** is the partition of the domain D into maximal regions where a specific function f_i achieves the minimum. Each region is called a *cell* of the diagram, and the boundary between two cells is formed by the intersection points of the corresponding functions.

Definition 14.10 (Complexity of an Envelope). The **complexity** of a lower envelope is the number of maximal continuous pieces (arcs) that compose it. Each arc is a maximal connected subset of the domain on which a single function f_i achieves the pointwise minimum.

14.2.3 Theory

The complexity of the lower envelope depends on how many times any two functions can intersect.

1. **Lines:** Any two lines intersect at most once. The lower envelope of n lines is a convex chain with at most n segments.
2. **Line Segments:** Two segments can intersect at most once, but the envelope can have up to $O(n\alpha(n))$ segments due to the “visibility” of segments behind each other.
3. **Parabolas/Circles:** Two functions intersect at most twice. The envelope complexity is $O(n)$.
4. **General Curves:** If any two curves intersect at most s times, the complexity is bounded by the Davenport–Schinzel sequence $\lambda_s(n)$.

The lower envelope can be computed using a **Divide and Conquer** algorithm:

1. Divide the set of functions into two halves.
2. Recursively compute the lower envelope of each half.
3. Merge the two envelopes by finding their intersection points and keeping the lower segments.

Theorem 14.11 (Lower Envelope Complexity for Lines). *The lower envelope of n lines in $\mathbb{R}^2$ consists of at most n segments and can be computed in $O(n \log n)$ time.*

Proof. Any two distinct lines intersect in at most one point. By the DS sequence framework with $s = 1$, the envelope is a $DS(n, 1)$ sequence, which has length at most n . The divide-and-conquer algorithm solves the recurrence $T(n) = 2T(n/2) + O(n)$, giving $T(n) = O(n \log n)$. $\square$

Theorem 14.12 (Lower Envelope Complexity for Segments). *The lower envelope of n line segments in $\mathbb{R}^2$ has $O(n \alpha(n))$ complexity, where $\alpha(n)$ is the inverse Ackermann function, and can be computed in $O(n \alpha(n) \log n)$ time.*

Proof. Any two line segments intersect in at most one point (since they are linear), so the envelope is a DS sequence of order $s = 3$ (segments can cross, touch, or miss). The DS bound $\lambda_3(n) = O(n \alpha(n))$ gives the complexity upper bound. The divide-and-conquer algorithm merges two sub-envelopes in time proportional to their combined complexity times $\log n$, yielding $O(n \alpha(n) \log n)$ total time. See Edelsbrunner [Eds87, Chapter 4]. $\square$

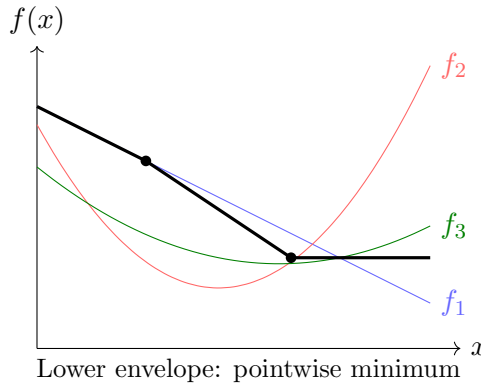


Figure 14.1: Lower envelope of three functions: the thick black curve is the pointwise minimum $\min\{f_1, f_2, f_3\}$.

Example 14.13 (Lower Envelope of Lines). Consider four lines: $f_1(x) = 2 - x$, $f_2(x) = -1 + 2x$, $f_3(x) = 3 - 0.5x$, $f_4(x) = x - 2$. The lower envelope is:

$$L(x) = \min(2 - x, -1 + 2x, 3 - 0.5x, x - 2)$$

On $(-\infty, 0)$: $f_4(x) = x - 2$ is the minimum. On $[0, 1]$: $f_1(x) = 2 - x$ is minimum. On $[1, 2]$: $f_2(x) = -1 + 2x$ is minimum. On $(2, \infty)$: $f_4(x) = x - 2$ is minimum again. The envelope has 4 pieces, which does not exceed $n = 4$.

14.2.4 Pseudocode

14.2.5 Parameter Selection

- **Function Type:** Linear, polynomial, or arbitrary continuous functions.
- **Intersection Solver:** A robust numerical or symbolic solver is needed to find where $f_i(x) = f_j(x)$.

14.2.6 Complexity

- **Complexity of Envelope:** $O(\lambda_s(n))$ where s is the number of intersections.
- **Construction Time:** $O(\lambda_s(n) \log n)$ using divide and conquer.
- **Space Complexity:** $O(\lambda_s(n))$ to store the resulting segments.

```
1: function COMPUTELOWERENVELOPE(functions,  $x_{\min}$ ,  $x_{\max}$ )
2:   if |functions| = 1 then
3:     return [(functions[0],  $x_{\min}$ ,  $x_{\max}$ )]
4:   end if
5:   mid  $\leftarrow$  |functions|/2
6:   left_env  $\leftarrow$  ComputeLowerEnvelope(functions[: mid],  $x_{\min}$ ,  $x_{\max}$ )
7:   right_env  $\leftarrow$  ComputeLowerEnvelope(functions[mid :],  $x_{\min}$ ,  $x_{\max}$ )
8:   merged  $\leftarrow$  []
9:   for segment_L in left_env do
10:    for segment_R in right_env do
11:      common_start  $\leftarrow$  max(segment_L.start, segment_R.start)
12:      common_end  $\leftarrow$  min(segment_L.end, segment_R.end)
13:      if common_start < common_end then
14:        intersections  $\leftarrow$  FindIntersections(segment_L.f, segment_R.f, common_start, common_end)
15:        merged.extend(PickLower(segment_L.f, segment_R.f, common_start, common_end, intersections))
16:      end if
17:    end for
18:  end for
19:  return Simplify(merged)
20: end function
```

14.2.7 Usages

- **Visibility:** Calculating the “horizon” of a set of mountains or the region visible from a point.
- **Motion Planning:** Finding the safest path by maximizing the distance to the nearest obstacle (lower envelope of distance functions).
- **Duality:** Solving problems like the “Width of a Point Set” or “Diameter” in the dual plane.
- **Dynamic Programming:** Optimizing decisions over time where costs are modeled as functions.
- **Statistics:** Calculating the “support” of a probability distribution.

14.2.8 Testing Methods and Edge Cases

- **Non-Intersecting Functions:** The envelope should simply be the single function that is globally minimum.
- **Vertical Functions:** Ensure the algorithm handles discontinuities or vertical lines.
- **Touching Functions:** Verify that tangencies (where functions touch but don’t cross) don’t create unnecessary segments.
- **Overlap:** Handle cases where two functions are identical over an interval.
- **Large n :** Verify the near-linear growth of the result size.

14.2.9 References

- Sharir, M., & Agarwal, P. K. (1995). “Davenport–Schinzel Sequences and Their Geometric Applications”. Cambridge University Press.

- Hershberger, J. (1989). “Finding the upper envelope of n line segments in $O(n \log n)$ time”. Information Processing Letters.
- Edelsbrunner, H. (1987). “Algorithms in Combinatorial Geometry”. Springer-Verlag.
- Wikipedia: [Lower envelope](#)

14.3 Levels in Arrangements and Half-Plane Discrepancy

14.3.1 Overview

In an arrangement of lines, every point not on a line has a well-defined *level*: the number of lines strictly below it. The set of points of a fixed level is a polygonal chain called the **k-level**, which generalizes the lower envelope (level 0) and the upper envelope (level $n - 1$). Levels are the bridge between arrangements and point sets: by duality, the vertices of the k -level of an arrangement of lines correspond exactly to the **k-sets** of the dual point set—subsets of size k separable from the rest by a line. Levels also control the **half-plane discrepancy** of a point set, a measure of how well a set of n points approximates the uniform distribution that arises, for example, in the supersampling analysis of ray tracing [SA95].

14.3.2 Definitions

Definition 14.14 (Level of a Point). In an arrangement $\mathcal{A}(\mathcal{L})$ of n lines, the *level* of a point p that lies on no line is the number of lines of $\mathcal{L}$ strictly below p . The k -level is the set of points whose level is exactly k ; it is an x -monotone polygonal chain. The *complexity* of the k -level is its number of vertices.

Definition 14.15 (k-Set). A k -set of a point set P is a subset $K \subseteq P$ with $|K| = k$ that can be separated from $P \setminus K$ by a line. Under point–line duality, the k -sets of P correspond one-to-one with the vertices of the k -th level of the arrangement of lines dual to P .

Definition 14.16 (Half-Plane Discrepancy). Let P be a set of n points inside a region B of area 1. The *half-plane discrepancy* of P is

$$D(P) = \max_{h \text{ half-plane}} \left| |P \cap h| - n \cdot \text{area}(h \cap B) \right|.$$

The discrepancy $D(n)$ is the minimum of $D(P)$ over all n -point sets P .

14.3.3 Theory

The lower envelope studied in Section 14.2 is precisely the 0-level, and the upper envelope is the $(n - 1)$ -level. The complexity of intermediate levels is the central open problem of the area; the best known results are as follows.

Theorem 14.17 (Complexity of the k-Level). *The complexity of the k -th level in an arrangement of n lines is $O(nk^{1/3})$ for every k [Dey98]; for the middle level ($k \approx n/2$) this is $O(n^{4/3})$. The best known lower bound is $n 2^{\Omega(\sqrt{\log k})}$.*

Proof sketch. The older bound $O(n\sqrt{k})$ of Lovász follows by a potential-function argument: sweep the arrangement and charge the $\Omega(\sqrt{k})$ left-turning vertices of the level. Dey’s improvement applies the crossing lemma to the union of k x -monotone concave chains that cover the region below the level, yielding the $O(nk^{1/3})$ bound [Dey98]. $\square$

The connection between levels and discrepancy is that a half-plane whose incidence count deviates strongly from the area expectation must contain many vertices of a level in the dual arrangement; hence a low-discrepancy point set forces every level to be simple.

Theorem 14.18 (Half-Plane Discrepancy of n Points). *The half-plane discrepancy of every set of n points is at least $cn^{1/4}$, and there exist n -point sets (such as suitable lattices) whose half-plane discrepancy is $O(n^{1/4})$. In particular $D(n) = \Theta(n^{1/4})$.*

Proof sketch. The lower bound is due to Beck and Alexander using Fourier-analytic (integral-geometric) techniques [Mat99]. The matching upper bound is due to Matoušek, who constructed point sets—related to scaled integer lattices—for which every half-plane counts the expected number of points up to an error of $O(n^{1/4})$ [Mat95, Mat99]. $\square$

14.3.4 Pseudocode

A simple incremental construction of the k -level of an arrangement of n lines uses the duality between levels and k -sets.

Algorithm 66 Compute the k -Level of a Line Arrangement

```

1: function COMPUTEKLEVEL( $lines, k$ )
2:    $\mathcal{A} \leftarrow \text{LineArrangement}(lines)$ 
3:    $level \leftarrow 0$ 
4:    $p \leftarrow$  point at  $x = -\infty$  on the lowest line of  $\mathcal{A}$ 
5:   while  $x \neq +\infty$  do
6:     if  $level = k$  then
7:       append the current edge to the output chain
8:     end if
9:     advance to the next vertex of  $\mathcal{A}$ 
10:     $level \leftarrow level \pm 1$  ▷ crossing a line
11:  end while
12:  return the output chain
13: end function

```

14.3.5 Parameter Selection

- **Level k :** choose $k = 0$ for the lower envelope and $k = n - 1$ for the upper envelope; intermediate k give the k -sets.
- **Duality:** for discrepancy problems, dualize the point set to lines and operate on levels of the arrangement.

14.3.6 Complexity

- **Level Complexity:** $O(nk^{1/3})$ vertices (Dey); the middle level has $O(n^{4/3})$ complexity.
- **Discrepancy:** $D(n) = \Theta(n^{1/4})$ for half-planes.
- **Construction Time:** $O(n^2)$ for a full arrangement; a single level can be traced in time proportional to its complexity plus the sorting of the lines.

14.3.7 Usages

- **Ray tracing:** bounding the discrepancy of sampling patterns for anti-aliased rendering.
- **k -sets:** counting the number of subsets of size k separable by a line (used in statistical classification and pattern recognition).

- **Geometric data analysis:** measuring how evenly a point set approximates a uniform distribution over a region.
- **Range counting:** levels dualize to half-plane range counting, connecting arrangements to the simplex range searching chapter.

14.3.8 Testing Methods and Edge Cases

- **Parallel lines:** the k -level may be empty or unbounded.
- **Multiple lines through one point:** perturb or handle coincident vertices consistently.
- **Grid construction:** verify that a $m \times m$ grid point set indeed achieves discrepancy $O(m^{1/2}) = O(n^{1/4})$.
- **Small n :** for $n = 4$, any set has a half-plane containing 3 of the points when all four are in convex position, matching the $n^{1/4}$ order.

14.3.9 References

- Dey, T. K. (1998). “Improved bounds on planar k -sets and related problems”. Discrete & Computational Geometry.
- Matoušek, J. (1995). “Tight upper bounds for the discrepancy of half-spaces”. Discrete & Computational Geometry.
- Matoušek, J. (1999). “Geometric Discrepancy: An Illustrated Guide”. Springer.
- Sharir, M., & Agarwal, P. K. (1995). “Davenport–Schinzel Sequences and Their Geometric Applications”. Cambridge University Press.

14.4 Union Volume Estimation

14.4.1 Overview

The union volume estimation problem, also known as **Klee’s measure problem**, involves calculating the total volume (or area in 2D) occupied by the union of n geometric objects (typically axis-aligned boxes). This is a fundamental challenge because the union can have a very complex topology even if the individual objects are simple. The problem was posed by Klee [Kle77], and significant algorithmic advances were made by Overmars and Yap [OY91] and Bringmann [Bri12]. While exact algorithms exist for boxes, estimation methods are often preferred for more complex shapes like spheres or non-convex meshes.

14.4.2 Definitions

Definition 14.19 (Union Measure). For a family of n measurable sets $\{O_1, \dots, O_n\}$ in $\mathbb{R}^d$, the **union measure** is

$$\mu\left(\bigcup_{i=1}^n O_i\right),$$

where μ denotes Lebesgue measure (length in 1D, area in 2D, volume in 3D).

Definition 14.20 (Monte Carlo Estimation). A **Monte Carlo estimator** for the volume of a region $\Omega \subset B$ (where B is a bounding box of known volume V_B) is

$$\hat{V} = V_B \cdot \frac{k}{N},$$

where N points are sampled uniformly at random from B and k of them fall inside Ω . The estimator is unbiased, with $\mathbb{E}[\hat{V}] = V(\Omega)$ and standard deviation $\sigma = V_B \sqrt{\frac{V(\Omega)(V_B - V(\Omega))}{N \cdot V_B^2}}$.

Definition 14.21 (Sweep-Line Algorithm). A **sweep-line algorithm** for union area computation processes events at all unique x -coordinates of vertical edges. Between consecutive events, the union area reduces to computing the union of 1D intervals along the y -axis.

14.4.3 Theory

Exact 2D Calculation (Sweep-Line)

For n axis-aligned rectangles, the exact area can be calculated in $O(n \log n)$ time:

1. Identify all unique x -coordinates of the vertical edges. These form a set of “strips.”
2. For each strip, find the total length of the union of 1D intervals along the y -axis (using a Segment Tree).
3. The area is the sum of (strip width $\times$ union length).

Monte Carlo Estimation (N -Dimensions)

For more complex or higher-dimensional shapes:

1. Enclose all objects in a large bounding box B with known volume V_B .
2. Sample N random points $P_1, \dots, P_N$ inside B .
3. For each point P_i , check if it lies inside **any** object O_j (using a spatial index like an R-tree for speed).
4. If k points are inside the union, the estimated volume is $\hat{V} = V_B \cdot \frac{k}{N}$.

Theorem 14.22 (Sweep-Line Union Complexity). *The area of the union of n axis-aligned rectangles in $\mathbb{R}^2$ can be computed in $O(n \log n)$ time and $O(n)$ space.*

Proof. The algorithm sweeps a vertical line from left to right across all $2n$ unique x -coordinates of vertical edges. At each event, it adds or removes a horizontal interval from a 1D interval set. The union length of the active intervals is maintained using a segment tree, supporting $O(\log n)$ insertions, deletions, and queries per event. With $O(n)$ events, the total time is $O(n \log n)$. Space is $O(n)$ for the interval set and event queue. See Klee [Kle77] for the original formulation. $\square$

Theorem 14.23 (Klee’s Measure Problem Bounds). *In d dimensions with $d \geq 3$, the exact measure of the union of n axis-aligned boxes can be computed in $O(n^{d/2})$ time for even d and $O(n^{(d+1)/2})$ time for odd d , improving upon the naive $O(n^d)$ approach.*

Proof. The result follows from the observation that in d dimensions, the complexity of the union of n boxes is $O(n^{\lfloor d/2 \rfloor})$ by the Davenport–Schinzel framework applied level by level. Overmars and Yap [OY91] achieved $O(n^{d/2})$ for even d , and Bringmann [Bri12] improved bounds for small fixed d . The algorithm uses a multi-level data structure that maintains partial unions across dimensions. $\square$

Example 14.24 (Monte Carlo Volume Estimation). Estimate the area of the union of two axis-aligned squares: $S_1 = [0, 2] \times [0, 2]$ (area 4) and $S_2 = [1, 3] \times [1, 3]$ (area 4), overlapping on $[1, 2] \times [1, 2]$ (area 1). The true union area is $4 + 4 - 1 = 7$.

Bounding box: $B = [0, 3] \times [0, 3]$, $V_B = 9$. Sample $N = 1000$ random points from B . Suppose $k = 778$ fall inside $S_1 \cup S_2$. Then $\hat{V} = 9 \cdot \frac{778}{1000} = 7.002$, which is close to the true value of 7.

14.4.4 Pseudocode

Monte Carlo Estimation

```

1: function ESTIMATEUNIONVOLUME(objects, num_samples)
2:   bbox  $\leftarrow$  Union([obj.bbox | obj  $\in$  objects])
3:   total_volume  $\leftarrow$  bbox.volume()
4:   index  $\leftarrow$  BuildRTree(objects)
5:   inside_count  $\leftarrow$  0
6:   for _ in range(num_samples) do
7:     p  $\leftarrow$  bbox.RandomPoint()
8:     if index.IsPointInsideAny(p) then
9:       inside_count  $\leftarrow$  inside_count + 1
10:    end if
11:  end for
12:  return total_volume  $\cdot$  (inside_count/num_samples)
13: end function

```

14.4.5 Parameter Selection

- **Sample Count (N):** Higher N leads to lower variance but increased computation. Accuracy improves as $1/\sqrt{N}$.
- **Bounding Box:** A “tight” bounding box reduces the number of samples that fall outside the union, improving efficiency.

14.4.6 Complexity

- **Exact 2D:** $O(n \log n)$ using sweep-line.
- **Exact 3D:** $O(n^{1.5})$ or $O(n \log n)$ with complex data structures.
- **Monte Carlo:** $O(N \cdot \log n)$ where N is samples and n is objects.

14.4.7 Usages

- **CAD/CAM:** Calculating the material required for a complex part built from boolean primitives.
- **Computational Biology:** Estimating the volume of a protein or a cellular structure modeled as a union of spheres (Van der Waals volume).
- **Robotics:** Calculating the total “reachable volume” of a robot arm.
- **GIS:** Determining the total area covered by a set of circular sensor ranges.
- **Computer Graphics:** Ambient occlusion and visibility calculations.

14.4.8 Testing Methods and Edge Cases

- **Disjoint Objects:** The union volume should be the sum of individual volumes.
- **Identical Objects:** The union volume should be equal to a single object’s volume.
- **Nested Objects:** Ensure that a small object inside a large one doesn’t add to the volume.

- **Analytical Shapes:** Test with spheres or boxes where the exact union volume can be calculated manually.
- **High Overlap:** Verify that the algorithm doesn't double-count overlapping regions.

14.4.9 References

- Klee, V. (1977). "Can the measure of $\cup[a_i, b_i]$ be computed in less than $O(n \log n)$ steps?". American Mathematical Monthly.
- Overmars, M. H., & Yap, C. K. (1991). "New upper bounds in Klee's measure problem". SIAM Journal on Computing.
- Bringmann, K. (2012). "New upper bounds for Klee's measure problem in small dimensions". Computational Geometry.
- Wikipedia: [Klee's measure problem](#)

Chapter 15

Geometric Duality

15.1 Point-Line Duality

Geometric duality is a systematic correspondence that swaps the roles of points and lines (in the plane) or, more generally, of points and hyperplanes (in $\mathbb{R}^d$), while *preserving incidence and order*. Its value is algorithmic: many problems that are hard for points become easy for lines, and vice versa, so a duality transform lets us solve a problem on whichever side of the correspondence is more convenient. The fundamental planar transform, developed in Section 2.14 of the mathematical foundations, maps the point $p = (a, b)$ to the non-vertical line

$$p^* : y = ax - b,$$

and maps a non-vertical line $\ell : y = mx + c$ to the point

$$\ell^* = (m, -c).$$

The choice of the *negative* intercept $-b$ is what makes the transform its own inverse up to the sign convention, so that “dualizing twice” returns the original object.

Definition 15.1 (Dual Transform). Let $p = (a, b)$ be a point and $\ell : y = mx + c$ a non-vertical line of the plane. The **dual line** of p is $p^* : y = ax - b$, and the **dual point** of ℓ is $\ell^* = (m, -c)$. The map $p \mapsto p^*$ on points and $\ell \mapsto \ell^*$ on lines is the **dual transform**. A property that refers to the original plane is said to hold in the **primal**; the transform maps primal objects to **dual** objects.

Theorem 15.2 (Involution). *The dual transform is an involution on the set of points together with non-vertical lines: for every point p and every non-vertical line ℓ ,*

$$(p^*)^* = p \quad \text{and} \quad (\ell^*)^* = \ell.$$

Proof. For a point $p = (a, b)$, the dual line is $p^* : y = ax - b$, i.e., the line of slope a and intercept $-b$. The dual of that line is the point $(a, -(-b)) = (a, b) = p$. For a line $\ell : y = mx + c$, the dual point is $(m, -c)$; its dual line is $y = mx - (-c) = mx + c = \ell$. $\square$

Example 15.3 (Basic Dual Pairs). The point $p = (2, 3)$ has dual line $p^* : y = 2x - 3$; the line $\ell : y = -x + 1$ has dual point $\ell^* = (-1, -1)$. Dually, the line through the dual points $(2, 3)$ and $(-1, -1)$ in the dual plane is $y = 2x - 3$ restricted to the two coordinates — the point $(-1, -1)$ lies on p^* because $-1 = 2(-1) - 3$, which is exactly the incidence property that Section 15.2 makes precise.

The transform as stated applies only to non-vertical lines and to points of finite coordinates. Vertical lines $\ell : x = c$ have no finite slope and are mapped, in the projective closure, to *points at infinity* of the dual plane; conversely a point with a prescribed dual line of the form $x = \text{const}$ corresponds, in the primal, to a direction. The standard workarounds are to rotate the whole configuration so that no relevant line is vertical, or to use the homogeneous-coordinate machinery of Section 2.13, where a vertical line is represented by a vector whose slope component vanishes and the dual point has an infinite coordinate. In this chapter we assume, unless stated otherwise, that the input is in general position: no two input points share an x -coordinate and no dual line is vertical.

Remark 15.4 (Conventions and Variants). There are several sign conventions for duality. The convention $p = (a, b) \mapsto (y = ax - b)$ used here has the property that the dual of a *line* stores its slope in the first coordinate and the negative of its intercept in the second; interchanging the sign of the intercept (e.g., $y = ax + b$) reverses the above/below relations and flips upper and lower hulls into one another. A different and equally useful family of transforms is the *normal-form* or *polar duality* $p = (a, b) \mapsto \{x : ax + by = 1\}$, which treats all lines symmetrically at the price of the point $(0, 0)$ being sent to infinity; the polar duality is the subject of Section 15.7. The choice of convention never changes the complexity of the resulting algorithms, only the direction in which inequalities and “above/below” relations are read.

15.2 Incidence Preservation

The defining property of the dual transform — the property that makes it a duality rather than a mere parameterization — is that it turns *point-line incidence* into itself.

Definition 15.5 (Incidence). A point p and a line ℓ are **incident** if p lies on ℓ . Two lines are **concurrent** if they share a point; a set of points is **collinear** if all of them lie on one line.

Theorem 15.6 (Incidence Preservation). *Let $p = (a, b)$ be a point and $\ell : y = mx + c$ a non-vertical line. Then*

$$p \in \ell \iff \ell^* \in p^*.$$

That is, the primal statement “the point p lies on the line ℓ ” is true if and only if the dual statement “the point ℓ^ lies on the line p^* ” is true.*

Proof. Both statements are the equation $b = ma + c$. Indeed, $p \in \ell$ means $b = ma + c$. The dual point is $\ell^* = (m, -c)$ and the dual line is $p^* : y = ax - b$; the incidence $\ell^* \in p^*$ means $-c = am - b$, which rearranges to exactly $b = ma + c$. $\square$

The power of incidence preservation is that it lifts to any configuration that is described by incidences. Two primal points p, q determine the line $\overline{pq}$; dually, the two dual lines p^*, q^* intersect at the single point $\overline{pq}^*$. A primal line through k points becomes a dual point common to k lines. This is the first of the two great translation rules of duality:

Corollary 15.7 (Collinearity and Concurrence). *A set $\{p_1, \dots, p_k\}$ of points is collinear in the primal if and only if the dual lines $\{p_1^*, \dots, p_k^*\}$ are concurrent in the dual. In particular, three points are collinear if and only if their dual lines are concurrent, i.e., pass through one common point.*

Example 15.8 (Detecting Collinearity by Hand). Consider the points $p_1 = (0, 0)$, $p_2 = (1, 1)$, and $p_3 = (2, 2)$. Their dual lines are $p_1^* : y = 0$, $p_2^* : y = x - 1$, and $p_3^* : y = 2x - 2$. The three dual lines all pass through the point $(1, 0)$, since $0 = 1 - 1$ and $0 = 2 \cdot 1 - 2$; hence they are concurrent, and by the corollary the primal points are collinear. Indeed they lie on $y = x$. The dual of that line is $\ell^* = (1, -0) = (1, 0)$, the concurrence point.

Two further remarks complete the incidence picture. First, the incidence theorem extends to *segments*: a primal segment $s = pq$ consists of all points of the segment, so its set of *lines crossing it* is, in the dual, the union of the dual lines of all points of s — a *double wedge* bounded by p^* and q^* , with the vertex $\overline{pq}^*$ at the crossing point of the two boundary lines. Two primal segments cross if and only if some primal line meets both, which by duality happens if and only if the two corresponding double wedges overlap. This translation is the basis of the duality reductions for segment intersection and for the red–blue problems of Section 5.11. Second, incidence alone already makes duality a tool for *verification*: any combinatorial statement that asserts “these points lie on these lines” has an exactly equivalent dual statement, so a checkable certificate in one plane is automatically a checkable certificate in the other.

15.3 Order Reversal

Incidence is preserved, but *order* is reversed in a precise sense, and this reversal is the second great translation rule of duality. We need two order facts: the local order of a point and a line, and the global order of a chain of points versus a chain of lines.

Lemma 15.9 (Point–Line Order). *Let $p = (a, b)$ be a point and $\ell : y = mx + c$ a non-vertical line. The point p lies above ℓ if and only if the dual point ℓ^* lies above the dual line p^* at $x = m$. Symmetrically, p lies below ℓ if and only if ℓ^* lies below p^* .*

Proof. The point p is above ℓ when $b > ma + c$. The dual point has y -coordinate $-c$, and the dual line evaluated at $x = m$ has value $am - b$; the point is above the line when $-c > am - b$, which rearranges to $b > ma + c$. The “below” statement is the same inequality reversed. $\square$

The lemma says that, for a fixed pair (p, ℓ) , the above/below relation is preserved by duality. The reversal appears once we compare *many* points against *many* lines, because the sign convention makes the dual value $a_i m - b_i$ the *negative* of the signed intercept $b_i - ma_i$.

Theorem 15.10 (Order Reversal). *Let P be a set of points in general position and fix a slope m . Sort the points of P by the signed distance $b_i - ma_i$ to the line of slope m through the origin, smallest first (the point “lowest” in the direction perpendicular to slope m comes first). Then the dual lines p_i^* , sorted by their value $a_i m - b_i$ at $x = m$, appear in exactly the reverse order, largest first. In particular, the point of P lowest in the direction of slope m corresponds to the dual line attaining the largest value at $x = m$, and vice versa.*

Proof. For each point $p_i = (a_i, b_i)$ let $c_i = b_i - ma_i$, the intercept of the line of slope m through p_i . The value of the dual line at $x = m$ is $a_i m - b_i = -c_i$. Sorting the points by c_i increasing is therefore the same as sorting the dual values $-c_i$ decreasing, which is exactly the claimed reversal. $\square$

The theorem is the mechanism behind every hull–envelope duality. At a fixed slope m , the lowest primal point is dual to the top of the dual arrangement at $x = m$; letting m range over all slopes, the chain of lowest points — the lower convex hull — traces the upper envelope of the dual lines, and the upper hull traces the lower envelope. We record the statement in the form that is used throughout the rest of the book.

Corollary 15.11 (Upper Hull and Lower Envelope). *For a point set P in general position, a point $p \in P$ is a vertex of the upper convex hull of P if and only if its dual line p^* appears on the lower envelope of the dual lines $\{q^* : q \in P\}$; the upper-hull vertices, read left to right, appear on the lower envelope in the reverse order. Symmetrically, the lower-hull vertices, read left to right, appear on the upper envelope of the dual lines in the same order.*

Proof. A point p is a vertex of the upper hull exactly when it is the highest point of P in the direction perpendicular to the slope of some supporting line of the upper hull. By Theorem 15.10, “highest in that direction” is dual to “largest value at $x = m$ ”, which means that p^* is the topmost line at $x = m$, i.e., a segment of the lower envelope. The reversal of the order follows because the slopes of the supporting lines of the upper hull increase from the leftmost to the rightmost vertex, while the envelope is read from left to right. The lower-hull statement is symmetric. $\square$

Example 15.12 (Envelope Duality on a Triangle). Let $P = \{(0, 0), (1, 1), (2, 0)\}$. The upper hull is the chain $(0, 0), (1, 1), (2, 0)$. The dual lines are $y = 0$, $y = x - 1$, and $y = 2x$. The lower envelope of these three lines is $y = 2x$ for $x < -1$, $y = x - 1$ for $-1 < x < 1$, and $y = 0$ for $x > 1$, i.e., the lines dual to $(2, 0), (1, 1), (0, 0)$ in that order — exactly the upper hull read from right to left, confirming the corollary. The vertical distance between the upper and lower envelopes at $x = 0$ is $0 - (-1) = 1$, which is the width of the triangle in the vertical direction; the general formula is derived in Section 15.8.

The order-reversal machinery is what connects duality to the theory of k -sets and levels. A *separating line* with exactly k points of P below it defines a k -set (Definition 14.15); by the lemma, a line ℓ with k points below it is dual to a point ℓ^* that has exactly k dual lines *below* it, i.e., a point on the k -level of the dual arrangement. The boundary of a k -set is therefore the dual of a k -level, and the whole family of k -sets of P is encoded in the single arrangement of the dual lines. Section 15.4 develops this correspondence, and Section 14.3 of the envelopes chapter studies the levels themselves.

15.4 Dual Arrangements

Given a set P of n points in general position, the dual lines $\{p^* : p \in P\}$ form an **arrangement** in the dual plane: the decomposition of the plane into vertices (intersection points of two dual lines), edges (the maximal open segments of dual lines between vertices), and faces (the open cells between consecutive edges). The arrangement is a single, global object that encodes *all* incidences and order relations among the points of P , and the theory of arrangements developed in the envelopes chapter (Section 14.3 and the level structure it introduces) applies verbatim.

Definition 15.13 (Dual Arrangement). Let P be a set of n points in general position in the primal plane. The **dual arrangement** $A(P) = A(\{p^* : p \in P\})$ is the planar arrangement formed by the n dual lines. Each face of $A(P)$ corresponds to a set of primal lines sharing the same “profile”: every primal line whose dual point lies in a given face has exactly k points of P below it, for a value of k that depends only on the face.

Definition 15.14 (Level of the Dual Arrangement). For an integer k , the **k -level** of the dual arrangement is the set of points of $A(P)$ that have exactly k dual lines strictly below them; it is a polygonal chain made of one segment per dual line. The 0-level is the lower envelope and the $(n - 1)$ -level is the upper envelope of the dual lines (Definition 14.7).

The correspondence between the primal configuration and the dual arrangement can be read off directly:

- each dual line p^* is the set of dual points corresponding to all primal lines through p ;
- each vertex $p^* \cap q^*$ is the dual point of the line $\overline{pq}$ through p and q ;
- the cells above (respectively below) a segment of p^* correspond to the primal lines passing above (respectively below) p ;
- the k -level separates the cells whose primal lines have k points below them from the cells whose primal lines have $k + 1$ points below them;

- by Corollary 15.7, k collinear points of the primal correspond to a vertex of the dual arrangement where k lines meet.

The arrangement is the complete *signature* of the point set: two point sets have combinatorially isomorphic dual arrangements if and only if all their order and incidence relations agree, which is precisely the notion of an order type. This makes arrangements the right data structure for problems in which many points and many lines interact simultaneously.

The complexity and algorithmic questions about arrangements are classical and have simple answers for lines.

Theorem 15.15 (Arrangement Complexity). *The arrangement of n lines in the plane has at most $n(n-1)/2$ vertices, n^2 edges, and $n(n+1)/2 + 1$ faces; all of these bounds are tight for lines in general position. The arrangement can be constructed in $O(n^2)$ time and stored in $O(n^2)$ space. The worst-case complexity of the k -level is $\Theta(n\sqrt{k+1})$.*

Proof sketch. Every pair of lines intersects at most once, giving the vertex bound; each line is split by the $n-1$ other lines into n edges, giving n^2 edges total; the face count follows from Euler’s formula. The $O(n^2)$ construction is a standard sweep of the arrangement; the zone-based refinement of the sweep is the classical algorithm (Edelsbrunner–O’Rourke–Seidel [EOS86]). The k -level bound is due to the theory of levels (Theorem 14.17 in the envelopes chapter): the upper bound is $\Theta(n\sqrt{k})$ via the “exponential decay” argument, and point sets for which the level has $\Omega(n\sqrt{k})$ vertices exist; the modern treatment appears in Edelsbrunner [Ede87]. $\square$

Theorem 15.16 (Zone of a Line). *In an arrangement of n lines, the zone of any line ℓ — the set of cells of the arrangement crossed by ℓ — has complexity $O(n)$, and the zone can be constructed in $O(n)$ time once the arrangement is available.*

Proof sketch. Add ℓ to the arrangement as the last line. Each edge of the zone is bounded by at most two vertices of the original arrangement, and a charging argument (each zone vertex is charged to a distinct vertex of the arrangement left of it on its level) shows that the zone has at most $O(n)$ vertices; hence it has $O(n)$ edges and faces. See [EOS86]. $\square$

In the dual, the zone theorem has a natural reading: any line of the primal crossing k of the cells of the dual arrangement corresponds to a line whose “order profile” (the sequence of points above it) changes $O(n)$ times as it moves. The zone theorem is what makes incremental constructions in arrangements efficient, and it is the structural reason that duality reductions for problems such as red–blue segment intersection (Section 5.11) run in near-linear time.

Finally, the dual arrangement viewpoint extends to the paraboloid lifting of Chapter 12: the arrangement of the dual lines of a point set is the 2D analogue of the arrangement of tangent planes to the paraboloid, and the Voronoi diagram is the projection of the upper envelope of that tangent-plane arrangement (Definition 12.23, Theorem 12.24, and Section 12.8). The dual arrangement is thus the common ancestor of the envelope, level, and Voronoi structures used throughout the book.

15.5 Convex Hulls through Duality

15.5.1 1. Overview

The order-reversal theorems of Section 15.3 turn the convex-hull problem into an envelope problem and back. Because the convex hull of n points can be computed in $O(n \log n)$ time (Chapter 4) and the lower/upper envelope of n lines can be computed in $O(n \log n)$ time (Chapter 14), duality gives a second family of algorithms for both problems, with the two complexities coinciding. The same correspondence, lifted one dimension to the paraboloid, is

the mechanism behind the duality of the Delaunay triangulation and the Voronoi diagram of Chapter 12: the “hull of lifted points” view of Section 12.8 is literally a hull–envelope duality in one dimension higher.

15.5.2 2. Definitions

Definition 15.17 (Upper and Lower Hull). Let P be a point set in general position. The **upper hull** of P is the chain of vertices of $\text{conv}(P)$ visible from above, ordered by increasing x -coordinate; the **lower hull** is the complementary chain ordered by increasing x -coordinate (Definition 4.7). Together they form the boundary of the convex hull.

Definition 15.18 (Envelope Pair). For a family of dual lines $L = \{p^* : p \in P\}$, the **lower envelope** L_{low} is the pointwise minimum of the lines and the **upper envelope** L_{up} their pointwise maximum (Definitions 14.7 and 14.8 of the envelopes chapter). Each envelope is a piecewise-linear convex chain.

15.5.3 3. Theory

The central theorem is the formal version of Corollary 15.11.

Theorem 15.19 (Hull–Envelope Duality). *Let P be a set of points in general position and let $L = \{p^* : p \in P\}$ be their dual lines. Then:*

- *the vertices of the upper hull of P , read left to right, are exactly the dual lines of the segments of the lower envelope of L , read right to left;*
- *the vertices of the lower hull of P , read left to right, are exactly the dual lines of the segments of the upper envelope of L , read left to right.*

In particular p is an upper-hull vertex if and only if p^ is on the lower envelope of the dual lines.*

Proof. A point p_i is an upper-hull vertex exactly when it maximizes the signed distance $b_i - ma_i$ for all supporting lines of slope m over an open interval of slopes. By Theorem 15.10, maximizing $b_i - ma_i$ is equivalent to minimizing the dual value $a_i m - b_i$ at $x = m$, so p_i^* is the line attaining the pointwise minimum over that interval of slopes — a segment of the lower envelope. The order reversal along the chains follows by the monotonicity of the supporting slopes along the upper hull. $\square$

Corollary 15.20 (Computational Equivalence). *The convex hull of n points can be computed from the lower and upper envelopes of the n dual lines, and the envelopes can be computed from the hull of the dual points, with no overhead beyond the $O(n)$ transform. Both problems therefore admit algorithms of the same asymptotic complexity.*

Proof. Theorem 15.19 gives both hull chains from both envelopes. Conversely, given the hull of the dual points $\{p^*\}$ as points, its upper and lower chains dualize back to the lower and upper envelopes of the primal dual lines by the same theorem applied to the point set $\{p^*\}$. $\square$

The same duality, in one higher dimension, underlies the Delaunay–Voronoi duality. Lifting each primal point $p = (x, y)$ to $p^+ = (x, y, x^2 + y^2)$ on the paraboloid, the Delaunay triangulation of P is the projection of the *lower* hull of the lifted points, and the Voronoi diagram is dual to it; equivalently the Voronoi diagram is the projection of the *upper envelope* of the tangent planes to the paraboloid at the lifted points. This is developed in full in Section 12.8 of Chapter 12, where the lifting theorem (Theorem 12.24) is proved by computing the vertical gap $\|x - p\|^2$ between the tangent plane and the paraboloid. The hull in that chapter is a hull in $\mathbb{R}^3$; the duality here is the same envelope duality as in Theorem 15.19, with planes replacing lines.

15.5.4 4. Pseudocode

The algorithm computes the hull of P by computing the two envelopes of the dual lines and reading off the vertices.

Algorithm 67 Convex Hull via Dual Envelopes

Input: a set P of n points in general position.

Output: the vertices of $\text{conv}(P)$ in counterclockwise order.

```

1: function CONVEXHULLBYDUALITY( $P$ )
2:    $L \leftarrow \emptyset$ 
3:   for each point  $p = (a, b) \in P$  do
4:     add the line  $y = ax - b$  to  $L$  ▷ dual line  $p^*$ 
5:   end for
6:    $\text{low} \leftarrow \text{LowerEnvelope}(L)$  ▷ recursive divide and conquer
7:    $\text{high} \leftarrow \text{UpperEnvelope}(L)$  ▷ negate and repeat
8:    $H_{\text{up}} \leftarrow$  dual lines of the segments of  $\text{low}$ , read right to left ▷ upper hull, left to right
9:    $H_{\text{low}} \leftarrow$  dual lines of the segments of  $\text{high}$ , read left to right ▷ lower hull, left to right
10:  return  $H_{\text{low}}$  followed by  $H_{\text{up}}$  minus its first and last vertices
11: end function

```

The subroutines `LowerEnvelope` and `UpperEnvelope` are the divide-and-conquer envelope algorithms of Chapter 14 (Algorithm 14.2.4); they run in $O(n \log n)$ time. The dual direction — computing the envelopes from a convex hull — uses any hull algorithm of Chapter 4 on the point set $\{p^*\}$ and reads off the chains as in the proof of Corollary 15.20.

15.5.5 5. Parameter Selection

- **Sign convention:** the transform $y = ax - b$ and its “upper hull to lower envelope” correspondence are tied together; switching to $y = ax + b$ exchanges upper and lower. Pick one convention and keep it consistent across the hull and envelope subroutines.
- **General position:** points with equal x -coordinates produce parallel dual lines, whose envelope has vertical segments; either perturb the input slightly, or break ties by a consistent rule in the hull and envelope algorithms.
- **Exact predicates:** the envelope merges and the hull computations both rest on orientation and ordering predicates; the robust predicates of Chapter 4 and the geometric-predicates chapter (Section 3.3 and its orientation test) should be used.

15.5.6 6. Complexity

- **Time:** $O(n \log n)$, the cost of the two envelope computations (each $O(n \log n)$) or, dually, of the hull computation on the dual points. This matches the direct hull algorithms of Chapter 4.
- **Space:** $O(n)$ for the envelopes and the resulting hull.

15.5.7 7. Usages

- **Envelope-based hulls:** when the input is naturally given as lines (visibility, kinetic configurations), computing the hull of the dual points is the efficient route.
- **Delaunay and Voronoi via lifting:** the same duality one dimension higher computes the Delaunay triangulation as the lower hull of lifted points and the Voronoi diagram as

the upper envelope of tangent planes (Section 12.8, Chapter 12); the connection to the farthest-point Voronoi diagram is made in Section 11.3 of Chapter 11.

- **Hull peeling:** onion peeling (Section 4.5 of Chapter 4) can be run in the dual by peeling levels of the arrangement, one level per hull layer.

15.5.8 8. Testing Methods and Edge Cases

- **Cross-check:** compare the output against a direct hull algorithm (e.g., monotone chain) on random and degenerate inputs.
- **Collinear and coincident points:** verify the envelope routines handle parallel and identical dual lines without producing zero-length chain segments.
- **Extreme coordinates:** points with large coordinates make the dual intercepts large; verify the predicates used in the envelope merge are robust.

15.5.9 9. References

- de Berg, M., et al. [dBCvKO08], Ch. 8 and Sec. 11.4.
- Preparata, F. P., Shamos, M. I. [PS85], Ch. 3 and Ch. 7.
- Edelsbrunner, H. [Ede87], Ch. 6.
- Chazelle, B., Guibas, L. J., Lee, D. T. [CGL85].

15.6 Half-Plane Intersection

15.6.1 1. Overview

The half-plane intersection problem asks for the intersection $R = \cap_i h_i$ of a set of n half-planes h_i ; the region R is empty, or a convex (possibly unbounded) polygon. This is the dual problem to the convex hull: by the order-reversal theorems of Section 15.3, the intersection of the lower half-planes of the dual lines is a convex region whose boundary is the lower envelope, and the lower envelope is the dual of the upper hull. Hence every algorithm for convex hulls yields an algorithm for half-plane intersection and conversely. The problem is also the geometric form of a linear program: testing whether R is nonempty is feasibility, and optimizing a linear function over R is optimization; both are treated in Section 15.7 and Chapter 17.

15.6.2 2. Definitions

Definition 15.21 (Half-Plane). A **half-plane** is a set of the form

$$h : y \leq mx + c \quad \text{or} \quad y \geq mx + c,$$

together with its boundary line $\ell : y = mx + c$. The **intersection region** of a family of half-planes is $R = \cap_i h_i$. The half-plane h is *lower* if it is of the form $y \leq mx + c$ and *upper* otherwise; half-planes with vertical boundary lines are handled by rotating the input.

Definition 15.22 (Region of Lines above a Point Set). Let P be a set of points and $L = \{p^* : p \in P\}$ the dual lines. The region

$$H = \{(u, v) : v \leq a_p u - b_p \text{ for all } (a_p, b_p) \in P\}$$

is the intersection of the lower half-planes of the dual lines. A dual point (u, v) belongs to H exactly when the primal line $y = ux - v$ has every point of P on or below it.

15.6.3 3. Theory

The duality between the two problems is exact.

Theorem 15.23 (Half-Plane Intersection is Dual to the Convex Hull). *Let P be a point set in general position and let H be the intersection of the lower half-planes of the dual lines $\{p^*\}$.*

- *The upper boundary of H is the lower envelope of the dual lines, which by Theorem 15.19 is the dual of the upper hull of P .*
- *More generally, let R be the intersection of a set of half-planes whose boundary lines are in general position. The edges of the lower chain of R correspond to the vertices of the upper convex hull of the dual points of the boundary lines of the lower half-planes; the edges of the upper chain of R correspond to the vertices of the lower convex hull of the dual points of the boundary lines of the upper half-planes.*
- *R is bounded if and only if both chains are nonempty and the two envelopes cross; R is empty if and only if the two chains do not cross.*

Proof. A dual point (u, v) is on the boundary of H when $v = \min_i(a_i u - b_i)$, i.e., when (u, v) lies on the lower envelope of the dual lines. Theorem 15.19 identifies the lines of the lower envelope with the upper-hull vertices of P , giving the first statement. For the general case, let R be cut into a lower chain (from the lower half-planes) and an upper chain (from the upper half-planes). The lower chain is the lower envelope of the boundary lines of the lower half-planes; applying the hull-envelope duality to the point set $\{\ell_i^*\}$ of the dual points of those boundary lines shows the lower envelope is the dual of the upper hull of $\{\ell_i^*\}$, and similarly for the upper chain. R is the set of points between the two chains over the x -interval where the lower chain is below the upper chain; the interval is empty exactly when the chains do not cross. $\square$

Corollary 15.24 (Hull and Half-Plane Intersection are Equivalent). *Computing the convex hull of n points and computing the intersection of n half-planes are computationally equivalent: each reduces to the other in linear time via the dual transform.*

Proof. By Theorem 15.23, the vertices of the intersection of half-planes are the duals of the vertices of the convex hull of the dual points, and conversely by Theorem 15.19. Both transforms cost $O(n)$. $\square$

This equivalence is the classical content of Chazelle–Guibas–Lee’s study of “the power of geometric duality” [CGL85]: every algorithm and every lower bound for convex hulls transfers, through duality, to half-plane intersection, and both problems are LP-feasibility problems in the sense of Section 15.7.

15.6.4 4. Pseudocode

The two hull computations are any $O(n \log n)$ hull algorithm of Chapter 4. An alternative that avoids duality entirely is the divide-and-conquer half-plane intersection of Chapter 17 (Section 17.2), which merges two convex polygons in linear time and has the same $O(n \log n)$ bound; the randomized incremental variant is analyzed in Section 15.7.

15.6.5 5. Parameter Selection

- **Orientation classes:** the algorithm separates lower and upper half-planes; half-planes with vertical boundary lines must be rotated out of the input first, or handled by a preliminary y -sweep.

Algorithm 68 Half-Plane Intersection via the Dual Hull

Input: n half-planes h_i in general position, each with boundary line $\ell_i : y = m_i x + c_i$ and an orientation (lower or upper).

Output: the convex region $R = \cap_i h_i$, or a certificate that R is empty.

```

1: function INTERSECTHALFPLANES( $\{h_i\}$ )
2:   low  $\leftarrow \emptyset$ ; high  $\leftarrow \emptyset$ 
3:   for each half-plane  $h_i$  do
4:      $q_i \leftarrow (m_i, -c_i)$   $\triangleright$  dual point of the boundary line
5:     if  $h_i$  is a lower half-plane ( $y \leq m_i x + c_i$ ) then
6:       add  $q_i$  to low
7:     else
8:       add  $q_i$  to high
9:     end if
10:  end for
11:   $C_{\text{up}} \leftarrow$  upper convex hull of the points of low  $\triangleright$  dual of the lower chain of  $R$ 
12:   $C_{\text{down}} \leftarrow$  lower convex hull of the points of high  $\triangleright$  dual of the upper chain of  $R$ 
13:  dualize the edges of  $C_{\text{up}}$  to the lower chain and the edges of  $C_{\text{down}}$  to the upper chain of
     $R$ 
14:   $R \leftarrow$  the region between the two chains over the  $x$ -interval where lower  $\leq$  upper
15:  if the interval is empty then
16:    return  $R = \emptyset$ 
17:  end if
18:  clip the chains at their crossing points and return  $R$ 
19: end function

```

- **Parallel boundary lines:** parallel lines are the degenerate case of general position; the hull of the dual points must tolerate coincident points (duplicate dual points).
- **Boundedness:** to obtain a bounded region, the input must contain both lower and upper half-planes in every relevant direction; otherwise R is an unbounded slab and the algorithm reports it as such.

15.6.6 6. Complexity

- **Time:** $O(n \log n)$ worst case, dominated by the two hull computations; the merging of the two chains is $O(n)$.
- **Space:** $O(n)$.
- **Lower bounds:** matching the convex hull, $\Omega(n \log n)$ comparisons are required to compute the intersection of n half-planes in the comparison model (by Corollary 15.24 and the hull lower bounds of Chapter 4).

15.6.7 7. Usages

- **Linear programming feasibility:** deciding whether $R \neq \emptyset$ and, if so, finding a point of R is the geometric form of LP feasibility (Section 15.7 and Section 17.1 of Chapter 17).
- **Visibility and kernels:** the kernel of a simple polygon is the intersection of half-planes defined by its edges; duality transfers the hull machinery to kernel computation.
- **Smallest enclosing circle:** the smallest enclosing circle of a point set can be computed through the farthest-point Voronoi diagram (Section 11.3), whose vertices are dual to half-plane constraints; the two views meet in the LP formulation of Section 15.7.

15.6.8 8. Testing Methods and Edge Cases

- **Empty intersection:** construct inputs whose two envelope chains do not cross (e.g., half-planes with no common point) and verify the empty certificate.
- **Unbounded regions:** parallel “open” configurations give an unbounded slab; verify the algorithm does not emit spurious vertices at infinity.
- **Cross-validation:** compare the output polygon against the divide-and-conquer intersection of Section 17.2 on random inputs.

15.6.9 9. References

- de Berg, M., et al. [dBCvKO08], Sec. 4.2 and Ch. 8.
- Preparata, F. P., Shamos, M. I. [PS85], Ch. 7.
- Chazelle, B., Guibas, L. J., Lee, D. T. [CGL85].

15.7 Linear Programming through Duality

15.7.1 1. Overview

A linear program (LP) asks for a point maximizing (or minimizing) a linear objective over a set of linear constraints. In the plane, both the constraints and the objective are geometric: the feasible region is the intersection of n half-planes, and the optimum of a bounded objective is a vertex of that region. Duality serves three distinct purposes in this setting. First, the *primal–dual* correspondence of linear programming gives certificates: an infeasibility or optimality claim is witnessed by a convex combination of the constraints, and in the plane these certificates are exactly the polar-duality statements of the point–line transform. Second, the randomized incremental algorithms of Seidel solve fixed-dimension LPs in expected linear time by maintaining the optimal vertex and repairing it only when a new constraint violates it. Third, the polar (point–hyperplane) duality extends the planar theory to all dimensions and is the language in which high-dimensional hulls and their certificates are phrased. This section develops these three uses; the algebraic LP theory and its dual program are treated in Chapter 17 (Section 17.4).

15.7.2 2. Definitions

Definition 15.25 (Geometric Linear Program). Let $H = \{h_1, \dots, h_n\}$ be half-planes and let $c \in \mathbb{R}^2$ be an objective direction. The **2D linear program** is

$$\text{minimize } c \cdot x \quad \text{subject to } x \in \bigcap_{i=1}^n h_i.$$

The program is **infeasible** if the intersection is empty, **unbounded** if the objective is unbounded below over the region, and otherwise has an optimal vertex x^* of the feasible polygon.

Definition 15.26 (Polar Body). Let $K \subseteq \mathbb{R}^d$ be a convex set containing the origin in its interior. The **polar** of K is

$$K^\circ = \{y \in \mathbb{R}^d : \langle x, y \rangle \leq 1 \text{ for all } x \in K\}.$$

For a finite set $S = \{s_1, \dots, s_n\}$, the polar of the polytope $\text{conv}(S)$ is the intersection of the half-spaces $\{y : \langle s_i, y \rangle \leq 1\}$.

The polar is the higher-dimensional analogue of the point–line transform: a point s is mapped to the half-space of the dual hyperplane $\{y : \langle s, y \rangle = 1\}$ containing the origin. Where the planar transform swaps points and lines, polarity swaps points and hyperplanes, and it inverts the face lattice.

15.7.3 3. Theory

The fundamental structural facts are the bipolar theorem, the Farkas separation statement, and the face-lattice inversion.

Theorem 15.27 (Bipolar Theorem). *Let K be a closed convex set containing the origin. Then $(K^\circ)^\circ = K$. If K is a polytope with the origin in its interior, then polarity is a bijection between the faces of K and the faces of K° that reverses dimension: a k -dimensional face of K corresponds to a $(d - 1 - k)$ -dimensional face of K° . In particular, the vertices of K correspond to the facets of K° .*

Proof sketch. The inclusion $K \subseteq (K^\circ)^\circ$ is immediate from the definition; the reverse inclusion follows from the hyperplane separation theorem (Theorem 2.9 of the foundations chapter): if $x \notin K$, a strictly separating hyperplane yields a dual vector y with $\langle x, y \rangle > 1 \geq \langle z, y \rangle$ for all $z \in K$, so $y \in K^\circ$ and $x \notin (K^\circ)^\circ$. The face-lattice statement follows because a face F of K is the set where a supporting hyperplane is tight, and the supporting hyperplane's normal is a vertex of K° whose polar facet is the dual of F ; the dimension count is the rank of the tight constraints. $\square$

Theorem 15.28 (Zero in a Convex Hull by Polarity). *Let $S = \{s_1, \dots, s_n\} \subseteq \mathbb{R}^d$. Then*

$$0 \in \text{conv}(S) \iff \bigcap_{i=1}^n \{x : \langle s_i, x \rangle \geq 1\} = \emptyset.$$

Equivalently, $0 \notin \text{conv}(S)$ if and only if there exists a point x with $\langle s_i, x \rangle \geq 1$ for all i , i.e., a hyperplane strictly separating 0 from $\text{conv}(S)$.

Proof. If $0 \in \text{conv}(S)$ then $0 = \sum_i \lambda_i s_i$ for convex weights $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$. For any x , multiplying $\langle s_i, x \rangle \geq 1$ by λ_i and summing gives $0 \geq 1$, a contradiction, so the intersection is empty. Conversely, if $0 \notin \text{conv}(S)$, the strong separation theorem gives a vector v and a scalar $\alpha > 0$ with $\langle v, s_i \rangle \geq \alpha$ for all i ; scaling, the point v/α satisfies the system, so the intersection is nonempty. $\square$

The theorem is the certificate machinery behind LP duality: an infeasible dual constraint system is witnessed by the convex combination (the dual variables λ_i), and the dual LP of Section 17.4 is exactly the search for such a combination. This is the sense in which polarity is the higher-dimensional dual transform, and it is the mechanism behind the high-dimensional hull and polarity algorithms of Chapter 16.

For the algorithmic side, the 2D LP has a simple randomized incremental solution.

Theorem 15.29 (Randomized Incremental LP). *The 2D linear program of Definition 15.25 can be solved in $O(n)$ expected time by Seidel's randomized incremental algorithm: insert the half-planes in random order, maintain the optimal vertex of the current intersection, and when the new half-plane violates the current optimum, solve a one-dimensional subproblem on the new boundary line. For any fixed dimension d , the algorithm runs in $O(d!n)$ expected time.*

Proof sketch. Maintain the feasible region $R_i = \bigcap_{j \leq i} h_{\pi(j)}$ and its optimal vertex x_i for a random permutation π . If the next half-plane $h_{\pi(i)}$ does not contain x_{i-1} , the new optimum x_i must lie on the boundary line of $h_{\pi(i)}$, so the problem reduces to a 1D LP on that line over the previous constraints — an interval, whose optimum is an endpoint. By backward analysis, the probability that $h_{\pi(i)}$ violates x_{i-1} is $O(1/i)$ (for fixed dimension d , the d constraints binding at the optimum of a random i -set are a uniformly random d -subset, so the new one binds with probability d/i). The expected total work is therefore $O(\sum_i d/i \cdot i + n) = O(n)$. $\square$

Algorithm 69 Seidel's Randomized Incremental 2D LP

Input: half-planes $H = \{h_1, \dots, h_n\}$ and an objective direction c .**Output:** an optimal point x^* over $\cap H$, or a certificate of infeasibility or unboundedness.

```

1: function SEIDELLP( $H, c$ )
2:    $\pi \leftarrow$  uniformly random permutation of  $\{1, \dots, n\}$ 
3:    $x^* \leftarrow$  an arbitrary feasible point of the initial bounding half-planes
4:   for  $i = 1, \dots, n$  do
5:     if  $x^*$  violates  $h_{\pi(i)}$  then
6:        $x^* \leftarrow \text{OptimumOnLine}(\partial h_{\pi(i)}, c, \{h_{\pi(1)}, \dots, h_{\pi(i-1)}\})$   $\triangleright$  1D LP on the boundary
       line
7:       if the 1D subproblem is infeasible then
8:         return infeasible
9:       end if
10:    end if
11:  end for
12:  return  $x^*$ 
13: end function
14: function OPTIMUMONLINE( $\ell, c, \text{constraints}$ )
15:    $[u, v] \leftarrow (-\infty, +\infty)$   $\triangleright$  interval of feasible points of  $\ell$ 
16:   for each half-plane  $h$  in constraints do
17:     clip  $[u, v]$  by the intersection of  $\ell$  with  $h$ 
18:   end for
19:   if  $u > v$  then
20:     return infeasible
21:   end if
22:   return the point of  $[u, v]$  minimizing  $c \cdot x$ 
23: end function

```

15.7.4 4. Pseudocode

The bounding half-planes of the initialization can be taken as four faraway constraints enclosing the input; they are also how unboundedness is detected (the interval $[u, v]$ stays unbounded in the direction of c).

15.7.5 5. Parameter Selection

- **Feasibility as a subproblem:** when only a point of R is needed, take c to be any direction and accept the returned optimum; alternatively run the same algorithm with $c = 0$ (any point of the intersection is optimal) to test feasibility.
- **Degenerate constraints:** a new half-plane containing the current optimum is skipped; coincident and parallel boundary lines must be deduplicated so that the 1D subproblem sees a clean interval.
- **Exact arithmetic:** the predicate “does x^* violate h ” and the interval clipping use orientation tests; the robust predicates of Chapter 4 should be used to avoid near-degenerate misclassification.

15.7.6 6. Complexity

- **Time:** $O(n)$ expected for fixed dimension d (e.g., $d = 2$); $O(n \log n)$ deterministic by divide-and-conquer half-plane intersection (Section 17.2); the expected number of violated constraints is $O(d \log n)$, each costing $O(nd)$ for the 1D repair.
- **Space:** $O(n)$.
- **Certificates:** infeasibility and optimality are certified by the convex combinations of Theorem 15.28, computable from the binding constraints; the certificate size is $O(d)$.

15.7.7 7. Usages

- **Feasibility and optimization:** the 2D LP solves feasibility of half-plane intersection and hence, by Theorem 15.23, convex-hull inclusion and separation queries in the primal (Section 17.1).
- **Smallest enclosing circle and Chebyshev center:** the smallest enclosing circle of a point set is a 2D LP over the (dual) constraints of the farthest-point Voronoi diagram (Section 11.3), and the largest empty circle centered in a region is the corresponding maximization.
- **High-dimensional certificates:** polar duality (Theorem 15.27) provides the certificates for high-dimensional hulls and for the LP duality treated in Chapter 17 (Section 17.4).

15.7.8 8. Testing Methods and Edge Cases

- **Infeasibility:** verify the algorithm reports infeasible on half-planes with empty intersection (e.g., $y \leq 0$ and $y \geq 1$).
- **Unboundedness:** an objective pointing into an open slab must return the unbounded certificate rather than a spurious optimum.
- **Degenerate vertices:** when the optimum is a vertex of degree > 2 (three half-planes through one point), the algorithm must still terminate with the correct value.

15.7.9 9. References

- de Berg, M., et al. [dBCvKO08], Sec. 4.4 and Ch. 4 (randomized LP).
- Preparata, F. P., Shamos, M. I. [PS85], Ch. 4.
- Matoušek, J. [Mat99] treats the discrepancy and approximation sides of the same duality.
- Chazelle, B., Guibas, L. J., Lee, D. T. [CGL85].

15.8 Applications to Optimization

Duality is not merely a reformulation technique: it reduces a family of optimization problems over point sets to structurally simpler problems over arrangements and envelopes, for which the near-linear algorithms of Chapter 14 apply. This section collects the classical applications.

Detecting collinearities and concurrences. By Corollary 15.7, a set of k collinear primal points is a set of k concurrent dual lines, i.e., a vertex of the dual arrangement of degree k . Counting or detecting collinear triples among n points therefore reduces to detecting vertices of the arrangement of the n dual lines at which three lines meet; in the dual arrangement these are the vertices of the arrangement lying on levels $k - 1$ to $k - 1$ measured from the bottom (Definition 15.14). The reduction is the subject of the classic “power of geometric duality” paper of Chazelle, Guibas, and Lee [CGL85], which uses duality to reduce several point-set problems — including the detection of collinearities, of intersections, and of certain optimization queries — to arrangement and envelope computations. In the algebraic sense, detecting collinear triples is a 3SUM-hard problem, and the dual formulation exposes exactly why: a three-line concurrence is the triple meeting condition of an arrangement vertex, and no subquadratic algorithm is known in the comparison model.

Computing envelopes and the width of a point set. The order-reversal theorem reduces envelope computation to convex hulls (Section 15.5) and gives duality a direct role in the width problem.

Theorem 15.30 (Width by Envelope Distance). *Let P be a point set, and let L_{low} and L_{up} be the lower and upper envelopes of the dual lines of P . The width of P in the direction perpendicular to the line of slope m is*

$$w(m) = \frac{L_{up}(m) - L_{low}(m)}{\sqrt{1 + m^2}}.$$

The minimum of $w(m)$ is attained at an x -coordinate of a vertex of the envelopes, i.e., at the slope of a line through two hull vertices.

Proof. The lowest and highest lines of slope m supporting $\text{conv}(P)$ have intercepts $c_{\min} = \min_i(b_i - ma_i)$ and $c_{\max} = \max_i(b_i - ma_i)$, and the width perpendicular to slope m is $(c_{\max} - c_{\min})/\sqrt{1 + m^2}$. By Theorem 15.10, the dual values at $x = m$ are $a_i m - b_i = -c_i$, so $L_{up}(m) = -c_{\min}$ and $L_{low}(m) = -c_{\max}$, giving the formula. Since both envelopes are piecewise linear and the denominator is fixed between consecutive arrangement vertices, the function w is minimized on a segment where it is the quotient of a linear by a convex function; its minimum over a segment is at an endpoint, which is a vertex of the envelopes, dual to a line through two hull vertices. $\square$

The theorem is the duality version of the rotating-calipers width computation: instead of rotating two parallel supporting lines around the hull, one reads the vertical gap between the

two envelopes of the dual lines and minimizes the scaled gap over the arrangement vertices. This is a concrete instance of the general phenomenon that envelope algorithms (Section 14.2 and the Davenport–Schinzel theory of Section 14.1 of Chapter 14) solve optimization problems whose objective is a minimum or maximum of linear functions.

Ham-sandwich cuts and level-set problems. The dual of a line that splits a point set is a point on a fixed level of the dual arrangement. Concretely, a line ℓ with exactly k points of P below it is dual to a point ℓ^* with exactly k dual lines below it, i.e., a point of the k -level (Section 14.3 and Definition 15.14). A *bisecting* line for a set of n points (the median in every direction) therefore corresponds to the *median level* of the dual arrangement, and the ham-sandwich theorem for two point sets A and B is a statement about the intersection of two median levels.

Theorem 15.31 (Ham-Sandwich via Median Levels). *Let A and B be two finite point sets with $|A|$ and $|B|$ even. The median levels of the dual arrangements of A and of B intersect, and a point of their intersection is the dual of a line that simultaneously bisects A and bisects B .*

Proof sketch. For each slope m , let $\ell_A(m)$ be the line of slope m with $|A|/2$ points of A below it; its dual point lies on the median level of the A -arrangement at $x = m$. The function $f(m) = b_A(m) - b_B(m)$, where $b_A(m)$ and $b_B(m)$ are the intercepts of the two bisecting lines, changes sign as m runs from $-\infty$ to $+\infty$ (at $m \rightarrow \pm\infty$ the two bisecting lines are ordered by the projections of A and B on the horizontal axis). By continuity, f has a zero, at which the bisecting lines coincide; that common line is the ham-sandwich cut, and its dual point lies on both median levels. $\square$

Example 15.32 (A Ham-Sandwich Cut by Duality). Let $A = \{(0,0), (2,2)\}$ and $B = \{(0,2), (2,0)\}$. The line $y = 1$ has one point of A below it $((0,0))$ and one point of B below it $((2,0))$, so it bisects both sets. Its dual point is $(0, -1)$. The dual lines of A are $y = 0$ and $y = 2x - 2$; at $x = 0$ exactly one of them ($y = 2x - 2$, value -2) lies below $(0, -1)$, so $(0, -1)$ is on the median level of A . The dual lines of B are $y = -2$ and $y = 2x$; at $x = 0$ the line $y = -2$ lies below $(0, -1)$, again exactly one line, so $(0, -1)$ is on the median level of B as well — the dual point is on both median levels, exactly as the theorem predicts.

The level-set view also underlies the theory of k -sets and discrepancy studied in Chapter 14: the k -set boundaries (Definition 14.15) are dual to the k -levels, and the half-plane discrepancy of a point set (Definition 14.16) is a statement about how far all the levels of the dual arrangement deviate from being evenly spaced. The discrepancy chapter of Matoušek [Mat99] develops the deep connections between levels, discrepancy, and the balance of point sets; the duality here is the geometric bridge between the two theories.

The power of geometric duality in optimization. The reductions of this section have a common structure: an optimization problem over a set of points is transformed into a problem over the arrangement of the dual lines, where envelope and level algorithms run in near-linear or near-quadratic time, and where certificates are read off from local structure (vertices, zone edges, level crossings) rather than from global searches. Chazelle, Guibas, and Lee [CGL85] formalized this scheme for a broad family of problems — collinearity detection, nearest and farthest neighbor statistics, convex-hull and intersection queries — establishing duality as one of the primary algorithmic paradigms of computational geometry, alongside the divide-and-conquer and sweeping paradigms of Chapters 4 and 5.

15.9 Exercises

1. Verify Theorem 15.2 and Theorem 15.6 algebraically for the points $(1, 2)$, $(3, -1)$ and the line $y = 2x - 1$: compute both duals and check all four incidence statements by hand.
2. Prove Lemma 15.9 in full detail, and show that if the sign of the intercept in the transform is changed (i.e., $p^* : y = ax + b$), the above/below relation is reversed instead of preserved.
3. Let P be a set of points in general position. Prove that a point $p \in P$ is a vertex of the upper hull of P if and only if p^* is on the lower envelope of the dual lines, and that the order of the upper-hull vertices is the reverse of the order of the envelope segments (Corollary 15.11).
4. Use Theorem 15.28 to show that the point $(0, 0)$ lies in the convex hull of $S = \{(1, 0), (0, 1), (-1, -1)\}$ but not in the convex hull of $S' = \{(1, 0), (0, 1)\}$, by writing down the dual systems and checking infeasibility directly.
5. Describe how the width of a point set is computed from the two envelopes of the dual lines, and prove that the minimizing slope corresponds to a vertex of the envelopes (Theorem 15.30). Compute the width of the triangle $\{(0, 0), (1, 1), (2, 0)\}$ this way.
6. Show that two segments in the primal intersect if and only if their dual double wedges overlap, and explain how this reduces red–blue segment intersection (Section 5.11) to an arrangement or envelope problem in the dual plane.
7. Implement Algorithm 68 (half-plane intersection via the dual hull) and cross-validate it against the divide-and-conquer intersection of Section 17.2 on random and degenerate inputs, including empty and unbounded cases.
8. Verify the ham-sandwich statement of Theorem 15.31 computationally: for random even-size sets A and B , compute the median levels of the two dual arrangements, find an intersection point, dualize it back, and check that the resulting line bisects both sets.

Part VI

Convexity and Geometric
Optimization

Chapter 16

Convex Polytopes

16.1 Convex Polytopes

A **convex polytope** is the d -dimensional generalization of a convex polygon: a bounded convex set whose boundary is made of finitely many flat pieces, the *facets*. Convex polytopes are the feasible regions of linear programs (Chapter 17), the cells of hyperplane arrangements, and the convex hulls built by Chapter 4 and used by the computational and polyhedral geometry of Chapters 19 and 30. In the plane a convex polytope is a convex polygon; in $\mathbb{R}^3$ it is a convex polyhedron, and the cube and the tetrahedron are the models to keep in mind throughout.

Definition 16.1 (Polytope). A **convex polytope** in $\mathbb{R}^d$ is a bounded convex set that is the convex hull of finitely many points (Section 2.5), equivalently the bounded intersection of finitely many closed halfspaces (Section 2.9); the equivalence is the Minkowski–Weyl theorem of the next section. A polytope of dimension d is a **d -polytope**.

Two complementary descriptions drive the subject. The **V-description** lists the extreme points (vertices); the **H-description** lists the defining linear inequalities (facets). This chapter develops the dictionary between them, the combinatorial structure they share (the *face lattice*), the extremal question of how many facets a polytope with n vertices can have, and the algorithms that compute hulls and enumerate vertices in three and higher dimensions. A running thread is **polarity**, the duality that interchanges vertices with facets and V- with H-descriptions (Section 16.10), closely related to the point–hyperplane duality of Chapter 15.

The subject has classical roots: Euler’s 1758 counting formula $V - E + F = 2$ [Eul58] predates the modern computational questions by two centuries, and the affine theorems of Carathéodory, Helly, and Radon (Sections 16.6–16.8) organize higher-dimensional convexity into three clean statements. Computationally, the three-dimensional hull is built in $O(n \log n)$ time, while the higher-dimensional theory is governed by the upper bound theorem: a d -polytope with n vertices has $\Theta(n^{\lfloor d/2 \rfloor})$ facets in the worst case (Section 16.3), separating the tractable three-dimensional case from the combinatorial explosion of dimension four and above.

16.2 V-Representations and H-Representations

Every polytope admits two canonical descriptions, each with its own data structure and its own algorithmic tasks. The **V-representation** specifies the polytope as the convex hull of a finite point set; the **H-representation** specifies it as the intersection of a finite set of halfspaces. The fundamental structural theorem states that the two coincide.

Definition 16.2 (V-Polytope and H-Polytope). A **V-polytope** is the convex hull $\text{conv}(S) = \{\sum_i \lambda_i p_i : p_i \in S, \lambda_i \geq 0, \sum_i \lambda_i = 1\}$ of a finite set $S \subset \mathbb{R}^d$ (Section 2.5). An **H-polytope** is a bounded set of the form $\{x : Ax \leq b\}$, a finite intersection of closed halfspaces (Section 2.9).

Theorem 16.3 (Minkowski–Weyl Theorem). *A subset of $\mathbb{R}^d$ is a V-polytope if and only if it is an H-polytope.*

The proof has two halves. The convex hull of finitely many points is the intersection of the halfspaces supporting it (Theorem 16.13), so it is an H-polytope. Conversely, every point of the bounded polyhedron $\{x : Ax \leq b\}$ is a convex combination of finitely many extreme points: move from x along a line to the boundary, recurse on the lower-dimensional faces, and the dimension drops at every step. The equivalence is treated in [Ede87, PS85].

Definition 16.4 (Extreme Point). A point p of a convex set C is **extreme** (a **vertex**) if it cannot be written as a nontrivial convex combination of two distinct points of C ; the vertex set is $\text{vert}(P)$.

The standard examples organize the picture. The **simplex** $\Delta_d = \text{conv}\{e_1, \dots, e_{d+1}\}$ has $d + 1$ vertices and $d + 1$ facets, the fewest possible. The **cube** $[0, 1]^d$ has 2^d vertices and $2d$ facets. The **cross-polytope** $\text{conv}\{\pm e_1, \dots, \pm e_d\}$ has $2d$ vertices and 2^d facets; it is the polar dual of the cube (Section 16.10). The **cyclic polytope**, the convex hull of points on the moment curve $\gamma(t) = (t, t^2, \dots, t^d)$, maximizes the number of faces among all d -polytopes with the same number of vertices (Theorem 16.10).

Each description has its own hard problem. Given a V-representation, computing the facets is **facet enumeration**; given an H-representation, computing the vertices is **vertex enumeration** (Definition 16.33); both are the same problem up to polarity and are treated in Section 16.11. Deciding whether a candidate point is a vertex, and evaluating a linear objective, are the tasks of Chapter 17. The point–hyperplane duality of Section 15.5 converts convex hull problems into halfplane intersection problems, a transformation whose power is analyzed in [CGL85].

16.3 Faces and Face Lattices

Definition 16.5 (Faces). A **face** of a polytope P is the intersection $P \cap H$ with a supporting hyperplane H (Section 16.4), together with P itself and the empty set. A face of dimension k is a **k -face**: 0-faces are **vertices**, 1-faces are **edges**, $(d - 2)$ -faces are **ridges**, and $(d - 1)$ -faces are **facets**. Faces other than P itself are **proper faces**.

Definition 16.6 (Face Lattice). The **face lattice** $\mathcal{L}(P)$ is the set of all faces of P ordered by inclusion, with the empty face at the bottom and P at the top.

Theorem 16.7 (Face Lattice Structure). *The face lattice of a d -polytope is a finite **graded lattice** of rank $d + 1$: the meet of two faces is their intersection, the join is the convex hull of their union, and the rank of a face is its dimension plus one. Every face is the join of its vertices.*

The face lattice is what graphics, mesh processing, and topological algorithms pass around implicitly: the half-edge and winged-edge data structures of Chapter 21 store the rank-two incidences of a three-dimensional polytope, and the Betti-number checks of Chapter 30 read the topological invariants of the boundary off the lattice. Because the boundary of a 3-polytope is a planar graph, the lattice is equivalently that planar graph together with its embedding; by Steinitz’s theorem the planar 3-connected graphs are exactly the graphs of convex polytopes in $\mathbb{R}^3$.

Definition 16.8 (Simple and Simplicial Polytopes). A d -polytope is **simplicial** if all its facets are simplices (in $\mathbb{R}^3$: all facets are triangles), and **simple** if every vertex has degree d . Polarity interchanges the two classes (Theorem 16.28).

16.3.1 Combinatorial complexity

How many facets can a polytope with n vertices have? In $\mathbb{R}^3$ the answer is linear — at most $2n - 4$ — by Euler's formula (Theorem 16.22). In higher dimensions the answer is polynomial in n with a dimension-dependent exponent, and the extremal shapes are the cyclic polytopes.

Definition 16.9 (Cyclic Polytope). The **cyclic polytope** $C(d, n)$ is the convex hull of n distinct points on the moment curve $\gamma(t) = (t, t^2, \dots, t^d)$. It is **neighborly**: every set of at most $\lfloor d/2 \rfloor$ vertices spans a face.

Theorem 16.10 (Upper Bound Theorem). A d -polytope with n vertices has $O(n^{\lfloor d/2 \rfloor})$ facets, and the cyclic polytope $C(d, n)$ attains the bound: $\Phi(d, n) = \Theta(n^{\lfloor d/2 \rfloor})$. By polarity, a d -polytope with n facets has at most $\Theta(n^{\lfloor d/2 \rfloor})$ vertices.

The upper bound theorem settles the *size* of every higher-dimensional hull, hence the output complexity of every hull algorithm (Section 16.11). In $\mathbb{R}^3$ the hull is linear in size, which is why three-dimensional hull algorithms (Chapter 19) run in $O(n \log n)$ time. In $\mathbb{R}^4$ and above the output size $\Theta(n^{\lfloor d/2 \rfloor})$ is superlinear, so no algorithm can be linear in the input size alone. The combinatorial complexity of polytopes and arrangement cells, and its role in discrepancy lower bounds, are treated in [Mat99] and [Ede87].

The dual of a triangulated surface. The boundary of a simplicial 3-polytope is a triangulated sphere, and its dual graph — one node per triangular facet, edges across shared edges — is again planar.

Example 16.11 (Dual of a Triangulated Surface). A triangulated surface that bounds a convex polytope with V vertices has $F = 2V - 4$ triangles and $E = 3V - 6$ edges (Theorem 16.25). Its dual graph has F nodes, each of degree 3, and E edges; it is exactly the 1-skeleton of the *polar* polytope, which is simple (Theorem 16.28). Duality swaps vertices and facets, and $F = 2V - 4$ is the identity behind the linear-size storage of triangular meshes in Chapter 30.1.

16.4 Supporting Hyperplanes

Definition 16.12 (Supporting Hyperplane). A hyperplane $H = \{x : \langle x, u \rangle = c\}$ **supports** a polytope P if $\langle x, u \rangle \leq c$ for all $x \in P$ and equality holds for at least one point of P . Every face of P other than the empty face is exposed by a supporting hyperplane.

Theorem 16.13 (Supporting Hyperplane Theorem). Let P be a polytope and u any direction. The linear functional $x \mapsto \langle x, u \rangle$ attains its maximum on P in the face $F_P(u) = \{x \in P : \langle x, u \rangle = h_P(u)\}$, where

$$h_P(u) = \max_{x \in P} \langle x, u \rangle$$

is the **support function** of P in direction u . Every boundary point of P lies on a supporting hyperplane, and a face is a vertex iff it is the unique maximizer in some direction.

The support function h_P is exactly the support function of Section 18.4, and it packages the whole polytope: $P = \{x : \langle x, u \rangle \leq h_P(u) \text{ for all } u\}$ is the H-representation re-parametrized by directions. Evaluating $h_P(u)$ is the oracle behind the support-function algorithms of Chapter 18, and for an H-polytope it is a linear program, solved by the methods of Chapter 17.

The supporting hyperplanes organize the ambient space into the **normal fan** of the polytope.

Definition 16.14 (Normal Cone and Normal Fan). The **normal cone** of a face F of P is the cone of directions u for which $F_P(u) = F$. The **normal fan** $\mathcal{N}(P)$ is the collection of the normal cones of all faces; it is a complete fan of polyhedral cones whose union is $\mathbb{R}^d$, ordered by inclusion in exactly the reverse of the face lattice.

The normal fan is where polytopes meet arrangements and duality: each full-dimensional normal cone is a cell of the arrangement of the facet hyperplanes of the *polar* polytope (Section 16.10), and the point–hyperplane duality of Section 2.14 and Chapter 15 realizes the normal fan as a section of a hyperplane arrangement — the mechanism behind the duality-based hull and Voronoi algorithms [Ede87, CGL85]. For a convex polygon the normal fan is the subdivision of the plane by the vertex outward normals; for the cube $[0, 1]^d$ it is the arrangement of coordinate hyperplanes.

16.5 Separating Hyperplane Theorems

Separation theorems assert that disjoint convex sets can be told apart by a hyperplane; they are the geometric form of duality and the source of certificates in optimization.

Theorem 16.15 (Strict Separation). *Let $A, B \subset \mathbb{R}^d$ be disjoint compact convex sets. Then there is a vector u and a scalar c with $\langle a, u \rangle < c < \langle b, u \rangle$ for all $a \in A$ and $b \in B$: a **strictly separating hyperplane**.*

Proof. The distance $\text{dist}(A, B) = \min_{a \in A, b \in B} \|a - b\|$ is attained at (a^*, b^*) because the sets are compact (Section 18.3). Set $u = b^* - a^*$ and $c = \langle (a^* + b^*)/2, u \rangle$. If some $a \in A$ satisfied $\langle a, u \rangle \geq c$, the segment from a^* to a , which lies in A by convexity, would approach b^* more closely than a^* does, contradicting the minimality of $\|a^* - b^*\|$; symmetrically $\langle b, u \rangle \leq c$ is impossible for $b \in B$. $\square$

The affine separation theorem (Theorem 2.9 of Section 2.10) is the general formulation for arbitrary convex sets; strictness holds here because polytopes are compact. Separation underlies the separating axis theorem (Theorem 18.3): for two disjoint polytopes the separating hyperplane can be tilted until it supports both, yielding the finite set of candidate axes. It is also the certificate machinery of linear programming, made explicit by the Farkas lemma.

Theorem 16.16 (Farkas Lemma). *Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Exactly one of the following two systems is solvable:*

1. $Ax \leq b$ for some $x \in \mathbb{R}^n$;
2. $y \geq 0$, $y^\top A = 0$, and $y^\top b < 0$ for some $y \in \mathbb{R}^m$.

Farkas' lemma [Far02] is the infeasibility certificate for linear systems: when $Ax \leq b$ has no solution, the nonnegative combination y proves it, since multiplying the system by $y \geq 0$ gives $0 = y^\top Ax \leq y^\top b < 0$, a contradiction. It follows from strict separation applied to the polyhedron $\{Ax \leq b\}$ and the origin, and it is the primal–dual bridge behind the duality theory of Section 17.4. Section 16.11 returns to the lemma as the pivot test underlying polytope traversal and vertex enumeration.

16.6 Carathéodory's Theorem

Carathéodory's theorem bounds how many points of a set are needed to represent any point of its convex hull: the dimension d , not the size of the set, is all that matters.

Theorem 16.17 (Carathéodory). *Let $S \subset \mathbb{R}^d$ and let $x \in \text{conv}(S)$. Then x lies in the convex hull of at most $d + 1$ points of S .*

Proof. Write $x = \sum_{i=1}^k \lambda_i p_i$ with $p_i \in S$, $\lambda_i > 0$, $\sum \lambda_i = 1$, and k minimal. If $k > d + 1$, the points are affinely dependent (Section 2.4): there are μ_i , not all zero, with $\sum \mu_i p_i = 0$ and $\sum \mu_i = 0$. For small ε the coefficients $\lambda_i + \varepsilon \mu_i$ still sum to 1, are nonnegative, and represent the same point; choosing the largest $|\varepsilon|$ keeping them nonnegative kills at least one positive coefficient, contradicting minimality. Hence $k \leq d + 1$. $\square$

The bound is tight: the center of the simplex spanned by $d + 1$ affinely independent points requires all of them. The version stated here is also recorded as Theorem 2.7 of the mathematical background chapter.

The algorithmic content is that convex hulls are determined by their $(d + 1)$ -subsets: every point of $\text{conv}(S)$ lies in a d -simplex spanned by $d + 1$ points of S . This is why all hull algorithms of Section 16.11 restrict attention to $(d + 1)$ -tuples and $(d - 1)$ -faces: a facet of $\text{conv}(S)$ is spanned by d points of S and is a facet exactly when all other points lie on one side of its hyperplane. Carathéodory's theorem also supplies the $d + 1$ in Helly's and Radon's theorems (Sections 16.7 and 16.8), the two pillars of the local theory of higher-dimensional convexity.

16.7 Helly's Theorem

Theorem 16.18 (Helly). *Let $\mathcal{F} = \{C_1, \dots, C_m\}$ be a finite family of convex sets in $\mathbb{R}^d$. If every subfamily of $d + 1$ sets has nonempty intersection, then the entire family has nonempty intersection.*

Proof. Induct on the number of sets, assuming every proper subfamily intersects; then each $D_j = \bigcap_{i \neq j} C_i$ is nonempty, so choose $p_j \in D_j$. If $m \geq d + 2$, Radon's theorem (Section 16.8) gives a Radon partition $A \cup B$ of $\{p_1, \dots, p_m\}$ with $x \in \text{conv}(A) \cap \text{conv}(B)$. Every point of A lies in all C_i except its own index; by convexity $\text{conv}(A)$ lies in the intersection of all C_i except those indexed by A , and the analogous claim holds for B . The two subfamilies together cover all of $\mathcal{F}$, so x lies in every C_i . $\square$

The proof shows the precise relation of the companion theorems: Helly's $d + 1$ is Radon's $d + 2$ shifted by one, and Radon's theorem rests on affine dependence, the same fact underlying Carathéodory's bound. Helly's theorem is the finite combinatorial counterpart of the separation theorems: the family has empty intersection iff some $d + 1$ of its members already do, so infeasibility of a convex system is always witnessed by a small subsystem — at most $d + 1$ constraints. This is the geometry behind linear-programming infeasibility certificates (Section 16.5) and the low-dimensional witness principle of Chapter 17.

Example 16.19 (Helly in the Plane). For $d = 1$, Helly's theorem says that a family of intervals on the line with pairwise intersection has total intersection. For $d = 2$, a family of convex figures in the plane any three of which share a point share a common point — the form used to bound witnesses in planar piercing problems.

16.8 Radon's Theorem

Theorem 16.20 (Radon). *Any set of $d + 2$ points in $\mathbb{R}^d$ can be partitioned into two disjoint sets A and B such that $\text{conv}(A) \cap \text{conv}(B) \neq \emptyset$.*

Proof. The $d + 2$ points are affinely dependent (Section 2.4): there are α_i , not all zero, with $\sum \alpha_i p_i = 0$ and $\sum \alpha_i = 0$. Let A collect the points with $\alpha_i > 0$ and B the rest; both are nonempty. Then $\sum_{i \in A} \alpha_i = -\sum_{j \in B} \alpha_j = \sigma > 0$ and

$$x = \sum_{i \in A} \frac{\alpha_i}{\sigma} p_i = \sum_{j \in B} \frac{-\alpha_j}{\sigma} p_j$$

is a common point of $\text{conv}(A)$ and $\text{conv}(B)$. $\square$

Definition 16.21 (Radon Partition). A partition of a set of $d + 2$ points in $\mathbb{R}^d$ into two parts with intersecting convex hulls is a **Radon partition**, and the common point of the two hulls is a **Radon point** of the set.

Radon's theorem is sharp: $d + 1$ affinely independent points form a simplex, whose vertices cannot be separated into parts with intersecting hulls. In the plane, four points are either in convex position, the crossing diagonals giving the Radon partition, or one lies inside the triangle of the other three, the Radon point being that point itself. The partition read along the moment curve is the combinatorial seed of the neighborliness of cyclic polytopes (Definition 16.9).

Radon partitions are the local combinatorial step of higher-dimensional algorithms. In the randomized incremental hull of Section 16.11, the conflicted facets of a new point in $\mathbb{R}^3$ form a topological disk whose boundary (the horizon) is a single cycle — a planar Radon statement about the facets the point sees. Radon's theorem is also the engine of the proof of Helly's theorem above and underlies the perturbation arguments that put higher-dimensional inputs into general position.

16.9 Euler–Poincaré Relations

The oldest theorem of polytope combinatorics counts the faces of a convex polytope in $\mathbb{R}^3$.

Theorem 16.22 (Euler's Formula). *For every convex polytope in $\mathbb{R}^3$ with V vertices, E edges, and F facets,*

$$V - E + F = 2.$$

Euler announced the formula in 1758 [Eul58]; the modern view [Poi95] sees the boundary of the polytope as a triangulated sphere and $V - E + F$ as its Euler characteristic (compare Section 2.15 and Chapter 30). The d -dimensional generalization counts all dimensions at once.

Definition 16.23 (Face Vector). The **face vector** (f-vector) of a d -polytope is the tuple $(f_0, f_1, \dots, f_{d-1})$, where f_k is the number of k -faces.

Theorem 16.24 (Euler–Poincaré Formula). *For every d -polytope,*

$$\sum_{k=0}^{d-1} (-1)^k f_k = 1 + (-1)^{d-1} = \chi(S^{d-1}),$$

the Euler characteristic of the $(d - 1)$ -sphere, the boundary of the polytope.

For $d = 3$ this reads $f_0 - f_1 + f_2 = 2$; for $d = 2$ (a polygon) it reads $f_0 = f_1$; for $d = 4$ it reads $f_0 - f_1 + f_2 - f_3 = 0$. The formula is proved by induction on the dimension: attaching a new vertex to the boundary triangulation changes the alternating sum by 0 because the boundary is a triangulated sphere. It is the most useful check in higher-dimensional computing: every face lattice, mesh, and hull satisfies it, and violations signal exactly the degenerate inputs that robust implementations must reject.

For a **simplicial** 3-polytope, $3F = 2E$ (each facet has three edges, each edge borders two facets); for a **simple** one, $3V = 2E$. Substituting into Euler's formula gives the sharp counting bound.

Theorem 16.25 (Facets of a 3-Polytope). *A convex polytope in $\mathbb{R}^3$ with V vertices has $F \leq 2V - 4$ facets, with equality iff all facets are triangles; dually, a simple polytope with F facets has $V = 2F - 4$ vertices. A triangulated surface with V vertices has $E = 3V - 6$ edges and $F = 2V - 4$ triangular faces.*

These identities are why triangular meshes consume linear storage (Chapter 30.1): the edge and face counts of a triangular surface are determined by its vertex count up to constants. The dimensional generalization is the upper bound theorem (Theorem 16.10), whose linear three-dimensional case is exactly $F \leq 2V - 4$.

16.10 Polarity

Polarity is the geometric duality that interchanges the two descriptions of a polytope: the polar of a V-polytope is an H-polytope, and vertices and facets are exchanged along with them.

Definition 16.26 (Polar). For a set $P \subset \mathbb{R}^d$ with $0 \in P$, the **polar** of P is

$$P^\circ = \{y \in \mathbb{R}^d : \langle x, y \rangle \leq 1 \text{ for all } x \in P\}.$$

Theorem 16.27 (Polar Duality). *If P is a full-dimensional polytope with $0 \in \text{int } P$, then P° is again a full-dimensional polytope with $0 \in \text{int } P^\circ$, and polarity is an involution: $(P^\circ)^\circ = P$.*

The theorem follows from the Minkowski–Weyl theorem: if $P = \text{conv}\{v_1, \dots, v_n\}$, then $P^\circ = \{y : \langle v_i, y \rangle \leq 1 \text{ for all } i\}$, an H-polytope, and applying the step again returns P by separation (Section 16.5). Polarity therefore converts every V-representation into an H-representation and back, which makes it the geometric engine of facet and vertex enumeration (Section 16.11). It is related to, but distinct from, the projective point–hyperplane duality of Section 2.14 and Chapter 15: both interchange points and hyperplanes, but polarity is an involution on polytopes that preserves incidence, whereas projective duality reverses order. The computational power of the projective form is analyzed in [CGL85].

Theorem 16.28 (Polarity Reverses the Face Lattice). *Let P be a full-dimensional polytope with $0 \in \text{int } P$. Polarity induces an inclusion-reversing bijection between the faces of P and of P° : the polar of a k -face F of P is the $(d - 1 - k)$ -face*

$$\hat{F} = \{y \in P^\circ : \langle x, y \rangle = 1 \text{ for all } x \in F\}.$$

In particular, facets of P correspond to vertices of P° , and vertices of P to facets of P° .

The dimension reversal $k \leftrightarrow d - 1 - k$ is why a simplicial polytope becomes a simple one under polarity (Definition 16.8), and the normal fan of P is the face fan of P° — the normal cone of F is the cone over the polar face $\hat{F}$ — so the normal-fan picture of Section 16.4 and the polarity picture coincide. The polar of the cube is the cross-polytope, the polar of a simplex is a simplex, and the polar of the icosahedron is the dodecahedron.

Example 16.29 (Polar of the Cube). Let $P = [-1, 1]^3$. Then $P^\circ = \{y : |y_1| + |y_2| + |y_3| \leq 1\}$ is the octahedron with vertices $\pm e_1, \pm e_2, \pm e_3$: the eight vertices of P become the eight facets of P° , and the six facets of P become the six vertices of P° .

16.11 Computing Higher-Dimensional Hulls

16.11.1 1. Overview

This section treats the computation of convex hulls in $\mathbb{R}^3$ and above, together with the dual traversal problems of vertex and facet enumeration. Three families of algorithms are central.

Divide and conquer. Preparata and Hong [PH77] split the point set by a plane, build the two subhulls recursively, and glue them by a “belt” of triangles wrapping around the union, giving a deterministic $O(n \log n)$ three-dimensional hull algorithm.

Randomized incremental. Add the points in random order, maintaining a conflict graph between points and current facets; a new point either lies inside the hull or destroys a connected patch of facets that is replaced by new triangles. The paradigm is due to Clarkson and Shor [CS89] and Seidel [Sei91], who established the expected $O(n \log n)$ bound in three dimensions; the same machinery extends to fixed dimension $d \geq 4$, where its complexity is governed by the upper bound theorem.

Output-sensitive. Gift wrapping [CK70] revolves a plane around an edge to generate the next facet, in $O(nh)$ time for a hull of h facets; Chan’s algorithm [Cha96] achieves the optimal $O(n \log h)$ by wrapping hulls of random samples.

The dual computational problem is **vertex enumeration**: given the H-representation, list the vertices. Traversing the adjacency graph of vertices by pivoting — moving along an edge to improve a linear objective — is the combinatorial face of the simplex method (Chapter 17), and its correctness rests on the Farkas lemma (Theorem 16.16) and the Fourier–Motzkin elimination theorem (Theorem 16.36). The section closes with these connections, which unify polytope theory with linear programming.

16.11.2 2. Definitions

Definition 16.30 (Adjacency Graph). The **adjacency graph** (1-skeleton) of a polytope is the graph whose vertices are the vertices of P and whose edges are the 1-faces of P . For a simple polytope the graph is d -regular, and for a 3-polytope it is planar and 3-connected (Steinitz’s theorem).

Definition 16.31 (Conflict and Conflict Graph). A point p **conflicts** with a facet f of $\text{conv}(S)$ if p lies strictly outside the closed halfspace bounded by f that contains $\text{conv}(S)$. The **conflict graph** stores, for every facet, the input points conflicting with it, and for every point, the conflicting facets.

Definition 16.32 (Horizon). Let p be a point outside the current hull. The **horizon** of p is the cycle of edges bounding the union of the facets that conflict with p ; the facets spanned by p and the horizon edges are the new facets created by inserting p .

Definition 16.33 (Vertex Enumeration). **Vertex enumeration** is the problem of listing all vertices of the H-polytope $P = \{x : Ax \leq b\}$; **facet enumeration** is the dual problem of listing all facets of $P = \text{conv}(S)$. By polarity (Section 16.10) the two problems are equivalent.

The algorithms assume **general position**: no four input points are coplanar in $\mathbb{R}^3$ (no $d+1$ in $\mathbb{R}^d$), so every facet is a simplex spanned by exactly d points. Degenerate inputs are handled by symbolic perturbation or by the exact predicates of Section 3.4.

16.11.3 3. Theory

Divide and conquer in 3D. Preparata–Hong [PH77] recursively computes the hulls H_1, H_2 of the two halves, then merges them by finding the two supporting planes that touch both hulls, tracing the “visible” band of edges on each, and connecting the bands by a belt of triangular facets. Each merge runs in $O(n)$ time because the hulls are convex and the bands are found by walking, giving the recurrence $T(n) = 2T(n/2) + O(n)$ and $T(n) = O(n \log n)$. Correctness rests on the observation that every facet of the merged hull is either a facet of a subhull or a new triangle spanning the two bands, validated by the orientation tests of Section 19.2.

Randomized incremental. The randomized incremental algorithm (Algorithm 70) inserts the points in a random order and maintains the conflict graph. When p_i is inserted, its conflicting facets are deleted and replaced by the fan of triangles over its horizon; the key geometric fact is that the conflicting facets form a topological disk in $\mathbb{R}^3$, so the horizon is a single cycle. The analysis is by **backward analysis**: a facet created at step i survives until the insertion of a point conflicting with it, and summing the expected creation times bounds the total work.

Theorem 16.34 (Randomized Incremental Hull, 3D). *For n points in general position in $\mathbb{R}^3$, the randomized incremental algorithm constructs the convex hull in expected $O(n \log n)$ time and $O(n)$ space, and the expected number of facets created over the run is $O(n)$.*

The accounting is in [CS89, dBCvKO08]. The same scheme works in higher dimensions, where it is governed by the upper bound theorem.

Theorem 16.35 (Randomized Incremental Hull, Higher Dimensions). *For points in general position in fixed dimension $d \geq 4$, the randomized incremental algorithm runs in expected $O(n^{\lfloor d/2 \rfloor} \log n)$ time, and this is optimal in the worst case: the output hull can have $\Theta(n^{\lfloor d/2 \rfloor})$ facets (Theorem 16.10). Chazelle’s deterministic algorithm achieves $O(n^{\lfloor d/2 \rfloor} + n \log n)$ [Cha93].*

Output-sensitive algorithms. When the hull is small, both of the above are wasteful. **Gift wrapping** [CK70] (Algorithm 71) generates the facets one at a time by pivoting a supporting plane around each facet edge, choosing the point minimizing the angle with the current facet; each facet costs $O(n)$, giving $O(nh)$ time. Chan’s algorithm [Cha96] computes the hulls of random samples of size h and wraps their union, iterating on the estimate of h , in $O(n \log h)$ time, the optimal output-sensitive bound (the planar version is Section 4.3, Algorithm 4).

Traversal and vertex enumeration. Given an H-representation, the vertices can be visited by traversing the adjacency graph: from a vertex, each admissible pivoting direction leads across an edge to an adjacent vertex, and the simplex method of Chapter 17 performs exactly this traversal under a linear objective. Two classical exact tools are complementary. **Fourier–Motzkin elimination** eliminates one variable at a time by combining every pair of constraints with opposite coefficient signs, deciding feasibility and projecting polytopes exactly.

Theorem 16.36 (Fourier–Motzkin Elimination). *Projecting the polyhedron $P = \{x : Ax \leq b\} \subset \mathbb{R}^d$ onto $\mathbb{R}^{d-1}$ (deleting one coordinate) yields the polyhedron defined by the linear system produced by Algorithm 72. In particular P is feasible iff the eliminated system is feasible, so feasibility is decided by iterated elimination.*

The **double description** method [MRTT53] maintains a V-and-H pair (V, H) for the same polytope, inserting halfspaces one at a time and adding the vertices each new halfspace cuts off; the Farkas lemma certifies that a candidate point is a new vertex. For output-polynomial guarantees in the worst case, the **reverse search** of Avis and Fukuda [AF92] walks the adjacency graph in a deterministic cycle-free order (each vertex points to its unique parent), enumerating every vertex in time polynomial in the input size per vertex. Together, the Farkas lemma and Fourier–Motzkin elimination are the polytopal form of linear-programming duality: feasibility, infeasibility certificates, and vertex description are one story, told in the language of Section 17.2.

16.11.4 4. Pseudocode

16.11.5 5. Parameter Selection

- **Random permutation:** correctness is independent of the order, but the expected time depends on randomness; a fixed seed gives reproducible runs, and adversarial orders are the failure mode that randomization avoids.
- **General position:** coplanar points break the “horizon is a cycle” and “facet is a triangle” invariants; symbolic perturbation restores the assumptions without moving the output.
- **Conflict maintenance:** the full conflict graph uses expected $O(n)$ space in 3D and is the standard choice; alternatively, locate each new point inside a facet by walking (Section 19.3), trading space for the point-location cost.
- **Output-sensitivity:** for hulls small compared with the input, gift wrapping or Chan’s algorithm should be selected on the basis of an estimate of h , the number of output facets.
- **Exact arithmetic:** orientation and incircle-style tests (Sections 19.2 and 3.4) must be exact or certified; a wrong orientation sign converts a valid facet into a non-facet and corrupts the entire output.

Algorithm 70 Randomized Incremental 3D Convex Hull

Input: A point set $S \subset \mathbb{R}^3$ in general position.**Output:** The facets of $\text{conv}(S)$, oriented so normals point outward.

```

1: function INCREMENTALHULL( $S$ )
2:   choose a random permutation  $p_1, \dots, p_n$  of  $S$ 
3:    $\mathcal{F} \leftarrow$  the facets of the hull of the first 4 points (a tetrahedron)
4:   build the conflict lists of all points against  $\mathcal{F}$ 
5:   for  $i = 5$  to  $n$  do
6:      $\mathcal{C} \leftarrow$  facets of  $\mathcal{F}$  conflicting with  $p_i$ 
7:     if  $\mathcal{C}$  is empty then
8:       continue  $\triangleright p_i$  lies inside the current hull
9:     end if
10:    delete  $\mathcal{C}$  from  $\mathcal{F}$ ; trace the horizon cycle bounding  $\mathcal{C}$ 
11:    for each horizon edge  $e$  do
12:      add the triangle  $\text{conv}(\{p_i\} \cup e)$  to  $\mathcal{F}$ 
13:    end for
14:    update the conflict lists of the remaining points against the new facets
15:  end for
16:  return  $\mathcal{F}$ 
17: end function

```

Algorithm 71 Gift Wrapping in 3D

Input: A point set $S \subset \mathbb{R}^3$ in general position and a facet f_0 of $\text{conv}(S)$.**Output:** All facets of $\text{conv}(S)$.

```

1: function GIFTWRAP( $S, f_0$ )
2:    $\mathcal{F} \leftarrow \emptyset$ ;  $Q \leftarrow \{f_0\}$ ; mark  $f_0$  discovered
3:   while  $Q$  is not empty do
4:     pop  $f$  from  $Q$ ; add  $f$  to  $\mathcal{F}$ 
5:     for each edge  $e$  of  $f$  do
6:       if  $e$  is shared with a discovered facet then
7:         continue  $\triangleright$  the edge is already wrapped
8:       end if
9:        $H \leftarrow$  plane through  $e$  and the third vertex of  $f$ 
10:       $p \leftarrow$  point of  $S$  minimizing the angle of  $\text{conv}(\{e\} \cup \{p\})$  with  $H$ 
11:       $g \leftarrow \text{conv}(\{e\} \cup \{p\})$ ; mark  $g$  discovered; push  $g$  onto  $Q$ 
12:    end for
13:  end while
14:  return  $\mathcal{F}$ 
15: end function

```

Algorithm 72 Fourier–Motzkin Elimination of One Variable**Input:** A system $Ax \leq b$ in variables $x_1, \dots, x_d$ and an index j .**Output:** A system in the remaining $d - 1$ variables whose solutions are exactly the projections of the solutions of $Ax \leq b$.

```

1: function FOURIERMOTZKIN( $A, b, j$ )
2:   partition the rows by the sign of  $\alpha_i = A_{ij}$ , the coefficient of  $x_j$ 
3:    $L \leftarrow$  rows with  $\alpha_i > 0$ ;  $E \leftarrow$  rows with  $\alpha_i = 0$ ;  $U \leftarrow$  rows with  $\alpha_i < 0$ 
4:   for each pair  $(\ell, u)$  of a row from  $L$  and a row from  $U$  do
5:     add the scaled sum of the two rows that cancels  $x_j$ 
6:   end for
7:   return the system of all rows of  $E$  plus the combination rows
8: end function

```

16.11.6 6. Complexity

- **Preparata–Hong divide and conquer:** $O(n \log n)$ time, $O(n)$ space, deterministic, in $\mathbb{R}^3$.
- **Randomized incremental:** expected $O(n \log n)$ time and $O(n)$ space in $\mathbb{R}^3$ (Theorem 16.34); expected $O(n^{\lfloor d/2 \rfloor} \log n)$ time in fixed dimension $d \geq 4$ (Theorem 16.35).
- **Gift wrapping:** $O(nh)$ time for a hull with h facets, $O(n + h)$ space.
- **Chan:** $O(n \log h)$ time, the optimal output-sensitive bound [Cha96].
- **Vertex enumeration:** output-polynomial by reverse search [AF92] — each vertex in time polynomial in the input size; Fourier–Motzkin decides feasibility exactly but may square the constraint count per elimination step (Theorem 16.36).

The lower bounds are information-theoretic: the output hull has up to $\Theta(n^{\lfloor d/2 \rfloor})$ facets (Theorem 16.10), so every correct algorithm in dimension $d \geq 4$ takes $\Omega(n^{\lfloor d/2 \rfloor})$ time in the worst case, which the algorithms above attain.

16.11.7 7. Usages

- **Mesh generation and geometric computing:** the three-dimensional hull is the boundary representation of convex meshes and of the Delaunay and Voronoi structures of Chapter 12; the algorithms here are the ancestors of the incremental polyhedral constructions of Chapter 19.
- **Linear programming and optimization:** vertex enumeration and the Farkas/Fourier–Motzkin machinery are the geometric skeleton of the simplex method and of the duality certificates of Chapter 17 (Section 17.2).
- **Higher-dimensional data:** hulls and support functions of high-dimensional point clouds are primitives of the statistics and learning pipelines of Chapter 39.
- **Visibility and duality:** point–hyperplane duality turns convex hull problems into halfplane intersection problems (Section 15.5), so the algorithms here solve both.

16.11.8 8. Testing Methods and Edge Cases

- **Euler check:** verify $V - E + F = 2$ on every output hull (Theorem 16.22); for a triangulated hull, verify $F = 2V - 4$. Any violation signals a duplicate or a mis-oriented facet.

- **Dual agreement:** build the polar of the output and verify the face-lattice reversal of Theorem 16.28, then run vertex enumeration on the polar to recover the input vertices.
- **Algorithm agreement:** compare the facet sets produced by the randomized incremental and gift-wrapping algorithms on the same inputs; they must coincide up to orientation and order.
- **Degenerate inputs:** collinear and coplanar sets, duplicate points, and points on the hull boundary must be absorbed by the general-position convention and must not create zero-area facets.
- **Conflict consistency:** after every insertion, the removed facets must be exactly the conflict set of p_i ; assert that the horizon is a single cycle and that no new facet conflicts with an unprocessed point that does not see it.
- **Stress dimensions:** in $\mathbb{R}^4$, hulls of points on the moment curve must realize the upper bound $\Theta(n^{\lfloor d/2 \rfloor})$ facets, testing the output-size handling of the higher-dimensional machinery.

16.11.9 9. References

The divide-and-conquer hull is due to Preparata and Hong [PH77], with the full treatment in [PS85]; the randomized incremental paradigm and its analysis to Clarkson and Shor [CS89] and Seidel [Sei91], with the textbook presentation in [dBCvKO08]. Gift wrapping is due to Chand and Kapur [CK70], and Chan’s output-sensitive algorithm to Chan [Cha96]. The combinatorial background — the upper bound theorem and the structure of face lattices — is in [Ede87, Mat99], and the deterministic optimal algorithm in fixed dimension in [Cha93]. Vertex enumeration by double description is due to Motzkin et al. [MRTT53], its output-polynomial reformulation to Avis and Fukuda [AF92], and the Farkas lemma to Farkas [Far02]; its linear-programming consequences are the subject of Chapter 17.

16.12 Exercises

1. Prove the Minkowski–Weyl theorem in the plane: show that a bounded intersection of halfplanes is the convex hull of its vertices, and conversely that the convex hull of a finite planar point set is a bounded intersection of halfplanes.
2. Using Euler’s formula, prove that a 3-polytope has $F \leq 2V - 4$ facets, with equality iff all facets are triangles, and derive the dual statement $V = 2F - 4$ for simple polytopes.
3. State and prove Radon’s theorem for four points in the plane explicitly: the four points either form a convex quadrilateral, whose crossing diagonals give the Radon partition, or have one point inside the triangle of the other three.
4. Compute the polar of the cube $[-1, 1]^3$ and verify (i) that it is the octahedron, (ii) that $(P^\circ)^\circ = P$, and (iii) the face-lattice reversal of Theorem 16.28 by matching facets of P to vertices of P° .
5. Prove Carathéodory’s theorem by the minimal-convex-combination argument, and show the bound $d + 1$ is tight by exhibiting a point of a d -simplex that cannot be represented with fewer points.
6. Use Helly’s theorem to prove that a finite family of convex sets in $\mathbb{R}^d$ has empty intersection iff some $d + 1$ members already do, and derive from this the size of the infeasibility certificate for a linear system in $\mathbb{R}^2$.

7. Simulate Algorithm 70 by hand on six points in general position in $\mathbb{R}^3$, inserting them in a fixed order, and verify that the conflicted facets form a single cycle (the horizon) and that $V - E + F = 2$ holds after every step.
8. Implement Algorithm 72 for $d = 2$ and use it to decide feasibility of a system of halfplanes; compare your answers with the halfplane intersection algorithm of Section 17.2.

Chapter 17

Linear Programming in Geometry

Linear programming provides a common language for geometric optimization. Many geometric problems ask for a point satisfying linear inequalities while maximizing or minimizing a linear objective. In two dimensions, the feasible region is an intersection of half-planes and therefore a convex polygon.

17.1 Geometric Linear Programs

A linear program has the form

$$\text{maximize } c^T x \quad \text{subject to} \quad Ax \leq b.$$

Each row of $Ax \leq b$ defines a half-plane. The feasible set is convex, and when it is non-empty and bounded, an optimum occurs at a vertex of the feasible polytope. In two dimensions this reduces optimization to a convex polygon; in three dimensions it reduces to a convex polyhedron.

Applications include smallest enclosing regions, collision separation, supporting-line queries, facility placement, and motion-planning constraints.

17.2 Half-Plane Intersection

For an oriented line through points p and q , the left half-plane is

$$H(p, q) = \{x \in \mathbb{R}^2 : \text{orient2d}(p, q, x) \geq 0\}.$$

The intersection of a finite collection of half-planes is a convex polygon, possibly empty or unbounded. A bounding box can be added when a finite output polygon is required.

One robust approach sorts the boundary lines by direction and maintains a deque of candidate lines. Whenever the intersection of the last two lines is outside the next half-plane, the last line is removed. A symmetric check is performed at the front of the deque.

After sorting, the deque algorithm takes $O(n)$ time and $O(n)$ space. Parallel or coincident boundary lines require a consistent rule: retain the more restrictive half-plane when their feasible sides agree, and report an empty intersection when they conflict.

17.3 Low-Dimensional Linear Programming

In fixed dimension, randomized incremental algorithms solve linear programs efficiently by processing constraints in random order. When a new constraint violates the current optimum, the optimum must lie on that constraint's boundary, reducing the problem dimension by one.

Algorithm 73 Half-Plane Intersection

```
1: function HALFPLANEINTERSECTION( $H$ )
2:   Sort half-planes by boundary direction
3:    $D \leftarrow$  empty deque
4:   for  $h$  in  $H$  do
5:     while  $|D| \geq 2$  and intersection of the last two lines is outside  $h$  do
6:       Remove the last half-plane from  $D$ 
7:     end while
8:     while  $|D| \geq 2$  and intersection of the first two lines is outside  $h$  do
9:       Remove the first half-plane from  $D$ 
10:    end while
11:    Append  $h$  to  $D$ 
12:  end for
13:  while  $|D| \geq 3$  and the last intersection is outside the first half-plane do
14:    Remove the last half-plane from  $D$ 
15:  end while
16:  while  $|D| \geq 3$  and the first intersection is outside the last half-plane do
17:    Remove the first half-plane from  $D$ 
18:  end while
19:  return polygon formed by consecutive boundary intersections
20: end function
```

This gives expected linear time for two-dimensional linear programming and expected linear time for fixed-dimensional LP under the usual boundedness and constant-dimension assumptions.

The same boundary-reduction idea appears in smallest-enclosing-circle algorithms, support-point queries, and randomized convex-hull construction. It is therefore a useful bridge between predicates, convexity, optimization, and geometric data structures.

17.4 Duality and Applications

Point-line duality maps a point (a, b) to a line such as $y = ax - b$. Under a consistent duality transform, incidence and above/below relationships are preserved or reversed in a controlled way. This converts convex-hull and half-plane questions into upper- or lower-envelope questions.

Linear constraints also describe collision-free configuration-space regions, visibility restrictions, separating planes, and feasible placement regions. Numerical implementations should use the robust orientation predicates from Chapter 3 when testing boundary intersections.

Chapter 18

Intersection and Distance of Convex Objects

18.1 Convex Polygon Intersection

18.1.1 1. Overview

Given two convex polygons P and Q in the plane, the intersection problem asks whether $P \cap Q$ is empty and, if not, returns the convex polygon $P \cap Q$. Because the inputs are convex, the intersection is either empty or a convex polygon with at most $|P| + |Q|$ vertices, and it can be computed in linear time. The general (non-convex) polygon intersection problem is harder and is handled by the segment-intersection and overlay machinery of Chapter 6; convexity is what makes this chapter's results possible.

18.1.2 2. Definitions

Definition 18.1 (Polygon Intersection). The **intersection** of two convex polygons P and Q is the set $P \cap Q$, which is empty or a convex polygon. Its vertices are either vertices of P or Q inside the other polygon, or crossing points of the two boundaries.

18.1.3 3. Theory

Two classic facts underlie the linear-time algorithms. First, since the boundaries of two convex polygons cross at most $|P| + |Q|$ times, the intersection polygon has $O(|P| + |Q|)$ vertices. Second, the two boundaries can be merged by a plane sweep in a single pass because each boundary is sorted around its hull; this is the basis of the $O(n)$ algorithm of Shamos [PS85]. The distance version of the same problem — finding the closest points of two convex polygons — is solved in $O(\log n)$ after linear preprocessing by the Dobkin–Kirkpatrick hierarchy [DK83, DK85], and in $O(n)$ by rotating calipers [Tou83].

18.1.4 4. Pseudocode

18.1.5 5. Parameter Selection

- **Orientation convention:** both polygons must be traversed in the same orientation (counterclockwise); a clockwise input must be reversed first.
- **Degeneracies:** touching edges (one vertex on an edge of the other) and coincident edges must be handled by the exact predicate discipline of Section 3.4, and reported once.

Algorithm 74 Intersection of Two Convex Polygons**Input:** Two convex polygons P, Q with vertices in counterclockwise order.**Output:** The convex polygon $P \cap Q$, or empty.

```

1: function CONVEXPOLYGONINTERSECTION( $P, Q$ )
2:    $R \leftarrow \emptyset$ ;  $i \leftarrow 0$ ;  $j \leftarrow 0$ ;  $n \leftarrow |P|$ ;  $m \leftarrow |Q|$ 
3:   while  $i < n$  or  $j < m$  do
4:     test edge  $e_P = (P_i, P_{i+1})$  against edge  $e_Q = (Q_j, Q_{j+1})$  using orient()
5:     add crossing points of  $e_P, e_Q$  to  $R$ 
6:     advance the edge whose supporting line the other polygon is “ahead of” ( $i$  or  $j$ )
7:     add the endpoint that lies inside the other polygon to  $R$ 
8:   end while
9:   output the vertices of  $R$  in order as the intersection polygon
10: end function

```

18.1.6 6. Complexity

- **Time:** $O(n + m)$.
- **Space:** $O(n + m)$ for the output (which has at most $n + m$ vertices).

18.1.7 7. Usages

- **Boolean geometry:** union/intersection/difference of convex parts (Chapter 6).
- **Windowing and culling:** computing the visible overlap of two convex windows.

18.1.8 8. Testing Methods and Edge Cases

- **Separation:** disjoint polygons with large and small gaps.
- **Containment:** $Q \subset P$, $P \subset Q$, and identical polygons.
- **Boundary contact:** polygons sharing an edge, a vertex, or a vertex on an edge.

18.1.9 9. References

The linear intersection algorithm is due to Shamos (1978 thesis) and is presented in [PS85]; the logarithmic distance and intersection queries follow from the Dobkin–Kirkpatrick hierarchy [DK83, DK85] and the rotating-calipers technique of Toussaint [Tou83].

18.2 Separating Axis Theorem**18.2.1 1. Overview**

The **separating axis theorem** (SAT) gives a finite test for whether two convex polygons (in 2D) or convex polyhedra (in 3D) intersect: two convex sets are disjoint exactly when there is an axis such that their projections onto that axis are disjoint. For polygons and polyhedra, only a small finite set of axes — the face normals and, in 3D, the cross products of edge directions — must be tested. SAT is the fastest intersection test for boxes and for convex primitives in physics engines, and it underlies the separating plane reasoning of Section 2.10.

18.2.2 2. Definitions

Definition 18.2 (Projection Interval). The **projection** of a convex set C onto an axis u is the interval $[\min_{x \in C} \langle x, u \rangle, \max_{x \in C} \langle x, u \rangle]$. Two sets are **separated by axis** u if their projection intervals are disjoint.

18.2.3 3. Theory

Theorem 18.3 (Separating Axis Theorem). *Two convex polygons in the plane are disjoint iff their projections are disjoint on some axis perpendicular to one of their edges. Two convex polyhedra in $\mathbb{R}^3$ are disjoint iff their projections are disjoint on some axis perpendicular to one of their faces, or parallel to the cross product of one edge of each polyhedron.*

The theorem follows from the hyperplane-separation theorem of Section 2.10: the separating line (or plane) can be rotated until it supports both sets, at which point it is parallel to an edge (or a face, or an edge-edge cross product). For two axis-aligned boxes, the candidate axes are the three coordinate axes, which makes the box test three interval comparisons.

18.2.4 4. Pseudocode

Algorithm 75 Separating Axis Test for Two Convex Polygons

Input: Convex polygons P, Q .

Output: True iff $P \cap Q = \emptyset$.

```

1: function SATINTERSECT( $P, Q$ )
2:   for each edge  $e$  of  $P$  and each edge  $e$  of  $Q$  do
3:      $u \leftarrow$  unit normal of  $e$ 
4:      $I_P \leftarrow$  projection of  $P$  onto  $u$ ;  $I_Q \leftarrow$  projection of  $Q$  onto  $u$ 
5:     if  $I_P \cap I_Q = \emptyset$  then
6:       return FALSE
7:     end if
8:   end for
9:   return TRUE
10: end function
```

18.2.5 5. Parameter Selection

- **Test order:** in practice, short-circuit on the most discriminating axis first (e.g., the bounding-box axes), which exits early for most disjoint pairs.
- **3D:** the candidate set is all face normals plus all edge-pair cross products; for boxes this is 6 normals (only 3 distinct) plus 9 cross products, but only 3 are distinct.

18.2.6 6. Complexity

- **Time:** $O(n + m)$ with n, m the numbers of vertices; for boxes, $O(1)$.
- **Space:** $O(1)$.

18.2.7 7. Usages

- **Physics engines:** box-box and box-convex tests in real-time physics use SAT because of its tiny constant and early exit.
- **Culling:** as a fast filter before the precise (GJK) test of Section 18.6.

18.2.8 8. Testing Methods and Edge Cases

- **Touching configurations:** verify that touching (projections sharing an endpoint) is treated consistently as intersecting or not.
- **Degenerate polygons:** collinear vertices produce zero-length edges and must be filtered before computing normals.

18.2.9 9. References

SAT is folklore in the game-physics community and is developed carefully in [Eri04, vdB04]; its theoretical basis is the separation theorem of Section 2.10 and [dBCvKO08].

18.3 Distance between Convex Sets

The **distance** between two convex sets A and B is $\text{dist}(A, B) = \min_{a \in A, b \in B} \|a - b\|$. The minimum is attained because the sets are closed; the minimizers are the **closest points** of A and B . For convex polygons in the plane, the closest pair is found in $O(n)$ time by rotating calipers [Tou83], and in $O(\log n)$ time after $O(n)$ preprocessing by the Dobkin–Kirkpatrick method [DK85]. For convex polyhedra in $\mathbb{R}^3$ the same hierarchy gives an $O(\log^2 n)$ distance query, while the iterative GJK algorithm of Section 18.6 computes the distance by support-function evaluations without any preprocessing.

Distance computations between convex sets are the primitive of collision response (how far must the objects move apart?), of motion planning (Chapter 26), and of the minimum-separation checks in Section 18.7. Two structural facts are used throughout:

- the distance between A and B equals the distance from the origin to the Minkowski difference $A \ominus B$ defined in Section 18.5;
- the closest points are found by projecting onto the supporting plane of the nearest feature (vertex, edge, or face), which is exactly the geometric intuition exploited by GJK and by the Gilbert–Johnson–Keerthi family of algorithms.

18.4 Support Functions

Definition 18.4 (Support Function and Support Point). For a convex set C and a direction u , the **support point** $s_C(u)$ is a point of C maximizing $\langle x, u \rangle$ over $x \in C$. The **support function** $h_C(u) = \langle s_C(u), u \rangle$ is the maximum of the linear functional in direction u .

Support points are the fundamental oracle for convex intersection and distance algorithms, because many questions about C reduce to evaluating s_C in a few directions and s_C is easy to compute for every convex primitive:

- **box:** $s(u) = (\text{sign}(u_x)L_x, \dots)$, one comparison per coordinate;
- **polygon:** $O(\log n)$ by binary search over the edge directions, or $O(n)$ by scanning;
- **sphere/ellipsoid:** $s(u) = c + Ru/\|u\|$;
- **Minkowski sum:** $s_{A \oplus B}(u) = s_A(u) + s_B(u)$.

The last identity is crucial: the support point of a Minkowski sum is the sum of the support points of the summands, which is how GJK computes support on the difference object $A \ominus B = A \oplus (-B)$ without ever building it explicitly (Section 18.5). Support functions also make the distance-to-origin queries of Section 18.3 well defined for any convex body, including those with smooth boundary, and they are the basis of the Dobkin–Kirkpatrick hierarchy [DK83].

18.5 Minkowski Difference

The **Minkowski difference** (or Minkowski sum of A with $-B$) is

$$A \ominus B = A \oplus (-B) = \{a - b : a \in A, b \in B\}.$$

The Minkowski difference reformulates every intersection and distance question about A and B as a question about the origin and $A \ominus B$:

Theorem 18.5 (Minkowski Difference Reductions). *For convex sets A, B :*

- $A \cap B \neq \emptyset$ iff $\mathbf{0} \in A \ominus B$;
- $\text{dist}(A, B) = \text{dist}(\mathbf{0}, A \ominus B)$;
- if $\mathbf{0} \notin A \ominus B$, the penetration depth of Section 18.7 is the distance from the origin to the boundary of $A \ominus B$.

If A and B are convex, so is $A \ominus B$. The boundary of $A \ominus B$ is made of translated copies of features of A and B (vertices, edges, faces), which is what GJK's simplex search exploits: the closest point of $A \ominus B$ to the origin lies on a simplex whose vertices are support points of $A \ominus B$, and that simplex is found by iteratively adding the support point in the direction of the current closest point and discarding redundant vertices. The support-point construction of Section 18.4 makes every evaluation of $A \ominus B$ cheap: $s_{A \ominus B}(u) = s_A(u) - s_B(-u)$. This is the computational heart of GJK.

18.6 Gilbert–Johnson–Keerthi Algorithm

18.6.1 1. Overview

The **Gilbert–Johnson–Keerthi (GJK) algorithm** [GJK88] computes the distance between two convex sets A and B , or decides that they intersect, using only support-function evaluations. Because it never materializes the Minkowski difference, it runs in near-constant time for simple primitives (boxes, spheres, convex hulls of a few points) and is the standard narrow-phase test for convex objects in real-time physics engines [Eri04, vdB04].

18.6.2 2. Definitions

Definition 18.6 (Simplex in Minkowski Space). A **simplex** $\sigma \subseteq A \ominus B$ is a set of up to $d + 1$ support points $\{s(u_1), \dots, s(u_k)\}$, $k \leq d + 1$, generated by the algorithm. The origin's location relative to $\text{conv}(\sigma)$ determines whether A and B intersect.

18.6.3 3. Theory

GJK iterates the following scheme. Start with an arbitrary support point of $A \ominus B$ in some direction. At each step, let v be the point of $\text{conv}(\sigma)$ closest to the origin. If $v = \mathbf{0}$, then $\mathbf{0} \in A \ominus B$ and the objects intersect. Otherwise compute $w = s_{A \ominus B}(-v)$, the support point in the direction opposite the current closest point. If $\langle w, -v \rangle \leq \langle v, -v \rangle$ — equivalently, if the origin is closer to $\mathbf{0}$ than w in direction $-v$ — then no point of $A \ominus B$ is closer to the origin than v , and the distance is $\|v\|$. Otherwise add w to the simplex, remove any vertex not needed to keep the origin in the affine hull of the simplex, and repeat.

Theorem 18.7 (GJK Termination). *The GJK iteration terminates: the distance of the current closest point to the origin strictly decreases at every step, and the stopping criterion is met when the support point in the test direction does not improve on the current closest point. In $\mathbb{R}^3$ the simplex has at most four vertices, so each iteration is $O(1)$.*

The number of iterations is small in practice (typically a few dozen for polyhedra); the worst case is bounded by the number of features but is not polynomial in the input size, which is why GJK is used as an iterative numerical algorithm rather than analyzed to an exact worst-case bound.

18.6.4 4. Pseudocode

Algorithm 76 GJK Distance / Intersection Test

Input: Convex sets A, B with support functions.

Output: Distance $\text{dist}(A, B)$ and closest points, or “intersect”.

```

1: function GJK( $A, B$ )
2:    $d \leftarrow (1, 0, 0)$ ;  $S \leftarrow \{s_{A \oplus B}(d)\}$ 
3:    $v \leftarrow$  closest point of  $\text{conv}(S)$  to origin
4:   while true do
5:     if  $\|v\| < \varepsilon$  then
6:       return INTERSECT
7:     end if
8:      $w \leftarrow s_{A \oplus B}(-v)$ 
9:     if  $\langle w, -v \rangle \leq \langle v, v \rangle$  then
10:      return  $\|v\|$  (and the features supporting  $v$ )
11:    end if
12:     $S \leftarrow S \cup \{w\}$ , reduced to the minimal simplex containing the origin in its affine hull
13:     $v \leftarrow$  closest point of  $\text{conv}(S)$  to origin
14:  end while
15: end function

```

18.6.5 5. Parameter Selection

- **Tolerance ε :** the termination threshold trades accuracy against iterations; for boolean queries a tolerance of 10^{-9} in double precision is typical, and the convergence is quadratic once the simplex is right (the closest point approaches the origin).
- **Initial direction:** any nonzero direction works; using the relative displacement of the centers of mass speeds up convergence.
- **Caching:** caching the last support feature (warm starting) reduces iterations dramatically for nearby queries between the same objects across time steps.

18.6.6 6. Complexity

- **Per iteration:** $O(1)$ support evaluations (each $O(1)$ for simple primitives, $O(\log n)$ for a polygon, $O(n)$ for a general polyhedron), plus a constant-size closest-point computation on the simplex.
- **Iterations:** a small constant in practice; the theoretical worst case is larger but does not arise on typical inputs.

18.6.7 7. Usages

- **Real-time physics:** the narrow phase of collision detection for convex bodies in physics engines such as PhysX, Havok, and Bullet [Eri04, vdB04].

- **Robotics:** distance queries in configuration spaces (Chapter 26).
- **Haptics and surgical simulation:** fast contact queries for force feedback.

18.6.8 8. Testing Methods and Edge Cases

- **Touching objects:** verify the zero-distance threshold distinguishes touching from overlapping consistently.
- **Deep penetration:** GJK with an origin strictly inside $A \ominus B$ returns “intersect” but no depth; the penetration depth of Section 18.7 requires EPA.
- **Support correctness:** the support oracle must be exact on flat faces (all vertices of a face are support points); the choice of representative affects the closest-point computation.

18.6.9 9. References

GJK was introduced in [GJK88]; the standard engineering treatments with implementation details are [Eri04, vdB04].

18.7 Penetration Depth

18.7.1 1. Overview

When two convex objects overlap, the **penetration depth** measures how deeply: the minimum translation that separates them. For convex A, B , the penetration depth is the distance from the origin to the boundary of the Minkowski difference $A \ominus B$ (Theorem 18.5), and the separating translation is the closest boundary point. The **expanding polytope algorithm (EPA)** computes this distance by iteratively expanding a polytope inside $A \ominus B$ toward the boundary point closest to the origin [vdB04].

18.7.2 2. Definitions

Definition 18.8 (Penetration Depth). The **penetration depth** of overlapping convex sets A, B is $\min\{\|t\| : (A + t) \cap B = \emptyset\}$, the length of the smallest translation that separates them. The **penetration vector** is the translation achieving the minimum.

18.7.3 3. Theory

By Theorem 18.5, the penetration depth equals $\text{dist}(\mathbf{0}, \partial(A \ominus B))$. Since the boundary of $A \ominus B$ is composed of facets whose support directions are the feature normals, the closest boundary point is found by expanding a simplex (from GJK) into a polytope: maintain the polytope’s facets, find the facet closest to the origin, push the polytope outward at the support point in the facet’s normal direction, and repeat until the closest facet converges. EPA is correct because the polytope is always contained in $A \ominus B$ and expands toward the closest boundary point.

18.7.4 4. Pseudocode

18.7.5 5. Parameter Selection

- **Initial simplex:** EPA starts from the GJK simplex containing the origin; the polytope quality determines convergence speed.
- **Convergence threshold:** the ε on the facet support controls accuracy versus iterations; for game physics a value near 10^{-4} of the object scale is common.

Algorithm 77 Expanding Polytope Algorithm (Penetration Depth)**Input:** Simplex S from GJK with origin in $\text{conv}(S) \subseteq A \ominus B$.**Output:** Penetration vector of A, B .

```

1: function EPA( $A, B, S$ )
2:   build a polytope  $P$  (triangulation of the boundary of  $S$ )
3:   while true do
4:      $f \leftarrow$  facet of  $P$  closest to the origin;  $n \leftarrow$  normal of  $f$ 
5:      $w \leftarrow s_{A \ominus B}(n)$ 
6:     if  $\langle w, n \rangle - \langle \text{point of } f, n \rangle < \varepsilon$  then
7:       return  $\langle \text{point of } f, n \rangle \cdot n$  ▷ penetration vector
8:     end if
9:     add  $w$  to  $P$ ; retriangulate the visible facets
10:  end while
11: end function

```

18.7.6 6. Complexity

- **Time:** $O(1)$ per iteration plus $O(1)$ support evaluations; the polytope grows linearly in the number of iterations, and the iteration count is a small constant in practice.
- **Space:** $O(k)$ for the polytope with k facets.

18.7.7 7. Usages

- **Collision response:** computing the separating impulse magnitude in physics engines.
- **Contact manifolds:** the penetration vector defines the contact normal for solver-based dynamics.

18.7.8 8. Testing Methods and Edge Cases

- **Deep overlaps:** verify convergence when the origin is far inside $A \ominus B$.
- **Flat faces:** a facet parallel to a face of the difference gives a unique closest point; test boxes and degenerate (flat) convex bodies.
- **Consistency:** for small overlaps, the penetration vector must agree with the vector computed by a reference translation test.

18.7.9 9. References

EPA is described in the game-physics literature [vdB04, Eri04]; its Minkowski-difference foundation is the classical theory of convex sets (Section 2.10).

18.8 Collision Detection

Collision detection in an interactive or physical system is organized in two phases. The **broad phase** prunes pairs of objects that cannot possibly touch using cheap bounding volumes: each object is wrapped in an axis-aligned bounding box (AABB), and pairs whose boxes do not overlap are discarded. The canonical broad-phase algorithms are:

Sweep and prune sort the boxes by one coordinate and slide a window, testing only pairs within the window; for moderately sized scenes this is near-linear because the window is small [Eri04].

Boundary volume hierarchies each object stores a tree of bounding volumes (boxes or spheres); overlapping subtrees are descended recursively, giving output-sensitive behavior for complex objects (Chapter 10).

Uniform and spatial hashing objects are bucketed into cells of a grid; only pairs sharing a cell are tested, which suits uniformly distributed scenes.

The **narrow phase** then tests the surviving pairs exactly. For convex pairs, the narrow phase uses SAT (Section 18.2) for boxes and GJK (Section 18.6) for general convex bodies, followed by EPA when penetration is needed (Section 18.7). The overall cost is dominated by the broad phase, so the engineering advice is to make the bounding volumes tight, keep the broad-phase prune aggressive, and spend the narrow-phase budget only on pairs that truly overlap. The spatial data structures of Chapter 10 and the sweep techniques of Chapter 5 provide the primitives.

18.9 Continuous Collision Detection

18.9.1 1. Overview

Continuous collision detection (CCD) asks not only whether two objects overlap at the current time, but whether they overlap at *any* time in an interval — and, if so, the first time of contact (the **time of impact**, TOI). Discrete tests at the endpoints of the interval can miss a collision that occurs between the sampled instants (tunneling), so CCD computes the swept behavior.

18.9.2 2. Definitions

Definition 18.9 (Time of Impact). For bodies moving on prescribed trajectories during $[0, 1]$, the **time of impact** is the smallest t at which two bodies touch.

18.9.3 3. Theory

The standard CCD method for convex objects is **conservative advancement**: at the current time, compute the distance δ between the two bodies (with GJK, Section 18.6) and bound the relative speed $v_{\max}$ over the interval; then no collision can occur before the objects close the distance, so the time is advanced by $\delta/v_{\max}$ and the process repeats until the step is below a tolerance. Conservative advancement is correct as long as the speed bound is valid, which makes it the default for convex bodies. An alternative for polyhedra is the exact swept-volume/feature-interval method: subdivide time at the events where the separating feature changes, and test each interval algebraically [Eri04].

18.9.4 4. Pseudocode

18.9.5 5. Parameter Selection

- **Speed bound**: a global Lipschitz bound on the relative motion makes the algorithm conservative (never misses a collision) but may step slowly; a local bound on the closest features is tighter.
- **Tolerance**: ε controls the accuracy of the reported TOI; the cost grows logarithmically with $1/\varepsilon$.

Algorithm 78 Conservative Advancement (Time of Impact)

Input: Convex bodies $A(t), B(t)$, trajectories over $[0, 1]$, tolerance ε .**Output:** The time of impact t (or no collision).

```
1: function CONSERVATIVEADVANCEMENT( $A, B$ )
2:    $t \leftarrow 0$ 
3:   while  $t \leq 1$  do
4:      $\delta \leftarrow \text{GJK}(A(t), B(t))$ 
5:     if  $\delta \leq \varepsilon$  then
6:       return  $t$ 
7:     end if
8:      $v_{\max} \leftarrow$  bound on relative speed of closest features over  $[t, 1]$ 
9:      $t \leftarrow \min(1, t + \delta/v_{\max})$ 
10:  end while
11:  return NOCOLLISION
12: end function
```

18.9.6 6. Complexity

- **Time:** $O(I \cdot T)$ where T is the GJK cost and I the number of advancement steps, which is $O(\log(\text{scale}/\varepsilon))$ for linearly separated trajectories.

18.9.7 7. Usages

- **Games and physics:** preventing tunneling of fast bullets and thin objects.
- **Robotics:** verifying that a moving manipulator path is collision-free (Chapter 26).

18.9.8 8. Testing Methods and Edge Cases

- **Tunneling regression:** a fast small object passing through a thin wall must be reported, not missed.
- **Graze and rest contact:** contact that lasts exactly at an instant (tangent) must be returned consistently.
- **Consistency:** compare TOI against fine-grained discrete sampling on random motions.

18.9.9 9. References

CCD and conservative advancement are treated in [Eri04, vdB04].

18.10 Applications in Robotics and Simulation

The intersection and distance algorithms of this chapter power the collision layer of three industries:

Game and simulation physics Real-time physics engines — NVIDIA PhysX, Havok, and Bullet — run SAT, GJK, and EPA millions of times per second to maintain non-penetrating rigid-body simulations; the algorithms of this chapter (broad-phase bounding hierarchies, SAT for boxes, GJK for convex hulls, EPA for penetration, and conservative advancement for CCD) are the exact components they ship [Eri04, vdB04].

Robotics Motion planners compute distances between the robot’s links and its environment using GJK-based distance queries over convex decompositions of non-convex links, and the configuration-space techniques of Chapter 26 rely on these queries for feasibility tests. Collision-free programming for industrial arms is a direct consumer.

CAD and CAM Clearance analysis, toolpath validation, and assembly feasibility checks are distance and penetration queries between convex primitives and convex hulls of tooling geometry; the closest-feature results are reused for tolerancing and clearance reporting.

In each setting the convex intersection chapter’s contribution is the same: a small set of primitives (support oracles, projection tests, and simplex geometry) from which the entire collision pipeline is assembled, with the exactness and robustness requirements of Sections 1.10–1.12.

18.11 Exercises

1. Prove the separating axis theorem for two convex polygons from the hyperplane-separation theorem of Section 2.10.
2. Show that the projection interval of a convex polygon onto an axis is determined by its support function values at the axis and its negation.
3. Verify the Minkowski-difference reductions of Theorem 18.5 for two boxes, one contained in the other.
4. Implement GJK for two polygons in the plane and verify it against a brute-force distance computation on random inputs.
5. Prove that in $\mathbb{R}^2$ the GJK simplex has at most three vertices, and that the stopping test $\langle w, -v \rangle \leq \langle v, v \rangle$ is necessary and sufficient.
6. Explain why GJK alone cannot report penetration depth, and construct an input where the origin lies strictly inside $A \ominus B$.
7. Adapt the separating axis test to axis-aligned boxes in $\mathbb{R}^3$ and show it requires only three interval comparisons per candidate axis.
8. Simulate a tunneling failure: sample two trajectories at their endpoints and show a missed collision that conservative advancement (Section 18.9) detects.

18.12 Enclosure and Convex Distance

18.13 Welzl’s Algorithm

18.13.1 Overview

Welzl’s algorithm is a randomized algorithm for finding the smallest enclosing circle (the minimum-area circle that contains all points in a set). Originally described by Emo Welzl in 1991 [Wel91], it is a brilliant application of the linear programming paradigm to a geometric problem. The algorithm builds on earlier linear-time results for fixed-dimension linear programming by Megiddo [Meg83]. It achieves an expected $O(n)$ time complexity, making it highly efficient for point sets in 2D.

18.13.2 Definitions

Definition 18.10 (Smallest Enclosing Circle). The **smallest enclosing circle (SEC)** of a point set $P \subset \mathbb{R}^2$ is the unique circle of minimum radius that contains every point of P . The SEC is also called the *minimum covering circle* or *minidisk*.

Definition 18.11 (Support Points). The **support points** of the SEC are the points of P that lie on the boundary of the SEC. The SEC is uniquely determined by at most three support points: a single point (the center), two points (the circle having them as diameter), or three non-collinear points (the circumcircle).

18.13.3 Theory

The algorithm is based on a recursive reduction of the point set. The key insight is that if a new point P_i is already inside the SEC of the previous $i - 1$ points, then the SEC remains the same. If P_i is outside, it **MUST** lie on the boundary of the new SEC. This allows us to recursively solve the problem with P_i fixed as a “boundary point.”

The recursion continues until we have a set of 0, 1, 2, or 3 points that must lie on the boundary. These “fixed” points uniquely define the circle (0: none, 1: point as center, 2: segment as diameter, 3: circumcircle).

Theorem 18.12 (Expected Linear Time). *Welzl’s algorithm computes the smallest enclosing circle of n points in $\mathbb{R}^2$ in expected $O(n)$ time.*

Proof. The proof uses a backwards analysis argument. Fix a random permutation π of the n points. Consider the moment when the algorithm first inserts a point $p_{\pi(i)}$ onto the boundary (i.e., the recursion rejects the current disk and adds $p_{\pi(i)}$ to R). By backwards analysis, conditioning on this event, $p_{\pi(i)}$ is equally likely to be any of the i points seen so far. The expected number of boundary insertions across all n points is $\sum_{i=1}^n \frac{3}{i} = O(\log n)$ (since at most 3 points are ever on the boundary and each insertion reduces the remaining budget by 1). The total work is therefore $O(n + \log n) = O(n)$. See [Wel91] and Megiddo [Meg83] for the full analysis. $\square$

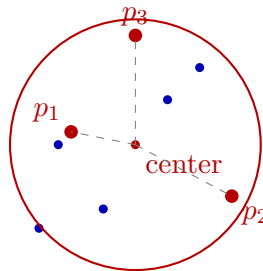


Figure 18.1: Welzl’s algorithm: the smallest enclosing circle is defined by at most 3 support points (red).

Example 18.13 (Welzl Recursion Trace). Consider points $P = \{(0, 0), (4, 0), (2, 3), (2, 1)\}$ processed in order:

1. Recurse with $R = \{\}$: pop $(2, 1)$, compute SEC of remaining 3 points. The SEC of $\{(0, 0), (4, 0), (2, 3)\}$ is the circumcircle: center $(2, 0.833)$, radius ≈ 2.404 .
2. Check $(2, 1)$: distance from center is $\approx 0.167 < 2.404$ —inside! Return this SEC.

The algorithm finishes in one pass because $(2, 1)$ happens to be interior to the circumcircle of the other three.

18.13.4 Pseudocode

```

1: function WELZL( $P$ ,  $R = []$ )
2:   if  $|P| = 0$  or  $|R| = 3$  then
3:     return CircleFromBoundary( $R$ )
4:   end if
5:    $p \leftarrow P.\text{pop}()$ 
6:    $D \leftarrow \text{Welzl}(P.\text{copy}(), R)$ 
7:   if  $p$  is in  $D$  then
8:     return  $D$ 
9:   end if
10:  return Welzl( $P.\text{copy}(), R + [p]$ )
11: end function

```

▷ To start:

```

12: random.shuffle(points)
13: SEC  $\leftarrow$  Welzl(points)

```

18.13.5 Parameter Selection

- **Randomization:** Shuffling the input points at the beginning is crucial. Without randomization, structured input (like points on a line) can cause $O(n^2)$ worst-case performance.
- **Numerical Precision:** Circumcircle calculations for 3 points can be sensitive to collinearity. An epsilon should be used to check if 3 points are nearly on a line.

18.13.6 Complexity

- **Time Complexity:** $O(n)$ expected. Although the recursion depth is n , the probability that a point falls outside the current circle decreases rapidly, leading to the $O(n)$ bound.
- **Space Complexity:** $O(n)$ for the recursion stack and the copies of the point set.

18.13.7 Usages

- **Collision Detection:** Generating a bounding sphere for an object as a fast first-pass check [Eri04].
- **Facility Location:** Finding the center for a new resource (e.g., a cell tower) to cover all clients with minimum range.
- **Cluster Analysis:** Determining the spatial extent of a group of points.
- **Camera Control:** Finding the minimum zoom/pan to keep all characters in a scene visible.

18.13.8 Testing Methods and Edge Cases

- **Collinear Points:** The SEC should have its diameter equal to the distance between the two furthest points.
- **Duplicate Points:** Handled naturally by the inclusion check.
- **Small Sets:** Verify with 0, 1, and 2 points.

- **Symmetry:** Regular polygons should result in a circle centered at the polygon’s centroid.
- **Redundant Points:** Verify that a million points inside a circle don’t change the SEC defined by 3 points on the boundary.

18.13.9 References

- Welzl, E. (1991). “Smallest enclosing disks (balls and ellipsoids)”. Lecture Notes in Computer Science.
- Megiddo, N. (1983). “Linear-time algorithms for linear programming in $\mathbb{R}^3$ and related problems”. SIAM Journal on Computing.
- Wikipedia: [Smallest-circle problem](#)

18.14 Convex Polygon Distance

18.14.1 Overview

The convex polygon distance problem involves finding the minimum Euclidean distance between two non-intersecting convex polygons P and Q . This is a fundamental operation in collision detection and motion planning. While a naive $O(n \cdot m)$ approach checks all pairs of edges, more efficient algorithms leverage the convexity of the shapes to achieve $O(n + m)$ or even $O(\log n + \log m)$ performance. The rotating calipers approach was popularized by Toussaint [Tou83], and the GJK algorithm was introduced by Gilbert, Johnson, and Keerthi [GJK88]. Edelsbrunner [Ede85] provides further complexity bounds.

18.14.2 Definitions

Definition 18.14 (Euclidean Distance between Polygons). The **Euclidean distance** between two polygons P and Q is

$$\text{dist}(P, Q) = \min_{p \in P, q \in Q} \|p - q\|_2.$$

For non-intersecting convex polygons, this minimum is achieved between two vertices or a vertex and an edge.

Definition 18.15 (Separating Axis). A **separating axis** is a line ℓ such that the orthogonal projections of two convex shapes P and Q onto ℓ do not overlap. By the separating axis theorem, two convex shapes do not intersect if and only if a separating axis exists.

Definition 18.16 (Minkowski Difference). The **Minkowski difference** of two sets $P, Q \subset \mathbb{R}^2$ is

$$P \ominus Q = \{p - q \mid p \in P, q \in Q\}.$$

For convex polygons with n and m vertices, $P \ominus Q$ is a convex polygon with at most $n + m$ vertices. The distance $\text{dist}(P, Q)$ equals the distance from the origin to $P \ominus Q$.

Definition 18.17 (GJK Algorithm). The **Gilbert–Johnson–Keerthi (GJK) algorithm** computes the distance between two convex shapes by iteratively constructing a simplex inside the Minkowski difference $P \ominus Q$ that converges toward the origin. If the origin lies inside $P \ominus Q$, the shapes intersect.

18.14.3 Theory

The most common approach for linear-time distance calculation is based on the **Rotating Calipers** method.

1. Find the pair of points (one on each polygon) that minimize the distance. For non-intersecting convex polygons, this minimum distance occurs between two vertices, or a vertex and an edge.
2. Use the rotating calipers technique to “crawl” around both polygons simultaneously. Because the polygons are convex, the distance function is unimodal along the boundary, allowing us to find the global minimum in a single pass.

Another powerful approach is the **GJK Algorithm**:

1. Compute the **Minkowski Difference** $D = P \ominus Q$.
2. The distance between P and Q is the distance from the origin $(0,0)$ to the set D .
3. GJK iteratively builds a simplex inside D that gets closer and closer to the origin.

Theorem 18.18 (Convex Distance Lower Bound). *For two convex polygons P with n vertices and Q with m vertices (both in convex position), the minimum Euclidean distance can be computed in $O(n + m)$ time using rotating calipers, and in $O(\log n + \log m)$ time using binary search if the polygons have been preprocessed.*

Proof. The $O(n + m)$ bound follows from the rotating calipers observation: at each step, at least one caliper advances to the next vertex, and the total number of advances is at most $n + m$. The distance between the two caliper positions is evaluated at each step, and the minimum is returned. The $O(\log n + \log m)$ bound follows from the fact that for each edge of one polygon, the distance to the other polygon is a unimodal function along its boundary, which can be minimized by binary search. $\square$

18.14.4 Pseudocode

Rotating Calipers for Distance

```

1: function CONVEXDISTANCE( $P, Q$ )
2:    $i \leftarrow \text{index\_of\_min\_y}(P)$ 
3:    $j \leftarrow \text{index\_of\_max\_y}(Q)$ 
4:    $\text{min\_dist} \leftarrow \infty$ 
5:   for  $\_$  in  $\text{range}(|P| + |Q|)$  do
6:      $d \leftarrow \text{Distance}(P[i], Q[j])$ 
7:      $\text{min\_dist} \leftarrow \min(\text{min\_dist}, d)$ 
8:      $\text{angle\_P} \leftarrow \text{AngleOfEdge}(P, i)$ 
9:      $\text{angle\_Q} \leftarrow \text{AngleOfEdge}(Q, j)$ 
10:    if  $\text{angle\_P} < \text{angle\_Q}$  then
11:       $i \leftarrow (i + 1) \bmod |P|$ 
12:    else
13:       $j \leftarrow (j + 1) \bmod |Q|$ 
14:    end if
15:  end for
16:  return  $\text{min\_dist}$ 
17: end function

```

Example 18.19 (Rotating Calipers Walkthrough). Consider $P = \{(0, 0), (3, 0), (3, 2), (0, 2)\}$ (a 3×2 rectangle) and $Q = \{(5, 1), (7, 1), (7, 3), (5, 3)\}$ (a 2×2 rectangle). Both are CCW.

1. Start: $i = \text{index of min } y \text{ in } P = 0$ (point $(0, 0)$), $j = \text{index of max } y \text{ in } Q = 2$ (point $(7, 3)$).
2. Distance from $(0, 0)$ to $(7, 3)$: $\sqrt{49 + 9} = \sqrt{58} \approx 7.62$. Advance i (edge $P[0] \rightarrow P[1]$ has angle 0° , smaller than Q 's edge).
3. Distance from $(3, 0)$ to $(7, 3)$: $\sqrt{16 + 9} = 5.0$. Advance j .
4. Continue until both calipers complete a full cycle. The minimum distance is 2.0 (from edge $x = 3$ of P to edge $x = 5$ of Q).

18.14.5 Parameter Selection

- **Polygon Orientation:** Polygons must be consistently oriented (e.g., both CCW).
- **Initial Points:** Starting at extreme points ensures the calipers are correctly synchronized.

18.14.6 Complexity

- **Time Complexity:** $O(n + m)$ using rotating calipers or GJK. Binary search methods can achieve $O(\log n + \log m)$ for preprocessed polygons.
- **Space Complexity:** $O(1)$ auxiliary space if the polygons are already stored.

18.14.7 Usages

- **Collision Avoidance:** Maintaining a “safety buffer” between a robot and obstacles.
- **Physics Simulation:** Calculating the time of impact or the depth of penetration.
- **Tolerance Analysis:** Determining if two mechanical parts will fit together with a required clearance.
- **UI Layout:** Calculating the spacing between complex geometric buttons or containers.

18.14.8 Testing Methods and Edge Cases

- **Intersecting Polygons:** The distance should be 0. (GJK handles this by detecting that the origin is inside the Minkowski difference).
- **Parallel Edges:** The minimum distance might be achieved along an entire segment.
- **Points and Lines:** Test with “polygons” that are actually single points or line segments.
- **Very Thin Polygons:** Ensure numerical stability for nearly collinear vertices.
- **Vast Scales:** Precision check for polygons that are very far apart.

18.14.9 References

- Toussaint, G. T. (1983). “Solving geometric problems with the rotating calipers”. Proceedings of MELECON '83.
- Gilbert, E. G., Johnson, D. W., & Keerthi, S. S. (1988). “A fast procedure for computing the distance between complex objects in three-dimensional space”. IEEE Journal on Robotics and Automation.
- Edelsbrunner, H. (1985). “Computing the extreme distances between two convex polygons”. Journal of Algorithms.
- Wikipedia: [Gilbert–Johnson–Keerthi distance algorithm](#)

Part VII

Curves, Surfaces, and Three-Dimensional Geometry

Chapter 19

Computational Geometry in Three Dimensions

The convex hull of a finite set of points in $\mathbb{R}^3$ is the smallest convex polyhedron containing them. It is the three-dimensional analogue of the planar convex hull and is a foundation for collision detection, visibility, solid modeling, meshing, and geometric optimization.

19.1 Polyhedral Representation

A polyhedron can be represented by its vertices, edges, and polygonal faces, or by half-spaces of the form $a_i^\top x \leq b_i$. A valid boundary representation records face cycles and the adjacency between faces and edges. For a closed connected convex polyhedron, Euler's formula is

$$V - E + F = 2.$$

The boundary orientation must be consistent so that all outward normals point in the same direction.

19.2 Three-Dimensional Predicates

The signed volume test

$$\text{orient3d}(A, B, C, D) = \det \begin{pmatrix} (B - A)^\top \\ (C - A)^\top \\ (D - A)^\top \end{pmatrix}$$

determines which side of the oriented plane ABC contains D . A positive value indicates one side, a negative value the other, and zero coplanarity. Robust evaluation is essential: an incorrect sign can create a non-manifold polyhedron or reverse a face during hull construction.

19.3 Incremental Three-Dimensional Hulls

An incremental hull algorithm starts with a non-degenerate tetrahedron and inserts points one at a time. For each point, it finds the faces visible from that point, removes those faces, extracts the horizon cycle between visible and invisible faces, and connects the new point to the horizon edges.

The visible-face search can be accelerated with a conflict graph storing the points visible from each face. With suitable randomized ordering, expected running time is $O(n \log n + k)$ for n input points and k output features; the worst case is output-sensitive and can be quadratic for poorly ordered inputs.

Algorithm 79 Incremental Three-Dimensional Convex Hull

```
1: function CONVEXHULL3D( $P$ )
2:   Select four non-coplanar points and initialize tetrahedron  $H$ 
3:   for each remaining point  $p$  do
4:      $V \leftarrow$  faces of  $H$  visible from  $p$ 
5:     if  $V$  is non-empty then
6:        $R \leftarrow$  boundary edges between visible and invisible faces
7:       Remove  $V$  from  $H$ 
8:       for each edge  $(u, v)$  in  $R$  do
9:         Add face  $(u, v, p)$  with consistent outward orientation
10:      end for
11:    end if
12:  end for
13:  return  $H$ 
14: end function
```

19.4 Degeneracies and Validation

Implementations must define behavior for coplanar points, duplicate points, collinear subsets, cospherical configurations, and points lying on existing faces. Symbolic perturbation or exact predicates can make the combinatorial result deterministic. A resulting hull should be checked for positive face orientation, edge incidence, connectedness, Euler's formula, and containment of every input point.

19.5 Applications

Three-dimensional convex hulls provide collision proxies, bounding volumes, support mappings for GJK, visibility boundaries, polyhedral approximations, and input domains for tetrahedral and hexahedral mesh generation.

Chapter 20

Curves and Surfaces

20.1 Surface Modeling and Mesh Generation

20.2 Bézier Surfaces

20.2.1 Overview

A Bézier surface is a smooth surface used in computer graphics and computer-aided design (CAD) that is controlled by a rectangular grid of points, called control points [B68, Far02]. It is a 2D extension of the Bézier curve. The surface is defined by a pair of parameters (u, v) , both in the range $[0, 1]$. Evaluating a Bézier surface means calculating the 3D coordinates (x, y, z) corresponding to a given (u, v) coordinate.

20.2.2 Definitions

Definition 20.1 (Control Points). A grid of $m + 1$ by $n + 1$ points P_{ij} in 3D space that define the shape of the Bézier surface.

Definition 20.2 (Bernstein Polynomials). The basis functions $B_{i,n}(t) = \binom{n}{i} t^i (1 - t)^{n-i}$ used to weight the control points in Bézier surface evaluation.

Definition 20.3 (Tensor Product). The method of forming a surface by multiplying two orthogonal Bézier curves, i.e., $Q(u, v) = \sum_{i,j} B_{i,m}(u) B_{j,n}(v) P_{ij}$.

Definition 20.4 (de Casteljau's Algorithm). A recursive algorithm for evaluating Bézier curves and surfaces that is numerically stable and geometrically intuitive.

20.2.3 Theory

A Bézier surface point $Q(u, v)$ is calculated using the formula:

$$Q(u, v) = \sum_{i=0}^m \sum_{j=0}^n B_{i,m}(u) B_{j,n}(v) P_{i,j}$$

The most efficient way to evaluate this is by using **de Casteljau's algorithm** [Far02] in two steps:

1. Evaluate $m + 1$ separate Bézier curves in the v -direction using parameter v . This produces a temporary set of control points $P'_0, \dots, P'_m$ that depend only on v .
2. Evaluate the single Bézier curve defined by P'_i using parameter u .

This “tensor product” approach effectively reduces the surface evaluation to several simpler curve evaluations.

Example 20.5 (Bicubic Bézier Surface Evaluation). Consider a 4×4 bicubic Bézier surface (degree $m = n = 3$) with 16 control points forming a 4×4 grid. To evaluate at $(u, v) = (0.25, 0.75)$:

1. For each of the 4 rows, apply de Casteljau’s algorithm in the v -direction with $v = 0.75$, producing 4 intermediate points P'_0, P'_1, P'_2, P'_3 .
2. Apply de Casteljau’s algorithm to $P'_0, \dots, P'_3$ in the u -direction with $u = 0.25$, yielding the final 3D point $Q(0.25, 0.75)$.

Each de Casteljau step requires $O(m)$ linear interpolations, so the total cost is $O(m \cdot n)$ per evaluation.

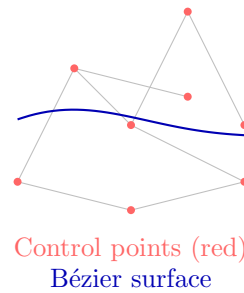


Figure 20.1: Bézier surface: a tensor-product surface defined by a control net of points.

20.2.4 Pseudocode

20.2.5 Parameter Selection

- **Parameter Range:** Parameters u, v must be strictly within $[0, 1]$.
- **Order:** The number of control points is $(m + 1) \times (n + 1)$. Typically $m = 3, n = 3$ for bicubic surfaces.

20.2.6 Complexity

- **Time Complexity:** $O(m \cdot n^2 + m^2)$ using the two-step de Casteljau approach. For bicubic surfaces ($m = 3, n = 3$), this is a small constant.
- **Space Complexity:** $O(m \cdot n)$ to store the control point grid.

20.2.7 Usages

- **CAD (Computer Aided Design):** Modeling car bodies, aircraft wings, and consumer products.
- **Animation:** Smooth character skinning and facial deformations.
- **TrueType/PostScript Fonts:** While fonts are 2D curves, they can be interpreted as 1D surfaces.
- **Geometric Design:** Creating smooth transitions (blends) between different parts of a model.
- **Fluid Simulation:** Representing the surface of a liquid with a set of smooth patches.

Algorithm 80 Evaluate Bézier Surface

```

1: function EVALUATEBEZIERSURFACE(control_points, u, v)
2:    $m \leftarrow \text{num\_rows} - 1$ 
3:    $n \leftarrow \text{num\_cols} - 1$  ▷ Reduce in the  $v$  direction
4:   temp_points  $\leftarrow \emptyset$ 
5:   for  $i \leftarrow 0$  to  $m$  do
6:     row  $\leftarrow \text{control\_points}[i]$ 
7:     EvaluateBezierCurve(row, v)
8:     temp_points.append( $p'$ )
9:   end for ▷ Reduce in the  $u$  direction
10:  return EvaluateBezierCurve(temp_points, u)
11: end function
12: function EVALUATEBEZIERCURVE(points, t)
13:  current  $\leftarrow \text{points}[:, ]$ 
14:  while  $|\text{current}| > 1$  do
15:    next_level  $\leftarrow \emptyset$ 
16:    for  $i \leftarrow 0$  to  $|\text{current}| - 2$  do
17:       $p \leftarrow (1 - t) \cdot \text{current}[i] + t \cdot \text{current}[i + 1]$ 
18:      next_level.append( $p$ )
19:    end for
20:    current  $\leftarrow \text{next\_level}$ 
21:  end while
22:  return current[0]
23: end function

```

20.2.8 Testing Methods and Edge Cases

- **Corners:** Verify that $Q(0, 0), Q(0, 1), Q(1, 0), Q(1, 1)$ exactly match the four corner control points.
- **Flat Surface:** If all control points lie on a plane, the entire evaluated surface must lie on that plane.
- **Degree Elevation:** Increasing the number of control points without changing the surface shape.
- **Precision:** Ensure the surface remains smooth and doesn't "break" at $u, v \approx 0.5$.
- **Continuity:** Checking that adjacent Bézier patches meet smoothly (G1 or C1 continuity).

20.2.9 References

- Bézier, P. [B68] "How Renault Uses Numerical Control for Car Body Design and Tooling". SAE paper.
- Farin, G. [Far02] "Curves and Surfaces for CAGD: A Practical Guide". Morgan Kaufmann.
- Piegl, L., & Tiller, W. [PT97] "The NURBS Book". Springer.
- Wikipedia: [Bézier surface](#)

20.3 NURBS Surfaces

20.3.1 Overview

NURBS (Non-Uniform Rational B-Splines) are a mathematical model commonly used in computer graphics and computer-aided design (CAD) for generating and representing curves and surfaces [PT97]. NURBS surfaces provide a highly flexible and unified way to represent both analytic shapes (like spheres, cones, and cylinders) and complex, free-form organic shapes. Evaluating a NURBS surface involves calculating the 3D coordinates (x, y, z) at a given parameter pair (u, v) within the surface's domain.

20.3.2 Definitions

Definition 20.6 (Knot Vector). A non-decreasing sequence of real numbers $t_0 \leq t_1 \leq \dots \leq t_{n+p+1}$ that determines how the control points affect the NURBS surface.

Definition 20.7 (NURBS Weights). A scalar $w_{ij} > 0$ associated with each control point P_{ij} . A higher weight “pulls” the surface closer to that control point.

Definition 20.8 (B-Spline Basis Functions). Piecewise polynomial functions $N_{i,p}$ of degree p defined over a knot vector via the Cox-de Boor recursion formula.

Definition 20.9 (Rational Surface). The term “rational” refers to the use of weights and a division step, allowing for the exact representation of conic sections (circles, ellipses).

20.3.3 Theory

A point on a NURBS surface $S(u, v)$ is defined as:

$$S(u, v) = \frac{\sum_{i=0}^n \sum_{j=0}^m N_{i,p}(u) N_{j,q}(v) w_{i,j} P_{i,j}}{\sum_{i=0}^n \sum_{j=0}^m N_{i,p}(u) N_{j,q}(v) w_{i,j}}$$

where $P_{i,j}$ are the control points, $w_{i,j}$ are weights, and $N_{i,p}, N_{j,q}$ are the B-spline basis functions of degrees p and q .

The calculation typically follows these steps:

1. **Locate Knots:** Find the indices of the knots u and v in their respective knot vectors.
2. **Evaluate Basis Functions:** Compute the values of all non-zero basis functions at u and v using the Cox-de Boor recursion formula.
3. **Summation:** Compute the weighted sum of the control points and the sum of the weights.
4. **Division:** Divide the weighted point sum by the total weight to find the final 3D point.

Example 20.10 (NURBS Circle Representation). A circle of radius r can be exactly represented by a single quadratic NURBS curve ($p = 2$) with 9 control points and a specific knot vector. The key insight is that the weights w_i for the control points at the cardinal directions are $\cos(\pi/4) = \sqrt{2}/2$, while the weights for the diagonal control points are 1. This rational weighting is what allows NURBS to represent conic sections exactly, which polynomial Bézier curves cannot do.

20.3.4 Pseudocode

20.3.5 Parameter Selection

- **Degree (p, q) :** Usually $p = 3, q = 3$ (bicubic) for smooth modeling.
- **Knot Vector Type:** **Clamped** knot vectors (where the first and last $p + 1$ knots are identical) ensure the surface passes exactly through its corner control points.

Algorithm 81 Evaluate NURBS Surface

```

1: function EVALUATENURBSSURFACE(control_points, weights, knots_u, knots_v, p, q, u, v)
2:   FindKnotSpan(n,p,u,knots_u)
3:   FindKnotSpan(m,q,v,knots_v)
4:   BasisFunctions(span_u,u,p,knots_u)
5:   BasisFunctions(span_v,v,q,knots_v)
6:   sum_wp ← 0
7:   sum_w ← 0
8:   for l ← 0 to q do
9:     temp ← 0
10:    temp_w ← 0
11:    for k ← 0 to p do
12:      idx_u ← span_u - p + k
13:      idx_v ← span_v - q + l
14:      w ← weights[idx_u][idx_v]
15:      p_val ← control_points[idx_u][idx_v]
16:      sum_wp ← sum_wp +  $N_u[k] \cdot N_v[l] \cdot w \cdot p_{\text{val}}$ 
17:      sum_w ← sum_w +  $N_u[k] \cdot N_v[l] \cdot w$ 
18:    end for
19:  end for
20:  return sum_wp / sum_w
21: end function

```

20.3.6 Complexity

- **Time Complexity:** $O(p^2 + q^2)$ to evaluate the basis functions, plus $O(p \cdot q)$ for the final summation. This is extremely efficient for low degrees.
- **Space Complexity:** $O(n \cdot m)$ to store the control point and weight grids.

20.3.7 Usages

- **Industrial CAD:** Standard data format (IGES, STEP) for exchanging 3D models between different engineering software (SolidWorks, Rhino, CATIA).
- **Reverse Engineering:** Fitting smooth surfaces to scanned point cloud data.
- **Scientific Visualization:** Smoothly interpolating scattered spatial data (e.g., bathymetry or atmospheric pressure).
- **High-End Animation:** Rendering complex character models (Pixar's RenderMan uses NURBS extensively).
- **Naval Architecture:** Designing boat hulls for optimal fluid flow.

20.3.8 Testing Methods and Edge Cases

- **Analytic Shapes:** Verify that a NURBS surface with appropriate weights can exactly represent a sphere or a cylinder.
- **Weights of 1:** If all weights are 1.0, the surface must behave identically to a non-rational B-spline surface.
- **Boundary Conditions:** Verify that the surface interpolates its corner control points (for clamped knots).

- **Numerical Stability:** Ensure the denominator (total weight) does not approach zero.
- **Derivative Calculation:** Verify that the surface normals (calculated from u and v derivatives) are smooth and consistent.

20.3.9 References

- Piegl, L., & Tiller, W. [PT97] “The NURBS Book”. Springer.
- Rogers, D. F. (2001). “An Introduction to NURBS: With Historical Perspective”. Morgan Kaufmann.
- Farin, G. [Far02] “Curves and Surfaces for CAGD: A Practical Guide”. Morgan Kaufmann.
- Wikipedia: [Non-uniform rational B-spline](#)

20.4 Mesh Voxelization

20.4.1 Overview

Mesh voxelization is the process of converting a 3D surface mesh (typically composed of triangles) into a volumetric representation called a **voxel grid** [AM01,NT03]. A voxel (volume element) is the 3D equivalent of a pixel. Voxelization is used to simplify complex geometry, perform volumetric analysis, and prepare models for physics simulations or 3D printing. The process can result in a “binary” voxel grid (occupied vs. empty) or a “solid” voxelization (distinguishing interior from exterior).

20.4.2 Definitions

Definition 20.11 (Voxel). A unit of volume on a regular 3D grid.

Definition 20.12 (Conservative Voxelization). A method that ensures a voxel is marked as occupied if any part of the triangle passes through it.

Definition 20.13 (Parity Rule). A technique for solid voxelization where a voxel’s “insideness” is determined by counting intersections of a ray with the mesh boundary using the even-odd rule.

20.4.3 Theory

The most common approach to surface voxelization is based on **triangle-box intersection** tests [AM01].

1. Define a 3D grid that encloses the mesh’s bounding box.
2. For each triangle in the mesh:
 - Determine the range of voxels it might intersect (its AABB in grid coordinates).
 - For every voxel in that range, perform a rigorous triangle-box intersection test (e.g., using the Akenine-Möller algorithm based on the Separating Axis Theorem).
3. To perform **solid voxelization** (filling the interior):
 - After surface voxelization, use a flood-fill algorithm starting from the grid boundaries to identify “outside” voxels [NT03].
 - Everything not reachable from the outside is “inside.”

- Alternatively, use ray-casting along each grid line and apply the even-odd rule.

Example 20.14 (Voxelizing a Cube). Consider a unit cube (8 vertices, 12 triangular faces) voxelized at resolution $8 \times 8 \times 8$. The bounding box is $[0, 1]^3$ with voxel size $1/8$. Each of the 12 triangles is tested against its AABB in grid coordinates. Surface voxels on each face are marked. Then flood-fill from a corner voxel identifies all voxels on the exterior. The remaining interior voxels form a $6 \times 6 \times 6$ solid block of 216 inside voxels (the boundary voxels separate inside from outside).

20.4.4 Pseudocode

Algorithm 82 Voxelize Mesh

```

1: function VoxelizeMesh(mesh, resolution)
2:   bbox  $\leftarrow$  mesh.GetBoundingBox()
3:   voxel_size  $\leftarrow$  bbox.size.max() / resolution
4:   grid  $\leftarrow$  array[resolution]3 initialized to EMPTY ▷ Surface Voxelization
5:   for each triangle in mesh.faces do
6:     WorldToGrid(triangle.min, bbox, voxel_size)
7:     WorldToGrid(triangle.max, bbox, voxel_size)
8:     for  $x \leftarrow \text{min}_v.x$  to  $\text{max}_v.x$  do
9:       for  $y \leftarrow \text{min}_v.y$  to  $\text{max}_v.y$  do
10:        for  $z \leftarrow \text{min}_v.z$  to  $\text{max}_v.z$  do
11:          GetVoxelBox( $x, y, z$ , bbox, voxel_size)
12:          if TriangleBoxIntersect(triangle, voxel_box) then
13:            grid[ $x$ ][ $y$ ][ $z$ ]  $\leftarrow$  SURFACE
14:          end if
15:        end for
16:      end for
17:    end for
18:  ▷ Solid Voxelization (Optional)
19:  FloodFill(grid, start_pos=(0,0,0), target = EMPTY, replacement = OUTSIDE)
20:  for ( $x, y, z$ ) in grid do
21:    if grid[ $x$ ][ $y$ ][ $z$ ] = EMPTY then
22:      grid[ $x$ ][ $y$ ][ $z$ ]  $\leftarrow$  INSIDE
23:    end if
24:  end for
25:  return grid
26: end function

```

20.4.5 Parameter Selection

- **Resolution:** Higher resolution captures more detail but increases memory consumption cubically ($O(N^3)$).
- **Intersection Strategy:** Conservative voxelization is preferred for tasks like collision detection to avoid “gaps” in the surface.

20.4.6 Complexity

- **Time Complexity:** $O(F \cdot K^3)$ where F is the face count and K is the local grid size per triangle. For solid filling, $O(N^3)$ where N is the resolution.

- **Space Complexity:** $O(N^3)$ to store the 3D grid. Sparse representations like **OpenVDB** or **Octrees** can reduce this significantly.

20.4.7 Usages

- **3D Printing:** Slicing software voxelizes models to generate toolpaths and infill patterns.
- **Physics Simulation:** Converting complex meshes into voxels for fluid dynamics (Lattice Boltzmann Method) or smoke simulation.
- **Collision Detection:** Using a voxel grid as a spatial index for fast intersection queries.
- **Medical Imaging:** Modeling organs or bones from surface scans for surgical planning.
- **Game Development:** Destructible environments (e.g., Minecraft or Teardown) and GPU-based global illumination.

20.4.8 Testing Methods and Edge Cases

- **Watertightness:** Verify that solid voxelization produces no “leaks” for a closed mesh.
- **Thin Features:** Ensure triangles thinner than a voxel are still captured (conservative test).
- **Non-Manifold Mesh:** Test how the solid filler handles meshes with holes (usually results in the entire model being marked as outside).
- **Numerical Precision:** Triangle-box tests are sensitive to floating-point errors; use robust epsilon values.
- **Memory Limit:** Check performance and stability at high resolutions (e.g., 1024^3).

20.4.9 References

- Akenine-Möller, T. [AM01] “Fast 3D Triangle-Box Overlap Testing”. Journal of Graphics Tools.
- Nooruddin, F. S., & Turk, G. [NT03] “Simplification and Repair of Polygonal Models Using Volumetric Techniques”. IEEE TVCG.
- Schwarz, M., & Seidel, H. P. (2010). “Fast Parallel Surface and Solid Voxelization on GPUs”. ACM TOG.
- Wikipedia: [Voxel](#)

20.5 Winding-Filtered Tetrahedral Meshing

20.5.1 Overview

Winding-Filtered Tetrahedral Meshing is a robust technique for generating a volumetric tetrahedral mesh from a surface mesh, even if the surface mesh is “dirty” (i.e., contains self-intersections or holes) [H⁺18].

Traditional tetrahedralization algorithms (like constrained Delaunay) often require a clean, manifold surface. In contrast, winding-filtered meshing uses the Generalized Winding Number (GWN) to classify tetrahedra as interior or exterior, which is robust even for non-manifold inputs.

20.5.2 Implementation Details

In `CompGeom`, the `WindingFilteredTetMesher` uses the following process (inspired by `fTetWild` [H⁺18]):

1. **Bounding Box Expansion:** Create a large super-tetrahedron containing the entire surface mesh.
2. **Point Insertion:** Add surface vertices and a dense grid of internal Steiner points to the tetrahedralization.
3. **Delaunay Tetrahedralization:** Construct a global Delaunay tetrahedralization of the point set.
4. **Winding-Based Filtering:** For each tetrahedron, compute the Generalized Winding Number of its centroid with respect to the original surface mesh.
5. **Output:** Retain only tetrahedra where the $\text{GWN} > 0.5$ (indicating the interior of the domain).

Example 20.15 (Winding Number Classification). Consider a closed tetrahedron surface mesh. For a point inside the tetrahedron, the generalized winding number is 1.0 (the surface wraps around it exactly once). For a point outside, the winding number is 0.0. The threshold of 0.5 cleanly separates interior from exterior. Even if the surface has a small gap (non-watertight), the winding number degrades gracefully, correctly classifying most interior tetrahedra.

20.5.3 Usage

Algorithm 83 Winding-Filtered Tet Meshing

```

1: from compgeom.mesh.volume.tetmesh.winding_filtered_mesher import
   WindingFilteredTetMesher
2: import numpy as np
3:                                     ▷ surface vertices and faces as numpy arrays
4: vertices ← np.array([[0, 0, 0], [1, 0, 0], [0, 1, 0], [0, 0, 1]])
5: faces ← np.array([[0, 1, 2], [0, 2, 3], [0, 3, 1], [1, 2, 3]])
6:
7: WindingFilteredTetMesher(vertices, faces)
8: tet_mesh ← mesher.mesh(refinement_factor=1.0)
```

20.5.4 References

- Hu, Y., et al. “fTetWild: Robust Tetrahedral Meshing in the Wild.” *ACM Transactions on Graphics (SIGGRAPH)*, 2018.
- Jacobson, A., et al. “Robust Inside-Outside Segmentation using Generalized Winding Numbers.” *ACM Transactions on Graphics (SIGGRAPH)*, 2013.

20.6 Mesh Refinement

20.6.1 Overview

Mesh refinement is the process of increasing the resolution of a mesh by subdividing its elements (faces and edges) into smaller ones. The goal is to improve the mesh’s quality, accuracy for

numerical simulations, or visual smoothness [Loo87, CC78]. Refinement can be **global** (applied to the entire mesh) or **local** (applied only to specific regions of interest). Common algorithms include Loop subdivision (for triangles) and Catmull-Clark subdivision (for quads).

20.6.2 Definitions

Definition 20.16 (Subdivision). Splitting an existing face into multiple smaller faces.

Definition 20.17 (Edge Split). Inserting a new vertex at the midpoint of an edge and splitting the edge into two.

Definition 20.18 (Face Split). Connecting a new vertex at the centroid of a face to its original vertices.

Definition 20.19 (Adaptive Refinement). Refinement that occurs only where a specific error metric exceeds a threshold.

Definition 20.20 (Stencil). A weighted average formula used to determine the position of new and original vertices after subdivision.

20.6.3 Theory

Most refinement schemes follow a two-step process:

1. **Topological Split:** The mesh connectivity is modified to create more faces (e.g., 1-to-4 triangle split).
2. **Smoothing (Geometric Update):** The positions of the new and existing vertices are adjusted based on their neighbors to achieve a specific level of smoothness (e.g., C^1 or C^2 continuity).

For example, in **Loop Subdivision** [Loo87]:

- Each triangle is split into four smaller triangles.
- New “edge vertices” are positioned using a weighted average of the endpoints and the opposite vertices of the two incident faces.
- Original “even vertices” are repositioned based on their neighbors to maintain surface smoothness.

For **Catmull-Clark Subdivision** [CC78]:

- Each quad is split into four smaller quads by inserting vertices at face centers and edge midpoints.
- The limiting surface is C^2 everywhere except at extraordinary vertices (where C^1 continuity holds).

Example 20.21 (1-to-4 Triangle Refinement). Start with a single equilateral triangle with vertices A , B , C . After one level of 1-to-4 refinement:

- Three new edge midpoints are created: M_{AB} , M_{BC} , M_{CA} .
- Four new triangles are formed: (A, M_{AB}, M_{CA}) , (B, M_{BC}, M_{AB}) , (C, M_{CA}, M_{BC}) , (M_{AB}, M_{BC}, M_{CA}) .
- The number of faces goes from 1 to 4, edges from 3 to 9, and vertices from 3 to 6.
- The Euler characteristic is preserved: $\chi = 6 - 9 + 4 = 1$ (disk), same as the original $\chi = 3 - 3 + 1 = 1$.

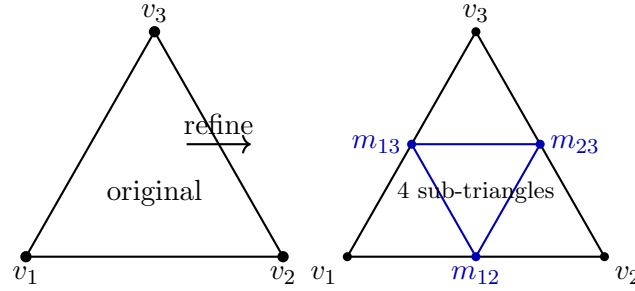


Figure 20.2: 1-to-4 triangle refinement: each edge is split at its midpoint, producing four congruent sub-triangles.

Algorithm 84 Refine Mesh

```

1: function REFINEMESH(mesh)
2:   new_vertices  $\leftarrow$  mesh.vertices[:]
3:   new_faces  $\leftarrow \emptyset$ 
4:   edge_midpoints  $\leftarrow \{\}$  ▷ Create new vertices at every edge midpoint
5:   for each edge in mesh.edges do
6:      $(v_1, v_2) \leftarrow$  edge
7:     mid_p  $\leftarrow$  (mesh.vertices[v1] + mesh.vertices[v2]) / 2.0
8:     edge_midpoints[sorted(edge)]  $\leftarrow$  [new_vertices]
9:     new_vertices.append(mid_p)
10:  end for ▷ Create four new triangles for every original face
11:  for each face in mesh.faces do
12:     $(v_1, v_2, v_3) \leftarrow$  face
13:    m1  $\leftarrow$  edge_midpoints[sorted(v1, v2)]
14:    m2  $\leftarrow$  edge_midpoints[sorted(v2, v3)]
15:    m3  $\leftarrow$  edge_midpoints[sorted(v3, v1)]
16:    new_faces.append((v1, m1, m3))
17:    new_faces.append((v2, m2, m1))
18:    new_faces.append((v3, m3, m2))
19:    new_faces.append((m1, m2, m3))
20:  end for
21:  return Mesh(new_vertices, new_faces)
22: end function

```

20.6.4 Pseudocode: Simple 1-to-4 Triangle Refinement

20.6.5 Parameter Selection

- **Refinement Level:** The number of iterations. Each iteration of 1-to-4 split quadruples the number of faces.
- **Metric:** For adaptive refinement, common metrics include curvature, gradient of a simulated field (e.g., stress or temperature), or edge length.

20.6.6 Complexity

- **Time Complexity:** $O(F \cdot 4^k)$ where F is the original face count and k is the refinement level.
- **Space Complexity:** $O(F \cdot 4^k)$ to store the refined mesh.

20.6.7 Usages

- **Computer Graphics:** Generating high-fidelity models from low-poly base meshes (e.g., characters in movies or games).
- **Finite Element Method (FEM):** Improving solution accuracy in regions with high stress or fluid turbulence.
- **3D Printing:** Increasing the resolution of a mesh to ensure smooth curved surfaces after slicing.
- **Terrain Modeling:** Adding detail to a coarse elevation grid where the landscape is complex.
- **Surface Reconstruction:** Improving the density of a mesh fitted to a sparse point cloud.

20.6.8 Testing Methods and Edge Cases

- **Consistency:** Verify that the refined mesh occupies the exact same volume as the original (for linear refinement).
- **Manifold Property:** Ensure refinement doesn't introduce non-manifold edges or vertices.
- **Boundaries:** Verify that boundary edges are correctly split and handled (often boundary rules are different from interior rules).
- **Watertightness:** Check that no gaps are introduced between adjacent sub-triangles.
- **High Valence:** Test on meshes with “poles” (vertices with many incident edges).

20.6.9 References

- Loop, C. [Loo87] “Smooth subdivision surfaces based on triangle meshes”. Master’s thesis, University of Utah.
- Catmull, E., & Clark, J. [CC78] “Recursively generated B-spline surfaces on arbitrary topological meshes”. Computer-Aided Design.
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing”. CRC Press.
- Wikipedia: [Subdivision surface](#)

20.7 Approximate Convex Decomposition

20.7.1 Overview

Approximate Convex Decomposition (ACD) is the process of partitioning a 3D surface mesh into a set of nearly convex parts [WLW22]. Unlike exact convex decomposition, which can produce an enormous number of tiny, thin pieces for complex organic shapes, ACD allows for a controllable level of “concavity tolerance.” This results in a small number of visually meaningful parts that are easy to handle in physics engines and motion planning. CoACD (Collision-oriented ACD) is a modern, state-of-the-art implementation that prioritizes accuracy and performance for collision detection.

20.7.2 Definitions

Definition 20.22 (Concavity). A measure of how much a shape deviates from being convex. It is often measured as the maximum distance from the surface to its convex hull.

Definition 20.23 (Convex Decomposition). A collection of parts whose union matches the original shape and whose interiors are disjoint.

Definition 20.24 (Cutting Plane). A plane used to split a non-convex part into two smaller, more convex parts.

20.7.3 Theory

CoACD operates by recursively splitting the mesh using a **best-fit cutting plane** [WLW22].

1. **Voxelization:** The input mesh is first converted into a high-resolution voxel grid to simplify the geometry and handle potential topological defects (like holes or self-intersections).
2. **Concavity Estimation:** For each part, the maximum distance from the surface to its convex hull is calculated. If this distance is below the **threshold**, the part is accepted as “approximately convex.”
3. **Plane Selection:** If a part is too concave, the algorithm searches for a cutting plane that minimizes the sum of concavities of the two resulting parts. This is often done by sampling candidate planes from the mesh’s dual graph or using a manifold-aware search.
4. **Merging:** After decomposition, adjacent parts that are nearly convex together can be merged to further reduce the total part count.

Example 20.25 (CoACD of an L-Shape). Consider an L-shaped polyhedron. Its convex hull is much larger than the original shape, so the concavity is large. CoACD with threshold 0.05 might split it along the inner corner, producing two rectangular prisms (each with small concavity relative to its own convex hull). With a larger threshold (e.g., 0.5), the algorithm might accept the L-shape as a single part, since the concavity is below the threshold.

20.7.4 Pseudocode

20.7.5 Parameter Selection

- **Concavity Threshold:** The most important parameter. A smaller threshold (e.g., 0.01) produces many accurate parts; a larger threshold (e.g., 0.1) produces a few coarse parts.
- **Resolution:** The voxel grid resolution (e.g., 50–100 voxels per side). Higher resolution is slower but more accurate.
- **Max Parts:** A hard limit on the total number of pieces to prevent excessive fragmentation.

Algorithm 85 CoACD

```
1: function CoACD(mesh, threshold, max_parts)
2:   Voxelize(mesh)
3:   parts  $\leftarrow$  [v_mesh]
4:   result  $\leftarrow$   $\emptyset$ 
5:   while parts  $\neq$   $\emptyset$  do
6:     current  $\leftarrow$  parts.pop()
7:     CalculateConcavity(current)
8:     if concavity  $\leq$  threshold or |result|  $\geq$  max_parts then
9:       result.append(current)
10:    else
11:      FindBestCuttingPlane(current)
12:      Split(current, best_plane)
13:      parts.extend([ $P_1$ ,  $P_2$ ])
14:    end if
15:  end while
16:  MergeSmallParts(result, threshold)
17:  return MapToOriginalSurface(final_parts, mesh)
18: end function
```

20.7.6 Complexity

- **Time Complexity:** $O(K \cdot N \log N)$ where N is the number of voxels/vertices and K is the number of resulting parts. The plane search is the most expensive step.
- **Space Complexity:** $O(N)$ to store the voxel grid and the sub-meshes.

20.7.7 Usages

- **Physics Simulation (Collision Shapes):** Modern engines like PhysX, Bullet, and Havok work much faster with a set of convex hulls than with a single complex triangle mesh.
- **Robotics:** Path planning for grippers or mobile robots, where obstacles are simplified into convex hulls for fast distance queries.
- **Computer Vision:** Segmenting a complex scanned object into its functional components (e.g., a chair into legs, seat, and back).
- **Animation:** Generating ragdoll physics rigs for characters.

20.7.8 Testing Methods and Edge Cases

- **Perfectly Convex Mesh:** The algorithm should return the original mesh as a single part.
- **Non-Manifold Geometry:** Verify that the voxelization step correctly “heals” the mesh.
- **Sharp Spikes:** These create high local concavity; ensure the threshold correctly identifies them.
- **Thin Structures:** Very thin regions (like a piece of paper) can be tricky for voxelization; check for “leaking” or vanishing parts.
- **Holes:** Ensure the decomposition doesn’t try to “fill” holes unless intended.

20.7.9 References

- Wei, X., Liu, J., & Wang, J. [WLW22] “CoACD: Collision-oriented Approximate Convex Decomposition”. SIGGRAPH.
- Mamou, K., & Ghorbel, F. (2009). “A Simple and Efficient Approach for 3D Mesh Approximate Convex Decomposition”. SEGE.
- Lien, J. M., & Amato, N. M. (2007). “Approximate Convex Decomposition of Polyhedra”. ACM TOG.
- [GitHub: CoACD repository](#)

20.8 Triangle to Quad Conversion

20.8.1 Overview

Triangle to quad conversion is the process of transforming a triangle-only mesh into a quad-dominant or pure quadrilateral mesh [R⁺12, BLP⁺13]. Quadrilateral meshes are often preferred in computer graphics and numerical simulations (like Finite Element Analysis) because they follow principal curvature directions better and provide more natural coordinate systems for texture mapping and subdivision.

20.8.2 Definitions

Definition 20.26 (Dual Graph (Triangle Mesh)). A graph where each triangle is a node and edges connect nodes representing triangles that share an edge.

Definition 20.27 (Perfect Matching). A set of edges in the dual graph such that every node is incident to exactly one edge. In our context, this corresponds to a pairing of every triangle into a quad.

Definition 20.28 (Quad-Dominant Mesh). A mesh containing mostly quads but some triangles.

Definition 20.29 (Pure Quad Mesh). A mesh where every face has exactly four vertices.

20.8.3 Theory

The most common approach to triangle-to-quad conversion is based on **pairing adjacent triangles**.

1. Represent the mesh as a dual graph.
2. Assign a weight to each edge in the dual graph based on the “quality” of the potential quad (e.g., how close it is to a square or a rectangle, flatness, etc.).
3. Find a maximum weight matching in this graph.
4. Each matched pair of triangles is merged into a quadrilateral by removing their shared edge.
5. Unmatched triangles remain as triangles in the final mesh.

To achieve a **pure quad mesh**, more aggressive techniques are needed, such as:

- **Catmull-Clark Subdivision**: Every triangle is split into three quads by adding a vertex at the centroid and midpoints of edges. This always results in a pure quad mesh but significantly increases the face count.
- **Q-Morph**: A front-advancing algorithm that builds quads from the boundary inward.

20.8.4 Pseudocode: Greedy Triangle Merging

Algorithm 86 Greedy Triangle to Quad

```
1: function GREEDYTRIANGLETOQUAD(mesh)
2:   BuildDualGraph(mesh)
3:   CalculateQuadQualityWeights(dual_graph, mesh)
4:   sorted_pairs  $\leftarrow$  sort(weights, key = weights, reverse = True)
5:   matched_triangles  $\leftarrow \emptyset$ 
6:   quads  $\leftarrow \emptyset$ 
7:   for ( $T_1, T_2$ ) in sorted_pairs do
8:     if  $T_1 \notin$  matched_triangles and  $T_2 \notin$  matched_triangles then
9:       MergeTriangles( $T_1, T_2$ , mesh)
10:      quads.append(new_quad)
11:      matched_triangles.add( $T_1$ )
12:      matched_triangles.add( $T_2$ )
13:    end if
14:  end for
15:  remaining_tris  $\leftarrow [T \in \text{mesh.faces} \mid T \notin \text{matched\_triangles}]$ 
16:  return Mesh(mesh.vertices, quads+remaining_tris)
17: end function
```

20.8.5 Parameter Selection

- **Quality Metric:** A common metric is the ratio of the quad's diagonals or the deviation of its internal angles from 90° .
- **Matching Algorithm:** Greedy matching is fast; Blossom's algorithm finds the global maximum weight matching but is much slower.

20.8.6 Complexity

- **Graph Construction:** $O(F)$ where F is the number of faces.
- **Greedy Matching:** $O(F \log F)$ for sorting the dual edges.
- **Blossom Algorithm:** $O(E \cdot V^2)$, which is $O(F^3)$ for meshes.

20.8.7 Usages

- **Finite Element Method (FEM):** Quads are often more accurate for structural and thermal analysis.
- **Animation:** Quad meshes deform more predictably during character rigging and skinning.
- **Subdivision Modeling:** Catmull-Clark subdivision is the industry standard for creating smooth surfaces from quad cages.
- **Texture Mapping:** Providing a more natural UV grid for 2D textures.

20.8.8 Testing Methods and Edge Cases

- **Planar Mesh:** Merging two triangles into a quad should preserve the flatness.

- **Non-Convex Pairs:** Merging two triangles into a concave quad should be avoided or flagged (using the quality metric).
- **Singularities:** Identify “extraordinary vertices” (those not shared by 4 quads).
- **Watertightness:** Ensure no holes are created at the boundaries of merged quads.
- **Manifoldness:** The merging process must preserve the manifold property of the mesh.

20.8.9 References

- Remacle, J. F., et al. [R⁺12] “A Frontal Delaunay Quad Mesh Generator”. International Journal for Numerical Methods in Engineering.
- Owen, S. J. (1998). “A Survey of Unstructured Mesh Generation Technology”. IMR.
- Bommes, D., Lévy, B., Pietroni, N., Puppo, E., Silva, C., Tarini, M., & Zayer, R. [BLP⁺13] “State of the Art in Quad-Meshing”. Eurographics STAR.
- Wikipedia: [Types of mesh](#)

20.9 Voxel Grid Point Simplification

20.9.1 Overview

Voxel Grid Decimation is a point cloud simplification algorithm that reduces the number of points in a dataset while maintaining its spatial distribution [BKP⁺10]. It works by partitioning the space into a regular grid of “voxels” (3D pixels) and replacing all points within each voxel with a single representative point. This is an extremely efficient way to downsample large point clouds, such as those generated by LiDAR or photogrammetry.

20.9.2 Definitions

- **Voxel:** A volume element representing a value on a regular grid in three-dimensional space.
- **Decimation:** The process of reducing the sampling rate of a signal or data set.
- **Grid Hashing:** Mapping a 2D or 3D coordinate (x, y, z) to an integer triple (i, j, k) based on a fixed cell size, used as a key for fast lookups.
- **Representative Point:** The point chosen to represent a voxel; typically the first point found, the centroid of all points in the voxel, or the point closest to the voxel’s center.

20.9.3 Theory

The algorithm discretizes the continuous space inhabited by the point cloud. By defining a voxel size L , any point (x, y, z) can be assigned to a voxel with indices $(\lfloor x/L \rfloor, \lfloor y/L \rfloor, \lfloor z/L \rfloor)$.

In the simplest form, the first point that falls into a previously empty voxel is kept, and all subsequent points falling into the same voxel are discarded. More sophisticated versions calculate the average position (centroid) of all points within the voxel to better preserve the underlying surface geometry. This process results in a point cloud where no two points are closer than the voxel size (approximately), leading to a uniform density.

Algorithm 87 Voxel Grid Simplify

```
1: function VOXELGRIDSIMPLIFY(points, voxel_size)
2:   grid  $\leftarrow$  hash_map()
3:   simplified_points  $\leftarrow \emptyset$ 
4:   for each  $p$  in points do
5:     gx  $\leftarrow \lfloor p.x / \text{voxel\_size} \rfloor$ 
6:     gy  $\leftarrow \lfloor p.y / \text{voxel\_size} \rfloor$ 
7:     gz  $\leftarrow \lfloor p.z / \text{voxel\_size} \rfloor$ 
8:     key  $\leftarrow$  (gx, gy, gz)
9:     if key  $\notin$  grid then
10:      if NoPointTooCloseInNeighbors( $p$ , key, grid, voxel_size) then
11:        grid[key]  $\leftarrow p$ 
12:        simplified_points.append( $p$ )
13:      end if
14:    end if
15:  end for
16:  return simplified_points
17: end function
```

20.9.4 Pseudocode**20.9.5 Parameter Selection**

- **Voxel Size (L):** Often determined by a **ratio** of the bounding box diagonal (e.g., $L = 0.01 \times \text{diagonal}$). A larger L results in more aggressive simplification.
- **Protected Points:** Specific points (like corners or edges) can be flagged by ID to be excluded from decimation, ensuring critical features are preserved.

20.9.6 Complexity

- **Time Complexity:** $O(n)$, where n is the number of points. Hash map insertions and lookups are $O(1)$ on average.
- **Space Complexity:** $O(n)$ in the worst case (if every point ends up in a different voxel), but typically much less as points are decimated.

20.9.7 Usages

- **LiDAR Processing:** Reducing millions of points to a manageable number for terrain modeling.
- **Real-time Rendering:** Generating Levels of Detail (LOD) for 3D environments.
- **Collision Detection:** Simplifying complex meshes into point-representative volumes for faster intersection tests.
- **Robotics:** Downsampling sensor data for simultaneous localization and mapping (SLAM).

20.9.8 Testing Methods and Edge Cases

- **Uniform Points:** A perfectly regular grid of points should be decimated predictably.
- **High-Density Clusters:** Ensure that thousands of points in a single voxel are correctly reduced to one.

- **Voxel Size > Bounding Box:** Should result in a single point (the first one processed).
- **Voxel Size ≈ 0 :** Should return the original point cloud.
- **Precision:** Handle floating-point errors at voxel boundaries.

20.9.9 References

- PCL (Point Cloud Library) Documentation: `pcl::VoxelGrid`.
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing”. CRC Press.
- Wikipedia: [Point cloud](#)

20.10 Surface Reconstruction

20.10.1 Overview

Surface reconstruction is the process of recovering a continuous surface representation from a discrete set of sampled points in three-dimensional space. This problem arises in numerous applications: 3D scanning devices such as LiDAR and structured-light scanners produce dense point clouds that must be converted into watertight meshes for downstream processing. Photogrammetry pipelines generate point clouds from photographs. Medical imaging yields volumetric point data that benefits from surface extraction. The two principal families of reconstruction algorithms are *local* methods, which grow a mesh directly from the point set by pivoting a probing ball across the surface, and *global* methods, which recover an implicit function whose iso-surface is then extracted.

20.10.2 Definitions

Definition 20.30 (Surface Reconstruction). Given a set $P = \{p_1, \dots, p_n\} \subset \mathbb{R}^3$ sampled from an unknown surface S , the surface reconstruction problem is to compute a mesh M that approximates S in a topologically and geometrically faithful manner.

Definition 20.31 (Ball Pivoting Algorithm). A local mesh-growing algorithm introduced by Bernardini et al. [BMR⁺99] that reconstructs a triangle mesh from a point cloud by rolling a ball of radius ρ over the point set. The ball pivots around the boundary edges of the current mesh, seeking new triangles.

Definition 20.32 (Empty Ball Property). A ball of radius ρ centered at c satisfies the *empty ball property* with respect to a triangle $\triangle(p_i, p_j, p_k)$ if no point of $P \setminus \{p_i, p_j, p_k\}$ lies strictly inside the ball, and the ball passes through p_i , p_j , and p_k .

Definition 20.33 (Seed Triangle). A triangle $\triangle(p_i, p_j, p_k)$ formed by three points of P such that a ball of radius ρ rests on the three vertices and contains no other point of P in its interior. The seed triangle initiates the mesh-growing process.

Definition 20.34 (Boundary Edge Front). The set of boundary edges $\mathcal{F}$ maintained during ball pivoting. A boundary edge is an edge of the current mesh shared by exactly one triangle. The algorithm iteratively pivots around edges in $\mathcal{F}$ to discover new triangles.

Definition 20.35 (Poisson Equation). Given a vector field $\mathbf{V}$ defined over a domain Ω , the Poisson equation seeks a scalar function ϕ satisfying $\Delta\phi = \nabla \cdot \mathbf{V}$, where Δ is the Laplacian operator.

Definition 20.36 (Indicator Function). A binary function $\chi_S : \mathbb{R}^3 \rightarrow \{0, 1\}$ that equals 1 inside the solid whose boundary is the surface S and 0 outside. Recovering χ_S from oriented point samples is the goal of Poisson reconstruction.

Definition 20.37 (Marching Cubes). An algorithm for extracting an iso-surface from a scalar field sampled on a regular grid. Each grid cell is classified into one of 256 topological cases, and triangulated accordingly [LC87].

20.10.3 Theory

Ball Pivoting Algorithm

The BPA proceeds in three phases: seed selection, pivoting, and front propagation.

Seed Triangle Finding. The algorithm searches for three points $p_i, p_j, p_k \in P$ such that a ball of radius ρ can be placed on the three vertices without enclosing any other point. In the implementation (`BallPivotingReconstructor`), a triple is accepted when (i) the circumradius r of $\triangle(p_i, p_j, p_k)$ satisfies $r \leq \rho$, and (ii) the ball centered at the computed center c contains no point of P other than the three vertices.

Ball Center Computation. Given three non-collinear points p_1, p_2, p_3 with circumradius $r < \rho$, the ball center lies on the line perpendicular to the triangle plane through its circumcenter m . Setting $h = \sqrt{\rho^2 - r^2}$, the center is $c = m + h \hat{n}$, where $\hat{n}$ is the unit normal to the triangle plane oriented consistently with the surface orientation.

Pivoting Around a Boundary Edge. Given a boundary edge (v_1, v_2) with an existing triangle $\triangle(v_1, v_2, v_{\text{opp}})$, the ball is rotated around the edge until it touches a new point p_k . The pivot point is selected as the candidate minimizing the rotation angle such that the new ball on (v_1, v_2, p_k) satisfies the empty ball property. This guarantees a Delaunay-like local optimality.

Front Propagation. Each new triangle $\triangle(v_1, p_k, v_2)$ adds three edges to the mesh. Edges that are shared by exactly one triangle become boundary edges and are appended to the front $\mathcal{F}$. The algorithm terminates when the front is empty.

Poisson Surface Reconstruction

Vector Field Construction. Given P with oriented normals $\{n_i\}$, the algorithm rasterizes the normals into a regular grid of resolution res^3 , computing an averaged normal vector $\mathbf{V}(g)$ at each grid cell g from the points that fall within it.

Divergence Computation. The divergence of $\mathbf{V}$ is approximated on the grid using finite differences:

$$\nabla \cdot \mathbf{V} \approx \frac{V_x(i+1, j, k) - V_x(i, j, k)}{\Delta x} + \frac{V_y(i, j+1, k) - V_y(i, j, k)}{\Delta y} + \frac{V_z(i, j, k+1) - V_z(i, j, k)}{\Delta z}.$$

Solving the Poisson Equation. The discrete Laplacian $\Delta\phi = \nabla \cdot \mathbf{V}$ is assembled as a sparse linear system $L\phi = \text{div}(\mathbf{V})$ and solved via a sparse direct solver. Boundary conditions are set to Dirichlet ($\phi = 0$) on the grid boundary.

Iso-Surface Extraction. The isovalue τ is chosen as the median of ϕ evaluated at the input point positions. Marching cubes [LC87] then extracts the triangulated iso-surface $\{x : \phi(x) = \tau\}$.

Theorem 20.38 (BPA Topology Recovery). *Let S be a compact surface without boundary, and let P be an ε -sample of S , meaning every point $q \in S$ has some $p_i \in P$ with $\|q - p_i\| \leq \varepsilon \cdot \text{med}(S)$, where $\text{med}(S)$ is the medial axis distance. If $\rho \geq \varepsilon \cdot \text{med}(S)$ and the sampling is sufficiently dense, then the Ball Pivoting Algorithm with ball radius ρ recovers a mesh homeomorphic to S .*

Proof Sketch. By the ε -sampling condition, every point on S is within distance $\varepsilon \cdot \text{med}(S)$ of a sample. The medial axis guarantees that the local feature size controls the sampling density needed for topological correctness. The ball of radius ρ is large enough to bridge adjacent samples while small enough to avoid puncturing through thin features of S . The empty ball property ensures that each pivoting step selects a point lying on the tangent plane of the local surface patch, preserving the local Delaunay property. The front propagation covers S without introducing topological artifacts because the ε -sample condition prevents the ball from “jumping” across surface features. A complete proof requires showing that the front never creates handles or tunnels, which follows from the feature-size argument of Amenta and Bern [ABE98]. $\square$

Example 20.39 (Ball Pivoting on Four Points). Consider four points forming a tetrahedron: $p_0 = (0, 0, 0)$, $p_1 = (1, 0, 0)$, $p_2 = (0, 1, 0)$, $p_3 = (0, 0, 1)$, with ball radius $\rho = 1.2$.

1. **Seed triangle:** The circumradius of $\triangle(p_0, p_1, p_2)$ is $\frac{\sqrt{2}}{2} \approx 0.707 < \rho$. The ball center is at $c = (\frac{1}{2}, \frac{1}{2}, h)$ with $h = \sqrt{\rho^2 - 0.5} \approx 0.872$. The empty ball check confirms that p_3 lies outside (distance to $c \approx 0.866 > \rho$). The seed triangle $\triangle(p_0, p_1, p_2)$ is accepted.
2. **Pivoting edge (p_0, p_1) :** The ball pivots around (p_0, p_1) starting from p_2 . The candidate p_3 yields a new ball center with circumradius of $\triangle(p_0, p_1, p_3) = \frac{\sqrt{2}}{2}$, same empty ball check. The triangle $\triangle(p_0, p_1, p_3)$ is added.
3. The process continues until all boundary edges are covered. The result is four triangles forming the tetrahedron surface.

20.10.4 Pseudocode

20.10.5 Parameter Selection

- **Ball Radius (ρ):** The most critical parameter for BPA. Typically set to $\rho \approx 2\text{--}5 \times$ the average nearest-neighbor distance. Too small a radius results in holes in the reconstruction; too large a radius causes the ball to bridge across concavities and produce an over-smoothed surface.
- **Grid Resolution (Poisson):** Controls the spatial fidelity of the implicit function. Higher resolution yields finer detail but increases memory and solve time as $O(\text{res}^3)$. A typical starting value is $\text{res} = 32$, increased to 64–128 for fine detail.
- **Isovalue (Poisson):** Set as the median of the implicit function ϕ evaluated at the input points. This places the extracted surface “on” the samples.

20.10.6 Complexity

- **BPA Time Complexity:** In the worst case $O(n^2)$ per seed search and $O(n)$ per pivot step. With spatial hashing for the empty ball test, the average-case pivoting step is $O(k \log n)$ where k is the number of candidate points, yielding an overall $O(n \cdot k \log n)$ reconstruction.
- **BPA Space Complexity:** $O(n + t)$ where n is the number of points and t is the number of output triangles.

Algorithm 88 Ball Pivoting Algorithm Reconstruction

```

1: function RECONSTRUCT(points,  $\rho$ )
2:   triangles  $\leftarrow \emptyset$ 
3:   front  $\leftarrow \emptyset$  ▷ Boundary edge front
4:   used  $\leftarrow \emptyset$ 
5:   edge_count  $\leftarrow$  empty map ▷ Edge  $\rightarrow$  triangle count
6:   seed  $\leftarrow$  FindSeedTriangle(points,  $\rho$ )
7:   if seed = null then
8:     return triangles
9:   end if
10:  AddTriangle(triangles, front, edge_count, seed)
11:  while front  $\neq \emptyset$  do
12:     $(v_1, v_2, v_{\text{opp}}) \leftarrow$  front.pop(0)
13:     $e \leftarrow$  sort( $v_1, v_2$ )
14:    if edge_count[e]  $\geq 2$  then
15:      continue ▷ Interior edge
16:    end if
17:     $p_k \leftarrow$  PivotAroundEdge( $v_1, v_2, v_{\text{opp}}$ , points,  $\rho$ )
18:    if  $p_k \neq \text{null}$  then
19:      AddTriangle(triangles, front, edge_count,  $(v_1, p_k, v_2)$ )
20:    end if
21:  end while
22:  return triangles
23: end function
24: function FINDSEEDTRIANGLE(points,  $\rho$ )
25:   for  $i \leftarrow 0$  to  $n - 1$  do
26:     for  $j \leftarrow i + 1$  to  $n - 1$  do
27:       if  $\|p_i - p_j\|^2 > 4\rho^2$  then continue
28:       end if
29:       for  $k \leftarrow j + 1$  to  $n - 1$  do
30:          $c \leftarrow$  BallCenter( $p_i, p_j, p_k, \rho$ )
31:         if  $c \neq \text{null}$  and IsEmptyBall( $c, \{i, j, k\}$ , points,  $\rho$ ) then
32:           return  $(i, j, k)$ 
33:         end if
34:       end for
35:     end for
36:   end for
37:   return null
38: end function
39: function PIVOTAROUNDEDGE( $v_1, v_2, v_{\text{opp}}$ , points,  $\rho$ )
40:   best  $\leftarrow$  null, min_angle  $\leftarrow \infty$ 
41:    $c_{\text{old}} \leftarrow$  BallCenter( $p_{v_1}, p_{v_2}, p_{v_{\text{opp}}}, \rho$ )
42:   mid  $\leftarrow \frac{1}{2}(p_{v_1} + p_{v_2})$ 
43:   for  $k \leftarrow 0$  to  $n - 1$  do
44:     if  $k = v_1$  or  $k = v_2$  or  $k = v_{\text{opp}}$  then continue
45:     end if
46:     if  $\|p_k - \text{mid}\|^2 > 4\rho^2$  then continue
47:     end if
48:      $c_{\text{new}} \leftarrow$  BallCenter( $p_{v_1}, p_{v_2}, p_k, \rho$ )
49:     if  $c_{\text{new}} = \text{null}$  then continue
50:     end if
51:     if not IsEmptyBall( $c_{\text{new}}, \{v_1, v_2, k\}$ , points,  $\rho$ ) then continue
52:     end if
53:      $\alpha \leftarrow$  PivotAngle(mid,  $p_{v_2} - p_{v_1}$ ,  $c_{\text{old}}, c_{\text{new}}$ )
54:     if  $\alpha < \text{min\_angle}$  then
55:       min_angle  $\leftarrow \alpha$ , best  $\leftarrow k$ 
56:     end if

```


- **Poisson Time Complexity:** $O(N^3 \log N)$ where $N = \text{res}$, dominated by the sparse solve of the Laplacian system on the N^3 grid.
- **Poisson Space Complexity:** $O(N^3)$ for the grid and sparse matrix.

20.10.7 Usages

- **3D Scanning:** Reconstructing object surfaces from structured-light or time-of-flight scanner data.
- **LiDAR Processing:** Generating terrain and building meshes from aerial or terrestrial LiDAR point clouds.
- **Photogrammetry:** Converting multi-view stereo point clouds into watertight 3D models.
- **Cultural Heritage:** Digital preservation of sculptures, buildings, and archaeological sites.
- **Medical Imaging:** Extracting organ surfaces from volumetric scans (CT, MRI) for surgical planning.

20.10.8 Testing Methods and Edge Cases

- **Sphere Sampling:** Reconstructing a sphere from uniformly sampled points should yield a mesh homeomorphic to S^2 with correct genus.
- **Sparse Points:** Verify graceful degradation when the point cloud is too sparse for the chosen ball radius.
- **Collinear Points:** The ball center computation must return `null` for degenerate collinear triples.
- **Duplicate Points:** Duplicates should not cause infinite loops or degenerate triangles.
- **Isolated Components:** Point clouds with disconnected parts should each be reconstructed independently.
- **Poisson with Noisy Normals:** Incorrect normals lead to inverted or distorted iso-surfaces; test with synthetic noise.

20.10.9 References

- Bernardini, F., Mittleman, J., Rushmeier, H., Silva, C., & Taubin, G. [BMR⁺99] “The Ball-Pivoting Algorithm for Surface Reconstruction”. IEEE TVCG.
- Kazhdan, M., Bolitho, M., & Hoppe, H. [KBH06] “Poisson Surface Reconstruction”. Eurographics Symposium on Geometry Processing.
- Amenta, N., & Bern, M. [ABE98] “Surface Reconstruction by Voronoi Filtering”. Discrete & Computational Geometry.
- Lorensen, W. E., & Cline, H. E. [LC87] “Marching Cubes: A High Resolution 3D Surface Construction Algorithm”. ACM SIGGRAPH.

20.11 α -Shapes

20.11.1 Overview

An α -shape is a family of geometric structures that generalizes the convex hull of a point set, providing a notion of “concave hull” or “crust” that captures the underlying shape at multiple scales. Introduced by Edelsbrunner, Kirkpatrick, and Seidel [ES83], α -shapes are obtained by filtering the simplices of a Delaunay triangulation according to a radius parameter α . As α varies from small to large values, the α -shape transitions from a collection of disconnected components to the full convex hull. This multi-scale property makes α -shapes valuable for extracting meaningful boundary geometry from point clouds in both 2D and 3D.

20.11.2 Definitions

Definition 20.40 (α -Shape). Given a finite point set $P \subset \mathbb{R}^d$ and a real parameter $\alpha > 0$, the α -shape of P is the polyhedral complex obtained by retaining every Delaunay simplex (edge in 2D, triangle in 2D, tetrahedron in 3D) whose *circumradius* is strictly less than α .

Definition 20.41 (Concave Hull). A non-convex boundary that tightly encloses a point set, capturing concavities that the convex hull omits. The α -shape boundary is a canonical example of a concave hull.

Definition 20.42 (Rolling Ball). A conceptual construction in which a ball of radius α is “rolled” along the exterior of the point set. Portions of the boundary that the ball cannot reach (because it is too large to fit into concavities) are excluded from the α -shape.

Definition 20.43 (Circumradius). The circumradius R of a simplex is the radius of its circumscribed sphere. For a triangle with side lengths a, b, c and area A , the circumradius is $R = \frac{abc}{4A}$. For a tetrahedron, the circumradius is the radius of the unique sphere passing through all four vertices.

Definition 20.44 (α -Complex). The α -complex is the abstract simplicial complex consisting of all Delaunay simplices with circumradius less than α . The α -shape is the geometric realization of this complex.

20.11.3 Theory

2D α -Shapes

Delaunay Triangulation. The algorithm begins by computing the Delaunay triangulation of the input point set P . In 2D, this yields a triangulation of the convex hull in which every triangle has the empty circumcircle property.

Circumradius Filtering. For each Delaunay triangle T_i with vertices (p_a, p_b, p_c) , the circumradius R_i is computed as $R_i = \frac{abc}{4A}$, where a, b, c are the side lengths and A is the area of T_i . If $R_i < \alpha$, the triangle is *accepted*; otherwise, it is *rejected*.

Boundary Edge Extraction. The α -shape boundary consists of all edges that belong to exactly one accepted triangle. Symmetric difference logic achieves this: each edge is added to a set when encountered in an accepted triangle; if the edge is encountered a second time, it is removed (it is an interior edge).

3D α -Shapes

In 3D, the Delaunay triangulation yields tetrahedra. Each tetrahedron is accepted or rejected based on its circumradius. The boundary of the α -shape consists of the triangular faces shared by exactly one accepted tetrahedron, again computed via symmetric difference.

Relationship to the Convex Hull. As $\alpha \rightarrow \infty$, every Delaunay simplex has circumradius less than α , so the α -shape equals the convex hull. As $\alpha \rightarrow 0$, only simplices with vanishing circumradius survive, eventually yielding only the original point set.

Theorem 20.45 (α -Shape Topology Recovery). *Let $S \subset \mathbb{R}^d$ be a compact set with a well-defined boundary, and let P be an ε -sample of S with ε sufficiently small relative to the local feature size of S . Then there exists $\alpha > 0$ such that the boundary of the α -shape of P is homeomorphic to ∂S .*

Proof Sketch. The proof relies on the stability of Delaunay triangulations under perturbation (the *smoothed analysis* framework of Amenta, Bern, and Eppstein [ABE98]). For sufficiently dense sampling (ε small enough), the Delaunay triangulation of P contains simplices whose circumradii approximate the local feature size of S . Choosing α between the maximum circumradius of simplices that cross the medial axis and the minimum circumradius of simplices that lie on the boundary surface ensures that exactly the boundary-crossing simplices are retained. The resulting α -complex is then a topological ball (or the appropriate surface type), establishing the homeomorphism. The key geometric argument is that the ε -sample condition prevents the Delaunay triangulation from introducing spurious features at scales smaller than ε . $\square$

Example 20.46 (α -Shape of Five Points in 2D). Consider five points in $\mathbb{R}^2$: $p_0 = (0, 0)$, $p_1 = (2, 0)$, $p_2 = (2, 2)$, $p_3 = (0, 2)$, $p_4 = (1, 1)$. The Delaunay triangulation produces four triangles: $T_1 = \triangle(p_0, p_1, p_4)$, $T_2 = \triangle(p_1, p_2, p_4)$, $T_3 = \triangle(p_2, p_3, p_4)$, $T_4 = \triangle(p_3, p_0, p_4)$.

1. For $\alpha = 1.5$: The circumradius of each T_i is $R = \frac{\sqrt{2}}{2} \approx 0.707 < 1.5$, so all four triangles are accepted. The boundary is the outer square (p_0, p_1, p_2, p_3) , which equals the convex hull.
2. For $\alpha = 0.5$: The circumradius $R \approx 0.707 > 0.5$, so all triangles are rejected. The α -shape degenerates to the point set alone.
3. For $\alpha = 1.0$: All four triangles are accepted (since $0.707 < 1.0$). The α -shape boundary is the square, same as the convex hull in this case.

Now consider a sixth point $p_5 = (3, 1)$ outside the square. With $\alpha = 1.0$, the Delaunay triangle $\triangle(p_1, p_2, p_5)$ has circumradius $R = 1.0$. Since $R = \alpha$ and the filter uses strict inequality, this triangle is rejected, creating a concavity in the α -shape boundary. This illustrates how α -shapes capture concavities that the convex hull does not.

20.11.4 Pseudocode

20.11.5 Parameter Selection

- **Alpha (α):** The radius of the rolling ball. For uniform point distributions, a good starting value is $\alpha \approx 2\text{--}3 \times$ the average nearest-neighbor distance. Smaller α values reveal finer concavities; larger values smooth them out. Setting $\alpha = \infty$ recovers the convex hull.
- **Adaptive Selection:** In practice, α can be chosen adaptively based on local sampling density, using the local feature size estimated from the ε -sampling condition.

Algorithm 89 α -Shape Computation (2D)

```
1: function COMPUTE2D(points,  $\alpha$ )
2:   mesh  $\leftarrow$  DelaunayTriangulation(points)
3:   boundary  $\leftarrow$  empty set  $\triangleright$  Symmetric difference of edges
4:   for each triangle  $T$  in mesh do
5:      $(v_0, v_1, v_2) \leftarrow T$ .vertices
6:      $a \leftarrow \|p_{v_1} - p_{v_0}\|$ ,  $b \leftarrow \|p_{v_2} - p_{v_1}\|$ ,  $c \leftarrow \|p_{v_0} - p_{v_2}\|$ 
7:      $s \leftarrow \frac{a+b+c}{2}$ 
8:      $A \leftarrow \sqrt{\max(0, s(s-a)(s-b)(s-c))}$ 
9:     if  $A < 10^{-12}$  then
10:       continue
11:     end if  $\triangleright$  Degenerate triangle
12:      $R \leftarrow \frac{abc}{4A}$ 
13:     if  $R < \alpha$  then
14:       for  $i \leftarrow 0$  to 2 do
15:          $e \leftarrow \text{sort}(v_i, v_{(i+1) \bmod 3})$ 
16:         if  $e \in \text{boundary}$  then
17:           boundary.remove( $e$ )
18:         else
19:           boundary.insert( $e$ )
20:         end if
21:       end for
22:     end if
23:   end for
24:   return boundary
25: end function
```

20.11.6 Complexity

- **Time Complexity:** $O(n \log n)$, dominated by the Delaunay triangulation in 2D (or $O(n^2)$ in the worst case for 3D Delaunay). The circumradius filtering and boundary extraction are $O(n)$.
- **Space Complexity:** $O(n)$ to store the Delaunay triangulation and the boundary edge set.

20.11.7 Usages

- **Molecular Biology:** Computing the solvent-accessible surface and packing defects of protein structures.
- **Geographic Information Systems:** Extracting coastline or terrain boundaries from scattered elevation points.
- **Material Science:** Analyzing the grain structure and porosity of polycrystalline materials.
- **Point Cloud Processing:** Providing a concave hull alternative to convex hull for bounding irregular point distributions.
- **Robotics:** Generating obstacle boundaries from sensor point clouds with configurable tightness.

20.11.8 Testing Methods and Edge Cases

- **Convex Hull Limit:** For $\alpha \rightarrow \infty$, the α -shape must equal the convex hull of the point set.
- **Degenerate Triangles:** Collinear triples produce zero-area triangles; the circumradius computation must handle this gracefully.
- **Duplicate Points:** Identical points create degenerate simplices that must be filtered or handled.
- **Uniform Grid:** A regular grid of points should produce an α -shape whose boundary closely approximates the bounding rectangle.
- **Ring Topology:** Points on a circle should yield an α -shape that is a polygon approximating the circle for appropriate α .
- **3D Boundary Consistency:** In 3D, the boundary of the α -shape must form a closed manifold (no dangling faces).

20.11.9 References

- Edelsbrunner, H., Kirkpatrick, D., & Seidel, R. [ES83] “On the Shape of a Set of Points in the Plane”. IEEE Transactions on Information Theory.
- Amenta, N., Bern, M., & Eppstein, D. [ABE98] “The Crust and the β -Skeleton: Combinatorial Curve Reconstruction”. Graphical Models and Image Processing.
- Edelsbrunner, H. & Mücke, E. P. [EM94a] “Three-Dimensional Alpha Shapes”. ACM Transactions on Graphics.
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing”. CRC Press.

20.12 Surface Parameterization and Geometry

20.13 BFF Parameterization

20.13.1 1. Overview

Boundary First Flattening (BFF) is a state-of-the-art method for **conformal parameterization** of surface meshes, introduced by Keenan Crane et al. in 2017 [CWW17]. Unlike traditional methods like Harmonic Mapping or LSCM, BFF allows the user to prescribe the target boundary curvature, providing unmatched control over the flattening process (e.g., flattening to a disk, a rectangle, or a specific shape).

20.13.2 2. Definitions

Definition 20.47 (Conformal Mapping). A mapping that locally preserves angles between curves on a surface.

Definition 20.48 (Laplacian Matrix). A discrete representation of the Laplace-Beltrami operator, typically constructed using cotangent weights.

Definition 20.49 (Scale Factor). A scalar field u on the mesh representing the local scaling needed to achieve the target metric under a conformal transformation $g' = e^{2u}g$.

Definition 20.50 (Geodesic Curvature). The curvature of a curve on a surface, relative to the surface itself.

20.13.3 3. Theory

BFF is rooted in the **Yamabe equation**, which describes how the metric changes under a conformal transformation $g' = e^{2u}g$ [CWW17]. The target geodesic curvature κ' on the boundary relates to the original curvature κ and the normal derivative of the scale factor u by:

$$\kappa' = e^{-u} \left(\kappa + \frac{\partial u}{\partial n} \right)$$

BFF solves this by:

1. Determining boundary scale factors u that satisfy the target curvature κ' .
2. Extending u to the interior via a Dirichlet problem.
3. Integrating the flattened boundary and extending the UV coordinates harmonically to the interior.

Example 20.51 (BFF Disk Flattening). Consider a hemisphere mesh with a circular boundary. To flatten it to a disk, BFF sets the target geodesic curvature to $\kappa' = 2\pi/n$ at each of the n boundary vertices (uniform curvature). The boundary system $L_{bb} \cdot u_b = \kappa' - \kappa$ is solved for the boundary scale factors, which encode how much each boundary edge must shrink or stretch. The interior scale factors are then determined by the Dirichlet problem, and the final UV coordinates are obtained by integrating the scaled edge lengths along the boundary and extending harmonically to the interior.

Algorithm 90 BFF Parameterization

```

1: function BFFPARAMETERIZATION( $M$ )
2:   Build the cotangent Laplacian matrix  $L$  for mesh  $M$ 
3:   Identify the boundary loop of  $M$ 
4:   Compute the current geodesic curvature  $\kappa$  at each boundary vertex
5:   Define the target curvature  $\kappa_{\text{target}}$  (e.g.,  $2\pi/n$  for disk)
6:   Solve the boundary system for scale factors  $u_b$ :  $L_{bb} \cdot u_b = \kappa_{\text{target}} - \kappa$ 
7:   Solve the Dirichlet problem for interior scale factors  $u_i$ :  $L_{ii} \cdot u_i = -L_{ib} \cdot u_b$ 
8:   Integrate boundary coordinates  $uv_b$  using  $l_{\text{flat}} = l_{\text{orig}} \cdot \exp\left(\frac{u_i + u_j}{2}\right)$ 
9:   Extend  $uv_b$  to the interior via harmonic mapping:  $L_{ii} \cdot uv_i = -L_{ib} \cdot uv_b$ 
10:  return  $UV$  coordinates for each vertex
11: end function

```

20.13.4 4. Pseudocode**20.13.5 5. Parameter Selection**

- **Target Curvature:** For a disk, set to $2\pi/n$. For a rectangle, set to $\pi/2$ at four vertices and 0 elsewhere.
- **Solver:** Sparse direct solvers (like SuperLU) are recommended for solving the Laplacian systems.

20.13.6 6. Complexity

- **Construction:** $O(N)$ for building the Laplacian.
- **Solving:** $O(N^{1.5})$ to $O(N^2)$ depending on mesh connectivity and sparse solver performance.

20.13.7 7. Usages

Implemented in `compgeom.mesh.surface.parameterization_bff.BFFParameterizer`. Used for UV unwrapping, texture mapping, and surface remeshing.

20.13.8 8. Testing Methods and Edge Cases

- **Testing:** Verify local angle preservation (conformal property). Check boundary curvature error.
- **Edge Cases:** Non-manifold meshes, meshes with no boundary (unsupported), and self-intersecting UV outputs.

20.13.9 9. References

- Crane, K., Weischedel, C., & Wardetzky, M. [CWW17] “Boundary first flattening.” ACM TOG.
- Wikipedia: Conformal mapping.

20.14 Harmonic Mapping**20.14.1 1. Overview**

Mesh Topology Transfer is the process of adapting a source mesh’s connectivity (its faces and edges) to a new geometric boundary. This is achieved using Harmonic Mapping, a technique

that positions interior vertices to minimize a “stretching” energy (Dirichlet energy). The result is a new mesh that shares the same topology as the source but fits the geometry of the target domain with minimal distortion.

20.14.2 2. Definitions

Definition 20.52 (Topology). The connectivity of the mesh (which vertices form which faces), independent of their spatial coordinates.

Definition 20.53 (Harmonic Map). A mapping between geometric domains that satisfies the Laplace equation $\Delta f = 0$.

Definition 20.54 (Barycentric Mapping). A discrete version of harmonic mapping where every interior vertex is the weighted average of its neighbors.

Definition 20.55 (Dirichlet Energy). A measure of how much a mapping “stretches” the domain; harmonic maps are the minimizers of this energy functional.

20.14.3 3. Theory

Based on Tutte’s Embedding Theorem [Tut63], a valid planar embedding of a graph can be found by fixing its boundary to a convex polygon and solving for the interior positions such that each vertex is at the centroid of its neighbors.

Theorem 20.56 (Tutte’s Embedding Theorem). *Let G be a planar graph with a distinguished boundary cycle mapped to a convex polygon. If every interior vertex is placed at the convex combination of its neighbors with strictly positive weights, the resulting embedding is a planar embedding with no edge crossings.*

Proof. Tutte’s proof [Tut63] uses a variational argument. The embedding minimizes the Dirichlet energy $E = \sum_{(i,j) \in E} w_{ij} \|p_i - p_j\|^2$ subject to the boundary constraints. Since the domain is convex and the weights are positive, the energy functional is strictly convex on the interior variables, guaranteeing a unique minimum. At the minimum, each interior vertex satisfies $\nabla_{p_i} E = 0$, which gives $p_i = \frac{1}{\sum_j w_{ij}} \sum_j w_{ij} p_j$ —exactly the weighted average. The convexity of the target polygon, combined with the maximum principle for harmonic functions, ensures that no edge crossings occur. $\square$

Mathematically, for each interior vertex i , we solve:

$$V_i = \frac{1}{\sum_{j \in N(i)} w_{ij}} \sum_{j \in N(i)} w_{ij} V_j$$

Where w_{ij} are weights (often $w_{ij} = 1$ for simple barycentric mapping or cotangent weights for more accurate harmonic maps). This forms a system of linear equations $LV = B$, where L is the Laplacian matrix and B contains boundary constraints.

Example 20.57 (Harmonic Mapping of a Square). Consider a 4×4 grid of vertices (a planar quad mesh) with the boundary vertices fixed to a unit square. Using uniform weights ($w_{ij} = 1$), each interior vertex is positioned at the arithmetic mean of its 4 neighbors. After convergence (typically 50–100 Gauss-Seidel iterations), the interior vertices distribute uniformly, producing a smooth parameterization of the square domain. This is the discrete harmonic map—the unique minimizer of the Dirichlet energy.

Algorithm 91 Transfer Topology

```

1: function TRANSFERTOPOLOGY(source_mesh, target_boundary)
2:   source_boundary  $\leftarrow$  GetBoundaryCycle(source_mesh)
3:   MapBoundary(source_boundary, target_boundary) ▷ Using arc-length
4:   target_centroid  $\leftarrow$  Centroid(target_boundary)
5:   for  $v \in \text{InteriorVertices}(\textit{source\_mesh})$  do
6:      $v.\textit{position} \leftarrow \textit{target\_centroid}$ 
7:   end for
8:   max_delta  $\leftarrow \infty$ 
9:   while max_delta >  $\epsilon$  do
10:    max_delta  $\leftarrow 0$ 
11:    for  $v \in \text{InteriorVertices}(\textit{source\_mesh})$  do
12:      old_pos  $\leftarrow v.\textit{position}$ 
13:      neighbors  $\leftarrow \text{GetNeighbors}(v, \textit{source\_mesh})$ 
14:       $v.\textit{position} \leftarrow \frac{\sum n.\textit{position}}{|\textit{neighbors}|}$ 
15:      max_delta  $\leftarrow \max(\textit{max\_delta}, \text{dist}(\textit{old\_pos}, v.\textit{position}))$ 
16:    end for
17:  end while
18:  return CreateMesh(new_positions, source_mesh.faces)
19: end function

```

20.14.4 4. Pseudocode**20.14.5 5. Parameter Selection**

- **Iteration Count:** Typically 100–1000 iterations for the Gauss-Seidel solver, or use a direct sparse solver for large meshes.
- **Boundary Mapping:** Usually based on normalized arc-length to prevent the boundary vertices from “bunching up” in corners.

20.14.6 6. Complexity

- **Time Complexity:** $O(n \cdot i)$ for the iterative solver (where n is vertices, i is iterations). $O(n \log n)$ or $O(n^{1.5})$ for direct sparse solvers.
- **Space Complexity:** $O(n + e)$ to store the mesh connectivity and vertex positions.

20.14.7 7. Usages

- **Shape Morphing:** Smoothly transitioning one 3D model into another with the same topology.
- **Texture Mapping:** Unwrapping a 3D surface onto a 2D plane (UV mapping).
- **Remeshing:** Transferring a high-quality triangulation onto a newly defined boundary.
- **Surface Reconstruction:** Mapping a template mesh onto scanned point cloud data.

20.14.8 8. Testing Methods and Edge Cases

- **Non-Convex Boundaries:** Tutte’s theorem only guarantees no edge crossings for convex boundaries. For non-convex boundaries, self-intersections (flipped triangles) may occur.

- **Disconnected Components:** The algorithm must be applied to each connected component of the mesh separately.
- **High Valence Vertices:** Vertices with many neighbors may converge more slowly.
- **Mesheres with Holes:** Requires mapping multiple boundary cycles (e.g., one to the outer boundary, others to internal “islands”).

20.14.9 9. References

- Tutte, W. T. [Tut63] “How to draw a graph.” Proceedings of the London Mathematical Society.
- Floater, M. S. (1997). “Parametrization and smooth approximation of surface triangulations.” Computer Aided Geometric Design.
- Pinkall, U., & Polthier, K. (1993). “Computing discrete minimal surfaces and their conjugates.” Computing.
- Wikipedia: Tutte embedding.

20.15 Functional Maps

20.15.1 1. Overview

Functional maps are a powerful framework for finding correspondences between 3D shapes [OBGS⁺12]. Instead of mapping points directly to points, which is a combinatorial problem, functional maps represent the correspondence as a linear operator between **functions** defined on the surfaces. This transformation from a point-based to a functional representation turns the non-convex matching problem into a much simpler linear algebra problem in a reduced basis.

20.15.2 2. Definitions

Definition 20.58 (Basis Functions). A set of smooth functions $\{\phi_i\}$ defined on a surface, typically the eigenfunctions of the Laplace-Beltrami operator.

Definition 20.59 (Functional Map). A matrix C that maps coefficients of a function in shape A ’s basis to coefficients in shape B ’s basis.

Definition 20.60 (Correspondence). A mapping Π from vertices of shape A to vertices of shape B .

Definition 20.61 (Commutativity). The property that a functional map should commute with other geometric operators (like the Laplacian), i.e., $C\Delta_A \approx \Delta_B C$.

20.15.3 3. Theory

Given two shapes M and N , we compute the first k eigenfunctions of their respective Laplacians: $\Phi = [\phi_1, \dots, \phi_k]$ and $\Psi = [\psi_1, \dots, \psi_k]$. Any function f on surface M can be approximated as $f \approx \Phi \mathbf{a}$.

A correspondence between M and N can be represented by a matrix C such that if f is a function on M and g is its corresponding function on N , then $\mathbf{b} = C\mathbf{a}$, where $\mathbf{b}$ are the coefficients of g in basis Ψ .

The matrix C is found by solving an optimization problem:

$$\min_C \|CA - B\|^2 + \alpha \|C\Delta_M - \Delta_N C\|^2$$

where A and B are coefficients of “descriptor functions” (like curvature or HKS [CL06]) that should be preserved under the mapping.

Theorem 20.62 (Functional Map from Pointwise Map). *Given a pointwise correspondence $\Pi : M \rightarrow N$ between two shapes, the induced functional map is $C = \Psi^\dagger \Pi^\top \Phi$, where $\dagger$ denotes the pseudoinverse.*

Proof. For a function f on M with coefficient vector $\mathbf{a} = \Phi^\dagger f$, the pullback $\Pi_* f = f \circ \Pi^{-1}$ has coefficients $\mathbf{b} = \Psi^\dagger (f \circ \Pi^{-1})$. Substituting $f = \Phi \mathbf{a}$, we get $\mathbf{b} = \Psi^\dagger \Pi^\top \Phi \mathbf{a}$, so $C = \Psi^\dagger \Pi^\top \Phi$. $\square$

Example 20.63 (Functional Map for Shape Matching). Consider two discretizations of a sphere (shapes A and B) with a 90° rotation between them. The first k eigenfunctions of the Laplacian on both spheres are the spherical harmonics. The functional map C corresponding to the rotation is a $k \times k$ block-diagonal matrix where each block is a rotation matrix acting on the harmonic subspace of a given frequency. With $k = 50$ basis functions, C captures the rotation precisely, and the pointwise correspondence can be recovered by comparing the columns of ΨC with the columns of Φ .

20.15.4 4. Pseudocode

Algorithm 92 Compute Functional Map

```

1: function COMPUTEFUNCTIONALMAP(mesh_A, mesh_B, num_basis = 50)
2:    $L_A, M_A \leftarrow \text{ComputeLaplacianAndMass}(\text{mesh\_A})$ 
3:    $L_B, M_B \leftarrow \text{ComputeLaplacianAndMass}(\text{mesh\_B})$ 
4:    $\text{evals\_A}, \Phi \leftarrow \text{eigsh}(L_A, M_A, k = \text{num\_basis})$ 
5:    $\text{evals\_B}, \Psi \leftarrow \text{eigsh}(L_B, M_B, k = \text{num\_basis})$ 
6:    $\text{desc\_A} \leftarrow \text{ComputeHKS}(\text{mesh\_A}, \Phi, \text{evals\_A})$ 
7:    $\text{desc\_B} \leftarrow \text{ComputeHKS}(\text{mesh\_B}, \Psi, \text{evals\_B})$ 
8:    $A\_coeffs \leftarrow \Phi^\top \cdot M_A \cdot \text{desc\_A}$ 
9:    $B\_coeffs \leftarrow \Psi^\top \cdot M_B \cdot \text{desc\_B}$ 
10:   $C \leftarrow \text{SolveLeastSquares}(A\_coeffs^\top, B\_coeffs^\top)^\top$ 
11:   $C \leftarrow \text{RefineMap}(C, \Phi, \Psi)$  ▷ Optional ICP refinement
12:  return  $C$ 
13: end function

```

20.15.5 5. Parameter Selection

- **Number of Basis Functions (k):** Typically 30–100. More basis functions capture more detail but increase computation and noise sensitivity.
- **Descriptors:** High-quality descriptors (like HKS, WKS, or SHOT) are critical for a good initial map.
- **Regularization:** Weighting the Laplacian commutativity term helps ensure the map is “smooth” and “geometric.”

20.15.6 6. Complexity

- **Eigen-decomposition:** $O(N^{1.5})$ for sparse Laplacian matrices.
- **Map Calculation:** $O(k^3 + k^2 \cdot D)$ where D is the number of descriptors.
- **Point Recovery:** $O(N \cdot k)$ using nearest neighbor search in the spectral embedding.

20.15.7 7. Usages

- **Shape Matching:** Finding how one 3D character corresponds to another for animation transfer.
- **Statistical Shape Analysis:** Building a “mean shape” from a collection of scanned objects.
- **Texture Transfer:** Mapping a texture from one model to another with a different topology.
- **Symmetry Detection:** Finding self-correspondences within a single shape.
- **Shape Interpolation:** Generating intermediate shapes between two models.

20.15.8 8. Testing Methods and Edge Cases

- **Identical Shapes:** The functional map C should be an identity matrix.
- **Isometries:** For shapes that are bent but not stretched (like a human in different poses), C should be nearly orthogonal ($C^\top C \approx I$).
- **Partial Matching:** Test how the algorithm handles shapes where parts are missing (requires more advanced “Partial Functional Maps” logic).
- **Topology Changes:** Verify that functional maps can still find a reasonable matching even if one shape has a different genus.
- **Low Resolution:** Ensure the basis is stable even on coarse meshes.

20.15.9 9. References

- Ovsjanikov, M., Ben-Chen, M., Solomon, J., Butscher, A., & Guibas, L. [OBCS⁺12] “Functional maps: a flexible representation of maps between shapes.” ACM TOG.
- Solomon, J., Nguyen, A., Butscher, A., Guibas, L., & Ovsjanikov, M. (2016). “Soft maps between surfaces.” Computer Graphics Forum.
- Ovsjanikov, M., et al. (2017). “Computing and Processing Correspondences with Functional Maps.” SIGGRAPH Course.
- Wikipedia: Spectral shape analysis.

20.16 Diffusion Distance

Diffusion distance is an intrinsic metric on a Riemannian manifold (or its discrete mesh representation) that measures the connectivity of points through a diffusion process [CL06].

20.16.1 Overview

Unlike Euclidean distance, which ignores the underlying shape, or Geodesic distance, which is sensitive to small topological changes (like “short circuits”), diffusion distance is robust to noise and captures the global structure of the shape. It is based on the Heat Kernel $K_t(x, y)$, which represents the probability of a random walker moving from x to y in time t .

The diffusion distance $d_t(x, y)$ is defined as:

$$d_t^2(x, y) = \int_M |K_t(x, u) - K_t(y, u)|^2 du$$

Definition 20.64 (Diffusion Distance). Given a manifold M and a time parameter $t > 0$, the diffusion distance between points $x, y \in M$ is $d_t(x, y) = \left(\int_M |K_t(x, u) - K_t(y, u)|^2 du \right)^{1/2}$, where K_t is the heat kernel at time t .

20.16.2 Spectral Representation

In the discrete setting, diffusion distance can be computed efficiently using the eigenvalues λ_i and eigenvectors ϕ_i of the Laplace-Beltrami operator:

$$d_t^2(x, y) = \sum_{i=1}^k e^{-2\lambda_i t} (\phi_i(x) - \phi_i(y))^2$$

This allows for a **diffusion embedding** where each vertex x is mapped to a point in $\mathbb{R}^k$:

$$\Psi_t(x) = \left(e^{-\lambda_1 t} \phi_1(x), e^{-\lambda_2 t} \phi_2(x), \dots, e^{-\lambda_k t} \phi_k(x) \right)$$

The Euclidean distance in this embedding space is exactly the diffusion distance.

20.16.3 Usage

Algorithm 93 Compute Diffusion Distance

```

1: function COMPUTEDIFFUSIONDISTANCE(mesh, i, j, t)
2:   Compute eigenvalues  $\lambda$  and eigenvectors  $\phi$  of the Laplace-Beltrami operator
3:    $d^2 \leftarrow \sum_k e^{-2\lambda_k t} (\phi_k(i) - \phi_k(j))^2$ 
4:   return  $\sqrt{d^2}$ 
5: end function
6:
7: function COMPUTEDIFFUSIONEMBEDDING(mesh, t, k)
8:   Compute eigenvalues  $\lambda$  and eigenvectors  $\phi$  of the Laplace-Beltrami operator
9:   for each vertex  $x$  do
10:     $\Psi_t(x) \leftarrow (e^{-\lambda_1 t} \phi_1(x), \dots, e^{-\lambda_k t} \phi_k(x))$ 
11:   end for
12:   return  $\Psi_t$ 
13: end function

```

20.16.4 References

- Coifman, R. R., & Lafon, S. [CL06] “Diffusion maps.” Applied and Computational Harmonic Analysis.
- Bronstein, A. M., et al. “Numerical Geometry of Non-Rigid Shapes.” Springer, 2008.

20.17 Mean Curvature Flow

20.17.1 1. Overview

Polygonal Mean Curvature Flow is a geometric process used to smooth the boundaries of a polygon [GH86, Gra87]. It is a discrete version of the curve-shortening flow, where each point on a curve moves in the direction of its curvature vector. Over time, this process eliminates high-frequency noise and sharp corners, causing the polygon to become smoother and more circular.

20.17.2 2. Definitions

Definition 20.65 (Mean Curvature (Curves)). In the context of curves, this is the rate at which the tangent direction changes along the curve, i.e., $\kappa = \frac{d\theta}{ds}$ where θ is the tangent angle and s is arc length.

Definition 20.66 (Discrete Laplacian). An operator that approximates the second derivative of the vertex positions. For a vertex V_i in a polygon, $L_i = V_{i-1} - 2V_i + V_{i+1}$.

Definition 20.67 (Time Step). A parameter Δt controlling the magnitude of displacement in each iteration. Must satisfy $\Delta t < 0.5$ for numerical stability.

20.17.3 3. Theory

The flow is based on the diffusion equation (heat equation) applied to geometry: $\frac{\partial P}{\partial t} = \Delta P$. In a discrete polygon, moving a vertex V_i by its Laplacian is equivalent to moving it toward the midpoint of its two neighbors (V_{i-1} and V_{i+1}). This local averaging effect reduces the “sharpness” of vertices.

Theorem 20.68 (Gage–Hamilton–Grayson Theorem). *Any simple closed plane curve evolving under curve-shortening flow (mean curvature flow) becomes convex in finite time and subsequently shrinks to a round point.*

Proof. The proof proceeds in two stages. Gage and Hamilton (1986) [GH86] showed that any simple closed curve with non-negative curvature evolves to become convex. Grayson (1987) [Gra87] completed the argument by showing that if a curve is not initially convex, the flow eventually makes it convex, after which the Gage–Hamilton result applies. The key ingredient is the monotonicity of the width of the curve under the flow and the maximum principle applied to the curvature. $\square$

A known property of this flow is that any simple closed curve will eventually become convex and then shrink to a circular point. To use this for smoothing without losing the shape entirely, the process is usually limited to a few iterations or includes a scaling step to preserve the original perimeter or area.

Example 20.69 (Mean Curvature Flow on a Star Shape). Consider a 5-pointed star polygon with 10 vertices (5 tips, 5 valleys). At each iteration with $\Delta t = 0.1$:

- **Step 0:** The tips have large negative curvature (sharp outward corners) and the valleys have large positive curvature.
- **Step 1:** Tips move inward (toward midpoints of their neighbors), valleys move outward. The star becomes less pointy.
- **Step 5:** The polygon is noticeably smoother. The concavities are reduced.
- **Step 20:** The polygon is nearly convex and approximately elliptical.

After enough iterations (without size preservation), it would shrink to a point. With perimeter preservation, it converges to a smooth, near-circular shape.

20.17.4 4. Pseudocode

20.17.5 5. Parameter Selection

- **Time Step (Δt):** Typically set between 0.01 and 0.2. Values above 0.5 can lead to numerical instability where the polygon “explodes.”

Algorithm 94 Mean Curvature Flow

```

1: function MEANCURVATUREFLOW(polygon, iterations, dt, preserve_size)
2:    $n \leftarrow \text{len}(\text{polygon})$ 
3:   current  $\leftarrow$  polygon
4:   original_perimeter  $\leftarrow$  CalculatePerimeter(polygon)
5:   for  $k = 1$  to iterations do
6:     next_gen  $\leftarrow$  []
7:     for  $i = 0$  to  $n - 1$  do
8:        $L_x \leftarrow \text{current}[i-1].x - 2 \cdot \text{current}[i].x + \text{current}[i+1].x$ 
9:        $L_y \leftarrow \text{current}[i-1].y - 2 \cdot \text{current}[i].y + \text{current}[i+1].y$ 
10:       $\text{new\_x} \leftarrow \text{current}[i].x + dt \cdot L_x$ 
11:       $\text{new\_y} \leftarrow \text{current}[i].y + dt \cdot L_y$ 
12:      append(next_gen, Point(new_x, new_y))
13:    end for
14:    if preserve_size then
15:      current_perimeter  $\leftarrow$  CalculatePerimeter(next_gen)
16:      scale  $\leftarrow$  original_perimeter / current_perimeter
17:      next_gen  $\leftarrow$  Scale(next_gen, scale)
18:    end if
19:    current  $\leftarrow$  next_gen
20:  end for
21:  return current
22: end function

```

- **Iterations:** Depends on the level of noise. Usually 10–100 steps are sufficient for smoothing.
- **Preservation:** `keep_perimeter` or `keep_area` prevents the polygon from shrinking to a point.

20.17.6 6. Complexity

- **Time Complexity:** $O(n \cdot i)$, where n is the number of vertices and i is the number of iterations.
- **Space Complexity:** $O(n)$ to store the vertex positions.

20.17.7 7. Usages

- **Image Processing:** Smoothing the results of contour extraction.
- **GIS:** Generalizing map boundaries for different zoom levels.
- **Computer Graphics:** Creating organic-looking shapes from blocky initial geometry.
- **CAD:** Removing artifacts from digitized or scanned blueprints.

20.17.8 8. Testing Methods and Edge Cases

- **Jagged Input:** Verify that “zig-zag” patterns are flattened.
- **Stability:** Test with large Δt to identify the breaking point.
- **Self-intersection:** Check if the flow resolves or creates new self-intersections.
- **Collinear Vertices:** Verify that straight edges remain straight.

- **Drift:** Ensure the centroid remains relatively stationary if re-centering is applied.

20.17.9 9. References

- Gage, M., & Hamilton, R. S. [GH86] “The heat equation shrinking convex plane curves.” *Journal of Differential Geometry*.
- Grayson, M. A. [Gra87] “The heat equation shrinks embedded plane curves to round points.” *Journal of Differential Geometry*.
- Desbrun, M., Meyer, M., Schröder, P., & Barr, A. H. (1999). “Implicit fairing of irregular meshes using diffusion and curvature flow.” *SIGGRAPH*.
- Wikipedia: Curve-shortening flow.

20.18 Angle-Bounded Remeshing

Angle-bounded remeshing is a surface mesh refinement process that improves the quality of a triangular mesh by enforcing specific bounds on the internal angles of all triangles.

20.18.1 Overview

High-quality triangular meshes are essential for accurate physical simulations [BKP⁺10]. Small or large angles can lead to numerical instability. This remeshing technique (inspired by TriWild) iteratively modifies the mesh topology and geometry to improve the overall quality, targeting a minimum angle of 30° and a maximum angle of 120° .

20.18.2 Implementation Details

The `AngleBoundedRemesher` in `CompGeom` uses a sequence of local mesh operations:

1. **Edge Splits:** Long edges are split to reduce triangle size.
2. **Edge Collapses:** Short edges are collapsed to remove small or sliver triangles.
3. **Edge Flips:** Edges are flipped to maximize the minimum angle in the local quadrilateral.
4. **Vertex Smoothing:** Vertices are moved to their local centroids (Laplacian smoothing) to create more uniform distributions.

20.18.3 Usage

Algorithm 95 Angle-Bounded Remesh

```
1: function ANGLEBOUNDEDREMESH(mesh, min_angle, max_angle, iterations)
2:   for k = 1 to iterations do
3:     SplitLongEdges(mesh, max_angle)
4:     CollapseShortEdges(mesh, min_angle)
5:     FlipEdges(mesh, min_angle, max_angle)
6:     SmoothVertices(mesh)
7:   end for
8:   return mesh
9: end function
```

20.18.4 References

- Hu, Y., et al. “TriWild: Robust Remeshing with Angle Bounds.” ACM Transactions on Graphics (SIGGRAPH), 2021.
- Botsch, M., et al. “Polygon Mesh Processing.” CRC Press, 2010.

20.19 Affine Heat Flow

20.19.1 1. Overview

Affine heat flow is a geometric smoothing process for polygons and curves that is invariant under affine transformations (rotation, scaling, translation, and shearing). While standard heat flow (curve shortening flow) tends to evolve shapes toward circles, affine heat flow evolves them toward ellipses. It is used in image processing and computer vision for shape simplification and denoising where the affine structure of the object must be preserved.

20.19.2 2. Definitions

Definition 20.70 (Affine Invariance). A property where the result of the flow on a transformed shape is the same as the transformation applied to the flow result.

Definition 20.71 (Euclidean Curvature). The standard rate of change of the tangent direction: $\kappa = d\theta/ds$.

Definition 20.72 (Affine Curvature). A measure of curvature that remains unchanged under affine maps, defined as $\kappa_a = \kappa^{1/3}$ for planar curves.

20.19.3 3. Theory

Standard curve shortening flow is defined by $\frac{\partial P}{\partial t} = \kappa \mathbf{N}$.

Affine heat flow is defined by $\frac{\partial P}{\partial t} = \kappa^{1/3} \mathbf{N}$.

This specific power of 1/3 is what provides the affine invariance. In the discrete case (for polygons), the flow is often implemented by iteratively updating vertex positions using a weighted average of their neighbors, but with weights that account for the local area or affine metric.

A common discrete approximation for a vertex V_i with neighbors V_{i-1} and V_{i+1} is:

$$V_i^{\text{new}} = V_i + \Delta t \cdot \text{Area}(V_{i-1}, V_i, V_{i+1})^{1/3} \cdot \mathbf{N}_i$$

where $\mathbf{N}_i$ is the discrete normal vector.

Example 20.73 (Affine Heat Flow on a Sheared Square). Consider a square with vertices $(0,0), (1,0), (1,1), (0,1)$ that is sheared by the transformation $(x,y) \mapsto (x + 0.5y, y)$, giving vertices $(0,0), (1,0), (1.5,1), (0.5,1)$. Under standard curve-shortening flow, this shape would evolve toward a circle. Under affine heat flow, it evolves toward an ellipse whose eccentricity matches the affine structure of the original sheared square. If we then “un-shear” the result, we recover the same shape as applying affine heat flow directly to the original square.

20.19.4 4. Pseudocode

20.19.5 5. Parameter Selection

- **Time Step (dt):** Must be small enough to ensure stability (CFL condition).
- **Iterations:** More iterations result in smoother, more simplified shapes.
- **Area Threshold:** Small areas (near-collinear vertices) can lead to numerical instability; an epsilon threshold is often used.

Algorithm 96 Affine Polygon Smoothing

```
1: function AFFINEPOLYGONSMOOTHING(polygon, dt, iterations)
2:   current_vertices  $\leftarrow$  polygon.vertices
3:   for  $\_ = 1$  to iterations do
4:     new_vertices  $\leftarrow$  []
5:     n  $\leftarrow$  len(current_vertices)
6:     for i = 0 to n - 1 do
7:       vprev  $\leftarrow$  current_vertices[(i - 1) mod n]
8:       vcurr  $\leftarrow$  current_vertices[i]
9:       vnext  $\leftarrow$  current_vertices[(i + 1) mod n]
10:      edge1  $\leftarrow$  vcurr - vprev
11:      edge2  $\leftarrow$  vnext - vcurr
12:      area  $\leftarrow$  0.5 · |cross_product(edge1, edge2)|
13:      normal  $\leftarrow$  Normalize (Perpendicular(edge2) + Perpendicular(edge1))
14:      displacement  $\leftarrow$  area1/3 · normal · dt
15:      append(new_vertices, vcurr + displacement)
16:    end for
17:    current_vertices  $\leftarrow$  new_vertices
18:  end for
19:  return Polygon(current_vertices)
20: end function
```

20.19.6 6. Complexity

- **Time Complexity:** $O(I \cdot n)$ where I is iterations and n is vertex count.
- **Space Complexity:** $O(n)$ to store the vertex positions.

20.19.7 7. Usages

- **Shape Recognition:** Normalizing shapes to a “canonical” form before matching in a database.
- **Computer Vision:** Preprocessing object silhouettes to remove pixelation or noise while preserving geometric structure.
- **Cartography:** Simplifying coastlines or boundaries for small-scale maps.
- **Generative Design:** Creating organic, smooth shapes from rough polyline sketches.
- **Image Analysis:** Analyzing the “affine scale space” of an image.

20.19.8 8. Testing Methods and Edge Cases

- **Ellipses:** An ellipse should remain an ellipse under affine heat flow (only its size should change).
- **Squares:** A square should gradually round its corners and evolve toward an ellipse.
- **Affine Symmetry:** Verify that shearing a polygon, smoothing it, and then “un-shearing” it gives the same result as smoothing the original polygon.
- **Self-Intersections:** Ensure the flow doesn’t cause the polygon to cross itself (though this is theoretically possible for non-convex shapes).
- **Zero Area:** Handle perfectly collinear triplets of vertices.

20.19.9 9. References

- Sapiro, G., & Tannenbaum, A. (1993). “Affine invariant scale-space.” *International Journal of Computer Vision*.
- Olver, P. J., Sapiro, G., & Tannenbaum, A. (1994). “Affine invariant geometric evolutions of planar curves.” *SIAM Journal on Applied Mathematics*.
- Angenent, S., Sapiro, G., & Tannenbaum, A. (1998). “On the affine heat equation for non-convex curves.” *Journal of the American Mathematical Society*.
- Wikipedia: Affine shape adaptation.

20.20 Geodesic Distance

20.20.1 1. Overview

Finding the shortest path between two points on a 3D surface mesh (the geodesic distance) is a fundamental problem in geometric modeling [MMP87, CWW13b, Set96]. Unlike Euclidean distance, which ignores the surface, the geodesic distance must follow the “terrain” of the mesh. While the Fast Marching Method (FMM) is the most famous numerical approach, other algorithms like the MMP algorithm (exact) and the Heat Method (highly efficient) provide different trade-offs between speed and accuracy.

20.20.2 2. Definitions

Definition 20.74 (Geodesic). The shortest path between two points on a surface, restricted to stay on that surface.

Definition 20.75 (MMP Algorithm). An exact algorithm (Mitchell, Mount, Papadimitriou [MMP87]) that generalizes Dijkstra’s algorithm to continuous surfaces by propagating “windows” along edges.

Definition 20.76 (Heat Method). A modern method (Crane et al. [CWW13b]) that approximates geodesics by observing the flow of heat from a source point.

Definition 20.77 (Edge-Walking). A simple approximation that treats the mesh as a graph and uses Dijkstra’s algorithm on edges, which usually provides poor results as it cannot “cut” across faces.

20.20.3 3. Theory

MMP Algorithm (Exact)

The MMP algorithm (Mitchell, Mount, and Papadimitriou [MMP87]) works by propagating “intervals” of potential paths across the surface. When a wavefront reaches an edge, it creates a new set of intervals for the next triangle. This results in an exact polyhedral geodesic but is mathematically complex to implement.

The Heat Method (Fast Approximation)

The Heat Method [CWW13b], introduced by Keenan Crane et al. (2013), uses a three-step process based on the heat equation:

1. **Heat Flow:** Solve the heat equation $\dot{u} = \Delta u$ for a short time t , with a source at the starting vertex.

2. **Gradient Normalization:** Calculate the gradient of the heat field ∇u and normalize it to have unit length ($X = -\nabla u / |\nabla u|$).
3. **Poisson Solve:** Solve the Poisson equation $\Delta\phi = \nabla \cdot X$. The resulting scalar field ϕ is the geodesic distance to the source.

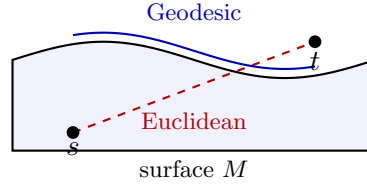


Figure 20.3: Geodesic vs. Euclidean distance on a curved surface. The geodesic (blue) follows the terrain.

Example 20.78 (Geodesic on a Cylinder). Consider a cylinder of radius r and height h . The geodesic between two points on opposite sides of the cylinder does not travel “through” the cylinder but wraps around it. If the two points are at the same height but separated by angle π (opposite sides), the geodesic distance is πr (half the circumference), not $2r$ (the Euclidean distance through the interior). The Heat Method computes this by solving the heat equation with a point source, normalizing the gradient, and solving a Poisson equation.

20.20.4 4. Pseudocode

Heat Method

Algorithm 97 Compute Geodesic via Heat Method

```

1: function COMPUTEGEODESICHEAT(mesh, source_vertex)
2:    $L \leftarrow \text{ComputeCotangentLaplacian}(\text{mesh})$ 
3:    $A \leftarrow \text{ComputeVertexAreas}(\text{mesh})$ 
4:    $t \leftarrow \text{mean\_edge\_length}(\text{mesh})^2$ 
5:    $u \leftarrow \text{SolveLinearSystem}(A - t \cdot L, \text{source\_indicator})$ 
6:    $X \leftarrow []$ 
7:   for  $\text{face} \in \text{mesh.faces}$  do
8:      $\text{grad\_}u \leftarrow \text{ComputeFaceGradient}(\text{face}, u)$ 
9:      $\text{append}(X, -\text{grad\_}u / \text{norm}(\text{grad\_}u))$ 
10:  end for
11:   $\text{div\_}X \leftarrow \text{ComputeDivergence}(X, \text{mesh})$ 
12:   $\phi \leftarrow \text{SolveLinearSystem}(L, \text{div\_}X)$ 
13:  return  $\phi - \phi[\text{source\_vertex}]$  ▷ Shift so source is 0
14: end function

```

20.20.5 5. Parameter Selection

- **Time Step (t):** In the Heat Method, t is typically set to the square of the average edge length. Too small t leads to numerical noise; too large t leads to smoothing errors.
- **Solver:** Sparse direct solvers (like SuperLU or CHOLMOD) are essential for the Laplacian systems.

20.20.6 6. Complexity

- **MMP Algorithm:** $O(n^2)$ in the worst case, but $O(n \log n)$ in practice.
- **Heat Method:** $O(n)$ after an initial $O(n^{1.5})$ matrix factorization. Subsequent queries from different sources are extremely fast.

20.20.7 7. Usages

- **Surface Parameterization:** Normalizing distances for texture mapping.
- **Mesh Segmentation:** Grouping vertices based on their distance from a set of “seeds.”
- **Shape Matching:** Using geodesic “fingerprints” to compare 3D models.
- **Texture Synthesis:** Propagation of patterns along a surface.
- **Medicine:** Measuring distances along the folds of the brain or surface of an organ.

20.20.8 8. Testing Methods and Edge Cases

- **Flat Mesh:** The results should match the Euclidean distance exactly.
- **Sphere:** Geodesics between any two points should follow “Great Circles.”
- **Concave Regions:** Verify that the path correctly “bends” into valleys rather than jumping across them.
- **Mesh Resolution:** Test how the accuracy of the Heat Method improves as the mesh is refined.
- **Boundaries:** Ensure the algorithm handles the edges of a surface without “leaking” or crashing.

20.20.9 9. References

- Mitchell, J. S. B., Mount, D. M., & Papadimitriou, C. H. [MMP87] “The discrete geodesic problem.” SIAM Journal on Computing.
- Crane, K., Weischedel, C., & Wardetzky, M. [CWW13b] “Geodesics in Heat: A New Approach to Computing Distance Based on Heat Flow.” ACM TOG.
- Surazhsky, V., Surazhsky, O., Kirsanov, D., Gortler, S. J., & Hoppe, H. (2005). “Fast exact and approximate geodesics on meshes.” SIGGRAPH.
- Wikipedia: Geodesic.

20.21 Medial Axis Transform

20.21.1 1. Overview

The Medial Axis of a shape is the set of all points within the shape that have more than one closest point on the shape’s boundary. Informally, it can be thought of as the “skeleton” or “midline” of the shape. Originally proposed by Harry Blum in 1967 [Blu67] as a tool for biological shape recognition, the Medial Axis Transform (MAT) is now a cornerstone of shape analysis, computer vision, and geometric modeling.

20.21.2 2. Definitions

Definition 20.79 (Medial Axis). The locus of the centers of all maximal inscribed circles that touch the boundary in at least two points.

Definition 20.80 (Maximal Inscribed Circle). A circle that is entirely contained within the shape and is not contained in any other such circle.

Definition 20.81 (Medial Axis Transform). The Medial Axis combined with the radius function (the radius of the maximal inscribed circle at each point), which together form a complete representation of the shape.

Definition 20.82 (Bifurcation Point). A point on the medial axis where the skeleton branches into multiple directions.

20.21.3 3. Theory

The Medial Axis can be understood through the **grassfire analogy** [Blu67]: if the boundary of a shape (represented by dry grass) is set on fire simultaneously at all points, the fire will propagate inward at a constant speed. The points where the fire fronts from different parts of the boundary meet and extinguish each other form the medial axis.

Mathematically, the medial axis of a polygon is composed of segments of **straight lines** (bisectors of two edges) and **parabolic arcs** (bisectors of an edge and a vertex). For 2D shapes, the medial axis is a 1D graph structure that topologically summarizes the 2D region.

Example 20.83 (Medial Axis of a Rectangle). Consider a rectangle with vertices $(0, 0)$, $(4, 0)$, $(4, 2)$, $(0, 2)$. The medial axis consists of a horizontal line segment from $(1, 1)$ to $(3, 1)$ (the bisector of the top and bottom edges) and two diagonal segments connecting the endpoints to the corners. Specifically, from $(1, 1)$ the axis branches to $(0, 0)$ and $(0, 2)$, and from $(3, 1)$ it branches to $(4, 0)$ and $(4, 2)$. The resulting skeleton is shaped like an “I” with bifurcation points at $(1, 1)$ and $(3, 1)$.

20.21.4 4. Pseudocode

Algorithm 98 Medial Axis

```

1: function MEDIALAXIS(polygon)
2:    $VD \leftarrow \text{Voronoi}(\text{polygon.edges}, \text{polygon.vertices})$ 
3:    $\text{skeleton\_edges} \leftarrow []$ 
4:   for  $\text{edge} \in VD.\text{edges}$  do
5:     if  $\text{IsInside}(\text{edge}, \text{polygon})$  then
6:        $\text{append}(\text{skeleton\_edges}, \text{edge})$ 
7:     end if
8:   end for
9:    $\text{pruned\_skeleton} \leftarrow \text{Prune}(\text{skeleton\_edges}, \text{threshold})$ 
10:  return  $\text{pruned\_skeleton}$ 
11: end function
12:
13: function ISINSIDE( $\text{edge}, \text{polygon}$ )
14:  return  $\text{PointInPolygon}(\text{edge.midpoint}, \text{polygon})$ 
15: end function

```

20.21.5 5. Parameter Selection

- **Pruning Threshold:** A crucial parameter that determines which branches are kept. Without pruning, even a tiny bump on the boundary creates a long branch in the medial axis.
- **Voronoi Precision:** Robustness is key, especially when dealing with nearly collinear segments or sharp corners.

20.21.6 6. Complexity

- **Time Complexity:** $O(n \log n)$ for a polygon with n vertices, leveraging the fact that the medial axis is a subset of the Voronoi diagram of the polygon's edges and vertices.
- **Space Complexity:** $O(n)$ to store the resulting graph structure.

20.21.7 7. Usages

- **Character Recognition (OCR):** Extracting the “bone” structure of letters for identification.
- **Shape Simplification:** Representing complex 2D shapes by their 1D skeletons for faster processing.
- **Motion Planning:** Finding the safest path for a robot (the medial axis is as far away from the walls as possible).
- **Mesh Generation:** Using the skeleton to guide the placement of elements in a structured grid.
- **Animation:** Creating skeletal rigs for 2D character deformation.

20.21.8 8. Testing Methods and Edge Cases

- **Convex Polygons:** The medial axis is purely composed of straight line segments.
- **Concave Polygons:** Contains parabolic segments where a vertex is the closest point to one side.
- **Noise on Boundary:** Verify that pruning correctly handles small irregularities without destroying the main structure.
- **Circular Shapes:** The medial axis of a perfect circle is a single point at its center.
- **Parallel Edges:** The medial axis is exactly halfway between the edges.

20.21.9 9. References

- Blum, H. [Blu67] “A Transformation for Extracting New Descriptors of Shape.” Models for the Perception of Speech and Visual Form.
- Chin, F., Snoeyink, J., & Wang, C. A. (1999). “Finding the Medial Axis of a Simple Polygon in Linear Time.” Discrete & Computational Geometry.
- Wikipedia: Medial axis.

20.22 Straight Skeleton

20.22.1 1. Overview

The straight skeleton of a polygon is a 1D skeleton structure topologically similar to the medial axis, but composed entirely of straight line segments. It is formed by a “wavefront propagation” process where the edges of the polygon move inward parallel to themselves at a constant speed. The paths traced by the vertices of the shrinking polygon form the straight skeleton.

20.22.2 2. Definitions

Definition 20.84 (Straight Skeleton). The locus of vertices as the edges of a polygon move inward at a uniform speed.

Definition 20.85 (Wavefront). The shrinking polygon boundary at any point in time during the propagation.

Definition 20.86 (Bisector). The ray originating from a polygon vertex that bisects the angle formed by its two incident edges.

Definition 20.87 (Edge Event). An event where an edge shrinks to zero length and disappears, causing its two neighboring vertices to merge into one.

Definition 20.88 (Split Event). An event where a vertex moves into and splits an edge of the wavefront into two separate parts.

20.22.3 3. Theory

Unlike the medial axis, which can contain curved (parabolic) segments, the straight skeleton is piecewise linear because each point on the skeleton is the intersection of two straight-line edge trajectories.

The construction of a straight skeleton is typically modeled as a **kinetic process**. We keep track of the time until the next “event” (either an edge disappearing or an edge being split). As events occur, the topology of the wavefront changes, and new events are calculated. This process continues until the entire polygon has collapsed.

Example 20.89 (Straight Skeleton of a Triangle). Consider a triangle with vertices A , B , C . As the three edges move inward at unit speed, the three bisectors meet at a single point (the incenter). The straight skeleton is a star with three edges connecting the incenter to the three original vertices. There are three edge events (one per original edge disappearing) but no split events. The time to collapse is r/v where r is the inradius and v is the wavefront speed.

20.22.4 4. Pseudocode

20.22.5 5. Parameter Selection

- **Precision:** Highly sensitive to numerical precision during event time calculations, especially for nearly parallel edges.
- **Degeneracies:** Multiple events occurring at the exact same time (e.g., in a square or a regular hexagon) must be handled carefully.

Algorithm 99 Straight Skeleton

```
1: function STRAIGHTSKELETON(polygon)
2:   events  $\leftarrow$  PriorityQueue()
3:   for  $v \in \text{polygon.vertices}$  do
4:     events.push(CalculateNextEvent( $v, \text{polygon}$ ))
5:   end for
6:   skeleton_edges  $\leftarrow$  []
7:   while  $\neg \text{events.empty}()$  do
8:     event  $\leftarrow$  events.pop()
9:     if event.type = "EDGE" then
10:      HandleEdgeEvent(event, skeleton_edges, events)
11:     else if event.type = "SPLIT" then
12:      HandleSplitEvent(event, skeleton_edges, events)
13:     end if
14:     UpdateNeighbors(event, events)
15:   end while
16:   return skeleton_edges
17: end function
```

20.22.6 6. Complexity

- **Time Complexity:** $O(n \log n)$ for simple polygons, using more advanced data structures (like kinetic priority queues). A simpler $O(n^2)$ approach is often implemented.
- **Space Complexity:** $O(n)$ to store the polygon wavefront and the resulting skeleton edges.

20.22.7 7. Usages

- **Roof Construction:** Defining the ridges and valleys of a roof for a given house footprint.
- **Corridor Generation:** Creating paths in architectural layouts.
- **Offsetting:** Generating inward/outward offsets of polygons (mitered offsets).
- **Computer Graphics:** Creating “puffy” 3D effects from 2D shapes (Beveling).
- **GIS:** Generating centerlines of road networks.

20.22.8 8. Testing Methods and Edge Cases

- **Convex Polygons:** In this case, the straight skeleton and the medial axis are topologically similar, though the straight skeleton remains linear.
- **Concave Polygons:** Split events are common and must be handled to ensure the skeleton correctly reflects the topology.
- **Holes:** The straight skeleton can be extended to polygons with holes, where the outer boundary moves in and the holes move out.
- **Collinear Edges:** Edges that are almost in a straight line can lead to events far into the future (numerical instability).
- **Symmetry:** Regular polygons should result in a perfectly symmetric skeleton meeting at a single point.

20.22.9 9. References

- Aichholzer, O., Aurenhammer, F., Alberts, D., & Gärtner, B. (1995). “A novel type of skeleton for polygons.” *Journal of Universal Computer Science*.
- Eppstein, D., & Erickson, J. (1999). “Raising Roofs, Formulating Foundations, and Dissecting Designs: Techniques for Discrete Dirichlet Problems.” *Discrete & Computational Geometry*.
- Wikipedia: Straight skeleton.

20.23 Mesh Cut Loops

20.23.1 1. Overview

Mesh cut loops (or cut graphs) are sets of edges on a surface mesh that, when removed, transform the mesh into a single topological disk (a genus-0 surface with a single boundary). Identifying these loops is a prerequisite for many surface parameterization algorithms (like BFF [CWW17] or LSCM), as these algorithms require the surface to be homeomorphic to a disk before they can flatten it into 2D UV space.

20.23.2 2. Definitions

Definition 20.90 (Genus). The number g of “handles” on a surface. A mesh with genus g requires $2g$ non-separating loops to be cut into a disk.

Definition 20.91 (Fundamental Loops). A basis for the first homology group of the surface; these loops “wrap around” the handles or holes.

Definition 20.92 (Cut Graph). A set of edges such that the remaining mesh is a topological disk.

Definition 20.93 (Tree-Cotree Decomposition). A technique for constructing a cut graph using a spanning tree of the primal mesh and a spanning tree of the dual mesh.

20.23.3 3. Theory

For a closed manifold mesh with genus g , the Euler characteristic is $\chi = V - E + F = 2 - 2g$. A cut graph that reduces this surface to a disk must contain $2g$ independent cycles that meet at a single vertex (forming a “wedge of circles”).

A robust method for finding a cut graph is the **Tree-Cotree Decomposition**:

1. Compute a **Spanning Tree** T of the vertices and edges of the mesh.
2. Compute a **Dual Spanning Tree** T^* of the faces and dual edges, but only using dual edges whose primal counterparts are NOT in T .
3. The edges that are in neither T nor T^* form the **Cut Graph**.

This algorithm ensures that the cut graph is minimal and that cutting along it results in exactly one connected component with no handles.

Example 20.94 (Cut Graph on a Torus). A torus has genus $g = 1$, so its Euler characteristic is $\chi = 0$. The tree-cotree decomposition produces a cut graph with exactly $2g = 2$ independent cycles. These two cycles correspond to the meridian (wrapping around the tube) and the longitude (wrapping through the hole). Cutting along both cycles opens the torus into a single rectangle (topological disk), which can then be parameterized.

20.23.4 4. Pseudocode

Algorithm 100 Find Cut Graph

```

1: function FINDCUTGRAPH(mesh)
2:   primal_tree  $\leftarrow$  ComputeSpanningTree(mesh.vertices, mesh.edges)
3:   candidate_dual  $\leftarrow$  [e | e  $\in$  mesh.edges  $\wedge$  e  $\notin$  primal_tree]
4:   dual_tree  $\leftarrow$  ComputeDualSpanningTree(mesh.faces, candidate_dual)
5:   cut_graph_edges  $\leftarrow$  []
6:   for e  $\in$  mesh.edges do
7:     if e  $\notin$  primal_tree  $\wedge$  e_dual(e)  $\notin$  dual_tree then
8:       append(cut_graph_edges, e)
9:     end if
10:  end for
11:  return cut_graph_edges
12: end function

```

20.23.5 5. Parameter Selection

- **Root Selection:** The choice of root for the primal and dual trees affects the shape of the cut graph but not its topological property.
- **Edge Weights:** Using weights (e.g., edge lengths) in the spanning tree calculation (Kruskal’s algorithm) can result in “shorter” or more “geometric” cut paths.

20.23.6 6. Complexity

- **Time Complexity:** $O(E \log E)$ or $O(E)$ depending on the spanning tree algorithm used.
- **Space Complexity:** $O(E)$ to store the primal and dual tree structures.

20.23.7 7. Usages

- **Surface Parameterization (UV Unwrapping):** Cutting a torus or a complex organic model so it can be laid flat in 2D.
- **Texture Mapping:** Defining the “seams” of a 3D model where the texture coordinates are discontinuous.
- **Mesh Morphing:** Aligning the topology of two different meshes before interpolating between them.
- **Computational Topology:** Calculating the homology or Betti numbers of a surface.
- **Geometry Compression:** Cutting a mesh into a single spanning triangle strip or a predictable layout.

20.23.8 8. Testing Methods and Edge Cases

- **Genus-0 Sphere:** The cut graph should be empty (or a single point/edge if a boundary is forced).
- **Torus (Genus-1):** The cut graph should consist of two loops (meridian and longitude) meeting at a vertex.

- **Multiple Components:** Ensure the algorithm handles disconnected parts of the mesh correctly.
- **Boundary Handling:** If the mesh already has a boundary, the cut graph only needs to “connect” the boundary to the handles.
- **Watertight Check:** After cutting, the resulting mesh should have exactly one connected component and its Euler characteristic should be 1.

20.23.9 9. References

- Erickson, J., & Whittlesey, K. (2005). “Greedy optimal homotopy and homology generators.” SODA.
- Gu, X., & Yau, S. T. (2002). “Computing conformal structures of surfaces.” Communications in Information and Systems.
- Eppstein, D. (2003). “Dynamic generators of topologically distinct cycles.” Proceedings of the 15th Canadian Conference on Computational Geometry.
- Wikipedia: Cut graph.

20.24 Mesh Tunnel Loops

20.24.1 1. Overview

Mesh tunnel loops (or handle loops) are non-separating cycles on a surface mesh that “wrap around” the handles of the surface. For a surface of genus g , there are $2g$ such fundamental loops. Identifying these loops is essential for computational topology, mesh editing (e.g., “closing” a handle), and understanding the global connectivity of complex shapes. Tunnel loops are often computed as part of a homology basis.

20.24.2 2. Definitions

Definition 20.95 (Handle). A topological feature equivalent to a bridge or a donut hole.

Definition 20.96 (Tunnel Loop). A cycle that goes “through” a handle.

Definition 20.97 (Handle Loop). A cycle that goes “around” a handle (orthogonal to the tunnel loop).

Definition 20.98 (Non-Separating Cycle). A cycle that, when removed, does not split the surface into two disconnected components.

Definition 20.99 (Homology Basis). A minimal set of cycles that generate all possible non-contractible loops on the surface.

20.24.3 3. Theory

Tunnel and handle loops come in pairs for each handle of a manifold mesh. The most common algorithm for finding them is based on the **Tree-Cotree Decomposition**:

1. Compute a **Primal Spanning Tree** T of the vertices and edges.
2. Compute a **Dual Spanning Tree** T^* of the faces and dual edges, using only dual edges whose primal edges are NOT in T .

3. The remaining edges $L = E \setminus (T \cup T^*)$ are the **generator edges**.
4. Each generator edge $e \in L$ completes exactly one cycle in $T \cup \{e\}$. These cycles form a basis for the first homology group $H_1(M)$.

For a mesh with genus g , there will be exactly $2g$ generator edges, each defining one fundamental loop.

20.24.4 4. Pseudocode

Algorithm 101 Find Tunnel Loops

```

1: function FINDTUNNELLOOPS(mesh)
2:   primal_tree  $\leftarrow$  ComputeSpanningTree(mesh.vertices, mesh.edges)
3:   candidate_dual  $\leftarrow$  [e_dual(e) | e  $\in$  mesh.edges  $\wedge$  e  $\notin$  primal_tree]
4:   dual_tree  $\leftarrow$  ComputeDualSpanningTree(mesh.faces, candidate_dual)
5:   generators  $\leftarrow$  []
6:   for e  $\in$  mesh.edges do
7:     if e  $\notin$  primal_tree  $\wedge$  e_dual(e)  $\notin$  dual_tree then
8:       append(generators, e)
9:     end if
10:  end for
11:  loops  $\leftarrow$  []
12:  for gen_edge  $\in$  generators do
13:    path  $\leftarrow$  primal_tree.GetPath(gen_edge.v1, gen_edge.v2)
14:    append(loops, path + [gen_edge])
15:  end for
16:  return loops
17: end function

```

20.24.5 5. Parameter Selection

- **Weighting:** Using edge lengths as weights in the spanning tree (Kruskal’s algorithm) ensures that the resulting loops are “short” and follow the geometric features of the handle.
- **Basis Type:** The tree-cotree method produces a **homology basis**. More advanced algorithms can produce a **canonical basis** (where loops meet at a single vertex) or **greedy optimal basis** (shortest possible loops).

20.24.6 6. Complexity

- **Time Complexity:** $O(E \log E)$ or $O(E \log V)$ for the spanning tree construction.
- **Space Complexity:** $O(E)$ to store the primal and dual trees.

20.24.7 7. Usages

- **Mesh Simplification:** Removing handles to reduce the genus of a mesh (e.g., making a model “watertight” for 3D printing).
- **Shape Analysis:** Calculating Betti numbers to distinguish between different topological classes.
- **Texture Mapping:** Using handle loops as natural boundaries for UV charts.

- **Animation:** Constraining deformations to preserve the topological structure of a character (e.g., not “merging” fingers).
- **Medical Imaging:** Identifying and measuring the size of holes or passages in anatomical structures (like blood vessels or lung airways).

20.24.8 8. Testing Methods and Edge Cases

- **Sphere (Genus-0):** The algorithm should find zero tunnel loops.
- **Torus (Genus-1):** Should find exactly two loops (the “donut” hole and the “tube” wrap).
- **Double Torus (Genus-2):** Should find exactly four loops.
- **Open Mesh (Boundary):** Loops that wrap around a boundary (hole) should be handled correctly (they are part of the relative homology).
- **Disconnected Mesh:** The algorithm must be applied to each connected component independently.

20.24.9 9. References

- Erickson, J., & Whittlesey, K. (2005). “Greedy optimal homotopy and homology generators.” SODA.
- Dey, T. K., Sun, J., & Wang, Y. (2010). “Approximating Cycles in a Shortest Homology Basis.” *Discrete & Computational Geometry*.
- Gu, X., & Yau, S. T. (2008). “Computational Conformal Geometry.” International Press.
- Wikipedia: Homology (mathematics).

20.25 Iterative Closest Point (ICP) Registration

20.25.1 1. Overview

Point cloud registration is the problem of aligning two or more point sets into a common coordinate frame. This is a fundamental task in 3D scanning, *Simultaneous Localization and Mapping* (SLAM), multi-view reconstruction, and robotics, where measurements from different sensors or viewpoints must be fused into a single coherent model [BM92]. The **Iterative Closest Point** (ICP) algorithm, introduced by Besl and McKay [BM92] and independently by Chen and Medioni [CM92], is the most widely used method for rigid registration. ICP iteratively establishes correspondences between points in the source and target clouds, then computes the rigid transformation that best aligns them, converging to a local minimum of the mean squared error (MSE) objective.

20.25.2 2. Definitions

Definition 20.100 (Point Cloud Registration). Given a source point set $P = \{p_1, \dots, p_n\}$ and a target point set $Q = \{q_1, \dots, q_m\}$ in $\mathbb{R}^3$, registration is the problem of finding a rigid transformation (R, t) with $R \in SO(3)$ and $t \in \mathbb{R}^3$ that minimizes the distance between corresponding points.

Definition 20.101 (Rigid Transformation). A mapping $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ of the form $T(x) = Rx + t$, where R is an orthogonal matrix with $\det(R) = 1$ (i.e., $R \in SO(3)$) and t is a translation vector. Rigid transformations preserve Euclidean distances: $\|T(x) - T(y)\| = \|x - y\|$ for all $x, y \in \mathbb{R}^3$.

Definition 20.102 (Closest Point Correspondence). For each source point $p_i \in P$, the closest point in the target set Q is defined as $q_j = \arg \min_{q \in Q} \|p_i - q\|$. The index map $j(i)$ encodes this correspondence.

Definition 20.103 (Mean Squared Error Objective). The ICP objective is to find the transformation (R, t) that minimizes:

$$E(R, t) = \frac{1}{n} \sum_{i=1}^n \|Rp_i + t - q_{j(i)}\|^2$$

where $j(i)$ denotes the index of the closest target point to p_i .

20.25.3 3. Theory

The ICP algorithm alternates between two steps: (1) assigning correspondences by finding closest points, and (2) computing the optimal rigid transformation that minimizes the MSE for those correspondences.

Closest Point Assignment

Given the current estimate of the source points, each source point p_i is matched to its nearest neighbor $q_{j(i)}$ in the target set Q . This nearest-neighbor query is accelerated using a **KD-tree** [BM92], reducing the per-iteration cost from $O(nm)$ to $O(n \log m)$.

Optimal Rotation via SVD

Once correspondences are established, the centroids of the source and matched target points are computed:

$$\mu_P = \frac{1}{n} \sum_{i=1}^n p_i, \quad \mu_Q = \frac{1}{n} \sum_{i=1}^n q_{j(i)}.$$

The points are centered by subtracting their respective centroids:

$$\tilde{p}_i = p_i - \mu_P, \quad \tilde{q}_i = q_{j(i)} - \mu_Q.$$

The **cross-covariance matrix** H is then formed:

$$H = \sum_{i=1}^n \tilde{p}_i \tilde{q}_i^T = \tilde{P}^T \tilde{Q}$$

where $\tilde{P}$ and $\tilde{Q}$ are the $n \times 3$ centered point matrices.

Applying the **Singular Value Decomposition** (SVD) to H :

$$H = U \Sigma V^T$$

where U and V are 3×3 orthogonal matrices and Σ is a diagonal matrix of singular values.

The optimal rotation is:

$$R = V^T U^T$$

Theorem 20.104 (Reflection Handling). *If $\det(R) < 0$, the solution introduces a reflection (improper rotation). To enforce a proper rotation ($\det(R) = 1$), the last column of V (equivalently, the last row of V^T) is negated before recomputing R .*

After computing R , the optimal translation is:

$$t = \mu_Q - R \mu_P$$

Iterative Refinement

The source points are updated: $p_i \leftarrow Rp_i + t$, and the process repeats. The transformation is accumulated across iterations into a single 4×4 homogeneous matrix. Convergence is detected when the change in mean error between successive iterations falls below a specified `tolerance`.

Theorem 20.105 (ICP Convergence). *The ICP algorithm monotonically decreases the MSE objective $E(R, t)$ at each iteration. Since $E \geq 0$ is bounded below, the sequence of objective values converges to a local minimum.*

Proof. At iteration k , the closest point assignment step finds correspondences $j_k(i)$ that minimize the distance for each p_i independently:

$$\sum_{i=1}^n \|p_i - q_{j_k(i)}\|^2 \leq \sum_{i=1}^n \|p_i - q_{j_{k-1}(i)}\|^2$$

since $j_k(i)$ is the nearest neighbor of p_i in Q . The SVD step then finds the rotation R_k and translation t_k that minimize $\sum_{i=1}^n \|R_k p_i + t_k - q_{j_k(i)}\|^2$ over all rigid transformations. Therefore:

$$E_k = \frac{1}{n} \sum_{i=1}^n \|R_k p_i + t_k - q_{j_k(i)}\|^2 \leq \frac{1}{n} \sum_{i=1}^n \|p_i - q_{j_k(i)}\|^2 \leq \frac{1}{n} \sum_{i=1}^n \|R_{k-1} p_i^{(k-1)} + t_{k-1} - q_{j_{k-1}(i)}\|^2 = E_{k-1}.$$

The sequence $\{E_k\}$ is non-increasing and bounded below by zero, so it converges. The limit is a *local* minimum because each step is greedy and no global search over correspondences is performed. $\square$

Example 20.106 (Aligning Two Partially Overlapping Squares). Consider two overlapping unit squares in $\mathbb{R}^3$ lying in the xy -plane. The target Q has vertices at $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$, $(0, 1, 0)$. The source P is the same square rotated by 15° about the z -axis and translated by $(0.3, 0.1, 0)$:

$$R_{\text{true}} = \begin{pmatrix} \cos 15^\circ & -\sin 15^\circ & 0 \\ \sin 15^\circ & \cos 15^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad t_{\text{true}} = \begin{pmatrix} 0.3 \\ 0.1 \\ 0 \end{pmatrix}.$$

Iteration 1: Each source vertex is matched to its nearest target vertex. The centroids are computed: $\mu_P \approx (0.36, 0.19, 0)$, $\mu_Q = (0.5, 0.5, 0)$. After centering, the cross-covariance matrix

H is formed, and SVD yields $R_1 \approx \begin{pmatrix} 0.966 & 0.259 & 0 \\ -0.259 & 0.966 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, close to R_{true}^{-1} . The source points are

updated, reducing the MSE.

Iteration 2: With updated source positions, new correspondences are found. The rotation refinement is smaller. The mean error drops further.

After approximately 5–8 iterations (depending on sampling density), the algorithm converges with $\text{MSE} < 10^{-5}$, recovering a transformation close to $(R_{\text{true}}^{-1}, -R_{\text{true}}^{-1}t_{\text{true}})$.

20.25.4 4. Pseudocode

20.25.5 5. Parameter Selection

- `max.iterations` (k_{max}): Maximum number of ICP iterations. Typical values range from 10 to 50. For well-aligned inputs, convergence is achieved in fewer iterations. Setting this too low may prevent convergence; too high wastes computation.

Algorithm 102 Iterative Closest Point (ICP) Registration

```

1: function ICP( $P, Q, k_{\max}, \epsilon$ )
2:    $T \leftarrow I_4$  ▷ Accumulated  $4 \times 4$  homogeneous transform
3:    $e_{\text{prev}} \leftarrow \infty$ 
4:   Build KD-tree  $\mathcal{T}$  from target points  $Q$ 
5:   for  $\ell = 1$  to  $k_{\max}$  do
6:      $(d, j) \leftarrow \mathcal{T}.\text{Query}(P)$  ▷ Closest-point query;  $j_i$  is index of nearest target to  $p_i$ 
7:      $e \leftarrow \frac{1}{n} \sum_{i=1}^n d_i$  ▷ Mean distance
8:     if  $|e_{\text{prev}} - e| < \epsilon$  then
9:       break
10:    end if
11:     $e_{\text{prev}} \leftarrow e$ 
12:     $\mu_P \leftarrow \frac{1}{n} \sum_{i=1}^n p_i$ 
13:     $\mu_Q \leftarrow \frac{1}{n} \sum_{i=1}^n q_{j_i}$ 
14:     $H \leftarrow \sum_{i=1}^n (p_i - \mu_P)(q_{j_i} - \mu_Q)^T$ 
15:     $(U, \Sigma, V^T) \leftarrow \text{SVD}(H)$ 
16:     $R \leftarrow V^T U^T$ 
17:    if  $\det(R) < 0$  then
18:      Negate the last column of  $V$ 
19:       $R \leftarrow V^T U^T$ 
20:    end if
21:     $t \leftarrow \mu_Q - R \mu_P$ 
22:     $P \leftarrow R \cdot P + t$  ▷ Transform source points
23:    Update accumulated transform:  $T \leftarrow \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} \cdot T$ 
24:  end for
25:  return  $(P_{\text{aligned}}, T)$ 
26: end function

```

- **tolerance (ϵ):** Convergence threshold on the change in mean distance error between successive iterations. A value of 10^{-5} works well for most applications. For noisy data, a larger tolerance (e.g., 10^{-3}) avoids oscillation.
- **KD-tree Structure:** The KD-tree is built once over the target points and reused every iteration. For very large point clouds, approximate nearest-neighbor structures (e.g., random projection trees) can replace exact KD-tree queries.

20.25.6 6. Complexity

- **Time Complexity:** $O(k \cdot n \log n)$, where k is the number of iterations until convergence and $n = \max(|P|, |Q|)$. Each iteration involves an $O(n \log n)$ nearest-neighbor query (via KD-tree), $O(n)$ centroid computation, and an $O(n)$ SVD of a 3×3 matrix (constant time).
- **Space Complexity:** $O(n + m)$ to store the source and target point sets and the KD-tree.

20.25.7 7. Usages

- **3D Scanning:** Aligning multiple scans of an object taken from different viewpoints into a complete model.
- **SLAM:** Aligning successive LiDAR or depth-camera frames to track robot motion and build maps.
- **Multi-View Reconstruction:** Registering point clouds extracted from different camera views to form a dense 3D reconstruction.
- **Medical Imaging:** Aligning pre-operative CT/MRI models with intra-operative patient geometry for surgical navigation.
- **Quality Inspection:** Comparing a manufactured part's point cloud to its CAD model to detect deviations.

Implemented in `compgeom.mesh.surface.registration.MeshRegistration`.

20.25.8 8. Testing Methods and Edge Cases

- **Identity Transformation:** Source and target are identical; ICP should converge in one or zero iterations with zero error.
- **Known Rigid Transform:** Apply a known rotation and translation to generate the source, then verify that ICP recovers the inverse transformation within tolerance.
- **Partial Overlap:** Source and target share only a subset of points. ICP may converge to an incorrect alignment; testing should verify that the overlapping region is well-aligned.
- **Noise Robustness:** Add Gaussian noise to source points and verify that ICP still converges to a reasonable alignment.
- **Reflection Input:** A source that is a mirror image of the target. The reflection-handling logic ($\det(R) < 0$) should prevent convergence to a reflected solution.
- **Degenerate Configurations:** Coplanar or collinear point sets may yield ambiguous rotations. The algorithm should still produce a valid (if non-unique) alignment.
- **Empty or Mismatched Sets:** Empty source or target should raise an appropriate error.

20.25.9 9. References

- Besl, P. J., & McKay, N. D. [BM92] “A method for registration of 3-D shapes.” IEEE TPAMI, 14(2), 239–256.
- Chen, Y., & Medioni, G. [CM92] “Object modelling by registration of multiple range images.” Image and Vision Computing, 10(3), 145–155.
- Rusinkiewicz, S., & Levoy, M. (2001) “Efficient variants of the ICP algorithm.” 3DIM.
- Wikipedia: Iterative closest point.

Chapter 21

Half-Edge Data Structure

21.1 Half-Edge Representation

Almost every algorithm in this book operates on geometry that is combinatorial as well as numeric: a planar subdivision, a polygon with holes, or a polyhedral surface is a collection of vertices joined by edges into faces, and the algorithms repeatedly ask *adjacency questions* — which face lies across this edge, which edges bound this face, which edges meet at this vertex. The simplest representation, the **indexed face set** (a list of vertex coordinates plus, for each face, the indices of its vertices), answers none of these questions directly: finding the face on the other side of an edge requires scanning every face, and walking around a vertex requires reassembling the incident edges from all faces that mention the vertex. An indexed face set is a good *input format*; it is a poor *working structure* for algorithms that traverse the mesh.

Definition 21.1 (Planar Subdivision). A **planar subdivision** is a partition of a connected region of the plane (or of the entire plane) into vertices, edges, and faces by a straight-line planar embedding of a planar graph: edges meet only at vertices, and each bounded face is a polygonal region bounded by a cycle of edges. A **polyhedral surface** is the analogous 2-manifold embedded in $\mathbb{R}^3$, built from planar polygonal faces.

The **doubly connected edge list** (DCEL), introduced in the planar-subdivision chapter of de Berg et al. [dBCvKO08], and the **half-edge** structure of the polygon-mesh literature [BKP⁺10] are the same object under two names. The central idea is to abandon the undirected edge and store, for every edge, two directed **half-edges**, one for each side of the edge. Each half-edge owns a pointer to the face on its side and to the half-edge that precedes and follows it around that face. With this single change, every adjacency query becomes either $O(1)$ or a short walk, and local topological edits — splitting an edge, inserting a vertex, merging two faces — become $O(1)$ pointer surgery rather than global reconstruction.

The half-edge structure appears throughout this book. It is the substrate on which polygon booleans (Chapter 6), mesh-analysis validity checks (Chapter 30.1), Delaunay edge flips and insertion legalization (Chapter 12), walking point location (Chapter 8), and the topological counting of Chapter 30 are built. This chapter develops the representation itself: its elements (Section 21.2), its operators (Section 21.3), its treatment of boundaries and manifolds (Section 21.4), the adjacency and traversal queries it supports (Sections 21.5 and 21.6), its relation to the winged-edge, quad-edge, and half-face structures (Section 21.7), its storage cost (Section 21.8), its construction from unstructured input (Section 21.9), and its role in mesh-processing applications (Section 21.10).

21.2 Vertices, Half-Edges, and Faces

Definition 21.2 (Half-Edge). A **half-edge** is a directed edge of the subdivision. It records a start vertex (its *origin*), the face on its left (its *incident face*), and references to the next and previous half-edges in the boundary cycle of that face. Each undirected edge of the subdivision is realized by exactly two half-edges, one for each of its two sides.

Definition 21.3 (DCEL / Half-Edge Structure). A **doubly connected edge list** (equivalently a **half-edge structure**) over a planar subdivision or polyhedral surface consists of:

- a set of **vertices**, each storing coordinates and one incident half-edge;
- a set of **half-edges**, each storing an origin vertex, a twin half-edge, a next half-edge, a prev half-edge, and an incident face;
- a set of **faces**, each storing one boundary half-edge and, for a face with holes, the extra boundary cycles.

The half-edges of a face are linked by **next** into one closed cycle per boundary component.

Definition 21.4 (Boundary Cycle). The **boundary cycle** of a face f is the cyclic sequence of half-edges reached by repeatedly applying **next**, starting from any half-edge of f . For a face without holes this is a single cycle; the length of the cycle is the **face degree** of f , written $|f|$.

The orientation convention fixes the geometry-to-topology link: in the plane, half-edges run counterclockwise around bounded faces and clockwise around the unbounded outer face, so that the interior of every face lies consistently to the left (or right) of its directed boundary. On a closed polyhedral surface the same rule orients every face so that the outward normal is consistently on one side (see Section 21.4 and Chapter 19).

The half-edge data is redundant, and the redundancy is the structure's invariant: it can be checked after every edit, exactly as the adjacency invariants of a graph can be checked via the Handshaking Lemma [Die17].

Theorem 21.5 (Half-Edge Incidence Identity). *Let H be the number of half-edges of a subdivision with E edges, V vertices, and faces F (including the unbounded face in the planar case). Then*

$$H = 2E = \sum_{v \in V} \deg(v) = \sum_{f \in F} |f|,$$

where $\deg(v)$ counts the number of half-edges whose origin is v .

Proof. Each undirected edge is realized by exactly two half-edges, one per side, so $H = 2E$. Each half-edge has exactly one origin vertex, so summing the incident-half-edge count over all vertices counts every half-edge exactly once: $H = \sum_v \deg(v)$. Similarly each half-edge is traversed by exactly one face (it lies on the boundary of its incident face), and the face degrees count exactly the half-edges of that face, so $H = \sum_f |f|$. $\square$

Example 21.6 (A Single Triangle). The subdivision of a triangle has $V = 3$ vertices, $E = 3$ edges, and $F = 2$ faces (the bounded triangle and the unbounded outer face). Its $H = 2E = 6$ half-edges split into three pairs: the three CCW half-edges $(v_0, v_1), (v_1, v_2), (v_2, v_0)$ bound the triangle, and their twins bound the outer face clockwise. Hence $\sum_v \deg(v) = 2 + 2 + 2 = 6$ and $\sum_f |f| = 3 + 3 = 6$, matching Theorem 21.5.

Remark 21.7. For a closed polyhedral surface (a sphere-like mesh, no boundary), every edge has two sides and $H = 2E$ still holds, but there is no unbounded face. The incidence identity is dimension-independent: it is the combinatorial core of the Euler-characteristic machinery of Chapter 30.

21.3 Twin, Next, and Prev Operators

The three core operators are what make the half-edge structure a *navigational* structure rather than a static list.

Definition 21.8 (Twin Operator). The **twin** operator $\text{twin}(h)$ returns the half-edge of the same undirected edge that runs in the opposite direction and lies on the other side of the edge. It is an involution: $\text{twin}(h) = h'$ if and only if $\text{twin}(h') = h$, and $\text{twin}(\text{twin}(h)) = h$.

Definition 21.9 (Next and Prev Operators). The **next** operator $\text{next}(h)$ returns the half-edge that follows h in the boundary cycle of its face, and the **prev** operator $\text{prev}(h)$ returns the half-edge that precedes it. Both are defined within a face: next and prev are inverse to each other on the same face cycle.

Lemma 21.10 (Inverse on Face Cycles). *For every half-edge h , $\text{prev}(h) = \text{next}^{-1}(h)$; in particular $\text{next}(\text{prev}(h)) = h$ and $\text{prev}(\text{next}(h)) = h$.*

Proof. By definition both next and prev restrict to cyclic successor and predecessor permutations of the same face boundary cycle, and the successor permutation's inverse is the predecessor permutation. $\square$

Two derived queries are used constantly and are worth naming. The **origin** function $\text{org}(h)$ returns the start vertex of h ; the **target** is $\text{org}(\text{twin}(h))$, the vertex the half-edge points to. The **incident face** $\text{face}(h)$ returns the face whose boundary contains h . Every adjacency primitive in Sections 21.5 and 21.6 is a short composition of twin , next , prev , and the origin/face accessors.

The two compositions that matter most are those that cross between the *face* world and the *vertex* world. A face cycle is traced by iterating next ; a **vertex star** — the set of faces or edges incident to a vertex — is traced by the composition $\text{next} \circ \text{twin}$, which walks from one incident face of v to the next one around v .

Theorem 21.11 (Cycle Structure). *The next cycles partition the half-edges, one cycle for each boundary component of each face. The cycles of $\text{next} \circ \text{twin}$ partition the half-edges around vertices: each cycle visits exactly the half-edges whose origin is a fixed vertex, once each.*

Proof. Every half-edge has exactly one next and one prev , and the two are inverses (Lemma 21.10), so the functional graph of next decomposes into disjoint cycles; since each half-edge lies on exactly one face boundary, these are the face cycles. For the second claim, fix v and consider the map $h \mapsto \text{next}(\text{twin}(h))$ on the set S_v of half-edges with origin v . If h has origin v , then $\text{twin}(h)$ has origin $\text{org}(h)$'s neighbor w , and $\text{next}(\text{twin}(h))$ is the half-edge of the next face around v that shares v ; its origin is again v , so the map sends S_v into itself, and it is a bijection on the finite set S_v , hence a permutation whose cycles visit every half-edge of S_v . Distinct vertices have disjoint origin sets, so the cycles partition all half-edges. $\square$

Remark 21.12. Guibas and Stolfi [GS85] formalized this algebra for their **quad-edge** structure, in which the operators form a group algebra on each geometric edge; the half-edge structure is exactly the restriction of the quad-edge algebra to the primal side of a subdivision. We return to this comparison in Section 21.7.

21.4 Boundary and Manifold Handling

Most real meshes — a scanned surface, a finite-element domain, a terrain — are not closed polyhedra: they have a boundary. The half-edge structure must represent boundaries explicitly, because an edge that is incident to only one face has no second half-edge to pair with.

Definition 21.13 (Boundary Edge and Boundary Half-Edge). A **boundary edge** is an edge incident to exactly one face. The half-edge of a boundary edge that belongs to the incident face is a **boundary half-edge**; it has no twin among the half-edges of the other side, because no face exists there.

Definition 21.14 (Manifold Edge and Manifold Vertex). An edge is **edge-manifold** if it is incident to at most two faces; a vertex is **vertex-manifold** if its incident faces form a single connected fan (a topological disk, or half-disk at the boundary) rather than two or more cones meeting only at the vertex. A mesh is a **manifold mesh** if every edge and vertex is manifold. Non-manifold edges (three or more incident faces) cannot be represented faithfully by a structure in which every edge has exactly two sides.

The boundary behavior is forced by the twin involution:

Theorem 21.15 (Twin Boundary Dichotomy). *Let h be a half-edge of a manifold mesh. If h is an interior half-edge, then $\text{twin}(h)$ is a distinct half-edge belonging to the unique other face on the edge. If h is a boundary half-edge, then no such half-edge exists, and implementations either leave $\text{twin}(h)$ null or link it to a synthetic **dummy boundary half-edge**.*

Proof. By edge-manifoldness each edge has at most two incident faces. An interior half-edge's edge has exactly two incident faces, so it has exactly one partner half-edge on the other side, which is its twin. A boundary half-edge's edge has exactly one incident face, so no partner exists. $\square$

Boundary half-edges are not isolated: they chain together.

Theorem 21.16 (Boundary Cycles). *In a manifold mesh with boundary, the boundary half-edges that are not paired (i.e., those with null or dummy twins) decompose into a unique set of disjoint closed chains, each oriented consistently, one chain per **boundary component** of the mesh.*

Proof. Each boundary half-edge has exactly one next, which is again a boundary half-edge: walking forward along the boundary keeps the single incident face on one side, and the successor half-edge of that face can only leave the boundary when the boundary makes a closed turn, which by definition continues the chain. Since there are finitely many boundary half-edges and every boundary vertex has exactly one incoming and one outgoing boundary half-edge (vertex-manifoldness), the chains are disjoint cycles. Each cycle bounds one connected component of the boundary, and distinct cycles cannot belong to the same boundary component because the components are disjoint sets of vertices and edges. $\square$

The boundary cycle count b enters the Euler characteristic, the fundamental numerical invariant of a subdivision (Theorem 21.28 in Section 21.8). It also drives the boundary-detection routines of Chapter 30.1: with a half-edge structure, an edge is a boundary edge exactly when its twin is null (or dummy), a test that costs $O(1)$ and needs no hashing.

Example 21.17 (Square with a Hole). A square region with a square hole is a subdivision with two boundary components — the outer perimeter and the hole's perimeter — so $b = 2$, and every edge on either cycle is a boundary edge by Theorem 21.16. The region's single face is bounded by two cycles: its outer boundary runs counterclockwise and the hole's boundary runs clockwise, so the face interior lies consistently on one side of both; the structure stores both cycles in the face record (Definition 21.3).

Remark 21.18. Two degenerate situations defeat the manifold assumptions: **T-junctions** (a vertex lying on the interior of an edge of another face) and **pinched** (non-manifold) vertices where two fans meet only at the vertex. Both must be detected and repaired before a half-edge structure is built; the verification algorithms of Chapter 30.1 and the manifold concepts of Section 2.15 provide the tests.

21.5 Adjacency and Neighborhood Queries

21.5.1 1. Overview

The half-edge structure exists to answer adjacency queries. This section catalogues the standard queries a mesh algorithm asks, states their costs, and gives the pointer expressions that answer them. All of them reduce to a constant number of operator applications plus one cyclic walk, which is why mesh-analysis and mesh-editing algorithms are built on this structure [dBCvKO08, BKP⁺10].

21.5.2 2. Definitions

Definition 21.19 (Adjacency Query). An **adjacency query** takes one element of the subdivision (vertex, edge, half-edge, or face) and returns a related element or a list of related elements, such as “the face across edge e ” or “all edges incident to vertex v ”.

21.5.3 3. Theory

The cost of a query is governed by Theorem 21.11: constant compositions of twin, next, and prev step from one element to a unique neighbor in $O(1)$, while enumeration queries walk an entire cycle and cost the cycle length.

Theorem 21.20 (Query Complexity). *Each of the single-element adjacency queries (face across an edge, face of a half-edge, origin and target of a half-edge, twin, next, prev, boundary test) costs $O(1)$. Each enumeration query (faces or edges around a vertex, edges or vertices of a face, faces around a face) costs $O(1)$ per returned element, hence $O(\deg(v))$ or $O(|f|)$ in total.*

Proof. The single-element queries apply one or two of the operators defined in Section 21.3, each an $O(1)$ pointer dereference; the boundary test reads the twin field once. An enumeration query iterates the relevant cycle of Theorem 21.11, performing $O(1)$ work per element and visiting each element of the star or face exactly once, for $O(\deg(v))$ or $O(|f|)$ work. $\square$

21.5.4 4. Pseudocode

The neighborhood queries are one-line compositions of the operators; the following is the canonical enumeration of the faces incident to a vertex, using the cycle of Theorem 21.11.

Algorithm 103 Enumerate Faces Around a Vertex

Input: A half-edge structure and a vertex v .

Output: The list of faces incident to v .

```

1: function INCIDENTFACES( $v$ )
2:    $h \leftarrow v.\text{halfedge}$   $\triangleright$  any half-edge with origin  $v$ 
3:    $h_0 \leftarrow h$ ;  $\text{faces} \leftarrow []$ 
4:   loop
5:      $\text{faces.append}(\text{face}(h))$ 
6:      $h \leftarrow \text{twin}(\text{prev}(h))$ 
7:     if  $h = h_0$  then break
8:   end loop
9:   return  $\text{faces}$ 
10: end function
```

Remark 21.21. The step $h \leftarrow \text{twin}(\text{prev}(h))$ rotates the half-edge to the next face around v ; the equivalent step $h \leftarrow \text{next}(\text{twin}(h))$ rotates in the opposite direction. Both visit every incident face exactly once by Theorem 21.11.

21.5.5 5. Parameter Selection

- **Origin pointer:** storing the origin vertex per half-edge makes `org` free; a variant that stores only the *target* vertex changes the derived expressions but not their cost. Store whichever the dominant algorithms read most.
- **Null versus dummy twins:** a null twin halves the boundary test to one comparison but breaks code that assumes `twin(h)` is always valid; dummy boundary half-edges keep the twin involution total at the price of $2b$ extra half-edge records (one per boundary half-edge) and the explicit handling of the dummy's operators.
- **Prev pointer:** `prev` can be derived from `next` by storing the face cycles in circular arrays, but a direct pointer doubles the storage of the operator fields and halves the walking distance; most library implementations store both [BKP⁺10, Eri04].

21.5.6 6. Complexity

- **Per-query time:** $O(1)$ for single-element queries, $O(\deg(v))$ or $O(|f|)$ for enumerations (Theorem 21.20).
- **Space:** the operator fields dominate; see the detailed accounting in Section 21.8.

21.5.7 7. Usages

- **Point location by walking** (Chapter 8) advances from triangle to triangle by repeatedly answering “which face is across this edge?” in $O(1)$.
- **Delaunay legalization** (Chapter 12) tests an edge's two incident faces and relinks the four half-edges of a quadrilateral, both $O(1)$.
- **Mesh analysis** (Chapter 30.1) computes valence, boundary loops, and orientability by enumerating the vertex stars and face cycles of this section.

21.5.8 8. Testing Methods and Edge Cases

- **Invariant audit:** after building any half-edge structure, verify `twin(twin(h)) = h`, `next(prev(h)) = h`, and the counts of Theorem 21.5.
- **Star enumeration:** for every vertex, assert that `IncidentFaces` (Algorithm 103) returns each incident face exactly once and terminates.
- **Boundary vertices:** on an open disk, boundary vertices must yield the half-disk fan; the traversal must not leave the structure through a null twin.
- **Isolated and non-manifold cases:** isolated vertices have an empty star; non-manifold vertices break the single-cycle property and must be rejected at construction time (Section 21.9).

21.5.9 9. References

The DCEL treatment follows de Berg et al. [dBCvKO08]; the pointer expressions, the boundary conventions, and the storage trade-offs follow Botsch et al. [BKP⁺10] and the data-structure overview of Ericson [Eri04]. The graph-theoretic counting arguments use Diestel [Die17] and the dictionary organization of Cormen et al. [CLRS09].

21.6 Traversal Operations

21.6.1 1. Overview

Traversal is the algorithmic use of the navigation operators: walking the boundary of a face, walking the star of a vertex, and walking a boundary cycle. A traversal is any algorithm that visits the elements of a cycle of Theorem 21.11 in order. Traversals are the loops inside the mesh-analysis routines of Chapter 30.1 — valence counting, boundary detection, orientability propagation — and the per-vertex stencils of mesh processing [BKP⁺10].

21.6.2 2. Definitions

Definition 21.22 (Face Walk). A **face walk** from a half-edge h visits the half-edges $h, \text{next}(h), \text{next}^2(h), \dots$ until it returns to h . It visits exactly the boundary cycle of $\text{face}(h)$.

Definition 21.23 (Vertex Walk). A **vertex walk** from a half-edge h with origin v visits the half-edges $h, \text{next}(\text{twin}(h)), \text{next}(\text{twin}(\text{next}(\text{twin}(h)))) \dots$ until it returns to h ; by Theorem 21.11 this visits every half-edge with origin v exactly once.

21.6.3 3. Theory

Theorem 21.24 (Cycle Cover). *Every half-edge is visited by exactly one face walk and exactly one vertex walk, so the walks partition the half-edges into $\sum_f b_f$ face cycles (b_f boundary components of face f) and V vertex cycles.*

Proof. This is the combinatorial content of Theorem 21.11 restated: the next-permutation decomposes the half-edges into the face cycles, and the $\text{next} \circ \text{twin}$ permutation decomposes them into the vertex cycles. Each half-edge appears in exactly one cycle of each type. $\square$

21.6.4 4. Pseudocode

Algorithm 104 Walk a Face Boundary

Input: A half-edge h .

Output: The half-edge sequence of the boundary cycle of $\text{face}(h)$.

```

1: function WALKFACE( $h$ )
2:    $h_0 \leftarrow h$ ;  $cycle \leftarrow []$ 
3:   loop
4:      $cycle.append(h)$ 
5:      $h \leftarrow \text{next}(h)$ 
6:     if  $h = h_0$  then break
7:   end loop
8:   return  $cycle$ 
9: end function
```

Remark 21.25. To walk the boundary cycles of an open mesh, start at any unvisited boundary half-edge (twin null or dummy) and apply Algorithm 104; the returned cycle is exactly one boundary component (Theorem 21.16).

21.6.5 5. Parameter Selection

- **Direction:** walking with next traverses a face in the stored orientation; the opposite orientation is obtained by swapping next and prev .

Algorithm 105 Walk Around a Vertex

Input: A half-edge h and a flag *emit_faces*.**Output:** The incident faces (or incident edges) of $\text{org}(h)$ in cyclic order.

```
1: function WALKVERTEX( $h$ , emit_faces)
2:    $h_0 \leftarrow h$ ;  $\text{cycle} \leftarrow []$ 
3:   loop
4:     append  $\text{face}(h)$  if emit_faces, else append  $h$ 
5:      $h \leftarrow \text{twin}(\text{prev}(h))$ 
6:     if  $h = h_0$  then break
7:   end loop
8:   return  $\text{cycle}$ 
9: end function
```

- **Termination test:** comparing against the start half-edge is the only safe termination test; counting steps is a bug source when faces have varying degrees.
- **Boundary walk seeding:** boundary walks must be seeded from a boundary half-edge; seeds are cheaply found by keeping a single boundary half-edge per component (Section 21.4).

21.6.6 6. Complexity

- **Face walk:** $O(|f|)$ steps, $O(1)$ per step.
- **Vertex walk:** $O(\deg(v))$ steps, $O(1)$ per step.
- **Full traversal** of all faces or all vertices: $O(H) = O(V + E)$, because the walks partition the half-edges (Theorem 21.24) and each half-edge is touched $O(1)$ times.

21.6.7 7. Usages

- **Valence and degree computation:** the vertex walk of Algorithm 105 implements the degree loop of Chapter 30.1.
- **Boundary detection and hole filling:** boundary walks (Theorem 21.16) enumerate the loops to be filled or parameterized.
- **Orientation propagation:** the face walk carries the orientation consistency check of Section 21.4 across adjacent faces.
- **Geodesic and area queries:** walking the faces around a vertex accumulates the angle defect used in discrete curvature (Chapter 30.1).

21.6.8 8. Testing Methods and Edge Cases

- **Cycle closure:** assert every walk returns to its start half-edge; a walk that escapes indicates broken next/twin links.
- **Partition test:** run all walks and verify that every half-edge is visited exactly once (Theorem 21.24).
- **Boundary:** on an open mesh, a vertex walk must stop at a boundary half-edge and must visit each boundary face once.
- **Edge cases:** a 2-gon (two half-edges, degenerate face), a lone edge, and a mesh of one triangle are minimal inputs on which the termination test must still fire.

21.6.9 9. References

The traversal primitives and their use in mesh algorithms are described in Botsch et al. [BKP⁺10]; the walking queries are the workhorses of the triangulation walk in Chapter 8 and the Delaunay insertion of Chapter 12.

21.7 Comparison with Other Representations

The half-edge structure is one point on a spectrum of mesh representations differing in which adjacencies are stored explicitly and whether duality is exposed [Eri04, BKP⁺10].

Indexed face set Vertices plus per-face index lists. Minimum storage and the universal file format, but no adjacency: every neighbor query costs a scan, and local edits require global rebuilds. It is the *input* format for the construction pipeline of Section 21.9.

Winged edge The precursor of the half-edge structure, from the solid-modeling literature. Every undirected edge stores the two faces it borders and four half-edge links (the successors and predecessors of both halves). It answers the same queries as the half-edge structure but with more fields per edge and without the symmetric twin/next pairing; the half-edge structure is essentially a pointer-slimmed winged edge.

Half-edge / DCEL This chapter's structure: two half-edges per edge, each with origin, twin, next, prev, and face. Exactly the adjacency needed by the planar algorithms of this book, at roughly two to three times the storage of an indexed face set.

Quad edge The structure of Guibas and Stolfi [GS85] stores each geometric edge with *four* directed edges: the two primal half-edges and the two dual half-edges crossing the edge. Its algebra (operators rot, sym, flip, next) handles the Voronoi–Delaunay duality uniformly, supports subdivisions of arbitrary topology (including non-manifold edge incidences via edge rings), and degrades gracefully to a half-edge structure when only the primal is needed. It is the representation of choice when an algorithm must navigate the dual subdivision, as in Chapter 12.

Half-face The 3D analogue for volumetric cell complexes. A face of a 3D mesh is shared by up to two cells; a **half-face** is the restriction of a face to one incident volume, and the twin relation pairs half-faces across interior faces. Tetrahedral meshes implement the same idea more directly as four neighbor pointers per tetrahedron. Volumetric half-edge generalizations (the radial-edge family) additionally support non-manifold boundaries by cycling the faces around each edge.

Remark 21.26 (Choosing a Representation). As a rule of thumb: use an indexed face set for I/O and archival, the half-edge structure for 2D subdivisions and polyhedral *surfaces* on which the algorithms of Chapters 6, 12, and 30.1 run, the quad-edge structure when primal–dual duality is part of the algorithm, and half-faces or tetrahedral adjacency for volumetric meshes (Chapter 22). The structures are freely convertible in linear time in either direction (Section 21.9 gives the hard direction).

21.8 Incidence and Storage Complexity

A half-edge structure stores one record per vertex, per half-edge, and per face. With the fields of Definition 21.3, a half-edge holds the twin, next, and prev pointers, the origin vertex, and the incident face; a vertex holds its coordinates and one incident half-edge; a face holds one boundary half-edge (plus one per extra boundary cycle). The counts are exact:

Theorem 21.27 (Storage Counts). *For a subdivision with V vertices, E edges, and F faces, the half-edge structure stores exactly $H = 2E$ half-edge records, V vertex records, and F face records. The total storage is $O(V + E + F)$ and, for a fixed record size, $O(V + E)$.*

Proof. Every undirected edge contributes exactly two half-edges, one per side (Theorem 21.15); no other half-edges exist. Vertices and faces are stored one per element of the subdivision by Definition 21.3. For a connected planar subdivision $E = O(V)$ by the planar Euler formula of Theorem 21.28, and in general $E \leq 3V$ for planar triangulations (and for surface meshes of bounded genus), so the bound follows. $\square$

In a triangular mesh the numbers are explicit: $H = 2E$, and by the Handshaking Lemma for triangle meshes [Die17], $3F = 2E$, so $E \approx 3V$ and $H \approx 6V$ for large closed meshes. The half-edge structure therefore costs roughly 6 half-edge records plus 1 vertex record per vertex, against 3 face records and 1 vertex record for an indexed face set: a constant factor, not an asymptotic change. This is the standard price of $O(1)$ adjacency [BKP⁺10, Eri04].

The counting invariants of Theorem 21.5 are the mesh’s **topological checksum**: any inconsistency in the pointers changes one of the sums, so the identity

$$\sum_v \deg(v) = 2E = \sum_f |f|$$

catches almost every corrupted structure. The deepest such invariant is the Euler characteristic.

Theorem 21.28 (Euler–Poincaré Formula). *Let M be a connected 2-manifold mesh with V vertices, E edges, F faces, b boundary components, and genus g (see Section 2.15 and Chapter 30). Then*

$$V - E + F = 2 - 2g - b.$$

In particular, a closed surface of genus g has $V - E + F = 2 - 2g$, and a disk has $V - E + F = 1$ [Eul58, Poi95].

Proof. Sketch. Euler’s classical result [Eul58] gives $V - E + F = 2$ for convex polyhedra, which are closed surfaces of genus 0. Removing a disk (opening one boundary component) reduces F by 1, giving the $-b$ term; attaching a handle raises g by 1 and reduces $V - E + F$ by 2, by triangulating the added tube with V' new vertices and checking $V' - E' + F' = 0$. A full proof is the induction presented in Chapter 30.1 (Theorem 30.27) and the algebraic-topology formulation of Section 2.15. $\square$

Remark 21.29 (Planar Subdivisions). For a connected planar subdivision one conventionally counts the unbounded face as an extra face, which adds exactly 1 to F and yields the classical $V - E + F = 2$ of Euler’s planar formula [dBCvKO08]. The two conventions differ by exactly the outer face: the surface version above gives $V - E + F = 2 - b$ for a disk-like region with b boundary components, and adding the outer face turns this into $V - E + F = 2$. Keep the convention fixed when checking a mesh.

Example 21.30 (Checking a Mesh). A closed tetrahedron mesh has $V = 4$, $E = 6$, $F = 4$, hence $V - E + F = 2$, matching $g = 0$, $b = 0$. An open triangulated disk with $V = 6$, $E = 9$, $F = 4$ satisfies $V - E + F = 1 = 2 - b$, so $b = 1$: exactly one boundary component. If an edit corrupts the mesh, one of these counts changes and the invariant test fails; this is why Chapter 30.1 runs the Euler test as a validity check.

Remark 21.31. Because $V - E + F$ is a topological invariant, it is constant across *remeshing*: retriangulating a surface without changing its topology preserves $V - E + F$, which is how topology-preservation is tested in simplification and remeshing pipelines (Chapter 30.1 and Chapter 30).

21.9 Construction from Polygon Soup

21.9.1 1. Overview

Real inputs — an STL file, an OBJ export, the output of a reconstruction algorithm — arrive as an **polygon soup**: an unordered list of oriented polygon loops with no shared structure. Building a half-edge structure from such a soup is a two-step process: **vertex welding** merges coincident coordinates into shared vertex records, and **edge pairing** matches the half-edges of each face to the half-edges of its neighbors [BKP⁺10, dBCvKO08]. The same pipeline converts an indexed face set into the working structure used by every other chapter.

21.9.2 2. Definitions

Definition 21.32 (Polygon Soup). A **polygon soup** is a list of faces, each given by an oriented sequence of point coordinates (or of indices into an unshared coordinate list). There is no requirement that vertices, edges, or faces be shared between any two faces.

Definition 21.33 (Vertex Welding). **Vertex welding** replaces the points of a polygon soup by a set of welded vertex records: two soup points map to the same vertex if their distance is at most a tolerance ϵ (in practice, below the coordinate quantization of the file format).

21.9.3 3. Theory

The correctness condition for the pairing step is exact:

Theorem 21.34 (Pairing Condition). *Let the faces of a soup be consistently oriented. The unordered vertex pairs $\{u, v\}$ of the soup's edges are the undirected edges of a manifold mesh if and only if every pair is used exactly twice. In that case each bucket of size two contains one half-edge from each of the two adjacent faces, and these two half-edges are each other's twins. An edge used once is a boundary edge; an edge used three or more times is non-manifold and cannot be represented.*

Proof. A manifold mesh has every edge incident to exactly one or two faces (Definition 21.14). Each incidence of face f with edge $\{u, v\}$ contributes one half-edge of that face over $\{u, v\}$, so an edge appears in the soup once per incident face: once for a boundary edge, twice for an interior edge. Hence the bucket size of $\{u, v\}$ equals the number of incident faces. Two occurrences with the same face would duplicate a half-edge and are impossible for a well-formed face. If the faces are consistently oriented, the two half-edges of a size-two bucket traverse $\{u, v\}$ in opposite directions and are twins. Conversely, bucket sizes outside $\{1, 2\}$ witness a boundary-less non-manifold edge or a duplicated face. $\square$

Consistency of orientation is required for the conclusion; a soup with mixed orientations still welds and pairs, but the extracted faces have inconsistent normals, which the orientability pass of Chapter 30.1 then repairs.

21.9.4 4. Pseudocode

21.9.5 5. Parameter Selection

- **Welding tolerance:** choose ϵ slightly larger than the coordinate quantization (e.g., 10^{-6} relative units for single-precision STL), but small enough not to merge genuinely distinct vertices. Too large a tolerance collapses thin features.
- **Bucket key:** canonicalize each edge as $(\min(u, v), \max(u, v))$ so both orientations hash to the same bucket; a key on the oriented pair would fail to pair twins.

Algorithm 106 Weld Polygon-Soup Vertices

Input: A list P of points and a tolerance $\epsilon > 0$.**Output:** A map from soup points to welded vertex indices.

```

1: function WELDVERTICES( $P, \epsilon$ )
2:    $G \leftarrow$  uniform grid of cell size  $\epsilon$ ;  $map \leftarrow []$ ;  $k \leftarrow 0$ 
3:   for each point  $p \in P$  do
4:      $c \leftarrow$  grid cell of  $p$ ;  $found \leftarrow \text{None}$ 
5:     for each vertex  $q$  in  $c$  or in the 8 neighboring cells do
6:       if  $\|p - q\| \leq \epsilon$  then
7:          $found \leftarrow \text{index}(q)$ ; break
8:       end if
9:     end for
10:    if  $found \neq \text{None}$  then
11:       $map.append(found)$ 
12:    else
13:      insert  $p$  as vertex  $k$ ;  $map.append(k)$ ;  $k \leftarrow k + 1$ 
14:    end if
15:  end for
16:  return  $map$ 
17: end function

```

Algorithm 107 Build Half-Edge Structure from a Polygon Soup

Input: Welded, consistently oriented faces, each an index cycle.**Output:** A valid half-edge structure.

```

1: function BUILDDCEL( $faces$ )
2:   create one half-edge per directed face edge; set origin and face fields
3:    $buckets \leftarrow$  hash map on unordered vertex pairs
4:   for each half-edge  $h$  from  $u$  to  $v$  do
5:      $buckets[\{u, v\}].append(h)$ 
6:   end for
7:   for each bucket  $\{h_i, h_j\}$  do
8:     if  $|bucket| = 2$  then
9:        $twin(h_i) \leftarrow h_j$ ;  $twin(h_j) \leftarrow h_i$ 
10:    else if  $|bucket| = 1$  then
11:      leave the twin null (boundary half-edge)
12:    else
13:      error: non-manifold edge  $\{u, v\}$ 
14:    end if
15:  end for
16:  for each face do
17:    link its half-edges cyclically with next and prev
18:  end for
19:  extract boundary cycles and append a dummy boundary half-edge per cycle  $\triangleright$  optional
20:  return the structure
21: end function

```

- **Boundary half-edges:** keep twins null and fix the boundary convention before any traversal (Sections 21.4 and 21.6).

21.9.6 6. Complexity

- **Welding:** $O(n)$ expected time and space with the uniform grid of Algorithm 106, $O(n \log n)$ with a balanced-tree variant; each point touches a constant number of cells for fixed ϵ .
- **Pairing:** $O(n)$ expected, dominated by one hash insert and one lookup per half-edge.
- **Total:** $O(n)$ expected time and space for a soup with n directed edges; the output satisfies the storage bounds of Theorem 21.27.

21.9.7 7. Usages

- **File loading:** STL, OBJ, and PLY importers run exactly this pipeline [BKP⁺10, Eri04].
- **Scan reconstruction:** the triangle soup produced by surface-reconstruction algorithms is welded and paired before any analysis (Chapter 22 consumes such meshes).
- **Repair:** the pairing stage reports non-manifold edges and inconsistent orientations, which is the first repair pass of Chapter 30.1.

21.9.8 8. Testing Methods and Edge Cases

- **Closed mesh:** a welded tetrahedron or sphere must produce zero boundary half-edges and satisfy $V - E + F = 2$ (Theorem 21.28).
- **Open mesh:** a single triangle must yield one boundary cycle of three half-edges and $V - E + F = 1$.
- **Duplicate vertices:** faces whose coordinates are identical to 12 decimals must weld to a single shared vertex record.
- **Coordinate noise:** perturb the coordinates by 10^{-5} before welding and verify the tolerance merges them back.
- **Non-manifold soup:** an edge shared by three faces must raise the error branch of Algorithm 107.
- **Duplicate faces:** two identical triangles use each edge twice but generate two half-edge pairs with the same face on both sides; the structure must flag or reject this.

21.9.9 9. References

The welding-and-pairing pipeline follows Botsch et al. [BKP⁺10] and the file-import discussion of Ericson [Eri04]; the underlying hashing and graph-construction techniques are standard dictionary algorithms [CLRS09]. The DCEL construction for planar subdivisions is described by de Berg et al. [dBCvKO08].

21.10 Applications in Mesh Processing

The half-edge structure pays for its storage in the local, $O(1)$ topological edits called **Euler operations**. These are the primitives from which every topology-changing mesh edit — extruding, bridging, deleting a handle, subdividing — is assembled [BKP⁺10, dBCvKO08]. The classical set, drawn from the solid-modeling tradition, includes:

- **Make vertex, face, solid (MVFS)**: creates a new component with one vertex and one face.
- **Make edge, vertex (MEV)**: adds an edge from a vertex to a new vertex on a face.
- **Make edge, face (MEF)**: adds an edge inside a face, splitting it into two faces.
- The inverse operations **Kill edge, face (KEF)**, **Kill edge, vertex (KEV)**, and **Kill vertex, face, solid (KVFS)**.

Theorem 21.35 (Euler Operations Preserve the Invariant). *Each of MEV, MEF, KEV, and KEF changes (V, E, F) by one of $(1, 1, 0)$, $(0, 1, 1)$, $(-1, -1, 0)$, $(0, -1, -1)$, and therefore leaves $V - E + F$ unchanged. Every connected orientable manifold mesh can be built from a single MVFS by a sequence of MEV and MEF operations, and every edit that preserves the Euler characteristic of Theorem 21.28 is a sequence of these operations.*

Proof. The count changes follow directly from the operation definitions. For generation: a connected mesh’s dual graph is connected, so its faces can be ordered so that each new face shares an edge with a previous one; adding faces one at a time by MEF (after placing the first face’s vertices with MEV) constructs the mesh. Conversely, any sequence of the four operations preserves $V - E + F$ by direct inspection of the increments, and since these operations span the local relinking moves (edge split, edge collapse, face split, edge flip, vertex split), any topology-preserving edit decomposes into them. $\square$

The pointer surgery for an edge split, for example, creates two new half-edges, reassigns two twin pointers and the next/prev links of four neighbors, and never touches any other half-edge: $O(1)$ work with no global reindexing. This locality is what makes interactive mesh editing — the vertex-painting, extrusion, and Boolean workflows of modern modeling tools — practical on million-face meshes [BKP⁺10, Eri04].

Boolean operations on polyhedra (union, intersection, difference) combine the classification of Chapter 6 with half-edge surgery: intersecting faces are split along the intersection polygons, the faces inside each operand are classified, and the surviving pieces are stitched by pairing twin half-edges at the cut [Eri04]. The result is reassembled by the construction of Section 21.9 run on the subdivided soup, and its validity is verified with the Euler test (Theorem 21.28) and the manifold checks of Chapter 30.1.

Example 21.36 (Edge Flip in Delaunay Legalization). In a Delaunay triangulation, the legality test of Chapter 12 examines the two faces on either side of an interior edge; when the edge is illegal, it is replaced by the other diagonal of the quadrilateral. In half-edge terms the flip re-links four half-edges’ next/prev pointers and the two twin pairs: four pointer assignments per half-edge, $O(1)$ total, which is why incremental Delaunay insertion (Algorithm 107 supplies the structure it runs on) achieves expected $O(\log n)$ per point.

Beyond booleans and editing, the half-edge structure is the substrate of the traversal-heavy algorithms throughout this book and its companion chapters:

- **Mesh analysis** (Chapter 30.1): connected components, boundary detection, orientability, and manifold verification are exactly the walks and twin tests of Sections 21.5–21.6.
- **Polygon operations** (Chapter 6): polygon booleans and clipping resample the boundary cycles (Theorem 21.16) of the input subdivisions.
- **Delaunay triangulations** (Chapter 12): point insertion, legalization flips, and the walking point location of Chapter 8 are half-edge walks.

- **Computational topology** (Chapter 30): the Euler characteristic, Betti numbers, and homology computations read the incidence information encoded by the twin and face links, and Section 2.15 supplies the manifold background that the boundary machinery of Section 21.4 assumes.
- **Mesh generation** (Chapter 22): the output of a conforming mesher is delivered as a half-edge (or index-based) structure with the exact adjacency that refinement and adaptation need.

21.11 Exercises

1. Draw the half-edge structure of a square subdivided into two triangles by a diagonal. List all ten half-edges (five undirected edges, two half-edges each) with their origin, twin, next, and prev links, and verify $\sum_v \deg(v) = 2E = \sum_f |f|$ (Theorem 21.5).
2. Prove that in a half-edge structure, $\text{org}(h) = \text{org}(\text{twin}(\text{next}(h)))$ holds for every half-edge h , and interpret the equality in words.
3. Show that a face walk that counts steps instead of testing for the start half-edge can loop forever, and give a concrete small mesh where this happens.
4. Let T be a triangulated connected surface with V vertices, F triangular faces, E edges, B boundary edges, and b boundary components. Prove $3F = 2E - B$ (each triangle contributes three directed edge incidences; interior edges are counted twice, boundary edges once) and combine it with Theorem 21.28 to derive $F = 2V - 4 + 2b - B$. Check the formula on a square with one interior vertex ($V = 5$, $B = 4$, $b = 1$) and on a plain square.
5. Design a linear-time algorithm, using Algorithm 104, that lists the boundary components of an open mesh from the null-twin half-edges alone, and prove it visits every boundary edge exactly once (Theorem 21.16).
6. Convert the indexed face set of a tetrahedron into a half-edge structure by hand, following Algorithm 107, and confirm that every twin is paired and that $V - E + F = 2$.
7. Implement vertex welding with tolerance ϵ on a soup of noisy coordinates, then apply the Euler test ($V - E + F$) to the output and explain which failures the test can and cannot detect.
8. Give the half-edge pointer assignments for an edge flip in a triangulated quadrilateral, as in Example 21.36, and verify that only the four half-edges of the two faces change their links.

Chapter 22

Mesh Generation

22.1 Computational Meshes

A **computational mesh** is a discretization of a geometric domain into simple cells — triangles and quadrilaterals in the plane, tetrahedra, hexahedra, and prisms in space — on which numerical methods (finite elements, finite volumes, boundary elements) compute approximate solutions of partial differential equations. Mesh generation is the algorithmic discipline that constructs such meshes automatically from a geometric description of the domain: a polygon, a polyhedron, or a CAD boundary representation, possibly with prescribed mesh sizes near features and a quality requirement on the cells.

The quality of a mesh directly controls the accuracy and stability of the numerical solution: poorly shaped cells (very small angles, slivers) inflate the discretization error and the condition number of the assembled linear system. Mesh generation therefore balances three competing goals: **conformity** (the mesh reproduces the domain boundary and any prescribed features exactly), **quality** (every cell satisfies a geometric quality bound, e.g., a minimum angle), and **efficiency** (the number of cells is as small as the size field allows). These goals, their algorithmic resolutions, and their analysis are the subject of this chapter.

Two families of methods dominate practice. **Delaunay-based** methods (Chapter 12) build the mesh as a Delaunay triangulation refined by inserting points at circumcenters of bad cells; they dominate the triangulated meshes of scientific computing (Section 22.7). **Advancing-front** methods grow the mesh inward from the boundary (Section 22.10). Both are supported by the mesh-analysis machinery of Chapter 30.1, which verifies the validity and quality of the output.

22.2 Simplicial Complexes

Definition 22.1 (Simplicial Complex). A **simplicial complex** K is a finite collection of simplices (vertices, edges, triangles, tetrahedra) such that (i) every face of a simplex of K is in K , and (ii) the intersection of any two simplices of K is a face of both. The **underlying space** $|K|$ is the union of the simplices.

Simplicial complexes are the abstract combinatorial skeleton of a mesh: a triangular mesh is a 2D simplicial complex whose underlying space approximates the domain, and a tetrahedral mesh is a 3D simplicial complex. The axioms guarantee that cells meet only in shared faces — there are no hanging vertices or cracked edges — which is exactly the topological validity that finite-element assembly requires. Two meshes are related by **refinement** if one is obtained from the other by subdividing simplices; a **conforming** refinement keeps the complex a simplicial complex after every subdivision, which is what makes adaptive meshing (Section 22.11) possible.

The half-edge data structure of Chapter 21 stores a 2D simplicial complex with adjacency queries in $O(1)$; the combinatorial machinery of Chapter 30 (Euler characteristic, Betti numbers)

checks that a mesh is a manifold of the intended genus. These validity checks are run as part of mesh generation’s testing phase (Section 22.15).

22.3 Triangular and Tetrahedral Meshes

The two most common mesh types are the **triangular mesh** (2D, or a surface triangulation in 3D) and the **tetrahedral mesh** (3D volumetric). A triangular mesh of a polygonal domain conforms to the domain if every boundary edge of the domain is a union of mesh edges; a tetrahedral mesh of a polyhedron conforms if the boundary triangles exactly cover the domain’s faces. The mesh is **simplicial** if all cells are simplices, which is the case here; quadrilateral and hexahedral meshes require the different machinery of structured and block-structured methods and are not treated in this chapter.

The generation pipeline is the same for both dimensions:

1. **Input:** a domain described by its boundary (a polygon with holes, a CAD boundary representation, or a point cloud with boundary information);
2. **Initial triangulation:** a coarse conforming triangulation of the domain, typically a constrained Delaunay triangulation of the boundary (Section 22.6);
3. **Size field:** a function that prescribes the desired cell size at each point, derived from feature sizes, curvature, or a prescribed grading (Section 22.11);
4. **Refinement:** inserting points until every cell satisfies the size and quality requirements (Sections 22.7–22.9);
5. **Optimization:** smoothing and edge flips to improve quality beyond the guarantee (Section 22.12);
6. **Output:** the mesh in a standard format (e.g., Gmsh’s MSH format, or the half-edge/triangle-index structures of Chapter 21).

The Delaunay triangulation of Section 22.5 is the central building block of the most widely used pipeline, implemented in production meshers such as TetGen [Si15] and Gmsh [GR09].

22.4 Mesh Quality Measures

The **quality** of a triangle or tetrahedron measures how far the cell is from equilateral; the numerical stability of finite-element assembly degrades with poor quality. The standard measures are:

Minimum angle For a triangle, the minimum interior angle. The fundamental bound of this chapter is that every angle is at least 30° (Delaunay refinement with the smallest-angle rule), because the discretization error in the finite-element method is inversely proportional to $\sin(\text{minimal angle})$.

Aspect ratio The ratio of the circumradius to the inradius; 2 for an equilateral triangle and ∞ for a degenerate one. A bound on the aspect ratio is equivalent to a bound on the minimum angle.

Radius–edge ratio The ratio $\rho = R/\ell_{\min}$ of the circumradius R to the shortest edge $\ell_{\min}$. A triangle with ρ bounded by ρ^* has all angles bounded away from 0 and 180° ; this is the quantity that Ruppert’s algorithm controls directly (Section 22.8).

Sliver measure (3D) A tetrahedron whose four vertices are nearly coplanar but whose edges are long; slivers have small volume-to-edge ratios and pass the angle tests that fail triangles. Sliver exudation removes them by vertex perturbation (Section 22.12).

The relationships among measures matter for algorithm design: bounding ρ bounds all angles for triangles, but in 3D no single edge-based ratio bounds the sliver shape, which is why tetrahedral Delaunay refinement has weaker guarantees than the 2D theory. Quality is measured and reported by the mesh-analysis routines of Chapter 30.1, which compute histograms of these measures over the mesh.

22.5 Delaunay Mesh Generation

Delaunay mesh generation builds the mesh as (a constrained) **Delaunay triangulation** of the domain and its inserted points. Recall from Chapter 12 that a triangulation is Delaunay if every triangle’s circumcircle contains no point of the triangulation in its interior (the *empty-circumcircle property*, Section 12.2); equivalently, no edge is *illegal* and flipping illegal edges terminates at the Delaunay triangulation (Section 12.4). The Delaunay triangulation maximizes the minimum angle over all triangulations of the same point set, which makes it the natural starting point for quality mesh generation: it is already the best possible on the given points, and refinement only improves it further.

Mesh generation uses the Delaunay machinery in two ways:

- **As the initial mesh:** the boundary of the domain is triangulated by a constrained Delaunay triangulation (Section 22.6);
- **As the refinement engine:** inserting a new point into a Delaunay triangulation requires locating its enclosing triangle (by the walking methods of Section 8.8 or the history DAG of Section 8.6), splitting the triangle, and legalizing edges by flips. Each insertion is a local operation, so the mesh is updated in expected $O(\log n)$ time.

The theory of Section 12.6 (expected $O(n \log n)$ construction) and Section 12.12 (angle guarantees) therefore transfers directly to mesh generation, and the practical implementations in TetGen [Si15] and Gmsh [GR09] are direct descendants of these algorithms.

22.6 Constrained Delaunay Meshing

A mesh must reproduce the domain’s boundary exactly, but an ordinary Delaunay triangulation of the boundary vertices can contain edges that cross the boundary or lie outside the domain. The **constrained Delaunay triangulation (CDT)** fixes this: a set of constraint edges (the domain boundary and any prescribed internal features such as material interfaces) is forced into the triangulation, and every triangle must have the empty-circumcircle property *relative to the constraints*: its circumcircle contains no vertex that is *visible* from the triangle’s interior, where the line of sight must not cross a constraint. This is the definition developed in Section 12.10.

The CDT of the boundary provides the initial conforming mesh from which refinement proceeds. The constrained edges are the domain’s boundary segments, and the refinement process (Section 22.7) never removes them: points are inserted only in the interior of cells (or on constrained edges in a way that preserves the boundary, by splitting the edge into two constrained pieces). Guaranteeing that the boundary is reproduced *exactly* requires exact arithmetic on the input coordinates; floating-point snap is not acceptable, and the robust predicates of Chapter 3 are used throughout.

For a polyhedron, the 3D constrained Delaunay triangulation is considerably harder than the 2D CDT because the boundary is a surface triangulation and the empty-circumsphere condition

is more delicate; the practical resolution used by TetGen [Si15] is to start from a boundary conforming Delaunay triangulation of the input’s vertices and then enforce the boundary faces, using Shewchuk’s boundary-conforming algorithms [Si15].

22.7 Delaunay Refinement

22.7.1 1. Overview

Delaunay refinement is the process of improving a Delaunay triangulation by inserting a new point at the **circumcenter** of any cell that fails the size or quality requirements, and legalizing the affected cells. The two classical algorithms are Chew’s (Section 22.9) and Ruppert’s (Section 22.8), analyzed in the survey of Shewchuk [She02]; together they show that a Delaunay refinement process that inserts circumcenters of cells whose radius–edge ratio exceeds a threshold terminates and produces a mesh whose minimum angle is bounded by a constant — 30° being the canonical bound.

22.7.2 2. Definitions

Definition 22.2 (Skinny Triangle). A triangle is **skinny** if its radius–edge ratio $\rho = R/\ell_{\min}$ exceeds a threshold ρ^* (equivalently, if some angle is too small). Its **circumcenter** is a candidate insertion point.

22.7.3 3. Theory

The power of circumcenter insertion is a locality argument. Inserting a point at the circumcenter of a skinny triangle subdivides that triangle into three smaller triangles, each with radius–edge ratio at most that of the parent (the circumradius halves while the shortest edge stays bounded below). The danger is that the insertion point lies *outside* the domain (it is a boundary or a split of a constrained edge), in which case the point is inserted on the midpoint of the offending edge instead. The termination and quality proofs control the accumulation of short edges: Ruppert’s algorithm [Rup95] shows that if the threshold is $\rho^* < 2$ and every boundary segment is *encroached* (its diametral circle contains another vertex), then the process terminates and the output mesh has no angle smaller than $\arcsin(\sqrt{3}/(2\rho^*))$ — at least 20.7° for $\rho^* = 2$, and 30° for the variant that refines skinny and encroached cells together.

22.7.4 4. Pseudocode

22.7.5 5. Parameter Selection

- **Threshold ρ^* :** smaller ρ^* gives better angles but more cells; the practical sweet spot is $\rho^* = 2$ (angle bound 20.7°) to $2\sqrt{2}$.
- **Priority order:** processing encroached segments before circumcenters (Chew’s “split the worst first”) is required for the termination proof.
- **Size field:** refinement stops when cells satisfy both the quality bound and the size field of Section 22.11; the two criteria are combined into one priority.

22.7.6 6. Complexity

- **Time:** $O(n \log n)$ expected for a mesh of n cells, since each point insertion is an expected- $O(\log n)$ Delaunay update.
- **Size:** for a domain whose features are resolved, the number of cells is proportional to the size-field integral; the constants depend on the angle bound.

Algorithm 108 Delaunay Refinement (Ruppert–Chew style)**Input:** A constrained Delaunay triangulation of the domain, quality threshold ρ^* .**Output:** A conforming mesh with no triangle of radius–edge ratio $> \rho^*$.

```

1: function DELAUNAYREFINE( $\mathcal{T}$ ,  $\rho^*$ )
2:    $Q \leftarrow$  priority queue of skinny triangles and encroached boundary segments
3:   while  $Q$  is not empty do
4:     pop the highest-priority item
5:     if it is an encroached segment  $e$  then
6:       insert the midpoint of  $e$ ; split  $e$  into two constrained pieces; legalize
7:     else if the circumcenter  $c$  of skinny triangle  $t$  encroaches a segment  $e$  then
8:       insert the midpoint of  $e$  instead of  $c$ ; legalize
9:     else
10:      insert  $c$ ; legalize the affected edges by flips
11:    end if
12:    re-evaluate the quality of all affected cells and update  $Q$ 
13:  end while
14:  return  $\mathcal{T}$ 
15: end function

```

22.7.7 7. Usages

- **Finite-element meshing:** the default engine of production triangular and tetrahedral meshers (Section 22.13).
- **Guaranteed-quality requirements:** applications that must bound the worst element (e.g., FEM error control, computational fluid dynamics meshes) use Delaunay refinement precisely because of its angle guarantee.

22.7.8 8. Testing Methods and Edge Cases

- **Angle check:** after refinement, verify numerically that no angle is below the promised bound, using the quality measures of Section 22.4.
- **Encroachment:** deliberately construct an input where a circumcenter encroaches a boundary segment and verify the midpoint split.
- **Small features:** boundaries with features closer than the requested size must be resolved to their feature size first.

22.7.9 9. References

The refinement algorithms are due to Chew [Che89] and Ruppert [Rup95]; the complete analysis and engineering guidance are in Shewchuk's survey [She02] and thesis [She98], and the Delaunay background in Chapter 12.

22.8 Ruppert's Algorithm**22.8.1 1. Overview**

Ruppert's algorithm [Rup95] is the canonical Delaunay refinement method with a provable angle guarantee. It maintains a constrained Delaunay triangulation and refines it by inserting circumcenters of skinny triangles and midpoints of *encroached* boundary segments, in a priority

order that guarantees termination. The output has all angles at least a fixed bound (about 20.7° for the standard parameter) and respects the domain boundary exactly.

22.8.2 2. Definitions

Definition 22.3 (Encroachment). A boundary segment e is **encroached** if some vertex of the triangulation lies in the interior of the circle with diameter e . Inserting a vertex into the circle with diameter e can make the Delaunay triangulation lose e as an edge, so encroached segments must be split at their midpoint first.

22.8.3 3. Theory

The correctness rests on two lemmas proved in [Rup95, She02]:

- **Termination:** every inserted point is at distance at least $\ell_{\min} \cdot \rho^* / (\rho^* + 1)$ from every existing vertex (a separation bound), so the refinement inserts only finitely many points and the mesh size is bounded in terms of the smallest feature.
- **Quality:** after termination, no triangle has radius–edge ratio exceeding ρ^* ; for $\rho^* = 2$ this implies no angle smaller than $\arcsin(\sqrt{3}/4) \approx 20.7^\circ$. The “smallest angle” refinement variant (inserting the circumcenter with the rule that the new triangle angles are at least 30°) gives the 30° bound.

22.8.4 4. Pseudocode

The algorithm is the specialization of Algorithm 108 with the priority rule “always split an encroached segment before inserting any circumcenter”, i.e., the Chew–Ruppert order in which a skinny triangle is processed only if its circumcenter is not encroaching a segment.

22.8.5 5. Parameter Selection

- **Angle bound:** the angle guarantee improves with the choice of ρ^* ; the theory is cleanest at $\rho^* = 2$.
- **Size field:** Ruppert’s algorithm was originally analyzed without a size field; combining it with the graded size field of Section 22.11 preserves the angle bound and the linear size behavior.

22.8.6 6. Complexity

- **Time:** $O(n \log n)$ expected, dominated by the Delaunay insertions.
- **Size:** $O(n)$ cells, with constants that grow as the angle bound tightens.

22.8.7 7. Usages

- **Guaranteed-quality 2D meshing:** the reference algorithm implemented in Triangle and in the planar meshers of Gmsh [GR09, She02].
- **Conforming boundary meshes:** the exact boundary reproduction makes it the tool for FEM and CFD preprocessing (Section 22.13).

22.8.8 8. Testing Methods and Edge Cases

- **Angle histogram:** assert the promised minimum angle over thousands of random domains.
- **Boundary reproduction:** verify every input boundary segment is a union of output edges.
- **Encroachment regression:** construct domains whose boundary segments are longer than the local feature and verify midpoint splitting.

22.8.9 9. References

Ruppert [Rup95] introduced the algorithm; Shewchuk [She02] presents the complete proof of termination and the angle bounds, and the engineering details [She98].

22.9 Chew's Algorithm

22.9.1 1. Overview

Chew's algorithm [Che89] is the earliest Delaunay refinement method with a quality guarantee. It maintains a constrained Delaunay triangulation and repeatedly splits the **worst** triangle: the triangle with the largest circumradius–shortest-edge ratio is refined by inserting its circumcenter. Chew proved that this process converges to a mesh in which every angle is at least 30° when a careful tie-breaking is used; the algorithm is the prototype of the priority-queue refinement implemented in Algorithm 108.

22.9.2 2. Definitions

Definition 22.4 (Worst Triangle). The **worst triangle** of a triangulation is the triangle whose radius–edge ratio is maximal. Chew's algorithm always refines the worst triangle, resolving ties by a fixed rule (e.g., by edge length).

22.9.3 3. Theory

The worst-first rule is what makes the analysis simple: refining the worst triangle at every step guarantees that no triangle ever becomes worse than the current worst, so the worst ratio decreases monotonically toward the bound. Chew's proof shows the process terminates with no angle below 30° on inputs with well-separated features; Ruppert's refinement (Section 22.8) extended the guarantee to arbitrary inputs by adding the encroachment rule.

22.9.4 4. Pseudocode

The pseudocode is Algorithm 108 restricted to the single rule “pop the worst triangle t ; insert its circumcenter unless it encroaches a segment, in which case split the segment”.

22.9.5 5. Parameter Selection

- **Worst-first priority:** the priority is the radius–edge ratio; the deterministic tie-break matters for reproducibility.
- **Degenerate resolution:** when the circumcenter is ill-defined (nearly collinear cells), refine by the longest edge instead.

22.9.6 6. Complexity

- **Time:** $O(n \log n)$ expected with a heap-based priority queue of $O(n)$ triangles.
- **Size:** linear in the output mesh size.

22.9.7 7. Usages

- **Theoretic basis:** the “worst-first” scheme is the template for guaranteed-quality refinement in two and three dimensions.
- **Implementation default:** many meshers use worst-first with the 30° target as the practical default refinement policy.

22.9.8 8. Testing Methods and Edge Cases

- **Bound verification:** assert the 30° angle bound on structured and unstructured test domains.
- **Infinite loop guard:** verify that the priority queue drains on domains with tiny features (the separation bound of Section 22.8 is what guarantees termination).

22.9.9 9. References

Chew’s original result [Che89] and its placement in the refinement theory by Shewchuk [She02].

22.10 Advancing-Front Methods

Advancing-front methods generate the mesh by growing cells inward from the domain boundary: the boundary is first discretized into a set of front edges, and new triangles (or tetrahedra) are created by connecting front vertices, one cell at a time, with the front advancing into the domain until it collapses. The method is the main non-Delaunay approach and is favored in certain settings:

- **Boundary fidelity:** cells start at the boundary, so the mesh matches the input surface precisely and can align cells with anisotropic directions on the boundary.
- **Anisotropy:** the local cell shapes are controlled directly by the front construction, which suits anisotropic boundary-layer meshes in computational fluid dynamics.
- **No global structure:** no global Delaunay machinery is needed, which simplifies implementations for domains with many disconnected boundary components.

Its weaknesses are equally well known. The quality guarantee is local and empirical rather than provable; the front can “collapse” poorly near internal boundaries; and the element count is not controlled by a size field as cleanly as in Delaunay refinement. In practice the two families are often combined: an advancing-front method generates the boundary layer and the coarse interior, and a Delaunay refinement pass cleans up the interior cells (as in Gmsh’s default pipeline [GR09]). Simpler but robust alternatives for many applications are the distmesh-style physically motivated generators [PS04], which relax an initial triangulation under forces derived from the desired cell size, and the Delaunay refinement methods of Sections 22.7–22.9.

22.11 Adaptive Meshing

Adaptive meshing generates a mesh whose cell size varies over the domain according to a **size field**: small cells where the solution (or the geometry) changes rapidly, large cells where it is smooth. The size field is prescribed either a priori from geometric features (curvature, thin regions, feature edges) or a posteriori from a computed error estimator of the numerical solution — the loop that iterates “solve, estimate error, adapt mesh, solve again” is the **adaptive finite-element method**.

The two standard adaptation mechanisms are:

***h*-refinement** Subdividing cells where the size field demands smaller cells; the subdivision must be **conforming** (the result remains a simplicial complex, Section 22.2), which is enforced by propagating subdivisions across faces and, for tetrahedra, by the standard (newest-vertex or longest-edge) refinement templates.

Regeneration Rebuilding the mesh with the new size field, which is how the Delaunay refinement methods of Sections 22.7–22.9 adapt: the size field enters as a second stopping criterion, and the insertion of circumcenters proceeds until every cell satisfies both the quality and the size bounds.

The efficiency guarantee of adaptive Delaunay refinement is that the final mesh size is bounded by the size-field integral (a consequence of the separation bound), so the mesh does not overshoot the requested sizes even under rapid grading. Verifying that the computed mesh respects the size field is part of the testing protocol of Section 22.15.

22.12 Mesh Optimization

The refinement methods of Sections 22.7–22.9 guarantee a worst-case angle bound but not the best achievable quality. **Mesh optimization** improves the mesh beyond the guarantee by local operations that preserve the conforming boundary:

Edge flips In 2D, an interior edge shared by two triangles is flipped to the other diagonal when the flip increases the minimum angle (the Delaunay legality test of Section 12.4); repeatedly flipping illegal edges converges to the Delaunay triangulation.

Laplacian smoothing Each interior vertex is moved to the average of its neighbors’ positions; this usually improves the angles, but it can invert cells, so the move is accepted only if it improves a quality measure (constrained smoothing).

Optimization-based smoothing Each vertex is moved to the position maximizing the minimum angle (or minimizing the maximum radius–edge ratio) over its incident cells, solved by a small local nonlinear optimization; this is the strongest local smoother.

Sliver exudation (3D) Slivers (Section 22.4) are removed by perturbing their vertices slightly, keeping the Delaunay condition while destroying the near-coplanarity; sliver exudation is what turns a merely angle-bounded tetrahedral mesh into a numerically acceptable one [Si15].

Optimization is a postprocess: its moves must never violate the domain boundary (boundary vertices are fixed), must never invert a cell, and must be idempotent for reproducibility. The quality measures of Section 22.4 and the validity checks of Chapter 30.1 are the acceptance criteria.

22.13 Finite-Element Applications

The meshes built in this chapter are the discretization layer of the **finite-element method** (FEM) and the finite-volume and finite-difference methods. The FEM on a triangular mesh approximates a PDE solution by continuous piecewise-linear functions on the triangles; the accuracy of the approximation depends on the mesh through two mechanisms:

- the **interpolation error** is bounded in terms of the circumradius of each cell and the second derivatives of the true solution, so small cells (the size field) control accuracy;
- the **stability** is controlled by the minimal angle, which appears in the constants of the error bounds and in the condition number of the stiffness matrix, so the angle guarantee of Section 22.7 controls robustness.

This is the precise reason mesh generation and mesh analysis (Chapter 30.1) are inseparable: the quality measures of Section 22.4 are exactly the quantities that FEM analysis cites, and the adaptive loop of Section 22.11 drives the accuracy of industrial simulations. The production meshers used in FEM workflows — Gmsh [GR09] for mesh generation and visualization, and TetGen [Si15] for guaranteed-quality tetrahedral meshes — implement the algorithms of this chapter, and their interfaces are the de facto standards for exchanging meshes with solvers.

22.14 Exercises

1. Prove that the minimum angle of a triangle is a decreasing function of its radius–edge ratio, and give the exact relation.
2. Show that inserting the circumcenter of a triangle subdivides it into three triangles whose radius–edge ratios are at most the parent’s.
3. Define encroachment and prove that inserting a point strictly inside the diametral circle of a segment can destroy the segment’s presence in the Delaunay triangulation.
4. Simulate Ruppert’s algorithm by hand on a right triangle whose hypotenuse is a boundary segment longer than the other edges, and verify the termination rule.
5. Compare the angle guarantees of Ruppert’s and Chew’s algorithms and state which parameter choices produce a 30° bound.
6. Explain why the 3D Delaunay refinement analysis fails for slivers and how sliver exudation repairs the problem.
7. Implement a conforming adaptive h -refinement for a 2D mesh with a prescribed size field, and verify the conforming property (no hanging vertices) after each step.
8. Run a mesh-quality histogram (Chapter 30.1) on the output of Algorithm 108 for a random polygon and confirm the angle bound numerically.

22.15 Project: Adaptive Mesh Generator

Build an adaptive triangular mesh generator for planar domains:

1. **Input:** read a simple polygon with holes and a size field (a function of position, e.g., $h(x, y)$ small near features).
2. **Initial mesh:** compute the constrained Delaunay triangulation of the boundary (Section 22.6).

3. **Refinement:** implement Algorithm 108 with a priority queue over radius–edge ratio and size violation, inserting circumcenters and splitting encroached boundary segments.
4. **Quality controls:** verify every angle against the promised bound and every edge length against the size field, and report the histograms using the quality measures of Section 22.4.
5. **Optimization:** add constrained Laplacian smoothing (Section 22.12) and measure the quality improvement.
6. **Verification:** for a known domain (square, annulus), check the cell count against the size-field integral and check the mesh with the validity routines of Chapter 30.1.
7. **Visualization:** draw the mesh, color cells by radius–edge ratio, and animate the refinement steps.

The project mirrors the pipeline of Gmsh and TetGen at small scale and exercises the Delaunay, point-location, and mesh-analysis machinery of the corresponding chapters.

Chapter 23

Voxel Geometry and OpenVDB

Voxel geometry represents three-dimensional shape on a regular grid of volume elements, or voxels. Where a mesh stores geometry on its boundary, a voxel representation samples the volume itself, making topological queries such as inside/outside tests, boolean operations, and morphological changes trivial and robust. The cost is memory: a dense uniform grid grows as the cube of its resolution. Voxel geometry became practical at high resolution through *sparse* representations, most prominently the OpenVDB library and its hierarchical tree of fixed-size blocks [Mus13]. This chapter develops the dense and sparse representations, signed distance fields, and the conversions between voxel and mesh form. Mesh voxelization itself is treated in Section 20.4.

23.1 Voxel Grids

23.1.1 1. Overview

A **voxel grid** is the 3D analogue of a pixel image: a rectangular array of cubic cells aligned to an axis-aligned bounding box. Each cell either records a binary occupancy (occupied or empty) or stores a scalar, vector, or other attribute sampled at the cell center. Grids support constant-time random access and simple neighbor iteration, which makes them the substrate for most volumetric algorithms [AM01, NT03].

23.1.2 2. Definitions

Definition 23.1 (Voxel). A *voxel* is a volume element: the cubic cell centered at a grid point. The grid resolution (n_x, n_y, n_z) and the voxel size s determine the box that the grid represents.

Definition 23.2 (Binary and Attribute Grids). A *binary grid* stores occupancy per voxel (0 empty, 1 occupied). An *attribute grid* stores values such as a signed distance, density, or velocity field.

Definition 23.3 (Conservative Voxelization). A voxelization is *conservative* if a voxel is occupied whenever any part of the surface passes through it; otherwise it is *exact* (a voxel is occupied only when it is pierced by the surface) [NT03].

23.1.3 3. Theory

A dense grid of resolution n^3 costs n^3 entries and thus $O(n^3)$ bytes. The occupied surface of a triangle mesh passes through only $O(n^2)$ voxels, so for high resolution almost all storage is wasted on empty space. This observation motivates the sparse schemes of Section 23.3 and the hierarchical trees of Section 23.4. The resolution must balance aliasing (too coarse) against

memory and computation time (too fine); adaptive sampling that keeps dense cells near the surface and coarsens empty regions is the resolution used in production systems [Mus13].

23.1.4 4. Pseudo code

```
function IsINSIDE( $p$ , grid, occ)
     $c \leftarrow \text{ToCell}(p, \text{grid})$ 
    return occ[ $c$ ]
end function
```

23.1.5 5. Parameters Selections

- **Resolution:** choose n so that the smallest surface feature spans several voxels; alias errors appear below 3 voxels per feature.
- **Bounds:** an axis-aligned bounding box must contain the whole object; padding avoids boundary artifacts.
- **Data type:** one bit per voxel for occupancy, 8–32 bits per voxel for scalar attributes.

23.1.6 6. Complexity

Dense grids: $O(n^3)$ space, $O(1)$ access, $O(n^3)$ build. A surface of $O(n^2)$ cells can be voxelized in $O(n^2)$ time plus $O(n^3)$ for the grid itself.

23.1.7 7. Usages

- **Physics simulation:** grid-based fluids, collision and pressure fields.
- **Medical imaging:** CT and MRI produce data natively as grids.
- **Boolean and morphological operations:** union, intersection, erosion, and dilation are voxel-wise.

23.1.8 8. Testing methods and Edge cases

- **Surface vs. solid:** verify whether boundary voxels are classified as inside.
- **Resolution sensitivity:** re-run the pipeline at two resolutions and compare the extracted surfaces.
- **Boundary alignment:** voxels exactly on the box boundary must be handled consistently.

23.1.9 9. References

- Akenine-Möller, T. (2001). “Fast 3D triangle-box overlap testing”. *Journal of Graphics Tools*.
- Nooruddin, F. S., & Turk, G. (2003). “Simplification and repair of polygonal models using volumetric techniques”. *IEEE TVCG*.
- Wikipedia: [Voxel](#)

23.2 Signed Distance Fields

23.2.1 1. Overview

A **signed distance field** (SDF) stores, at every voxel, the signed distance to the nearest surface, negative inside and positive outside. The zero set of the field is the surface itself, so blending, boolean operations, and collision queries reduce to arithmetic on the sampled field [OF03].

23.2.2 2. Definitions

Definition 23.4 (Signed Distance Field). A *signed distance field* is a function $d : \mathbb{R}^3 \rightarrow \mathbb{R}$ with $|d(x)| = \text{dist}(x, \Gamma)$, where Γ is the surface, and $\text{sgn}(d)$ indicates the side of Γ .

Definition 23.5 (Narrow Band). A *narrow band* stores d only within w voxels of Γ (typically $w = 3$); outside the band the field is clamped to $\pm w s$. This bounds the stored volume to the neighborhood of the surface [OF03, Mus13].

23.2.3 3. Theory

An SDF is usually constructed either by computing the distance to a triangle mesh with a fast sweeping or fast marching method, or by reinitializing an arbitrary level-set function so that $|\nabla d| = 1$. Once it is a true signed distance, operations are exact up to the grid resolution: the surface is extracted as the zero level set, and distance queries are direct voxel lookups. Level-set evolution updates d with a velocity field, moving the surface while preserving its implicit character [OF03].

23.2.4 4. Pseudo code

```

function BUILDSDF( $\Gamma$ , grid,  $w$ )
    initialize voxels near  $\Gamma$  with exact distances
    propagate distances outward with fast sweeping
    clamp all values to  $[-w s, +w s]$ 
    return  $d$ 
end function

```

23.2.5 5. Parameters Selections

- **Band width** w : $w = 3$ is the standard OpenVDB default; larger bands smooth boolean operations at higher cost.
- **Renormalization**: enforce $|\nabla d| = 1$ periodically so the field stays a true distance.

23.2.6 6. Complexity

Fast sweeping computes the distance to a grid of N voxels in $O(N)$ passes of $O(N)$ work; reinitialization of the narrow band is $O(w n^2)$ per sweep for an n^3 grid.

23.2.7 7. Usages

- **Boolean operations**: min / max of SDFs implement union/intersection.
- **Collision and proximity**: distance queries become lookup + clamp.
- **Level-set surface evolution** for sculpting and fluid boundaries [OF03].

23.2.8 8. Testing methods and Edge cases

- **Accuracy:** compare against analytic distance for a sphere.
- **Sign consistency:** verify the inside/outside convention matches the mesh orientation.
- **Thin features:** features thinner than $2s$ may vanish from the field.

23.2.9 9. References

- Osher, S., & Fedkiw, R. (2003). “Level Set Methods and Dynamic Implicit Surfaces”. Springer.
- Wikipedia: [Signed distance function](#)

23.3 Sparse Voxel Representations

23.3.1 1. Overview

Sparse voxel representations store only occupied cells, typically in a **sparse voxel octree** (SVO): a tree that subdivides the box into octants, recursing only where occupied cells exist. Hierarchical sparse volumes such as OpenVDB push the idea further, storing dense fixed-size blocks at the leaves and summarized tiles in the interior [Mus13].

23.3.2 2. Definitions

Definition 23.6 (Sparse Voxel Octree). An *SVO* is an octree whose nodes store the geometry and material of the cells they represent; nodes whose subtree is empty are pruned. Resolution increases only where the object actually exists.

Definition 23.7 (Tile). A *tile* is an internal node value that summarizes a uniform region of the volume (for example a constant SDF value) without expanding it into children. Tiles are the key to storing large homogeneous regions cheaply [Mus13].

23.3.3 3. Theory

An SVO with n occupied voxels uses $O(n \log n)$ nodes and supports ray-marching and level-of-detail rendering directly on the tree. OpenVDB’s tree generalizes the octree to arbitrary branching and depth (up to five levels) and adds *active* states: every node or tile is either active (stores topology) or inactive (implicitly background), so constant regions are represented by single tiles instead of millions of voxels [Mus13].

23.3.4 4. Pseudo code

```
function INSERT(tree,  $c$ , value)
    descend to the leaf containing cell  $c$ , expanding empty nodes
    set the leaf voxel value and mark the voxel active
    return tree
end function
```

23.3.5 5. Parameters Selections

- **Branching factor and depth:** octrees use 8; OpenVDB uses 16-child levels with 8^3 leaves.
- **Tile threshold:** collapse a subtree into a tile when all children share a value.

23.3.6 6. Complexity

Insertion and lookup descend $O(\log n)$ levels (bounded by the fixed tree depth in OpenVDB). Memory is proportional to the number of active voxels and tiles, not the bounding-box volume.

23.3.7 7. Usages

- **High-resolution volume rendering:** SVO ray tracing of gigavoxel scenes.
- **Production VFX:** OpenVDB effects in film pipelines [Mus13].

23.3.8 8. Testing methods and Edge cases

- **Deep trees:** verify behavior at the maximum supported depth.
- **Empty regions:** inserting into a fully empty tree must create only the minimal path.
- **Tile/leaf transitions:** adjacent active and inactive regions must share correct boundaries.

23.3.9 9. References

- Museth, K. (2013). “VDB: High-resolution sparse volumes with dynamic topology”. ACM Transactions on Graphics.
- Wikipedia: [Sparse voxel octree](#)

23.4 OpenVDB: Hierarchical Sparse Volumes

23.4.1 1. Overview

OpenVDB is an open-source C++ library, originated at DreamWorks and now under the Academy Software Foundation, that implements a hierarchical, sparse, dynamic topology data structure for volumetric data [Mus13]. It is the de facto standard for production voxel geometry, supporting narrow-band signed distance fields, boolean operations, smoothing, morphing, and level-set tracking at billion-voxel effective resolutions.

23.4.2 2. Definitions

Definition 23.8 (VDB Tree). A *VDB tree* has at most five levels: a root node with a hash-mapped set of children, up to three internal levels, and leaf nodes. Leaves store dense $8 \times 8 \times 8$ blocks; internal nodes store the states of their children as fixed-size arrays, and may hold a constant value (a tile) instead of children [Mus13].

Definition 23.9 (Active State). A node, tile, or voxel is *active* if its value is explicitly set and participates in topology; inactive elements are implicitly equal to the background value. All tree operations respect and maintain these states.

23.4.3 3. Theory

Each level of the tree multiplies the coordinate space by 16 in each dimension, so five levels span coordinates of 2^{19} per axis (about 524,288 voxels) while a leaf stores only 512 voxels. Because interior levels can be represented by tiles, a large flat region costs the same as a single voxel. Random access descends at most four pointer hops, and topology is dynamic: voxels can be inserted or deleted in expected $O(1)$ amortized time through the hash-mapped root, which makes level-set evolution efficient even as the interface moves [Mus13].

23.4.4 4. Pseudo code

```
function GETVALUE(tree,  $x$ )  
   $n \leftarrow \text{root}(\text{tree})$   
  for level  $L = 1 \dots 5$  do  
     $i \leftarrow \text{coord2index}_L(x)$   
    if  $n$  holds a tile then  
      return tile value  
    else if  $n[i]$  is a child then  
       $n \leftarrow n[i]$   
    else  
      return background value  
    end if  
  end for  
  return leaf voxel value at  $x$   
end function
```

23.4.5 5. Parameters Selections

- **Background value:** choose the "empty" value (e.g., $\pm\infty$ for SDF grids) consistently; it determines inactive semantics.
- **Tree configuration:** leaf log-dimension 3 and internal log-dimension 4 are defaults; larger leaf blocks trade memory for iteration speed.
- **Narrow-band mode:** for SDF grids, band width 3 is standard [Mus13].

23.4.6 6. Complexity

Access is $O(1)$ (at most five level hops, hash lookup at the root). Memory is proportional to active voxels and tiles; a 3D SDF of effective resolution 4096^3 typically requires only a few megabytes.

23.4.7 7. Usages

- **Visual effects:** level-set sculpting, liquid surfaces, and volumetric fire/smoke in film [Mus13].
- **Robotics and simulation:** signed distance collision and meshing pipelines.
- **Medical and geometry processing:** fast boolean operations on scanned geometry.

23.4.8 8. Testing methods and Edge cases

- **Bounding:** writing outside the current tree extent must grow the root's hash map correctly.
- **Tile collapse:** coalescing a region must not corrupt neighboring active voxels.
- **Level-set reinitialization:** verify the band stays $|\nabla d| = 1$ after many operations.

23.4.9 9. References

- Museth, K. (2013). "VDB: High-resolution sparse volumes with dynamic topology". ACM Transactions on Graphics.
- OpenVDB: <https://www.openvdb.org>
- Wikipedia: [OpenVDB](#)

23.5 Voxel-to-Mesh Conversion

23.5.1 1. Overview

Converting a voxel field back into a mesh recovers an explicit surface from an implicit or occupancy representation. The standard algorithm is **marching cubes**, which extracts the zero level set of an SDF (or the boundary of an occupancy grid) by triangulating each cell independently [LC87].

23.5.2 2. Definitions

Definition 23.10 (Marching Cubes). *Marching cubes* enumerates, for every grid cell, the 2^8 cases determined by which of the eight corner values are below the isovalue, and emits the corresponding polygon from a precomputed case table.

Definition 23.11 (Surface Nets). *Surface nets* place one vertex per crossing cell and connect adjacent vertices into a quad mesh, relaxing the vertices toward the surface; the result avoids some of marching cubes' topology artifacts [Gib98].

23.5.3 3. Theory

Marching cubes produces watertight surfaces for well-sampled SDFs but has ambiguous cases (15 base cases, 256 with symmetry and inversion) that must be resolved consistently to avoid cracks. Because the surface is the zero set of a sampled field, the mesh resolution is bounded by the grid resolution, and features thinner than one voxel are lost. Surface nets produce smoother results with fewer triangles at the same resolution [Gib98].

23.5.4 4. Pseudo code

```

function MARCHINGCUBES( $d, \rho$ )
   $M \leftarrow$  empty mesh
  for each cell with corners  $c_0 \dots c_7$  do
    case  $\leftarrow \sum_i 2^i [d(c_i) < \rho]$ 
    emit triangles of case from the lookup table
  end for
  return  $M$ 
end function

```

23.5.5 5. Parameters Selections

- **Iso-value ρ :** 0 for a signed distance field; 0.5 for a binary grid.
- **Vertex placement:** linear interpolation on edges vs. central placement in the cell.
- **Ambiguity resolution:** use the asymptotic-decider criterion for consistent topology.

23.5.6 6. Complexity

Marching cubes runs in $O(N)$ time for N cells and outputs $O(N)$ triangles; output is proportional to the surface area, not the volume.

23.5.7 7. Usages

- **Surface extraction** from CT/MRI scans.
- **Mesh generation** from SDF sculpting and level-set simulations.
- **Mesh repair**: voxelize a broken mesh, then extract a clean, watertight one.

23.5.8 8. Testing methods and Edge cases

- **Watertightness**: verify no cracks at cell boundaries (ambiguity cases).
- **Normals**: orient triangles consistently from the field gradient.
- **Empty grids**: fields with no sign change must produce no triangles.

23.5.9 9. References

- Lorensen, W. E., & Cline, H. E. (1987). “Marching cubes: A high resolution 3D surface construction algorithm”. Computer Graphics (SIGGRAPH).
- Gibson, S. F. F. (1998). “Constrained elastic surface nets”. MICCAI.
- Wikipedia: [Marching cubes](#)

23.6 Applications

- **Visual effects**: volumetric fluids, explosions, and organic sculpting in production using OpenVDB [Mus13].
- **Autonomous driving and robotics**: occupancy grids and signed-distance collision maps for perception and planning.
- **Medical imaging**: segmentation, surface extraction, and morphometry from volumetric scans [LC87].
- **Computational fabrication**: voxel models drive 3D printing and material assignment.

Part VIII

Motion, Visibility, and Robotics

Chapter 24

Visibility

24.1 Visibility Problems

The geometric notion of **visibility** formalizes the everyday question “what can be seen from here?”: two points see each other when the straight segment joining them is not blocked by intervening obstacles. Visibility is at the heart of a surprising fraction of computational geometry. Guarding an art gallery, rendering a scene, guiding a robot, and finding the shortest route around a building all reduce, in one way or another, to questions about which points see which other points.

Definition 24.1 (Mutual Visibility). Let $\mathcal{O}$ be a finite set of pairwise-disjoint closed polygonal obstacles, or a simple polygon P regarded as the only obstacle (so the free space is P itself). Two points p and q that do not lie in the interior of an obstacle are **mutually visible** if the open segment between them does not intersect the interior of any obstacle; for a polygon, this is $\overline{pq} \subseteq P$.

Visibility comes in several strengths, because the objects whose visibility is of interest are usually segments, edges, or regions rather than bare points.

Definition 24.2 (Point, Edge, and Strong Visibility). A point p **sees** a segment e if it sees at least one point of e . The point p **strongly** (or **fully**) sees e if it sees every point of e . Two segments e_1 and e_2 are mutually visible if some point of e_1 sees some point of e_2 , and strongly visible if every point of e_1 sees every point of e_2 .

Guarding uses point visibility (every polygon point must be seen by a guard), rendering needs the first surface hit along each viewing ray, and shortest paths use edge visibility between obstacle vertices. The chapter develops these in turn:

- **Visibility inside polygons** (Section 24.2): the visibility relation within a simple polygon and its kernel;
- **Visibility polygons** (Sections 24.3–24.4): the region seen from a point and the algorithms that compute it;
- **Guarding** (Section 24.5): how many points suffice to see all of a polygon (the art gallery theorem);
- **Visibility graphs** (Section 24.6): the graph of mutually visible obstacle vertices, its complexity, and its computation;
- **Shortest and geodesic paths** (Sections 24.7–24.8): visible-edge chains in the plane and inside polygons;

- **Applications** (Section 24.9): hidden surface removal and visibility roadmaps.

The chapter connects to the rest of the book: the art gallery theorem is developed fully in Chapter 25, visibility roadmaps are a basic tool of Chapter 26, the rendering algorithms reuse the machinery of Chapter 35, the envelopes of Chapter 14 describe a surface’s silhouette from a viewpoint, and the shortest-path results sit beside those of Chapter 24.11.

24.2 Visibility inside Polygons

Fix a simple polygon P with n vertices, its boundary oriented counterclockwise. Inside P , visibility is the relation of Definition 24.1: two points are visible when their connecting segment lies in P . The relation is reflexive and symmetric but *not* transitive — a point around a corner may see two points that do not see each other — which is why visibility is geometrically interesting and algorithmically nontrivial.

A central structural object is the set of points from which *everything* is visible.

Definition 24.3 (Kernel). The **kernel** $\ker(P)$ of a simple polygon P is the set of points of P that see every point of P . A polygon whose kernel is nonempty is called **star-shaped**.

Theorem 24.4 (Kernel Structure). *The kernel of a simple polygon P is the intersection of the closed half-planes to the left of every directed edge of P (edges oriented counterclockwise). It can be computed in $O(n)$ time, and P is star-shaped if and only if its kernel is nonempty.*

Proof. A point p sees an edge e of P if and only if p lies in the half-plane determined by e that is on the interior side of P ; if p saw the interior side of every edge it would see the entire polygon, since P is the intersection of those half-planes locally along the boundary. The kernel is therefore the intersection of the n interior half-planes. Intersecting half-planes in the plane takes $O(n \log n)$ time in general; for a polygon the linear-time kernel algorithm of Lee and Preparata [LP79] exploits the cyclic order of the edges and runs in $O(n)$. $\square$

The kernel is the “point-visibility” extremum: it is the largest set of guards of size one, and star-shaped polygons are exactly the polygons that a single guard can cover. The art gallery theorem of Section 24.5 asks how many points are needed when one is not enough.

Between the kernel (one point) and all points lies the vertex-pair relation: two vertices of P are visible if the segment between them is a **diagonal** of P , lying inside P and meeting the boundary only at its endpoints. Diagonals are the edges of every triangulation, the backbone of visibility computation, because visibility questions inside a polygon can be answered by walking across its triangles.

Theorem 24.5 (Triangulation). *Every simple polygon with n vertices has a triangulation with exactly $n - 2$ triangles and $n - 3$ diagonals. Such a triangulation can be computed in $O(n \log n)$ time by the monotone-decomposition method (Chapter 7) and in linear time by Chazelle’s algorithm [Cha91].*

Testing whether two vertices are mutually visible reduces to checking that the segment between them is a diagonal: it must not cross any boundary edge (the segment-intersection machinery of Chapter 5) and must locally lie inside P at both endpoints, a test using the orientation predicates of Chapter 3. A simple $O(n)$ -per-pair check follows, so the visibility relation on the vertices is computable in $O(n^2)$ after triangulation.

24.3 Visibility Polygons

Definition 24.6 (Visibility Polygon). Let p be a point of a simple polygon P (or a point in the free space bounded by a set of obstacles). The **visibility polygon** of p , written $VP(p)$, is the set of points q that p sees:

$$VP(p) = \{q \in P : \overline{pq} \subseteq P\}.$$

The visibility polygon is the exact region a viewer sees and a guard covers; it is the query object underlying the guarding and navigation problems of the next sections, and its structure is highly constrained.

Theorem 24.7 (Structure of the Visibility Polygon). *Let P be a simple polygon and $p \in P$ a point. Then $VP(p)$ is a simple polygon, star-shaped with p in its kernel. Its boundary consists of two kinds of edges:*

- **boundary pieces:** maximal portions of edges of P that are visible from p ;
- **windows:** segments not lying on the boundary of P .

Every window is an open segment whose inner endpoint is a reflex vertex v of P , which lies on the ray from p through v , and which extends from v to the first point of the boundary of P met by that ray.

Proof. The set $VP(p)$ is star-shaped by definition: for every $q \in VP(p)$ the segment $\overline{pq}$ is contained in $VP(p)$, so p lies in its kernel. The boundary of the region seen from p consists of points at which the segment from p just grazes an obstacle. Along an edge of P the visible part is a contiguous interval, since the boundary between two visible points of one edge is straight and unblocked. Where the visible boundary leaves ∂P , it does so at a reflex vertex v (at a convex vertex the ray from p continues to hug the boundary); the departing edge lies on the ray from p through v and extends until that ray hits ∂P again, beyond which $\overline{pq}$ is blocked. These are exactly the window edges of the statement. $\square$

Remark 24.8. If the obstacles are polygons with holes, or arbitrary disjoint segments, then $VP(p)$ remains a star-shaped (hence simply connected) region: every visible point is joined to p by a segment inside the region. The boundary now includes windows through reflex vertices of the obstacle polygons, and also segments joining the boundary of a hole to itself, but the star-shaped structure and the window characterization persist.

Example 24.9 (Visibility in a Staircase Polygon). In a rectangular “staircase” of two steps, place the viewer p at the bottom-left corner of the lower step. The visibility polygon contains the two horizontal boundary segments directly ahead, the vertical walls opposite, and the portion of the upper step’s floor not occluded by the lower step’s wall. The occlusion ends at a window: a segment through the reflex corner atop that wall, running along the ray from p through that corner until it hits the far wall, beyond which the upper floor is visible again — the prototypical “see around a corner.”

24.4 Computing Visibility from a Point

24.4.1 1. Overview

Two algorithms dominate the computation of $VP(p)$:

- **Lee’s algorithm** [Lee83] computes $VP(p)$ for a simple polygon in $O(n)$ time by processing the vertices in the rotational order induced by the polygon and maintaining a stack of currently visible vertices; it is the historical first optimal algorithm.

- **The rotational sweep** sweeps a ray around p and records the closest obstacle along each direction. It runs in $O(n \log n)$ time and, unlike Lee's algorithm, generalizes to polygons with holes and to arbitrary sets of disjoint segments.

Guibas, Hershberger, Leven, Sharir, and Tarjan [GHL⁺87] showed that after triangulation the visibility polygons of *all* vertices of a simple polygon can be computed in $O(n)$ total time, and Laszlo [Las96] gives an implementer-oriented account. This section presents the rotational sweep and how Lee's algorithm sharpens it to linear time for simple polygons.

24.4.2 2. Definitions

Definition 24.10 (Radial Order). The **radial order** of a set of points around p is their cyclic order by polar angle around p . Two points at the same angle are ordered by distance from p .

Definition 24.11 (Front). During a sweep of a ray rotating about p , the **front** is the obstacle segment that intersects the ray at the smallest distance from p , i.e., the segment currently occluding the view along the ray. The front changes only at **events**, the angles at which the ray passes an endpoint of an obstacle segment.

The window edges of Theorem 24.7 are the records of front changes: when the front jumps discontinuously from one segment to another at a reflex vertex, the jump is a window.

24.4.3 3. Theory

The rotational sweep is correct because visibility is radial: q is visible from p if and only if no obstacle lies strictly between p and q on the ray through q , so in every angular direction the boundary of $VP(p)$ is the closest obstacle point. Between consecutive event angles the set of segments crossed by the ray is fixed, so the visible boundary is one straight piece of the front, clipped to that interval. At an event the front may change; if the hit point jumps from a to b , the segment $\overline{ab}$ is a window (through a reflex vertex, by Theorem 24.7). Sweeping all $O(n)$ events and emitting the visible pieces and windows therefore yields exactly $VP(p)$.

Lee's linear-time algorithm [Lee83] removes the sorting step using two structural facts about simple polygons: (i) the radial order of the vertices around an interior point is obtainable in linear time without sorting, because the polygon's edges restrict how that order can change; and (ii) each polygon edge contributes only a bounded number of changes to the stack of visible vertices. Processing vertices in that order with a stack reproduces the sweep's output in $O(n)$ total time: the correctness transfers from the sweep, and only the bound is sharpened.

24.4.4 4. Pseudocode

24.4.5 5. Parameter Selection

- **General position:** the sweep assumes no two events coincide and no event lies exactly at the sweep start angle. Coincident angles are ordered by distance; the case of the ray passing exactly through a vertex is resolved by a fixed convention (treat the vertex as reached by the ray, then reprocessed after updating the front).
- **Tie-breaking:** when several vertices share an angle, they must be processed in distance order from p ; the closest reflex vertex is the one that can create a window.
- **Robust predicates:** all front updates and hit-point computations are orientation and intersection tests; they must use the exact predicates of Chapter 3 to avoid inconsistent front states.

Algorithm 109 Rotational Sweep for the Visibility Polygon

Input: A point p and the edges of a simple polygon (or of disjoint obstacle polygons), in general position.

Output: The visibility polygon $VP(p)$ as a cyclic sequence of boundary points.

```

1: function VP_SWEEP( $p$ , edges of the scene)
2:    $E \leftarrow$  all edge endpoints sorted by angle around  $p$  (ties by distance)
3:    $\ell \leftarrow$  the front segment at the initial sweep angle
4:    $B \leftarrow \emptyset$   $\triangleright$  visible boundary, emitted in angular order
5:   for event vertex  $v \in E$ , in angular order do
6:      $\theta \leftarrow$  angle of  $v$ 
7:      $a \leftarrow$  hit point of the front  $\ell$  with the ray at angle  $\theta$ 
8:     update the front: insert the segment(s) incident to  $v$ , delete the segment whose
       angular interval ended
9:      $b \leftarrow$  hit point of the updated front with the ray at angle  $\theta$ 
10:    if  $a \neq b$  then
11:      emit the window  $\overline{ab}$  (passing through a reflex vertex)
12:    end if
13:    emit the visible piece of the front between angle  $\theta$  and the next event
14:  end for
15:  emit the final front piece and close the polygon
16:  return  $B$ 
17: end function

```

24.4.6 6. Complexity

- **Time:** $O(n \log n)$ for the rotational sweep, dominated by the radial sort of n endpoints; $O(n)$ for Lee's algorithm on a simple polygon [Lee83].
- **Space:** $O(n)$ for both algorithms (the sorted event list and the output polygon).
- **Lower bound:** $\Omega(n)$ trivially, since the output may have $\Theta(n)$ vertices, so Lee's algorithm is optimal for simple polygons.

24.4.7 7. Usages

- **Guarding:** $VP(p)$ is the region covered by a guard at p (Section 24.5).
- **Visibility graph:** computing the visibility graph of a polygon's vertices reduces to computing $VP(v)$ for every vertex v (Section 24.6); the vertices visible from v are exactly the polygon vertices on the boundary of $VP(v)$.
- **Sensor models:** in robotics and games, the visibility polygon is the exact field of view of a sensor or camera, used for coverage, surveillance, and line-of-sight queries.

24.4.8 8. Testing Methods and Edge Cases

- **Brute-force reference:** sample points densely in a bounding box and compare against a brute-force visible test; every sampled visible point must lie in the computed $VP(p)$ and conversely.
- **Star-shaped check:** verify that p lies in the kernel of the computed polygon (Section 24.2) and that no two edges of the output cross.

- **Window audit:** every non-boundary edge of the output must be a segment through a reflex vertex of P on the ray from p (Theorem 24.7); the number of windows agrees with the number of such reflex vertices up to $O(1)$.
- **Degenerate inputs:** points on the boundary of P (where $VP(p)$ degenerates to a wedge), collinear vertices, and nested holes must be exercised.

24.4.9 9. References

Lee’s linear-time algorithm is the classic reference [Lee83]; the linear-time treatment of visibility and shortest paths from a triangulation, together with the radial-order lemma for simple polygons, is in [GHL⁺87]. Textbook presentations appear in [O’R87, BCKO08, Las96], and the robust implementation issues follow Chapter 3.

24.5 Guarding Polygons

The **art gallery problem** asks: how many points (guards) inside a polygon are needed so that every point of the polygon is seen by at least one of them? A guard at p covers exactly $VP(p)$ (Section 24.3), so the question is a covering problem for the family of visibility polygons.

Theorem 24.12 (Art Gallery Theorem). *Every simple polygon with n vertices can be guarded by at most $\lfloor n/3 \rfloor$ guards, and this bound is tight: for every n there are polygons that cannot be guarded by fewer than $\lfloor n/3 \rfloor$ guards.*

Proof. The upper bound is Fisk’s elegant proof [Fis78]. Triangulate the polygon (Theorem 24.5). The triangulation graph is 3-colorable: its dual graph is a tree, so one can 3-color the vertices by walking the dual tree, each new triangle forced to receive the third color on its new vertex. Of the three color classes, one contains at most $\lfloor n/3 \rfloor$ vertices. Place guards at the vertices of that class. Every triangle has exactly one vertex of each color, so every triangle contains a guard at one of its vertices; since a triangle is convex, that guard sees the entire triangle, and every point of the polygon lies in some triangle. The polygon is therefore guarded by at most $\lfloor n/3 \rfloor$ guards. $\square$

The lower bound is a **comb polygon**: a long thin rectangle with $\lfloor n/3 \rfloor$ triangular “teeth” cut into one side so that the teeth are separated by narrow necks. A single guard can see points in at most one tooth plus the adjacent corridor, so at least one guard per tooth is required; each tooth costs three vertices, giving the $\lfloor n/3 \rfloor$ lower bound. The theorem is Chvatal’s [Chv75]; the full history and the algorithmic extensions are in Chapter 25.

Remark 24.13. Variants change the picture. **Vertex guards** (restricted to vertices) satisfy the same $\lfloor n/3 \rfloor$ bound, since Fisk’s proof already produces vertex guards. For polygons with h holes the bound is $\lfloor (n + 2h)/3 \rfloor$, and for orthogonal polygons $\lfloor n/4 \rfloor$. Computing the *minimum* number of guards is NP-hard even for simple polygons, so the worst-case bound is the practical substitute for exact optimization.

The guarding problem is the point-coverage version of set cover, with the visibility polygons of Section 24.3 as the sets to be covered; Chapter 25 develops the theorem, its variants, and the hardness results in depth.

24.6 Visibility Graphs

24.6.1 1. Overview

The **visibility graph** of a set of obstacles records which obstacle vertices see each other. Its paths are exactly the candidate shortest paths among the obstacles: a path that changes direction

only at obstacle corners moves in a straight line between consecutive vertices precisely when those vertices are mutually visible. The visibility graph therefore turns the continuous shortest-path problem into a discrete graph problem (Section 24.7). This section studies the base case of pairwise-disjoint segments, as analyzed in [BCKO08].

24.6.2 2. Definitions

Definition 24.14 (Visibility Graph). Let $\mathcal{S}$ be a set of pairwise-disjoint closed line segments (the obstacles). The **visibility graph** of $\mathcal{S}$ has a vertex for every segment endpoint and an edge between two vertices u, v if u and v are mutually visible: the open segment $\overline{uv}$ crosses no segment of $\mathcal{S}$. For polygonal obstacles one takes the obstacle vertices as vertices and requires the open segment to avoid all obstacle interiors.

Definition 24.15 (Bitangent). A **bitangent** (or visibility edge) of a set of disjoint convex obstacles is a segment that is mutually visible to the two obstacles it touches at its endpoints. In the segment-obstacle setting the bitangents are exactly the visibility-graph edges, since every visibility edge is tangent to the segments at both endpoints.

24.6.3 3. Theory

Theorem 24.16 (Complexity of the Visibility Graph). *The visibility graph of n pairwise-disjoint segments has $O(n^2)$ edges, and the bound is tight: configurations with $O(n^2)$ mutual-visibility pairs exist. The graph can be computed in $O(n^2 \log n)$ time by a per-vertex rotational sweep, in $O(n^2)$ time with more care, and in $O(n \log n + k)$ time with an output-sensitive algorithm, where k is the number of visibility edges [GM91, OW92].*

The rotational sweep analyzes visibility from one vertex v at a time. Sort the other endpoints around v and rotate a ray, maintaining the front (Definition 24.11), the closest segment crossing the ray. When the ray points at a vertex u , the open segment $\overline{vu}$ is crossed by an obstacle if and only if the front intersects it: a closer blocker would be the front, and a farther blocker would be beyond it. Hence u is visible from v exactly when the front does not cross $\overline{vu}$. The sweep costs $O(n \log n)$ per vertex, giving $O(n^2 \log n)$; the output-sensitive methods of [GM91, OW92] spend time proportional to the number of edges actually reported.

For a simple polygon, the visibility graph is obtained by computing $VP(v)$ for every vertex v (Section 24.4): the vertices visible from v are exactly the polygon vertices on the boundary of $VP(v)$. Since each visibility polygon has $O(n)$ boundary vertices, all n computations take $O(n^2)$ time, matching the worst-case number of edges. The bitangent structure connects to the envelopes of Chapter 14: the visibility edges from a vertex are the vertices of the lower envelope of the obstacle segments as seen from that vertex.

24.6.4 4. Pseudocode

24.6.5 5. Parameter Selection

- **General position:** assume no two endpoints share an angle from v and no endpoint lies on another segment; ties and collinearities are resolved by the fixed conventions of Chapter 3.
- **Distinct obstacles:** input segments must be pairwise disjoint; touching segments are split at their contact points first.
- **Output size:** the graph may be dense; applications should build only the edges they need (e.g., the reduced visibility graph keeping only bitangent edges between reflex vertices plus tangent edges to convex vertices) when k is large.

Algorithm 110 Rotational Sweep for the Visibility Graph

Input: Pairwise-disjoint obstacle segments $\mathcal{S}$ in general position.**Output:** The visibility graph $G = (V, E)$.

```

1: function VISIBILITYGRAPH( $\mathcal{S}$ )
2:    $V \leftarrow$  all segment endpoints;  $E \leftarrow \emptyset$ 
3:   for vertex  $v \in V$  do
4:      $W \leftarrow$  the other endpoints of  $V$ , sorted by angle around  $v$ 
5:      $\ell \leftarrow$  the front segment at the initial sweep angle from  $v$ 
6:     for endpoint  $u \in W$ , in radial order do
7:       if the front  $\ell$  does not cross the open segment  $\overline{vu}$  then
8:          $E \leftarrow E \cup \{(v, u)\}$   $\triangleright u$  is visible from  $v$ 
9:       end if
10:      update  $\ell$  as the ray passes the endpoints of the segments incident to  $u$ 
11:    end for
12:  end for
13:  return  $G$ 
14: end function

```

24.6.6 6. Complexity

- **Time:** $O(n^2 \log n)$ for Algorithm 110 ($O(n \log n)$ per vertex); $O(n \log n + k)$ output-sensitive [GM91, OW92].
- **Space:** $O(n + k)$ for the graph and auxiliary structures.
- **Lower bound:** $\Omega(n^2)$ time in the worst case, since the output can have $\Theta(n^2)$ edges.

24.6.7 7. Usages

- **Shortest paths:** the shortest obstacle-avoiding path is a visibility-graph path (Section 24.7).
- **Motion planning:** visibility graphs are the classical **roadmaps** for a point robot moving among polygonal obstacles; Chapter 26 uses them as the simplest complete roadmap.
- **Navigation meshes:** reduced visibility graphs (bitangents between reflex vertices) form the skeleton of navigation meshes for agents in games and robotics.

24.6.8 8. Testing Methods and Edge Cases

- **Brute-force verification:** for small n , compute all pairs (u, v) by a direct segment-intersection test against every obstacle and require the same edge set.
- **Edge validity:** every reported edge must be visible, and no omitted pair may be visible, on random and on adversarial inputs.
- **Tangency:** endpoints lying on the same ray from v , and segments whose endpoints are collinear, must produce a consistent edge set under the fixed tie-breaking rule.
- **Dense instances:** a ring of short segments with a central vertex should produce a near-complete visibility graph; verify the count against $\binom{2n}{2}$.

24.6.9 9. References

The visibility graph and its rotational-sweep computation are presented in [BCKO08]; the optimal output-sensitive algorithms are due to Ghosh and Mount [GM91] and to Overmars and Welzl [OW92]; the use of visibility graphs in planning dates to the configuration-space approach of Lozano-Perez [LP83] and is treated in [O'R87, SA95].

24.7 Shortest Paths among Obstacles

The shortest-path problem among obstacles is: given two points s, t and pairwise-disjoint polygonal obstacles, find the shortest path from s to t avoiding the obstacle interiors — the “rubber band” path pulled tight between s and t . Its combinatorial structure is the central fact that makes visibility graphs useful.

Theorem 24.17 (Shortest Path Structure). *Let s, t be points and $\mathcal{O}$ a set of pairwise-disjoint polygonal obstacles. The shortest obstacle-avoiding path from s to t is a simple polygonal chain; every interior vertex of the chain is a vertex of an obstacle, and every two consecutive vertices of the chain are mutually visible. Consequently the shortest path is a path in the visibility graph of the obstacle vertices augmented with s and t .*

Proof. A shortest path avoids obstacle interiors. Any subpath that runs through free space with no obstacle corner between its endpoints can be replaced by the straight segment between its endpoints, which shortens it; hence every “kink” occurs at an obstacle vertex, and the path is a polygonal chain through obstacle vertices. If two consecutive vertices u, v of the chain were not mutually visible, $\overline{uv}$ would cross an obstacle and the path could be locally detoured to a shorter one, contradicting optimality. Hence consecutive vertices are visible, and the chain is a path in the visibility graph of the vertices plus s and t . $\square$

The theorem reduces the continuous problem to a shortest-path problem in a graph with $O(n)$ vertices and $O(n^2)$ edges:

Algorithm 111 Shortest Path among Polygonal Obstacles

Input: Points s, t and pairwise-disjoint polygonal obstacles $\mathcal{O}$.

Output: The shortest obstacle-avoiding path from s to t .

- 1: **function** SHORTESTPATH($s, t, \mathcal{O}$)
 - 2: $G \leftarrow$ visibility graph of $\mathcal{O}$ (Algorithm 110)
 - 3: add s and t to G with their visible edges
 - 4: run Dijkstra’s algorithm on G with edge weights equal to Euclidean lengths
 - 5: **return** the resulting s - t path
 - 6: **end function**
-

With $O(n^2)$ edges the Dijkstra run costs $O(n^2 \log n)$; with the output-sensitive visibility graph of Theorem 24.16 the total is $O(n \log n + k)$. The general Euclidean shortest-path problem and its rectilinear and weighted variants are discussed in Chapter 24.11; the visibility graph remains the reference approach whenever the shortest path bends only at obstacle corners [LP83, BCKO08].

Remark 24.18. Three refinements are standard. (i) The **reduced** visibility graph (bitangent edges between reflex vertices plus tangents to convex vertices) still contains the shortest path and is much sparser in practice. (ii) For a fixed start and many targets, the graph is processed once and each query is a Dijkstra-like exploration. (iii) For a robot of nonzero size, the obstacles are first inflated to a configuration space (Section 24.9).

24.8 Geodesic Paths

When the path is required to stay inside a simple polygon, the shortest path is a **geodesic path**: the shortest curve in P joining two points of P . The geodesic is the planar analogue of the great-circle distance on a surface, and it defines the **geodesic distance** $d_P(s, t)$, the length of the geodesic.

Definition 24.19 (Geodesic). For points s, t in a simple polygon P , the **geodesic** $\pi(s, t)$ is the shortest path from s to t whose points all lie in P , and $d_P(s, t)$ is its length.

Theorem 24.20 (Structure and Computation of Geodesics). *The geodesic $\pi(s, t)$ is unique and simple, and all its interior vertices are reflex vertices of P . Given a triangulation of P , the geodesic can be computed in $O(n)$ time.*

Proof. Uniqueness follows because the free space is simply connected: any two distinct shortest paths of equal length between the same endpoints enclose a region of P of positive area, and a slight shortening within that region contradicts optimality. As in Theorem 24.17, every bend of the geodesic must lie at a reflex vertex of P : at an interior bend the path could be shortcut, and at a convex boundary vertex the path could be pulled into the interior of P . The geodesic crosses exactly the triangles on the unique path between the triangles of s and t in the dual tree of the triangulation, and the linear-time algorithms of [GHL⁺87] walk this chain while maintaining a **funnel**: the shortest path to the current boundary of the visited region, which has the shape of a funnel with the source s at the apex. $\square$

The funnel algorithm processes the triangle chain one at a time. Each step adds a triangle contributing one new vertex; the funnel's two boundary chains (the shortest paths from s to the two ends of the path front) are updated by a visibility test of which vertex lies inside the current triangle. When the two chains cross, the apex advances to the crossing point and the funnel resets — the geodesic's turning point moves forward. Each triangle is processed once, so the total cost is linear after locating s and t in the triangulation (Chapter 8).

Algorithm 112 Shortest Path in a Triangulated Polygon (Funnel Algorithm)

Input: A simple polygon P with a triangulation T , and points s, t in P .

Output: The geodesic path from s to t .

```

1: function FUNNELSHORTESTPATH( $P, T, s, t$ )
2:   locate the triangles  $\Delta_s \ni s$  and  $\Delta_t \ni t$  in  $T$ 
3:    $\Delta_s, \Delta_1, \dots, \Delta_k, \Delta_t \leftarrow$  the triangle chain along the dual tree of  $T$ 
4:    $F \leftarrow$  funnel with apex  $s$  and the two boundary chains along the current path front
5:   for triangle  $\Delta$  in the chain do
6:     add the new vertex of  $\Delta$ ; update the two funnel chains
7:     if the two chains cross at a point  $c$  then
8:       advance the apex to  $c$  and reset the funnel
9:     end if
10:  end for
11:  return the shortest path read off the final funnel
12: end function

```

Example 24.21 (Two Geodesics Bound a Funnel). Fix a source s and a reflex vertex v of P . The two geodesics $\pi(s, v)$ together with the boundary of P between them enclose a funnel-shaped region; every geodesic from s to a point t in that region follows one of the two paths until the funnel narrows, then crosses to the other side. This structure is the organizing principle of [GHL⁺87] and of data structures answering geodesic-distance queries in logarithmic time after $O(n \log n)$ preprocessing.

The geodesic notion extends to polyhedral surfaces, where a shortest path is a sequence of straight segments across faces meeting only at surface-reflex vertices. The **discrete geodesic problem** [MMP87] computes such paths by continuous Dijkstra simulation, together with the polygon case in Chapter 24.11; visibility enters because the path’s straight segments must lie in a mutually visible region.

24.9 Applications

Visibility is a lens through which several mature application areas are best understood. Three stand out: rendering, motion planning, and robotics.

Hidden Surface Removal and Rendering

The **hidden surface removal** problem is to draw a 3D scene so that only the surfaces visible from the viewpoint appear. It is, per pixel, a point-visibility problem: the pixel’s color comes from the *first* surface hit along the viewing ray through the pixel. The two classical solutions are:

The painter’s algorithm [NNS72] sorts the polygons by depth and draws them back-to-front, letting near polygons overpaint far ones. It fails when no consistent depth order exists (cyclically overlapping polygons), requiring the polygons to be split.

The z-buffer [Cat74] performs the depth comparison per pixel: a fragment is written only if its depth beats the stored depth, so the buffer holds exactly the per-pixel first hit at $O(1)$ per fragment — the hardware-standard algorithm of modern graphics pipelines.

Binary space partitions [FKN80, NAT90] build a recursive partition of space by planes; any viewpoint then has a correct back-to-front order from a single tree walk, giving the painter’s algorithm its missing order.

Geometrically, the visible surface points are the **lower envelope** of the surfaces as seen from the viewpoint: a surface point is visible exactly when it lies on that envelope. This ties rendering to the envelope machinery of Chapter 14 and, on the engineering side, to the spatial structures of [Eri04] used for occlusion culling; Chapter 35 uses these algorithms as its rendering core.

Motion Planning and Visibility Roadmaps

The configuration-space approach of Lozano-Perez [LP83] reduces motion planning for a robot with geometry to point planning for its *configuration* among inflated obstacles (the **C-obstacles**). For a point robot among polygonal obstacles the visibility graph is a complete roadmap: every shortest path is a visibility-graph path (Theorem 24.17), so planning reduces to the shortest-path search of Algorithm 111. Visibility roadmaps remain the reference “exact” planner for low-dimensional configuration spaces, and visibility is also the connection criterion in sampling-based planners. Chapter 26 develops the roadmap approach and its sampling-based successors.

Robotics, Coverage, and Simulation

Guarding (Section 24.5) is the coverage question behind sensor placement: placing cameras, guards, or beacons so that every point of a domain is seen by at least one sensor, with the art gallery bound as worst-case guarantee (optimal placement is NP-hard, Remark 24.13). In navigation, the visibility polygon is the local sensor model of simultaneous localization and mapping, and the geodesic paths of Section 24.8 are the routes agents follow indoors. In games and simulation, reduced visibility graphs form navigation meshes, and the point-visibility and

ray-shooting primitives of this chapter underlie the collision and occlusion queries discussed in [Eri04].

24.10 Exercises

1. Show that mutual visibility is not transitive: construct three points p, q, r in a simple polygon with p and q visible, q and r visible, but p and r not visible. Prove that in a convex polygon the relation *is* transitive.
2. Prove that the kernel of a simple polygon is convex and is exactly the intersection of the half-planes of Definition 24.3, and give a polygon whose kernel is a single point.
3. Prove Theorem 24.7 in detail: show that every window of $VP(p)$ passes through a reflex vertex of P and lies on the ray from p through that vertex.
4. Carry out Fisk's proof of Theorem 24.12: prove that the triangulation graph of a simple polygon is 3-colorable by induction on the dual tree, and derive the $\lfloor n/3 \rfloor$ bound.
5. Design a comb polygon on n vertices and prove that it cannot be guarded with fewer than $\lfloor n/3 \rfloor$ guards.
6. Prove Theorem 24.17: show that every kink of the shortest obstacle-avoiding path occurs at an obstacle vertex and that consecutive vertices of the path are mutually visible.
7. Compute by hand the visibility polygon of a point in a staircase polygon, and verify that the number of windows equals the number of reflex vertices visible on windows, as guaranteed by Theorem 24.7.
8. Implement the rotational sweep of Algorithm 109 for a simple polygon and validate it against a brute-force visibility test on random polygons, checking in particular the window audit described in Section 24.4.

24.11 Distance Transforms and Shortest Paths

24.12 Fast Marching Method

24.12.1 Overview

The Fast Marching Method (FMM) is a numerical technique for solving the **Eikonal equation**, a non-linear first-order partial differential equation that describes the evolution of a wave front. It is used to compute the shortest travel time (or distance) from a source point to all other points in a domain, even when the “speed” of travel varies across space. It can be viewed as a continuous version of Dijkstra's algorithm [Set96, Tsi95].

24.12.2 Definitions

Definition 24.22 (Eikonal Equation). The Eikonal equation $|\nabla u(x)| = F(x)$ relates the arrival time $u(x)$ of a wave front to the reciprocal of the speed $F(x)$ at each point x in the domain. For a uniform speed $F(x) \equiv 1$, the solution $u(x)$ is the Euclidean distance from the source.

Definition 24.23 (Isotropic Speed). An isotropic speed $F(x)$ is one that depends only on the spatial location, not on the direction of travel. This contrasts with anisotropic speeds that depend on the gradient direction, as encountered in crystal growth or seismic wave propagation.

Definition 24.24 (Upwind Scheme). An upwind numerical discretization computes the solution at a point using values from “upstream” locations where the solution is already smaller (earlier in time). This ensures that information propagates causally, from regions of smaller u to regions of larger u .

24.12.3 Theory

FMM simulates the propagation of a front by moving from points where the travel time is already known to points where it is unknown. To maintain efficiency, points are categorized into three sets:

1. **Accepted:** Points where the time is correctly calculated and fixed.
2. **Trial:** Points on the boundary of the Accepted set; their times are estimated and kept in a priority queue.
3. **Far:** Points that have not been reached yet.

The key to FMM is the **upwind discretization** of the gradient $|\nabla u|$. In 2D, on a Cartesian grid with spacing h , the discretization is:

$$\max(D_{ij}^{-x}u, -D_{ij}^{+x}u, 0)^2 + \max(D_{ij}^{-y}u, -D_{ij}^{+y}u, 0)^2 = F_{ij}^2$$

where $D_{ij}^{-x}u = (u_{i-1,j} - u_{i,j})/h$ and $D_{ij}^{+x}u = (u_{i+1,j} - u_{i,j})/h$ are backward and forward finite differences. The $\max(\cdot, 0)$ operators ensure that only neighbors with smaller arrival times contribute, enforcing the upwind condition.

Theorem 24.25 (FMM Correctness). *For an isotropic speed field $F > 0$ and a single source point, the Fast Marching Method Algorithm 113 computes the viscosity solution of the Eikonal equation Theorem 24.22. That is, at convergence $U[p]$ equals the shortest travel time from the source to every grid point p .*

Proof. The proof proceeds by induction on the order in which points are moved from Trial to Accepted. Let p be the point popped from the priority queue with minimum value $U[p]$. All Accepted neighbors of p have $U[q] \leq U[p]$ by the priority queue invariant. Since FMM processes points in non-decreasing order of U , no later Accepted point can provide a shorter path to p . The upwind update then solves the quadratic Eikonal stencil using only Accepted or equal-valued neighbors, yielding the correct viscosity solution by the comparison principle for first-order PDEs. $\square$

24.12.4 Algorithm

Example 24.26 (FMM on a Uniform Grid). Consider a 5×5 grid with constant speed $F = 1$ and source at the center pixel $(2, 2)$. FMM proceeds as follows: initially $U(2, 2) = 0$ and the four neighbors are Trial with $U = 1$. The algorithm pops one of them (say $(1, 2)$) to Accepted, then updates its FAR neighbors. After processing, the distance field approximates concentric diamonds (the ℓ^∞ -ball) with values $1, \sqrt{2}, 2, \sqrt{5}, \dots$ radiating outward, converging to the true Euclidean distance field as the grid is refined.

24.12.5 Parameter Selections

- **Grid Resolution:** Finer grids produce more accurate wave fronts but require more memory and time.

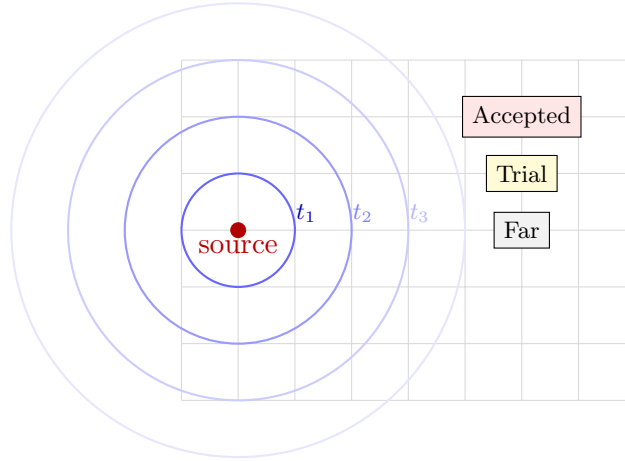


Figure 24.1: Fast Marching Method: the wavefront propagates from the source. Points are categorized as Accepted, Trial, or Far.

Algorithm 113 Fast Marching Method

```

1: function FASTMARCHING(source, speed_field)
2:    $U \leftarrow \text{Array}(\infty)$ 
3:    $\text{status} \leftarrow \text{Array}(FAR)$ 
4:    $U[\text{source}] \leftarrow 0$ 
5:    $\text{status}[\text{source}] \leftarrow ACCEPTED$ 
6:    $\text{trial\_queue} \leftarrow \text{PriorityQueue}()$ 
7:   for neighbor  $\in$  GetNeighbors(source) do
8:      $U[\text{neighbor}] \leftarrow \text{CalculateEikonal}(\text{neighbor}, U, \text{speed\_field})$ 
9:      $\text{status}[\text{neighbor}] \leftarrow TRIAL$ 
10:     $\text{trial\_queue.push}(\text{neighbor}, U[\text{neighbor}])$ 
11:   end for
12:   while not  $\text{trial\_queue.empty}()$  do
13:      $p \leftarrow \text{trial\_queue.pop\_min}()$ 
14:      $\text{status}[p] \leftarrow ACCEPTED$ 
15:     for neighbor  $\in$  GetNeighbors( $p$ ) do
16:       if  $\text{status}[\text{neighbor}] = ACCEPTED$  then continue
17:       end if
18:        $\text{old\_val} \leftarrow U[\text{neighbor}]$ 
19:        $\text{new\_val} \leftarrow \text{CalculateEikonal}(\text{neighbor}, U, \text{speed\_field})$ 
20:       if  $\text{status}[\text{neighbor}] = FAR$  then
21:          $U[\text{neighbor}] \leftarrow \text{new\_val}$ 
22:          $\text{status}[\text{neighbor}] \leftarrow TRIAL$ 
23:          $\text{trial\_queue.push}(\text{neighbor}, \text{new\_val})$ 
24:       else if  $\text{new\_val} < \text{old\_val}$  then
25:          $U[\text{neighbor}] \leftarrow \text{new\_val}$ 
26:          $\text{trial\_queue.update\_priority}(\text{neighbor}, \text{new\_val})$ 
27:       end if
28:     end for
29:   end while
30:   return  $U$ 
31: end function

```

- **Speed Function (F):** Can be constant (1.0 for Euclidean distance) or varying (to simulate terrain or obstacles).
- **Solver Order:** First-order FMM is standard; higher-order versions use more neighbors to achieve better accuracy.

24.12.6 Complexity

Theorem 24.27 (FMM Complexity). *The Fast Marching Method Algorithm 113 runs in $O(N \log N)$ time and $O(N)$ space, where N is the number of grid points.*

Proof. Each of the N grid points is moved from Far to Trial to Accepted exactly once, contributing a constant number of neighbor updates. Each update involves a constant-size quadratic solve and a priority queue operation (insert, decrease-key, or pop). A binary heap gives $O(\log N)$ per operation, yielding $O(N \log N)$ total. Space is $O(N)$ for the arrays U , $status$, and the heap. $\square$

24.12.7 Usages

- **Geodesic Distance on Meshes:** Computing distances along a surface (e.g., finding the shortest path over a mountain).
- **Image Segmentation:** Active contours (snakes) that expand to find object boundaries.
- **Robotics:** Path planning in complex environments with different “terrains” (e.g., water vs. land).
- **Medical Imaging:** Bone or organ segmentation in CT/MRI scans.
- **Seismology:** Simulating earthquake wave propagation.

24.12.8 Testing Methods and Edge Cases

- **Constant Speed:** The resulting time field should be exactly proportional to the Euclidean distance from the source.
- **Obstacles:** Set speed to zero (or $F \rightarrow \infty$) at obstacles; the front should flow around them.
- **Varying Speed:** Use a simple linear speed gradient to verify that the front bends (refraction) correctly according to Snell’s law.
- **Grid Boundaries:** Ensure the algorithm handles the edges of the domain without crashing.
- **Multiple Sources:** Initialize multiple points to 0; the result should be a Voronoi-like partition of arrival times.

24.12.9 References

- Sethian, J. A. (1996). “A fast marching level set method for monotonically advancing fronts.” *Proceedings of the National Academy of Sciences* [Set96].
- Tsitsiklis, J. N. (1995). “Efficient algorithms for globally optimal trajectories.” *IEEE Transactions on Automatic Control* [Tsi95].
- Kimmel, R., & Sethian, J. A. (1998). “Computing geodesic paths on manifolds.” *Proceedings of the National Academy of Sciences* [KS98].
- Wikipedia: Fast marching method.

24.13 Distance Map

24.13.1 Overview

A distance map (or distance transform) is a representation of a digital image or geometric scene where each point stores its distance to the nearest boundary or obstacle. For a binary image, the distance map calculates the distance from every background pixel to the closest foreground pixel. It is a fundamental tool in image processing, computer vision, and motion planning [RP66].

24.13.2 Definitions

Definition 24.28 (Distance Transform). A Distance Transform (DT) is an operator applied to binary images that produces a grayscale image where each pixel value encodes its distance to the nearest foreground (object) pixel under a chosen metric d .

Definition 24.29 (Signed Distance Function). A Signed Distance Function (SDF) is a variation of the distance transform where points inside a shape receive negative distances and points outside receive positive distances: $u(x) = \pm \min_{y \in \partial\Omega} \|x - y\|$, with the sign indicating inside/outside.

Definition 24.30 (Euclidean Distance Map). A Euclidean Distance Map (EDM) is a map where distances are computed using the true Euclidean metric $d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$, as opposed to chamfer approximations.

Definition 24.31 (Chamfer Distance). A Chamfer distance is an approximation of Euclidean distance using integer-valued masks applied in two passes. Common variants include the 3-4 chamfer (weights 3 for diagonals, 4 for axis-aligned) and the 5-7-11 chamfer for higher accuracy [RP66].

24.13.3 Theory

The most efficient algorithms for computing the Euclidean distance map are based on **separable filters** or **two-pass scanning** [MRH02, FH12].

1. **Chamfer Matching (Two-Pass)**: A simple algorithm that passes a small mask over the image twice (top-to-bottom and bottom-to-top). Each pixel is updated using the values of its neighbors plus the mask weight.
2. **Meijster's Algorithm (Separable)**: A more accurate $O(N)$ algorithm that decomposes the 2D problem into 1D problems. First, it computes the distance transform for each row. Then, it uses the row results to compute the final 2D distances using a parabolic lower envelope approach.
3. **Fast Marching Method** Algorithm 113: Can also be used to build a distance map by treating the boundary as a wave front propagating at unit speed.

The key insight behind Meijster's and Felzenszwalb's linear-time algorithms is the observation that the squared Euclidean distance transform can be decomposed as:

$$D^2(p) = \min_{q \in \text{foreground}} [(p_x - q_x)^2 + (p_y - q_y)^2] = \min_{q_y} \left[(p_y - q_y)^2 + \min_{q_x} (p_x - q_x)^2 \right]$$

The inner minimization over q_x can be solved independently for each row in $O(N)$ time using a left-to-right and right-to-left sweep. The outer minimization reduces to computing the lower envelope of a family of parabolas, which can also be done in linear time [FH12].

Theorem 24.32 (Linear-Time Distance Transform). *The Euclidean distance transform of an N -pixel binary image can be computed in $O(N)$ time and $O(N)$ space using the separable algorithm of Meijster [MRH02] or Felzenszwalb and Huttenlocher [FH12].*

Proof. The row-wise pass performs two linear scans (forward and backward) over each of the H rows, each scan touching W pixels, for $O(WH) = O(N)$ total. The column-wise parabolic envelope computation processes each of the W columns independently; in each column, the lower envelope of H parabolas is computed by the linear-time algorithm of Felzenszwalb and Huttenlocher, yielding $O(WH) = O(N)$ total. The overall complexity is therefore $O(N)$ in both time and space. $\square$

24.13.4 Algorithm: Meijster’s Algorithm (1D Pass)

Algorithm 114 Meijster’s Distance Transform

```

1: function DISTANCEMAP2D(binary_grid)
2:   width, height  $\leftarrow$  grid.size
3:   for  $y \leftarrow 0$  to  $height - 1$  do
4:     for  $x \leftarrow 0$  to  $width - 1$  do
5:       if grid[ $x$ ][ $y$ ] = FOREGROUND then
6:          $G[x][y] \leftarrow 0$ 
7:       else
8:          $G[x][y] \leftarrow \infty$ 
9:       end if
10:    end for
11:    for  $x \leftarrow 1$  to  $width - 1$  do
12:       $G[x][y] \leftarrow \min(G[x][y], G[x - 1][y] + 1)$ 
13:    end for
14:    for  $x \leftarrow width - 2$  downto  $0$  do
15:       $G[x][y] \leftarrow \min(G[x][y], G[x + 1][y] + 1)$ 
16:    end for
17:  end for
18:  final_map  $\leftarrow$  array[width][height]
19:  for  $x \leftarrow 0$  to  $width - 1$  do
20:    final_map[ $x$ ]  $\leftarrow$  ComputeParabolicEnvelope( $G[x]$ )
21:  end for
22:  return  $\sqrt{\textit{final\_map}}$ 
23: end function

```

Example 24.33 (Distance Transform of a 1D Signal). Consider the binary signal $[0, 0, 1, 0, 0, 0, 1, 0]$ where 1 denotes foreground. The row-wise forward/backward sweep gives $G = [0, 0, 0, 1, 2, 3, 0, 1]$. The parabolic envelope then yields the squared distances: $[0, 0, 0, 1, 1, 1, 0, 1]$, and the final Euclidean distance map is $[0, 0, 0, 1, 1, 1, 0, 1]$. Note that pixels equidistant from two foreground pixels are correctly assigned.

24.13.5 Parameter Selections

- **Metric:** Choose between Euclidean (accurate), Manhattan (fast), or Chebyshev (fast) based on the application’s needs.
- **Sign:** Decide if an SDF Theorem 24.29 is needed (requires identifying “inside” vs. “outside”).

24.13.6 Complexity

- **Time Complexity:** $O(N)$, where N is the total number of pixels/grid points, using Meijster’s Algorithm 114 or Felzenszwalb’s algorithm Theorem 24.32.
- **Space Complexity:** $O(N)$ to store the distance values.

24.13.7 Usages

- **Robotics:** Path planning using the “Artificial Potential Field” method (moving away from high-distance regions).
- **Image Processing:** Skeletonization (medial axis extraction) and morphological thinning.
- **Computer Graphics:** Rendering anti-aliased text and icons using Signed Distance Fields (SDF fonts).
- **Computer Vision:** Template matching and object recognition (Chamfer matching).
- **Medical Imaging:** Calculating the thickness of tissues or the distance between organs.

24.13.8 Testing Methods and Edge Cases

- **Single Point:** The distance map should look like a perfect radial gradient (concentric circles).
- **Empty Image:** All distances should be infinity (or a defined maximum).
- **Full Image:** All distances should be zero.
- **Thin Lines:** Verify that the algorithm correctly identifies the closest point even for 1-pixel-wide features.
- **Concave Shapes:** Verify that the “shadowing” effect of concavities is handled correctly.

24.13.9 References

- Rosenfeld, A., & Pfaltz, J. L. (1966). “Sequential operations in digital picture processing.” *Journal of the ACM* [RP66].
- Meijster, A., Roerdink, J. B., & Hesselink, W. H. (2002). “A general algorithm for computing distance transforms in linear time.” *Mathematical Morphology and its Applications to Image and Signal Processing* [MRH02].
- Felzenszwalb, P. F., & Huttenlocher, D. P. (2012). “Distance Transforms of Sampled Functions.” *Theory of Computing* [FH12].
- Wikipedia: Distance transform.

24.14 Dijkstra’s Algorithm

24.14.1 Overview

Dijkstra’s algorithm is a fundamental graph-search algorithm that finds the shortest paths between nodes in a graph. For a given source node, the algorithm finds the shortest distance to every other node. It is the discrete equivalent of the Fast Marching Method Section 24.12 and serves as the backbone of modern navigation and network routing systems [Dij59].

24.14.2 Definitions

Definition 24.34 (Graph). A weighted graph $G = (V, E)$ consists of a finite set of vertices V and a set of edges $E \subseteq V \times V$, each associated with a non-negative weight $w : E \rightarrow \mathbb{R}_{\geq 0}$ representing distance, cost, or time.

Definition 24.35 (Shortest Path). The shortest path from a source s to a vertex v in a weighted graph G is a path $\pi = (s = v_0, v_1, \dots, v_k = v)$ minimizing $w(\pi) = \sum_{i=0}^{k-1} w(v_i, v_{i+1})$.

Definition 24.36 (Relaxation). Relaxation is the process of testing whether a known path to a vertex v through an intermediate vertex u is shorter than the currently known best path to v . If $\text{dist}[u] + w(u, v) < \text{dist}[v]$, then $\text{dist}[v]$ is updated.

24.14.3 Theory

The algorithm operates by maintaining a set of “visited” nodes and a set of “unvisited” nodes. Initially, the distance to the source is 0, and the distance to all other nodes is infinity. In each step, the algorithm:

1. Selects the unvisited node with the smallest known distance.
2. “Relaxes” Theorem 24.36 all of its neighbors: for each neighbor, it checks if going through the current node provides a shorter path than the previously known best path.
3. Marks the current node as visited.

This greedy strategy works because the weights are non-negative; once a node is visited, its distance is guaranteed to be the shortest possible. Formally, this relies on the following principle.

Theorem 24.37 (Dijkstra Correctness). *When Dijkstra's algorithm Algorithm 115 terminates, for every vertex v , $\text{distances}[v]$ equals the length of the shortest path from source to v , provided all edge weights are non-negative.*

Proof. We prove by induction on the number of extractions from the priority queue. Let S denote the set of vertices that have been extracted (visited). **Base case:** $S = \{\text{source}\}$ and $\text{distances}[\text{source}] = 0$, which is correct. **Inductive step:** Assume all vertices in S have correct shortest-path distances. Let u be the next extracted vertex with $\text{distances}[u] = d$. We claim d is optimal. Any path from *source* to u not using vertices in S must cross the $S/\bar{S}$ boundary at some edge (s', t) with $s' \in S, t \notin S$. By non-negativity, $\text{distances}[t] \leq \text{distances}[s'] + w(s', t) \leq d$ since the priority queue extracted u with minimum key. Hence no alternative path can be shorter, and $\text{distances}[u] = d$ is optimal. After relaxing u 's neighbors, all vertices in $S \cup \{u\}$ maintain correct distances, completing the induction. $\square$

Remark 24.38. Dijkstra's algorithm is a special case of the Bellman–Ford algorithm restricted to non-negative weights. It can also be derived from the label-correcting framework of [CLRS09].

24.14.4 Algorithm

Example 24.39 (Dijkstra on a Small Graph). Consider the graph with vertices $\{A, B, C, D\}$ and edges: $A \rightarrow B$ (weight 1), $A \rightarrow C$ (weight 4), $B \rightarrow C$ (weight 2), $B \rightarrow D$ (weight 6), $C \rightarrow D$ (weight 3), with source A .

1. Initialize: $\text{dist} = \{A : 0, B : \infty, C : \infty, D : \infty\}$.
2. Extract A (dist 0). Relax neighbors: $\text{dist}[B] = 1, \text{dist}[C] = 4$.
3. Extract B (dist 1). Relax: $\text{dist}[C] = \min(4, 1 + 2) = 3, \text{dist}[D] = 1 + 6 = 7$.

Algorithm 115 Dijkstra's Shortest Path Algorithm

```

1: function DIJKSTRA(graph, source)
2:   distances  $\leftarrow \{v : \infty \mid v \in \text{graph.vertices}\}$ 
3:   previous  $\leftarrow \{v : \text{None} \mid v \in \text{graph.vertices}\}$ 
4:   distances[source]  $\leftarrow 0$ 
5:   pq  $\leftarrow$  PriorityQueue()
6:   pq.push(source, 0)
7:   while not pq.empty() do
8:     u, dist_u  $\leftarrow$  pq.pop_min()
9:     if dist_u > distances[u] then continue
10:    end if
11:    for (v, weight)  $\in$  graph.neighbors(u) do
12:      new_dist  $\leftarrow$  distances[u] + weight
13:      if new_dist < distances[v] then
14:        distances[v]  $\leftarrow$  new_dist
15:        previous[v]  $\leftarrow$  u
16:        pq.push(v, new_dist)
17:      end if
18:    end for
19:  end while
20:  return distances, previous
21: end function

```

4. Extract C (dist 3). Relax: $\text{dist}[D] = \min(7, 3 + 3) = 6$.

5. Extract D (dist 6). No unvisited neighbors.

Final distances: $\{A : 0, B : 1, C : 3, D : 6\}$. The shortest path $A \rightarrow B \rightarrow C \rightarrow D$ has total weight $1 + 2 + 3 = 6$.

24.14.5 Parameter Selections

- **Data Structure:** Using a **Fibonacci Heap** or **Binary Heap** for the priority queue is critical for performance. Binary heaps are most common in practice.
- **Edge Weights:** If any edge weight is negative, the algorithm fails; the Bellman–Ford algorithm should be used instead.

24.14.6 Complexity

Theorem 24.40 (Dijkstra Complexity). *With a binary heap, Dijkstra's algorithm Algorithm 115 runs in $O((E + V) \log V)$ time. With a Fibonacci heap, it runs in $O(E + V \log V)$ time. In both cases, space is $O(V + E)$.*

Proof. Each vertex is extracted from the priority queue at most once, contributing $O(\log V)$ per extraction with a binary heap. Each edge is relaxed at most once (when its tail is extracted), and each relaxation may trigger a decrease-key or insert operation costing $O(\log V)$. Total: $O(V \log V + E \log V) = O((V + E) \log V)$. With a Fibonacci heap, the decrease-key operation is $O(1)$ amortized, giving $O(V \log V + E) = O(E + V \log V)$. $\square$

24.14.7 Usages

- **GPS Navigation:** Finding the fastest route between two locations.

- **Network Routing:** Protocols like OSPF (Open Shortest Path First) use Dijkstra to route data packets across the internet.
- **Social Networks:** Finding “degrees of separation” between people.
- **Robotics:** Pathfinding for autonomous vehicles on a roadmap.
- **Game Development:** AI movement (often using A*, which is a heuristic-based extension of Dijkstra).

24.14.8 Testing Methods and Edge Cases

- **Disconnected Graphs:** Nodes that cannot be reached from the source should maintain a distance of infinity.
- **Single-Node Graph:** Distance to itself is 0.
- **Parallel Edges:** The algorithm should always select the edge with the minimum weight between two nodes.
- **Cycles:** Dijkstra correctly handles cycles as long as all weights are non-negative.
- **Path Reconstruction:** Ensure the **previous** pointers correctly trace the full path back to the source.

24.14.9 References

- Dijkstra, E. W. (1959). “A note on two problems in connexion with graphs.” *Numerische Mathematik* [Dij59].
- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). “Introduction to Algorithms.” MIT Press [CLRS09].
- Wikipedia: Dijkstra’s algorithm.

24.15 K-Geodesics

24.15.1 Overview

K-Geodesics is an adaptation of the K-Means clustering algorithm to the surface of a 3D mesh. Instead of using Euclidean distance to cluster points in 3D space, K-Geodesics uses **geodesic distance** (the shortest path along the surface). This allows for the segmentation of a mesh into regions that are intrinsically meaningful, such as the limbs of a character or the functional parts of a machine [Llo82].

24.15.2 Definitions

Definition 24.41 (Geodesic Centroid). The geodesic centroid (or Fréchet mean) of a set of vertices on a mesh is the vertex that minimizes the sum of squared geodesic distances to all other vertices in the set: $c^* = \arg \min_{c \in V} \sum_{v \in S} d_g(c, v)^2$, where d_g denotes geodesic distance.

Definition 24.42 (Geodesic Voronoi Region). The geodesic Voronoi region of center c_i is the set $V_i = \{v \in V \mid d_g(v, c_i) \leq d_g(v, c_j) \text{ for all } j \neq i\}$, partitioning the mesh vertices according to their nearest center under the geodesic metric.

24.15.3 Theory

The K-Geodesics algorithm follows the standard iterative process of Lloyd’s algorithm [Llo82]:

1. **Initialization:** Pick k initial seed vertices as centers.
2. **Assignment:** Assign every vertex on the mesh to the cluster of its nearest center using a multi-source **Fast Marching Method** Algorithm 113 or the **Heat Method**.
3. **Update:** For each cluster, find a new center. This is the “Fréchet mean” on the surface Theorem 24.41. Since finding the exact mean is expensive, it is often approximated by picking the vertex in the cluster that minimizes the average distance to all other cluster members.
4. **Repeat:** Continue until the centers no longer change or a maximum number of iterations is reached.

Theorem 24.43 (K-Geodesics Monotone Energy Decrease). *At each iteration of the K-Geodesics algorithm Algorithm 116, the total geodesic energy $E = \sum_{i=1}^k \sum_{v \in V_i} d_g(v, c_i)^2$ does not increase.*

Proof. In the assignment step, each vertex is reassigned to its nearest center, which can only decrease or maintain its contribution to the energy. In the update step, the centroid of each cluster (by definition Theorem 24.41) minimizes the sum of squared distances within that cluster. Hence both steps are non-increasing in energy, and their composition is non-increasing. Since the energy is bounded below by zero, the algorithm converges in a finite number of iterations for discrete meshes. $\square$

24.15.4 Algorithm

Algorithm 116 K-Geodesics Clustering

```

1: function KGEODESICS( $mesh, k$ )
2:    $centers \leftarrow \text{random.sample}(mesh.vertices, k)$ 
3:   while not converged do
4:      $(distances, labels) \leftarrow \text{MultiSourceGeodesic}(mesh, centers)$ 
5:      $new\_centers \leftarrow []$ 
6:     for  $i \leftarrow 0$  to  $k - 1$  do
7:        $cluster\_v \leftarrow \{v \mid labels[v] = i\}$ 
8:        $best\_v \leftarrow \text{FindLocalCentroid}(mesh, cluster\_v)$ 
9:        $new\_centers.append(best\_v)$ 
10:    end for
11:    if  $new\_centers = centers$  then break
12:    end if
13:     $centers \leftarrow new\_centers$ 
14:  end while
15:  return  $labels, centers$ 
16: end function

```

Example 24.44 (K-Geodesics on a Cylinder). Consider a cylinder mesh with $k = 2$ clusters and seed centers placed at the top and bottom rims. The assignment step computes geodesic distances from both centers. The Voronoi partition Theorem 24.42 divides the cylinder into a top half and a bottom half along the geodesic midline. After centroid updates, the two centers migrate toward the centroids of their respective halves. Euclidean K-Means would also produce this partition for a cylinder, but for a bent or folded shape, K-Geodesics would correctly separate geometrically close but geodesically distant regions (e.g., the two ends of a U-shaped mesh).

24.15.5 Parameter Selections

- **Number of Clusters (k):** Depends on the desired level of segmentation.
- **Distance Solver:** The **Heat Method** [CWW13b] is recommended for the assignment step because it is extremely fast after an initial factorization.
- **Initialization:** Using **K-Means++** logic (picking seeds that are far from each other) significantly improves convergence speed and result quality.

24.15.6 Complexity

- **Assignment:** $O(N)$ using the Heat Method or $O(N \log N)$ using Fast Marching.
- **Update:** $O(k \cdot N_{cluster})$ to find new centers.
- **Total:** $O(I \cdot N)$ where I is the number of iterations.

24.15.7 Usages

- **Mesh Segmentation:** Dividing a model into organic parts (e.g., body, head, arms).
- **Texture Atlas Generation:** Partitioning a complex mesh into k nearly-flat charts for UV unwrapping.
- **Shape Analysis:** Identifying repeating structural motifs on a surface.
- **Remote Sensing:** Clustering geographic regions based on travel time over terrain.
- **Remeshing:** Using the centers as a basis for generating a coarse, high-quality version of the mesh.

24.15.8 Testing Methods and Edge Cases

- **Convex Shapes:** For a sphere, the regions should look like a spherical Voronoi diagram.
- **Long Thin Regions:** Verify that the algorithm correctly segments limbs (which Euclidean K-Means might merge if they are physically close but geodesically far).
- **Number of Iterations:** Monitor the total “energy” (sum of distances) to ensure it decreases monotonically Theorem 24.43.
- **Empty Clusters:** Handle cases where a center becomes so isolated that no vertices are assigned to it.
- **Disconnected Components:** The algorithm should be applied to each component separately or handle infinity distances correctly.

24.15.9 References

- Peyré, G., & Cohen, L. D. (2006). “Geodesic remeshing using front propagation.” *International Journal of Computer Vision* [PC06].
- Crane, K., Weischedel, C., & Wardetzky, M. (2013). “Geodesics in Heat: A New Approach to Computing Distance Based on Heat Flow.” *ACM TOG* [CWW13b].
- Lloyd, S. (1982). “Least squares quantization in PCM.” *IEEE Transactions on Information Theory* [Llo82].
- Wikipedia: K-means clustering.

24.16 Fréchet Distance

24.16.1 Overview

The Fréchet distance is a measure of similarity between two curves (trajectories) that takes into account the location and ordering of points along the curves. It is often described using the **dog-leash analogy**: imagine a person walking along one curve and a dog walking along the other. The Fréchet distance is the minimum length of a leash needed to connect the two as they traverse their respective paths from start to finish. It is a more robust measure of curve similarity than the Hausdorff distance because it respects the parameterization of the curves [Fré06].

24.16.2 Definitions

Definition 24.45 (Discrete Fréchet Distance). The discrete Fréchet distance between two polygonal chains $P = (p_1, \dots, p_n)$ and $Q = (q_1, \dots, q_m)$ is

$$d_F(P, Q) = \min_{\alpha, \beta} \max_{t \in [0, 1]} \|P(\alpha(t)) - Q(\beta(t))\|$$

where $\alpha : [0, 1] \rightarrow [0, n-1]$ and $\beta : [0, 1] \rightarrow [0, m-1]$ are non-decreasing surjections (reparametrizations) [Fré06, AG95].

Definition 24.46 (Free Space Diagram). The free space diagram for two curves P and Q with threshold ϵ is the set $\mathcal{F}_\epsilon = \{(i, j) \mid \|p_i - q_j\| \leq \epsilon\} \subseteq [0, n] \times [0, m]$. The Fréchet distance is at most ϵ if and only if there exists a monotone path from $(0, 0)$ to (n, m) entirely within $\mathcal{F}_\epsilon$.

Definition 24.47 (Hausdorff Distance). The directed Hausdorff distance from P to Q is $h(P, Q) = \max_{p \in P} \min_{q \in Q} \|p - q\|$. The symmetric Hausdorff distance is $d_H(P, Q) = \max(h(P, Q), h(Q, P))$. Unlike the Fréchet distance, it does not account for the ordering of points along the curves.

24.16.3 Theory

For two curves P and Q of lengths n and m , the discrete Fréchet distance $d_F(P, Q)$ is the minimum cost of a “coupling” between the two sequences. This is typically solved using **dynamic programming**.

1. Let $dp[i][j]$ be the Fréchet distance between the prefix $P[0 \dots i]$ and $Q[0 \dots j]$.
2. The base case is $dp[0][0] = \text{dist}(P[0], Q[0])$.
3. The recurrence relation is: $dp[i][j] = \max(\text{dist}(P[i], Q[j]), \min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]))$.
4. The final result is $dp[n-1][m-1]$.

For continuous curves, the problem is solved by searching for a “monotone path” from $(0, 0)$ to $(1, 1)$ in the **Free Space Diagram** Theorem 24.46.

Theorem 24.48 (Fréchet DP Correctness). *The dynamic programming recurrence for the discrete Fréchet distance Theorem 24.45 correctly computes $d_F(P, Q)$ in $O(nm)$ time.*

Proof. The proof is by strong induction on $i + j$. The recurrence considers all valid reparametrizations: at step (i, j) , the coupling can advance on P only ($dp[i-1][j]$), on Q only ($dp[i][j-1]$), or on both ($dp[i-1][j-1]$). Taking the minimum of these three predecessors ensures we find the coupling that minimizes the maximum leash length. The outer max ensures the leash is long enough to accommodate the current pair. The base case $dp[0][0] = \|P[0] - Q[0]\|$ is correct by definition. By induction, $dp[i][j]$ equals the optimal Fréchet distance for prefixes $P[0..i]$ and $Q[0..j]$. $\square$

24.16.4 Algorithm

Algorithm 117 Discrete Fréchet Distance

```

1: function DISCRETEFRECHET( $P, Q$ )
2:    $n \leftarrow |P|, m \leftarrow |Q|$ 
3:    $dp \leftarrow \text{array}[n][m]$ , initialized to  $-1$ 
4:    $dp[0][0] \leftarrow \text{Distance}(P[0], Q[0])$ 
5:   for  $i \leftarrow 1$  to  $n - 1$  do
6:      $dp[i][0] \leftarrow \max(dp[i - 1][0], \text{Distance}(P[i], Q[0]))$ 
7:   end for
8:   for  $j \leftarrow 1$  to  $m - 1$  do
9:      $dp[0][j] \leftarrow \max(dp[0][j - 1], \text{Distance}(P[0], Q[j]))$ 
10:  end for
11:  for  $i \leftarrow 1$  to  $n - 1$  do
12:    for  $j \leftarrow 1$  to  $m - 1$  do
13:       $cost \leftarrow \text{Distance}(P[i], Q[j])$ 
14:       $dp[i][j] \leftarrow \max(cost, \min(dp[i - 1][j], dp[i - 1][j - 1], dp[i][j - 1]))$ 
15:    end for
16:  end for
17:  return  $dp[n - 1][m - 1]$ 
18: end function

```

Example 24.49 (Fréchet Distance of Two Triangular Paths). Consider $P = \{(0, 0), (1, 1), (2, 0)\}$ and $Q = \{(0, 0), (1, 2), (2, 0)\}$. The midpoint of P deviates by 1 unit, while Q 's midpoint deviates by 2 units. The discrete Fréchet distance is computed by the DP table:

- $dp[0][0] = 0$.
- $dp[1][0] = \|P_1 - Q_0\| = \sqrt{2}$; $dp[0][1] = \|P_0 - Q_1\| = \sqrt{5}$.
- $dp[1][1] = \max(\|P_1 - Q_1\|, \min(dp[0][1], dp[0][0], dp[1][0])) = \max(1, 0) = 1$.
- $dp[2][1] = \max(\|P_2 - Q_1\| = \sqrt{5}, \min(dp[1][1] = 1, dp[1][0] = \sqrt{2}, dp[2][0] = 2)) = \max(\sqrt{5}, 1) = \sqrt{5}$.
- $dp[2][2] = \max(\|P_2 - Q_2\| = 0, \min(dp[1][2], dp[1][1], dp[2][1])) = 0$.

Wait, let us recompute more carefully. Actually $dp[2][2] = \max(0, \min(dp[1][2], dp[1][1], dp[2][1]))$. We need $dp[1][2] = \max(\|P_1 - Q_2\| = \sqrt{2}, \min(dp[0][2], dp[0][1], dp[1][1]))$. Since $dp[1][1] = 1$, we get $dp[1][2] = \max(\sqrt{2}, 1) = \sqrt{2}$. Thus $dp[2][2] = \max(0, \min(\sqrt{2}, 1, \sqrt{5})) = 1$. The Fréchet distance is $\boxed{1}$, which is the leash length needed to connect the walkers at the midpoints of the two curves.

24.16.5 Parameter Selections

- **Distance Metric:** Euclidean distance is standard, but Manhattan or Chebyshev distances can also be used.
- **Normalization:** Curves should be normalized for scale or rotation if required by the application.

24.16.6 Complexity

- **Time Complexity:** $O(n \cdot m)$ for curves with n and m vertices, as each entry of the DP table Theorem 24.48 is computed in $O(1)$.
- **Space Complexity:** $O(n \cdot m)$ to store the DP table. Can be reduced to $O(\min(n, m))$ using a rolling array.

24.16.7 Usages

- **Trajectory Analysis:** Comparing the paths taken by animals, vehicles, or sports players.
- **Handwriting Recognition:** Comparing a drawn stroke to a template of letters.
- **Speech Recognition:** Time-aligning two audio signals (Dynamic Time Warping is a similar concept).
- **Bioinformatics:** Comparing the backbones of proteins or the shapes of molecules.
- **Computer Vision:** Shape matching for object tracking.

24.16.8 Testing Methods and Edge Cases

- **Identical Curves:** The distance should be 0.
- **Reversed Curves:** If one curve is the reverse of the other, the distance should be large (unlike Hausdorff distance).
- **Different Sampling Rates:** Test curves where one has many more vertices than the other but covers the same path.
- **Large Deviations:** A single outlier point in one curve should correctly increase the Fréchet distance.
- **Looping Curves:** Verify that the algorithm handles self-intersecting paths correctly.

24.16.9 References

- Fréchet, M. (1906). “Sur quelques points du calcul fonctionnel.” *Rendiconti del Circolo Matematico di Palermo* [Fré06].
- Alt, H., & Godau, M. (1995). “Computing the Fréchet distance between two polygonal curves.” *International Journal of Computational Geometry & Applications* [AG95].
- Eiter, T., & Mannila, H. (1994). “Computing discrete Fréchet distance.” Technical Report [EM94b].
- Wikipedia: Fréchet distance.

Chapter 25

Art Gallery Problem

The art gallery problem is a visibility question that has become one of the most studied problems in computational geometry: given the floor plan of an art gallery, modeled as a simple polygon, how many stationary guards are needed so that every point of the gallery is visible from at least one guard? Chvátal’s theorem gives a tight bound of $\lfloor n/3 \rfloor$ guards for an n -vertex polygon, and the structural connection between guarding, polygon triangulation, vertex coloring, and polygon kernels makes the problem a rich source of theory and algorithms. This chapter develops the classical results, the guard-placement algorithm, and the closely related notion of a polygon kernel.

25.1 Art Gallery Guard Placement

25.1.1 1. Overview

The **Art Gallery Problem** (AGP) is a classic problem in computational geometry that asks for the minimum number of guards required to monitor every point within a given art gallery (represented as a simple polygon) [O’R87]. A guard at a point P can “see” any point Q if the line segment PQ lies entirely within the polygon.

25.1.2 2. Definitions

Definition 25.1 (Art Gallery). An *art gallery* is a simple polygon P with n vertices.

Definition 25.2 (Visibility). A point Q is *visible* from a point P inside a polygon P if the line segment $\overline{PQ}$ lies entirely within the polygon: $\overline{PQ} \subseteq P$.

Definition 25.3 (Guard Set). A *guard set* is a set of points $\{G_1, G_2, \dots, G_k\}$ such that every point in P is visible from at least one G_i . The *guard number* $\gamma(P)$ is the minimum cardinality of a guard set.

Theorem 25.4 (Chvátal’s Art Gallery Theorem). For a simple polygon with n vertices, $\lfloor n/3 \rfloor$ guards are always sufficient and sometimes necessary to guard the polygon [Chw75].

Proof. The proof proceeds by **polygon triangulation and 3-coloring** [Fis78].

1. **Triangulate** the polygon into $n - 2$ triangles using an algorithm like ear clipping or monotone decomposition.
2. **3-color** the vertices of the resulting triangulation graph such that no two adjacent vertices share the same color. Since the dual graph of a triangulated simple polygon is a tree, the triangulation graph is chordal and therefore 3-colorable.

3. **Count** the number of vertices for each of the three colors. Since there are n vertices total, by the pigeonhole principle, the color with the minimum count has at most $\lfloor n/3 \rfloor$ vertices.
4. **Place** a guard at every vertex of this minimum-count color. Since every triangle in the triangulation must contain all three colors, every triangle is visible from at least one guard, and hence the entire polygon is guarded.

The bound is tight: Chvátal constructed “comb” polygons with $3k$ vertices that require exactly $k = \lfloor n/3 \rfloor$ guards [Chv75]. $\square$

Remark 25.5. For **orthogonal polygons** (all edges horizontal or vertical), the bound improves to $\lfloor n/4 \rfloor$ guards, proved by O’Rourke [O’R87].

Example 25.6 (Art Gallery with 6 Vertices). Consider the simple polygon with vertices $(0, 0), (4, 0), (4, 3), (2, 1), (0, 3)$ (5 vertices). After triangulation we have 3 triangles. A 3-coloring of the vertices yields color classes of sizes at most $\lfloor 5/3 \rfloor = 1$. Placing a single guard at a vertex of the minority color (e.g., at $(2, 1)$) guards the entire polygon.

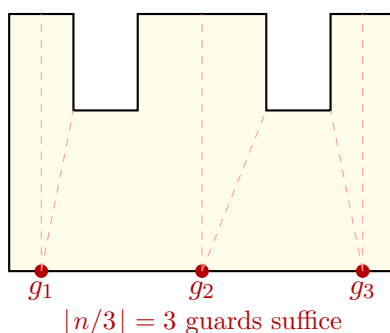


Figure 25.1: Art gallery problem: a comb polygon with $n = 12$ vertices requires $\lfloor 12/3 \rfloor = 4$ guards in the worst case.

25.1.3 4. Pseudo code

```

1: function PLACEGUARDS(polygon)                                ▷ 1. Triangulate the polygon
2:   triangles  $\leftarrow$  Triangulate(polygon)
                                                                ▷ 2. Build the triangulation graph
3:   graph  $\leftarrow$  BuildTriangulationGraph(triangles)
                                                                ▷ 3. 3-Color the graph (using DFS since it's a chordal graph)
4:   colors  $\leftarrow$  {v: None for v in polygon.vertices}
5:   DFS_Coloring(graph, start_vertex, colors)
                                                                ▷ 4. Partition vertices by color
6:   color_groups  $\leftarrow$  [], [], []
7:   for v, c in colors.items() do
8:     color_groups[c].append(v)
9:   end for
                                                                ▷ 5. Select the smallest group
10:  guard_positions  $\leftarrow$  min(color_groups, key=len)
11:  return guard_positions
12: end function

```

25.1.4 5. Parameters Selections

Definition 25.7 (Vertex vs. Point Guarding). *Vertex guarding* restricts guards to polygon vertices. *Point guarding* allows guards anywhere in the interior. Point guarding might reduce the total count in some cases, but the Chvátal bound $\lfloor n/3 \rfloor$ remains the same for worst-case bounds.

Definition 25.8 (Triangulation Algorithm Choice). The choice of triangulation (ear clipping vs. monotone decomposition) does not affect the 3-coloring result but does affect the $O(n \log n)$ vs. $O(n)$ construction time.

25.1.5 6. Complexity

Theorem 25.9 (Art Gallery Algorithm Complexity). *The guard placement algorithm runs in $O(n \log n)$ time using monotone decomposition for triangulation, followed by $O(n)$ for graph construction and 3-coloring. The space is $O(n)$.*

25.1.6 7. Usages

- **Physical Security:** Optimizing the placement of cameras or motion sensors in buildings.
- **Wireless Networking:** Determining the minimum number of Wi-Fi access points or cellular towers to cover a given region.
- **Robotics:** Path planning for a surveillance robot that needs to maintain visibility of its surroundings.
- **Environmental Monitoring:** Placing sensors to detect smoke or pollutants in complex indoor spaces.

25.1.7 8. Testing methods and Edge cases

- **Convex Polygons:** $n = 3, 4, 5$ all require only 1 guard ($\lfloor 5/3 \rfloor = 1$).
- **Comb Polygons:** “Comb” shapes (ortho-polygons) reach the upper bound of $\lfloor n/3 \rfloor$.
- **Star Polygons:** By definition, star polygons only require 1 guard (placed at the kernel).
- **Orthogonal Polygons:** For polygons where all edges are horizontal or vertical, the bound is tighter: $\lfloor n/4 \rfloor$ guards are sufficient (Hoffman’s Theorem).
- **Holes:** Polygons with holes require significantly more guards, and the problem becomes NP-hard.

25.1.8 9. References

- Chvátal, V. (1975). “A combinatorial theorem in plane geometry”. *Journal of Combinatorial Theory, Series B*.
- Fisk, S. (1978). “A short proof of Chvátal’s watchman theorem”. *Journal of Combinatorial Theory, Series B*.
- O’Rourke, J. (1987). “Art Gallery Theorems and Algorithms”. Oxford University Press.
- [Wikipedia: Art gallery problem](#)

25.2 Polygon Kernel and Star-Shaped Polygons

25.2.1 1. Overview

The **kernel** of a simple polygon is the set of points from which every point of the polygon is visible. A polygon whose kernel is non-empty is called **star-shaped**. Computing the kernel answers visibility questions such as “is there a point that sees the whole polygon?”, which arises in art gallery problems, motion planning for a point robot inside a polygon, and star-shaped polygonization [Las96].

25.2.2 2. Definitions

Definition 25.10 (Kernel of a Polygon). The *kernel* of a simple polygon P is

$$\ker(P) = \{p \in P \mid \text{the segment } \overline{pq} \subset P \text{ for all } q \in P\}.$$

Definition 25.11 (Star-Shaped Polygon). A polygon is *star-shaped* if it has a non-empty kernel: there exists a point p such that every point of the polygon is visible from p . The kernel need not be a single point; it is itself a (possibly degenerate) convex polygon.

25.2.3 3. Theory

Let P be given in counter-clockwise order. The kernel is the intersection of the closed left half-planes defined by the directed edges of P :

$$\ker(P) = \bigcap_i H(e_i), \quad H(e_i) = \{x \mid \text{orient2d}(v_i, v_{i+1}, x) \geq 0\}.$$

For a simple polygon this half-plane intersection is always convex and non-empty exactly when P is star-shaped. In general the kernel is computed by half-plane intersection in $O(n \log n)$ time; for the special case of polygon kernels there is a linear-time $O(n)$ algorithm using a deque, due to Lee and Preparata [LP79]. A point $p \in \ker(P)$ need only be checked against the vertices: p sees the entire polygon if and only if $\overline{pv_i} \subset P$ for every vertex v_i .

25.2.4 4. Pseudo code

```

function KERNEL( $P$ )
   $H \leftarrow \emptyset$ 
  for each directed edge  $(v_i, v_{i+1})$  of  $P$  do
    add the left half-plane of  $(v_i, v_{i+1})$  to  $H$ 
  end for
  return HalfPlaneIntersection( $H$ )
end function

```

25.2.5 5. Parameters Selections

- **Vertex order:** assume a consistent counter-clockwise ordering; reverse the half-plane side if the polygon is clockwise.
- **Exact predicates:** use robust orientation tests, since near-degenerate edges decide membership of kernel boundary points.
- **Collinear edges:** consecutive collinear edges define the same half-plane and may be merged.

25.2.6 6. Complexity

The general half-plane intersection approach runs in $O(n \log n)$ time and $O(n)$ space. The specialized deque algorithm of Lee and Preparata computes the kernel in optimal $O(n)$ time [LP79].

25.2.7 7. Usages

- **Art gallery:** the kernel identifies points from which a single guard observes the entire polygon.
- **Motion planning:** a point robot can navigate between any two points of a star-shaped region without leaving it.
- **Star-shaped polygonization:** connecting points to a common kernel point produces a star-shaped polygonization of a point set [Las96].

25.2.8 8. Testing methods and Edge cases

- **Convex polygon:** the kernel equals the polygon itself.
- **L-shaped polygon:** the kernel is the region common to both arms.
- **Spiral polygon:** the kernel is empty.
- **Clockwise input:** verify the half-plane orientation is flipped correctly.

25.2.9 9. References

- Laszlo, M. J. (1996). “Computational Geometry and Computer Graphics in C++”. Prentice Hall.
- Lee, D. T., & Preparata, F. P. (1979). “An optimal algorithm for finding the kernel of a polygon”. Journal of the ACM.
- Wikipedia: [Star-shaped polygon](#)

Chapter 26

Motion Planning

Motion planning asks whether an object can move from a start state to a goal state without colliding with obstacles. The central idea is to replace motion of a physical object by a path for a point in a higher-dimensional *configuration space*.

26.1 Configuration Space

A configuration records all degrees of freedom of a moving object. A point robot translating in the plane has configuration $q = (x, y) \in \mathbb{R}^2$. A rigid body translating and rotating in the plane has

$$q = (x, y, \theta) \in \mathbb{R}^2 \times S^1,$$

while a rigid body in three dimensions has six degrees of freedom, $q = (x, y, z, \alpha, \beta, \gamma) \in \mathbb{R}^3 \times SO(3)$.

The configuration space $\mathcal{C}$ contains all possible configurations. The obstacle region $\mathcal{C}_{\text{obs}}$ consists of configurations in which the robot intersects an obstacle. The free configuration space is

$$\mathcal{C}_{\text{free}} = \mathcal{C} \setminus \mathcal{C}_{\text{obs}}.$$

Motion planning then becomes the problem of finding a continuous path $\tau : [0, 1] \rightarrow \mathcal{C}_{\text{free}}$ with $\tau(0) = q_{\text{start}}$ and $\tau(1) = q_{\text{goal}}$.

26.2 Configuration-Space Obstacles

For a translating robot R and workspace obstacle O , collision-free translations can be computed using the Minkowski difference:

$$\mathcal{C}_{\text{obs}} = O \oplus (-R).$$

Thus a finite-sized robot can be replaced by a point, while the obstacles are expanded by the reflected robot. For a rotating robot, this construction is performed for each orientation, producing a slice of the higher-dimensional configuration-space obstacle.

Definition 26.1 (Free Path). A free path is a continuous map $\tau : [0, 1] \rightarrow \mathcal{C}_{\text{free}}$. It is *feasible* if it connects the start and goal configurations and satisfies any required constraints, such as joint limits or velocity bounds.

26.3 Exact Planning Methods

26.3.1 Cell Decomposition

Cell decomposition partitions $\mathcal{C}_{\text{free}}$ into simple cells whose interiors are either entirely free or entirely blocked. A connectivity graph is built by connecting adjacent free cells. A graph search gives a sequence of cells, after which a continuous path is constructed through them. Exact decompositions provide completeness, but their complexity grows rapidly with the dimension and algebraic complexity of the configuration space.

26.3.2 Visibility Graphs

For a polygonal point robot in the plane, a visibility graph uses the start, goal, and obstacle vertices as nodes. Two nodes are connected when the segment between them lies in free space. Dijkstra's algorithm or A* then finds a shortest collision-free path in the graph. Visibility graphs are exact for Euclidean shortest paths among polygonal obstacles.

26.4 Roadmaps and Sampling-Based Planning

Sampling-based planners avoid explicitly constructing high-dimensional obstacle boundaries. They sample configurations, reject samples in $\mathcal{C}_{\text{obs}}$, and connect nearby collision-free samples.

Algorithm 118 Probabilistic Roadmap Planner

```

1: function BUILDROADMAP( $\mathcal{C}$ ,  $N$ )
2:    $G \leftarrow$  empty graph
3:   for  $i = 1$  to  $N$  do
4:     Sample  $q$  from  $\mathcal{C}$ 
5:     if  $q \in \mathcal{C}_{\text{free}}$  then
6:       Add  $q$  to  $G$ 
7:       for nearby vertex  $p$  in  $G$  do
8:         if the local path from  $p$  to  $q$  is collision-free then
9:           Add edge  $(p, q)$  to  $G$ 
10:        end if
11:      end for
12:    end if
13:  end for
14:  return  $G$ 
15: end function

```

Probabilistic roadmaps (PRM) are effective for repeated queries in a fixed environment. Rapidly-exploring random trees (RRT) grow from the start toward unexplored regions and are useful for single-query planning and systems with complex dynamics. Both methods are probabilistically complete under standard sampling and local-planning assumptions, but they do not automatically return the shortest path.

26.5 Planning Constraints and Validation

Collision checking must test the entire local path, not only its endpoints. Practical planners also handle robot joint limits, self-collision, narrow passages, dynamic obstacles, clearance requirements, and kinodynamic limits on velocity and acceleration. A returned path should be checked independently, then shortened or smoothed only with collision tests after every modification.

26.6 Applications

Motion planning is used in robot manipulation, autonomous vehicles, computer animation, computer games, surgical navigation, warehouse automation, and molecular modeling. Configuration-space reasoning also connects polygon offsets, Minkowski operations, shortest paths, visibility, and geometric collision detection.

Part IX

Shape Analysis and Geometric Data

Chapter 27

Sampling

27.1 Poisson Disk Sampling

27.1.1 Overview

Poisson disk sampling is a technique for generating a set of points in a multidimensional space such that no two points are closer to each other than a specified minimum distance r . The resulting distribution has a “blue noise” characteristic: the points are tightly packed but avoid the structured look of a grid and the clumping found in pure random (white noise) sampling. This makes it ideal for computer graphics, procedural generation, and sensor placement [Coo86].

27.1.2 Definitions

Definition 27.1 (Minimum Distance (r)). The radius of an exclusion disk around each point. No two generated points may be closer than r to each other, ensuring a minimum spacing constraint.

Definition 27.2 (Blue Noise). A random distribution where low-frequency components are minimized in the power spectrum, resulting in a visually pleasing and uniform distribution. Formally, the Fourier transform of the point set has negligible energy below a cutoff frequency determined by the density [Coo86].

Definition 27.3 (Bridson’s Algorithm). A fast $O(n)$ algorithm for generating Poisson disk samples in any dimension, introduced by Robert Bridson [Bri07]. It uses a background grid for $O(1)$ proximity queries and an active list for incremental growth.

Definition 27.4 (Rejection Sampling). A naive approach where random points are generated and discarded if they fall within distance r of any existing point. The acceptance rate decreases rapidly as the domain fills, making it $O(n^2)$ in the worst case [LD08].

27.1.3 Theory

Bridson’s algorithm [Bri07] is the standard for efficient Poisson disk sampling:

1. **Initialize:** Place an initial point randomly in the domain and add it to an **active list**.
2. **Grid:** Initialize a background grid with cell size $r/\sqrt{d}$ (where d is dimension) to ensure each cell can contain at most one point. This guarantees that proximity checks only need to examine the 9 (in 2D) or 27 (in 3D) neighboring cells.
3. **Active Sampling:** While the active list is not empty:
 - Pick a random point P from the active list.

- Generate up to k candidates in an annulus $[r, 2r]$ around P .
- For each candidate Q , check if it is within distance r of any existing points (using the grid for $O(1)$ lookups).
- If Q is valid, add it to the grid and the active list.
- If no candidates are found after k trials, remove P from the active list.

Theorem 27.5 (Complexity of Bridson’s Algorithm). *Bridson’s algorithm generates n Poisson disk samples in expected $O(n)$ time and $O(n)$ space, assuming a fixed minimum distance r and dimension d .*

Proof. Each of the n points is processed at most once from the active list. For each active point, at most k candidates are generated, and each candidate requires $O(1)$ time for the grid-based neighborhood query (since at most 3^d cells need to be checked). When a candidate is accepted, the grid insertion is $O(1)$. A point is removed from the active list after k failed trials, so the total number of candidate tests is at most $kn = O(n)$. The grid stores at most $O(n)$ points in $O(n)$ cells. Thus the total expected time is $O(n)$ and space is $O(n)$ [Bri07]. $\square$

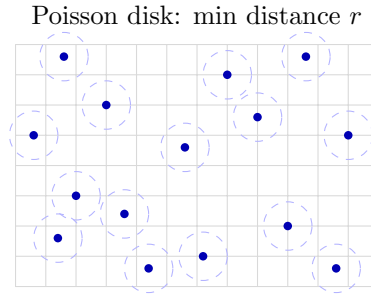


Figure 27.1: Poisson disk sampling: points are placed with minimum distance r between any pair.

Example 27.6 (Poisson Disk Sampling Trace on a 4×4 Grid). Consider a small domain $[0, 4] \times [0, 4]$ with $r = 1.5$ and $k = 8$.

1. Seed: Place $P_1 = (2.0, 2.0)$. Grid cell size = $1.5/\sqrt{2} \approx 1.06$, so the grid is 4×4 .
2. Active list: $[P_1]$. Pick P_1 , generate candidates around it. Suppose $Q_1 = (0.8, 1.2)$ passes the distance check ($|P_1 Q_1| \approx 1.44 < 1.5$ —rejected). Try again: $Q_2 = (3.3, 1.1)$, distance $\approx 1.35 < 1.5$ —rejected. After 8 failures, remove P_1 from active list.
3. Since only one point was placed and no new points accepted, the result is $\{(2.0, 2.0)\}$.

For a larger domain with more seed attempts, the algorithm fills the space more densely, with each new point guaranteed to be at least r from all others.

27.1.4 Pseudocode

27.1.5 Parameter Selection

- **Radius (r):** Determines the density of the points.
- **Trials (k):** Typically set to 30. Higher k results in tighter packing but increases computation time.
- **Variable Radius:** The algorithm can be adapted to use a varying $r(x, y)$ based on an importance map or image brightness.

Algorithm 119 Poisson Disk Sampling

```

function POISSONDISKSAMPLING(width, height, r, k=30)           ▷ 1. Initialize Grid
    cell_size  $\leftarrow r/\sqrt{2}$ 
    cols  $\leftarrow \lceil \text{width}/\text{cell\_size} \rceil$ 
    rows  $\leftarrow \lceil \text{height}/\text{cell\_size} \rceil$ 
    grid  $\leftarrow$  array[cols][rows] initialized to None

    start_p  $\leftarrow$  Point(random(width), random(height))           ▷ 2. Seed Point
    active_list  $\leftarrow$  [start_p]
    InsertIntoGrid(start_p, grid, cell_size)
    points  $\leftarrow$  [start_p]

    ▷ 3. Main Loop
    while active_list is not empty do
        p  $\leftarrow$  random.choice(active_list)
        found  $\leftarrow$  False
        for k in range(k) do                                     ▷ Generate candidate in annulus [r, 2r]
            q  $\leftarrow$  GenerateRandomInAnnulus(p, r, 2 · r)
            if IsInBounds(q, width, height) and IsValid(q, grid, r, cell_size) then
                points.append(q)
                active_list.append(q)
                InsertIntoGrid(q, grid, cell_size)
                found  $\leftarrow$  True
                break
            end if
        end for
        if  $\neg$ found then
            active_list.remove(p)
        end if
    end while
    return points
end function

```

27.1.6 Complexity

- **Time Complexity:** $O(n)$ where n is the number of points generated, thanks to the grid-based spatial lookup (Theorem 27.5).
- **Space Complexity:** $O(n)$ to store the points and the grid.

27.1.7 Usages

- **Computer Graphics:** Placing objects (like trees or grass) in a natural-looking distribution.
- **Rendering:** Anti-aliasing and jittered sampling for ray tracing.
- **Image Processing:** Halftoning and stippling (representing an image with dots).
- **Procedural Content Generation:** Generating city layouts or dungeon rooms.
- **Physical Simulation:** Initializing particles for fluid or granular simulations.

27.1.8 Testing Methods and Edge Cases

- **Distance Constraint:** Verify that for all pairs (P_i, P_j) , $\text{dist}(P_i, P_j) \geq r$.
- **Boundary Handling:** Ensure points are correctly generated near the edges of the domain.
- **Clumping:** Check the Fourier transform of the point set to verify the blue noise profile (lack of low-frequency spikes).
- **Empty Result:** If the domain is smaller than r , the algorithm should return at most one point.
- **Grid Overflow:** Ensure the grid coordinates are correctly clamped to the array bounds.

27.1.9 References

- Bridson, R. (2007). “Fast Poisson disk sampling in arbitrary dimensions”. SIGGRAPH Sketches.
- Cook, R. L. (1986). “Stochastic sampling in computer graphics”. ACM Transactions on Graphics.
- Lagae, A., & Dutré, P. (2008). “A comparison of methods for generating Poisson disk distributions”. Computer Graphics Forum.
- Wikipedia: Poisson disk sampling

27.2 Halton Points

27.2.1 Overview

Halton sequences are a type of low-discrepancy sequence used to generate points in a d -dimensional space. Unlike pseudo-random numbers, which can clump together, low-discrepancy sequences are designed to be “more uniform than random,” ensuring that every part of the domain is sampled evenly [Hal60]. Halton points are a generalization of the 1D van der Corput sequence [VdC35] to higher dimensions using different prime numbers as bases.

27.2.2 Definitions

Definition 27.7 (Low-Discrepancy Sequence). A sequence of points $\{x_i\}_{i=1}^N$ in $[0, 1]^d$ where the discrepancy D_N (a measure of non-uniformity) satisfies $D_N = O(N^{-1}(\log N)^d)$. This is near-optimal by the Schmidt theorem bound, and achieves faster convergence than pseudo-random sequences in Quasi-Monte Carlo integration [Nie92].

Definition 27.8 (Quasi-Monte Carlo (QMC)). Integration and simulation methods that use low-discrepancy sequences instead of random numbers. QMC achieves a convergence rate of $O(N^{-1})$ for integrands of bounded variation, compared to $O(N^{-1/2})$ for standard Monte Carlo [Hal60].

Definition 27.9 (Van der Corput Sequence). A 1D sequence in base b formed by reversing the digits of the base- b representation of integers. The n -th element is $\phi_b(n) = \sum_{k=0}^{\infty} a_k b^{-(k+1)}$ where $n = \sum_{k=0}^{\infty} a_k b^k$ is the base- b expansion of n [VdC35].

Definition 27.10 (Discrepancy). A mathematical measure of how “non-uniform” a set of points is. For a point set $P = \{x_1, \dots, x_N\}$ in $[0, 1]^d$, the star discrepancy is:

$$D_N^* = \sup_{\mathbf{b} \in [0, 1]^d} \left| \frac{|P \cap [0, \mathbf{b})|}{N} - \text{Vol}([0, \mathbf{b})) \right|.$$

27.2.3 Theory

The Halton sequence in d dimensions is constructed by using the first d prime numbers as bases $(p_1, p_2, \dots, p_d)$. The i -th point in the sequence is:

$$x_i = (\phi_{p_1}(i), \phi_{p_2}(i), \dots, \phi_{p_d}(i))$$

where $\phi_b(i)$ is the **radical inverse function** in base b .

To compute $\phi_b(i)$:

1. Write i in base b : $i = \sum a_k b^k$.
2. Reverse the digits and place them after a decimal point: $\phi_b(i) = \sum a_k b^{-(k+1)}$.

Because the prime bases are co-prime, the coordinates in different dimensions are effectively “de-correlated,” filling the d -dimensional unit hypercube uniformly.

Theorem 27.11 (Discrepancy Bound for Halton Sequences). *For the first N points of the d -dimensional Halton sequence, the star discrepancy satisfies:*

$$D_N^* = O\left(\frac{1}{N} \prod_{i=1}^d \log p_i \cdot \log N\right) = O\left(\frac{(\log N)^{d+1}}{N}\right).$$

This is significantly better than the $O(N^{-1/2})$ convergence of pseudo-random Monte Carlo [Hal60, Nie92].

Proof. The proof proceeds by the Koksma–Hlawka inequality, which bounds the integration error by the product of the discrepancy D_N^* and the variation of the integrand. For the Halton sequence, the discrepancy is bounded using the Erdős–Turá inequality on the digital sequence construction. Each coordinate ϕ_{p_i} is a van der Corput sequence in base p_i , with discrepancy $O(\log p_i/N)$. By the product property of digital sequences [Nie92], the multi-dimensional discrepancy is the product of the one-dimensional discrepancies, giving the stated bound. $\square$

Example 27.12 (First Five Halton Points in 2D). Using bases $p_1 = 2$ and $p_2 = 3$:

n	$\phi_2(n)$	$\phi_3(n)$	(x, y)
1	$0.1_2 = 0.5$	$0.1_3 = 0.\bar{3}$	$(0.500, 0.333)$
2	$0.01_2 = 0.25$	$0.2_3 = 0.\bar{6}$	$(0.250, 0.667)$
3	$0.11_2 = 0.75$	$0.01_3 = 0.1\bar{1}$	$(0.750, 0.111)$
4	$0.001_2 = 0.125$	$0.12_3 = 0.4\bar{4}$	$(0.125, 0.444)$
5	$0.101_2 = 0.625$	$0.22_3 = 0.\bar{7}$	$(0.625, 0.778)$

Notice how the points are evenly spread across the unit square with no clumping, unlike pseudo-random sampling.

27.2.4 Pseudocode

Algorithm 120 Halton Sequence Generator

```

function HALTONSEQUENCE(num_points, dimensions)
   $primes \leftarrow \text{GetFirstPrimes}(\text{dimensions})$ 
   $points \leftarrow []$ 
  for  $i$  in range(1, num_points + 1) do
     $point \leftarrow []$ 
    for  $d$  in range(dimensions) do
       $point.append(\text{RadicalInverse}(i, primes[d]))$ 
    end for
     $points.append(point)$ 
  end for
  return  $points$ 
end function

function RADICALINVERSE(n, base)
   $val \leftarrow 0$ 
   $inv\_base \leftarrow 1.0/base$ 
   $f \leftarrow inv\_base$ 
  while  $n > 0$  do
     $val \leftarrow val + (n \bmod base) \cdot f$ 
     $n \leftarrow \lfloor n/base \rfloor$ 
     $f \leftarrow f \cdot inv\_base$ 
  end while return  $val$ 
end function

```

27.2.5 Parameter Selection

- **Prime Bases:** Must use unique prime numbers for each dimension. Using non-primes or identical bases will lead to strong correlations (linear patterns).
- **Burn-in (Leaping):** For higher dimensions, the first few hundred points can show patterns. “Scrambled Halton” or skipping the initial points can improve uniformity [Nie92].

27.2.6 Complexity

- **Time Complexity:** $O(N \cdot D \cdot \log_b N)$ to generate N points in D dimensions. Since b is small, this is very fast.
- **Space Complexity:** $O(D)$ to store the current point (or $O(N \cdot D)$ to store the full sequence).

27.2.7 Usages

- **Quasi-Monte Carlo Integration:** Estimating high-dimensional integrals with faster convergence ($O(1/N)$) than standard Monte Carlo ($O(1/\sqrt{N})$).
- **Computer Graphics:** Sampling light sources and pixels for ray tracing and path tracing.
- **Physics Simulation:** Initializing particles in a way that minimizes clumping.
- **Global Optimization:** Sampling the search space for black-box optimization algorithms.
- **Financial Modeling:** Pricing complex derivatives where high-dimensional integrals are involved.

27.2.8 Testing Methods and Edge Cases

- **Range Check:** All coordinates must be in the interval $[0, 1)$.
- **Uniformity:** Test using the “Chi-Squared” test or by calculating the actual discrepancy of the set.
- **Dimension Correlation:** Plot 2D projections of high-dimensional sequences (e.g., dim 10 vs dim 11) to check for “ghosting” patterns.
- **Large N :** Ensure the radical inverse function doesn’t lose precision for very large indices.
- **Base Selection:** Verify that using the same base for two dimensions results in points lying on a single line.

27.2.9 References

- Halton, J. H. (1960). “On the efficiency of certain quasi-random sequences of points in evaluating multi-dimensional integrals”. *Numerische Mathematik*.
- Niederreiter, H. (1992). “Random Number Generation and Quasi-Monte Carlo Methods”. SIAM.
- Van der Corput, J. G. (1935). “Verteilungsfunktionen”. *Proc. Akad. Wet. Amsterdam*.
- Wikipedia: Halton sequence

27.3 Uniform Random Point Generation

27.3.1 Overview

Generating points uniformly at random inside a basic shape—square, rectangle, circle, or sphere—is a building block for Monte Carlo simulation, rendering, and randomized algorithms. Two families of methods are standard: **rejection sampling**, which works for any shape, and exact **transform** methods that map uniform variables onto the shape without waste [Dev86]. For the square and rectangle the task is trivial (independent coordinates); for the circle and sphere a naive coordinate-based approach is biased, so the Jacobian of the mapping must be corrected.

27.3.2 Definitions

Definition 27.13 (Uniform Point Set). A point set is *uniform* in a region $\Omega \subseteq \mathbb{R}^d$ if the number of points in every measurable subset $A \subseteq \Omega$ is proportional to $\text{Vol}(A)$. Equivalently, the points are drawn from the uniform measure on Ω .

Definition 27.14 (Rejection Sampling). To sample uniformly in a bounded region Ω , sample candidate points uniformly in a bounding box $B \supseteq \Omega$ and *accept* those that lie inside Ω . The acceptance rate is $\text{Vol}(\Omega)/\text{Vol}(B)$.

27.3.3 Theory

For the axis-aligned rectangle $[x_1, x_2] \times [y_1, y_2]$, two independent uniform coordinates $u, v \sim \text{Unif}[0, 1]$ give the exact uniform point $(x_1 + u\Delta x, y_1 + v\Delta y)$.

For the disk of radius R , uniform coordinates in $[0, 1]^2$ must be transformed with the radial density of a disk: radii cluster near the center if drawn uniformly, so the inverse of the cumulative radial measure is used.

Theorem 27.15 (Uniform Disk and Sphere Sampling). *Let $u, v, w \sim \text{Unif}[0, 1]$ be independent. Then*

- $(\sqrt{u}R \cos 2\pi v, \sqrt{u}R \sin 2\pi v)$ is uniform in the disk of radius R ;
- $u^{1/3}R$ times a unit direction (drawn by Marsaglia's method below) is uniform in the ball of radius R ;
- $(2x\sqrt{1-x^2-y^2}, 2y\sqrt{1-x^2-y^2}, 1-2(x^2+y^2))$ for (x, y) uniform in the unit disk is uniform on the unit sphere [Mar72].

Proof sketch. The first claim follows from the Jacobian of the polar map: the density of a disk is uniform in the area element $r dr d\theta$, so the radius must be distributed as $\sqrt{u}$. The last claim is Marsaglia's method: a uniform point in the unit disk, lifted to the sphere by the inverse stereographic mapping scaled by $r \mapsto (2\sqrt{1-r^2}, 1-2r^2)$, preserves the uniform area measure [Mar72]. $\square$

Rejection sampling's acceptance rate is the volume ratio: $\pi/4 \approx 0.785$ for a disk in its bounding square, $\pi/6 \approx 0.524$ for a ball in its bounding cube. The expected number of trials to accept n points is n/rate .

For the *boundary* of a polygon the uniform measure is arc length, not area. A point sampled uniformly on the boundary is obtained by choosing an edge with probability proportional to its length and then sampling uniformly along that edge; equivalently, invert a uniform variate on the cumulative perimeter length (arc-length parametrization). For the rectangle $[x_1, x_2] \times [y_1, y_2]$ with width $w = x_2 - x_1$ and height $h = y_2 - y_1$, the edge probabilities are w, h, w, h over the total perimeter $2(w + h)$, so longer sides receive proportionally more points and no corner or side is over- or under-sampled.

27.3.4 Pseudocode

27.3.5 Parameter Selection

- **Rejection vs. Transform:** use rejection for arbitrary shapes (polygons, curved regions); use the exact transforms for disk/ball/sphere to avoid waste.
- **Seed:** fix the RNG seed for reproducibility in tests.

Algorithm 121 Uniform Random Points in Basic Shapes

```

function UNIFORMINRECTANGLE(x1, x2, y1, y2, n)
    return  $[(x_1 + u \cdot (x_2 - x_1), y_1 + v \cdot (y_2 - y_1)) \text{ for } (u, v) \text{ in Rand2D}(n)]$ 
end function

function UNIFORMINCIRCLE(cx, cy, R, n)
    return  $[(cx + \sqrt{u}R \cos 2\pi v, cy + \sqrt{u}R \sin 2\pi v) \text{ for } (u, v) \text{ in Rand2D}(n)]$ 
end function

function UNIFORMONSPHERE(n)
    pts  $\leftarrow []$ 
    while |pts| < n do
        x  $\leftarrow$  Unif[-1, 1], y  $\leftarrow$  Unif[-1, 1]
        if  $x^2 + y^2 < 1$  then ▷ reject points outside the unit disk
            s  $\leftarrow \sqrt{1 - x^2 - y^2}$ 
            pts.append((2xs, 2ys, 1 - 2(x2 + y2)))
        end if
    end while
    return pts
end function

function UNIFORMONRECTANGLEBOUNDARY(x1, x2, y1, y2, n)
    w  $\leftarrow x_2 - x_1$ , h  $\leftarrow y_2 - y_1$ 
    perim  $\leftarrow 2 \cdot (w + h)$ 
    pts  $\leftarrow []$ 
    for i in range(n) do
        s  $\leftarrow$  Unif[0, perim] ▷ arc length along the boundary
        if s < w then
            p  $\leftarrow (x_1 + s, y_1)$ 
        else if s < w + h then
            p  $\leftarrow (x_2, y_1 + (s - w))$ 
        else if s < 2w + h then
            p  $\leftarrow (x_2 - (s - w - h), y_2)$ 
        else
            p  $\leftarrow (x_1, y_2 - (s - 2w - h))$ 
        end if
        pts.append(p)
    end for
    return pts
end function

function UNIFORMONPOLYGONBOUNDARY(polygon, n)
    L  $\leftarrow$  edge lengths of polygon ▷ edges in CCW order
    cum  $\leftarrow$  cumulative sums of L
    pts  $\leftarrow []$ 
    for i in range(n) do
        s  $\leftarrow$  Unif[0, cum[last]]
        j  $\leftarrow$  first index with cum[j] > s ▷ binary search
        t  $\leftarrow (s - cum[j - 1]) / L[j]$ 
        p  $\leftarrow (1 - t) \cdot v_j + t \cdot v_{j+1}$ 
        pts.append(p)
    end for
    return pts
end function

```

27.3.6 Complexity

- **Transform Methods:** $O(n)$ time and space; no waste.
- **Rejection Sampling:** expected $O(n/\text{rate})$ time; worst-case unbounded, but the geometric distribution bound applies.
- **Boundary Sampling:** $O(n)$ for the rectangle (constant work per point); $O(n \log m)$ for a general m -gon via binary search on the cumulative perimeter lengths.

27.3.7 Usages

- **Monte Carlo Integration:** estimating areas and volumes (hit-or-miss sampling).
- **Rendering:** sampling light sources and area lights.
- **Initialization:** seeding particle and agent simulations.
- **Randomized Geometry:** point-in-polygon testing and random inputs for testing geometric algorithms.

27.3.8 Testing Methods and Edge Cases

- **Uniformity:** use a Chi-squared test on a sub-grid of the shape, or a 1D marginal (e.g., the x -marginal of a disk is uniform).
- **Boundary:** points on the boundary should be rare; include or exclude boundary deterministically.
- **Radius Distribution:** verify the disk histogram of r/R is linear (density of r proportional to r).
- **Zero-area / degenerate shapes:** a rectangle with $x_1 = x_2$ or $R = 0$ must return the boundary or an empty set deterministically.
- **Boundary density:** for a $w \times h$ rectangle, the number of points on the two horizontal sides must be in ratio $w : h$ (not $1 : 1$); points on the boundary must never fall inside the shape.

27.3.9 References

- Devroye, L. (1986). “Non-Uniform Random Variate Generation”. Springer.
- Marsaglia, G. (1972). “Choosing a point from the surface of a sphere”. The Annals of Mathematical Statistics.

27.4 Random Line Segment Generation

27.4.1 Overview

Generating random line segments in 2D or 3D is used for testing segment intersection and arrangement algorithms, for simulating sensor and cable networks, and for constructing random inputs [dBCvKO08]. The natural model samples the two endpoints independently from a uniform distribution over the domain, so that every segment is equally likely up to the position of its two endpoints. An alternative model fixes the segment length and samples a random direction, which is the right model for fibers, cracks, and insect paths.

27.4.2 Definitions

Definition 27.16 (Random Segment Model). A segment in $\mathbb{R}^d$ is a pair of endpoints a, b . In the *independent-endpoint model*, a and b are drawn independently from a uniform distribution over the domain Ω . In the *fixed-length model*, a length ℓ , a midpoint (or base point), and a uniform direction determine the segment.

27.4.3 Theory

The independent-endpoint model is exact: because each endpoint is uniform in Ω , the pair (a, b) is uniform in $\Omega \times \Omega$. A fixed length requires a uniform random direction, which in 2D is the angle $\theta \sim \text{Unif}[0, 2\pi)$ and in 3D a unit vector sampled uniformly on the sphere. A cheap 3D method picks a point in the unit ball and normalizes it (rejection of the z -axis bias of naive polar coordinates), or uses Marsaglia's method of Section 27.3.

Theorem 27.17 (Uniform Endpoints in a Box). *If a, b are independent and uniform in a box $\prod_i [x_i, y_i]$, then the endpoints and the segment are uniformly distributed over all segments whose endpoints lie in the box. The marginal density of the segment midpoint is not uniform: it concentrates near the center, since midpoints arise from many endpoint pairs.*

27.4.4 Pseudocode

Algorithm 122 Random Line Segment Generation

```

function RANDOMSEGMENT2D(x1, x2, y1, y2, n)
    return [RandomSegmentInBox( $x_1, x_2, y_1, y_2$ ) for  $_$  in range( $n$ )]
end function

function RANDOMSEGMENTINBOX(x1, x2, y1, y2, n)
     $a \leftarrow \text{UniformInRectangle}(x_1, x_2, y_1, y_2, 1)[0]$ 
     $b \leftarrow \text{UniformInRectangle}(x_1, x_2, y_1, y_2, 1)[0]$ 
    return ( $a, b$ )
end function

function RANDOMFIXEDLENGTHSEGMENT2D(cx, cy, ell, n)
    return [(( $cx, cy$ ), ( $cx + \ell \cos \theta, cy + \ell \sin \theta$ )) for  $\theta \sim \text{Unif}[0, 2\pi)$  in range( $n$ )]
end function

function RANDOMSEGMENT3D(x1, x2, y1, y2, z1, z2, n)
     $pts \leftarrow []$ 
    while  $|pts| < 2n$  do
         $u, v, w \sim \text{Unif}[0, 1)$ 
         $p \leftarrow (x_1 + u(x_2 - x_1), y_1 + v(y_2 - y_1), z_1 + w(z_2 - z_1))$ 
         $pts.append(p)$ 
    end while
    return [( $pts[2i], pts[2i + 1]$ ) for  $i$  in range( $n$ )]
end function

```

27.4.5 Parameter Selection

- **Endpoint model vs. fixed length:** choose the independent endpoint model for intersection and arrangement stress tests; fixed length for fiber and crack models.

- **Seed:** fix the RNG seed for reproducible random inputs.

27.4.6 Complexity

- **Time:** $O(n)$ to generate n segments.
- **Space:** $O(n)$ to store the segments; $O(1)$ streaming.

27.4.7 Usages

- **Testing segment intersection** algorithms with large random inputs.
- **Simulating sensor links**, cables, and cracks.
- **Arrangement** experiments and **sweep-line** stress tests.

27.4.8 Testing Methods and Edge Cases

- **Endpoint Range:** verify all endpoints lie in the box.
- **Count:** verify exactly n segments are returned.
- **Distinctness:** for the independent model, degenerate zero-length segments are possible but should be rare; decide whether to keep or reject them.
- **3D Direction:** for fixed length in 3D, verify the direction vectors have norm ℓ and are isotropically distributed.

27.4.9 References

- Devroye, L. (1986). “Non-Uniform Random Variate Generation”. Springer.
- de Berg, M., Cheong, O., van Kreveld, M., & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer.

27.5 Random Polygon Generation

27.5.1 Overview

Generating random convex and concave (simple) polygons in 2D provides stress tests for polygon algorithms: triangulation, point-in-polygon, convex partitioning, and boolean operations [dBCvKO08]. The two standard constructions are **angle-sorted points on a curve** for convex polygons, and **angular sorting around a center** for concave polygons. Both run in $O(n \log n)$ time and are easy to implement correctly.

27.5.2 Definitions

Definition 27.18 (Random Convex Polygon). A convex polygon with n vertices sampled by sorting n angles uniformly in $[0, 2\pi)$ and placing the corresponding points on the unit circle (or an ellipse/randomly scaled radius).

Definition 27.19 (Random Simple Polygon). A simple (non-self-intersecting) polygon built by ordering points by their polar angle around their centroid, giving a star-shaped polygon; a further random radial perturbation can introduce reflex (concave) vertices while preserving simplicity.

27.5.3 Theory

Sorting points by polar angle around any point inside their convex hull produces a simple polygon: the edges of the star-shaped polygon do not cross. For the convex case, points on a strictly convex curve (a circle or ellipse) sorted by angle give a convex polygon. The polygon is convex if and only if every interior angle is less than π ; an $O(n)$ check of the turn direction at each vertex certifies this.

Theorem 27.20 (Star-Shaped Polygonization). *If $P = \{p_1, \dots, p_n\}$ are in general position and c is a point in their convex hull, then the polygon obtained by connecting the points sorted by polar angle around c is simple and star-shaped with kernel containing c [dBCvKO08].*

Proof sketch. Every edge of the sorted polygon bounds a triangle with apex c that lies inside the polygon; adjacent triangles meet only along shared edges, so the polygon is simple and every point is visible from c . $\square$

27.5.4 Pseudocode

Algorithm 123 Random Polygon Generation

```

function RANDOMCONVEXPOLYGON( $n$ )
   $\theta \leftarrow \text{sort}(\text{Unif}[0, 2\pi) \text{ for } \_ \text{ in range}(n))$ 
   $r \leftarrow \text{Unif}[0.5, 1.0) \text{ for } \_ \text{ in range}(n)$ 
  return  $[(r_i \cos \theta_i, r_i \sin \theta_i) \text{ for } i \text{ in range}(n)]$ 
end function

function RANDOMSIMPLEPOLYGON( $n$ )
   $p \leftarrow \text{RandomConvexPolygon}(n)$ 
   $c \leftarrow \text{mean}(p)$ 
   $q \leftarrow \text{sort}(p, \text{key} = \text{angle around } c)$ 
  for  $i$  in  $\text{range}(n)$  do
    if reflex vertex at  $q_i$  then                                 $\triangleright$  flip to concave by radial move
       $q_i \leftarrow c + (1 + \varepsilon_i) \cdot (q_i - c)$ 
    end if
  end for
  return  $q$ 
end function

```

27.5.5 Parameter Selection

- **Radius range:** keep $r \in [0.5, 1.0)$ so points stay away from the origin and the polygon is not degenerate.
- **Perturbation:** small ε_i moves in the radial direction create reflex vertices without destroying simplicity; large moves can cause self-intersection.

27.5.6 Complexity

- **Time:** $O(n \log n)$ for sorting the angles; $O(n)$ for the perturbation and the convexity check.
- **Space:** $O(n)$.

27.5.7 Usages

- **Testing triangulation, point-in-polygon, and boolean operations** on large random inputs.
- **Benchmarking** convex partitioning (Keil–Snoeyink) and art gallery algorithms on concave polygons.

27.5.8 Testing Methods and Edge Cases

- **Simplicity**: verify no two edges intersect (except adjacent edges at shared vertices).
- **Convexity**: for the convex generator, check all interior angles $< \pi$ (positive turn direction at every vertex).
- **Vertex Count**: the polygon has exactly n distinct vertices.
- **Degenerate cases**: reject duplicate angles and points at the origin to keep the polygon simple.

27.5.9 References

- de Berg, M., Cheong, O., van Kreveld, M., & Overmars, M. (2008). “Computational Geometry: Algorithms and Applications”. Springer.
- Devroye, L. (1986). “Non-Uniform Random Variate Generation”. Springer.

27.6 Random Walk

27.6.1 Overview

A random walk is a mathematical object, known as a stochastic or random process, that describes a path consisting of a succession of random steps on some mathematical space such as the integers or a 2D/3D grid. In computational geometry and physics, random walks are used to model Brownian motion, diffusion, and to approximate the solutions of various partial differential equations (PDEs) through Monte Carlo methods [Pea05].

27.6.2 Definitions

Definition 27.21 (Step Size (ℓ)). The distance moved in a single step of the random walk. This may be fixed or drawn from a distribution (e.g., Gaussian).

Definition 27.22 (Lattice Walk). A random walk restricted to a discrete grid (e.g., integer coordinates). At each step, the walker moves to one of the neighboring lattice points with prescribed probabilities.

Definition 27.23 (Isotropic Walk). A walk where every direction is equally likely. In 2D, the step direction θ is uniformly distributed on $[0, 2\pi)$.

Definition 27.24 (Mean Squared Displacement (MSD)). A measure of how far the walker deviates from its starting point over time. For a simple d -dimensional random walk with step size ℓ , $\text{MSD}(n) = \langle |\mathbf{r}_n - \mathbf{r}_0|^2 \rangle = d \cdot \ell^2 \cdot n$, so $\text{MSD} \propto t$ [Spi76].

27.6.3 Theory

The theory of random walks is closely related to the diffusion equation $\frac{\partial \phi}{\partial t} = D \nabla^2 \phi$ [Spi76].

1. **1D Walk:** A walker starts at 0 and moves +1 or -1 with probability 1/2. After n steps, the expected position is 0, but the standard deviation is $\sqrt{n}$.
2. **2D/3D Walk:** The walker moves in a random direction. Polya's Theorem [P21] states that a random walk on a 2D lattice is “recurrent” (it will return to the start with probability 1), while in 3D it is “transient” (it may never return).
3. **Walk on Meshes:** A random walk on a mesh jumps from a vertex to one of its adjacent vertices. The probability of choosing a neighbor can be uniform (1/degree) or weighted by edge length or cotangent weights to simulate specific geometric properties.

Theorem 27.25 (Polya's Recurrence Theorem). *A simple random walk on the $\mathbb{Z}^d$ lattice is recurrent (returns to the origin with probability 1) for $d = 1$ and $d = 2$, and transient (returns with probability less than 1) for $d \geq 3$. Specifically, the return probability in $d = 3$ is approximately 0.3405 [P21].*

Proof. The probability of return is related to the Green's function $G(\mathbf{0}) = \sum_{n=0}^{\infty} P_n(\mathbf{0} \rightarrow \mathbf{0})$, which is the expected number of visits to the origin. The walk is recurrent if and only if $G(\mathbf{0}) = \infty$. For a d -dimensional walk, the return probability at step $2n$ is:

$$P_{2n}(\mathbf{0} \rightarrow \mathbf{0}) \sim \frac{C_d}{n^{d/2}}$$

for some constant $C_d > 0$. The series $\sum_{n=1}^{\infty} n^{-d/2}$ converges if and only if $d/2 > 1$, i.e., $d > 2$. Therefore, $G(\mathbf{0}) = \infty$ for $d \leq 2$ (recurrent) and $G(\mathbf{0}) < \infty$ for $d \geq 3$ (transient) [Spi76]. $\square$

Example 27.26 (1D Random Walk: Expected Position After 100 Steps). Consider a 1D random walk starting at position 0, taking steps of +1 or -1 with equal probability. After $n = 100$ steps:

- Expected position: $\mathbb{E}[X_{100}] = 0$.
- Standard deviation: $\sigma = \sqrt{100} = 10$.
- Expected squared displacement: $\mathbb{E}[X_{100}^2] = 100$.

This means that while the walker is equally likely to be at any position from -100 to +100, most outcomes will cluster within about 10 units of the origin.

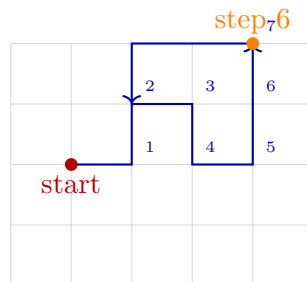


Figure 27.2: Random walk on a grid: at each step, move to a uniformly random neighboring vertex.

Algorithm 124 2D Random Walk

```
function RANDOMWALK2D(start_point, num_steps, step_size)
    current  $\leftarrow$  start_point
    path  $\leftarrow$  [current]
    for i in range(num_steps) do                                      $\triangleright$  1. Choose random direction
         $\theta \leftarrow \text{random.uniform}(0, 2\pi)$ 
                                                                     $\triangleright$  2. Update position
        dx  $\leftarrow$  step_size  $\cdot$  cos( $\theta$ )
        dy  $\leftarrow$  step_size  $\cdot$  sin( $\theta$ )
        current  $\leftarrow$  Point(current.x + dx, current.y + dy)
        path.append(current)
    end for
    return path
end function
```

27.6.4 Pseudocode

Simple 2D Random Walk (Continuous)

Random Walk on a Mesh

Algorithm 125 Random Walk on a Mesh

```
function MESHRANDOMWALK(mesh, start_v, num_steps)
    current_v  $\leftarrow$  start_v
    visited_v  $\leftarrow$  [current_v]
    for i in range(num_steps) do
        neighbors  $\leftarrow$  mesh.GetNeighbors(current_v)            $\triangleright$  1. Choose a random neighbor
        current_v  $\leftarrow$  random.choice(neighbors)
        visited_v.append(current_v)
    end for
    return visited_v
end function
```

27.6.5 Parameter Selection

- **Step Size:** Constant step size vs. variable (Gaussian) step size.
- **Connectivity:** On a grid, 4-connectivity (von Neumann) vs. 8-connectivity (Moore).
- **Weights:** In weighted random walks, the probability of a step is proportional to some value (e.g., $e^{-\Delta E/kT}$).

27.6.6 Complexity

- **Time Complexity:** $O(n)$ where n is the number of steps.
- **Space Complexity:** $O(n)$ to store the full path, or $O(1)$ if only the final position is needed.

27.6.7 Usages

- **Monte Carlo Simulation:** Estimating areas or volumes of complex shapes (Hit-or-Miss Monte Carlo).

- **Mesh Analysis:** Using random walks to calculate “closeness” between vertices or to partition meshes.
- **Generative Art:** Creating organic, branching patterns (e.g., Diffusion-Limited Aggregation).
- **Network Science:** Identifying communities in a graph or calculating PageRank.
- **Finance:** Modeling stock price movements (Geometric Brownian Motion).
- **Physics:** Simulating the movement of gas particles or the propagation of heat.

27.6.8 Testing Methods and Edge Cases

- **Mean Position:** Verify that after many trials, the average final position of an isotropic walk is the starting point.
- **Scaling Law:** Verify that the distance from the start grows as $\sqrt{n}$ on average.
- **Boundaries:** Test how the walk behaves when it hits a wall (e.g., reflection vs. absorption).
- **Mesh Boundaries:** Ensure the walker doesn’t “fall off” the edge of an open mesh.
- **Looping:** Observe how often the walker visits the same vertex in 2D vs. 3D.

27.6.9 References

- Pearson, K. (1905). “The Problem of the Random Walk”. *Nature*.
- Polya, G. (1921). “Über eine Aufgabe der Wahrscheinlichkeitsrechnung betreffend die Irrfahrt im Straßennetz”. *Mathematische Annalen*.
- Spitzer, F. (1976). “Principles of Random Walk”. Springer.
- Wikipedia: Random walk

27.7 Self-Avoiding Walk

27.7.1 Overview

A self-avoiding walk (SAW) is a sequence of moves on a lattice (such as a 2D or 3D grid) that does not visit the same point more than once. Unlike a simple random walk, which can cross its own path, a SAW must satisfy the constraint of self-exclusion. This exclusion makes SAWs significantly more difficult to analyze and simulate, as the “future” path depends on the entire “history” of the walk [MS96].

27.7.2 Definitions

Definition 27.27 (Lattice). A regular grid of points (e.g., square, hexagonal, cubic). The lattice determines the set of allowed moves at each step.

Definition 27.28 (Length (n)). The number of steps in the walk. The walk consists of $n + 1$ vertices and n edges.

Definition 27.29 (Connective Constant (μ)). A lattice-specific constant describing the exponential growth rate of the number c_n of possible SAWs of length n : $c_n \sim A\mu^n n^{\gamma-1}$ as $n \rightarrow \infty$, where γ is a universal critical exponent depending only on the dimension [MS96].

Definition 27.30 (End-to-End Distance). The Euclidean distance between the starting point and the final point of the walk. For a SAW of length n , the mean end-to-end distance scales as $\langle R^2 \rangle \sim n^{2\nu}$ where ν is the Flory exponent ($\nu \approx 3/4$ in $d = 2$) [Flo53].

27.7.3 Theory

The number of simple random walks of length n on a d -dimensional lattice is exactly $(2d)^n$. For SAWs, there is no simple formula. The number of SAWs c_n is known to behave as $c_n \sim A\mu^n n^{\gamma-1}$, where μ depends on the lattice and γ is a “universal” exponent that depends only on the dimension [MS96].

Generating SAWs is a challenge due to **attrition**: if you build a walk step-by-step randomly, you often get “trapped” where all neighboring points have already been visited. This leads to several simulation strategies:

1. **Simple Sampling**: Generate random walks and discard them if they intersect themselves. (Extremely inefficient for large n).
2. **Pivot Algorithm**: Start with a valid SAW and apply a symmetry operation (rotation or reflection) to a portion of the walk. If the result is still self-avoiding, accept it. This is the most efficient known method for generating long SAWs.
3. **Rosenbluth Method**: A weighted sampling method that avoids immediate self-intersections but suffers from diminishing weights.

Theorem 27.31 (Connective Constant Bounds). *For the square lattice $\mathbb{Z}^2$, the connective constant satisfies $\mu \leq 2 + \sqrt{2} \approx 3.414$ (the Madras–Slade bound). Exact values are known only for self-avoiding walks on certain special lattices. For the hexagonal lattice, $\mu = \sqrt{2 + \sqrt{2}} \approx 1.848$ [MS96].*

Proof. The upper bound $\mu \leq 2 + \sqrt{2}$ is proved using the lace expansion technique, which decomposes the SAW into a random walk plus correction terms that account for self-avoidance. The key identity relates the two-point function of the SAW to that of the simple random walk through a renormalization argument. For the hexagonal lattice, the exact value follows from a combinatorial argument using the correspondence between SAWs and the zero-field Ising model at the critical temperature [MS96]. $\square$

Example 27.32 (Number of SAWs on $\mathbb{Z}^2$ for Small n). The exact counts c_n of self-avoiding walks on the square lattice starting from the origin:

n	c_n
1	4
2	12
3	36
4	100
5	284
6	780

From these values, one can estimate $\mu \approx c_{n+1}/c_n \rightarrow \mu \approx 2.64$ as n grows, converging to the true value $\mu \approx 2.638$.

27.7.4 Pseudocode

Simple Recursive SAW Generator

Note that Algorithm 126 uses exhaustive backtracking. For large n , the **pivot algorithm** is far more efficient, achieving near- $O(n)$ amortized cost per sample [MS96].

Algorithm 126 Generate Self-Avoiding Walk (Recursive)

```

function GENERATESAW(path, n, lattice)
  if len(path) = n then return path
  end if
  current ← path[-1]
  neighbors ← lattice.GetNeighbors(current)
  random.shuffle(neighbors)
  for all neighbor ∈ neighbors do
    if neighbor ∉ path then
      result ← GenerateSAW(path + [neighbor], n, lattice)
      if result ≠ None then return result
    end if
  end if
  end for
  return None
end function

```

▷ Shuffle neighbors to explore randomly

▷ 1. Recursive attempt

▷ 2. Dead end (Backtrack) **return** None

27.7.5 Parameter Selection

- **Lattice Type:** Square (4 neighbors) and Hexagonal (3 neighbors) are common in 2D.
- **Algorithm:** Use the **Pivot Algorithm** for generating very long SAWs (thousands of steps) as it is much faster than simple recursion or sampling.

27.7.6 Complexity

- **Simple Recursion:** $O(\mu^n)$ in the worst case, as it is an exhaustive search.
- **Pivot Algorithm:** $O(n)$ or even $O(n^{0.5})$ per successful update, making it feasible for $n \approx 10^6$.
- **Space Complexity:** $O(n)$ to store the coordinates of the walk.

27.7.7 Usages

- **Polymer Physics:** SAWs are the standard model for “linear polymers” in a good solvent, where the exclusion principle represents the “excluded volume” of atoms [Flo53].
- **Proteins:** Modeling the backbone of a protein during folding.
- **Network Science:** Analyzing paths in social or communication networks where nodes shouldn’t be repeated.
- **Material Science:** Studying the distribution of chains in a porous medium.
- **Combinatorics:** A fundamental problem in counting and probability.

27.7.8 Testing Methods and Edge Cases

- **No Intersections:** Verify that all points in the generated path are unique.
- **Connectivity:** Ensure each point in the path is a valid neighbor of the previous point on the lattice.

- **Connective Constant:** Verify that for small n , the number of generated SAWs matches known exact counts (Theorem [27.32](#)).
- **Lattice Boundaries:** If the walk is confined to a box, ensure it handles the boundaries correctly.
- **Trap Detection:** Verify the algorithm correctly backtracks or terminates when the walker is surrounded by visited points.

27.7.9 References

- Madras, N., & Slade, G. (1996). “The Self-Avoiding Walk”. Birkhäuser.
- Flory, P. J. (1953). “Principles of Polymer Chemistry”. Cornell University Press.
- Kennedy, T. (2002). “A faster implementation of the pivot algorithm for self-avoiding walks”. Journal of Statistical Physics.
- Wikipedia: Self-avoiding walk

Chapter 28

Shape Representation and Reconstruction

28.1 Shape Representation

A **shape** in this chapter is a surface: a compact connected two-dimensional manifold embedded in $\mathbb{R}^3$, or, for terrain modeling, a height function over a planar region. **Shape representation** describes such a surface with a data structure supporting the queries and algorithms of the surrounding chapters: rendering, intersection, simplification, parameterization, simulation, and — in the reverse direction — reconstruction from measurements. Representations fall into two families. **Discrete representations** store a finite set of primitives: a point cloud stores an unorganized set of samples with no connectivity, while a mesh stores a piecewise-linear surface on a simplicial complex (Section 22.2 in Chapter 22). **Continuous representations** store a mathematical description: a parametric patch, an implicit equation $f(x, y, z) = 0$, a signed distance function (Section 28.5), or a level set of a scalar field (Section 28.6).

- **Point clouds** are the raw output of 3D scanning and photogrammetry; they carry no topology, which must be re-discovered (Section 28.3).
- **Meshes** add topology: a triangle mesh is a 2D simplicial complex whose underlying space approximates the surface, stored with the half-edge structure of Chapter 21 and validated by Chapter 30.1.
- **Parametric representations** (Section 28.2) map a planar domain onto the surface; they are native to CAD but do not easily guarantee watertightness.
- **Implicit representations** (Sections 28.5 and 28.7) describe the surface as the zero set of a function; they are watertight by construction and make inside/outside and boolean operations trivial.

The choice follows the operations that come after it: a point cloud suffices for display and nearest-neighbor queries but not for finite-element assembly; an implicit function answers inside/outside queries and supports boolean CSG but resists direct manipulation; a mesh supports refinement, smoothing, and simulation but must be kept manifold and intersection-free. Because real acquisitions are point clouds, the central task of this chapter is the conversion from point data to a faithful, topologically correct surface.

28.2 Implicit and Parametric Representations

A **parametric surface** is the image $x(D)$ of a map $x: D \rightarrow \mathbb{R}^3$ from a planar parameter domain D (typically a set of patches). An **implicit surface** is the zero set of a function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$,

$S = \{x : f(x) = 0\}$. The two are dual in a practical sense. A parametric surface is rendered and meshed directly (sample D on a grid and connect the samples), and its points are enumerated cheaply; but the union or intersection of two patches requires numerical root finding, and inside/outside is ill-defined without an orientation convention. An implicit surface gives the test $f(x) \leq 0$ for free, its boolean union is simply $\min(f_1, f_2)$ and its intersection $\max(f_1, f_2)$, and watertightness is automatic; the cost is that meshing requires a polygonization such as the marching cubes algorithm of Section 28.6 [LC87].

CAD systems work overwhelmingly with parametric patches: tensor-product Bézier surfaces and their rational extension, non-uniform rational B-splines (NURBS), are the standard primitives of the industry [Far02, PT97]. A NURBS surface of degree (p, q) is

$$x(u, v) = \frac{\sum_i \sum_j w_{ij} P_{ij} N_{i,p}(u) N_{j,q}(v)}{\sum_i \sum_j w_{ij} N_{i,p}(u) N_{j,q}(v)},$$

where P_{ij} are control points, w_{ij} weights, and $N_{i,p}$ the B-spline basis functions. The parametrization is the main asset: a designer controls the shape through a coarse control polygon, and exact refinement (knot insertion) produces dense meshes. The main liability is that a NURBS model is watertight only by explicit agreement of the boundary curves of neighboring patches.

Implicit representations arise naturally from measurement rather than design: the signed distance functions of Section 28.5 and the indicator functions of Section 28.7 are both obtained from point clouds by fitting a function to the data, and both are regular enough (Lipschitz, often smooth) to be numerically well behaved; isosurface extraction (Section 28.6) turns them into meshes. The two families meet in parameterization: a triangle mesh has no intrinsic parameter domain, and Chapter 20.12 constructs one by mapping the mesh to a planar domain, after which patch-based methods apply.

28.3 Reconstruction from Point Clouds

28.3.1 The Problem

The **surface reconstruction problem** is: given a finite point set $P \subset \mathbb{R}^3$ sampled from an unknown surface S , compute a mesh $\hat{S}$ topologically and geometrically close to S . The problem is ill posed — a finite sample is consistent with infinitely many surfaces — so any solution must be guided by an a priori model: that S is smooth and that P samples it densely enough relative to its features. The sampling conditions quantify density in terms of the geometry of S , so that a sphere and a thin tube of similar sampling density receive similar treatments.

Definition 28.1 (Local Feature Size). Let S be a surface and let M be its medial axis (Definition 20.79 in Chapter 20.12). The **local feature size** of S at $x \in S$ is the distance from x to M : $\text{lfs}(x) = \text{dist}(x, M)$. It is a positive continuous function on S and is 1-Lipschitz: $\text{lfs}(x) \leq \text{lfs}(y) + \text{dist}(x, y)$.

Definition 28.2 (ε -Sample). A point set $P \subset S$ is an ε -**sample** of S if every $x \in S$ is within distance $\varepsilon \cdot \text{lfs}(x)$ of a point of P , i.e. $\sup_{x \in S} \min_{p \in P} \text{dist}(x, p) / \text{lfs}(x) \leq \varepsilon$.

Definition 28.3 (r -Sample). A point set $P \subset S$ is an r -**sample** of S if every $x \in S$ is within distance $r \cdot \text{lfs}(p)$ of the sample p nearest to x : $\sup_{x \in S} \min_{p \in P} \text{dist}(x, p) / \text{lfs}(p) \leq r$.

By the Lipschitz property the two conditions are equivalent up to a factor $1 + O(\varepsilon)$: an ε -sample is an r -sample for $r = \varepsilon / (1 - \varepsilon)$, and conversely. Both require more points where the surface is thin or curved (small lfs) and fewer on flat regions — the same feature-adaptive philosophy as the size-field meshing of Section 22.4 — in contrast to the uniform-density samples of the discrepancy theory of [Mat99].

28.3.2 Voronoi-Based Sampling Guarantees

The sampling conditions translate into strong statements about the Voronoi diagram of P (Chapter 11) and its dual Delaunay complex (Chapter 12). The key object is the restricted Delaunay complex, whose dual Voronoi faces meet the surface.

Definition 28.4 (Restricted Delaunay Complex). A Delaunay simplex σ of P belongs to the **restricted Delaunay complex** $\text{Del}(P, S)$ if its dual Voronoi face $V(\sigma)$ intersects S . It is a subcomplex of the Delaunay complex of P .

Theorem 28.5 (Sampling Guarantee). *Let S be a compact smooth surface and P an ε -sample of S . There is a constant ε_0 such that for every $\varepsilon \leq \varepsilon_0$ the restricted Delaunay complex $\text{Del}(P, S)$ is a two-dimensional manifold complex homeomorphic to S and within Hausdorff distance $O(\varepsilon \cdot \max_S \text{lfs})$ of S .*

Proof sketch. The proof has three steps. First, the poles of P (Section 28.4) approximate the surface normals: the two farthest Voronoi vertices of each cell lie on opposite sides of S at distance $\Omega(\text{lfs})$ from it. Second, every simplex of $\text{Del}(P, S)$ has circumradius $O(\varepsilon \cdot \text{lfs})$, so its vertices lie in a thickening of S of that width, strictly between the two sheets of the medial axis. Third, projecting the restricted triangles onto S along the normals is a homeomorphism for ε below ε_0 ; the classical proofs establish $\varepsilon_0 = 0.06$ for the crust and cocone filters [AB99, ACDL00]. $\square$

The theorem is the foundation of all Delaunay-based reconstruction: the surface is *already present* as a subcomplex of the Delaunay complex, and the algorithms of Section 28.4 differ only in how they select it. The same argument with the r -sample condition gives the provably good surface meshing theory of Boissonnat and Oudot [BO05]: for $r \leq 0.09$ the restricted Delaunay complex of an r -sample is homeomorphic to the surface and refinable with quality guarantees (Section 12.12, [She02]).

28.3.3 Delaunay-Based Meshing Approaches

The Voronoi and Delaunay structures organize reconstruction into a four-stage pipeline:

1. **Triangulation:** compute the Delaunay complex of P , a tetrahedralization built by randomized incremental insertion in expected $O(n \log n)$ time (worst-case size $\Theta(n^2)$ in $\mathbb{R}^3$, Theorem 12.41).
2. **Filtering:** select the triangles plausibly on S — by empty-sphere radius (alpha complex, Definition 12.38), pole orientation (crust), cocone angle, or the ball-pivoting empty-ball rule (Section 28.4).
3. **Cleaning:** remove non-manifold edges, delete small spurious components, and fill holes, using the topological checks of Chapter 30.
4. **Refinement:** re-mesh the output with the guaranteed-quality Delaunay refinement of Section 12.12 [She02].

For terrain data the pipeline simplifies to two dimensions: samples of a height function $z = h(x, y)$ are projected to the plane, the 2D Delaunay triangulation of the projection is the triangulated irregular network (TIN) — the natural choice by the maximin angle property (Chapter 12) — and rivers and breaklines enter as constraints of the constrained Delaunay triangulation, as in Chapter 22.

28.3.4 Complexity

Reconstruction is dominated by the Delaunay tetrahedralization of stage 1: expected $O(n \log n)$ time and $O(n)$ space for well-distributed samples, and $O(n^2)$ worst case in $\mathbb{R}^3$ (Theorem 12.41); filtering, cleaning, and hole filling are linear in the $O(n)$ triangles. Noise and outliers are handled only partially by the filter-based methods of the next section; the least-squares methods of Sections 28.5 and 28.7 are the robust alternatives.

28.4 Surface Reconstruction

28.4.1 1. Overview

This section develops the four classical Delaunay-based algorithms that turn Theorem 28.5 into working software. The *alpha-shape* family [EKS83, EM94a] selects triangles by an empty-sphere radius; the *crust* [ABE98, AB99] selects triangles through the poles of the Voronoi cells, whose directions approximate the surface normals; the *cocone* [ACDL00] sharpens the crust filter to a thin cone around the pole direction and adds manifold checks that produce a watertight output; and the *ball-pivoting algorithm* [BMR⁺99] grows a mesh directly from the points, without building the Delaunay complex, by pivoting a ball of fixed radius around the front. All four share the filter-and-clean pipeline of Section 28.3.

28.4.2 2. Definitions

Definition 28.6 (Pole and Pole Vector). A **pole** of a sample p is a vertex of the Voronoi cell $V(p)$ farthest from p ; when $V(p)$ is unbounded the pole is taken on the far side of the surface, at distance $\geq \text{lfs}(p)$ from p . The **pole vector** $v_p = p^+ - p$, where p^+ is the pole, approximates the inward normal of S at p up to an angle $O(\varepsilon)$.

Definition 28.7 (Cocone). Let $\theta \in (0, \pi/2)$, typically $\theta = \arcsin(\varepsilon)$. The **cocone** of p with pole vector v_p is the subset of the Voronoi cell

$$C(p) = \left\{ x \in V(p) : \frac{(x - p) \cdot v_p}{\|x - p\|} \geq \sin \theta \right\},$$

a thin double cone around the pole direction containing the surface near p .

Definition 28.8 (Alpha Shape). The **alpha shape** of P at $\alpha \geq 0$ is the union of the simplices of the alpha complex at α (Definition 12.38): those Delaunay simplices whose smallest empty circumsphere has radius at most α , with all their faces. As α grows from 0 to ∞ the alpha shape interpolates from the point set to the convex hull of P [EKS83]; the family is studied in Section 20.11 of Chapter 20.1.

Definition 28.9 (Empty-Ball Property). A ball of radius ρ through three points of P is *empty* if its interior contains no point of P . The ball-pivoting algorithm accepts a triangle only if some empty ball of radius ρ passes through its three vertices.

28.4.3 3. Theory

Theorem 28.10 (Crust and Cocone Guarantee). *Let S be a smooth surface and P an ε -sample of S with $\varepsilon \leq 0.06$. The crust of P [AB99] and the cocone of P [ACDL00] each contain a set of triangles forming a surface homeomorphic to S and within Hausdorff distance $O(\varepsilon \cdot \text{lfs})$ of S . The cocone output is additionally watertight, and every cocone triangle has circumradius $O(\varepsilon \cdot \text{lfs})$ and makes an angle $O(\varepsilon)$ with the tangent plane of S .*

Proof sketch. By Theorem 28.5 the restricted Delaunay complex $\text{Del}(P, S)$ is homeomorphic to S . For the crust one shows that a Delaunay triangle of $P \cup \{\text{poles}\}$ with all three vertices in P is a triangle of the restricted complex, and conversely. For the cocone one shows the circumcenter of every restricted triangle lies in $C(p)$ for each of its vertices, because the pole direction makes an angle $O(\varepsilon)$ with the normal. The umbrella property (every interior vertex has one cycle of incident triangles) then yields the homeomorphism and the watertightness [ACDL00, Dey07]. $\square$

Theorem 28.11 (Ball-Pivoting Correctness). *Let S be a watertight surface and P a sample of S such that (i) no feature of S of diameter smaller than ρ separates two of its points (the ball of radius ρ never crosses the medial axis) and (ii) the ball can always be pivoted from any two adjacent samples to a third sample. Then the ball-pivoting algorithm [BMR⁺99] produces a triangle mesh homeomorphic to S .*

Proof sketch. Condition (i) keeps the pivoted ball from “jumping” across a thin feature, so the front advances without self-intersection; condition (ii) keeps the front from stalling in gaps. Under both, each step adds a triangle whose interior contains no other point of P , the front propagates over all of S by connectivity, and the output is a triangulation of the same topological type as S [BMR⁺99]. $\square$

The alpha-shape family has a corresponding guarantee in the r -sample model: for $r \leq 0.09$ some value of α makes the alpha complex contain exactly the restricted Delaunay complex, so the alpha shape reconstructs the surface [BO05]. In practice α is chosen by inspection: too small fragments the surface, too large fills the concavities of the convex hull.

Noise and outliers. The guarantees assume an ideal sample lying on S ; real scans perturb each point with noise and add outliers away from S . Delaunay-based filters are fragile under both: an outlier is a legitimate Delaunay vertex whose triangles may survive the filter, and noise inflates circumradii, breaking the empty-sphere conditions. The remedies are pre-processing (project each point onto the local tangent plane fitted to its nearest neighbors; prune outliers by nearest-neighbor statistics) and post-processing (the implicit least-squares methods of Sections 28.5 and 28.7 smooth by construction). Among the Delaunay methods the cocone is the most robust, because its manifold checks remove outlier-induced non-manifold structures [ACDL00, Dey07].

28.4.4 4. Pseudocode

Algorithm 127 Crust (Amenta–Bern)

Input: an ε -sample P of a smooth surface S .

Output: a triangle mesh homeomorphic to S .

```

1: function CRUST( $P$ )
2:   compute the Voronoi diagram of  $P$  (Chapter 11)
3:   for each sample  $p \in P$  do
4:      $p^+ \leftarrow$  pole of  $V(p)$ , the farthest Voronoi vertex of  $V(p)$ 
5:   end for
6:    $P' \leftarrow P \cup \{p^+ : p \in P\}$ 
7:   compute the Delaunay triangulation  $DT(P')$ 
8:   keep every triangle of  $DT(P')$  whose three vertices all lie in  $P$ 
9:   remove small connected components of the retained triangles
10:  return the retained triangles
11: end function

```

The 2D analogue of Algorithm 127, which reconstructs a planar curve, keeps the edges of $DT(P \cup \text{poles})$ whose endpoints both lie in P and provably recovers the curve [ABE98].

Algorithm 128 Cocone (Amenta–Choi–Dey–Leekha)

Input: an ε -sample P of a smooth surface S , angle $\theta = \arcsin(\varepsilon)$.**Output:** a watertight triangle mesh homeomorphic to S .

```

1: function COCONE( $P, \theta$ )
2:   compute the Voronoi diagram of  $P$ 
3:   for each sample  $p \in P$  do
4:      $p^+ \leftarrow$  pole of  $V(p)$ ;  $v_p \leftarrow$  unit pole vector
5:      $C(p) \leftarrow \{x \in V(p) : (x - p) \cdot v_p \geq \sin \theta \|x - p\|\}$ 
6:   end for
7:   keep every Delaunay triangle whose circumcenter lies in  $C(p)$  for some vertex  $p$ 
8:   remove non-manifold edges (umbrella property)
9:   fill small holes left by the filter
10:  return the cleaned triangle mesh
11: end function

```

Algorithm 129 Ball-Pivoting Algorithm (Bernardini et al.)

Input: a point set P , ball radius ρ , a spatial grid for neighbor queries.**Output:** a triangle mesh of the surface sampled by P .

```

1: function BALLPIVOT( $P, \rho$ )
2:    $G \leftarrow$  uniform grid over  $P$  for neighbor queries
3:   find a seed triangle  $(a, b, c)$  whose three points admit an empty ball of radius  $\rho$ 
4:   front  $\leftarrow$  the three oriented edges of the seed triangle
5:   while front is not empty do
6:     pop an oriented edge  $(a, b)$  from the front
7:     pivot: rotate the ball of radius  $\rho$  through  $a$  and  $b$  around the edge  $(a, b)$  until it
        touches a point of  $P$ 
8:     if a point  $c$  is touched and the ball through  $a, b, c$  is empty then
9:       add triangle  $(a, b, c)$  to the mesh
10:      push the two new oriented edges of  $(a, b, c)$  onto the front
11:    else
12:      mark  $(a, b)$  as a boundary edge of the mesh
13:    end if
14:  end while
15:  return the triangle mesh
16: end function

```

28.4.5 5. Parameter Selection

- **Alpha:** must exceed the sample spacing (so triangles span real patches) but stay below the feature size (so the shape does not bridge concavities); sweep α and stop at the smallest value for which the output is a manifold of the expected genus (Chapter 30).
- **Ball radius ρ :** bounded below by the maximum distance between adjacent samples (or the front stalls) and above by the minimum feature size (or the ball jumps across a thin feature). Run several passes with increasing ρ to fill holes.
- **Cocone angle θ :** the theory calls for $\theta = \arcsin(\varepsilon)$ with ε estimated from the maximum sample gap relative to the feature size; widen θ slightly to tolerate noise.
- **Cleaning thresholds:** the sizes of removed components and of filled holes trade recall (more true surface) against precision (fewer artifacts).

28.4.6 6. Complexity

- **Crust and cocone:** dominated by the Delaunay tetrahedralization (expected $O(n \log n)$, worst case $O(n^2)$ in $\mathbb{R}^3$, Theorem 12.41); pole computation and filtering are linear passes over the $O(n)$ cells and triangles.
- **Ball pivoting:** with a grid giving $O(1)$ expected neighbor queries, each pivot inspects $O(1)$ candidates and the total is $O(n)$ for a surface of n samples [BMR⁺99].
- **Space:** $O(n)$.

28.4.7 7. Usages

- **Reverse engineering:** scanned parts are reconstructed with the crust, cocone, or ball pivoting, then refined and smoothed for CAD; this is the front end of metrology and of the surface reconstruction of Section 28.8.
- **Scientific computing:** reconstructed surfaces seed the mesh-generation pipeline of Chapter 22, whose refinement (Section 12.12) produces quality meshes for simulation.
- **Multiscale analysis:** the alpha shape provides a one-parameter family of reconstructions, the reference for evaluating the other filters.

28.4.8 8. Testing Methods and Edge Cases

- **Synthetic verification:** sample a sphere and a torus analytically, reconstruct, and verify homeomorphism via the Euler characteristic and Betti numbers (Chapter 30) and the Hausdorff error against the known surface.
- **Non-uniform sampling:** density varying by two orders of magnitude defeats ball pivoting (one ρ cannot fit all) but is handled by the feature-adaptive filters; verify the cocone and crust degrade only where the ε -condition is violated.
- **Boundary, open surfaces, and sharp features:** the crust and ball pivoting assume a closed surface, so on open patches boundary edges must be reported, not filled as holes; near edges of zero local feature size the ε -sample model fails, and the feature edges must survive rather than be rounded away.
- **Noise and outliers:** add Gaussian noise and outliers, and verify that normal estimation and outlier pruning keep the reconstruction valid; nearly flat regions should produce no surviving slivers.

28.4.9 9. References

- Edelsbrunner, H., Kirkpatrick, D. G., Seidel, R. [EKS83], alpha shapes.
- Edelsbrunner, H., Mücke, E. P. [EM94a], 3D alpha shapes.
- Amenta, N., Bern, M., Eppstein, D. [ABE98], the curve crust.
- Amenta, N., Bern, M. [AB99], the surface crust.
- Amenta, N., Choi, S., Dey, T. K., Leekha, N. [ACDL00], the cocone.
- Bernardini, F., Mittleman, J., Rushmeier, H., Silva, C., Taubin, G. [BMR⁺99], ball pivoting.
- Dey, T. K. [Dey07], curve and surface reconstruction.

28.5 Signed Distance Functions

Definition 28.12 (Signed Distance Function). Let $\Omega \subset \mathbb{R}^3$ be a compact region with boundary $\partial\Omega = S$. The **signed distance function** of Ω is

$$\phi(x) = \begin{cases} \text{dist}(x, S) & \text{if } x \in \Omega, \\ -\text{dist}(x, S) & \text{if } x \notin \Omega. \end{cases}$$

The surface is the zero level set $\phi^{-1}(0)$; the interior of Ω is $\phi > 0$ and the exterior $\phi < 0$ (the sign convention varies).

The signed distance function is the canonical implicit representation because of three properties. It is 1-Lipschitz and satisfies the **Eikonal equation** $|\nabla\phi| = 1$ almost everywhere, so its gradients are unit vectors along the shortest paths to the surface. It is a solid representation: the sign test $\phi(x) > 0$ answers inside/outside queries in $O(1)$ time and boolean operations are min and max. And the distance field itself is the data many applications need — collision detection, closest-point queries, and the front propagation of Section 28.6 all run on ϕ .

Signed distance functions are computed from every input type:

From a mesh The distance to a triangle mesh is the minimum over vertices, edges, and faces; the fast marching method of Section 28.6 propagates it through a voxel grid in $O(N \log N)$ time for N voxels, with the sign decided once per voxel by a ray-crossing or winding-number test.

From a grid A binary volume (a segmented CT or MRI scan, or a rasterized solid) is converted to a signed distance by a linear-time Euclidean distance transform over the grid.

From a point cloud Oriented points p_i with normals n_i define a local tangent-plane field; ϕ is the signed distance to the plane through the nearest sample (moving least squares), the bridge between the raw scans of Section 28.3 and the implicit world.

Once ϕ is available, a mesh is extracted as the isosurface $\phi^{-1}(0)$ by marching cubes [LC87]; conversely, signed distance fields are computed from geometry and stored on a grid or octree to answer physical queries (penetration, proximity, closest point) directly — the standard interface between CAD geometry and the rendering, robotics, and simulation stacks of this book.

28.6 Level Sets and Front Propagation

Definition 28.13 (Level Set). For a function $\phi: \mathbb{R}^3 \rightarrow \mathbb{R}$ and a value c , the **level set** of ϕ at c is $L_c = \{x : \phi(x) = c\}$. When ϕ is a signed distance function, the level sets L_c , $c \neq 0$, are parallel surfaces to $S = L_0$.

Level-set methods represent the surface as a level set of an evolving function and move the surface by evolving the function, not by moving points on it [OF03]. The evolution is governed by the **level-set equation**

$$\phi_t + F |\nabla \phi| = 0,$$

where F is a speed function in the normal direction, solved numerically with upwind finite differences that keep the front sharp. The advantage over explicit surface evolution is topological: a level set can split and merge (a contracting figure-eight curve splits into two components), which a parametrization cannot do without re-meshing. Level-set evolution drives shape carving, curvature-driven fairing ($F = -\kappa$), and segmentation of medical volumes [OF03, Set96].

When the speed F is everywhere of one sign, the front moves monotonically and the arrival time $T(x)$ satisfies the *Eikonal equation* $|\nabla T| = 1/F$ with $T = 0$ on the initial front — a static problem solved once by the fast marching method [Set96].

Theorem 28.14 (Fast Marching). *Let Γ be an initial front and $F > 0$ a speed on a grid of N points. The fast marching method computes the arrival time of $|\nabla T| = 1/F$ at every grid point in $O(N \log N)$ time using a heap-ordered front.*

Proof sketch. The method maintains a heap of candidate points ordered by current arrival time. The smallest candidate x is accepted: by monotonicity of the front no later candidate can lower its time. The accepted point updates its grid neighbors by solving the local Eikonal equation with a first-order upwind scheme. Each of the N points is accepted once and updated a constant number of times, giving N heap operations at $O(\log N)$ each. $\square$

Example 28.15 (Arrival Time as Distance). With speed $F \equiv 1$, the arrival time $T(x)$ is exactly the Euclidean distance from x to Γ ; fast marching with $F = 1$ is how Section 28.5 propagates distances into a volume grid, and the same scheme on a surface mesh yields geodesic distances.

The level-set and fast marching ideas connect the whole chapter: the signed distance functions of Section 28.5 are the $F = 1$ arrival-time fields; their isosurfaces are extracted by **marching cubes** [LC87], which constructs surface triangles in each grid cell for the 15 configurations of how a level set crosses the cell and resolves their topological ambiguities; and the Poisson reconstruction of Section 28.7 produces a field whose isosurface is the output mesh.

28.7 Poisson Surface Reconstruction

28.7.1 1. Overview

Poisson surface reconstruction [KBH06] is an implicit method: it turns an oriented point cloud into a watertight surface by solving a global Poisson equation rather than by filtering a Delaunay complex. The input is a set of points P with approximate outward normals N ; the output is a smooth indicator function whose isosurface approximates the surface. Because the method fits one smooth field to *all* the data, it is far more robust to noise than the Delaunay-based filters of Section 28.4, and it fills holes and produces watertight output automatically — the de facto standard for reconstructing watertight models from real scans.

28.7.2 2. Definitions

Definition 28.16 (Indicator Function). For a surface $S = \partial\Omega$, the **indicator function** of Ω is $\chi_\Omega(x) = 1$ for $x \in \Omega$ and 0 otherwise. Reconstructing S is equivalent to reconstructing the indicator: S is the level set $\chi = \tau$ for a threshold τ between the two sides.

Definition 28.17 (Oriented Point Set). An **oriented point set** is a set of pairs (p_i, n_i) where p_i is a sample and n_i a unit vector approximating the outward normal of S at p_i . Normals are estimated by principal component analysis of the k nearest neighbors (the smallest eigenvector of the covariance matrix), with signs made globally consistent.

28.7.3 3. Theory

The method exploits a distributional identity. For a smooth surface S with outward normal n and surface measure dA ,

$$\nabla \chi_\Omega = -n(x) \delta_S(x),$$

where δ_S is the surface Dirac measure: the gradient of the indicator is a vector field supported on S and pointing inward. The oriented samples (p_i, n_i) discretize this field; convolving with a smoothing kernel F on the octree of the samples gives the vector field $V(x) = \sum_i n_i F(x - p_i)$, and the identity $\nabla \chi * F = V$ says the smoothed indicator solves a Poisson equation.

Theorem 28.18 (Poisson Reconstruction). *Let (P, N) be an oriented point set sampling a smooth surface S , and let $V = \sum_i n_i F(x - p_i)$ be the induced vector field for a smoothing kernel F . Then the indicator function χ of S satisfies, in the sense of distributions,*

$$\Delta \chi = \nabla \cdot V,$$

and solving this equation and extracting the isosurface at the average of χ at the sample points yields a surface approximating S to within the smoothing scale of F [KBH06].

Proof sketch. Taking the divergence of both sides of $\nabla(\chi * F) = V$ gives $\Delta(\chi * F) = \nabla \cdot V$. Convolution commutes with differentiation and F approximates the identity, so $\chi * F$ is a smoothed indicator function whose isosurface at the threshold $\tau = \chi * F(p_i)$ — the mean value at the samples — lies within one smoothing radius of S ; the threshold interpolates the value of χ across S because the samples lie on the boundary. $\square$

The practical system [KBH06] replaces the uniform grid by an octree: samples and normals live in the leaves, the kernel is a trilinear hat function scaled per cell, and the Poisson equation is solved by a multigrid solver on the tree. The later *screened* variant adds the constraint $\chi(p_i) = 0$ at the samples, keeping the surface near the data; both share the theory above.

28.7.4 4. Pseudocode

28.7.5 5. Parameter Selection

- **Octree depth d :** controls resolution. Doubling d halves the cell size; the practical range is $d = 8$ to 12 for typical scans, chosen so the smallest cell matches the sample spacing. Larger d captures more detail but amplifies noise and memory use.
- **Isovalue threshold:** the default τ , the mean of χ at the samples, is robust; it can be adjusted when the sampling is biased.
- **Normal estimation:** the k -nearest-neighbor radius for normal estimation trades sharpness against robustness; k grows with the noise level.
- **Smoothing/screening:** the screened variant's weight controls fidelity to the data versus smoothness; raise it for dense scans, lower it for sparse or very noisy data.

Algorithm 130 Poisson Surface Reconstruction**Input:** oriented points (p_i, n_i) , octree depth d .**Output:** a watertight triangle mesh approximating the sampled surface.

```

1: function POISSONRECONSTRUCT( $(P, N), d$ )
2:   build an octree  $\mathcal{O}$  of depth  $d$ , subdivided so each sample lies in a distinct leaf
3:    $F \leftarrow$  trilinear hat function on the octree cells
4:    $V(x) \leftarrow \sum_i n_i F(x - p_i)$  ▷ smoothed normal field
5:   solve  $\Delta\chi = \nabla \cdot V$  on  $\mathcal{O}$  by a multigrid solver
6:    $\tau \leftarrow \frac{1}{|P|} \sum_i \chi(p_i)$ 
7:   extract the isosurface  $\{x : \chi(x) = \tau\}$  by marching cubes on  $\mathcal{O}$ 
8:   return the extracted triangle mesh
9: end function

```

28.7.6 6. Complexity

- **Time:** $O(N)$ for N octree nodes with the multigrid solver of [KBH06], where N is proportional to the number of samples at a fixed depth; isosurface extraction is linear in the number of output triangles.
- **Space:** $O(N)$.

28.7.7 7. Usages

- **Watertight reconstruction:** turning laser and photogrammetric scans of objects, buildings, and terrain into closed models for 3D printing, inspection, and rendering.
- **Noisy data:** the smooth least-squares fit tolerates noise that destroys the Delaunay-based filters of Section 28.4; it is the method of choice when the point set is not an ε -sample.
- **Implicit interface:** the field output plugs directly into the signed distance framework of Section 28.5 and the learning-based pipelines of Chapter 40.

28.7.8 8. Testing Methods and Edge Cases

- **Watertightness:** verify the output is closed (every edge shared by exactly two triangles) with the Euler characteristic of the expected genus, using Chapter 30.
- **Error measurement:** for synthetic surfaces, report the Hausdorff distance to the true surface; the error should decrease with depth until the sample spacing is resolved.
- **Normals orientation:** inconsistent normal signs give a self-canceling field and an empty or inverted reconstruction; test with a scan of known orientation.
- **Depth sensitivity:** a depth sweep should stabilize above a minimum depth; instability at high depth indicates noise amplification.
- **Holes and sparse regions:** large gaps should still reconstruct watertight; verify the surface across a gap is smooth with no cusps.

28.7.9 9. References

- Kazhdan, M., Bolitho, M., Hoppe, H. [KBH06], Poisson surface reconstruction.
- Osher, S., Fedkiw, R. [OF03], level-set methods and implicit surfaces.
- Lorensen, W. E., Cline, H. E. [LC87], marching cubes.

28.8 Applications

28.8.1 Scanning and Reverse Engineering

3D scanning systems — laser triangulation, structured light, time-of-flight lidar, and photogrammetry — produce point clouds whose density, accuracy, and coverage vary widely. The reconstruction pipeline of Sections 28.3 and 28.4 is their common back end: overlapping scans are registered into one frame (the methods of [BM92, CM92]), outliers are pruned, normals are estimated, and a surface is fitted. Industrial reverse engineering then converts the mesh into a parametric CAD model using the patch machinery of Section 28.2. The algorithm follows the data: ball pivoting [BMR⁺99] is fastest and fits uniformly sampled mechanical parts; the co-cone [ACDL00] and the crust [AB99] reconstruct provably correct surfaces from feature-adaptive samples; Poisson reconstruction [KBH06] is the robust default for noisy or incomplete scans.

28.8.2 Surface Reconstruction

In medical imaging, segmented CT and MRI volumes are converted to signed distance functions (Section 28.5) and polygonized by marching cubes [LC87] into the anatomical surfaces used for planning, simulation, and implant design; the level-set machinery of Section 28.6 drives the segmentation itself. Tomographic and lidar acquisitions of fossils, terrain, and biological specimens follow the same pipeline: the sampling condition of Section 28.3 tells the operator the density needed to resolve a feature of a given size, and the topological checks of Chapter 30 certify the output.

28.8.3 Terrain Modeling

Airborne lidar produces dense height samples of the ground and of buildings. Terrain reconstruction is the two-dimensional specialization: the samples are projected to the plane, outliers (vegetation, vehicles) are removed, and the Delaunay triangulation of the projection becomes the triangulated irregular network (TIN) backing digital elevation models, contouring, and visibility analysis [dBCvKO08]. Breaklines and rivers enter as constraints of the constrained Delaunay triangulation, and the meshing machinery of Chapter 22 refines the TIN for flood and watershed simulation; see also Section 11.5.

28.8.4 Practical Tradeoffs

- **Fidelity versus watertightness:** the Delaunay-based filters follow the data closely but leave holes and artifacts; the implicit methods fill holes at the cost of smoothing sharp features. Watertight output is mandatory for 3D printing, CFD, and FEM.
- **Speed:** ball pivoting is $O(n)$ expected and fastest on uniform scans; the Delaunay pipeline costs $O(n \log n)$ expected with $O(n^2)$ worst case (Theorem 12.41); Poisson reconstruction is linear in the octree size with a larger constant.
- **Robustness:** the implicit least-squares methods (Sections 28.5 and 28.7) degrade gracefully under noise; the Delaunay methods degrade quickly unless the noise is removed first.
- **Representation downstream:** reconstructed meshes feed the simulation stacks of Chapter 22, distance fields feed the collision and robotics algorithms, and all three standard input representations feed the geometry-aware machine-learning pipelines of Chapter 40.

28.9 Exercises

1. Let S be a surface and P an ε -sample of S . Prove from the Lipschitz property of the local feature size (Definition 28.1) that P is an r -sample for $r = \varepsilon/(1 - \varepsilon)$, and give an example where the two conditions differ by a constant factor.
2. Compute the medial axis and the local feature size of a rectangle (compare Example 20.83) and of an annulus, and describe the distribution of sample density an ε -sample must have at the corners, on the long edges, and near the inner hole.
3. For samples of the unit circle, compute the Voronoi diagram, the poles of each cell, and the pole vectors by hand; verify that each pole vector points to the center and that the 2D crust (Algorithm 127 restricted to edges) recovers the convex hull of the samples.
4. Simulate the ball-pivoting algorithm (Algorithm 129) by hand on eight points evenly spaced around a circle, with a ball radius equal to the chord length, and count the pivoting steps; repeat with a smaller ball and explain where the front stalls.
5. Let P be an ε -sample and $\theta = \arcsin(\varepsilon)$ the cocone angle. Show that the cocone $C(p)$ of Definition 28.7 contains the ball of radius $\varepsilon \cdot \text{lfs}(p)$ centered at p , and explain why this guarantees that the circumcenter of every triangle of the restricted Delaunay complex lies in the cocone of its vertices.
6. Implement the signed distance function of a triangle mesh on a voxel grid (Section 28.5) using fast marching (Theorem 28.14), and verify the Eikonal condition $|\nabla\phi| = 1$ away from the surface.
7. Implement the level-set evolution $\phi_t + F|\nabla\phi| = 0$ for $F \equiv 1$ (distance) and for $F = -\kappa$ (mean-curvature flow), and verify that a non-convex curve splits into two components under the second flow, demonstrating why level sets handle topology change.
8. Reconstruct the same synthetic point cloud of a sphere, a torus, and a bumpy surface with (i) the alpha complex, (ii) the cocone, and (iii) Poisson reconstruction; add Gaussian noise of increasing amplitude and report for each method the watertightness, the genus, and the Hausdorff distance to the true surface.

Chapter 29

Geometric Matching

29.1 Comparing Geometric Objects

Geometric matching is the algorithmic discipline of deciding how well two geometric objects agree, and of finding the *transformation* that makes them agree. The objects can be point sets, polygonal curves, polygonal regions, or surfaces; the transformation is usually a rigid motion, a similarity, or an affine map; and the agreement is measured by a distance between the transformed object and its target. The three ingredients — objects, transformations, and measures — determine a family of problems with very different algorithmic flavors, unified by a single question: how much computation does it take to align one shape with another?

Definition 29.1 (Matching Instance). A **matching instance** is a triple $(\mathcal{A}, \mathcal{B}, \mathcal{T})$, where $\mathcal{A}$ and $\mathcal{B}$ are geometric objects (point sets, curves, or polygons) and $\mathcal{T}$ is a family of admissible transformations T of the plane or of space. A solution is a transformation $T \in \mathcal{T}$ together with a correspondence π between the features of $T(\mathcal{A})$ and those of $\mathcal{B}$, minimizing a prescribed distance $d(T(\mathcal{A}), \mathcal{B})$.

The correspondence π is usually a bipartite matching between the features (points, vertices, segments) of the two objects. When both objects are point sets of equal size and π must be a bijection, the problem is a **bipartite matching problem in the geometric plane**; the two classical objective functions are the **minimum-weight matching** (Definition 29.10), which minimizes the sum of the matched distances, and the **bottleneck matching** (Definition 29.11), which minimizes the largest matched distance. When the correspondence is free to omit parts of either set, the problem is **partial matching**, discussed in Section 29.5.

A useful taxonomy organizes the field along three axes:

- **Exact versus approximate:** *exact* matching decides whether two objects are congruent (Section 29.7); *approximate* matching minimizes a distance such as the Hausdorff distance (Section 29.2) or the Fréchet distance (Section 29.3).
- **Transformation class:** rigid motions (translations, rotations, reflections), similarities (rigid motions plus scaling), and affine maps. Larger classes make matching easier but the decision less discriminative; Sections 29.7 and 29.9 develop this trade-off.
- **Structure of the objects:** point sets admit the combinatorial machinery of matchings and of arrangements (Sections 29.5 and 29.6); curves require order-respecting measures such as the Fréchet distance and its discrete variant (Sections 29.3 and 29.4).

The chapter proceeds from measures to algorithms. Sections 29.2–29.4 define the two principal similarity measures for shapes and curves. Sections 29.5 and 29.6 develop matching over point sets, including partial matching, outlier handling, the assignment problem, and the

Hungarian algorithm [Kuh55, CLRS09]. Sections 29.7 and 29.9 treat exact congruence and the model-based geometric hashing scheme. The iterative closest point algorithm of Section 29.8 and the applications of Section 29.10 close the chapter. Throughout, the robustness questions raised in Section 2.11 of the mathematical foundations and the nearest-neighbor machinery of Chapter 13 recur as the computational primitives.

29.2 Hausdorff Distance

The **Hausdorff distance** is the most important measure of proximity between two sets, and it is the canonical criterion for approximate point-set matching: it measures the largest distance a point of either set has to travel to meet the other set.

Definition 29.2 (Directed Hausdorff Distance). Let P and Q be two nonempty compact sets of $\mathbb{R}^d$. The **directed Hausdorff distance** from P to Q is

$$h(P, Q) = \max_{p \in P} \min_{q \in Q} \|p - q\|.$$

Definition 29.3 (Undirected Hausdorff Distance). The **Hausdorff distance** between P and Q is the maximum of the two directed distances:

$$H(P, Q) = \max\{h(P, Q), h(Q, P)\}.$$

Theorem 29.4 (Hausdorff Distance is a Metric). *Restricted to nonempty compact sets, H is a metric: it is nonnegative, symmetric, vanishes exactly on equal sets, and satisfies the triangle inequality $H(P, R) \leq H(P, Q) + H(Q, R)$.*

Proof. Symmetry and nonnegativity are immediate. If $H(P, Q) = 0$ then every point of P is at distance 0 from Q , so $P \subseteq Q$; symmetrically $Q \subseteq P$, hence $P = Q$. For the triangle inequality, fix $p \in P$ and let $q \in Q$ attain $\min_q \|p - q\|$; then for any $r \in R$,

$$\|p - r\| \leq \|p - q\| + \|q - r\|,$$

and taking the minimum over $r \in R$ and then the maximum over $p \in P$ gives $h(P, R) \leq h(P, Q) + h(Q, R)$; the symmetric inequality is identical and H satisfies the triangle inequality. $\square$

For point sets the directed distance is computable from the nearest-neighbor structure of the target set.

Theorem 29.5 (Computing the Hausdorff Distance). *Let P and Q be point sets with $|P| = n$ and $|Q| = m$. The directed distance $h(P, Q)$ can be computed in $O((n+m) \log m)$ time by constructing the Voronoi diagram of Q and performing a nearest-neighbor query for each $p \in P$ (Section 13.2 of Chapter 13); the undirected distance $H(P, Q)$ therefore takes $O((n+m) \log(n+m))$ time.*

Proof. The Voronoi diagram of Q can be constructed in $O(m \log m)$ time and supports nearest-neighbor queries in $O(\log m)$ time each (Chapter 13 and [PS85, dBCvKO08]). Querying each point of P costs $O(n \log m)$ in total, giving the bound. For the undirected distance the roles of P and Q are swapped and the maximum of the two directed values is taken. $\square$

When a transformation is allowed, the **minimum Hausdorff matching** problem asks for the transformation minimizing the distance after transformation.

Definition 29.6 (Minimum Hausdorff Matching). Let P, Q be point sets and $\mathcal{T}$ a family of transformations. The **minimum Hausdorff distance** is

$$\min_{T \in \mathcal{T}} H(T(P), Q).$$

A transformation attaining the minimum is a **min-Hausdorff matching** of P into Q .

For translations, the objective function is minimized on an arrangement. Fix T to be a translation by the vector t . For each $p \in P$ the directed distance to Q is

$$\phi_p(t) = \min_{q \in Q} \|p + t - q\|,$$

which is the distance from the translated point $p + t$ to its nearest neighbor in Q . The maximum over p of the functions ϕ_p is an *upper envelope of Voronoi surfaces* [AG00], and the minimum Hausdorff distance under translations is the minimum value of this envelope. The envelope is piecewise defined over the arrangement of the bisectors of the pairs (p, q) [AG00]:

Theorem 29.7 (Min-Hausdorff under Translations). *For point sets P and Q with $|P| = n$ and $|Q| = m$, the minimum of $H(t+P, Q)$ over translations t is attained at a vertex of the arrangement induced by the distance functions ϕ_p , and the optimum can be found in $O((nm)^2 \log(nm))$ time in the worst case.*

Proof sketch. Each function ϕ_p is the pointwise minimum of the m functions $\|p + t - q\|$, each of which is a translate of the origin-centered distance function; the maximum over p is an upper envelope whose breakpoints lie on bisectors of the form $\|p + t - q\| = \|p' + t - q'\|$. Over any cell of the arrangement of these $O(nm)$ bisectors the envelope is a fixed linear distance function, so the minimum over each cell is at a cell vertex; a sweep over the arrangement evaluates all vertices. See [AG00] for the exact construction and the refined bounds for convex polygons. $\square$

Remark 29.8 (Measures for matching). The Hausdorff distance does *not* respect the ordering of points on curves, which is why curve matching uses the Fréchet distance of Section 29.3. It is also sensitive to outliers, since a single stray point of P dominates $h(P, Q)$; the partial matching measures of Section 29.5 repair this.

29.3 Fréchet Distance

The Hausdorff distance compares curves as *sets* of points and ignores their parameterization. The **Fréchet distance** measures instead how far two curves are when a point moves along one while a leash, held at the other, stays taut: the distance is the minimal leash length over all monotone ways of walking both curves from start to finish [Fré06]. Formally, for polygonal curves $P = (p_1, \dots, p_n)$ and $Q = (q_1, \dots, q_m)$, the Fréchet distance is

$$d_F(P, Q) = \min_{\alpha, \beta \in [0,1]^{[0,1]}} \max_t \|P(\alpha(t)) - Q(\beta(t))\|,$$

where the minimum is over all continuous monotone reparameterizations α, β of the two curves.

The Fréchet distance is the correct similarity measure for trajectories, GPS tracks, and any ordered data, precisely because it enforces that the correspondence between the two curves is order-preserving: two curves that run in opposite directions are far apart even if their point sets coincide. The full algorithmic treatment — the dog-leash analogy, the free-space diagram, the decision procedure, and the $O(nm \log nm)$ algorithms for polygonal curves [AG95] — is developed in Section 24.16 of Chapter 24.11, to which we refer. This chapter mentions the measure only to place it in the matching landscape: for curves it plays the role that the Hausdorff distance plays for point sets, and it is the natural objective for the ordered point matching of Section 29.5.

29.4 Discrete Fréchet Distance

For polygonal curves given by their vertex sequences, the continuous Fréchet distance can be approximated by its **discrete** variant, in which the walker and the dog may jump only between

consecutive vertices. The discrete Fréchet distance is defined by monotone couplings of the two vertex sequences [EM94b]; see Definition 24.45 of Chapter 24.11 for the formal statement.

Definition 29.9 (Discrete Fréchet Distance, Grid Form). Let $P = (p_1, \dots, p_n)$ and $Q = (q_1, \dots, q_m)$ be polygonal curves. A **coupling** is a monotone path in the $n \times m$ grid from $(1, 1)$ to (n, m) moving only right, down, or diagonally down-right. The **cost** of a coupling $\{(i_k, j_k)\}_k$ is $\max_k \|p_{i_k} - q_{j_k}\|$, and the **discrete Fréchet distance** is the minimum cost over all couplings.

The grid formulation yields a dynamic program. Let $D(i, j)$ be the discrete Fréchet distance between the prefixes $(p_1, \dots, p_i)$ and $(q_1, \dots, q_j)$. The recurrence is

$$D(i, j) = \max(\|p_i - q_j\|, \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\}),$$

with $D(1, 1) = \|p_1 - q_1\|$, which computes the distance in $O(nm)$ time and space by filling the grid one diagonal at a time; the coupling achieving the optimum is recovered by tracing the min-edges backward. The decision version — whether the distance is at most ε — is the reachability question in the ε -version of the grid, in which a cell is *free* when $\|p_i - q_j\| \leq \varepsilon$, exactly the free-space diagram of Section 24.16. This is the point where the discrete and continuous theories meet: the discrete distance bounds the continuous one from above, and the free-space diagram turns both into grid reachability problems [AG95, EM94b].

29.5 Shape Matching

Shape matching is the general problem of deciding, for two shapes $\mathcal{A}$ and $\mathcal{B}$, how similar they are and finding the aligning transformation. Point-set shape matching subsumes three decisions: which transformation family to admit, which distance to minimize, and whether all points of one set must be matched. This section treats the distance and the partial-matching dimensions; the transformation dimension is developed in Sections 29.7–29.9.

The two standard objectives for point-set matching are the minimum-weight matching and the bottleneck matching of the underlying bipartite distance graph.

Definition 29.10 (Minimum-Weight Matching). Let P and Q be point sets with $|P| = |Q| = n$. The **minimum-weight matching** is a bijection $\pi: P \rightarrow Q$ minimizing

$$\sum_{p \in P} \|p - \pi(p)\|.$$

Definition 29.11 (Bottleneck Matching). The **bottleneck matching** of two point sets of equal size is a bijection π minimizing the maximum matched distance

$$\max_{p \in P} \|p - \pi(p)\|.$$

Equivalently, it is the smallest radius r for which the bipartite graph of pairs (p, q) with $\|p - q\| \leq r$ admits a perfect matching.

The bottleneck version admits a clean decision procedure: sort all $O(n^2)$ pairwise distances, and find the smallest threshold r for which the graph G_r has a perfect matching, by a binary search over the sorted distances combined with a bipartite matching algorithm [PS85, CLRS09]. The minimum-weight version is the assignment problem of Section 29.6, solved by the Hungarian algorithm in $O(n^3)$ time [Kuh55, CLRS09]. When the points of P and Q are *ordered* (polygonal curves, sequences of landmarks), the bijection is constrained to be order preserving and the problem becomes the sequence matching solved by dynamic programming, with the Fréchet distance of Sections 29.3 and 29.4 as the natural cost [AG95, EM94b].

Real data rarely consists of two cleanly aligned point sets: there are **outliers** (points that belong to one set but have no counterpart in the other) and **clutter**. Partial matching handles this by allowing the matching to skip points.

Definition 29.12 (Partial Matching and Partial Hausdorff). Let P and Q be point sets. A **partial matching** matches a subset $P' \subseteq P$ to Q ; points of $P \setminus P'$ are **outliers** and pay no cost. The **partial directed Hausdorff distance** $h_k(P, Q)$ is the k -th smallest of the distances $\min_{q \in Q} \|p - q\|$ over $p \in P$: all but the worst $|P| - k$ outliers are ignored.

For the partial distance $h_k(P, Q)$, the outlier sensitivity of the Hausdorff distance disappears: a set with $|P| - k$ outliers has the same partial distance as if the outliers were removed. The partial variants of the minimum-weight and bottleneck matchings, in which a prescribed number of pairs are dropped, are solved by the same assignment machinery with dummy rows and columns (Section 29.6).

Matching also connects to the theory of arrangements. Fix a translation t and consider the pairs $(p, q) \in P \times Q$; the translation t maps p within distance r of q exactly when t lies in the disk $D(p, q, r) = \{t : \|p + t - q\| \leq r\}$. A perfect matching of radius r exists exactly when the translation graph defined by these disks has a perfect matching. Varying r and t sweeps an arrangement-like structure — the same bisector arrangement that underlies Theorem 29.7 — so the min-Hausdorff, bottleneck, and min-weight problems for translations are all reducible to searches over the cells of one arrangement [AG00]. This is the sense in which “matching via the arrangement” unifies the algorithms of this section with the level and envelope machinery of Chapter 14 (Section 14.3).

29.6 Point-Set Registration

Point-set registration is the workhorse of the applications: given two point clouds representing the same scene — two scans of a surface, two satellite images, two MRI volumes — find the transformation that aligns them, using the pointwise correspondence as an intermediate structure.

Definition 29.13 (Registration). Let P and Q be point sets and $\mathcal{T}$ a family of transformations. A **registration** of P into Q is a transformation $T \in \mathcal{T}$ together with a partial matching π of $T(P)$ into Q minimizing a cost such as

$$\sum_p \|T(p) - \pi(p)\|^2 \quad \text{or} \quad \max_p \|T(p) - \pi(p)\|,$$

over all admissible T and all matchings π .

The optimization over T and π jointly is what makes registration hard: each is easy once the other is fixed. Given the correspondence π , the best rigid transformation T is computed by the least-squares alignment of Section 29.7; given T , the best correspondence is the nearest-neighbor matching of each transformed point, i.e., a *minimum-weight matching* in the bipartite distance graph. The two optimizations are intertwined, and every practical algorithm — the iterative closest point method of Section 29.8 and its many variants — alternates between the two, an instance of alternating-optimization heuristics.

When the correspondence is a bijection and the cost is additive, the correspondence problem is the classical **assignment problem**.

Definition 29.14 (Assignment Problem). Given a square matrix $C = (c_{ij})$ of costs $c_{ij} = \|p_i - q_j\|^p$, the **assignment problem** asks for a permutation π of $\{1, \dots, n\}$ minimizing $\sum_i c_{i\pi(i)}$.

Theorem 29.15 (Hungarian Algorithm). *The assignment problem is solvable in $O(n^3)$ time and $O(n^2)$ space by the **Hungarian algorithm** (Kuhn–Munkres), which maintains a set of dual variables and finds a perfect matching among the zero-cost edges of the transformed cost matrix [Kuh55, CLRS09].*

Algorithm 131 Hungarian Algorithm for the Assignment Problem**Input:** an $n \times n$ cost matrix $C = (c_{ij})$.**Output:** a permutation π minimizing $\sum_i c_{i\pi(i)}$.

```

1: function HUNGARIAN( $C$ )
2:   subtract the row minimum from each row, then the column minimum from each column
3:    $\pi \leftarrow \emptyset$  ▷ partial assignment of zero-cost edges
4:   while  $\pi$  is not a perfect assignment do
5:     mark the rows and columns reachable by alternating zero-cost paths from the unas-
       signed rows
6:      $\Delta \leftarrow \min\{c_{ij} : i \text{ unmarked}, j \text{ marked}\}$ 
7:     subtract  $\Delta$  from the unmarked rows; add  $\Delta$  to the marked columns
8:     augment  $\pi$  along a zero-cost augmenting path, if one exists
9:   end while
10:  return  $\pi$ 
11: end function

```

In geometric settings the costs $c_{ij} = \|p_i - q_j\|$ are Euclidean distances, which gives the problem exploitable structure: the closest-pair and nearest-neighbor machinery of Chapter 13 can prune the dense cost matrix, and greedy nearest-neighbor matching provides the excellent initialization that makes the alternating algorithms of Section 29.8 converge fast in practice. The bottleneck variant of the assignment problem (minimize $\max_i c_{i\pi(i)}$, cf. Definition 29.11) is solved by binary search over thresholds with a bipartite matching feasibility test, in time $O(n^{2.5} \log n)$ for dense graphs [CLRS09]. Both the min-weight and the bottleneck assignment problems are used as the correspondence step of registration pipelines.

29.7 Rigid Transformations

Exact matching asks whether two point sets are *congruent*: whether one can be moved onto the other by an isometry. The isometries of the plane are the translations, rotations, and reflections (together with their compositions), and they form the **Euclidean group** $E(2)$; in space the analogous group is $E(3)$.

Definition 29.16 (Rigid Transformation and Isometry). A map $T: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is **rigid** if it preserves all distances: $\|T(x) - T(y)\| = \|x - y\|$ for all x, y . The set of rigid transformations is the group of **isometries**. Each isometry is a composition of an orthogonal map and a translation; it is **orientation-preserving** (a rotation composed with a translation) or **orientation-reversing** (composed with a reflection).

Two point sets are congruent if there is an isometry mapping one onto the other.

Definition 29.17 (Congruence). Point sets P and Q are **congruent** if there is an isometry T with $T(P) = Q$ as sets. If T is restricted to translations, the appropriate notion is that of *translational congruence*; adding rotations gives *rotational congruence*.

Deciding congruence is fast because isometries preserve distances. The classical approach sorts the points by distance to the centroid and matches invariants:

Theorem 29.18 (Congruence Testing of Point Sets). *Two point sets P and Q of size n can be tested for congruence in $O(n \log n)$ time: after translating both centroids to the origin, the multiset of squared distances $\{\|p\|^2 : p \in P\}$ is an invariant, and the angular order of the points on their convex hulls (Section 4.5) resolves the remaining rotations and reflections.*

Proof sketch. Any isometry fixing the centroid and preserving distances must map the points of P to the points of Q at equal distances from the origin, so the sorted distance lists must agree. The points at the maximum distance (the outer hull vertices) are permuted by the isometry, and the cyclic order of the neighbors on the hull fixes the rotation angle; comparing the resulting cyclic signatures decides congruence. The sorting dominates, so the bound is $O(n \log n)$ [Ata85, Hig86]. $\square$

Symmetries are the congruence tests where $P = Q$: a *symmetry* of a point set is an isometry mapping the set onto itself. Detecting all symmetries of a planar point set, and the rotational symmetries of a polygon, are solved by the same invariant machinery, with optimal $O(n \log n)$ algorithms [Ata85, WWV85, Hig86].

The exact congruences serve two purposes in this chapter. First, they are the decision problem underlying all approximate matching: an approximate matching with distance 0 is a congruence, so the complexity of exact matching is a lower benchmark for the approximate algorithms. Second, the least-squares alignment used inside the iterative closest point algorithm (Section 29.8) is itself an exercise in rigid transformations: given a correspondence, the best rigid map is computed from the singular value decomposition of the cross-covariance of the matched points, a closed-form computation that is the linear-algebra core of every registration pipeline.

29.8 Iterative Closest Point

29.8.1 1. Overview

Iterative Closest Point (ICP) is the dominant algorithm for registering two point clouds under a rigid transformation. It alternates two cheap steps — compute the closest-point correspondences under the current transform, then compute the rigid transform that best aligns the correspondences — and iterates to convergence. First proposed by Besl and McKay [BM92] and developed for range-image modeling by Chen and Medioni [CM92], ICP is the standard engine behind surface reconstruction from range scans (Section 28.3) and object registration in computer vision.

29.8.2 2. Definitions

Definition 29.19 (ICP Correspondence). Let P be a **source** point set and Q a **target** point set. Given a transform T , the **ICP correspondence** matches each $p \in P$ to its nearest neighbor in $T(Q)$ (or in Q , depending on the convention), i.e., to a point $\pi(p)$ minimizing $\|T(p) - \pi(p)\|$.

Definition 29.20 (Alignment Error). The **least-squares alignment error** of a transform T with respect to a correspondence π is

$$E(T, \pi) = \sum_{p \in P} \|T(p) - \pi(p)\|^2.$$

29.8.3 3. Theory

The convergence of ICP rests on the fact that each half of the alternation can only decrease the error.

Theorem 29.21 (Monotone Convergence of ICP). *Let E_k be the alignment error after iteration k of the closest-point step followed by the least-squares rigid alignment step. Then the sequence E_k is nonincreasing, and the algorithm converges to a local minimum of the registration objective [BM92].*

Proof sketch. In the closest-point step the correspondence is recomputed so that each matched target point is at least as close as before, so the error with the *old* transform cannot increase; in the alignment step the rigid transform minimizing the error for the *fixed* correspondence cannot increase it either. The error sequence is thus nonincreasing and bounded below, hence convergent. The limit is a local, not global, optimum, which is why ICP initialization and its variants matter (Section 5). $\square$

29.8.4 4. Pseudocode

Algorithm 132 Iterative Closest Point

Input: source point set P , target point set Q , initial transform T_0 .

Output: a rigid transform T approximately minimizing the registration error.

```

1: function ICP( $P, Q, T_0$ )
2:    $T \leftarrow T_0$ 
3:   repeat
4:     build a kd-tree or Voronoi query structure for  $Q$ 
5:     for each  $p \in P$  do
6:        $\pi(p) \leftarrow$  nearest neighbor of  $T(p)$  in  $Q$  ▷ closest-point step
7:     end for
8:      $T \leftarrow$  least-squares rigid alignment of  $\{(p, \pi(p))\}$  ▷ via the singular value decomposition
9:   until the decrease in alignment error is below a tolerance
10:  return  $T$ 
11: end function

```

29.8.5 5. Parameter Selection

- **Initialization:** ICP converges only locally (Theorem 29.21), so T_0 must already be close; coarse registration by matching invariants (centroids, principal axes, geometric hashing of Section 29.9) is standard.
- **Distance threshold:** pairs farther than a threshold are rejected as outliers before the alignment step; a threshold of a few multiples of the target resolution is typical.
- **Point selection:** uniform downsampling, or sampling the most salient points (high curvature), speeds the algorithm up with little loss of accuracy.
- **Query structure:** the nearest-neighbor queries dominate, so the kd-tree of Section 10.1 (Chapter 10) or the Voronoi structure of Chapter 13 is essential.
- **Point-to-plane variants:** replacing the point-to-point cost by the distance to the tangent plane at $\pi(p)$ (Chen–Medioni [CM92]) yields faster convergence on smooth surfaces.

29.8.6 6. Complexity

- **Per iteration:** $O(|P| \log |Q|)$ for the nearest-neighbor queries with a kd-tree (Section 9.3), plus $O(|P|)$ for the closed-form alignment via the singular value decomposition.
- **Iterations:** the number of iterations until convergence is data dependent (often tens), so the total cost is $O(k |P| \log |Q|)$ with k the iteration count.

29.8.7 7. Usages

- **Range-image registration:** aligning consecutive scans of a surface into a single model [BM92, CM92, dBCvKO08] (Section 28.3).
- **Computer vision:** object tracking, surgical registration, and the rigid part of many simultaneous localization and mapping pipelines (Section 29.10).
- **Global optimization:** as the local refinement engine inside global registration schemes that add loops to the pairwise alignments.

29.8.8 8. Testing Methods and Edge Cases

- **Synthetic alignment:** generate Q as a known rigid image of P with noise, run ICP, and verify the recovered transform.
- **Local minima:** test with adversarial initializations far from the optimum and verify the algorithm stops at a local, not global, optimum.
- **Outliers and partial overlap:** ICP should be tested when only a fraction of the points overlap, verifying the threshold rejection.
- **Symmetry:** objects with rotational symmetries can converge to an alternative valid alignment; verify the error, not the parameters.
- **Degenerate clouds:** collinear or coplanar point sets make the alignment step ill-conditioned; tests should exercise the SVD-based fallback.

29.8.9 9. References

- Besl, P. J., McKay, N. D. [BM92] introduced ICP and proved its monotone convergence.
- Chen, Y., Medioni, G. [CM92] developed the point-to-plane variant for range images.
- Alt, H., Guibas, L. J. [AG00] survey registration within the broader matching literature.

29.9 Geometric Hashing

29.9.1 1. Overview

Geometric hashing [WR97] is a model-based object recognition scheme that finds an object in a cluttered scene without explicitly searching the transformation space. The idea is to preprocess a library of models so that *invariant* properties of small point configurations are stored in a hash table, and then to recognize a scene by voting: every minimal “basis” configuration of scene points casts votes for the models whose invariants match. The scheme is famous for being transformation-invariant (rigid, similarity, or affine, depending on the invariant used) and robust to clutter and partial occlusion, at the price of a statistical trade-off between the hash bin size and the false-positive rate.

29.9.2 2. Definitions

Definition 29.22 (Geometric Hashing Scheme). A **geometric hashing scheme** consists of

- a family of **bases**: ordered tuples of points defining a coordinate frame (two points for the planar rigid/similarity case, three for the affine case);

- a **normalizing map** F_B that maps the basis B onto a canonical frame (e.g., the unit segment or unit triangle);
- a **hash function** that quantizes the normalized coordinates of the remaining points.

29.9.3 3. Theory

The correctness of geometric hashing rests on invariance: if the scene is a rigid (or similarity, or affine) image of a model, then normalizing by the corresponding basis maps the rest of the model to the same canonical coordinates regardless of the transformation, so the same hash entries are reproduced. A scene basis votes for exactly those model bases whose canonical coordinates coincide under the quantization.

Theorem 29.23 (Voting Correctness). *If the scene contains a subset congruent to a model under the admissible transformations, then the basis realizing that congruence accumulates the full set of votes of the model, one per matched point, and any threshold below the model size separates it from spurious matches with high probability when the bins are chosen randomly.*

Proof sketch. For the true basis, normalization is exact up to quantization, so every scene point that is the image of a model point hashes to the model’s bin; the number of votes equals the number of matched points. A random model point hashes into a given bin with probability proportional to the bin area, so the vote count of a wrong model is a binomial random variable whose mean is small when the bins are fine. Choosing the bin size and threshold to separate the means gives the claim; see [WR97, AG00]. $\square$

29.9.4 4. Pseudocode

29.9.5 5. Parameter Selection

- **Basis size:** two points for rigid and similarity matching, three for affine matching [WR97]; the basis is the “minimal index” that determines the transformation class.
- **Bin size:** the quantization of the canonical coordinates sets the tolerance of the matching; too coarse a bin merges distinct configurations (false positives), too fine a bin rejects valid matches.
- **Voting threshold τ :** must exceed the expected vote count of a wrong model-basis pair; τ is set from the target false-positive rate by the binomial analysis of Theorem 29.23.

29.9.6 6. Complexity

- **Preprocessing:** for a model with n points and basis size b , there are $O(n^b)$ bases and $O(n)$ remaining points each, so $O(n^{b+1})$ expected hash entries per model.
- **Recognition:** for a scene with m points, $O(m^{b+1})$ hash lookups in the worst case; with balanced hash tables each lookup is expected $O(1)$.
- **Space:** the hash table stores $O(\sum_i n_i^{b+1})$ entries.

29.9.7 7. Usages

- **Model-based object recognition:** recognizing known objects in cluttered images despite occlusion and noise [WR97].
- **Coarse registration:** geometric hashing provides the transformation-invariant initialization that ICP (Section 29.8) refines, combining global search with local precision.

Algorithm 133 Geometric Hashing (Recognition)

Input: a library of models $M_1, \dots, M_K$; a scene point set S ; a basis size b (2 for rigid/similarity, 3 for affine).

Output: the models present in S and the aligning transformations.

```

1: function GEOMETRICHASHINGPREPROCESS( $M_1, \dots, M_K$ )
2:   HT  $\leftarrow$  empty hash table
3:   for each model  $M_i$  do
4:     for each ordered basis  $B$  of  $M_i$  do
5:        $F \leftarrow$  normalizing map of  $B$ 
6:       for each remaining model point  $p$  do
7:         HT[quant( $F(p)$ )]  $\leftarrow$  HT[.]  $\cup \{(i, B)\}$ 
8:       end for
9:     end for
10:  end for
11:  return HT
12: end function
13: function GEOMETRICHASHINGRECOGNIZE( $S$ , HT,  $\tau$ )
14:  votes  $\leftarrow$  counter over model–basis pairs
15:  for each ordered basis  $B'$  of  $S$  do
16:     $F' \leftarrow$  normalizing map of  $B'$ 
17:    for each remaining scene point  $p$  do
18:      increment votes[ $(i, B)$ ] for all  $(i, B)$  in HT[quant( $F'(p)$ )]
19:    end for
20:    report the pairs  $(i, B)$  with votes  $\geq \tau$  and the aligning transform  $F'^{-1} \circ F$ 
21:  end for
22: end function

```

- **Biometry:** fingerprint and face recognition pipelines use hashed invariant point configurations.

29.9.8 8. Testing Methods and Edge Cases

- **Occlusion:** hide a fraction of the model points in the scene and verify that the vote count tracks the visible fraction.
- **Clutter:** add random points to the scene and measure the false-positive rate against the binomial prediction of Theorem 29.23.
- **Transformation classes:** test rigid, similarity, and affine scenes to verify the correct basis size and invariant.
- **Degenerate bases:** collinear or coincident basis points give degenerate frames; they must be rejected during preprocessing.
- **Bin sensitivity:** sweep the bin size and verify the precision–recall trade-off is monotone as predicted.

29.9.9 9. References

- Wolfson, H. J., Rigoutsos, I. [WR97] give the standard overview of geometric hashing, its invariants, and its applications.
- Alt, H., Guibas, L. J. [AG00] place geometric hashing in the general theory of transformation-invariant matching.
- The invariant and symmetry machinery of Section 29.7 provides the underlying congruence theory.

29.10 Applications in Computer Vision

Geometric matching is the computational core of shape alignment and registration across vision, medical imaging, and geographic information systems. The tools of this chapter — Hausdorff and Fréchet distances, min-weight and bottleneck matchings, the Hungarian algorithm, ICP, and geometric hashing — each serve a distinct application profile.

Shape alignment in vision takes a two-dimensional template and seeks its occurrence in an image. The template and the image feature points are matched by a min-weight matching or by a min-Hausdorff matching under rigid and similarity transformations (Sections 29.5 and 29.2); the Hausdorff criterion is favored when the image is cluttered, and the order-respecting Fréchet distance (Section 29.3) is used for contour and trajectory matching. When the transformation class is large or the object can be occluded, the voting scheme of geometric hashing (Section 29.9) replaces the exhaustive search.

Registration pipelines [BM92, CM92, AG00] align point clouds acquired at different times, from different sensors, or from different viewpoints: range scans of an object are merged into a single model (Section 28.3 of Chapter 28), successive frames are stitched into maps, and medical volumes from different modalities are brought into a common coordinate frame. The workhorse is ICP (Section 29.8) initialized by invariant matching or geometric hashing, with the assignment problem of Section 29.6 providing the correspondence step in feature-based pipelines. In all of these applications the geometric setting matters: the nearest-neighbor queries inside every iteration are answered by the kd-trees and Voronoi structures of Chapters 10 and 13, and robustness to outliers is provided by the partial matching measures of Section 29.5.

Geographic information systems (Chapter 36) use the same machinery to align maps and imagery: registering a satellite image to a vector map, conflation of road networks from different sources, and change detection between surveys. Here the transformations are usually rigid or similarity transforms of image patches, the features are detected road crossings, building corners, or coastlines, and the matchings are the min-Hausdorff and bottleneck matchings of Sections 29.2–29.5; the Fréchet distance (Section 29.3 and Chapter 24.11) is the measure of choice for aligning the ordered sequences of points along roads and rivers.

Across all applications two recurring principles emerge. First, *initialization and refinement* split the problem: a robust global method (matching invariants, geometric hashing) produces a coarse alignment that a local method (ICP) refines. Second, the *measure determines the algorithm*: outlier-prone data demands the partial and bottleneck measures, ordered data demands the Fréchet family, and clean, fully overlapping data can use the fastest least-squares and min-weight machinery.

29.11 Exercises

1. Prove Theorem 29.4 in full detail, including the triangle inequality for the undirected Hausdorff distance.
2. Implement the computation of the Hausdorff distance between two point sets (Theorem 29.5) using a kd-tree (Section 10.1) and verify the running time on random inputs.
3. Show that the partial Hausdorff distance $h_k(P, Q)$ (Definition 29.12) equals the directed Hausdorff distance of the subset of P obtained by deleting the $|P| - k$ points with the largest nearest-neighbor distances, and use this to give an $O((n + m) \log m)$ algorithm for it.
4. Implement the Hungarian algorithm (Algorithm 131) for the assignment problem and verify it against brute force for $n \leq 8$ with random cost matrices. Then apply it to the Euclidean bottleneck matching (Definition 29.11) via binary search over the pairwise distances.
5. Prove that the discrete Fréchet distance grid recurrence (Section 29.4) is correct, and show how the free-space diagram decides the ε -reachability question.
6. Simulate the iterative closest point algorithm (Algorithm 132) on two congruent copies of a small point set under a known rotation and translation, and verify monotone convergence of the alignment error (Theorem 29.21).
7. For a point set with a rotational symmetry, show that the min-Hausdorff distance of Section 29.2 over rotations is attained at an angle that is a multiple of the symmetry's angle, and test the claim with geometric hashing (Section 29.9).
8. Compare, experimentally, the Hausdorff, bottleneck, and min-weight matching criteria on a synthetic pair of point sets with 10% outliers (Definition 29.12): which criterion is most robust, and by how much does the running time of the Hungarian algorithm (Algorithm 131) grow with n ?

Chapter 30

Computational Topology

30.1 Mesh Analysis and Topology

30.2 Mesh Connected Components

30.2.1 1. Overview

Finding the connected components of a mesh is the process of partitioning the mesh into sets of faces or vertices that are reachable from each other through edge connectivity. A single 3D file often contains multiple disjoint geometric objects (e.g., a table and four separate chairs). Identifying these components is a standard preprocessing step for mesh analysis [BKP⁺10], repair, and selective editing.

30.2.2 2. Definitions

Definition 30.1 (Connected Component). A maximal subgraph of a mesh where any two nodes (vertices or faces) are connected by a path of adjacency relations.

Definition 30.2 (Adjacency). In a mesh, two faces are *edge-adjacent* if they share an edge. Two vertices are adjacent if they share an edge.

Definition 30.3 (Graph Traversal). The process of visiting all reachable elements in a graph starting from a seed point, following a systematic exploration strategy such as breadth-first or depth-first search.

30.2.3 3. Theory

The problem of finding connected components in a mesh is equivalent to finding the connected components of its dual graph (where nodes are faces) or its primal graph (where nodes are vertices).

Theorem 30.4 (Component Count). *Let M be a mesh with n vertices and m edges. The connected components of M can be computed in $O(n + m)$ time using either breadth-first search or depth-first search.*

Proof. Each vertex and edge is visited at most once by BFS or DFS. The algorithm maintains a set of unvisited vertices, and each vertex is removed from this set exactly once. For each vertex, all its incident edges are examined, contributing $O(\deg(v))$ work. Summing over all vertices gives $\sum_v O(\deg(v)) = O(m)$ by the Handshaking Lemma [Die17]. Combined with the $O(n)$ initialization, the total is $O(n + m)$. $\square$

Standard Breadth-First Search (BFS) or Depth-First Search (DFS) algorithms are used:

1. Initialize a set of unvisited faces.
2. While there are unvisited faces:
 - Pick an unvisited face as a “seed.”
 - Start a new component.
 - Perform a BFS/DFS to find all faces reachable from the seed via shared edges.
 - Mark all reached faces as visited and add them to the current component.

For manifold meshes, face-connectivity is usually sufficient. For non-manifold or point cloud data, vertex-connectivity might be required.

Example 30.5 (Connected Components). Consider a mesh consisting of a tetrahedron (4 vertices, 4 faces) and a disconnected triangle (3 vertices, 1 face). The algorithm starts at an arbitrary face of the tetrahedron and discovers all 4 faces via adjacency. It then picks the lone triangle as a new seed, finding 1 face. The result is two components: one with 4 faces and one with 1 face. The Euler characteristic of the first component is $\chi_1 = 4 - 6 + 4 = 2$ (sphere-like), and the second is $\chi_2 = 3 - 3 + 1 = 1$ (disk).

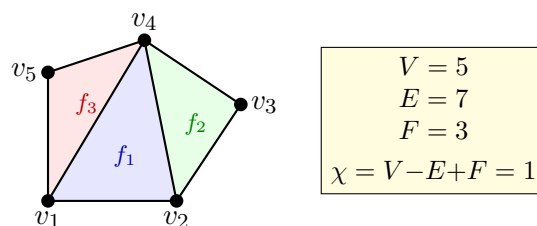


Figure 30.1: A triangle mesh with vertices v_i , edges, and faces f_j . The Euler characteristic $\chi = V - E + F$ is a topological invariant.

30.2.4 4. Pseudocode

30.2.5 5. Parameter Selection

- **Connectivity Type:** **Edge-connectivity** (faces sharing an edge) is standard. **Vertex-connectivity** (faces sharing only a vertex) is “weaker” and will result in fewer, larger components.
- **Data Structure:** Using a Half-Edge [GS85] structure or an Adjacency List makes finding neighbors $O(1)$, leading to optimal performance.

30.2.6 6. Complexity

- **Time Complexity:** $O(F)$ where F is the number of faces, as each face and each adjacency relationship is visited a constant number of times.
- **Space Complexity:** $O(F)$ to store the visited status and the resulting component lists.

30.2.7 7. Usages

- **Mesh Repair:** Identifying and removing small “floating” noise components (e.g., isolated triangles) from a scan.

Algorithm 134 Find Mesh Components

```

1: function FINDMESHCOMPONENTS(mesh)
2:   unvisited  $\leftarrow$  set(range(num_faces))
3:   components  $\leftarrow$  []
4:   while unvisited do
5:     seed  $\leftarrow$  unvisited.pop()
6:     current_component  $\leftarrow$  [seed]
7:     queue  $\leftarrow$  [seed]
8:     while queue do
9:       f  $\leftarrow$  queue.pop(0)
10:      for neighbor  $\in$  GetAdjacentFaces(f, mesh) do
11:        if neighbor  $\in$  unvisited then
12:          unvisited.remove(neighbor)
13:          current_component.append(neighbor)
14:          queue.append(neighbor)
15:        end if
16:      end for
17:    end while
18:    components.append(current_component)
19:  end while
20:  return components
21: end function

```

- **Selective Editing:** Selecting an entire object by clicking on a single triangle (e.g., “Select Linked” in Blender).
- **3D Printing:** Ensuring that a model consists of a single connected “shell” or identifying internal voids.
- **Rigging and Animation:** Automatically grouping parts of a character (e.g., separating the body from the clothing or eyes).
- **Physics Simulation:** Treating each connected component as a separate rigid body for collision and dynamics.

30.2.8 8. Testing Methods and Edge Cases

- **Single Watertight Object:** Should return exactly one component containing all faces.
- **Disconnected Objects:** Verify that multiple separate objects are correctly identified as distinct components.
- **Points/Lines:** Ensure the algorithm handles “degenerate” components like isolated vertices or edges.
- **Non-Manifold Edges:** Verify that connectivity is correctly traced even if three or more faces share an edge.
- **Large Meshes:** Test performance on meshes with millions of triangles to ensure linear scaling.

30.2.9 9. References

- Hopcroft, J., & Tarjan, R. (1973). “Efficient algorithms for graph manipulation.” Communications of the ACM.
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing.” CRC Press.
- Wikipedia: Connected component (graph theory).

30.3 Mesh Boundary Detection

30.3.1 1. Overview

Mesh boundary detection is the process of identifying vertices and edges that form the boundary of a surface mesh [BKP⁺10]. In a manifold mesh, every edge is shared by exactly two faces. Boundary edges are those that are part of only one face. Identifying these boundaries is essential for surface parameterization, texture mapping, and mesh editing.

30.3.2 2. Definitions

Definition 30.6 (Surface Mesh). A collection of vertices, edges, and faces that represent a 2-dimensional surface embedded in 3-dimensional space.

Definition 30.7 (Boundary Edge). An edge that is incident to exactly one face. In a half-edge data structure [GS85], this corresponds to a half-edge whose `twin` is null.

Definition 30.8 (Boundary Vertex). A vertex that is incident to at least one boundary edge.

Definition 30.9 (Boundary Loop). A connected cycle of boundary edges and vertices that define a hole or the perimeter of the mesh.

30.3.3 3. Theory

The fundamental principle of mesh boundary detection is counting the face-edge incidence. For a given mesh M with F faces:

1. Iterate through all faces.
2. For each face, extract its oriented edges (e.g., $(v_1, v_2), (v_2, v_3), (v_3, v_1)$).
3. Store the occurrence of each edge in a hash map, ignoring the orientation (i.e., (v_i, v_j) is the same as (v_j, v_i)).
4. Edges that appear only **once** in the total count are boundary edges.

Theorem 30.10 (Boundary Edge Characterization). *In a closed 2-manifold mesh, every edge is shared by exactly two faces. An edge e is a boundary edge if and only if it is incident to exactly one face.*

Proof. By definition of a 2-manifold with boundary, each edge is incident to either one face (boundary edge) or two faces (interior edge). The edge count algorithm enumerates all face-edge incidences. An interior edge appears twice (once per adjacent face), while a boundary edge appears exactly once. Hence, edges with count equal to 1 are precisely the boundary edges. $\square$

Once the boundary edges are identified, they can be chained together by matching endpoints to form one or more boundary loops.

Example 30.11 (Boundary Detection). Consider a single triangle with vertices (v_0, v_1, v_2) and edges $\{(v_0, v_1), (v_1, v_2), (v_2, v_0)\}$. The edge count map stores each edge with count 1. All three edges are boundary edges, and chaining them produces a single boundary loop (v_0, v_1, v_2) . For a tetrahedron (4 triangles), every edge is shared by exactly 2 faces, so the count is always 2 and no boundary edges are found.

30.3.4 4. Pseudocode

Algorithm 135 Detect Mesh Boundaries

```

1: function DETECTMESHBOUNDARIES(mesh)
2:   edge_counts  $\leftarrow \{\}$ 
3:   for face  $\in$  mesh.faces do
4:     for i = 0  $\rightarrow$  len(face) - 1 do
5:        $v_1, v_2 \leftarrow \text{SortedPair}(\text{face}[i], \text{face}[(i + 1) \bmod \text{len}(\text{face})])$ 
6:       edge_counts[ $(v_1, v_2)$ ]  $\leftarrow$  edge_counts.get( $(v_1, v_2)$ , 0) + 1
7:     end for
8:   end for
9:   boundary_edges  $\leftarrow [e \mid e, c \in \text{edge\_counts.items}() \text{ if } c = 1]$ 
10:  loops  $\leftarrow []$ 
11:  visited_edges  $\leftarrow \{\}$ 
12:  for start_edge  $\in$  boundary_edges do
13:    if start_edge  $\in$  visited_edges then
14:      continue
15:    end if
16:    current_loop  $\leftarrow [\text{start\_edge}[0], \text{start\_edge}[1]]$ 
17:    visited_edges.add(start_edge)
18:    while True do
19:      next_edge  $\leftarrow \text{FindNextBoundaryEdge}(\text{current\_loop}[-1], \text{boundary\_edges}, \text{visited\_edges})$ 
20:      if next_edge = None or next_edge = start_edge then
21:        break
22:      end if
23:      next_v  $\leftarrow \text{next\_edge}[1]$  if next_edge[0] = current_loop[-1] else next_edge[0]
24:      current_loop.append(next_v)
25:      visited_edges.add(next_edge)
26:    end while
27:    loops.append(current_loop)
28:  end for
29:  return loops
30: end function

```

30.3.5 5. Parameter Selection

- **Mesh Representation:** Using a Half-Edge [GS85] data structure makes boundary detection trivial, as boundary edges are typically stored as “null” half-edges or explicit boundary objects.
- **Sorting:** When using an indexed face set, sorting the vertex IDs for each edge (e.g., $(\min(u, v), \max(u, v))$) ensures that the count correctly accounts for both orientations of the same geometric edge.

30.3.6 6. Complexity

- **Time Complexity:** $O(F)$, where F is the number of faces in the mesh. Chaining the loops takes $O(E_{\text{boundary}})$.
- **Space Complexity:** $O(E)$ to store the edge counts.

30.3.7 7. Usages

- **Surface Parameterization (UV Mapping):** BFF and Harmonic Mapping require the mesh boundary to be identified and mapped to a 2D shape.
- **Mesh Repair:** Identifying and filling holes in scanned 3D models.
- **Texture Blending:** Applying effects specifically at the boundaries of different surface materials.
- **Physical Simulation:** Applying boundary conditions (like fixing a boundary in place) for Finite Element analysis.

30.3.8 8. Testing Methods and Edge Cases

- **Watertight Mesh:** A perfectly closed mesh (like a sphere) should return zero boundary edges.
- **Open Sheet:** A mesh like a flat square should return a single boundary loop containing all perimeter edges.
- **Non-Manifold Edges:** Edges shared by more than two faces (count > 2) should be flagged as errors or handled separately.
- **Duplicate Faces:** Two faces covering the same three vertices will result in edge counts of 2, making them appear “watertight” even if they are isolated.
- **Multiple Holes:** Verify that each hole is correctly identified as a separate loop.

30.3.9 9. References

- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing.” CRC Press.
- Guibas, L. J., & Stolfi, J. [GS85] “Primitives for the manipulation of general subdivisions.” ACM Transactions on Graphics.
- Wikipedia: Boundary (topology).

30.4 Mesh Orientability

30.4.1 1. Overview

Mesh orientability is a property of a surface mesh that determines whether a consistent “outside” and “inside” (or a consistent normal direction) can be defined for the entire surface. An orientable surface allows every face to be oriented so that any two adjacent faces share an edge with opposite orientations. Non-orientable surfaces, such as the Möbius strip or the Klein bottle, cannot be assigned a consistent orientation [BKP⁺10].

30.4.2 2. Definitions

Definition 30.12 (Face Orientation). The order in which the vertices of a face are listed (e.g., (v_1, v_2, v_3)), which determines the face normal using the right-hand rule.

Definition 30.13 (Orientable Surface). A surface where face orientations can be made consistent: for every edge shared by faces F_1 and F_2 , the orientations of F_1 and F_2 must traverse the edge in opposite directions.

Definition 30.14 (Inconsistent Edge). An edge where both incident faces traverse the edge in the same direction (e.g., both use (u, v)).

Definition 30.15 (Two-Sided Surface). An orientable manifold mesh effectively has two sides (e.g., the inside and outside of a sphere).

30.4.3 3. Theory

The process of orienting a mesh is a graph-search problem.

1. Represent the mesh as a dual graph where each node is a face and edges connect adjacent faces (those sharing an edge).
2. Select an initial face and assign it an arbitrary orientation (e.g., CCW).
3. Propagate this orientation to its neighbors: if face F_1 uses edge (u, v) , then its neighbor F_2 **must** use edge (v, u) .
4. If at any point we encounter a neighbor that has already been oriented in a way that conflicts with its current neighbor, the mesh is **non-orientable**.

Theorem 30.16 (Orientability Characterization). *A connected 2-manifold surface S is orientable if and only if it does not contain a Möbius strip as a sub-surface. Equivalently, S is orientable if and only if every edge is traversed by its two incident faces in opposite directions.*

Proof. If S is orientable, we can assign consistent face orientations, so every edge is traversed in opposite directions by its two incident faces. Conversely, if every edge is traversed in opposite directions, the BFS propagation never encounters a conflict, and we obtain a consistent global orientation. The non-orientable case requires the existence of an orientation-reversing cycle, which is topologically equivalent to a Möbius strip. $\square$

Mathematically, a manifold surface is orientable if and only if it does not contain a Möbius strip as a sub-surface.

Example 30.17 (Orienting a Triangle Mesh). Consider a tetrahedron with faces $F_1 = (v_0, v_1, v_2)$, $F_2 = (v_0, v_2, v_3)$, $F_3 = (v_0, v_3, v_1)$, $F_4 = (v_1, v_3, v_2)$. Starting at F_1 with CCW orientation, edge (v_0, v_2) is traversed from v_0 to v_2 . Face F_2 must then traverse this edge as (v_2, v_0) , which is satisfied by (v_0, v_2, v_3) . Propagating to F_3 via edge (v_0, v_3) , we require (v_3, v_0) , satisfied by (v_0, v_3, v_1) . All four faces receive consistent orientations, confirming the tetrahedron is orientable.

30.4.4 4. Pseudocode

30.4.5 5. Parameter Selection

- **Start Face:** Any face can be used as a starting point. For non-closed meshes, starting at a boundary can be helpful.
- **Components:** Orientation must be checked and performed independently for each connected component of the mesh.

Algorithm 136 Orient Mesh

```
1: function ORIENTMESH(mesh)
2:   oriented_faces  $\leftarrow \{\}$ 
3:   face_directions  $\leftarrow \{\}$ 
4:   for start_face  $\in$  mesh.faces do
5:     if start_face  $\in$  oriented_faces then
6:       continue
7:     end if
8:     queue  $\leftarrow [start\_face]$ 
9:     oriented_faces.add(start_face)
10:    while queue do
11:      current_face  $\leftarrow queue.pop(0)$ 
12:      for (neighbor_face, shared_edge)  $\in$  GetNeighbors(current_face, mesh) do
13:        if neighbor_face  $\notin$  oriented_faces then
14:          new_orientation  $\leftarrow$  MatchOrientation(current_face, neighbor_face, shared_edge)
15:          face_directions[neighbor_face]  $\leftarrow$  new_orientation
16:          oriented_faces.add(neighbor_face)
17:          queue.append(neighbor_face)
18:        else
19:          if  $\neg$ IsConsistent(current_face, neighbor_face, shared_edge) then
20:            return False, “Mesh is non-orientable (Möbius strip topology)”
21:          end if
22:        end if
23:      end for
24:    end while
25:  end for
26:  return True, face_directions
27: end function

28: function ISCONSISTENT(f1, f2, edge)
29:   u, v  $\leftarrow$  GetEdgeVertices(f1, edge)
30:   u', v'  $\leftarrow$  GetEdgeVertices(f2, edge)
31:   return (u = v')  $\wedge$  (v = u')
32: end function
```

30.4.6 6. Complexity

- **Time Complexity:** $O(F)$, where F is the number of faces, as each face is visited exactly once in the BFS.
- **Space Complexity:** $O(F)$ to store the face orientations and the BFS queue.

30.4.7 7. Usages

- **Rendering:** Backface culling and lighting calculations depend on correct, consistent normals.
- **Slicing for 3D Printing:** Determining what is “solid” vs “void” requires consistent orientations.
- **Surface Offsetting:** Offsetting a surface requires moving vertices in a consistent normal direction.
- **Fluid Simulation:** Pressure forces must be applied consistently along the surface normals.

30.4.8 8. Testing Methods and Edge Cases

- **Möbius Strip:** A classic non-orientable manifold; the algorithm should detect the inconsistency.
- **Klein Bottle:** A closed non-orientable surface (though it must self-intersect in 3D).
- **Watertight Sphere:** Orienting one face should correctly orient the entire sphere.
- **Disconnected Mesh:** Ensure the algorithm handles multiple separate objects in the same file.
- **Non-Manifold Mesh:** Orientability is generally defined for manifolds; non-manifold edges (shared by 3+ faces) make orientability ambiguous.

30.4.9 9. References

- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing.” CRC Press.
- Guibas, L. J., & Stolfi, J. [GS85] “Primitives for the manipulation of general subdivisions.” ACM Transactions on Graphics.
- Wikipedia: Orientability.

30.5 Manifold Verification

30.5.1 1. Overview

Manifold verification is the process of checking whether a 3D surface mesh satisfies the topological constraints of a 2nd-order manifold. A manifold surface is one that “locally looks like a flat plane” at every point. Manifold meshes are highly desirable in geometric modeling because they have well-defined orientations, consistent normals, and are compatible with standard data structures like the half-edge mesh [GS85].

30.5.2 2. Definitions

Definition 30.18 (Edge Manifold). An edge is *edge-manifold* if it is shared by exactly two faces (for a closed surface) or one face (for a boundary edge).

Definition 30.19 (Vertex Manifold). A vertex is *vertex-manifold* if its surrounding faces form a single, connected “fan” or “disk.”

Definition 30.20 (Watertight). A closed manifold mesh with no holes: every edge has exactly two incident faces. Equivalently, the mesh has no boundary edges.

Definition 30.21 (Euler Characteristic). For a manifold mesh with V vertices, E edges, and F faces, the Euler characteristic is $\chi = V - E + F$. For a sphere-like surface, $\chi = 2$.

30.5.3 3. Theory

To verify that a mesh is a manifold, we check three primary conditions [BKP⁺10]:

1. **Edge Condition:** Every edge must be incident to either one or two faces. Edges with three or more incident faces (non-manifold edges) are topologically ambiguous.
2. **Vertex Condition:** For each vertex, the set of faces sharing that vertex must be edge-connected. If a vertex belongs to two separate “cones” that only touch at the vertex itself, it is non-manifold.
3. **Face Consistency:** For each edge, the orientation of its two incident faces must be compatible (e.g., if one face uses (u, v) , the other must use (v, u)).

Theorem 30.22 (Manifold Verification Correctness). *A mesh M is a 2-manifold if and only if it satisfies the edge condition, the vertex condition, and the face consistency condition simultaneously.*

Proof. ($\Rightarrow$) If M is a 2-manifold, then by definition every point has a neighborhood homeomorphic to a half-plane (boundary) or full plane (interior). This implies each edge has at most two incident faces (edge condition), each vertex has a single fan of faces (vertex condition), and the orientations are globally consistent (face consistency).

($\Leftarrow$) If all three conditions hold, the BFS orientation propagation succeeds globally (face consistency). The edge condition ensures no edge has more than two faces. The vertex condition ensures each vertex has a single connected neighborhood. Together, these guarantee that every point has the required local topological structure. $\square$

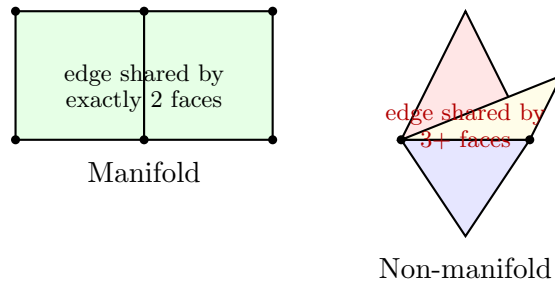


Figure 30.2: (Left) Manifold mesh: every edge is shared by exactly two faces. (Right) Non-manifold: an edge shared by three faces violates the manifold property.

Algorithm 137 Is Manifold

```

1: function ISMANIFOLD(mesh)
2:   edge_faces  $\leftarrow$  GetEdgeFaceMap(mesh)
3:   for (edge, faces)  $\in$  edge_faces.items() do
4:     if len(faces) > 2 then
5:       return False, “Non-manifold edge detected”
6:     end if
7:   end for
8:   for vertex  $\in$  mesh.vertices do
9:     neighbor_faces  $\leftarrow$  GetFacesSharingVertex(vertex, mesh)
10:    if  $\neg$ IsConnected(neighbor_faces, vertex) then
11:      return False, “Non-manifold vertex detected”
12:    end if
13:  end for
14:  if  $\neg$ HasConsistentOrientation(mesh) then
15:    return False, “Inconsistent face orientations”
16:  end if
17:  return True, “Mesh is manifold”
18: end function

19: function ISCONNECTED(faces, vertex)
20:   components  $\leftarrow$  FindFaceComponents(faces, vertex)
21:   return len(components) = 1
22: end function

```

30.5.4 4. Pseudocode**30.5.5** 5. Parameter Selection

- **Tolerance:** Floating-point precision should be considered when identifying “duplicate” vertices that should have been merged.
- **Boundary Handling:** A manifold with a boundary is still a manifold; the condition for boundary edges (1 face) and boundary vertices (faces form a single half-disk) must be handled.

30.5.6 6. Complexity

- **Time Complexity:** $O(F)$, where F is the number of faces, to build the edge-face map and check vertex connectivity.
- **Space Complexity:** $O(E + F)$ to store the connectivity information.

30.5.7 7. Usages

- **3D Printing:** Slicing software requires watertight, manifold meshes to correctly identify “inside” vs. “outside” volumes.
- **Physical Simulation:** Finite Element Method (FEM) and Fluid Dynamics simulations often fail on non-manifold geometry.
- **Mesh Boolean Operations:** Algorithms like Union/Intersection are much more robust on manifold meshes.

- **Surface Parameterization:** Algorithms like BFF or LSCM rely on a consistent half-edge structure, which requires a manifold mesh.

30.5.8 8. Testing Methods and Edge Cases

- **T-Junctions:** A vertex lying on an edge of another face (without sharing it) is geometrically non-manifold.
- **Pinch Points:** Two separate mesh components touching at a single vertex.
- **Self-Intersections:** A mesh can be topologically manifold while still being geometrically self-intersecting (this is usually checked separately).
- **Isolated Vertices:** Vertices with zero incident faces.
- **Inconsistent Normals:** A “Möbius strip” style mesh where orientations cannot be made consistent.

30.5.9 9. References

- Guibas, L. J., & Stolfi, J. [GS85] “Primitives for the manipulation of general subdivisions.” ACM Transactions on Graphics.
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing.” CRC Press.
- Wikipedia: Manifold.

30.6 Euler Characteristic

30.6.1 1. Overview

The Euler characteristic (χ) is a topological invariant, a number that describes a topological space’s shape or structure regardless of the way it is bent or twisted. For a 3D surface mesh (a polyhedral surface), it relates the number of vertices (V), edges (E), and faces (F). It is a fundamental tool for classifying surfaces and verifying the topological integrity of meshes [Eul58, Poi95].

30.6.2 2. Definitions

Definition 30.23 (Euler Formula). For a convex polyhedron (topologically equivalent to a sphere), the Euler formula states that $\chi = V - E + F = 2$.

Definition 30.24 (Genus). The genus g of a surface is the number of “handles” or “holes” through a surface (e.g., a torus has $g = 1$, a sphere has $g = 0$).

Definition 30.25 (Boundary Components). The number b of connected boundary loops in a mesh.

Definition 30.26 (Topological Invariant). A property that remains unchanged under continuous deformation (homeomorphism).

30.6.3 3. Theory

For a general connected orientable surface, the Euler characteristic is related to its genus and boundary count by the formula:

$$\chi = V - E + F = 2 - 2g - b$$

Theorem 30.27 (Euler–Poincaré Formula). *For a connected orientable 2-manifold with genus g and b boundary components, the Euler characteristic satisfies $\chi = V - E + F = 2 - 2g - b$. In particular, for a closed surface ($b = 0$), $\chi = 2 - 2g$.*

Proof. We prove this by induction on the genus g . For $g = 0$ (a sphere), the classical Euler formula gives $\chi = 2$. We establish the formula for $g = 0, b > 0$ by noting that each boundary component is obtained by removing a disk, which reduces χ by 1. Hence $\chi = 2 - b$.

For the inductive step, suppose the formula holds for genus g . To obtain a surface of genus $g+1$, we remove two disks from the surface and attach a cylinder (tube) between the resulting boundary circles. Removing two disks decreases χ by 2, and attaching the cylinder (which has $\chi = 0$) does not change χ . The genus increases by 1. Thus $\chi_{g+1} = \chi_g - 2 = (2 - 2g) - 2 = 2 - 2(g + 1)$, completing the induction. $\square$

This allows us to determine the global topology of a mesh simply by counting its elements:

- **Sphere:** $V - E + F = 2$ ($g = 0, b = 0$).
- **Disk:** $V - E + F = 1$ ($g = 0, b = 1$).
- **Torus:** $V - E + F = 0$ ($g = 1, b = 0$).
- **Cylinder:** $V - E + F = 0$ ($g = 0, b = 2$).

The Gauss–Bonnet theorem further relates the Euler characteristic to the total curvature of the surface: $\int_M K dA = 2\pi\chi$. In the discrete case, this means the sum of angle defects at all vertices is exactly $2\pi\chi$, where the angle defect at vertex v is defined as $\delta(v) = 2\pi - \sum_i \theta_i$, with θ_i being the angles of the faces incident to v .

Example 30.28 (Euler Characteristic Computations). 1. **Tetrahedron:** $V = 4, E = 6, F = 4$. Thus $\chi = 4 - 6 + 4 = 2$, confirming it is homeomorphic to a sphere ($g = 0$).

2. **Cube:** $V = 8, E = 12, F = 6$. Thus $\chi = 8 - 12 + 6 = 2$.

3. **Torus (mesh approximation):** A torus mesh with V vertices satisfies $V - E + F = 0$, regardless of resolution. For instance, a coarse torus with $V = 16, E = 32, F = 16$ gives $\chi = 0$.

4. **Two disjoint spheres:** $V = 8, E = 12, F = 8$ (two tetrahedra). $\chi = 8 - 12 + 8 = 4 = 2 \times 2$, consistent with the formula for $C = 2$ components.

30.6.4 4. Pseudocode

30.6.5 5. Parameter Selection

- **Element Counting:** Ensure that duplicate vertices or edges are correctly merged before counting, as they will artificially deflate/inflate χ .
- **Component Handling:** If the mesh has C disconnected components, the formula becomes $\chi = V - E + F = 2C - 2g - b$.

Algorithm 138 Calculate Euler Characteristic

```
1: function CALCULATEEULERCHARACTERISTIC(mesh)
2:    $V \leftarrow \text{len}(\text{mesh.vertices})$ 
3:    $F \leftarrow \text{len}(\text{mesh.faces})$ 
4:    $\text{edges} \leftarrow \{\}$ 
5:   for  $\text{face} \in \text{mesh.faces}$  do
6:     for  $i = 0 \rightarrow \text{len}(\text{face}) - 1$  do
7:        $v_1 \leftarrow \text{face}[i]$ 
8:        $v_2 \leftarrow \text{face}[(i + 1) \bmod \text{len}(\text{face})]$ 
9:        $\text{edges.add}(\text{tuple}(\text{sorted}((v_1, v_2))))$ 
10:    end for
11:  end for
12:   $E \leftarrow \text{len}(\text{edges})$ 
13:   $\chi \leftarrow V - E + F$ 
14:  return  $\chi$ 
15: end function
```

30.6.6 6. Complexity

- **Time Complexity:** $O(F)$, where F is the number of faces, to iterate through the mesh and identify unique edges.
- **Space Complexity:** $O(E)$ to store the set of unique edges.

30.6.7 7. Usages

- **Mesh Verification:** Detecting if a mesh has unexpected holes or disconnected parts.
- **Surface Classification:** Automatically determining if a scanned object is a “donut” (torus) or a “ball” (sphere).
- **Topology-Preserving Operations:** Ensuring that operations like mesh simplification or remeshing do not change the number of holes in the model.
- **Shape Matching:** Using χ as a fast, first-pass filter to distinguish between objects with different topologies.
- **Theoretical Physics:** Used in string theory and condensed matter physics to classify manifold structures.

30.6.8 8. Testing Methods and Edge Cases

- **Platonic Solids:** Verify that the Tetrahedron, Cube, Octahedron, Dodecahedron, and Icosahedron all yield $\chi = 2$.
- **Open Meshes:** A single triangle has $V = 3, E = 3, F = 1 \Rightarrow \chi = 1$ (a disk).
- **Disconnected Parts:** A mesh consisting of two separate spheres should have $\chi = 4$.
- **Non-Orientable Surfaces:** For a Möbius strip, $\chi = 0$ ($V - E + F = 0$, $g_{\text{non-orient}} = 1, b = 1$).
- **Non-Manifold Edges:** Ensure the edge counting logic correctly handles edges shared by 3+ faces (though the standard formula is defined for 2-manifolds).

30.6.9 9. References

- Euler, L. [Eul58] “Elementa doctrinae solidorum.” Novi Commentarii Academiae Scientiarum Petropolitanae.
- Poincaré, H. [Poi95] “Analysis situs.” Journal de l’École Polytechnique.
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing.” CRC Press.
- Wikipedia: Euler characteristic.

30.7 Node Degree and Valence

30.7.1 1. Overview

Node degree calculation is the process of counting the number of edges incident to each vertex (node) in a graph or mesh. In the context of 3D meshes, vertex degree (often called **valence**) is a key indicator of mesh quality and topological regularity. Vertices with unusual degrees (extraordinary vertices) can cause artifacts in subdivision and smoothing algorithms [BKP⁺10].

30.7.2 2. Definitions

Definition 30.29 (Valence (Degree)). The number of edges connected to a vertex v , denoted $\deg(v)$.

Definition 30.30 (Regular Vertex). A vertex with the “ideal” degree for its mesh type: degree 6 for triangle meshes, degree 4 for quad meshes.

Definition 30.31 (Extraordinary Vertex). A vertex with a degree other than the regular degree. Also called a singularity.

Definition 30.32 (Average Degree). The sum of all degrees divided by the number of vertices. For a closed triangle mesh, the average degree approaches 6 as $V \rightarrow \infty$.

30.7.3 3. Theory

Theorem 30.33 (Handshaking Lemma). *The sum of all vertex degrees in a mesh is equal to twice the number of edges: $\sum_{i=1}^V \deg(v_i) = 2E$.*

Proof. Each edge (u, v) contributes exactly 1 to $\deg(u)$ and 1 to $\deg(v)$. Summing over all edges, the total contribution to the degree sum is $2E$. $\square$

For a closed 2-manifold triangle mesh, the average degree is related to the Euler characteristic χ :

$$\bar{d} = 6 - \frac{6\chi}{V}$$

Theorem 30.34 (Average Degree of Triangle Mesh). *For a closed 2-manifold triangle mesh with V vertices, E edges, and Euler characteristic χ , the average vertex degree is $\bar{d} = 6 - 6\chi/V$.*

Proof. By the Handshaking Lemma [Die17], $\bar{d} = 2E/V$. For a triangle mesh, each face has 3 edges and each edge is shared by 2 faces, so $3F = 2E$, giving $F = 2E/3$. Substituting into Euler’s formula $\chi = V - E + F$:

$$\chi = V - E + \frac{2E}{3} = V - \frac{E}{3}$$

Hence $E = 3(V - \chi)$, and $\bar{d} = 2E/V = 6(V - \chi)/V = 6 - 6\chi/V$. $\square$

As the number of vertices $V \rightarrow \infty$, the average degree approaches 6.

In a Half-Edge [GS85] data structure, calculating the degree involves “circulating” around the vertex using the `next` and `twin` pointers until the starting half-edge is reached again.

Example 30.35 (Valence Distribution). Consider an icosahedron: $V = 12$, $E = 30$, $F = 20$. Every vertex has degree 5 (all vertices are regular for this polyhedron). The average degree is $\bar{d} = 2 \times 30/12 = 5$, which matches the formula $\bar{d} = 6 - 6(2)/12 = 5$. In a refined triangle mesh with genus 0 and many vertices, the average degree approaches 6, but the 12 original icosahedral vertices remain extraordinary with degree 5.

30.7.4 4. Pseudocode

Using an Adjacency List:

Algorithm 139 Calculate Vertex Degrees

```

1: function CALCULATEDEGREES(mesh)
2:   degrees  $\leftarrow$   $[0] \times \text{num\_vertices}$ 
3:   for  $(v_1, v_2) \in \text{mesh.edges}$  do
4:     degrees[ $v_1$ ]  $\leftarrow$  degrees[ $v_1$ ] + 1
5:     degrees[ $v_2$ ]  $\leftarrow$  degrees[ $v_2$ ] + 1
6:   end for
7:   return degrees
8: end function
```

Using a Half-Edge Structure:

Algorithm 140 Get Vertex Degree

```

1: function GETVERTEXDEGREE(mesh, v)
2:   count  $\leftarrow$  0
3:   start_he  $\leftarrow$  v.halfedge
4:   curr_he  $\leftarrow$  start_he
5:   loop
6:     count  $\leftarrow$  count + 1
7:     curr_he  $\leftarrow$  curr_he.twin.next
8:     if curr_he = start_he then
9:       break
10:    end if
11:  end loop
12:  return count
13: end function
```

30.7.5 5. Parameter Selection

- **Boundary Handling:** Vertices on the boundary of an open mesh naturally have lower degrees (typically 2 or 3). They should be flagged or handled separately in quality metrics.
- **Directed vs. Undirected:** For general graphs, degrees are split into **In-Degree** and **Out-Degree**. For standard surface meshes, edges are considered undirected.

30.7.6 6. Complexity

- **Time Complexity:** $O(E)$ to iterate through all edges, or $O(V \cdot \bar{d})$ to circulate around each vertex. Both are linear in the size of the mesh.

- **Space Complexity:** $O(V)$ to store the degree values.

30.7.7 7. Usages

- **Mesh Quality Analysis:** Identifying regions with poor connectivity that might cause simulation errors.
- **Subdivision Surfaces:** Stencils for Loop [Loo87] or Catmull-Clark [CC78] subdivision change based on the vertex degree to maintain smoothness.
- **Mesh Simplification:** Algorithms like Quadratic Error Metrics (QEM) often prioritize removing vertices based on their degree and geometric impact.
- **Network Science:** Identifying “hubs” (nodes with very high degrees) in social or infrastructure graphs.
- **Topology Verification:** Checking if a mesh is “regular” (e.g., all vertices have degree 6).

30.7.8 8. Testing Methods and Edge Cases

- **Simple Polyhedra:** Verify that all vertices of a cube have degree 3, and all vertices of an icosahedron have degree 5.
- **Regular Grid:** A flat triangle grid should have interior vertices of degree 6.
- **Boundary Vertices:** Test on an open square; corner vertices should have degree 2 and edge vertices degree 3.
- **Non-Manifold Vertices:** Verify the count when multiple “fans” of triangles meet at a single vertex.
- **Isolated Vertices:** Ensure vertices with zero edges are handled (degree 0).

30.7.9 9. References

- Diestel, R. [Die17] “Graph Theory.” Springer.
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing.” CRC Press.
- Wikipedia: Degree (graph theory).

30.8 Mesh Coloring

30.8.1 1. Overview

Mesh coloring is the process of assigning labels (colors) to the elements of a mesh (typically vertices or faces) such that no two adjacent elements share the same label. This is a direct application of graph coloring to the connectivity of a mesh. While it is often used for visual aesthetics, its primary technical purpose is to identify independent sets of elements that can be processed in parallel without race conditions.

30.8.2 2. Definitions

Definition 30.36 (*k*-Coloring). An assignment of k colors to vertices such that adjacent vertices have different colors.

Definition 30.37 (Vertex Graph). A graph where vertices of the mesh are nodes and edges of the mesh are graph edges.

Definition 30.38 (Dual Graph (Face Graph)). A graph where each face of the mesh is a node and edges connect faces that share an edge.

Definition 30.39 (Chromatic Number). The chromatic number $\chi(G)$ is the minimum number of colors needed to color a graph G .

30.8.3 3. Theory

For a 2D surface mesh (which is essentially a planar or surface-embedded graph), several key theorems apply:

Theorem 30.40 (Four-Color Theorem). *Any planar graph can be colored with at most 4 colors [WP67]. Since most surface meshes are locally planar, 4 to 6 colors are typically sufficient.*

Proof. The Four-Color Theorem was originally proved by Appel and Haken (1976) using computer-assisted case analysis, and later given a simpler (though still computer-assisted) proof by Robertson, Sanders, Seymour, and Thomas (1997). The key idea is that every planar graph has a vertex of degree at most 5, which allows an inductive argument: remove such a vertex, color the smaller graph by induction, then restore the vertex using one of the 4 colors not used by its at-most-5 neighbors (by the pigeonhole principle, at least one color is available among 4). The hard part is handling the finite set of unavoidable configurations. $\square$

Theorem 30.41 (Brooks' Theorem). *The chromatic number is at most the maximum vertex degree Δ , unless the graph is a complete graph or an odd cycle.*

The most common algorithm is the **Greedy Coloring** approach:

1. Iterate through vertices in a specific order.
2. For the current vertex, check the colors of its already-colored neighbors.
3. Assign the smallest available color that is not used by any neighbor.

Example 30.42 (Greedy Coloring Trace). Consider a triangle mesh with vertices v_0, v_1, v_2, v_3 forming two triangles (v_0, v_1, v_2) and (v_1, v_2, v_3) . Ordering by descending degree: v_1 (deg 4), v_2 (deg 4), v_0 (deg 2), v_3 (deg 2). Color v_1 with 0 (no colored neighbors). Color v_2 with 1 (neighbor v_1 uses 0). Color v_0 with 1 (neighbors v_1 uses 0, v_2 uses 1; but v_0 is only adjacent to v_1 and v_2). Wait, v_0 is adjacent to both v_1 and v_2 in triangle (v_0, v_1, v_2) , so its neighbors use colors $\{0, 1\}$; assign color 2. Color v_3 similarly with color 2 (neighbors v_1 and v_2 use $\{0, 1\}$). Total: 3 colors, matching the fact that the chromatic number of a triangle mesh is at most 4 by Theorem 30.40.

30.8.4 4. Pseudocode

30.8.5 5. Parameter Selection

- **Ordering Strategy:** Sorting vertices by degree (Largest First) often results in fewer total colors than a random ordering.
- **Saturation Degree (DSATUR):** A more advanced strategy that colors vertices with the most uniquely colored neighbors first.

Algorithm 141 Color Mesh

```

1: function COLORMESH(mesh, strategy = “GREEDY”)
2:   colors  $\leftarrow \{v : -1 \mid v \in \text{mesh.vertices}\}$ 
3:   ordered_vertices  $\leftarrow \text{SortByDegree}(\text{mesh.vertices}, \text{descending} = \text{True})$ 
4:   for v  $\in$  ordered_vertices do
5:     neighbor_colors  $\leftarrow \{\}$ 
6:     for neighbor  $\in$  GetNeighbors(v, mesh) do
7:       if colors[neighbor]  $\neq -1$  then
8:         neighbor_colors.add(colors[neighbor])
9:       end if
10:    end for
11:    color  $\leftarrow 0$ 
12:    while color  $\in$  neighbor_colors do
13:      color  $\leftarrow$  color + 1
14:    end while
15:    colors[v]  $\leftarrow$  color
16:  end for
17:  return colors
18: end function

```

30.8.6 6. Complexity

- **Time Complexity:** $O(V \cdot \Delta)$, where V is the number of vertices and Δ is the average vertex degree (typically $\Delta \approx 6$ for triangular meshes).
- **Space Complexity:** $O(V)$ to store the assigned color for each vertex.

30.8.7 7. Usages

- **Parallel Computing (GPGPU):** When performing operations like mesh smoothing (Mean Curvature Flow [GH86, Gra87]) on a GPU, vertices of the same color can be updated simultaneously because they don’t share data with each other.
- **Implicit Solvers:** Partitioning the Laplacian matrix into independent blocks for faster parallel solves.
- **Visualizing Mesh Structure:** Highlighting different regions or patches for debugging.
- **Texture Mapping:** Assigning different texture atlases to different parts of a mesh to avoid overlaps.

30.8.8 8. Testing Methods and Edge Cases

- **Adjacency Check:** After coloring, iterate through every edge (u, v) and verify that $\text{color}[u] \neq \text{color}[v]$.
- **Color Count:** Verify that the total number of colors used is reasonably small (e.g., < 10 for typical meshes).
- **Disconnected Components:** Ensure the algorithm handles multiple separate mesh objects correctly.
- **High-Valence Vertices:** Check performance and color count on meshes with “star” patterns (vertices with many neighbors).

- **Face Coloring:** Test by applying the same logic to the dual graph (coloring faces instead of vertices).

30.8.9 9. References

- Welsh, D. J. A., & Powell, M. B. [WP67] “An upper bound for the chromatic number of a graph and its application to timetabling problems.” The Computer Journal.
- Brélaz, D. (1979). “New methods to color the vertices of a graph.” Communications of the ACM.
- Wikipedia: Graph coloring.

30.9 Mesh Reordering

30.9.1 1. Overview

Mesh reordering is the process of permuting the indices of the vertices (and faces) of a mesh to improve the performance of numerical solvers and data access patterns. The most common objective is to minimize the **bandwidth** of the sparse matrices derived from the mesh (like the Laplacian). Reducing the bandwidth ensures that non-zero entries are clustered close to the main diagonal, leading to faster computations and more efficient CPU cache usage [CM69].

30.9.2 2. Definitions

Definition 30.43 (Bandwidth). For a matrix A , the bandwidth is the maximum value of $|i - j|$ such that $A_{ij} \neq 0$.

Definition 30.44 (Permutation). A mapping P that reassigns each vertex index i to a new index $j = P(i)$.

Definition 30.45 (Profile). The total number of zeros within the band of the matrix; smaller profiles are generally better.

Definition 30.46 (Adjacency Graph). The graph where vertices of the mesh are nodes and edges represent connections between them.

30.9.3 3. Theory

The most widely used reordering technique is the Reverse Cuthill–McKee (RCM) algorithm [CM69]. It is a refinement of a breadth-first search (BFS) that carefully chooses the starting node and the order of neighbors.

1. Find a “pseudo-peripheral” vertex (one that is far from other vertices, often by using a heuristic).
2. Perform a BFS starting from this vertex.
3. In each step of the BFS, sort the neighbors by their degree (number of connections) in ascending order.
4. Once all vertices are visited, reverse the resulting permutation. Reversing is key because it significantly reduces the matrix profile compared to the standard CM order.

Example 30.47 (RCM Reordering). Consider a path graph $v_0 - v_1 - v_2 - v_3 - v_4$. Starting BFS from the peripheral vertex v_0 : the permutation is $[v_0, v_1, v_2, v_3, v_4]$. Reversing gives $[v_4, v_3, v_2, v_1, v_0]$. The bandwidth of the adjacency matrix is 1 in both cases (since it is already optimal for a path). However, for a more complex mesh like a 2D grid, RCM produces a bandwidth of $O(\sqrt{N})$ rather than $O(N)$.

30.9.4 4. Pseudocode

Algorithm 142 RCM Reorder

```

1: function RCMREORDER(mesh)
2:    $v \leftarrow \text{FindPseudoPeripheralVertex}(\textit{mesh})$ 
3:    $R \leftarrow [v]$ 
4:    $\textit{visited} \leftarrow \{v\}$ 
5:   for  $\textit{current\_v} \in R$  do
6:      $\textit{neighbors} \leftarrow \text{GetNeighbors}(\textit{current\_v}, \textit{mesh})$ 
7:      $\textit{sorted\_neighbors} \leftarrow \text{sorted}([n \mid n \in \textit{neighbors} \text{ if } n \notin \textit{visited}], \textit{key} = \lambda n : \text{len}(\text{GetNeighbors}(n, \textit{mesh})))$ 
8:     for  $n \in \textit{sorted\_neighbors}$  do
9:        $R.\text{append}(n)$ 
10:       $\textit{visited}.\text{add}(n)$ 
11:    end for
12:  end for
13:  return  $\text{list}(\text{reversed}(R))$ 
14: end function

```

30.9.5 5. Parameter Selection

- **Starting Vertex:** The choice of the first vertex is critical. A bad starting point leads to high bandwidth. Heuristics like the Gibbs–Poole–Stockmeyer (GPS) algorithm are often used to find a good starting point.
- **Metric:** While bandwidth reduction is the most common goal, other orderings (like Nested Dissection) are better for parallel solvers or multifrontal methods.

30.9.6 6. Complexity

- **Time Complexity:** $O(V \cdot \Delta \log \Delta)$, where V is vertices and Δ is average degree. Since $\Delta \approx 6$ for meshes, this is essentially $O(V)$.
- **Space Complexity:** $O(V + E)$ to store the adjacency list and the visit status.

30.9.7 7. Usages

- **Sparse Linear Solvers:** Essential preprocessing step for Cholesky or LU factorization of Laplacian systems.
- **FEM Simulations:** Reducing memory access time and improving cache hit rates in structural or fluid simulations.
- **Data Compression:** Reordered meshes often have more predictable connectivity, allowing for better compression (e.g., in the PLY or OBJ formats).
- **Rendering:** Improving the vertex cache hit rate during triangle strip or fan rendering.

30.9.8 8. Testing Methods and Edge Cases

- **Bandwidth Calculation:** Compare the bandwidth of the Laplacian matrix before and after reordering. A successful RCM should show a significant reduction.
- **Graph with Disconnected Components:** The algorithm must be applied to each component separately and concatenated.
- **Highly Regular Mesh:** A perfect grid should be reordered into a diagonal-like pattern.
- **Very Large Mesh:** Ensure the algorithm handles millions of vertices without excessive memory overhead.
- **Degenerate Mesh:** Test on meshes with very high-valence vertices (where bandwidth reduction is harder).

30.9.9 9. References

- Cuthill, E., & McKee, J. [CM69] “Reducing the bandwidth of sparse symmetric matrices.” ACM National Conference.
- George, A. (1971). “Computer implementation of the finite element method.” Stanford Technical Report.
- Liu, W. H., & Sherman, A. H. (1976). “Comparative analysis of the Cuthill–McKee and the reverse Cuthill–McKee algorithms for sparse matrices.” SIAM Journal on Numerical Analysis.
- Wikipedia: Cuthill–McKee algorithm.

30.10 Mesh Rigidity

30.10.1 1. Overview

Mesh rigidity is the study of whether a framework of vertices and edges (a mesh) can be continuously deformed while preserving the lengths of all its edges. A mesh is **rigid** if the only possible motions are rigid body motions (translation and rotation). If it can be deformed in other ways, it is **flexible**. Rigidity analysis is critical for structural engineering, robotics, and molecular biology.

30.10.2 2. Definitions

Definition 30.48 (Bar-and-Joint Framework). A graph where edges represent rigid bars and vertices represent flexible joints.

Definition 30.49 (Rigidity Matrix). A matrix R that relates the velocities of vertices to the rates of change of edge lengths. For each edge (i, j) , the corresponding row contains $(P_i - P_j)$ at columns $3i$ through $3i + 2$ and $(P_j - P_i)$ at columns $3j$ through $3j + 2$.

Definition 30.50 (Infinitesimal Rigidity). A linear approximation where a mesh is rigid if the only vertex velocities that preserve edge lengths (to first order) are rigid body motions.

Definition 30.51 (Degrees of Freedom). The dimension of the kernel of the rigidity matrix. For a rigid 3D framework, $\text{DOF} = 6$ (3 translation + 3 rotation).

30.10.3 3. Theory

The rigidity of a mesh is determined by the **Rigidity Matrix** R . For each edge (i, j) , there is a row in R containing the vector $(P_i - P_j)$ at columns corresponding to P_i and $(P_j - P_i)$ at columns for P_j .

A framework is infinitesimally rigid if the rank of R is $3V - 6$ (in 3D) or $2V - 3$ (in 2D).

- If $\text{rank}(R) < 3V - 6$, the mesh has internal mechanisms (it is flexible).
- If $\text{rank}(R) = 3V - 6$, the mesh is rigid.
- If the number of edges $E > 3V - 6$, the mesh is **redundantly rigid** or over-constrained.

Theorem 30.52 (Maxwell's Counting Rule / Laman's Theorem in 2D). *A graph $G = (V, E)$ in 2D is generically rigid if and only if $E = 2V - 3$ and for every subgraph $H = (V', E')$ with $|V'| \geq 2$, $E' \leq 2|V'| - 3$.*

Proof. The necessity direction is straightforward: if $E > 2|V'| - 3$ for some subgraph, then that subgraph has too few constraints relative to its vertices, so it must be flexible. The sufficiency direction (the harder direction) was proved by Laman [Lam70] using a combinatorial characterization based on matroid theory. The key insight is that the generic rigidity matroid on a graph G coincides with the sparsity matroid defined by the counting condition, which can be checked efficiently using the pebble game algorithm. $\square$

Maxwell's Counting Rule provides a combinatorial necessary condition: a graph must have at least $3V - 6$ edges to be rigid in 3D, but this is not sufficient (as the edges might be poorly distributed).

Example 30.53 (Rigidity Analysis). 1. **Tetrahedron:** $V = 4$, $E = 6$, $3V - 6 = 6$. The rigidity matrix has rank 6, so the tetrahedron is rigid. It is the simplest rigid 3D structure.

2. **Square with no diagonal:** $V = 4$, $E = 4$, $3V - 6 = 6$. Since $E < 3V - 6$, the square is under-constrained and flexible (it can be deformed into a rhombus). Adding one diagonal gives $E = 5 < 6$, still flexible. Adding both diagonals gives $E = 6 = 3V - 6$, and the resulting framework is rigid.

3. **Icosahedron:** $V = 12$, $E = 30$, $3V - 6 = 30$. Exactly the right count, and the icosahedron is rigid.

30.10.4 4. Pseudocode

30.10.5 5. Parameter Selection

- **Numerical Tolerance:** Since rank calculation is sensitive to noise, a threshold based on the machine epsilon or the smallest non-zero singular value must be used.
- **Generic vs. Specific Position:** A mesh might be rigid in general but flexible in a specific degenerate configuration (e.g., all vertices on a line). Generic rigidity depends only on the graph topology.

30.10.6 6. Complexity

- **Matrix Construction:** $O(E)$.
- **Rank Calculation:** $O(E \cdot V^2)$ using SVD. For large meshes, sparse solvers or **Pebble Game** algorithms (for combinatorial rigidity) are used.

Algorithm 143 Analyze Mesh Rigidity

```
1: function ANALYZEMESHRIGIDITY(mesh)
2:    $V \leftarrow \text{len}(\text{mesh.vertices})$ 
3:    $E \leftarrow \text{len}(\text{mesh.edges})$ 
4:   if  $E < 3 \cdot V - 6$  then
5:     return “Flexible (under-constrained)”
6:   end if
7:    $R \leftarrow \text{SparseMatrix}(E, 3 \cdot V)$ 
8:   for (row_idx, edge)  $\in \text{enumerate}(\text{mesh.edges})$  do
9:      $i, j \leftarrow \text{edge}$ 
10:     $p_i \leftarrow \text{mesh.vertices}[i]$ 
11:     $p_j \leftarrow \text{mesh.vertices}[j]$ 
12:     $\text{diff} \leftarrow p_i - p_j$ 
13:     $R[\text{row\_idx}, 3i : 3i + 3] \leftarrow \text{diff}$ 
14:     $R[\text{row\_idx}, 3j : 3j + 3] \leftarrow -\text{diff}$ 
15:   end for
16:    $\text{rank} \leftarrow \text{CalculateRank}(R)$ 
17:    $\text{dof} \leftarrow 3 \cdot V - \text{rank}$ 
18:   if  $\text{dof} = 6$  then
19:     return “Rigid”
20:   else
21:     return “Flexible with  $\text{dof} - 6$  internal mechanisms”
22:   end if
23: end function
```

30.10.7 7. Usages

- **Structural Engineering:** Verifying that a bridge or building truss is stable and won’t collapse.
- **Robotics:** Analyzing the workspace and mobility of parallel manipulators.
- **Molecular Biology:** Studying the flexibility of protein structures to understand how they bind to other molecules.
- **Sensor Networks:** Determining if a set of distance measurements is sufficient to uniquely locate all sensors (Localization).
- **Computer Graphics:** Constraining mesh deformations to be “as-rigid-as-possible” (ARAP) for realistic animation.

30.10.8 8. Testing Methods and Edge Cases

- **Tetrahedron:** The simplest rigid 3D structure ($V = 4, E = 6, 3V - 6 = 6$).
- **Square:** A 2D square with 4 edges is flexible ($V = 4, E = 4, 2V - 3 = 5$ required); adding a diagonal makes it rigid.
- **Collinear Vertices:** Verify that the algorithm detects flexibility when vertices are aligned, even if the graph is combinatorially rigid.
- **Disconnected Mesh:** The rank check should reflect the sum of rigid body motions for each component (6 DOF per component).
- **Over-constrained Mesh:** Ensure the algorithm handles $E > 3V - 6$ correctly.

30.10.9 9. References

- Asimow, L., & Roth, B. (1978). “The rigidity of graphs.” Transactions of the American Mathematical Society.
- Laman, G. [Lam70] “On graphs and rigidity of plane skeletal structures.” Journal of Engineering Mathematics.
- Connelly, R. (1980). “The rigidity of polyhedra.” SIAM Journal on Mathematical Analysis.
- Wikipedia: Structural rigidity.

30.11 Mesh Decimation via Quadric Error Metrics

30.11.1 1. Overview

Mesh decimation is the process of reducing the number of faces (and vertices) in a triangle mesh while preserving its geometric shape as closely as possible. This is a fundamental operation in computer graphics and geometric computing. High-resolution meshes captured by 3D scanners or generated by subdivision surfaces often contain millions of triangles, far more than necessary for real-time rendering, transmission, or simulation. Mesh decimation enables Level-of-Detail (LOD) management, where coarser versions of a model are used when it occupies a small portion of the screen, and finer versions when it is viewed up close. Other critical applications include reducing the cost of Finite Element Method (FEM) simulations, accelerating ray tracing, and lowering memory and bandwidth requirements for networked 3D applications.

The Quadric Error Metric (QEM) method, introduced by Garland and Heckbert [GH97], is one of the most widely used decimation algorithms. It proceeds by iteratively collapsing edges, where each collapse merges two vertices into one. The key insight is that the geometric error introduced by an edge collapse can be efficiently approximated using 4×4 symmetric matrices called *quadrics*, enabling near-optimal vertex placement at each step.

30.11.2 2. Definitions

Definition 30.54 (Edge Collapse). An edge collapse is an operation that merges two adjacent vertices u and v into a single vertex $\bar{v}$, removing the edge (u, v) and all faces incident to it. Formally, the map $u \mapsto \bar{v}$, $v \mapsto \bar{v}$ is applied to all faces, and degenerate faces (those with fewer than three distinct vertices) are discarded.

Definition 30.55 (Quadric Error Metric). Given a vertex v , its quadric matrix Q_v is a 4×4 symmetric matrix that summarizes the squared distances from v to a set of planes. For a point $\bar{v}$ in homogeneous coordinates $\bar{v}_h = (\bar{v}_x, \bar{v}_y, \bar{v}_z, 1)^\top$, the quadric error is $\Delta(\bar{v}) = \bar{v}_h^\top Q_v \bar{v}_h$. The quadric for a single plane $\mathbf{p} = (a, b, c, d)^\top$ (where $a^2 + b^2 + c^2 = 1$) is $K_{\mathbf{p}} = \mathbf{p} \mathbf{p}^\top$.

Definition 30.56 (Link Condition). The link of a vertex v , denoted $\text{Link}(v)$, is the set of edges connecting the neighbors of v that are not incident to v itself. An edge (u, v) satisfies the link condition if $\text{Link}(u) \cap \text{Link}(v) \subseteq \text{Link}(u, v)$, where $\text{Link}(u, v)$ is the set of edges in the link of both u and v that are not the edge (u, v) itself. This condition ensures that the local topology is consistent after the collapse.

Definition 30.57 (Manifold Preservation). A decimation step *preserves manifoldness* if, after the edge collapse, every edge remains incident to at most two faces and every vertex retains a single connected fan of incident faces.

30.11.3 3. Theory

The QEM algorithm assigns each vertex v a 4×4 symmetric matrix Q_v that encodes the sum of squared distances from v to the planes of all faces incident to v .

Initial Quadric Computation

For each face with unit outward normal $\mathbf{n} = (n_x, n_y, n_z)^\top$ and offset $d = -\mathbf{n} \cdot \mathbf{v}_0$ (where $\mathbf{v}_0$ is any vertex of the face), define the plane vector $\mathbf{p} = (n_x, n_y, n_z, d)^\top$. The fundamental quadric of this face is:

$$K_{\mathbf{p}} = \mathbf{p} \mathbf{p}^\top = \begin{pmatrix} a^2 & ab & ac & ad \\ ab & b^2 & bc & bd \\ ac & bc & c^2 & cd \\ ad & bd & cd & d^2 \end{pmatrix}$$

The quadric of a vertex v is the sum of the fundamental quadrics of all faces incident to v :

$$Q_v = \sum_{\mathbf{p} \in \text{Faces}(v)} K_{\mathbf{p}}$$

Boundary Penalty Quadrics

Mesh boundaries require special treatment to prevent edges from collapsing inward and creating visual artifacts or topological anomalies. For each boundary edge (u, v) , a boundary penalty quadric is constructed as follows. Let $\mathbf{f}$ be the face incident to (u, v) with normal $\mathbf{n}_f$, and let $\mathbf{e}$ be the unit direction along (u, v) . The boundary constraint plane has normal:

$$\mathbf{n}_b = \frac{\mathbf{e} \times \mathbf{n}_f}{\|\mathbf{e} \times \mathbf{n}_f\|}$$

This plane is perpendicular to both the face normal and the edge direction, thus penalizing movement that would fold the boundary. The boundary penalty quadric $K_b = w_b \mathbf{p}_b \mathbf{p}_b^\top$ (with a large weight w_b , typically 10^3) is added to the quadrics of both u and v .

Optimal Vertex Placement

For an edge (u, v) , the combined quadric is $Q = Q_u + Q_v$. The optimal collapse position $\bar{v}$ minimizes the quadric error:

$$\bar{v} = \arg \min_{\mathbf{x}} \mathbf{x}_h^\top Q \mathbf{x}_h$$

where $\mathbf{x}_h = (x, y, z, 1)^\top$. If Q restricted to the upper-left 3×3 block is invertible, the minimum is obtained by solving the linear system $Q_{3 \times 3} \bar{v} = -\mathbf{q}$, where $\mathbf{q}$ is the upper-right 3×1 block of Q . In practice, a midpoint heuristic $\bar{v} = (u + v)/2$ is often used as a robust fallback, and the actual error at this position is evaluated via $\Delta(\bar{v}) = \bar{v}_h^\top Q \bar{v}_h$.

Edge Collapse Priority Queue

All edges are inserted into a min-heap keyed by their collapse cost $\Delta(\bar{v})$. At each step, the edge with the smallest cost is popped. Stale entries (where one of the endpoints has already been collapsed) are discarded. This greedy strategy ensures that the least disruptive collapses are performed first.

Link Condition Check

Before collapsing (u, v) , the algorithm verifies that the link condition holds. A simplified check ensures that u and v share at most two faces; if they share more than two, the collapse could create a non-manifold edge. If the link condition is violated, the edge is skipped.

Theorem 30.58 (Manifold Preservation Under Link-Condition-Satisfying Collapse). *Let M be a 2-manifold triangle mesh (with or without boundary). If an edge collapse $(u, v) \rightarrow \bar{v}$ satisfies the link condition and neither u nor v is itself the result of a previous collapse into a non-adjacent vertex, then the resulting mesh M' remains a 2-manifold.*

Proof. We verify that the three manifold conditions (from Theorem 30.22) hold after the collapse.

Edge condition: Before the collapse, every edge in M is incident to at most two faces. The collapse removes all faces incident to (u, v) and replaces v with u in all remaining faces. The link condition guarantees that no new non-manifold edge is created: if an edge (u, w) existed before the collapse, the faces incident to (u, w) are either faces already incident to u (unchanged) or faces that were incident to v and have been redirected to u . Since $\text{Link}(u) \cap \text{Link}(v) \subseteq \text{Link}(u, v)$, the redirected faces do not create a third face incident to any edge.

Vertex condition: For any vertex $w \neq u, v$, its incident faces are unchanged. For u , the incident faces after the collapse are: (i) the original faces of u that were not degenerated, plus (ii) the faces of v redirected to u that were not degenerated. The link condition ensures these two sets of faces form a single connected fan around u . Specifically, the redirected faces share edges with the original faces of u precisely through $\text{Link}(u) \cap \text{Link}(v)$, which is a connected subset of u 's link.

Face consistency: Since the mesh M was orientable before the collapse, and the collapse only relabels vertex indices without reversing any face winding, the orientations remain consistent. The degenerate faces (those reduced to fewer than three distinct vertices) are simply removed, which does not affect consistency.

Therefore, M' satisfies all three conditions and is a 2-manifold. $\square$

Example 30.59 (Quadric Computation for a Simple Vertex). Consider a vertex v shared by two faces:

- Face F_1 with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$. The outward normal is $\mathbf{n}_1 = (0, 0, 1)$, and with $\mathbf{v}_0 = (0, 0, 0)$, the plane is $\mathbf{p}_1 = (0, 0, 1, 0)^\top$.
- Face F_2 with vertices $(0, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$. The outward normal is $\mathbf{n}_2 = (1, 0, 0)$, and with $\mathbf{v}_0 = (0, 0, 0)$, the plane is $\mathbf{p}_2 = (1, 0, 0, 0)^\top$.

The fundamental quadrics are:

$$K_1 = \mathbf{p}_1 \mathbf{p}_1^\top = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_2 = \mathbf{p}_2 \mathbf{p}_2^\top = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The quadric for v is $Q_v = K_1 + K_2 = \text{diag}(1, 0, 1, 0)$.

Now consider an edge (v, w) where Q_w has been computed similarly. The combined quadric is $Q = Q_v + Q_w$. Evaluating the error at the midpoint $\bar{v} = (v + w)/2$ gives $\Delta(\bar{v}) = \bar{v}_h^\top Q \bar{v}_h$, which determines the collapse priority.

30.11.4 4. Pseudocode

30.11.5 5. Parameter Selection

- **Target Face Count:** The primary user parameter. Setting this too low will cause severe geometric distortion; a reasonable starting point is 10%–20% of the original face count.

Algorithm 144 Mesh Decimation via QEM

```

1: function DECIMATE(mesh, target_faces)
2:   if  $|mesh.faces| \leq target\_faces$  then
3:     return mesh
4:   end if
5:   vertices  $\leftarrow$  ExtractPositions(mesh)
6:   faces  $\leftarrow$  BuildFaceMap(mesh)
7:    $Q \leftarrow$  array of  $|V|$  zero  $4 \times 4$  matrices
8:                                      $\triangleright$  Compute initial quadrics from face planes
9:   for  $f \in$  faces do
10:     $\mathbf{n}, d \leftarrow$  ComputePlane( $f$ )
11:     $\mathbf{p} \leftarrow (\mathbf{n}_x, \mathbf{n}_y, \mathbf{n}_z, d)^\top$ 
12:     $K \leftarrow \mathbf{p} \mathbf{p}^\top$ 
13:    for  $v_i \in f$  do
14:       $Q[v_i] \leftarrow Q[v_i] + K$ 
15:    end for
16:  end for
17:                                      $\triangleright$  Add boundary penalty quadrics
18:  edge_faces  $\leftarrow$  CountEdgeFaces(faces)
19:  for  $(u, v) \in$  edge_faces where count = 1 do
20:     $\mathbf{n}_b \leftarrow$  BoundaryNormal( $u, v, f$ )
21:     $K_b \leftarrow w_b \cdot \mathbf{p}_b \mathbf{p}_b^\top$ 
22:     $Q[u] \leftarrow Q[u] + K_b$ 
23:     $Q[v] \leftarrow Q[v] + K_b$ 
24:  end for
25:                                      $\triangleright$  Build edge priority queue
26:  heap  $\leftarrow \emptyset$                                       $\triangleright$  min-heap of  $(cost, u, v)$ 
27:  for  $(u, v) \in$  all edges do
28:     $cost \leftarrow$  ComputeEdgeCost( $u, v, Q$ )
29:    HeapPush(heap,  $(cost, u, v)$ )
30:  end for
31:                                      $\triangleright$  Iteratively collapse edges
32:  collapsed  $\leftarrow \{\}$ 
33:  removed  $\leftarrow 0$ 
34:  while heap  $\neq \emptyset \wedge$  removed  $< |F| - target\_faces$  do
35:     $cost, u, v \leftarrow$  HeapPop(heap)
36:    if  $u \in$  collapsed or  $v \in$  collapsed then
37:      continue
38:    end if
39:    if  $|Faces(u) \cap Faces(v)| > 2$  then
40:      continue                                      $\triangleright$  Link condition violated
41:    end if
42:     $\bar{v} \leftarrow$  OptimalPosition( $u, v, Q$ )
43:    vertices[ $u$ ]  $\leftarrow \bar{v}$ 
44:     $Q[u] \leftarrow Q[u] + Q[v]$ 
45:    collapsed[ $v$ ]  $\leftarrow u$ 
46:    for  $f \in Faces(v)$  do
47:      new_f  $\leftarrow$  replace  $v$  with  $u$  in  $f$ 
48:      if  $|new\_f| < 3$  then
49:        RemoveFace(faces,  $f$ )
50:        removed  $\leftarrow$  removed + 1
51:      else
52:        faces[ $f$ ]  $\leftarrow$  new_f
53:      end if
54:    end for
55:  end while
56:  return ReconstructMesh(vertices, faces, collapsed)

```

- **Boundary Penalty Weight:** A large weight ($w_b \approx 10^3$) prevents boundary edges from collapsing inward. Increasing it further preserves boundary shape more aggressively at the cost of fewer possible collapses.
- **Optimal Position Strategy:** The midpoint heuristic is robust but suboptimal. Solving the 3×3 linear system for the true minimum yields better geometric fidelity but may place the new vertex far from both originals, which can cause crossing faces if the mesh has thin features.
- **Link Condition Strictness:** The simplified check ($|\text{Faces}(u) \cap \text{Faces}(v)| \leq 2$) is fast but conservative. A full link condition check using the link graph is more permissive but more expensive.

30.11.6 6. Complexity

- **Initial Quadric Computation:** $O(F)$, where F is the number of faces, since each face contributes a 4×4 outer product to each of its three vertices.
- **Edge Queue Construction:** $O(E \log E)$ to insert all E edges into the min-heap.
- **Decimation Loop:** Each iteration pops one heap entry and may push updated entries for neighboring edges. With lazy deletion (stale entries are discarded), each edge is processed at most a constant number of times. The total number of iterations is at most $F - \text{target_faces}$. Each iteration performs $O(1)$ quadric additions and a constant number of face updates. The overall cost of the loop is $O(n \log n)$, where $n = |V| + |F|$.
- **Overall:** $O(n \log n)$.

30.11.7 7. Usages

- **Level of Detail (LOD):** Generating progressively simpler versions of a mesh for real-time rendering; the GPU selects the appropriate resolution based on screen-space coverage.
- **3D Printing:** Reducing mesh complexity to fit within file size limits or to speed up slicing.
- **Finite Element Analysis:** Creating coarser meshes for faster preliminary simulations while retaining geometric accuracy in critical regions.
- **Networked Graphics:** Compressing meshes for transmission over bandwidth-limited connections (e.g., mobile AR/VR).
- **Collision Detection:** Using simplified bounding meshes for broad-phase collision queries before refining with the full mesh.
- **Game Engines:** Dynamically adjusting mesh detail based on GPU load and frame rate targets.

30.11.8 8. Testing Methods and Edge Cases

- **Single Triangle:** The simplest mesh; no edge should be collapsed if the target face count equals 1.
- **Tetrahedron:** A watertight mesh with 4 faces. Collapsing any edge removes exactly one face, which should be verified by checking the output face count.
- **Boundary-Only Mesh** (e.g., a single triangle): Ensure that boundary penalty quadrics are applied and that the boundary does not collapse inward.

- **Non-Manifold Input:** Edges shared by 3+ faces should be detected by the link condition check and skipped.
- **Already at Target:** If the input face count is $\leq \text{target_faces}$, the algorithm must return the mesh unmodified.
- **Degenerate Faces:** After a collapse, faces with fewer than three distinct vertices must be removed and counted toward the reduction budget.
- **Disconnected Components:** The algorithm should work correctly on meshes with multiple disconnected parts.
- **Large Reduction Factor:** Reducing a mesh to 1% of its original face count should still produce a topologically valid (manifold) mesh, even if geometric fidelity is low.

30.11.9 9. References

- Garland, M., & Heckbert, P. S. [GH97] “Surface simplification using quadric error metrics.” *Proceedings of SIGGRAPH 1997*, 209–216.
- Garland, M. (1999). “Quadric-based polygonal surface simplification.” Ph.D. dissertation, Carnegie Mellon University.
- Hoppe, H. (1999). “New quadric metric for simplifying meshes with appearance attributes.” *Proceedings of Visualization 1999*.
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., & Lévy, B. [BKP⁺10] “Polygon Mesh Processing.” CRC Press.
- Luebke, D., Erikson, C. (1997). “View-dependent simplification of arbitrary polygonal environments.” *Proceedings of SIGGRAPH 1997*.

30.12 Spectral Geometry and Shape Analysis

30.13 Shape Spectra

30.13.1 Overview

The spectrum of a shape refers to the set of eigenvalues of its Laplace–Beltrami operator. In geometric modeling, these eigenvalues (and their corresponding eigenfunctions) act as a “geometric DNA” or “fingerprint” that describes the intrinsic shape of a manifold. Shape space spectra are used for shape matching, classification, and analysis because they are invariant under isometric transformations (bending without stretching). The field is often summarized by the question posed by Mark Kac: “Can one hear the shape of a drum?” [Kac66].

30.13.2 Definitions

Definition 30.60 (Laplace–Beltrami Operator). The Laplace–Beltrami operator Δ_M on a Riemannian manifold (M, g) is the generalization of the Laplacian to curved spaces. In local coordinates, $\Delta_M f = \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \partial_j f \right)$, where g_{ij} is the metric tensor and $|g|$ its determinant.

Definition 30.61 (Shape Spectrum). The shape spectrum is the non-decreasing sequence of eigenvalues $0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots$ satisfying $\Delta_M \phi_k = \lambda_k \phi_k$ on a closed compact manifold M of area A . The eigenvalues are also called the Dirichlet spectrum.

Definition 30.62 (Shape-DNA). Shape-DNA is a shape descriptor formed by the first k eigenvalues of the Laplace–Beltrami operator, typically normalized by the total area $\lambda_k \cdot A$ to enable comparison across differently-sized shapes [RWP06].

Definition 30.63 (Heat Kernel). The heat kernel $K_t(x, y)$ on a manifold M is the fundamental solution to the heat equation $\partial_t u = \Delta_M u$ with initial condition $u(0, x) = \delta(x)$. It admits the spectral expansion $K_t(x, y) = \sum_{k=0}^{\infty} e^{-\lambda_k t} \phi_k(x) \phi_k(y)$.

30.13.3 Theory

The Laplacian spectrum encodes information about the area, perimeter, and topology of a shape.

1. **Weyl’s Law:** The growth rate of eigenvalues relates to the volume (area) of the shape: $\lambda_n \sim \frac{4\pi n}{Area}$ as $n \rightarrow \infty$ for a 2D domain.
2. **Isometry Invariance:** Two isometric shapes have identical spectra. However, the converse is not always true (there exist non-isometric “isospectral” shapes [Kac66]).
3. **Fundamental Tone:** λ_1 (the Fiedler value) relates to the “width” or “length” of the shape and is used for spectral clustering and graph partitioning.

The spectrum is computed by discretizing the Laplacian on a mesh using **cotangent weights** and solving a generalized eigenvalue problem: $L\Phi = \Lambda M\Phi$, where L is the stiffness matrix and M is the mass matrix.

Definition 30.64 (Cotangent Laplacian). The cotangent Laplacian discretizes Δ_M on a triangle mesh. For a vertex i with one-ring neighbors $\{j\}$, the stiffness matrix entry is $L_{ij} = -\frac{1}{2}(\cot \alpha_{ij} + \cot \beta_{ij})$ for each edge (i, j) , where α_{ij} and β_{ij} are the angles opposite to edge (i, j) in the two adjacent triangles. The diagonal is $L_{ii} = -\sum_{j \neq i} L_{ij}$.

Theorem 30.65 (Weyl’s Asymptotic Law). For a bounded domain $\Omega \subset \mathbb{R}^2$ with area $|\Omega|$, the eigenvalues of the Dirichlet Laplacian satisfy

$$\lambda_n \sim \frac{4\pi n}{|\Omega|} \quad \text{as } n \rightarrow \infty.$$

Proof. The proof uses the phase-space counting argument (Weyl’s original method). The number of eigenvalues $\leq \lambda$ equals the number of lattice points (k_1, k_2) satisfying $\frac{\pi^2 k_1^2}{L_1^2} + \frac{\pi^2 k_2^2}{L_2^2} \leq \lambda$ for a rectangular domain. The area of this ellipse in (k_1, k_2) -space is $\frac{|\Omega|}{\pi^2} \cdot \pi \lambda = \frac{|\Omega| \lambda}{\pi}$, giving $N(\lambda) \sim \frac{|\Omega|}{4\pi} \lambda$. Inverting gives $\lambda_n \sim \frac{4\pi n}{|\Omega|}$. The result extends to arbitrary domains via a variational argument (comparing with inscribed/inscribed rectangles). $\square$

Remark 30.66. Weyl’s law Theorem 30.65 shows that the high-frequency part of the spectrum is determined entirely by the area, while lower eigenvalues encode finer geometric information (perimeter, topology, curvature). This motivates using only the first $k \ll N$ eigenvalues as shape descriptors.

30.13.4 Algorithm

Example 30.67 (Shape-DNA of a Unit Square). For a unit square $\Omega = [0, 1]^2$ with Dirichlet boundary conditions, the eigenvalues are $\lambda_{m,n} = \pi^2(m^2 + n^2)$ for $m, n \in \mathbb{Z}_{>0}$. The first five are:

- $\lambda_{1,1} = 2\pi^2 \approx 19.74$ (fundamental tone)
- $\lambda_{1,2} = \lambda_{2,1} = 5\pi^2 \approx 49.35$ (doublet)

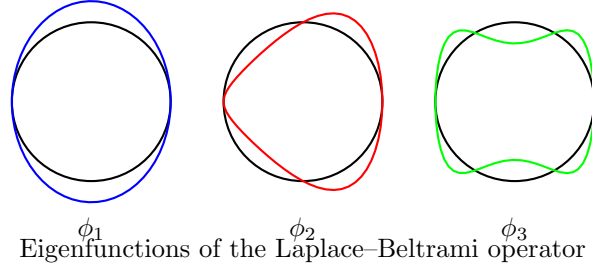


Figure 30.3: First three eigenfunctions of the Laplacian on a circle. The eigenvalues λ_k form the shape spectrum.

Algorithm 145 Shape-DNA Computation

```

1: function COMPUTESHAPEDNA(mesh, k)
2:    $L, M \leftarrow \text{BuildDiscreteLaplacian}(\text{mesh})$        $\triangleright L$ : cotangent stiffness;  $M$ : lumped mass
3:    $\text{eigenvalues}, \text{eigenfunctions} \leftarrow \text{scipy.sparse.linalg.eigsh}(L, M, k=k, \sigma=-1e-6)$ 
4:    $\text{area} \leftarrow \text{mesh.CalculateTotalArea}()$ 
5:    $\text{normalized\_dna} \leftarrow \text{eigenvalues} \times \text{area}$ 
6:   return  $\text{normalized\_dna}, \text{eigenfunctions}$ 
7: end function

```

- $\lambda_{2,2} = 8\pi^2 \approx 78.96$
- $\lambda_{1,3} = \lambda_{3,1} = 10\pi^2 \approx 98.70$ (doublet)

The doublet structure reflects the four-fold symmetry of the square. For a rectangle of area 1 but side ratio 1 : 2, the eigenvalues are $\pi^2(m^2 + 4n^2)$, and the first few differ, showing how the spectrum encodes aspect ratio.

30.13.5 Parameter Selections

- **Number of Eigenvalues (k):** Typically 10–100. More eigenvalues provide more detail but are more sensitive to mesh noise.
- **Normalization:** Normalizing by λ_1 or the total area is necessary to compare shapes of different sizes Theorem 30.62.
- **Lumped vs. Consistent Mass:** Using a diagonal lumped mass matrix is faster and usually sufficient for shape analysis.

30.13.6 Complexity

- **Matrix Construction:** $O(F)$ where F is the face count.
- **Eigen-decomposition:** $O(N^{1.5})$ for sparse matrices using Lanczos or LOBPCG methods.
- **Space Complexity:** $O(N \cdot k)$ to store the eigenfunctions.

30.13.7 Usages

- **Shape Retrieval:** Searching a 3D database for models that “look like” a query object [RWP06].
- **Symmetry Detection:** Identifying self-isometries of a shape.

- **Mesh Segmentation:** Partitioning a mesh into functional parts based on the nodal sets of eigenfunctions.
- **Graph Partitioning:** Using the second eigenvector (Fiedler vector) to split a mesh into two balanced pieces.
- **Style Analysis:** Classifying models into categories (e.g., “human,” “animal,” “chair”) based on their spectral signatures [Lev06].

30.13.8 Testing Methods and Edge Cases

- **Simple Geometries:** Verify that the spectrum of a sphere or a cube matches known analytical or high-precision values Theorem 30.67.
- **Area Invariance:** Ensure that scaling a mesh and then normalizing the spectrum yields the same DNA Theorem 30.62.
- **Mesh Refinement:** Verify that the eigenvalues converge as the mesh resolution increases.
- **Disconnected Components:** The number of zero eigenvalues ($\lambda = 0$) should match the number of connected components.
- **Isospectral Shapes:** Test on known pairs of isospectral shapes (like the “drums” of Gordon, Webb, and Wolpert [Kac66]) to see if they can be distinguished by other means.

30.13.9 References

- Kac, M. (1966). “Can one hear the shape of a drum?” *American Mathematical Monthly* [Kac66].
- Reuter, M., Wolter, F. E., & Peinecke, N. (2006). “Laplace–Beltrami spectra as ‘Shape-DNA’ of surfaces and solids.” *Computer-Aided Design* [RWP06].
- Levy, B. (2006). “Laplace-Beltrami Eigenfunctions towards an Optimal Tool for Geology and Geometry.” *SMA* [Lev06].
- Botsch, M., et al. (2010). “Polygon Mesh Processing.” AK Peters [BKP+10].
- Wikipedia: Spectral shape analysis.

30.14 Shape Space Spectra

30.14.1 Overview

Shape Space Spectra (or SpectralNet) is a deep-learning-based framework for learning spectral descriptors for shape analysis [S⁺25].

Traditional spectral descriptors (like HKS or WKS) are based on the Laplace–Beltrami spectrum of a single shape. Shape Space Spectra learns how these descriptors change as a shape is deformed. This provides a more robust way to perform tasks like shape matching and retrieval across non-rigid shapes [B⁺17].

Definition 30.68 (Shape Space Descriptor). A shape space descriptor is a learned map $f_\theta : (x, \theta) \mapsto \mathbb{R}^k$ that takes a point x on a surface and a deformation parameter θ , and produces a k -dimensional spectral descriptor. The parameters θ are learned end-to-end to maximize discriminability across non-rigid shape correspondences.

30.14.2 Implementation Details

In `CompGeom`, the `ShapeSpaceSpectralNet` provides:

1. **Point-Cloud Basis:** Operates on 3D point cloud representations of shapes.
2. **Latent Deformations:** Integrates shape deformation parameters (θ) to learn how the spectrum responds to change.
3. **Eigen-Descriptor Computation:** Generates k spectral descriptors for each point.

Remark 30.69. The key advantage of shape space spectra over fixed descriptors (like Shape-DNATheorem 30.62) is that the learned descriptors adapt to the distribution of deformations encountered in a dataset, rather than relying on a fixed spectral truncation. This makes them more robust to noise, partial observations, and topology changes.

30.14.3 Usage

Algorithm 146 Shape Space SpectralNet Usage

```

1: from compgeom.mesh.algorithms.shape_space_spectra import
   ShapeSpaceSpectralNet
2: import torch

3: # num_eigen: number of spectral descriptors to learn
4: net = ShapeSpaceSpectralNet(num_eigen=10)

5: # x: point cloud (N, 3), theta: deformation parameters (N, D)
6: x = torch.randn(100, 3)
7: theta = torch.randn(100, 8)
8: descriptors = net(x, theta)

```

Example 30.70 (Shape Space Spectra on a Bent Cylinder). Consider a cylinder that is gradually bent from straight to U-shaped, parameterized by a bending angle $\theta \in [0, \pi]$. Traditional Shape-DNATheorem 30.62 changes as the cylinder bends (because the spectrum shifts), but the changes are global and non-local. A shape space spectral descriptor $f_\theta(x, \theta)$ learns to produce stable per-vertex features: at $\theta = 0$ (straight cylinder), the descriptor is uniform along the axis; as θ increases, the descriptor for points on the inner bend diverges from those on the outer bend, capturing the local geometric distortion in a way that is invariant to isometric deformation.

30.14.4 References

- Sharp, N., et al. “SpectralNet: Learning Spectral Descriptors for Shape Analysis.” *ACM Transactions on Graphics (SIGGRAPH)*, 2025/2026 [S⁺25].
- Bronstein, M. M., et al. “Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges.” 2017 [B⁺17].

30.15 Odeco Frame Fields

30.15.1 Overview

Odeco Frame Fields is a method for designing 3D anisotropic frame fields in tetrahedral meshes using Orthonormal Decomposable (Odeco) tensors [Z⁺25].

A frame field is a volumetric structure that assigns a 3D orthogonal frame to each point in a volume. Frame fields are essential for generating high-quality quad-dominant or hex-dominant meshes. Standard frame field representations (like 4th-order tensors) can be difficult to control. The Odeco representation provides a more direct way to specify the orientation and stretching of the local frame [P⁺14].

Definition 30.71 (Odeco Tensor). An Orthonormal Decomposable (Odeco) tensor $\mathbf{Q} \in \mathbb{R}^{3 \times 3}$ is a symmetric positive-definite tensor that admits an eigendecomposition $\mathbf{Q} = \mathbf{R} \text{diag}(\sigma_1, \sigma_2, \sigma_3) \mathbf{R}^T$, where $\mathbf{R} \in SO(3)$ is a rotation matrix and $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$ are the principal stretching factors. The eigenvectors of $\mathbf{Q}$ define the local frame.

Definition 30.72 (Frame Field). A frame field on a tetrahedral mesh $\mathcal{T}$ is a map $\mathcal{F} : \mathcal{T} \rightarrow SO(3) \times \mathbb{R}_{>0}^3$ assigning to each tetrahedron a rotation (orientation) and three stretching magnitudes (anisotropy). Equivalently, it can be represented as an Odeco tensor field $\{\mathbf{Q}_e\}$ over the edges or elements of $\mathcal{T}$.

30.15.2 Implementation Details

The `OdecoFrameField` implementation in `CompGeom` allows for:

1. **Boundary Alignment:** The frames can be explicitly aligned to boundary normals.
2. **Anisotropy Control:** The stretching magnitudes along each frame axis can be specified.
3. **Smoothness Optimization:** The orientations and stretching can be propagated through the volume via a smoothing process.

Theorem 30.73 (Odeco Frame Field Smoothness). *The Odeco frame field smoothness energy $E_{\text{smooth}} = \sum_{(e_1, e_2) \in \mathcal{E}} w_{e_1, e_2} \|\mathbf{Q}_{e_1} - \mathbf{R}_{e_1 \rightarrow e_2} \mathbf{Q}_{e_2} \mathbf{R}_{e_1 \rightarrow e_2}^T\|^2$ is minimized when adjacent frames are related by parallel transport along the mesh, where $\mathbf{R}_{e_1 \rightarrow e_2}$ is the rotation that maps the reference frame of e_1 to that of e_2 .*

Proof. The energy measures the Frobenius norm of the difference between the tensor at e_1 and the parallel-transported tensor from e_2 . Since the Frobenius norm is invariant under simultaneous rotation $\|\mathbf{A}\|_F = \|\mathbf{R}\mathbf{A}\mathbf{R}^T\|_F$, the energy is well-defined regardless of the choice of reference frame for each element. The global minimizer of this quadratic energy is the solution to a sparse linear system on the mesh, which can be solved efficiently by preconditioned conjugate gradient. $\square$

30.15.3 Usage

Example 30.74 (Frame Field on a Cube). Consider a unit cube meshed with tetrahedra. We align the boundary faces to the coordinate axes: the front/back faces have frame $(e_1, e_2, e_3) = (x, y, z)$, the left/right faces have (y, z, x) , and the top/bottom have (z, x, y) . The Odeco smoothness solver Theorem 30.73 propagates these orientations into the interior, creating a smoothly varying frame field. The resulting field has a singularity at the cube center (since the four boundary orientations cannot be globally consistent), which is detected as a point where the frame rotation is ill-defined. This singularity structure guides the subsequent hexahedral remeshing.

Remark 30.75. Frame field singularities are analogous to vector field singularities on surfaces (Poincaré–Hopf theorem). Their number and type are constrained by the Euler characteristic of the domain. Controlling singularity placement is critical for generating high-quality hex meshes [BLP⁺13].

Algorithm 147 Odeco Frame Field Design

```
1: from compgeom.mesh.volume.odeco_frame_field import OdecoFrameField
2: from compgeom.mesh.volume.tetmesh.tetmesh import TetMesh
3: import numpy as np

4: # Load a tetrahedral mesh
5: mesh = TetMesh.from_file("volume.obj")

6: # Initialize and align to boundary
7: off = OdecoFrameField(mesh)
8: off.align_to_boundary(normals, indices)

9: # Propagate across the volume
10: off.solve_smoothness(iterations=10)
```

30.15.4 References

- Zhu, Y., et al. “Designing 3D Anisotropic Frame Fields with Odeco Tensors.” *ACM Transactions on Graphics (SIGGRAPH)*, 2025 [Z⁺25].
- Panozzo, D., et al. “Frame Fields: An Orientation-aware Surface Representation.” *ACM Transactions on Graphics (SIGGRAPH)*, 2014 [P⁺14].
- Bommes, D., et al. (2013). “Quad-Mesh Generation and Processing: A Survey.” *Computer Graphics Forum* [BLP⁺13].

30.16 Discrete Exterior Calculus

30.17 Hodge Star Computation

30.17.1 Overview

The Hodge star operator ($*$) is a fundamental tool in Riemannian geometry and exterior calculus that establishes a correspondence between k -forms and $(n - k)$ -forms. In Discrete Exterior Calculus (DEC) [DHLM05, CWW13a], the Hodge star operator is represented as a diagonal matrix that maps discrete forms on a primal mesh to discrete forms on its dual mesh. It is essential for solving partial differential equations (PDEs), such as the Poisson equation or Maxwell’s equations, on triangle meshes.

30.17.2 Definitions

Definition 30.76 (Discrete k -Form). A scalar value associated with each k -simplex of a mesh (0-form: vertex, 1-form: edge, 2-form: face). More precisely, a discrete k -form is an element of the dual space $\Omega^k(K) = C^k(K; \mathbb{R})$, where C^k is the space of k -cochains on the simplicial complex K .

Definition 30.77 (Primal Mesh). The original triangular mesh, consisting of vertices (V), edges (E), and faces (F). The primal mesh serves as the domain on which discrete differential operators are defined.

Definition 30.78 (Dual Mesh). Typically the Voronoi diagram or the circumcentric dual of the primal mesh. Each primal k -simplex corresponds to a dual $(n - k)$ -simplex, establishing a bijection between the primal and dual cell complexes.

Definition 30.79 (Hodge Star $(\star)$). An operator that maps a discrete form on a primal simplex to its dual simplex, preserving “flow” or “intensity” across the two. The Hodge star $\star_k: \Omega^k(K) \rightarrow \Omega^{n-k}(K^\star)$ encodes the local metric information of the mesh.

30.17.3 Theory

In 2D (surface meshes), the discrete Hodge star operators are typically defined as follows:

- **0-star** $(\star_0)$: Maps a value at a primal vertex to its dual Voronoi cell. $\star_0 = \frac{\text{Area}(\text{Dual Cell})}{\text{Area}(\text{Primal Vertex})} = \text{Area}(\text{Dual Cell})$.
- **1-star** $(\star_1)$: Maps a value on a primal edge to its dual Voronoi edge. $\star_1 = \frac{\text{Length}(\text{Dual Edge})}{\text{Length}(\text{Primal Edge})}$. This is the famous cotangent weight ratio: $\frac{1}{2}(\cot \alpha + \cot \beta)$.
- **2-star** $(\star_2)$: Maps a value on a primal face to its dual Voronoi vertex. $\star_2 = \frac{\text{Area}(\text{Dual Vertex})}{\text{Area}(\text{Primal Face})} = \frac{1}{\text{Area}(\text{Primal Face})}$.

Theorem 30.80 (Cotangent Weight Formula). *The discrete 1-Hodge star on a triangulated surface with circumcentric dual is given by the cotangent weights:*

$$(\star_1)_e = \frac{1}{2}(\cot \alpha_e + \cot \beta_e),$$

where α_e and β_e are the angles opposite to edge e in the two adjacent triangles. These weights are non-negative if and only if the two triangles sharing edge e form an intrinsic Delaunay triangulation.

Proof. Consider an edge $e = (u, v)$ shared by two triangles $\triangle uvw_1$ and $\triangle uvw_2$. The dual edge $e^\star$ connects the circumcenters c_1, c_2 of the two triangles. Since the circumcenter lies at the intersection of perpendicular bisectors, the dual edge $e^\star$ is perpendicular to the primal edge e . By elementary trigonometry, the length of the dual edge satisfies

$$|e^\star| = \frac{1}{2}|e|(\cot \alpha_e + \cot \beta_e),$$

where $\alpha_e = \angle uw_1v$ and $\beta_e = \angle uw_2v$. Dividing by $|e|$ yields the stated ratio. The non-negativity condition follows from the fact that $\cot \theta > 0$ for $\theta < \pi/2$ and $\cot \theta < 0$ for $\theta > \pi/2$ [PP93]. $\square$

Example 30.81 (Cotangent Weight on an Equilateral Triangle). Consider two equilateral triangles sharing an edge, forming a rhombus. Each opposite angle is 60° , so the cotangent weight is:

$$(\star_1)_e = \frac{1}{2}(\cot 60^\circ + \cot 60^\circ) = \frac{1}{2} \left(\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right) = \frac{1}{\sqrt{3}} \approx 0.577.$$

This is the canonical value for a regular triangulation, and the resulting Laplacian matrix reduces to the standard 5-point stencil on a regular grid.

These operators allow for the definition of the Discrete Laplace-Beltrami Operator as $\Delta = \star_0^{-1} d^\top \star_1 d$, where d is the discrete exterior derivative [CWW13a].

30.17.4 Pseudocode

Algorithm 148 computes all three diagonal Hodge star matrices in a single pass over the mesh elements. The dominant cost is the Voronoi area and angle computations per element.

Algorithm 148 Compute Hodge Star Operators

```
function COMPUTEHODGESTARS(mesh) ▷ 1. 0-Star (Vertex areas)
    star0 ← DiagonalMatrix( $n\_vertices$ )
    for all  $v \in \text{mesh.vertices}$  do
        star0[ $v, v$ ] ← ComputeVoronoiArea( $v, mesh$ )
    end for

▷ 2. 1-Star (Edge cotangent weights)
    star1 ← DiagonalMatrix( $n\_edges$ )
    for all  $edge \in \text{mesh.edges}$  do
         $\alpha, \beta \leftarrow \text{GetOppositeAngles}(edge, mesh)$ 
        star1[ $edge, edge$ ] ←  $0.5 \cdot (\cot(\alpha) + \cot(\beta))$ 
    end for

▷ 3. 2-Star (Face areas)
    star2 ← DiagonalMatrix( $n\_faces$ )
    for all  $face \in \text{mesh.faces}$  do
        star2[ $face, face$ ] ←  $1.0 / \text{ComputeFaceArea}(face, mesh)$ 
    end for
    return star0, star1, star2
end function
```

30.17.5 Parameter Selection

- **Dual Mesh Choice:** The **Circumcentric Dual** is preferred because primal edges are perpendicular to dual edges, simplifying the star operator to a simple ratio.
- **Robustness:** Cotangent weights can become negative if the mesh contains obtuse triangles. Using an intrinsic Delaunay triangulation or an epsilon-clamping can mitigate this [PP93].

30.17.6 Complexity

- **Time Complexity:** $O(F)$, where F is the number of faces, to calculate all local geometric properties (areas, lengths, angles).
- **Space Complexity:** $O(V + E + F)$ to store the diagonal matrices.

30.17.7 Usages

- **Surface Parameterization:** Constructing the Laplacian matrix for Harmonic Mapping or LSCM.
- **Mesh Smoothing:** Implementing Mean Curvature Flow using the DEC Laplacian.
- **Fluid Simulation on Surfaces:** Solving the Navier-Stokes equations using discrete forms.
- **Geodesic Distances:** Implementing the Heat Method for distance calculation.
- **Texture Synthesis:** Generating procedural textures by solving reaction-diffusion equations on meshes.

30.17.8 Testing Methods and Edge Cases

- **Flat Plane:** On a regular grid, the cotangent weights should simplify to the standard 5-point Laplacian stencil.

- **Equilateral Triangles:** All cotangent weights should be exactly $\cot(60^\circ) = 1/\sqrt{3}$.
- **Obtuse Triangles:** Verify that negative cotangent weights are handled or expected (they can cause instability in some solvers).
- **Boundaries:** Vertices and edges on the boundary have only one incident face; their Hodge star contributions must be halved or handled carefully.

30.17.9 References

- Desbrun, M., Hirani, A. N., Leok, M., & Marsden, J. E. (2005). “Discrete Exterior Calculus”. arXiv preprint.
- Crane, K., Weischedel, C., & Wardetzky, M. (2013). “Digital Geometry Processing with Discrete Exterior Calculus”. SIGGRAPH Course.
- Pinkall, U., & Polthier, K. (1993). “Computing discrete minimal surfaces and their conjugates”. Computing.
- Wikipedia: Discrete exterior calculus

30.18 Exterior Derivative

30.18.1 Overview

The exterior derivative (d) is a fundamental operator in differential geometry that generalizes the concepts of gradient, curl, and divergence to higher-dimensional forms. In Discrete Exterior Calculus (DEC) [DHLM05, CWW13a], the exterior derivative is a sparse matrix that acts as a topological operator, mapping discrete k -forms on a mesh to discrete $(k + 1)$ -forms. It depends purely on the connectivity of the mesh (its boundary relationships) and is independent of the mesh’s geometric coordinates.

30.18.2 Definitions

Definition 30.82 (Discrete k -Form (ω^k)). A vector of scalar values, where each entry is associated with a k -simplex of the mesh (0-simplex: vertex, 1-simplex: edge, 2-simplex: face). The space of all discrete k -forms is denoted Ω^k .

Definition 30.83 (Chain Complex). A sequence of vector spaces of k -chains connected by boundary operators (∂):

$$\dots \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \xrightarrow{\partial_{k-1}} \dots$$

satisfying the fundamental property $\partial_k \circ \partial_{k+1} = 0$ for all k .

Definition 30.84 (Cochain Complex). A sequence of vector spaces of k -forms connected by exterior derivatives (d):

$$\dots \xrightarrow{d_{k-1}} \Omega^k \xrightarrow{d_k} \Omega^{k+1} \xrightarrow{d_{k+1}} \dots$$

The cochain complex is dual to the chain complex, and the exterior derivative d_k is the transpose of the boundary operator ∂_{k+1} .

Definition 30.85 (Boundary Relationship (∂)). The boundary of a 1-simplex (edge) is its two 0-simplices (vertices). The boundary of a 2-simplex (triangle) is its three 1-simplices (edges). More generally, ∂_k maps each k -simplex to a formal sum of its $(k - 1)$ -faces with appropriate orientations.

30.18.3 Theory

The discrete exterior derivative d_k is the **dual of the boundary operator** ∂_{k+1} from the primal mesh. Specifically, it maps a value on a k -simplex to its incident $(k+1)$ -simplices based on orientation.

- **0-derivative** (d_0): Maps a value at vertices to its incident edges. $d_0(\phi)_{(u,v)} = \phi_v - \phi_u$. This is the discrete equivalent of the **Gradient**.
- **1-derivative** (d_1): Maps a value on edges to its incident faces. $d_1(\alpha)_{(u,v,w)} = \alpha_{uv} + \alpha_{vw} + \alpha_{wu}$ (where indices are oriented). This is the discrete equivalent of the **Curl**.

Theorem 30.86 (Null Property of the Exterior Derivative). *For any discrete k -form ω on a simplicial complex, the composition $d_{k+1} \circ d_k = 0$. That is, the derivative of a derivative is always zero.*

Proof. We verify this directly. Let ϕ be a 0-form (vertex values). Then $(d_0\phi)_{(u,v)} = \phi_v - \phi_u$. Applying d_1 to this 1-form yields, for a face (u, v, w) :

$$(d_1 d_0 \phi)_{(u,v,w)} = (\phi_v - \phi_u) + (\phi_w - \phi_v) + (\phi_u - \phi_w) = 0.$$

This holds for every face, so $d_1 d_0 = 0$ as a matrix product. In higher dimensions, the argument is analogous: each $(k+2)$ -simplex has exactly two $(k+1)$ -faces that contain each k -face, and their contributions cancel due to opposite orientations. This is the discrete version of the identities $\nabla \times (\nabla \phi) = 0$ and $\nabla \cdot (\nabla \times A) = 0$ [DHLM05]. $\square$

Example 30.87 (Exterior Derivative on a Single Triangle). Let a triangle have vertices u, v, w with a 0-form $\phi(u) = 1, \phi(v) = 3, \phi(w) = 0$. The 1-form $d_0\phi$ assigns values to oriented edges:

$$(d_0\phi)_{uv} = 3 - 1 = 2, \quad (d_0\phi)_{vw} = 0 - 3 = -3, \quad (d_0\phi)_{wu} = 1 - 0 = 1.$$

Applying d_1 :

$$(d_1 d_0 \phi)_{uvw} = 2 + (-3) + 1 = 0,$$

confirming the null property $d_1 \circ d_0 = 0$.

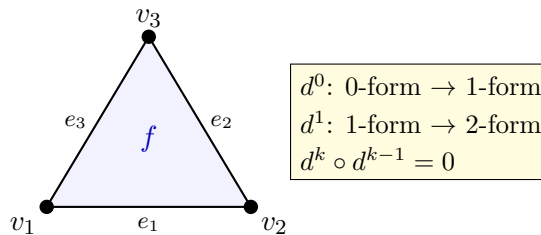


Figure 30.4: Discrete exterior derivative: d^0 maps vertex values to edge differences; d^1 maps edge values to face circulations.

30.18.4 Pseudocode

Note that Algorithm 149 requires only the mesh connectivity (vertex-edge and edge-face incidence), not any geometric information such as vertex positions or edge lengths.

Algorithm 149 Compute Exterior Derivatives

```

function COMPUTEEXTERIORDERIVATIVES(mesh)                                ▷ 1. d0 (0-forms to 1-forms)
     $d0 \leftarrow \text{SparseMatrix}(n\_edges, n\_vertices)$ 
    for all  $edge \in \text{mesh.edges}$  do
         $u, v \leftarrow edge.start\_vertex, edge.end\_vertex$ 
         $d0[edge\_id, v] \leftarrow 1$ 
         $d0[edge\_id, u] \leftarrow -1$ 
    end for

                                                                                         ▷ 2. d1 (1-forms to 2-forms)
     $d1 \leftarrow \text{SparseMatrix}(n\_faces, n\_edges)$ 
    for all  $face \in \text{mesh.faces}$  do
        for all  $(edge\_id, orientation) \in \text{GetOrientedEdges}(face)$  do
             $d1[face\_id, edge\_id] \leftarrow orientation$ 
        end for
    end for
    return  $d0, d1$ 
end function

```

30.18.5 Parameter Selection

- **Orientation:** All simplices must be consistently oriented. Primal edges are typically oriented by vertex ID (e.g., $u \rightarrow v$ if $u < v$). Primal faces are usually CCW.
- **Sparse Representation:** Since each simplex has only a few neighbors, d matrices are extremely sparse; CSR or CSC formats are essential for performance.

30.18.6 Complexity

- **Time Complexity:** $O(F)$, where F is the number of faces, as we iterate through each face and edge relationship once.
- **Space Complexity:** $O(E + F)$ to store the sparse matrix entries.

30.18.7 Usages

- **Vector Field Processing:** Decomposing vector fields into curl-free and divergence-free components (Hodge Decomposition).
- **Surface Parameterization:** Representing gradients of scalar functions used in UV mapping.
- **Fluid Simulation:** Enforcing divergence-free conditions on flow velocities.
- **Solving PDEs:** Formulating the Poisson or Heat equation using the DEC Laplacian $\Delta = \star^{-1} d^\top \star d$.
- **Topology Analysis:** Computing Betti numbers or identifying mesh “holes” using homology.

30.18.8 Testing Methods and Edge Cases

- **d o d Property:** Verify that $d_1 \times d_0$ results in a matrix where all entries are near zero (floating point) or exactly zero (integer).

- **Single Triangle:** On a single triangle, the sum of edge values around the boundary must match the face derivative.
- **Disconnected Mesh:** The matrices should naturally decompose into block-diagonal structures for each component.
- **Boundaries:** Ensure that orientation is handled correctly for edges on the mesh boundary.

30.18.9 References

- Desbrun, M., Hirani, A. N., Leok, M., & Marsden, J. E. (2005). “Discrete Exterior Calculus”. arXiv preprint.
- Crane, K., Weischedel, C., & Wardetzky, M. (2013). “Digital Geometry Processing with Discrete Exterior Calculus”. SIGGRAPH Course.
- Wikipedia: Exterior derivative

30.19 Hodge Decomposition

30.19.1 Overview

The Hodge-Helmholtz Decomposition (often simply Hodge Decomposition) is a fundamental theorem in vector calculus and differential geometry that states any vector field can be uniquely decomposed into three orthogonal components: a curl-free component (the gradient of a scalar potential), a divergence-free component (the curl of a vector potential), and a harmonic component (representing flow around topological “holes”). In the context of 2D surface meshes, it provides a powerful way to analyze and manipulate tangent vector fields [CWW13a, TLHD03].

30.19.2 Definitions

Definition 30.88 (Exact Component (ω_{exact})). The gradient of a scalar function α , $\omega_{\text{exact}} = d\alpha$. These are **curl-free**, meaning $d(\omega_{\text{exact}}) = d(d\alpha) = 0$ by Theorem 30.86. In $\mathbb{R}^3$ notation, $\omega_{\text{exact}} = \nabla\alpha$.

Definition 30.89 (Co-exact Component ($\omega_{\text{co-exact}}$)). The “rotational” part, represented as $\omega_{\text{co-exact}} = \star d\beta$. These are **divergence-free**, meaning $\delta(\omega_{\text{co-exact}}) = 0$, where $\delta = \star^{-1}d^\top\star$ is the codifferential. In $\mathbb{R}^3$ notation, $\omega_{\text{co-exact}} = \nabla \times \mathbf{A}$ for some vector potential $\mathbf{A}$.

Definition 30.90 (Harmonic Component (ω_{harmonic})). A field that is both curl-free and divergence-free, $d\omega = \delta\omega = 0$. These depend on the topology of the mesh—specifically, on the Betti numbers of the surface. On a closed surface of genus g , there are exactly $2g$ linearly independent harmonic 1-forms [PP03].

30.19.3 Theory

In Discrete Exterior Calculus (DEC), the decomposition is performed by solving two separate Poisson equations [CWW13a]:

1. **Gradient Part:** Minimize the energy $\|\omega - d\alpha\|^2$ to find the scalar potential α . This leads to the linear system $\Delta\alpha = \delta\omega$, where $\delta = \star^{-1}d^\top\star$ is the codifferential (divergence).
2. **Curl Part:** Minimize the energy $\|\omega - \delta\beta\|^2$ to find the dual potential β . This leads to the linear system $\Delta\beta = d\omega$, where d is the curl.
3. **Harmonic Part:** The remainder $\gamma = \omega - d\alpha - \delta\beta$ is the harmonic component.

Orthogonality between these components is guaranteed by the $d \circ d = 0$ property (Theorem 30.86) and the definition of the Hodge star.

Theorem 30.91 (Helmholtz-Hodge Decomposition). *For any 1-form ω on a compact, oriented Riemannian 2-manifold, there exist unique forms α (0-form), β (2-form), and γ (harmonic 1-form) such that:*

$$\omega = d\alpha + \star d\beta + \gamma,$$

where $d\alpha$ is exact (curl-free), $\star d\beta$ is co-exact (divergence-free), and γ is harmonic. Moreover, the three components are mutually orthogonal under the L^2 inner product:

$$\langle d\alpha, \star d\beta \rangle = \langle d\alpha, \gamma \rangle = \langle \star d\beta, \gamma \rangle = 0.$$

Proof. We construct the decomposition explicitly. Given a 1-form ω :

Step 1 (Exact part): Solve $\Delta\alpha = \delta\omega$ for α , where $\Delta = d^\top \star_1^{-1} d + d_1^\top \star_2 d_1^{-1} \dots$ is the discrete Laplacian. Define $\omega_{\text{exact}} = d\alpha$. Then $\delta(\omega - d\alpha) = \delta\omega - \Delta\alpha = 0$.

Step 2 (Co-exact part): Solve $\Delta\beta = d(\omega - d\alpha)$ for β . Define $\omega_{\text{co-exact}} = \star d\beta$. By Theorem 30.86, $d(\star d\beta) = 0$, so the co-exact part is divergence-free.

Step 3 (Harmonic part): Set $\gamma = \omega - d\alpha - \star d\beta$. By construction, $d\gamma = d\omega - d(d\alpha) - d(\star d\beta) = d\omega - 0 - 0 = d\omega - d\omega = 0$, and similarly $\delta\gamma = 0$.

Orthogonality follows from $\langle d\alpha, \star d\beta \rangle = \langle \alpha, d^\top \star d\beta \rangle = \langle \alpha, \Delta\beta \rangle$ and the self-adjointness of Δ [TLHD03]. $\square$

Example 30.92 (Hodge Decomposition on a Torus). Consider a uniform vector field ω flowing around the major circumference of a torus (genus $g = 1$). Since the field has no local variation, both $d\omega$ and $\delta\omega$ are zero, meaning ω is itself harmonic: $\omega = \gamma$. The exact and co-exact components are both zero. This demonstrates that harmonic components capture global topological flow—they cannot be expressed as local gradients or curls [PP03].

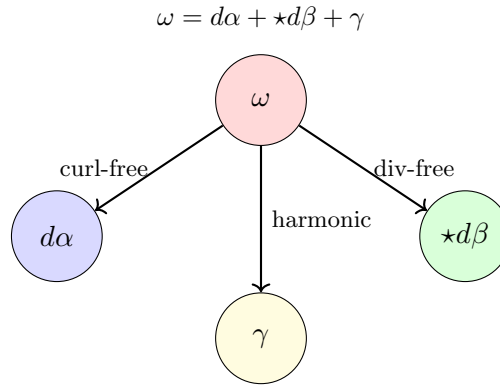


Figure 30.5: Hodge decomposition: any 1-form ω splits uniquely into exact, co-exact, and harmonic components.

30.19.4 Pseudocode

As shown in Algorithm 150, the algorithm relies on Algorithms 148 and 149 as subroutines and requires solving two sparse linear systems.

30.19.5 Parameter Selection

- **Boundary Conditions:** For meshes with boundaries, the decomposition is not unique unless boundary conditions are specified (e.g., $d\alpha$ is normal or tangent to the boundary).

Algorithm 150 Hodge Decomposition of a 1-Form

```

function HODGEDecompose(omega, mesh)                                ▷ 1. Compute discrete operators
    d0, d1 ← ComputeExteriorDerivatives(mesh)                        ▷ Algorithm 149
    star0, star1, star2 ← ComputeHodgeStars(mesh)                    ▷ Algorithm 148
    ▷ 2. Find Exact Component ( $d\alpha$ )

    Laplacian_0 ←  $d0^\top \cdot star1 \cdot d0$ 
    rhs_exact ←  $d0^\top \cdot star1 \cdot omega$ 
     $\alpha$  ← SolveSparse(Laplacian_0, rhs_exact)
    exact_part ←  $d0 \cdot \alpha$ 

    ▷ 3. Find Co-exact Component ( $\star d\beta$ )

    Laplacian_1 ←  $d1 \cdot star1^{-1} \cdot d1^\top$ 
    rhs_coexact ←  $d1 \cdot omega$ 
     $\hat{\beta}$  ← SolveSparse(Laplacian_1, rhs_coexact)
    coexact_part ←  $star1^{-1} \cdot d1^\top \star 2^{-1} \cdot \hat{\beta}$ 

    ▷ 4. Harmonic Part

    harmonic_part ←  $omega - exact\_part - coexact\_part$ 
    return exact_part, coexact_part, harmonic_part
end function

```

- **Sparse Solver:** Cholesky factorization (via `scipy.sparse.linalg.factorized`) is recommended as the Laplacian matrices are symmetric and positive semi-definite.

30.19.6 Complexity

- **Time Complexity:** $O(E^{1.5})$ where E is the number of edges, dominated by the sparse linear solver performance.
- **Space Complexity:** $O(E)$ to store the discrete operators and the field vectors.

30.19.7 Usages

- **Vector Field Design:** Creating smooth vector fields for hair, fur, or pen-and-ink rendering by manipulating the potentials.
- **Surface Parameterization:** Constructing “conformal” maps by ensuring the gradient field of UV coordinates is harmonic.
- **Fluid Animation:** Extracting the rotational (co-exact) part of a flow for vortex visualization or removing divergence to maintain volume.
- **Shape Matching:** Using harmonic components as topological signatures for 3D models.
- **Meteorology:** Analyzing global wind patterns into geostrophic (exact) and ageostrophic components.

30.19.8 Testing Methods and Edge Cases

- **Pure Gradient Field:** If $\omega = d\phi$ for some random scalar ϕ , then the co-exact and harmonic parts should be nearly zero.
- **Zero-Genus Mesh:** On a mesh with the topology of a sphere (genus 0), the harmonic part should be exactly zero.
- **Topological Holes:** On a torus (genus 1), a non-zero harmonic component should appear as a field that “wraps” around the hole without having any divergence or curl.

- **Exactness Verification:** Verify that $d_1 \times \text{exact_part} \approx 0$ and $\delta \times \text{coexact_part} \approx 0$.

30.19.9 References

- Polthier, K., & Preuss, E. (2003). “Identifying vector field singularities using a discrete Hodge decomposition”. Visualization and Mathematics III.
- Tong, Y., Lombeyda, S., Hirani, A. N., & Desbrun, M. (2003). “Discrete Multiscale Vector Field Decomposition”. ACM TOG.
- Crane, K. (2013). “Digital Geometry Processing with Discrete Exterior Calculus”. SIGGRAPH Course.
- Wikipedia: Helmholtz decomposition

30.20 Discrete Morse Theory

30.20.1 Overview

Discrete Morse Theory, developed by Robin Forman [For98, For02], is a combinatorial version of the classical Morse theory used to study the topology of a manifold by analyzing the critical points of a smooth function defined on it. In the discrete setting, it provides a powerful framework for simplifying complex simplicial complexes (like meshes) while preserving their fundamental topological properties (homology). It is used for topological data analysis, mesh compression, and identifying “features” like ridges and valleys in geometric data [GNPB08].

30.20.2 Definitions

Definition 30.93 (Simplicial Complex (K)). A collection of simplices (vertices, edges, faces) such that every face of a simplex is also in K . A simplicial complex encodes both the combinatorial structure (which elements are connected) and the dimensional hierarchy of the mesh.

Definition 30.94 (Discrete Morse Function (f)). A function $f: K \rightarrow \mathbb{R}$ that assigns a real number to every simplex in K such that for each simplex σ :

1. $|\{\tau^{(p+1)} > \sigma : f(\tau) \leq f(\sigma)\}| \leq 1$,
2. $|\{\nu^{(p-1)} < \sigma : f(\nu) \geq f(\sigma)\}| \leq 1$.

That is, at most one “upper neighbor” has a value no larger, and at most one “lower neighbor” has a value no smaller. A simplex where both counts are zero is called **critical**.

Definition 30.95 (Gradient Vector Field). A pairing of simplices into “gradient arrows” ($\alpha^p < \beta^{p+1}$) where $f(\alpha) \geq f(\beta)$. Each non-critical simplex belongs to exactly one gradient pair, and each gradient pair connects simplices of adjacent dimensions.

Definition 30.96 (Critical Simplex). A simplex that is not part of any gradient pair. Critical simplices correspond to the topological features of the underlying space: 0-dimensional critical simplices (minima) correspond to connected components, 1-dimensional (saddles) to handles, and 2-dimensional (maxima) to voids.

Definition 30.97 (Morse Complex). A reduced complex formed only by the critical simplices and their connectivity. The Morse-Smale complex further incorporates the gradient flow lines between critical points, providing a complete decomposition of the manifold into cells.

30.20.3 Theory

The core idea of Discrete Morse Theory is the **Discrete Gradient Vector Field** [For98].

1. A simplex α^p can be paired with an incident simplex β^{p+1} of higher dimension.
2. If every simplex is paired, the complex can be “collapsed” to a single point without changing its homotopy type.
3. Any simplex that *cannot* be paired is a **critical simplex**.
4. The number of critical simplices of dimension p (denoted m_p) provides an upper bound on the p -th Betti number β_p (Morse Inequalities): $m_p \geq \beta_p$.

Theorem 30.98 (Morse-Euler Relation). *For any discrete Morse function f on a finite simplicial complex K , the alternating sum of the counts of critical simplices equals the Euler characteristic:*

$$\sum_{p=0}^n (-1)^p m_p = \chi(K),$$

where m_p is the number of critical p -simplices.

Proof. The proof follows from the Morse complex construction. Each critical simplex contributes $(-1)^p$ to the alternating sum. Since the Morse complex is homotopy equivalent to the original complex, its Euler characteristic equals $\chi(K)$. For a finite complex K with v vertices, e edges, and f faces, we have $\chi(K) = v - e + f$. The gradient pairing removes one p -simplex and one $(p+1)$ -simplex simultaneously, which does not change the alternating sum $v - e + f$. Therefore, the sum over critical simplices alone must equal $\chi(K)$ [For02]. $\square$

Example 30.99 (Morse Function on a Sphere). Consider a geodesic sphere with a height function $f(v) = z$ -coordinate. A “perfect” discrete Morse function on a sphere should yield exactly two critical simplices: one vertex (the global minimum at the south pole, $m_0 = 1$) and one face (the global maximum at the north pole, $m_2 = 1$). The Morse-Euler relation gives $m_0 - m_1 + m_2 = 1 - 0 + 1 = 2 = \chi(S^2)$, which is correct for a sphere.

By constructing an appropriate Morse function, one can extract a **1D skeleton** (the Morse-Smale complex) that summarizes the flow and connectivity of a scalar field (like elevation) over a surface [GNPB08].

30.20.4 Pseudocode

Greedy Construction of a Discrete Gradient Field

Algorithm 151 produces a valid discrete gradient field, but not necessarily one with the fewest critical points. More sophisticated algorithms [GNPB08] use cancellation operations to merge pairs of critical simplices and further simplify the Morse complex.

30.20.5 Parameter Selection

- **Initial Function:** The choice of the input scalar field (e.g., height, curvature, or random) determines which geometric features are identified as critical points.
- **Simplification Threshold:** Small “noise” critical points can be removed by canceling pairs of critical points (e.g., a small hill and a small saddle) using Morse cancellations.

Algorithm 151 Greedy Construction of Discrete Gradient Field

```

function GREEDYGRADIENTFIELD(mesh)
    unpaired  $\leftarrow$  set(mesh.all_simplices)
    gradient_pairs  $\leftarrow$  []
    critical_simplices  $\leftarrow$  []
    while unpaired is not empty do
         $\alpha \leftarrow$  FindSimplexWithOneNeighbor(unpaired)
        if  $\alpha \neq \text{None}$  then
             $\beta \leftarrow$  GetOnlyNeighbor( $\alpha$ , unpaired)
            gradient_pairs.append(( $\alpha$ ,  $\beta$ ))
            unpaired.remove( $\alpha$ )
            unpaired.remove( $\beta$ )
        else
             $\alpha \leftarrow$  unpaired.pop()
            critical_simplices.append( $\alpha$ )
        end if
    end while
    return gradient_pairs, critical_simplices
end function

```

30.20.6 Complexity

- **Time Complexity:** $O(N \log N)$ to sort simplices by function value, followed by $O(N)$ for the greedy pairing, where N is the total number of simplices.
- **Space Complexity:** $O(N)$ to store the simplices and the gradient field.

30.20.7 Usages

- **Topological Data Analysis (TDA):** Computing persistent homology and identifying stable structural features in noisy data.
- **Mesh Compression:** Reducing a complex mesh to a much smaller Morse complex that has the same topology.
- **Terrain Analysis:** Automatically identifying peaks (maximums), pits (minimums), and passes (saddles) in geographic maps.
- **Scientific Visualization:** Constructing Morse-Smale complexes to segment a volume based on the behavior of a physical field (e.g., velocity or pressure).
- **Feature Extraction:** Detecting ridge and valley lines on 3D models.

30.20.8 Testing Methods and Edge Cases

- **Sphere:** A perfect discrete Morse function on a sphere should yield exactly two critical simplices: one vertex (min) and one face (max).
- **Torus:** Should yield one vertex, two edges (loops), and one face.
- **Euler Characteristic:** Verify the Morse-Euler relation (Theorem 30.98): $\sum (-1)^p m_p = \chi$, where m_p is the number of critical p -simplices.
- **Noise Handling:** Test the algorithm on a noisy field to ensure it produces many small critical points, which can then be simplified.

- **Boundaries:** Ensure the theory correctly accounts for simplices on the boundary of the complex.

30.20.9 References

- Forman, R. (1998). “Discrete Morse Theory”. *Advances in Mathematics*.
- Forman, R. (2002). “A user’s guide to discrete Morse theory”. *Sém. Lothar. Combin.*
- Gyulassy, A., Natarajan, V., Pascucci, V., & Bremer, P. T. (2008). “A Practical Approach to Morse-Smale Complex Computation: Quasi-complexes and Geometric Methods”. *IEEE TVCG*.
- Wikipedia: Discrete Morse theory

Part X

Advanced Algorithmic Techniques

Chapter 31

Randomized Geometric Algorithms

31.1 Why Randomization Helps

Many geometric algorithms are fast on typical inputs yet slow on a small set of adversarial configurations. Deterministic algorithms must be correct on every input, so their analysis must cover the worst case. **Randomization** breaks this deadlock: the algorithm flips coins, and we analyze its *expected* behavior over those coin flips. Because the randomness lives inside the algorithm rather than in the input, the expectation is meaningful for *every* input.

Definition 31.1 (Expected Time). The **expected running time** of a randomized algorithm on input I is the expectation, over the algorithm's random choices, of its running time on I . An algorithm whose expected time is $T(n)$ on all inputs of size n runs in **expected time** $O(T(n))$.

Expected time differs from worst-case time (the deterministic guarantee) and from average-case analysis (which averages over inputs): it is an adversarial-input guarantee averaged over the algorithm's own randomness. Two classes of randomized algorithms use this randomness differently.

Definition 31.2 (Las Vegas Algorithm). A **Las Vegas algorithm** always produces a correct answer; only its running time is random. It can be restarted to enforce a hard time bound at the price of a tiny failure probability.

Definition 31.3 (Monte Carlo Algorithm). A **Monte Carlo algorithm** may produce an incorrect answer, but only with probability at most a reported bound δ ; its running time is typically deterministic.

Nearly all algorithms in this chapter are Las Vegas: the randomized incremental constructions are correct for every permutation, and randomness only controls how fast they run. Monte Carlo behavior appears in the derived sampling structures (ϵ -nets, Section 31.7), analyzed with the concentration results of Section 31.6. The classical references are the monographs of Motwani and Raghavan [MR95] and Mulmuley [Mul94].

Why does randomization help in geometry specifically? Three phenomena recur. First, many structures are *order dependent*: an incremental construction is fast for most insertion orders and slow only for a measure-zero set that an adversary forces; a random order averages over equally likely orders. Second, geometry is *sample friendly*: a small random sample resembles the whole input, so the sample can guide the computation (Section 31.5). Third, random orders are *easier to analyze* than the worst order, thanks to Seidel's backward analysis (Section 31.3). The payoff: randomized incremental constructions achieve the optimal expected bounds for Delaunay triangulations (Section 31.9), convex hulls (Section 31.8), and linear programming (Section 31.10), with simpler algorithms than their deterministic counterparts.

31.2 Randomized Incremental Construction

31.2.1 1. Overview

Randomized incremental construction (RIC) inserts the input objects one at a time in a uniformly random order and repairs the current geometric structure locally after each insertion. Because the order is random, the repair cost at step i is small in expectation, giving expected $O(n \log n)$ total time. The pattern covers the Delaunay triangulation (Section 31.9), the trapezoidal map of Chapter 8 (Sections 8.4–8.5), convex hulls (Section 31.8), and linear programming (Section 31.10); it is due to Clarkson and Shor [CS89], Seidel [Sei93], and Mulmuley [Mul94].

31.2.2 2. Definitions

Definition 31.4 (Incremental Construction). Let $\mathcal{F}(P)$ be a geometric structure on an object set P of size n . An **incremental construction** of $\mathcal{F}(P)$ fixes an insertion order $p_1, \dots, p_n$ and builds $\mathcal{F}(\{p_1, \dots, p_i\})$ for $i = 1, \dots, n$; it is **randomized** if the order is a uniformly random permutation.

The cost of one insertion splits into **locating** the new object (typically $O(\log i)$ via a history DAG, Section 8.6), **inserting** it (destroying and creating cells), and **updating** the conflict graph of Section 31.4. The insertion cost is proportional to the destroyed cells, which backward analysis (Section 31.3) controls.

31.2.3 3. Theory

The accounting identity of RIC is that total work is proportional to the number of cells ever created. If c_i is the number created by the i -th insertion and C is the final structure, then

$$\sum_{i=1}^n \mathbb{E}[c_i] = \sum_{T \in C} \sum_{i=1}^n \Pr[T \text{ is created by insertion } i].$$

Theorem 31.5 (Expected Cost of RIC). *Consider a randomized incremental construction in which every cell of the final structure is determined by a constant number d of inserted objects, and in which locating a new object costs $O(\log n)$ expected via a history DAG. Then the expected total time is $O(n \log n)$ and the expected space is $O(n)$ plus the size of the final structure.*

31.2.4 4. Pseudocode

31.2.5 5. Parameter Selection

- **Random order:** draw one permutation up front with a reproducible seed; multiple seeds are the standard variance-reduction tool for benchmarks.
- **Base case and location:** handle the first d objects directly (a big triangle in the Delaunay chapter); use a history DAG to preserve the $O(\log n)$ location guarantee.

31.2.6 6. Complexity

- **Time:** expected $O(n \log n)$ (Theorem 31.5); worst case $O(n^2)$ for adversarial orders.
- **Space:** expected $O(n)$ for the structure plus the history DAG.

Algorithm 152 Generic Randomized Incremental Construction

Input: A set S of geometric objects and a random source.**Output:** The structure $\mathcal{F}(S)$.

```

1: function RIC( $S$ )
2:    $\sigma \leftarrow$  uniformly random permutation of  $S$ 
3:    $\mathcal{F} \leftarrow$  structure on the first  $d$  objects of  $\sigma$ 
4:   for  $i = d + 1, \dots, |S|$  do
5:      $o \leftarrow \sigma[i]$ 
6:     locate  $o$  in  $\mathcal{F}$  (walk, history DAG, or conflict list)
7:     insert  $o$ : destroy the conflicting cells, create the replacements
8:     update the conflict graph for the created cells
9:   end for
10:  return  $\mathcal{F}$ 
11: end function

```

31.2.7 7. Usages

- Delaunay triangulation, $O(n \log n)$ expected (Section 31.9).
- Trapezoidal map and point location (Chapter 8).
- Linear programming and convex hulls (Sections 31.10, 31.8).

31.2.8 8. Testing Methods and Edge Cases

- **Permutation invariance:** the output must be identical for every random order; run several seeds and compare exactly.
- **Degenerate objects:** duplicates and collinear triples must not create zero-area cells; use the exact predicates of Chapter 3.
- **Reproducibility:** identical seeds must give identical runs.

31.2.9 9. References

The framework is due to Clarkson and Shor [CS89], Seidel [Sei93], and Mulmuley [Mul94]; textbook treatments are in de Berg et al. [dBCvKO08] (Ch. 9) and Motwani–Raghavan [MR95].

31.3 Backward Analysis

The most widely used tool for analyzing randomized incremental algorithms is Seidel’s **backward analysis** [Sei93]: instead of analyzing step i *forward*, analyze the random process *backwards*. The insertion of the points of P in a random order is, in reverse, a process that deletes points one at a time from a set whose elements are uniformly distributed among the remaining points, so conditioning on the remaining set is trivial: it is always a uniformly random subset of P .

Definition 31.6 (Backward Analysis). Let the points of P be inserted in a uniformly random order and let $\mathcal{F}_i$ be the structure after i insertions. A cell T of the final structure is **created at step** i if $T \in \mathcal{F}_i$ but $T \notin \mathcal{F}_{i-1}$. **Backward analysis** estimates $\Pr[T \text{ created at step } i]$ by conditioning on the reverse process, where the defining objects of T form a uniformly random d -tuple among the i remaining objects.

Theorem 31.7 (Seidel’s Backward-Analysis Lemma). *Let T be a cell of the final structure determined by exactly d objects of the input, where d is a constant. Inserting the input objects in a uniformly random order, the probability that T is created at step i is $O(d/i)$. Hence the expected number of cells created during step i is $O(1)$ when every final cell is determined by a constant number of objects.*

Proof sketch. In the reverse process, T is created (in forward time) at the moment its last defining object is deleted. Conditioning on T being present when i objects remain, the next deleted object is uniform among the i , and T is created only if that object is one of the d defining T : probability d/i . Summing over the $O(n)$ cells of the final structure, the expected number of creations per step is $O(1)$, and the total expected creations are $O(n)$. $\square$

Example 31.8 (Expected Creations in Delaunay). Each final Delaunay triangle is determined by its three vertices, so a fixed triangle is created at step i with probability $O(3/i)$ by Theorem 31.7. Summing over the $O(n)$ final triangles gives expected creation work $O(n)$; with $O(\log n)$ expected location per insertion through the history DAG (Section 8.6) this yields the $O(n \log n)$ expected construction time.

Backward analysis was introduced by Seidel for the trapezoidal map [Sei91], developed into a general methodology in his survey [Sei93], and used by Guibas, Knuth, and Sharir for the randomized incremental Delaunay triangulation [GKS92].

Remark 31.9. The same principle—analyze the moment a structure is *created* by looking at the reverse deletion process—applies verbatim to convex hulls (a hull edge is created when its last vertex is deleted) and to linear programming (a constraint becomes binding when it is deleted).

31.4 Conflict Graphs

31.4.1 1. Overview

A randomized incremental construction must know, when a new object is inserted, which cells it destroys. The **conflict graph** maintains this information explicitly, linking every uninserted object to the cells it would destroy, and is updated as cells are created. It was introduced in this form by Clarkson and Shor [CS89] and makes the RIC template of Section 31.2 concrete.

31.4.2 2. Definitions

Definition 31.10 (Conflict). An uninserted object o **conflicts** with a cell T of the current structure if inserting o would destroy T . The set of cells conflicting with o is its **conflict zone**.

Definition 31.11 (Conflict Graph). The **conflict graph** is a bipartite graph whose left vertices are the uninserted objects, whose right vertices are the cells of the current structure, and whose edges are exactly the conflicts.

For the Delaunay triangulation the conflict relation is the in-circle test of Definition 12.7 (Section 12.5); for convex hulls a point conflicts with the hull facets it lies outside (Section 31.8).

31.4.3 3. Theory

Theorem 31.12 (Correctness of the Conflict-Graph Update). *Maintaining the conflict graph through a randomized incremental construction is correct, provided (i) the conflicts of every newly created cell are computed by checking each uninserted object in its neighborhood, and (ii) destroyed cells and their edges are removed.*

Proof sketch. Induction over insertions. Inserting o destroys exactly the cells of its conflict zone (verified for each application, e.g., the cavity theorem of Section 12.5), so those cells are removed correctly. A cell T newly created by the insertion can be destroyed only by an object in the conflict zone of the destroyed cells spanning T ; checking those objects restores the exact adjacency of T . $\square$

The update cost is proportional to the sum over destroyed cells of their conflict-graph neighbors, which backward analysis bounds by $O(1)$ per created cell in expectation.

31.4.4 4. Pseudocode

Algorithm 153 Conflict-Graph Maintenance

Input: The current structure $\mathcal{F}$, the conflict graph G , an uninserted object o .

Output: $\mathcal{F}$ and G updated after inserting o .

```

1: function INSERTWITHCONFLICTS( $\mathcal{F}, G, o$ )
2:    $\mathcal{Z} \leftarrow$  conflict zone of  $o$  in  $G$ 
3:    $\mathcal{B} \leftarrow$  boundary of the cells in  $\mathcal{Z}$ 
4:   remove  $\mathcal{Z}$  and their edges from  $\mathcal{F}$  and  $G$ 
5:   create the replacement cells from  $o$  and  $\mathcal{B}$ 
6:   for each new cell  $T$  do
7:     for each  $q$  adjacent to a destroyed cell meeting  $T$  do
8:       if  $q$  conflicts with  $T$  then
9:         add edge  $(q, T)$  to  $G$ 
10:      end if
11:    end for
12:  end for
13: end function

```

31.4.5 5. Parameter Selection

- **Conflict predicate:** evaluated exactly with the robust predicates of Section 3.4 (in-circle for Delaunay, outside-facet for hulls).
- **Neighborhood pruning:** test only objects near the destroyed cells against new cells, keeping the update local.

31.4.6 6. Complexity

- **Time:** expected $O(1)$ amortized per created cell; total expected $O(n \log n)$ for the applications of this chapter.
- **Space:** expected $O(n)$ conflicts at any time.

31.4.7 7. Usages

- The Delaunay conflict graph of Section 12.6.
- Hull facets store the points outside them (Section 31.8).
- The vertical-decomposition counterpart in point location (Section 8.5).

31.4.8 8. Testing Methods and Edge Cases

- **Exactness check:** recompute the conflict zone of a random uninserted object by brute force after each insertion and compare.
- **Boundary ties and empty zones:** an object on a cell boundary conflicts with both adjacent cells; an object inside the hull has an empty zone and must be a no-op.

31.4.9 9. References

- Clarkson, K. L., Shor, P. W. [CS89].
- Mulmuley, K. [Mul94], Ch. 3, the theory of conflict maintenance.
- de Berg, M., et al. [dBCvKO08], Sec. 9.4.

31.5 Clarkson–Shor Technique

31.5.1 1. Overview

The **Clarkson–Shor technique** uses random sampling to bound the number of geometric configurations: a random sample of the input “resembles” the whole input, so the configurations determined by the sample bound the number at a related “level”. This yields the classical $O(nk)$ bound on the number of k -sets in the plane and the first optimal expected-time algorithms for convex hulls and Delaunay triangulations [CS89]. It is the probabilistic counterpart of the deterministic method of Chazelle and Friedman [CF90] (Section 31.6).

31.5.2 2. Definitions

Definition 31.13 (k -Set). Let P be a set of n points in the plane in general position. A k -set of P is a subset of size k separable by a half-plane (equivalently, cut off by a level- k line of the dual arrangement; see Definition 14.14).

Definition 31.14 (Configurations Determined by a Sample). Let $\mathcal{C}(P)$ be the set of **configurations** (cells) determined by P , each determined by a fixed constant number d of objects. For $R \subseteq P$ and $T \in \mathcal{C}(P)$, T is **crossing** R if it is determined by a set meeting both R and $P \setminus R$.

31.5.3 3. Theory

Theorem 31.15 (Clarkson–Shor Sampling Lemma). *Let $\mathcal{C}$ be a configuration space as in Definition 31.14, and let $g_r(m)$ be the maximum, over sets of m objects, of the expected number of configurations determined by a uniform random sample of size r . Then the number of configurations crossed by at most k objects is $O(g_{n/(k+1)}(n))$. In particular, the number of $(\leq k)$ -sets of an n -point planar set is $O(nk)$.*

Proof sketch. Let R be a uniform random r -subset of P and X the number of configurations it determines, so $\mathbb{E}[X] \leq g_r(n)$. A configuration T is determined by R exactly when R contains the d objects of T and none of the (at most k) objects crossing T , with probability

$$\binom{n-d-k}{r-d} / \binom{n}{r} \approx (r/n)^d (1-r/n)^k.$$

For $r = n/(k+1)$ both factors are $\Omega(1)$, so every “level- k ” configuration is counted with probability $\Omega(1)$; hence their number is $O(g_{n/(k+1)}(n))$. For planar point sets the sample’s hull and triangulation are linear, so $g_r(n) = O(r)$ and the count is $O(nk)$. $\square$

The combinatorial corollary is the $O(nk)$ bound on the k -level of a planar line arrangement for each k [CS89], the basis of the complexity analysis of arrangements (Chapter 14). Choosing the sample as the recursive input gives the expected-time bounds for hulls and Delaunay triangulations.

31.5.4 4. Pseudocode

Algorithm 154 Clarkson–Shor Sampling for Convex Hulls

Input: point set P in the plane.

Output: the convex hull of P .

```

1: function SAMPLEDHULL( $P$ )
2:   if  $|P|$  is small then
3:     return HullByAnyMethod( $P$ )
4:   end if
5:    $R \leftarrow$  uniform random sample of  $P$  of size  $\lceil |P|/2 \rceil$ 
6:    $H \leftarrow$  SampledHull( $R$ )
7:   find the points of  $P$  outside  $H$  by the walk of Section 8.8
8:   insert the outside points into  $H$  one at a time
9:   return the resulting hull
10: end function

```

31.5.5 5. Parameter Selection

- **Sample size:** $r = n/(k+1)$ balances the two factors in the probability; a half-and-half split is a convenient recursion choice.
- **Point location:** the walk locates the outside points in expected linear time.

31.5.6 6. Complexity

- **Combinatorial:** $O(nk)$ ($\leq k$)-sets and $O(nk)$ ($\leq k$)-levels of a planar arrangement.
- **Algorithmic:** expected $O(n \log n)$ for convex hulls and Voronoi diagrams [CS89, Cla93].

31.5.7 7. Usages

- The k -level complexity of arrangements (Chapter 14).
- Output-sensitive hull constructions (Section 31.8).
- Randomized closest-pair and Voronoi algorithms [Cla93].

31.5.8 8. Testing Methods and Edge Cases

- **Combinatorial verification:** enumerate k -sets on small random sets by brute force and compare counts with the $O(nk)$ bound.
- **Sampling bias:** verify sample uniformity by statistical tests when the random source is suspect.

31.5.9 9. References

- Clarkson, K. L., Shor, P. W. [CS89].
- Clarkson, K. L. [Cla93].
- Chazelle, B., Friedman, J. [CF90].

31.6 Random Sampling

Random sampling is the source of the randomness in this chapter: a small, uniformly chosen subset of the input is used as a surrogate for the whole. The justification is **concentration**: with high probability the relative frequencies of every combinatorial range in the sample match those of the input. This is the content of the Chernoff bound and its geometric refinements, and it is what makes randomized construction and ε -nets (Section 31.7) correct with high probability.

Definition 31.16 (Range Space). A **range space** is a pair $(\mathcal{X}, \mathcal{R})$ of a ground set $\mathcal{X}$ and a family of subsets $\mathcal{R} \subseteq 2^{\mathcal{X}}$, the **ranges**; the most important geometric examples are the half-plane ranges in the plane and half-space ranges in $\mathbb{R}^d$.

Definition 31.17 (VC Dimension). A set $S \subseteq \mathcal{X}$ is **shattered** by $\mathcal{R}$ if every subset of S equals $S \cap R$ for some range R . The **VC dimension** of the range space is the size of the largest shattered set. Half-planes in the plane have VC dimension 3; half-spaces in $\mathbb{R}^d$ have VC dimension $d + 1$ [HW87].

Theorem 31.18 (Chernoff Bound). Let $X_1, \dots, X_r$ be independent Bernoulli trials with $\Pr[X_j = 1] = p$ and $\bar{X} = (1/r) \sum_j X_j$. For every $\delta \in (0, 1)$,

$$\Pr[\bar{X} \leq (1 - \delta)p] \leq e^{-\delta^2 pr/2}, \quad \Pr[\bar{X} \geq (1 + \delta)p] \leq e^{-\delta^2 pr/3}.$$

Chernoff bounds control the number of “bad” events in a randomized construction (a union bound converts them into a failure probability δ at the cost of a $O(\log 1/\delta)$ factor in the sample size) and the deviation of sample statistics from their expectation. The geometric refinement is the **random sampling lemma**.

Theorem 31.19 (Random Sampling Lemma). Let $(\mathcal{X}, \mathcal{R})$ be a range space of VC dimension d and let R be a uniform random sample of r points of $\mathcal{X}$. With probability at least $1 - \delta$, simultaneously for all ranges $T \in \mathcal{R}$,

$$\left| \frac{|R \cap T|}{r} - \frac{|T|}{n} \right| \leq O\left(\sqrt{\frac{d}{r} \log \frac{r}{d}} + \sqrt{\frac{1}{r} \log \frac{1}{\delta}} \right).$$

This is the uniform bound of Haussler and Welzl [HW87].

The lemma is the bridge to the structures of the next section: an ε -approximation is a sample in which every range deviates by at most ε , and an ε -net is a sample intersecting every range of measure at least ε . Both have size independent of n for constant d , which is why they reduce large geometric problems to small random ones [HP11].

Derandomization recovers the same guarantees deterministically. The Chazelle–Friedman method [CF90] replaces the random sample by a *deterministic* sample built so that no range of a canonical family contains an extreme number of points, using a discrepancy bound; the derandomized ε -nets and the optimal deterministic convex-hull algorithm of Chazelle [Cha93] rest on this principle. The price is complexity and constants, which is why randomized sampling remains the default in implementations.

Remark 31.20. The discrepancy viewpoint is developed by Matoušek [Mat99]; the half-plane discrepancy of the sampling chapter (Definition 14.16) is exactly the deviation controlled by the sampling lemma.

31.7 ε -Nets

31.7.1 1. Overview

An ε -**net** is a small subset of the input that hits every large range. If a problem is solved correctly whenever the solution set is an ε -net of the input, then a tiny random sample suffices, independent of n . This is the engine of many approximation algorithms in geometric data structures (Chapter 9) and was introduced by Haussler and Welzl [HW87] to support range searching.

31.7.2 2. Definitions

Definition 31.21 (ε -Net). Let $(\mathcal{X}, \mathcal{R})$ be a range space and $P \subseteq \mathcal{X}$ finite. A subset $N \subseteq P$ is an ε -**net** of P if every range $R \in \mathcal{R}$ with $|R \cap P| \geq \varepsilon|P|$ contains a point of N .

Definition 31.22 (ε -Approximation). A subset $A \subseteq P$ is an ε -**approximation** of P if for every range $R \in \mathcal{R}$,

$$\left| \frac{|R \cap A|}{|A|} - \frac{|R \cap P|}{|P|} \right| \leq \varepsilon.$$

An ε -approximation is in particular an ε -net.

31.7.3 3. Theory

Theorem 31.23 (Small Nets by Random Sampling). Let $(\mathcal{X}, \mathcal{R})$ be a range space of VC dimension d . A uniform random sample N of size

$$r = O\left(\frac{d}{\varepsilon} \log \frac{d}{\varepsilon} + \frac{1}{\varepsilon} \log \frac{1}{\delta}\right)$$

is an ε -net of P with probability at least $1 - \delta$; a sample of size $O(\frac{d}{\varepsilon^2} \log \frac{1}{\delta})$ is an ε -approximation with the same probability [HW87].

Proof sketch. Fix a range R with $|R \cap P| \geq \varepsilon|P|$. The sample misses R with probability $(1 - \varepsilon)^r \leq e^{-\varepsilon r}$. The packing argument of Haussler and Welzl bounds the number of ranges relevant at this scale by $O(r^d)$, so the union bound gives failure probability $r^d e^{-\varepsilon r} \leq \delta$ for the stated size. The approximation claim follows from Theorem 31.19 and a similar union bound. $\square$

For half-plane ranges ($d = 3$) the bound is $O(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$, independent of n ; the deterministic construction of Matoušek [Mat90] achieves the same size in time $O(n \text{ poly}(1/\varepsilon))$.

31.7.4 4. Pseudocode

31.7.5 5. Parameter Selection

- ε : the sample size is governed by the $O(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$ term; ε trades accuracy against sample size.
- **Failure probability** δ : the $\frac{1}{\varepsilon} \log \frac{1}{\delta}$ term grows logarithmically, so very small failure probabilities are cheap.

31.7.6 6. Complexity

- **Size**: $O(\frac{d}{\varepsilon} \log \frac{d}{\varepsilon})$, independent of n .
- **Time**: $O(n + r \log r)$ randomized; $O(n \text{ poly}(1/\varepsilon))$ deterministic [Mat90].

Algorithm 155 Randomized ε -Net Construction

Input: point set P in the plane, parameter ε , failure probability δ .**Output:** An ε -net of P (half-plane ranges) with probability at least $1 - \delta$.

```
1: function EPSILONNET( $P, \varepsilon, \delta$ )
2:    $r \leftarrow \lceil \frac{3}{\varepsilon} \log \frac{3}{\varepsilon} + \frac{1}{\varepsilon} \log \frac{1}{\delta} \rceil$ 
3:    $N \leftarrow$  uniform random sample of  $P$  of size  $r$ 
4:   if some half-plane contains  $> \varepsilon|P|$  points and none of  $N$  then
5:     resample, or fall back to the construction of [Mat90]
6:   end if
7:   return  $N$ 
8: end function
```

31.7.7 7. Usages

- Approximate simplex range counting (Section 9.12) and range searching (Chapter 9).
- The coresets of Har-Peled and Mazumdar [HPM04] are ε -approximations under geometric distances.

31.7.8 8. Testing Methods and Edge Cases

- **Brute-force verification:** on small inputs enumerate all ranges and assert the net property holds for every one.
- **Small ε :** verify the $O(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$ growth of the net size empirically.

31.7.9 9. References

- Haussler, D., Welzl, E. [HW87], the original theorem.
- Matoušek, J. [Mat90], deterministic construction.
- Matoušek, J. [Mat99] and Har-Peled, S. [HP11].

31.8 Applications to Convex Hulls

31.8.1 1. Overview

The randomized incremental convex hull inserts the points in a random order and maintains the hull of the inserted prefix, using the conflict graph to find, for every point still outside, the hull edge it lies in front of. Backward analysis shows each hull edge is created with probability $O(2/i)$ at step i , giving expected $O(n \log n)$ total time, matching the optimal deterministic bound of Chan's algorithm (Theorem 4.11) and the methods of Chapter 4 (Section 4.4).

31.8.2 2. Definitions

Definition 31.24 (Hull Conflict). Let $\text{conv}(Q)$ be the convex hull of the inserted points Q . An uninserted point p **conflicts** with a hull edge e if p lies in the open half-plane on the outside of e ; the conflict zone of p is the contiguous chain of hull edges visible from p . Points inside the hull have an empty conflict zone.

31.8.3 3. Theory

Theorem 31.25 (Randomized Incremental Hull). *Inserting the points of a planar n -point set in a uniformly random order, maintaining the hull and its conflict graph, computes the convex hull in expected $O(n \log n)$ time and $O(n)$ space.*

Proof sketch. The total work is the number of hull edges ever created. A final hull edge is determined by its two endpoints, so by Theorem 31.7 with $d = 2$ it is created at step i with probability $O(2/i)$. The final hull has at most n edges, giving expected creations $\sum_i O(2/i) \cdot O(i) = O(n \log n)$; locating each inserted point through the conflict list costs $O(1)$ amortized. $\square$

In $\mathbb{R}^3$ the same argument gives expected $O(n \log n)$; the Clarkson–Shor analysis [CS89] yields the same bound with optimal constants for random inputs.

31.8.4 4. Pseudocode

Algorithm 156 Randomized Incremental Convex Hull

Input: point set P in the plane.

Output: the convex hull of P .

```

1: function RANDOMIZEDHULL( $P$ )
2:    $\sigma \leftarrow$  uniformly random permutation of  $P$ 
3:    $H \leftarrow$  convex hull of the first two points of  $\sigma$ 
4:   initialize the conflict graph of the remaining points
5:   for  $i = 3, \dots, n$  do
6:      $p \leftarrow \sigma[i]$ 
7:     if  $p$ 's conflict zone is empty then
8:       continue  $\triangleright p$  is inside  $H$ 
9:     else
10:      remove the visible chain of  $H$ ; add the two tangent edges from  $p$ 
11:      repartition the conflicts of the removed edges onto the two new edges
12:    end if
13:  end for
14:  return  $H$ 
15: end function

```

31.8.5 5. Parameter Selection

- **Initialization:** start from the first two points as a degenerate “hull”; collinear prefixes are handled by the tangency routine.
- **Tangents and predicates:** walk the hull from the visible chain to find the two tangents; use the robust orientation predicate of Chapter 3.

31.8.6 6. Complexity

- **Time:** expected $O(n \log n)$; worst case $O(n^2)$ for adversarial deterministic orders.
- **Space:** $O(n)$.

31.8.7 7. Usages

- A testing oracle for the deterministic hulls of Chapter 4 (Sections 4.1–4.4).
- Extends to $\mathbb{R}^3$ with the same conflict-graph machinery; linked to LP by point-line duality (Section 31.10).

31.8.8 8. Testing Methods and Edge Cases

- **Cross-validation:** compare with the monotone chain and Graham scan outputs for many permutations.
- **Degenerate inputs:** drop duplicates, tolerate collinear boundary points, and assign points on a hull edge a consistent conflict zone.

31.8.9 9. References

- Clarkson, K. L., Shor, P. W. [CS89].
- Seidel, R. [Sei90], convex hulls made easy.
- Chazelle, B. [Cha93], the optimal deterministic hull.
- de Berg, M., et al. [dBCvKO08], Sec. 11.2.

31.9 Applications to Delaunay Triangulations

31.9.1 1. Overview

The Delaunay triangulation is the flagship example of randomized incremental construction: inserting the points in a random order and repairing the triangulation by edge flips yields the expected $O(n \log n)$ construction of Section 12.6 (Theorem 12.18). This section re-derives that bound from the backward-analysis and conflict-graph machinery of this chapter, and discusses higher-dimensional behavior.

31.9.2 2. Definitions

Definition 31.26 (Delaunay Conflict). In the Delaunay setting the conflict relation of Definition 31.10 is the in-circle test (Definition 12.7): an uninserted point p conflicts with a triangle T if p lies strictly inside the circumcircle of T . The conflict zone of p is exactly the cavity of Section 12.5 (Definition 12.17).

31.9.3 3. Theory

Theorem 31.27 (Expected Delaunay Cost). *The randomized incremental construction of the Delaunay triangulation of an n -point set in the plane runs in expected $O(n \log n)$ time and expected $O(n)$ space; the expected number of triangles ever created is $O(n)$.*

Proof sketch. A final triangle is determined by its three vertices, so by Theorem 31.7 it is created at step i with probability $O(3/i)$ and the expected number of created triangles is $O(n)$ (Example 31.8). The expected conflict-graph size is $O(n)$ by the same backward argument, and point location through the history DAG (Section 8.6) costs expected $O(\log n)$ per insertion, giving the claimed time. $\square$

In $\mathbb{R}^d$ the Delaunay complex has worst-case size $\Theta(n^{\lceil d/2 \rceil})$, and the randomized incremental construction runs in expected $O(n^{\lceil d/2 \rceil})$ time (Section 12.13); the planar case is the only one with a clean $O(n \log n)$ bound.

31.9.4 4. Pseudocode

Algorithm 157 Randomized Incremental Delaunay (Walk Variant)

Input: point set P , an enclosing “big triangle”.

Output: the Delaunay triangulation of P .

```

1: function RANDOMIZEDDELAUNAYWALK( $P$ )
2:    $\sigma \leftarrow$  uniformly random permutation of  $P$ 
3:    $\mathcal{T} \leftarrow$  big triangle
4:   for  $i = 1, \dots, n$  do
5:      $p \leftarrow \sigma[i]$ 
6:      $T \leftarrow$  triangle of  $\mathcal{T}$  containing  $p$ , found by the visibility walk of Section 8.8
7:     split  $T$  into three triangles; flip illegal edges of the neighborhood (Section 12.6)
8:   end for
9:   remove triangles incident to the big triangle
10:  return  $\mathcal{T}$ 
11: end function

```

This is the version used in practice; the history-DAG version of Algorithm 46 (Section 12.6) replaces the walk by a guaranteed $O(\log n)$ search.

31.9.5 5. Parameter Selection

- **Walk vs. DAG:** the visibility walk (Section 8.8) is simpler and faster in practice but has no worst-case guarantee; use the DAG when a provable bound is needed.
- **Flip strategy:** the legalization pass of Section 12.4 repairs only the insertion neighborhood, keeping the per-insertion cost proportional to the created triangles.

31.9.6 6. Complexity

- **Time:** expected $O(n \log n)$ (Theorem 31.27); expected $O(n^{\lceil d/2 \rceil})$ in $\mathbb{R}^d$; worst case $O(n^2)$ without a DAG.
- **Space:** expected $O(n)$ (planar).

31.9.7 7. Usages

- The engine of Delaunay meshing (Chapter 22).
- A reference oracle for deterministic Delaunay algorithms (Section 12.6); the dual Voronoi diagram is obtained for free (Chapter 12).

31.9.8 8. Testing Methods and Edge Cases

- **Permutation tests:** the output must agree with a divide-and-conquer construction for every permutation.
- **Cocircularity and duplicates:** break cocircular ties consistently (Section 12.6) and skip coincident points with the exact predicate.

31.9.9 9. References

- Guibas, L., Knuth, D. E., Sharir, M. [GKS92].
- Clarkson, K. L. [Cla93], randomized closest-point and Voronoi constructions.
- Mulmuley, K. [Mul90]; de Berg, M., et al. [dBCvKO08], Sec. 9.4.

31.10 Applications to Linear Programming

31.10.1 1. Overview

Seidel's randomized algorithm [Sei90] solves a linear program in $\mathbb{R}^d$ with n constraints in expected $O(d!n)$ time, hence expected linear time in any fixed dimension. Its principle is the boundary reduction of Section 17.3: when a random constraint is violated by the current optimum, the new optimum must lie on the boundary of that constraint, reducing the dimension by one. The analysis is a canonical backward-analysis argument.

31.10.2 2. Definitions

Definition 31.28 (Linear Program). A **linear program** in $\mathbb{R}^d$ maximizes $c^\top x$ subject to a set H of n linear constraints $a_j^\top x \leq b_j$, each a half-space (Section 17.1 of Chapter 17). The **optimum** is the feasible point maximizing the objective.

31.10.3 3. Theory

Theorem 31.29 (Seidel's Algorithm). *For a linear program in $\mathbb{R}^d$ with n constraints in general position and a bounded feasible region, Seidel's randomized algorithm computes the optimum in expected $O(d!n)$ time; in particular, expected $O(n)$ time for $d = 2$ and every fixed d .*

Proof sketch. Let $E(n, d)$ be the expected time. The algorithm processes the constraints in random order, maintaining the optimum v of the prefix. With probability at most $d/(n - d + 1)$ the next constraint violates v (by backward analysis: the optimum of the remaining constraints is determined by at most d of them), giving the recurrence

$$E(n, d) \leq E(n - 1, d) + O(1) + \frac{d}{n} E(n - 1, d - 1),$$

whose solution is $E(n, d) = O(d!n)$. □

31.10.4 4. Pseudocode

31.10.5 5. Parameter Selection

- **Random order:** draw one permutation per recursion level; reusing it preserves the analysis.
- **Base case and predicates:** $d = 1$ is a sorted scan with infeasibility certificates; feasibility tests use the orientation predicates of Chapter 3.

31.10.6 6. Complexity

- **Time:** expected $O(d!n)$, i.e., $O(n)$ for fixed d ; worst case $O(nd)$.
- **Space:** $O(n)$.

Algorithm 158 Seidel's Randomized Linear Programming**Input:** constraints H in $\mathbb{R}^d$, objective c .**Output:** The optimum v , or a certificate of infeasibility/unboundedness.

```

1: function SEIDELLP( $H, d$ )
2:   if  $d = 1$  then
3:     return the extreme feasible point (a one-dimensional scan)
4:   end if
5:    $v \leftarrow$  a vertex of the feasible region of a constant-size subset of  $H$ 
6:    $\sigma \leftarrow$  uniformly random permutation of  $H$ 
7:   for  $i = 1, \dots, |H|$  do
8:      $h \leftarrow \sigma[i]$ 
9:     if  $v$  violates  $h$  then
10:                                      $\triangleright$  the optimum lies on the hyperplane of  $h$ 
11:        $v \leftarrow \text{SeidelLP}(\{h' \cap \partial h : h' \in H\}, d - 1)$ 
12:     end if
13:   end for
14:   return  $v$ 
15: end function

```

31.10.7 7. Usages

- Welzl's smallest-enclosing-circle algorithm [Wel91] is the LP-type specialization of the recursion, in expected $O(n)$ time.
- Support queries, largest empty circle, and collision separation in $\mathbb{R}^2, \mathbb{R}^3$ (Section 17.3); the implementation of choice up to dimension three, where Megiddo's [Meg83] has no simpler competitor.

31.10.8 8. Testing Methods and Edge Cases

- **Cross-validation:** compare the randomized optimum with Algorithm 73 (Chapter 17).
- **Infeasible, unbounded, and parallel constraints:** detect an empty intersection at dimension 1 and handle parallel half-planes in the base case.

31.10.9 9. References

- Seidel, R. [Sei90], the randomized LP and hull algorithm.
- Welzl, E. [Wel91], smallest enclosing disks.
- Megiddo, N. [Meg83], deterministic linear-time LP in $\mathbb{R}^3$.
- de Berg, M., et al. [dBCvKO08], Sec. 4.4.

31.11 Exercises

1. Define expected time, worst-case time, and average-case time, and give an input family on which the deterministic incremental Delaunay construction (Section 12.5) runs in quadratic time while its randomized version is expected linear.
2. Prove that the probability that a specific edge of the final Delaunay triangulation is created at step i of the randomized incremental construction is at most $3/i$ (Theorem 31.7), and use it to bound the expected number of triangles ever created by $O(n)$.

3. Derive the recurrence $E(n, d) \leq E(n-1, d) + O(1) + \frac{d}{n} E(n-1, d-1)$ for Seidel's linear-programming algorithm and solve it to obtain $E(n, d) = O(d! n)$.
4. Show that the number of k -sets of an n -point planar set is $O(nk)$ using the Clarkson–Shor sampling lemma (Theorem 31.15); identify where the planarity of the range space is used.
5. Prove that a uniform random sample of size $O(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon} + \frac{1}{\varepsilon} \log \frac{1}{\delta})$ of a planar point set is an ε -net for half-plane ranges with probability at least $1 - \delta$ (Theorem 31.23).
6. Explain why the conflict graph of a randomized incremental Delaunay construction has expected $O(n)$ edges, and describe how the graph is updated when a point inside the hull of the current points is inserted.
7. Contrast Las Vegas and Monte Carlo algorithms, and give one geometric algorithm of each type from this chapter; argue which class the randomized incremental algorithms belong to and why.
8. Describe how Chazelle–Friedman derandomization [CF90] replaces the random sample of the ε -net construction by a deterministic one, and state the price paid in constants and simplicity relative to Algorithm 155.

Chapter 32

Approximation Algorithms in Geometry

32.1 Why Approximation?

Many optimization problems that sound easy to state are computationally intractable. The traveling salesman problem asks for the shortest tour that visits every given point; the k -center problem asks for the k balls of smallest radius covering a set; k -median and k -means ask for k centers minimizing the sum of (squared) distances to the points they serve; and geometric covering and packing ask for the fewest shapes covering a set, or the largest number of non-overlapping shapes fitting in a container. Every one of these problems is NP-hard, and several remain NP-hard even for points in the plane, so exact polynomial-time algorithms are out of the question unless the celebrated conjecture $P \neq NP$ fails.

This chapter studies the alternative computational geometry offers: **approximation algorithms** that run in polynomial time and return solutions provably close to optimal. Two resources trade off against each other: the running time and the *approximation factor*, the worst-case ratio between the returned and the optimal value. At one end, a constant-factor approximation is cheap; at the other, a polynomial-time approximation scheme (PTAS) drives the ratio to $1 + \varepsilon$ for any fixed $\varepsilon > 0$ at a cost that grows with $1/\varepsilon$.

Geometry makes approximation easier than it is for arbitrary graphs, for three reasons. First, Euclidean distances satisfy the **triangle inequality**, which bounds the cost of shortcuts and local improvements. Second, **packing bounds** – a ball of radius R in $\mathbb{R}^d$ holds only boundedly many mutually separated points – keep quadtree-based structures under control; these power the Arora–Mitchell PTAS, approximate nearest neighbors, and well-separated pairs. Third, **small representatives exist**: a small sample or core-set often suffices for a near-optimal answer. The systematic theory is Har-Peled’s monograph [HP11]; the textbooks [dBCvKO08, CLRS09] cover the elementary material, the randomization tools are in Chapter 31, and the range-counting machinery in Chapter 9.

Sections 32.2–32.3 set up the vocabulary (approximation ratios, PTAS/FPTAS) and the sampling theorems (ε -nets and ε -approximations). Sections 32.4–32.8 build the data structures: core-sets, approximate nearest neighbors, locality-sensitive hashing, well-separated pairs, and spanners. Sections 32.9–32.11 apply them to clustering, k -center/ k -median, and minimum enclosing shapes, and Section 32.12 treats high dimensions. The chapter closes with exercises and a project on approximate nearest-neighbor search.

32.2 Geometric Approximation

Definition 32.1 (Approximation Ratio). An α -**approximation algorithm** for a minimization problem returns, on every instance I , a feasible solution with value $\text{val}(A(I)) \leq \alpha \cdot \text{OPT}(I)$, where $\text{OPT}(I)$ is the optimal value and $\alpha \geq 1$ is the **approximation ratio**. For maximization problems the inequality is reversed. An algorithm with ratio $1 + \varepsilon$ for every $\varepsilon > 0$ is a $(1 + \varepsilon)$ -**approximation**.

Definition 32.2 (PTAS and FPTAS). A problem admits a **polynomial-time approximation scheme** (PTAS) if for every fixed $\varepsilon > 0$ there is a polynomial-time algorithm computing a $(1 + \varepsilon)$ -approximation, the constant of the polynomial depending arbitrarily on ε . A **fully polynomial-time approximation scheme** (FPTAS) runs in time polynomial in both n and $1/\varepsilon$; a **randomized PTAS** succeeds with probability at least $1/2$ and is boosted by repetition. These are the standard definitions of [CLRS09].

Two facts bound what is possible. An FPTAS implies the problem is only weakly NP-hard; strongly NP-hard geometric problems therefore have PTASs at best, with running time exponential in $1/\varepsilon$. And some problems resist a PTAS entirely: k -center cannot be approximated within $2 - \delta$ unless $P = NP$ (Section 32.10), so the goal there is a tight constant factor.

Example 32.3 (Metric TSP). In a general metric, the Christofides algorithm (see [CLRS09]) finds a tour of length at most $3/2$ times optimal: it builds a minimum spanning tree, duplicates the odd-degree vertices to get an Eulerian multigraph, and shortcuts the Euler tour. The triangle inequality enters in the shortcut step. The Euclidean case is far better behaved: by Theorem 32.4 it has a PTAS.

The signature achievement of geometric approximation is the **quadtree-based PTAS for Euclidean TSP**, found independently by Arora [Aro98] and Mitchell [Mit99].

Theorem 32.4 (Arora’s PTAS for Euclidean TSP). *For every fixed integer $c \geq 1$, there is a randomized algorithm that computes a Euclidean TSP tour on n points whose length is within a factor $1 + 1/c$ of optimal, with probability at least $1/2$, in time $O(n(\log n)^{O(c)})$. Mitchell’s guillotine subdivision method gives a deterministic scheme in $n^{O(c)}$ time with a simpler proof.*

Proof sketch. The scheme has three steps. (i) **Quadtree dissection**: enclose the points in a square of side 2^L with $L = O(\log n)$ and split it into four equal quadrants until each cell holds at most one point. (ii) **Portals**: place $m = O(c \log n)$ equally spaced points on every cell edge and allow the tour to cross edges only at portals; a *patching lemma* – the triangle inequality plus a packing bound on how often a tour crosses a cell boundary – shows some portals-only tour is within $1 + O(1/c)$ of optimal. (iii) **Dynamic programming**: over the quadtree cells, keep for each cell and each portal pairing the best set of paths visiting all points inside the cell; with $4m$ portals per cell there are $2^{O(m)} = n^{O(c)}$ pairing patterns, giving total time $O(n(\log n)^{O(c)})$. Repeating over a random shift of the grid boosts the probability. The scheme extends to $\mathbb{R}^d$, other norms, and Steiner tree and k -TSP; see [Aro98, Mit99]. $\square$

Remark 32.5. The two ingredients – a hierarchical decomposition whose cells have bounded crossing number, and an exact program over it – recur throughout the chapter: they drive the core-set algorithms of Section 32.11, the well-separated pairs of Section 32.7, and the approximate nearest-neighbor structure of Section 32.5.

32.3 ε -Approximations

32.3.1 1. Overview

Range searching (Chapter 9) answers exact counting queries. ε -nets and ε -approximations approximate *all* such counts simultaneously with a single small random sample whose size

depends only on the range family, not on n . They are the probabilistic backbone of this chapter: core-sets (Section 32.4), randomized algorithms (Section 31.7), and partition-tree range structures all rest on the sampling theorems proved here.

32.3.2 2. Definitions

Definition 32.6 (Range Space). A **range space** is a pair $(X, \mathcal{R})$ of a finite ground set X and a family $\mathcal{R}$ of subsets (the **ranges**); for point sets in $\mathbb{R}^d$ the standard ranges are halfspaces, disks, rectangles, and simplices. The **VC dimension** $\text{VCdim}(\mathcal{R})$ is the size of the largest subset of X that is *shattered* by $\mathcal{R}$, i.e., whose every subset is realized as $R \cap Y$. Halfplanes in the plane have VC dimension 3; axis-parallel rectangles have VC dimension 4.

Definition 32.7 (ε -Net and ε -Approximation). A subset $N \subseteq X$ is an ε -**net** if every range R with $|R| \geq \varepsilon|X|$ contains a point of N . A subset A is an ε -**approximation** if for every $R \in \mathcal{R}$,

$$\left| \frac{|A \cap R|}{|A|} - \frac{|R|}{|X|} \right| \leq \varepsilon.$$

An ε -approximation is stronger than an ε -net: a range with an ε -fraction of the points is counted with positive empirical measure, hence hit. This is the notion introduced as Definition 9.7 in Chapter 9: nets serve *hitting* problems, approximations serve *counting*.

32.3.3 3. Theory

Theorem 32.8 (Haussler–Welzl). *Let $(X, \mathcal{R})$ have VC dimension d . A random sample of $m = O(\frac{d}{\varepsilon} \log \frac{1}{\varepsilon})$ points is an ε -net with probability at least $1/2$; an additive $O(\frac{1}{\varepsilon} \log \frac{1}{\delta})$ raises it to $1 - \delta$.*

The proof in [HW87] bounds the number of “essentially different” ranges by $O(m^d)$ via shattering, then applies a union bound: a range of size $\varepsilon|X|$ is missed with probability at most $(1 - \varepsilon)^m$.

Theorem 32.9 (ε -Approximations by Sampling). *For VC dimension d , a random sample of $m = O(\frac{d}{\varepsilon^2} \log \frac{d}{\varepsilon})$ is an ε -approximation with probability at least $1/2$. Halfspace ranges in $\mathbb{R}^d$ admit ε -approximations of size $O(\frac{d}{\varepsilon^2} \log \frac{1}{\varepsilon})$.*

The size is optimal up to logarithmic factors: variance forces $\Omega(1/\varepsilon^2)$, and the halfspace upper bounds follow from the discrepancy theory of [Mat95, Mat99]; a set of small discrepancy is exactly a good ε -approximation.

Remark 32.10 (Covering and Hitting). An ε -net is a **hitting set** for the ranges that are large relative to X . By duality this gives constant-factor approximations for many geometric covering problems in fixed dimension. Packing is different: geometric bin packing (Chapter 33, Section 33.2) is NP-hard (Theorem 33.12) yet admits approximation schemes of its own; see [dBCvKO08, HP11].

32.3.4 4. Pseudocode

Verifying the net property uses the range trees and partition trees of Chapter 9.

32.3.5 5. Parameter Selection

- ε : the sample grows like d/ε (nets) or d/ε^2 (approximations); use a net when only a witness is needed and an approximation when counts must be accurate.
- **VC dimension** d : intervals on a line have $d = 2$; halfplanes and disks in the plane $d = 3$; simplices in $\mathbb{R}^d$ grow linearly in d ([HW87, Mat99]).
- **Confidence**: repetition boosts the guarantee at logarithmic cost.

Algorithm 159 Random Sampling for ε -Nets**Input:** Range space $(X, \mathcal{R})$ of VC dimension d ; error ε ; confidence δ .**Output:** An ε -net N of X with probability at least $1 - \delta$.

```

1: function NETSAMPLE( $X, d, \varepsilon, \delta$ )
2:    $m \leftarrow \lceil \frac{c}{\varepsilon}(d \log \frac{1}{\varepsilon} + \log \frac{1}{\delta}) \rceil$ 
3:   draw  $m$  points of  $X$  independently and uniformly at random into  $N$ 
4:   for each range  $R$  with  $|R| \geq \varepsilon|X|$  do
5:     verify  $R \cap N \neq \emptyset$  ▷ range-counting step
6:   end for
7:   if some large range misses  $N$  then
8:     return NetSample( $X, d, \varepsilon, \delta$ ) ▷ resample
9:   end if
10:  return  $N$ 
11: end function

```

32.3.6 6. Complexity

- **Time:** $O(m)$ to draw the sample, plus a verification pass; with logarithmic range counting, $O(m \log^{O(1)} n)$ total.
- **Size:** independent of $n - O(\frac{d}{\varepsilon} \log \frac{1}{\varepsilon})$ and $O(\frac{d}{\varepsilon^2} \log \frac{d}{\varepsilon})$ respectively. This independence is what makes the rest of the chapter nearly input-size-free.

32.3.7 7. Usages

- **Range counting:** estimate any query count in constant time after sampling.
- **Hitting sets for covering:** Remark 32.10.
- **Discrepancy theory:** the deterministic counterpart ([Mat99]).
- **Core-sets:** sampling is the cheapest construction of the core-sets of Section 32.4.

32.3.8 8. Testing Methods and Edge Cases

- Verify the net property on adversarial ranges just above $\varepsilon|X|$ points, and at the exact threshold.
- Test duplicates (VC dimension can drop) and collinear configurations.
- For approximations, check the count error is at most ε on all tested ranges, not merely on average.

32.3.9 9. References

Haussler and Welzl's ε -net theorem [HW87], Matoušek's discrepancy constructions [Mat95, Mat99], and the textbook treatments [dBCvK008, HP11].

32.4 Coresets

A recurring theme is that a *small representative* of the input suffices for a near-optimal answer.

Definition 32.11 (Core-Set). Let f be a cost on point sets and $P \subset \mathbb{R}^d$ the input. A weighted subset $S \subseteq P$ (point s with weight $w(s) \geq 0$) is an ε -**core-set** for f if for every candidate solution C ,

$$(1 - \varepsilon) f_C(P) \leq f_C(S) \leq (1 + \varepsilon) f_C(P),$$

where f_C holds C fixed. Then any near-optimal solution on S is near-optimal on P .

Core-sets give a dimension-free speed-up: an exact algorithm – however slow – run on the m points of S yields a $(1 + \varepsilon)$ -approximation for P . The two constructions are random sampling (Section 32.3) and greedy farthest-point expansion; both produce sizes bounded in terms of ε and d but not n , which also enables streaming and distributed algorithms that merge core-sets.

Theorem 32.12 (Core-Sets for Minimum Enclosing Balls). *Let P have n points and minimum enclosing ball radius r^* . For every $\varepsilon \in (0, 1)$: (i) an ε -core-set of size $O(1/\sqrt{\varepsilon})$ exists, and this is tight [BC08]; (ii) the farthest-point procedure of Algorithm 165 finds a ball of radius $(1 + \varepsilon)r^*$ in time $O(dn/\varepsilon + \varepsilon^{-5})$ with a core-set of size at most $\lceil 2/\varepsilon \rceil$ [BC08]; (iii) Kumar, Mitchell, and Yildirim compute a $(1 + \varepsilon)$ -approximation in time linear in n and d for fixed ε , with core-sets of size $O(1/\varepsilon)$ [KMY03].*

For clustering, Har-Peled and Mazumdar [HPM04] showed k -means and k -median admit core-sets of size $O(k\varepsilon^{-d} \log n)$ and hence $(1 + \varepsilon)$ -approximations linear in n for fixed k, d, ε ; Section 32.10 develops the algorithms. The connection to randomization is direct: every core-set here is either a certified random sample (Theorem 32.9) or a greedy procedure that terminates after $O(1/\varepsilon)$ rounds (Algorithm 165). Section 32.12 returns to the striking fact that the ball core-set bound is *dimension-free*, which is why approximate minimum enclosing balls are tractable in high dimensions where the exact algorithms of Chapter 18.12 (Section 18.13) are not.

Remark 32.13. The word “coreset” also names the ε -approximation of a range space restricted to a query family (Definition 32.7); the distinction is only which cost is preserved.

32.5 Approximate Nearest Neighbors

32.5.1 1. Overview

The nearest-neighbor problem (Definition 13.7, Chapter 13) asks for the point of a preprocessed set P closest to a query q . Exact search is easy in low dimensions – the k d-tree of Chapter 10 (Section 10.1, Algorithm 14) answers queries in $O(\log n)$ expected time – but degrades exponentially with the dimension, and in high dimensions exact search suffers the curse of dimensionality (Section 39.9). **Approximate nearest neighbors** (ANN) relax the answer by a factor $c \geq 1$ and recover polynomial structure. This section develops the low-dimensional regime via balanced box decompositions [AMN⁺98]; Section 32.6 handles the high-dimensional regime.

32.5.2 2. Definitions

Definition 32.14 (c -Approximate Nearest Neighbor). For $c \geq 1$, a point $p' \in P$ is a c -**approximate nearest neighbor** of query q if $\|q - p'\| \leq c \min_{p \in P} \|q - p\|$. A structure supports c -**ANN queries** if it returns such a point for every query (cf. Section 39.4).

The relaxation makes search *output-sensitive*: the algorithm may stop once it can prove no unexplored point can beat the current answer by more than c , and cell-based pruning makes that proof cheap.

32.5.3 3. Theory

Theorem 32.15 (Balanced Box Decomposition). *The **balanced box-decomposition (BBD) tree** stores n points in $\mathbb{R}^d$ in $O(dn)$ space and $O(dn \log n)$ preprocessing time and answers c -ANN queries in $O(c^d \log n)$ time [AMN⁺98]; in fixed dimension the query time is $O(\log n)$, which is optimal in the worst case.*

The BBD tree is a balanced cousin of the compressed quadtree whose nodes are *boxes* (axis-parallel boxes with few extra dimensions), with the property that every node is either a grid cell or decomposes into boundedly many sub-boxes. Search (Algorithm 160) keeps a priority queue over boxes keyed by distance to the query and prunes a box once its distance exceeds the current answer divided by c ; packing bounds limit the number of expanded boxes to $O(c^d)$. This is the same dissection structure as Theorem 32.4, and it repairs the balance weakness of the kd -tree (Theorem 10.6).

32.5.4 4. Pseudocode

Algorithm 160 c -Approximate Nearest Neighbor via Box Decomposition

Input: A BBD tree $\mathcal{T}$ over P ; query point q ; factor $c \geq 1$.

Output: A point $p' \in P$ with $\|q - p'\| \leq c \min_{p \in P} \|q - p\|$.

```

1: function APPROXNN( $\mathcal{T}, q, c$ )
2:    $R \leftarrow +\infty$ ;  $p' \leftarrow$  any point of  $P$ 
3:   priority queue  $Q$  of boxes keyed by  $\text{dist}(q, \text{box})$ ; push root
4:   while  $Q$  is not empty do
5:     pop box  $B$  with minimal  $\text{dist}(q, B)$ 
6:     if  $\text{dist}(q, B) > R/c$  then
7:       break
8:     end if
9:     if  $B$  is a leaf then
10:      for each  $p \in B$  do
11:        if  $\|q - p\| < R$  then  $R \leftarrow \|q - p\|$ ;  $p' \leftarrow p$ 
12:      end if
13:    end for
14:    else
15:      push the children of  $B$  onto  $Q$ 
16:    end if
17:  end while
18:  return  $p'$ 
19: end function

```

The pruning test is where the approximation enters: a box at distance exceeding R/c cannot contain a point closer than R/c , good enough since R is the best found so far.

32.5.5 5. Parameter Selection

- c : query time grows like c^d ; choose the smallest factor the application tolerates ($c = 1$ gives no pruning and degenerates to exact search).
- **Box decomposition**: cap the extra dimensions per node and the leaf capacity to balance query and update cost.
- **Search order**: best-first by distance minimizes expanded boxes; depth-first uses less queue memory.

32.5.6 6. Complexity

- **Preprocessing:** $O(dn \log n)$ time, $O(dn)$ space.
- **Query:** $O(c^d \log n)$; the term c^d is why this is a *fixed-dimensional* structure – in high dimensions Section 32.6 is the better tool.

32.5.7 7. Usages

- **Registration and reconstruction:** closest-point queries feed the ICP-style matching of Chapter 29.
- **Robotics and graphics:** proximity queries and collision acceleration (Chapter 35).
- **Similarity search:** practical systems such as navigable small-world graphs [MY20] are heuristic cousins of the guarantees developed here.

32.5.8 8. Testing Methods and Edge Cases

- Compare against brute force on random sets and verify the ratio $\|q - p'\| / \|q - p^*\| \leq c$.
- Adversarial clusters (true neighbor just beyond the first expanded box) stress the pruning test; verify termination with exact predicates (Chapter 3).
- Queries far outside the data must trigger the break condition without scanning the tree.

32.5.9 9. References

The BBD tree and its optimal query time are due to Arya, Mount, Netanyahu, Silverman, and Wu [AMN⁺98]; the quadtree background is in Chapter 10 and the high-dimensional treatment in [HP11].

32.6 Locality-Sensitive Hashing

32.6.1 1. Overview

In high dimensions the BBD tree's $O(c^d \log n)$ query time of Theorem 32.15 is exponential in d . **Locality-sensitive hashing** (LSH) instead hashes points so that *near* points collide with high probability and *far* points with low probability, then answers queries by hashing and scanning colliding buckets. Indyk and Motwani's scheme [IM98] was the first to break the curse of dimensionality in the worst case with sublinear query time. The engine is the random-projection intuition of Chapter 39 (Section 39.6).

32.6.2 2. Definitions

Definition 32.16 ((r, cr, p_1, p_2) -Sensitive Family). A family $\mathcal{H}$ of hash functions from a metric space to a range U is (r, cr, p_1, p_2) -**sensitive** for $c > 1$, $p_1 > p_2$ if for all p, q :

$$d(p, q) \leq r \Rightarrow \Pr_h[h(p) = h(q)] \geq p_1, \quad d(p, q) \geq cr \Rightarrow \Pr_h[h(p) = h(q)] \leq p_2.$$

The ratio $\rho = \frac{\ln(1/p_1)}{\ln(1/p_2)}$ controls efficiency.

Single-coordinate projections give an $(r, cr, 1 - r/d, 1 - cr/d)$ -sensitive family on the Hamming cube; random hyperplanes serve Euclidean space.

32.6.3 3. Theory

Theorem 32.17 (Indyk–Motwani). *With $k = \log_{1/p_2} n$ and $L = n^\rho$, the structure of Algorithm 161 answers (c, r) -approximate near-neighbor queries in $O(n^\rho)$ expected time using $O(n^{1+\rho} + dn)$ space, with constant probability [IM98].*

The design amplifies: with k hashes composed, a far point collides with a given query in one table with probability at most $1/n$, so the union bound over the n points keeps expected far collisions at $O(1)$ per table; with $L = n^\rho$ independent tables, a near neighbor is missed with probability at most $(1 - p_1^k)^L \leq 1/2$. This mirrors the union-bound logic of Theorem 32.8, with a hash function in place of a range.

32.6.4 4. Pseudocode

Algorithm 161 Locality-Sensitive Hashing: Index and Query

Input: Data set P ; parameters k, L ; an (r, cr, p_1, p_2) -sensitive family $\mathcal{H}$.

Output: An index answering (c, r) -approximate near-neighbor queries.

```

1: function BUILDLSH( $P, k, L$ )
2:   for  $j = 1$  to  $L$  do
3:     pick  $g_j = (h_1, \dots, h_k)$ ,  $h_i \in \mathcal{H}$  independent
4:     build a hash table  $T_j$  storing  $g_j(p)$  for every  $p \in P$ 
5:   end for
6: end function
7: function QUERYLSH( $q$ )
8:    $S \leftarrow \emptyset$ 
9:   for  $j = 1$  to  $L$  do
10:     $S \leftarrow S \cup \{p \in P : g_j(p) = g_j(q)\}$ 
11:    if more than  $2L$  points collected then
12:      break
13:    end if
14:  end for
15:  return the point of  $S$  within distance  $cr$  of  $q$ , or null
16: end function

```

32.6.5 5. Parameter Selection

- $k = \log_{1/p_2} n$: balances far-point collisions (down with k) against near-point sensitivity (also down with k).
- $L = n^\rho$: gives constant success probability; a constant-factor increase boosts it to $1 - \delta$.
- **Radius r** : run the structure at geometrically growing radii $r, 2r, 4r, \dots$ to answer full ANN queries.

32.6.6 6. Complexity

- **Preprocessing**: $O(dnL)$ time to hash, $O(n^{1+\rho} + dn)$ space.
- **Query**: $O(dkL)$ to hash plus $O(n^\rho)$ expected scan – strictly sublinear in n for $\rho < 1$.

32.6.7 7. Usages

- **Similarity search:** near-duplicate detection and retrieval, and the shape matching of Chapter 29 (Section 29.9).
- **High-dimensional ANN:** with the dimension-reduction pipeline of Section 39.4.
- **Learning:** kernel pipelines where exact search is the bottleneck (Chapter 40).

32.6.8 8. Testing Methods and Edge Cases

- Measure empirical recall as a function of L and check it rises toward 1.
- Adversarial inputs (points clustered near the query radius) trigger the break condition; test the abort path.
- Test points exactly at radii r and cr ; the guarantee is probabilistic, so verify empirically.

32.6.9 9. References

Indyk and Motwani [IM98] introduced LSH and its analysis; the modern treatment and notation are in [HP11].

32.7 Well-Separated Pair Decomposition

32.7.1 1. Overview

n -body simulation, k -nearest-neighbor graphs, and spanner construction need information about all n^2 point pairs. Euclidean space admits a linear-size **well-separated pair decomposition** (WSPD): $O(n)$ pairs of subsets, each far apart relative to its size, that together cover every pair of points. Callahan and Kosaraju introduced the concept [CK95] and used it for k -nearest neighbors and n -body potential fields; the construction is quadtree-based.

32.7.2 2. Definitions

Definition 32.18 (Well-Separated Pair). Non-empty sets $A, B \subset \mathbb{R}^d$ are s -**well-separated** for $s > 2$ if

$$\max\{\text{diam}(A), \text{diam}(B)\} \leq s \cdot \text{dist}(A, B),$$

where $\text{dist}(A, B) = \min_{a \in A, b \in B} \|a - b\|$. A **well-separated pair decomposition** of P with parameter s is a set of s -well-separated pairs of subsets of P such that every unordered point pair of P lies in exactly one pair.

The contract: for any $a \in A, b \in B$, the distance $\|a - b\|$ is within $1 + 2/s$ of $\text{dist}(A, B)$, so smooth functions of the distances can be evaluated once per pair.

32.7.3 3. Theory

Theorem 32.19 (Size and Construction of WSPD). *For every $s > 2$ and every n -point set $P \subset \mathbb{R}^d$, a well-separated pair decomposition with parameter s containing $O(s^d n)$ pairs exists and is constructed in $O(s^d n \log n)$ time [CK95].*

The construction runs on the **split tree**, a compressed quadtree (Section 10.2, Theorem 10.16) whose nodes may split into children whose cells are pairwise separated. The recursion $\text{Wspd}(u, v)$ stops when the two node sets are s -well-separated and otherwise descends into the larger node. Separation forces each recursive thread to visit $O(1)$ nodes (packing bound), giving $O(s^d n)$ pairs.

32.7.4 4. Pseudocode

Algorithm 162 Well-Separated Pair Decomposition from a Split Tree

Input: A split tree for $P \subset \mathbb{R}^d$ with root T ; separation parameter $s > 2$.

Output: A set $\mathcal{D}$ of s -well-separated pairs covering all pairs of P .

```

1: function WSPD( $u, v$ )
2:   if  $u, v$  are  $s$ -well-separated then
3:     add  $(P_u, P_v)$  to  $\mathcal{D}$ 
4:   else if  $u$  and  $v$  are both leaves then
5:     add  $(P_u, P_v)$  to  $\mathcal{D}$ 
6:   else if  $\text{radius}(u) \geq \text{radius}(v)$  then
7:     for each child  $u'$  of  $u$  do
8:       Wspd( $u', v$ )
9:     end for
10:  else
11:    for each child  $v'$  of  $v$  do
12:      Wspd( $u, v'$ )
13:    end for
14:  end if
15: end function
16:  $\mathcal{D} \leftarrow \emptyset$ 
17: for each pair  $(u, v)$  of children of the root do
18:   Wspd( $u, v$ )
19: end for
20: return  $\mathcal{D}$ 

```

32.7.5 5. Parameter Selection

- s : larger s improves the approximation ($2/s$ error) but increases the pair count ($O(s^d n)$); $s = 4$ is the canonical choice.
- **Granularity**: single-point leaves keep the recursion shallow; skip long chains of degree-one cells.

32.7.6 6. Complexity

- **Time**: $O(s^d n \log n)$; **size**: $O(s^d n)$ pairs – linear in n for constant s, d , replacing the quadratic all-pairs scan.

32.7.7 7. Usages

- **n -body problems**: evaluate potentials per well-separated pair [CK95].
- **k -nearest neighbors**: report approximate nearest pairs (cf. Section 13.1).
- **Spanners**: the application of Section 32.8.

32.7.8 8. Testing Methods and Edge Cases

- Verify that every point pair appears in exactly one pair of $\mathcal{D}$, and spot-check the separation inequality.
- Clustered and collinear inputs stress the separation test; verify termination.

32.7.9 9. References

Callahan and Kosaraju [CK95] introduced the WSPD; the quadtree machinery is in Chapter 10 and the spanner applications in [HP11].

32.8 Approximate Distance Queries

Definition 32.20 (Geometric Spanner). A graph $G = (P, E)$ is a **t -spanner** of point set P if for all $p, q \in P$ the shortest-path length satisfies $d_G(p, q) \leq t\|p - q\|$; $t \geq 1$ is the **stretch factor**. A spanner replaces the complete Euclidean graph (quadratic) with $O(n)$ edges that preserve all distances up to t .

Spanners are the data structure for **approximate distance queries**: preprocess the points once, then answer any pairwise distance by a shortest-path computation on the small graph.

Theorem 32.21 (WSPD Spanner). *Connecting each pair of a well-separated pair decomposition with parameter $s \geq 4$ by one edge between arbitrary representatives yields a graph with $O(s^d n)$ edges that is a t -spanner for $t = \frac{s+4}{s-4}$ ($t \leq 3$ for $s = 8$) [CK95, HP11].*

Proof sketch. For a point pair (p, q) covered by the well-separated pair (A_i, B_i) with representatives a_i, b_i , s -separation bounds $\|p - a_i\|$, $\|a_i - b_i\|$, and $\|b_i - q\|$ each by $O(1/s)\|p - q\|$, so the path $p \rightarrow a_i \rightarrow b_i \rightarrow q$ has length at most $\frac{s+4}{s-4}\|p - q\|$; the triangle inequality is used once, which is why the constant is explicit. $\square$

Once a t -spanner is available, approximate shortest paths follow from Dijkstra's algorithm (Chapter 24.11, Section 24.14) in $O(n \log n)$ per query, or from an *approximate distance oracle* answering in $O(1)$. Refinements require *bounded-degree* and *light* spanners (degree and total edge weight constant times the MST weight) while keeping constant stretch; the classical theory is surveyed in [HP11].

32.9 Geometric Clustering

Clustering partitions a point set into k groups with similar members. The geometric setting makes similarity a distance objective, and the objective determines both the algorithm and its guarantee. The three classical objectives are:

1. **k -center**: cover P by k balls of minimum radius;
2. **k -median**: minimize the sum of point-to-nearest-center distances;
3. **k -means**: minimize the sum of *squared* point-to-center distances.

All three are NP-hard, even in the plane [Gon85], and all three are workhorses of data analysis, so approximation is essential. Formal definitions and algorithms are in Section 32.10.

The objectives differ in their treatment of outliers and cluster sizes. k -center is the most conservative: a single far point sets the radius, so it is robust to cluster shape but sensitive to noise. k -median averages, so a small well-separated cluster hardly moves the optimum. k -means squares, amplifying large clusters and aligning with variance-based models; it is the objective of Lloyd's algorithm (Section 32.10) that dominates practice (Chapter 40, Section 40.4).

The algorithmic landscape mirrors the objectives. For k -center a greedy farthest-point rule achieves the tight factor 2 (Theorem 32.24). For k -median and k -means, *Lloyd-style* iteration converges to a local optimum, and core-sets (Section 32.4) reduce the input to $O(k\varepsilon^{-d} \log n)$ points, yielding $(1 + \varepsilon)$ -approximations linear in n for fixed k, d, ε [HPM04]. Clustering is thus the cleanest demonstration that the three tools of this chapter – sampling, core-sets, and greedy/local search – compose into end-to-end guarantees. The number k of clusters is part of the input; the guarantees hold for every k , and the application chooses it.

32.10 k -Center and k -Median Problems

32.10.1 1. Overview

This section develops approximations for the objectives of Section 32.9. The centerpiece is the farthest-point algorithm of Gonzalez [Gon85], which achieves the optimal factor 2 for k -center. For k -median and k -means, a constant factor below that style of guarantee is impossible, but Lloyd-style iteration [Llo82] plus core-sets deliver $(1 + \varepsilon)$ -approximations in fixed dimension [HPM04].

32.10.2 2. Definitions

Definition 32.22 (k -Center). Given $P \subset \mathbb{R}^d$ and k , find k centers C minimizing the **radius** $\max_{p \in P} \text{dist}(p, C)$, where $\text{dist}(p, C) = \min_{c \in C} \|p - c\|$; the optimum is r^* .

Definition 32.23 (k -Median and k -Means). The **k -median** problem minimizes $\sum_{p \in P} \text{dist}(p, C)$ over all k -point center sets C ; **k -means** minimizes $\sum_{p \in P} \text{dist}(p, C)^2$. Both are NP-hard [Gon85].

32.10.3 3. Theory

The farthest-point rule picks the first center arbitrarily and each further center as the point of P farthest from the current centers.

Theorem 32.24 (Gonzalez's 2-Approximation). *The farthest-point algorithm (Algorithm 163) computes k centers of radius at most $2r^*$ in $O(nk)$ time. No polynomial-time algorithm achieves ratio below 2 unless $P = NP$ [Gon85].*

Proof. If the returned radius exceeded $2r^*$, the farthest point would be at distance more than $2r^*$ from all chosen centers, so by induction the algorithm would have found $k + 1$ points with mutual distances exceeding $2r^*$. But the optimal solution covers P with k balls of radius r^* , each of diameter at most $2r^*$ by the triangle inequality, so two such points cannot share a ball: they need $k + 1$ balls, a contradiction. The lower bound reduces planar dominating set, which is NP-complete; see [Gon85]. $\square$

For k -median and k -means, Lloyd's algorithm (Algorithm 164) alternates assigning points to the nearest center and moving each center to the centroid of its assigned points.

Theorem 32.25 (Monotonicity of Lloyd's Algorithm). *Every iteration of Lloyd's algorithm (Algorithm 164) does not increase the k -means objective, and the center sequence converges to a fixed point that can be arbitrarily worse than the global optimum [Llo82].*

Core-sets upgrade the heuristic into an approximation: reducing to $O(k\varepsilon^{-d} \log n)$ weighted points and searching over candidate centers gives a $(1 + \varepsilon)$ -approximation in time linear in n for fixed k, d, ε [HPM04]. In practice the best of both worlds is achieved by seeding Lloyd's algorithm with the Gonzalez centers.

32.10.4 4. Pseudocode

32.10.5 5. Parameter Selection

- k : fixed by the application (Section 32.9); the guarantees hold for all k .
- **Seeding**: Gonzalez centers make good seeds for Algorithm 164; randomized seeding rules reduce the dependence on the first arbitrary center.
- **Termination**: Lloyd can cycle in degenerate cases; fix a tolerance and a maximum iteration count.

Algorithm 163 Farthest-Point k -Center (Gonzalez)

Input: Point set $P \subset \mathbb{R}^d$; integer $k \geq 1$.**Output:** A set C of at most k centers with radius at most $2r^*$.

```
1: function GONZALEZKCENTER( $P, k$ )
2:   pick arbitrary  $p_0 \in P$ ;  $C \leftarrow \{p_0\}$ 
3:   for  $i = 1$  to  $k - 1$  do
4:     let  $p$  be a point of  $P$  maximizing  $\min_{c \in C} \|p - c\|$ 
5:      $C \leftarrow C \cup \{p\}$ 
6:   end for
7:   assign each  $p \in P$  to its nearest center in  $C$ 
8:   return  $C$ 
9: end function
```

Algorithm 164 Lloyd's Algorithm (k -Means Heuristic)

Input: Point set $P \subset \mathbb{R}^d$; integer $k \geq 1$; initial centers C .**Output:** A locally optimal set of k centers for the k -means objective.

```
1: function LLOYD( $P, C$ )
2:   repeat
3:      $C' \leftarrow C$ 
4:     for each  $p \in P$  do assign  $p$  to the nearest center of  $C'$ 
5:   end for
6:   for each non-empty cluster  $P_i$  do  $c_i \leftarrow \frac{1}{|P_i|} \sum_{p \in P_i} p$ 
7:   end for
8:   until no center changes
9:   return  $C$ 
10: end function
```

32.10.6 6. Complexity

- **Gonzalez:** $O(nk)$ time, $O(n)$ space.
- **Lloyd:** $O(nkt)$ for t iterations; t is not polynomially bounded in the worst case but is small in practice.
- **Core-set schemes:** $O(n)$ time plus a term exponential in k , d , and $1/\varepsilon$ [HPM04].

32.10.7 7. Usages

- **Facility location and quantization:** k -center models emergency-response placement; k -means is the Lloyd–Max quantizer of signal processing [Llo82].
- **Data analysis:** the default clustering engine of Chapter 40 (Section 40.4).
- **Preprocessing:** Gonzalez centers double as metric nets and coarse summaries in many pipelines.

32.10.8 8. Testing Methods and Edge Cases

- Compare the Gonzalez radius with the optimum (exact for small n) and verify the factor 2.
- Lloyd can empty clusters; fix a re-seeding convention. Test $k \geq n$ and coincident points as defined cases.
- Verify monotone objective decrease at every Lloyd iteration (Theorem 32.25).

32.10.9 9. References

Gonzalez’s farthest-point algorithm [Gon85], Lloyd’s quantization algorithm [Llo82], and the core-set based $(1 + \varepsilon)$ -approximations of Har-Peled and Mazumdar [HPM04]; the textbook treatment is [dBCvKO08, HP11].

32.11 Minimum Enclosing Shapes

Definition 32.26 (Minimum Enclosing Ball). The **minimum enclosing ball** (MEB) of $P \subset \mathbb{R}^d$ is the ball of minimum radius containing P ; in the plane it is the smallest enclosing circle (SEC) of Definition 18.10 (Chapter 18.12). Its radius is the 1-center value r^* .

The MEB is the $k = 1$ case of k -center (Definition 32.22), and it is exceptional: it can be solved exactly in expected linear time by Welzl’s randomized incremental algorithm (Chapter 18.12, Section 18.13, Algorithm 18.13.4, Theorem 18.12) [Wel91, Meg83]. The catch is dimension: the exact algorithm’s constant grows exponentially in d (the ball has up to $d + 1$ support points), so beyond a few dozen dimensions the problem is solved *approximately* by the core-sets of Section 32.4, whose size is independent of n and d .

The loop is the farthest-point expansion of Theorem 32.12: each round adds the point that best witnesses the current error and recomputes the exact ball on the small sample (cheap, since $|S| \leq \lceil 2/\varepsilon \rceil$), terminating once the inflated sample ball contains P . Termination is geometric: while the loop continues, every added point grows the sample ball’s radius by a constant factor, so only $O(1/\varepsilon)$ points are ever added. Badoiu and Clarkson proved the size and time bounds [BC08]; Kumar, Mitchell, and Yildirim sharpened the running time to linear in n and d for fixed ε [KMY03].

Algorithm 165 Core-Set Based $(1 + \varepsilon)$ -Approximate MEB**Input:** Point set $P \subset \mathbb{R}^d$; error $\varepsilon > 0$.**Output:** Center c and radius r with $P \subseteq B(c, (1 + \varepsilon)r)$ and $r \leq (1 + \varepsilon)r^*$.

```

1: function MEBCORESET( $P, \varepsilon$ )
2:   pick arbitrary  $c_1 \in P$ ;  $S \leftarrow \emptyset$ ;  $t \leftarrow 1$ 
3:   while  $P \not\subseteq B(c_t, (1 + \varepsilon)r_t)$  do
4:     let  $p_t$  be the point of  $P$  farthest from  $c_t$ ;  $r_t \leftarrow \|p_t - c_t\|$ 
5:      $S \leftarrow S \cup \{p_t\}$ 
6:      $(c_{t+1}, r_{t+1}) \leftarrow$  exact MEB of  $S$  ▷ via Welzl;  $|S| \leq 2/\varepsilon$ 
7:      $t \leftarrow t + 1$ 
8:   end while
9:   return  $(c_t, r_t)$ 
10: end function

```

Remark 32.27. The algorithm exploits the support-point theorem behind Welzl’s exact algorithm: the optimal ball is determined by at most $d + 1$ points on its boundary. Approximating the MEB is thus a case study in how an exact structural theorem converts into an approximation algorithm with mild loss.

32.12 High-Dimensional Problems

In high dimensions the low-dimensional structures of this chapter fail in the worst case: quadtree methods carry factors c^d or s^d , and the BBD tree’s $O(c^d \log n)$ query time is exponential in d . The curse of dimensionality (Section 39.9) is real. Three ideas survive it, and they are the three pillars built here:

1. **Dimension reduction:** the Johnson–Lindenstrauss lemma (Chapter 39, Section 39.6) embeds n points into $O(\log n)$ dimensions while preserving distances within $1 + \varepsilon$, and its random projections are the same ones that power LSH (Section 32.6).
2. **Core-sets:** the MEB admits core-sets whose size is independent of d (Theorem 32.12), so Section 32.11 runs in time linear in d . The solution can be summarized without the dimension, even though no space decomposition can tame $\mathbb{R}^d$; the discrepancy bounds of [Mat99] supply the matching sampling theory.
3. **Sublinear query structures:** LSH answers approximate nearest-neighbor queries in time n^ρ for $\rho < 1$ independent of d , and the pipeline of Section 39.5 feeds reduced data into Section 32.5; Section 39.4 surveys the interplay.

The lesson is a hierarchy. Problems with geometric structure (balls, halfspaces) have dimension-free core-set guarantees; problems with metric structure (nearest neighbors, distance queries) have sublinear-time solutions via projections and hashing; and purely combinatorial problems (covering, packing; Chapter 33) let the dimension enter the approximation ratio itself. The practical high-dimensional systems – the navigable small-world graphs of [MY20] – blend these ideas with engineering and remain heuristic; the theory of this chapter is the vocabulary in which their guarantees are stated.

32.13 Exercises

1. Prove Theorem 32.24 in full: show the chosen k centers have pairwise distances exceeding the returned radius if it exceeds $2r^*$, and conclude by the triangle inequality that the optimal balls cannot separate them.

2. Argue that no polynomial-time algorithm approximates k -center within a factor below 2: show a $(2 - \delta)$ -approximation would decide the planar dominating-set problem exactly (see [Gon85]).
3. Exhibit a range space (points on a line, intervals as ranges) where an ε -net of size $\Omega(\frac{1}{\varepsilon} \log \frac{1}{\varepsilon})$ is necessary, and describe the worst-case configuration.
4. Show that an ε -approximation is an ε -net: prove that if A is an ε -approximation of $(X, \mathcal{R})$, then every range with $|R| \geq \varepsilon|X|$ contains a point of A .
5. For Algorithm 165, prove that while the loop continues, the sample radius grows by a constant factor per round, and conclude $|S| \leq \lceil 2/\varepsilon \rceil$ (the bound of Theorem 32.12).
6. Derive the stretch bound of Theorem 32.21: for an s -well-separated pair with $s \geq 4$, bound $\|p - a_i\|$, $\|a_i - b_i\|$, and $\|b_i - q\|$ in terms of $\|p - q\|$ and assemble the path $p \rightarrow a_i \rightarrow b_i \rightarrow q$ to get $(s + 4)/(s - 4)$.
7. Implement Algorithm 160 and Algorithm 161 on the same random point sets; plot the empirical approximation factor and query time against $d = 2, 4, 8, 16$ and identify where LSH overtakes the tree.
8. Use a random sample of size $m = O(1/\varepsilon^2)$ (Theorem 32.9) to estimate the fraction of n points in the unit square inside a fixed disk; compare the empirical error with the bound of Definition 32.7 over 1000 trials.

32.14 Project: Approximate Nearest-Neighbor Search

Build a practical approximate nearest-neighbor search system and compare the guarantees of this chapter with engineering reality.

1. **Input:** generate random point sets (Chapter 27, Section 27.3) in dimensions $d = 2, 4, 8, 16$ with $n = 10^4$ to 10^6 , and load a real feature-vector data set if available.
2. **Exact baseline:** implement brute-force nearest neighbors to compute ground truth for 100 queries; this is the yardstick for ratio and recall.
3. **BBD tree:** implement the balanced box decomposition and the c -approximate query of Algorithm 160; verify the ratio bound on every query.
4. **LSH:** implement Algorithm 161 with random hyperplane projections; tune k and L and record the recall-versus-query-time curve.
5. **Comparison:** for each dimension report average approximation factor, 95th percentile, recall, and query time for both structures, and locate the crossover dimension.
6. **Engineering:** compare against a practical library (e.g., the navigable small-world graphs of [MY20]) on the high-dimensional data set.
7. **Verification:** plot the empirical tradeoff against Theorems 32.15 and 32.17, and explain discrepancies in terms of c , k , L , and bucket size.

The project exercises the sampling, space-decomposition, and hashing machinery of this chapter and of Chapters 10 and 39, and it surfaces the gap between worst-case guarantees and practical behavior that motivates the field.

Chapter 33

Packing Algorithms

33.1 Circle Packing

33.1.1 Overview

Circle packing is the study of placing non-overlapping circles within a given container (like a square or another circle) or on a surface (like a mesh) such that the total area of the circles is maximized or the density is optimized. In computational geometry, it is a classic problem in both optimization and discrete geometry, with applications ranging from material science to map labeling and data visualization. The theory of circle packing was developed extensively by Stephenson [Ste05], building on the foundational work of Koebe [Koe36] and the influential exposition of Thurston [Thu85].

33.1.2 Definitions

Definition 33.1 (Packing Density). The **packing density** of a configuration of n circles $\{C_1, \dots, C_n\}$ in a container Ω is the ratio

$$\delta = \frac{\sum_{i=1}^n \text{Area}(C_i)}{\text{Area}(\Omega)},$$

where the areas of C_i are the portions lying inside Ω .

Definition 33.2 (Optimal Packing). An **optimal packing** is a configuration that achieves the maximum possible density for a given number of circles or a given container size. For n congruent circles in the plane, the problem of determining the optimal arrangement is open for most values of $n > 7$.

Definition 33.3 (Circle Contact Graph). The **circle contact graph** of a packing is a graph $G = (V, E)$ where each node $v_i \in V$ corresponds to a circle C_i and an edge $(v_i, v_j) \in E$ exists if and only if circles C_i and C_j are tangent (touching but not overlapping).

Definition 33.4 (Hexagonal Packing). **Hexagonal packing** is the densest possible packing of congruent circles in an infinite 2D plane, achieving a density of $\delta = \frac{\pi}{2\sqrt{3}} \approx 0.9069$ (approximately 90.69%).

33.1.3 Theory

Finding the optimal packing of n circles in a container is a **non-convex optimization** problem. The most common approaches are:

1. **Greedy Placement**: Placing circles one by one at the best available position (e.g., using a “front-advancing” algorithm).

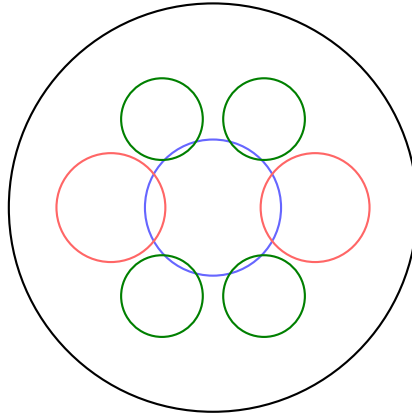
2. **Force-Directed Layout:** Treating circles as repelling particles and using a physics simulation to find an equilibrium state where they are packed tightly.
3. **Circle Packing on Surfaces:** Using **conformal mapping** and the Circle Packing Theorem to represent any planar graph as a set of tangent circles.
4. **Vortex-based Packing:** Packing circles into a spiral or vortex to fill a circular container.

For identical circles, the hexagonal lattice is optimal in 2D. For circles of different sizes (Apollonian gaskets), the packing can achieve 100% density in the limit.

Theorem 33.5 (Circle Packing Theorem). (*Koebe–Andreev–Thurston*) *Every connected planar graph G can be represented as the tangency graph of a circle packing in the plane (or on the sphere), and such a representation is unique up to Möbius transformations.*

Proof. The proof proceeds in two stages. First, Andreev’s theorem provides existence for triangulations using a discrete analogue of the Riemann mapping theorem. Second, Koebe’s original result [Koe36] establishes that every planar graph is a subgraph of a triangulation. Thurston [Thu85] gave an influential expository account combining these ideas with conformal mapping theory. See Stephenson [Ste05, Chapter 3] for a self-contained modern treatment. $\square$

Corollary 33.6. *The hexagonal lattice packing of congruent circles in $\mathbb{R}^2$ is optimal among all packings of congruent circles, with density $\frac{\pi}{2\sqrt{3}}$.*



Non-overlapping disks, maximally packed

Figure 33.1: Circle packing: placing non-overlapping disks of varying radii inside a bounding region.

Example 33.7 (Greedy Packing Trace). Consider packing circles of radii $r_1 = 1.0$, $r_2 = 0.8$, $r_3 = 0.6$ into a 4×3 rectangle, sorted in decreasing order:

1. Place C_1 at $(1.0, 1.0)$ —the first position that fits within the container boundary.
2. Place C_2 at $(2.8, 1.0)$ —the leftmost position with y -coordinate equal to the lowest feasible row, and $\text{dist}((2.8, 1.0), (1.0, 1.0)) = 1.8 = 1.0 + 0.8$ (tangent to C_1).
3. Place C_3 at $(1.0, 2.6)$ —directly above C_1 , with $\text{dist} = 1.6 = 1.0 + 0.6$ (tangent to C_1), and $\text{dist}((1.0, 2.6), (2.8, 1.0)) = 2.0 > 0.8 + 0.6$ (non-overlapping with C_2).

The final packing density is $\frac{\pi(1.0^2+0.8^2+0.6^2)}{12} = \frac{2.0\pi}{12} \approx 52.4\%$.

33.1.4 Pseudocode

Simple Greedy Packing (into a rectangle)

```

1: function GREEDYCIRCLEPACKING(container, radii)
2:   radii.sort(reverse = True)
3:   placed_circles  $\leftarrow$  []
4:   for  $r$  in radii do
5:     best_p  $\leftarrow$  FindCandidatePosition(container, placed_circles,  $r$ )
6:     if best_p  $\neq$  None then
7:       placed_circles.append(Circle(best_p,  $r$ ))
8:     end if
9:   end for
10:  return placed_circles
11: end function
12: function FINDCANDIDATEPOSITION(container, others,  $r$ )
13:  for  $p$  in SamplePoints(container) do
14:    if  $\neg$  IsWithin( $p, r$ , container) then
15:      continue
16:    end if
17:    is_overlapping  $\leftarrow$  False
18:    for  $c$  in others do
19:      if Distance( $p, c.p$ )  $< (r + c.r)$  then
20:        is_overlapping  $\leftarrow$  True
21:        break
22:      end if
23:    end for
24:    if  $\neg$  is_overlapping then
25:      return  $p$ 
26:    end if
27:  end for
28:  return None
29: end function

```

33.1.5 Parameter Selection

- **Radii Distribution:** Can be fixed (monodisperse) or varying (polydisperse).
- **Optimization Iterations:** In force-directed methods, more iterations lead to tighter packing but increase computation time.

33.1.6 Complexity

- **Greedy Approach:** $O(n^2)$ if each placement is checked against all previous circles. This can be reduced to $O(n \log n)$ using a spatial index like a quadtree.
- **Force-Directed:** $O(i \cdot n \log n)$ where i is the number of iterations and n is the number of circles.

33.1.7 Usages

- **Manufacturing:** Minimizing waste when cutting circular parts from a rectangular sheet of material.

- **Cartography:** Dorling Cartograms, where regions are represented by circles whose size is proportional to a variable (e.g., population).
- **Data Visualization:** Packing circles into categories to represent hierarchy (e.g., D3.js circle packing).
- **Material Science:** Modeling the structure of crystals or granular materials.
- **Logo Design and Art:** Generative art and “bubble” patterns.

33.1.8 Testing Methods and Edge Cases

- **Overlap Check:** Verify that no two circles have an intersection.
- **Boundary Check:** Ensure all circles are fully within the container.
- **Density Calculation:** Measure the final packing efficiency.
- **Single Circle:** Should be placed in the center of the container.
- **Radius > Container:** Should result in zero placed circles.
- **Varying Container Shapes:** Test packing into circles, triangles, or complex polygons.

33.1.9 References

- Stephenson, K. (2005). “Introduction to Circle Packing: The Theory of Discrete Analytic Functions”. Cambridge University Press.
- Koebe, P. (1936). “Kontaktprobleme der Konformen Abbildung”. *Berichte Sächs. Akad. Wiss. Leipzig*.
- Thurston, W. (1985). “The Finite Riemann Mapping Theorem”. *International Congress of Mathematicians*.
- Wikipedia: [Circle packing](#)

33.2 Rectangle Packing

33.2.1 Overview

Rectangle packing is the problem of arranging a set of rectangles of varying sizes within a larger container rectangle such that no two rectangles overlap and the total wasted space is minimized (or the density is maximized). This is a classic NP-hard optimization problem with wide-ranging applications in manufacturing, logistics, and computer graphics. Comprehensive surveys of 2D packing methods are given by Jylänki [Jyl10] and Lodi, Martello, and Monaci [LMM02].

33.2.2 Definitions

Definition 33.8 (Bin Packing). The **bin packing** problem asks: given a set of objects with prescribed sizes and a set of bins of fixed capacity, pack all objects into the minimum number of bins such that no two objects in the same bin overlap. The 2D version is strongly NP-hard.

Definition 33.9 (Strip Packing). The **strip packing** problem asks: given a set of rectangles and a container of fixed width W and unbounded height, arrange the rectangles (without rotation or with 90° rotation) to minimize the total height used. The decision version is NP-complete.

Definition 33.10 (Guillotine Cut). A **guillotine cut** is a straight-line cut that goes entirely from one edge of a rectangular piece to the opposite edge, dividing it into two smaller rectangles. A packing is *guillotine-cuttable* if it can be obtained by a sequence of such cuts.

Definition 33.11 (Skyline). A **skyline** representation tracks the upper boundary of the packed rectangles as a piecewise-constant function of x . New rectangles are placed at the lowest point on the skyline, and the skyline is updated accordingly.

33.2.3 Theory

Because rectangle packing is NP-hard, heuristic and metaheuristic algorithms are used in practice.

1. **Shelf Algorithms:** Rectangles are placed in “shelves” (rows). When a rectangle doesn’t fit in the current shelf, a new shelf is started above it. (e.g., Next-Fit Decreasing Height).
2. **Guillotine Algorithms:** The container is recursively split into two smaller rectangles using horizontal or vertical cuts.
3. **MaxRects:** Maintains a list of all maximal free rectangular areas. A new rectangle is placed into one of these areas, and the remaining free spaces are recalculated.
4. **Skyline Algorithms:** Tracks the top boundary of the packed rectangles. New rectangles are placed on the lowest part of the skyline.

Sorting the rectangles by height or area before packing (e.g., First-Fit Decreasing) significantly improves the performance of most heuristics.

Theorem 33.12 (NP-Hardness of Rectangle Packing). *The 2D bin packing problem and the 2D strip packing problem are both NP-hard in the strong sense.*

Proof. The NP-hardness of 2D bin packing follows by reduction from the 3-Partition problem, which is strongly NP-complete. Given a multiset $S = \{a_1, \dots, a_{3m}\}$ of integers with $\frac{B}{4} < a_i < \frac{B}{2}$ and $\sum a_i = mB$, construct $3m$ rectangles of sizes $a_i \times 1$ to be packed into m bins of size $B \times 1$. The constraint $\frac{B}{4} < a_i < \frac{B}{2}$ ensures exactly three rectangles must fit per bin, reducing 3-Partition to bin packing; see [LMM02, Section 8]. $\square$

Example 33.13 (MaxRects Trace). Consider packing rectangles $A = 3 \times 2$, $B = 2 \times 2$, $C = 1 \times 1$ into a 5×4 bin using Best Short Side Fit (BSSF):

1. Free list: $\{(0, 0, 5, 4)\}$. Sort: $A(3 \times 2)$, $B(2 \times 2)$, $C(1 \times 1)$ by decreasing height.
2. Place A at $(0, 0)$. Short-side fit: $\min(5 - 3, 4 - 2) = 2$. Split free rect $(0, 0, 5, 4)$:
 - Right remainder: $(3, 0, 2, 4)$
 - Top remainder: $(0, 2, 3, 2)$
3. Place B at $(3, 0)$ in free rect $(3, 0, 2, 4)$. Fit: $\min(0, 2) = 0$. Split:
 - Top remainder: $(3, 2, 2, 2)$
4. Place C at $(0, 2)$ in free rect $(0, 2, 3, 2)$. Fit: $\min(2, 1) = 1$.

Total density: $\frac{6+4+1}{20} = 55\%$.

```
1: function MAXRECTS(bin_width, bin_height, rectangles)
2:   free_rects  $\leftarrow$  [Rectangle(0, 0, bin_width, bin_height)]
3:   placed_rects  $\leftarrow$  []
4:   rectangles.sort(key =  $\lambda r$ :  $r.h$ , reverse = True)
5:   for  $r$  in rectangles do
6:     best_fit  $\leftarrow$  None
7:     min_short_side_fit  $\leftarrow$   $\infty$ 
8:     for  $f$  in free_rects do
9:       if  $f.w \geq r.w$  and  $f.h \geq r.h$  then
10:        fit  $\leftarrow$  min( $f.w - r.w$ ,  $f.h - r.h$ )
11:        if fit < min_short_side_fit then
12:          min_short_side_fit  $\leftarrow$  fit
13:          best_fit  $\leftarrow$   $f$ 
14:        end if
15:      end if
16:    end for
17:    if best_fit  $\neq$  None then
18:       $r.x, r.y \leftarrow$  best_fit.x, best_fit.y
19:      placed_rects.append( $r$ )
20:      new_free  $\leftarrow$  []
21:      for  $f$  in free_rects do
22:        new_free.extend(SplitFreeRect( $f, r$ ))
23:      end for
24:      free_rects  $\leftarrow$  PruneRedundant(new_free)
25:    end if
26:  end for
27:  return placed_rects
28: end function
```

33.2.4 Pseudocode

MaxRects (BSSF — Best Short Side Fit)

33.2.5 Parameter Selection

- **Rotation:** Allowing 90° rotation can significantly increase packing density.
- **Heuristic Choice:** **MaxRects** is generally considered the best-performing heuristic for 2D packing, while **Guillotine** is faster but less efficient.

33.2.6 Complexity

- **Heuristic Packing:** $O(n^2)$ for simple heuristics like First-Fit. MaxRects can be $O(n^3)$ in the worst case as the free list grows.
- **Optimal Packing:** Exponential $O(2^n)$, only feasible for small n (e.g., $n < 20$).

33.2.7 Usages

- **Manufacturing:** Cutting parts from a sheet of wood, metal, or fabric (Nesting).
- **Sprite Sheet Generation:** Packing many small game icons into a single large texture to improve GPU performance.

- **Logistics:** Loading pallets or shipping containers.
- **UI Layout:** Arranging windows or widgets in a tile-based interface.
- **VLSI Design:** Floorplanning of electronic components on a chip.

33.2.8 Testing Methods and Edge Cases

- **Overlap Check:** Verify that no two rectangles in the final layout intersect.
- **Container Boundary:** Ensure all rectangles are within the bin.
- **Density Calculation:** Calculate (Total Rectangle Area) / (Container Area).
- **Large Rectangles:** Test cases where one rectangle takes up most of the bin.
- **Thin Rectangles:** Test “sliver” rectangles that are hard to pack.
- **Rotation:** Verify that rotating a rectangle 90° is handled correctly by the splitting logic.

33.2.9 References

- Jylänki, J. (2010). “A Thousand Ways to Pack the Bin — A Practical Approach to Two-Dimensional Rectangle Bin Packing”. Technical Report.
- Lodi, A., Martello, S., & Monaci, M. (2002). “Two-dimensional packing problems: A survey”. European Journal of Operational Research.
- Wikipedia: [Packing problems](#)

33.3 Hopper Optimization

33.3.1 Overview

Hopper optimization is a specialized technique for packing non-convex polygons into a rectangular container (the “hopper”) such that the vertical height used is minimized. Unlike standard bin packing, which often uses bounding boxes, hopper optimization uses the exact geometry of the polygons, often leveraging the **No-Fit Polygon (NFP)** to find the tightest possible nesting. It is widely used in the textile, leather, and metal industries where material waste must be minimized. See Bennell and Oliveira [BO08] for a comprehensive tutorial and Burke et al. [BHKW07] for robust NFP computation.

33.3.2 Definitions

Definition 33.14 (Hopper). A **hopper** is a semi-infinite rectangular region $\{(x, y) \mid 0 \leq x \leq W, y \geq 0\}$ of fixed width W and unbounded height, into which polygons are placed from the top and allowed to slide down under gravity.

Definition 33.15 (No-Fit Polygon). The **No-Fit Polygon (NFP)** $\text{NFP}(A, B)$ of two convex polygons A (the *stationary* piece) and B (the *orbiting* piece) is the locus of all positions of a reference point on B such that B is translated to touch A without overlapping. The interior of $\text{NFP}(A, B)$ represents infeasible placements; the boundary represents tangent placements.

Definition 33.16 (Gravity Heuristic). The **gravity heuristic** places polygons into the hopper one at a time, allowing each polygon to “fall” as far down as possible (minimizing y), then as far left as possible (minimizing x), subject to non-overlap constraints with previously placed polygons and the hopper walls.

33.3.3 Theory

The core of hopper optimization is determining the **feasible placement** of a new polygon P_{new} relative to already placed polygons $\{P_1, \dots, P_k\}$.

1. **NFP Calculation:** For every pair of polygons (P_i, P_{new}) , the No-Fit Polygon $\text{NFP}_{i,\text{new}}$ is computed. This polygon defines the forbidden region for the reference point of P_{new} .
2. **Feasible Space:** The set of all possible positions for P_{new} is the area inside the hopper minus the union of all NFPs.
3. **Optimization:** The algorithm searches for a position in the feasible space that minimizes the current maximum height of the packing.

A common approach is the **Bottom-Left-Fill (BLF)** heuristic, which always places the next polygon at the lowest and then leftmost available position.

Theorem 33.17 (NFP Correctness). *For two convex polygons A with n vertices and B with m vertices, the NFP can be computed in $O(n + m)$ time. The NFP is itself a convex polygon with at most $n + m$ vertices, and a placement of B relative to A is valid if and only if the reference point of B lies outside $\text{int}(\text{NFP}(A, B))$.*

Proof. The NFP of two convex polygons is obtained by computing the Minkowski sum of A and the reflected polygon $-B$. Since the Minkowski sum of two convex polygons with n and m vertices is a convex polygon with at most $n + m$ vertices, and can be computed in $O(n + m)$ time by merging edge angle sequences, the result follows. A reference point placement is valid precisely when B does not overlap A , which corresponds to the reference point lying outside the interior of the Minkowski sum. $\square$

33.3.4 Pseudocode

```

1: function HOPPEROPTIMIZE(hopper_width, polygons)
2:   polygons.sort(key =  $\lambda p: p.\text{area}$ , reverse = True)
3:   placed_polygons  $\leftarrow$  []
4:   for  $p_{\text{new}}$  in polygons do
5:     best_pos  $\leftarrow$  FindLowestPosition( $p_{\text{new}}$ , placed_polygons, hopper_width)
6:      $p_{\text{new}}.\text{position} \leftarrow$  best_pos
7:     placed_polygons.append( $p_{\text{new}}$ )
8:   end for
9:   return GetMaxHeight(placed_polygons)
10: end function
11: function FINDLOWESTPOSITION( $p_{\text{new}}$ , placed, width)
12:   forbidden_region  $\leftarrow$  Union([ComputeNFP( $p_i, p_{\text{new}}$ ) |  $p_i \in$  placed])
13:   return  $\text{argmin}_{p \in \text{forbidden\_region.boundary}}(p.y, p.x)$ 
14: end function

```

33.3.5 Parameter Selection

- **Sorting Rule:** Decreasing area or decreasing height are standard.
- **Rotation:** Allowing discrete rotations (e.g., 0° , 90° , 180° , 270°) or continuous rotation significantly improves results but increases complexity.
- **NFP Precision:** For complex curved shapes, polygons are often simplified or sampled.

33.3.6 Complexity

- **NFP Calculation:** $O(n \cdot m)$ for two convex polygons with n and m vertices. For non-convex, it can be much higher.
- **Packing Heuristic:** $O(k^2)$ where k is the number of polygons, assuming NFP lookups are optimized.

33.3.7 Usages

- **Textile Industry:** Nesting clothing patterns onto a roll of fabric.
- **Sheet Metal Cutting:** Arranging mechanical parts on a metal plate for laser or waterjet cutting.
- **Leather Industry:** Packing parts onto irregular animal hides (which includes avoiding defects).
- **Shipbuilding:** Cutting large steel plates for hulls.
- **Furniture Manufacturing:** Maximizing the use of plywood or wood planks.

33.3.8 Testing Methods and Edge Cases

- **Convex vs. Non-Convex:** Verify that the algorithm handles interlocking U-shapes or L-shapes correctly.
- **Container Width:** Ensure no polygon extends beyond the hopper walls.
- **Overlap Check:** Rigorous verification that no two polygons intersect.
- **Hopper Height:** The primary metric to minimize.
- **Large Quantity:** Test with hundreds of small pieces to ensure stability.

33.3.9 References

- Bennell, J. A., & Oliveira, J. F. (2008). “The geometry of nesting problems: A tutorial”. European Journal of Operational Research.
- Burke, E. K., Hellier, R. S., Kendall, G., & Whitwell, G. (2007). “Complete and robust no-fit polygon generation for the irregular cutting and packing problem”. European Journal of Operational Research.
- Wikipedia: [Nesting \(process\)](#)

Chapter 34

Dynamic and Kinetic Geometry

34.1 Dynamic Geometric Problems

Every data structure in the preceding chapters was built for a *static* input: a set of points, segments, or polygons fixed once and for all. Real applications rarely respect that assumption: terrain vertices change as new survey data arrives, database entries are inserted and deleted, and the robots in a warehouse move continuously. This chapter studies geometric algorithms whose input *changes*, and it distinguishes three regimes:

Static The input is fixed; queries are answered against a structure built once (Chapter 9).

Dynamic The input changes by discrete insertions and deletions interleaved with queries (Sections 34.2–34.4).

Kinetic The input changes *continuously* along known trajectories, and the answer is a function of time maintained by *kinetic data structures* (Sections 34.6–34.10).

A dynamic structure is judged by its **update time**, **query time**, and **storage**. A kinetic structure is judged by the additional quantities of Section 34.6: how many *certificates* it holds, how fast it repairs a *certificate failure*, and how the number of failures compares with the number of times the answer changes. The unifying theme is **coherence**: between updates, or between small advances of time, the combinatorial answer barely changes, and a good structure changes only what it must. The dynamic results rest on the transform-and-conquer machinery of Overmars [Ove83]; the kinetic results rest on the kinetic data structure framework of Basch, Guibas, and Hershberger [BGH99, Gui98].

34.2 Dynamic Point Sets

The simplest dynamic geometric object is a set of points supporting **insert** and **delete** operations interleaved with queries. Most point-based structures — k -d trees, range trees, Voronoi diagrams — have dynamic versions, and the classical framework for deriving them is the **transform-and-conquer** method of Overmars [Ove83].

Definition 34.1 (Dynamic Point Structure). A **dynamic point structure** maintains a set $P \subseteq \mathbb{R}^d$ under $\text{INSERT}(p)$, $\text{DELETE}(p)$, and a class of queries. It is **semidynamic** if it supports only insertions and **fully dynamic** if it supports both.

The two basic design strategies are:

- **Partial rebuilding.** The **logarithmic method** partitions the points into $O(\log n)$ static structures of geometrically growing size: a new point enters the smallest structure, and

structures merge when their size doubles. A query runs against all levels, so the update cost is an $O(\log n)$ factor times the static construction cost and the query cost an $O(\log n)$ factor times the static query cost, both amortized. Section 9.10 analyzes this for range searching.

- **Balanced-tree dynamics.** The primary structure is kept balanced, and each rotation repairs only local secondary structures. A dynamic range tree costs $O(\log^2 n)$ per update in the plane (Section 9.10); a dynamic k -d tree (Section 10.1) inserts by leaf attachment and deletes by marking with periodic subtree rebuilds, at $O(\log n)$ amortized cost.

Two auxiliary problems recur. The **order maintenance** problem — maintain a sorted list so that the relative order of two elements is tested in $O(1)$ — is solved by the order-maintenance structure of Dietz and Sleator [DS87], which gives each element a monotone label from a dense order. The second is **dynamic point location**, answered across updates at expected $O(\log n)$ per insertion and $O(\log^2 n)$ per deletion (Section 8.9, [Ove83]). Textbook treatments are [Ove83, BCKO08] and the handbook chapters of [MS05b]; the priority search tree of Section 9.7 is the classic linear-storage structure for dynamic 3-sided queries.

34.3 Dynamic Convex Hulls

34.3.1 1. Overview

The dynamic convex hull problem maintains $\text{conv}(P)$ under insertions and deletions while answering queries about the hull. The two classical milestones are the $O(\log^2 n)$ -per-update structure of Overmars and van Leeuwen [OvL81] and the near-optimal structure of Brodal and Jacob [BJ02], whose total cost over any sequence of n updates is $O(n \log n)$. This section develops the Overmars–van Leeuwen approach and states the Brodal–Jacob improvement.

34.3.2 2. Definitions

Definition 34.2 (Dynamic Convex Hull). A **dynamic convex hull** structure maintains $\text{conv}(P)$ under $\text{INSERT}(p)$ and $\text{DELETE}(p)$, together with the queries of the following definition.

Definition 34.3 (Hull Queries). For a direction d , the **extreme-point query** asks for a vertex of $\text{conv}(P)$ maximizing $p \cdot d$; the **membership query** asks whether a given point lies inside the hull. Both are answered in $O(\log n)$ by binary search over the cyclic hull order (Chapter 4).

34.3.3 3. Theory

The Overmars–van Leeuwen structure maintains the upper hull and the lower hull separately, each as a balanced binary tree over the vertices in x -order. The key idea is the **bridge**: if a subtree’s vertices are split at a median x -coordinate, the subtree’s upper hull is the upper hull of the left part, one *bridge edge* connecting it to the upper hull of the right part, and the right part’s upper hull. A node stores its bridge, found in $O(\log n)$ time by a walk that advances a current vertex alternately on the two child chains.

Theorem 34.4 (Overmars–van Leeuwen). *The convex hull of an n -point set can be maintained under insertions and deletions in $O(\log^2 n)$ worst-case time per update, with extreme-point and membership queries in $O(\log n)$ time and $O(n)$ storage.*

Proof sketch. An update first rebalances the hull tree. The affected nodes lie on one root-to-leaf path, and at each such node the bridge is recomputed from the two child hull chains in $O(\log n)$ time; there are $O(\log n)$ nodes, giving $O(\log^2 n)$. A direction query walks one path, spending $O(1)$ per level following bridges, hence $O(\log n)$; the same search supports the membership test. $\square$

The structure is the canonical *two-level* dynamic structure, in the spirit of the multilevel structures of Section 9.5. Brodal and Jacob [BJ02] improved the update bound by a more carefully amortized, partition-based representation of the hull and its bridges:

Theorem 34.5 (Brodal–Jacob). *The convex hull of an n -point set can be maintained under insertions and deletions in $O(\log n)$ amortized time per update and $O(\log n)$ worst-case time per query, using $O(n)$ storage. Any sequence of n updates thus costs $O(n \log n)$ total, which is optimal up to logarithmic factors for comparison-based exact structures.*

34.3.4 4. Pseudocode

Algorithm 166 Dynamic Convex Hull Maintenance

Input: A dynamic point set P in general position.

Output: Insertions, deletions, and direction queries on $\text{conv}(P)$.

```

1: function DYNHULLINSERT( $p$ )
2:   insert  $p$  into the  $x$ -ordered trees of the upper and lower hulls
3:   for each node  $v$  on the rebalanced paths do
4:      $H[v] \leftarrow \text{BRIDGE}(\text{left}(v), \text{right}(v))$ 
5:   end for
6: end function
7: function DYNHULLDELETE( $p$ )
8:   delete  $p$  from both hull trees; rebalance and recompute bridges as above
9: end function
10: function EXTREMEPOINT( $d$ )
11:    $v \leftarrow$  root of the upper-hull tree
12:   while  $v$  is not a leaf do
13:     follow the bridge stored at  $v$  toward the maximum of the dot product with  $d$ 
14:   end while
15:   return the vertex reached; search the lower hull symmetrically for direction  $d$ 
16: end function

```

34.3.5 5. Parameter Selection

- **Balancing:** any balanced binary search tree works; the requirement is that an update touches only $O(\log n)$ nodes, so bridge recomputations stay on one root-to-leaf path.
- **Degenerate points:** vertical alignments, duplicate x -coordinates, and collinear hull vertices are handled by a symbolic perturbation (Section 1.11) or by storing, at each x , the interval of hull vertices at that x .

34.3.6 6. Complexity

- **Update:** $O(\log^2 n)$ worst case (Overmars–van Leeuwen), $O(\log n)$ amortized (Brodal–Jacob).
- **Query:** $O(\log n)$; **storage:** $O(n)$.
- **Total:** $O(n \log^2 n)$ and $O(n \log n)$ over n updates, respectively. A comparison-based lower bound of $\Omega(\log n)$ per operation makes the Brodal–Jacob bounds optimal up to constants.

34.3.7 7. Usages

- **Kinetic convex hulls:** the structure of Section 34.9 uses a dynamic hull as a subroutine when a point enters or leaves the hull.
- **Motion planning:** maintaining obstacle hulls under updates supports the queries of Chapter 26.
- **Onion peeling:** repeated extreme-point queries with dynamic deletion compute hull layers (Section 4.5) online.

34.3.8 8. Testing Methods and Edge Cases

- **Brute-force oracle:** after every update, recompute the hull statically and compare vertex sets over random update sequences.
- **Direction queries:** verify $\text{EXTREMEPOINT}(d)$ against an $O(n)$ scan over a grid of directions.
- **Degenerate hulls:** collinear points, duplicates, and two-point hulls must not break the bridge search.

34.3.9 9. References

The $O(\log^2 n)$ structure is due to Overmars and van Leeuwen [OvL81]; the dynamic-structure theory is [Ove83]. The $O(n \log n)$ total result is [BJ02]. Textbook treatments appear in [BCKO08, PS85] and [MS05b].

34.4 Dynamic Proximity Structures

Proximity queries — nearest neighbor, closest pair, Voronoi cells, Delaunay adjacency — are among the most important dynamic geometric problems. For **dynamic nearest neighbor** the standard options are:

- **Dynamic k -d tree:** leaf attachment with periodic rebuilds gives $O(\log n)$ amortized updates and the branch-and-bound query of Section 10.1 (Chapter 10).
- **Dynamic Delaunay triangulation:** since the nearest neighbor of a point is among the vertices of its Delaunay cell, the dynamic Delaunay structure of Section 7.4 (Chapter 12) answers nearest-neighbor queries in expected logarithmic time, with local-flip updates.
- **Reduction to hulls:** in the plane, nearest-neighbor queries lift to 3D convex hulls (Section 12.8); the dynamic 3D hull of Chan [Cha06] thereby yields a dynamic planar nearest-neighbor structure with polylogarithmic bounds.

The **dynamic closest pair** maintains the Delaunay triangulation together with a secondary structure reporting its shortest edge — the static reduction of Chapter 13, made dynamic by updating both levels at each insertion or deletion. Dynamic **range searching** is developed in Section 9.10 (Chapter 9), where the schemes of Section 34.2 yield the $O(\log^2 n)$ and $O(n^{1-1/d})$ trade-offs. When the motion is continuous, the same problems reappear in kinetic form and the Delaunay triangulation becomes the central tool (Sections 34.10–34.9); the surveys [Gui98, MS05b] collect the state of the art.

34.5 Moving Geometric Objects

Discrete updates are not the only way an input can change. In simulation, robotics, and tracking, objects are in *continuous motion*, and the questions are about time: when do two points cross, when do robots collide, and how does the nearest-neighbor structure evolve?

Definition 34.6 (Trajectory). A **trajectory** for an object is a function $p : [0, T] \rightarrow \mathbb{R}^d$ giving its position at time t . The motion is **linear** (constant velocity) if $p(t) = p_0 + vt$, and **algebraic** if each coordinate is a polynomial (or more generally an algebraic function) of t . The interval $[0, T]$ is the **time horizon**.

Example 34.7. For n points with linear trajectories, the relative order by x -coordinate changes only at the finitely many times two points have equal x -coordinate; between such times the order is fixed. Any quantity — sorted order, convex hull, closest pair — that depends on the configuration through order information is therefore *piecewise constant in time*, changing only at a discrete set of *event times* determined by the trajectories. This is the combinatorial core of kinetic geometry: continuous motion, discrete change.

Two strategies are available. The *sampling* strategy re-evaluates the answer at every step of a discretized clock; it costs a factor T/δ in the number of samples and can miss events between samples (the classic *tunneling* artifact of collision detection). The *event-driven* strategy computes, for each condition that guarantees the current answer, the exact future time at which it fails, schedules these times in a priority queue, and processes the earliest failure; it never misses an event but must compute failure times exactly, which is feasible only for algebraic trajectories (Section 34.7). When trajectories are known in advance (a “flight plan”), event-driven structures dominate; when they are revealed on the fly, a hybrid of sampling and event prediction is used. The kinetic data structure framework of Section 34.6 formalizes the event-driven strategy.

34.6 Kinetic Data Structures

The **kinetic data structure** (KDS) framework of Basch, Guibas, and Hershberger [BGH99, Gui98] maintains a combinatorial attribute of moving objects by reducing correctness to a set of small *certificates* whose validity guarantees the current answer, and scheduling the times at which the certificates fail.

Definition 34.8 (Kinetic Structure). A **kinetic data structure** for an attribute A of n moving objects consists of (i) a combinatorial representation of A correct for the current configuration, (ii) a set of **certificates** — boolean tests on the object positions — such that A is guaranteed correct as long as all certificates hold, and (iii) an **event queue** storing the predicted failure time of each certificate.

Definition 34.9 (Certificate). A **certificate** is a predicate $c(t)$ over the trajectories whose truth at time t is sufficient for the correctness of the maintained attribute at t . It **fails** at the smallest $t' > t$ with $c(t')$ false; this is its **failure time**.

Certificates are elementary geometric tests: order comparisons, orientation tests, or in-circle tests. Since trajectories are algebraic, each certificate function is a polynomial in t of constant degree, and its failure time is a root of that polynomial (at most degree four for an in-circle certificate under linear motion; Section 34.7).

Definition 34.10 (KDS Quality Measures). Let s be the number of certificates and n the number of objects.

- **Compactness:** $s = O(n)$.

- **Locality:** each object participates in $O(\log n)$ (ideally $O(1)$) certificates.
- **Responsiveness:** a certificate failure is repaired in polylogarithmic time.
- **Efficiency:** the number of certificate failures is within a polylogarithmic factor of the number of times the maintained attribute changes. The ratio separates *internal* events (failures that do not change the answer) from *external* events (changes of the answer).

Example 34.11 (Kinetic Tournament). Maintain the maximum of n points moving on a line. Arrange the points as leaves of a balanced tree; each internal node stores the larger child and a certificate $c(t)$: left winner $\geq$ right winner. As long as every certificate holds, the root stores the maximum. When a certificate fails, the loser has overtaken the winner: swap, reschedule the node's and its predecessors' certificates, and continue. The structure holds $O(n)$ certificates, each point is in $O(\log n)$ of them, and a failure is repaired in $O(\log n)$ time — compact, local, responsive. It is the canonical first KDS and the model for the kinetic heaps of Section 34.7.

Remark 34.12. The KDS framework is a *recipe*, not an algorithm: for a new attribute one designs the certificate set (a correctness proof), the repair procedure (a local update), and the failure-time computation (a root solver) [Gui98]. The following sections apply the recipe to sorting, convex hulls, closest pairs, and collision detection.

34.7 Certificates and Events

A KDS is only as good as its event scheduling. This section develops the algorithmic support: computing failure times, storing them, and repairing failures reliably.

Definition 34.13 (Kinetic Event). A **kinetic event** at time t is a set of one or more certificate failures at t . It is **external** if the maintained attribute changes and **internal** otherwise. The event queue stores the predicted failure times of all future certificates.

Computing failure times

For linear motion $p(t) = p_0 + vt$, an order or orientation certificate fails at a root of a degree-one polynomial, and an in-circle certificate (the determinant of the matrix with rows $(x_i, y_i, x_i^2 + y_i^2, 1)$) is a polynomial of degree at most four in t . All such roots are computed in $O(1)$ arithmetic operations, so every certificate schedules its failure in constant time. This is why KDS implementations are practical: every predicted failure time is the root of a low-degree polynomial.

Event scheduling with heaps

The event queue needs extract-min, insert, and decrease, each in $O(\log n)$; a binary heap delivers all three (Section B.4, [CLRS09]) and is the standard choice. The **kinetic heap** is a kinetic tournament over the keys to be minimized (here, the failure times), used to pick the earliest pending failure.

Definition 34.14 (Kinetic Heap). A **kinetic heap** is a kinetic tournament whose leaves hold the keys (failure times or attribute values). Each internal node stores a winner and a certificate that its key does not exceed its sibling's; the root holds the minimum at all times.

Repairing failures

When a certificate fails at time t , the repair has three steps: (i) *verify* the failure with the exact predicates of Section 3.4 — a floating-point evaluation at t can be inconclusive; (ii) *apply* the local combinatorial update (a swap, a flip, or a tangent recomputation); and (iii) *reschedule* the new certificates. Two issues dominate practice. First, **simultaneous events**: several certificates may fail together, and the queue must order them by a fixed secondary key or a symbolic perturbation must break the tie (Section 1.11). Second, **numerical drift**: a failure time computed in floating point is slightly wrong, so the certificate is tested at the scheduled time rather than trusted; if it has not failed yet, the failure time is recomputed and the event rescheduled (“re-prediction”) [Gui98].

Remark 34.15. The event machinery is not specific to kinetics: the same pattern — a priority queue of events, local repairs, exact predicates — drives the sweep-line algorithms of Chapter 5, where the event queue is the identical heap structure (Section 5.4). The difference is that sweep-line events are *discovered* from adjacency, while kinetic events are *predicted* from certificate roots.

34.8 Kinetic Sorting

34.8.1 1. Overview

Kinetic sorting maintains the sorted order of n points moving on a line, as a function of time. It is the simplest nontrivial KDS and the foundation of the kinetic hull and proximity structures, many of which reduce to maintaining one or more sorted orders (of points by a coordinate, or of directions by angle). The order can change $\Theta(n^2)$ times in the worst case even for constant-velocity motion.

34.8.2 2. Definitions

Definition 34.16 (Kinetic Sorted List). A **kinetic sorted list** (KSL) maintains n points moving on a line in sorted order by position. The certificates are the $n - 1$ adjacent-pair inequalities $c_i(t) : p_{\sigma(i)}(t) \leq p_{\sigma(i+1)}(t)$, where σ is the current order; an order change is exactly a certificate failure.

Definition 34.17 (Order Maintenance Label). An **order label** for each element is a value monotone in the sorted order, comparable in $O(1)$; the order-maintenance structures of [DS87] provide such labels with $O(1)$ amortized update cost.

34.8.3 3. Theory

The KSL is the textbook KDS [BGH99]: it holds $n - 1$ certificates (compact), each point participates in at most two (locality $O(1)$), and a failure swaps two points, changing only two adjacencies (responsiveness $O(1)$). Its limitation is the number of events.

Theorem 34.18 (Kinetic Sorted List Bounds). *The sorted order of n points on a line can change $\Theta(n^2)$ times in the worst case, even for constant-velocity motion. The kinetic sorted list maintains the order with $O(n)$ certificates and repairs each order change in $O(1)$ time plus $O(\log n)$ for the event queue.*

Proof. For the lower bound, take $n/2$ points far to the left moving right with velocity 1 and $n/2$ points on the right moving with velocity 0. Every moving–stationary pair crosses exactly once, reversing its relative order; there are $(n/2)^2 = \Theta(n^2)$ such reversals, each forcing an order change. For the upper bound, the certificates are the adjacent pairs; a failure is a swap changing

two adjacencies, and each new adjacency's failure time is the root of a degree-one polynomial, computed in $O(1)$ (Section 34.7). $\square$

For random or bounded-velocity motions the number of changes is much smaller, and the KSL is the standard component of kinetic nearest-neighbor and range structures.

34.8.4 4. Pseudocode

Algorithm 167 Kinetic Sorted List

Input: n points $p_1, \dots, p_n$ on a line with trajectories $p_i(t)$.

Output: The sorted order at all times in $[0, T]$.

```

1: function KINETICSORTEDLIST( $p_1, \dots, p_n$ )
2:   sort the points at  $t = 0$ ; let  $\sigma$  be the current order
3:    $Q \leftarrow$  empty kinetic heap
4:   for  $i = 1$  to  $n - 1$  do
5:     create certificate  $c_i$  for the pair  $(\sigma(i), \sigma(i + 1))$ 
6:      $Q.\text{insert}(\text{FailureTime}(c_i))$ 
7:   end for
8:   while  $Q$  is not empty and  $Q.\text{min}() \leq T$  do
9:      $t \leftarrow Q.\text{extract-min}()$ ; advance the clock to  $t$ 
10:    verify the failing pair exactly; swap  $\sigma(i)$  and  $\sigma(i + 1)$ 
11:    remove the two certificates of the swapped pair; create the two new ones
12:    push the new certificates' failure times into  $Q$ 
13:   end while
14:   return  $\sigma$  and the recorded change times
15: end function

```

34.8.5 5. Parameter Selection

- **Trajectory form:** the failure-time computation assumes polynomial trajectories; for arbitrary trajectories, roots are found numerically and events are re-predicted on verification (Section 34.7).
- **Simultaneous crossings:** order coincident failures by a fixed secondary key so the swap sequence is deterministic.
- **Order representation:** use order-maintenance labels ([DS87]) so other structures test order in $O(1)$.

34.8.6 6. Complexity

- **Certificates:** $n - 1$ (compact); locality $O(1)$.
- **Responsiveness:** $O(\log n)$ per order change.
- **Total:** $O((n^2 + k) \log n)$ worst case for k order changes, with $k \leq \binom{n}{2}$; $\Theta(n^2)$ events already occur for the two-velocity construction of Theorem 34.18.

34.8.7 7. Usages

- **Kinetic nearest neighbor:** sorted orders by each coordinate give candidate certificates for proximity structures (Section 34.4).

- **Kinetic convex hulls:** the angular order of points around a moving vantage point drives the hull structure of Section 34.9.
- **Event-driven simulation:** any simulation maintaining a set of threshold-crossing times uses a kinetic sorted list over them.

34.8.8 8. Testing Methods and Edge Cases

- **Oracle:** sort the positions by brute force at a dense grid of times and at every event time; compare with the maintained order.
- **Event validation:** verify with exact predicates that the scheduled pair truly crossed and that no pair crossed earlier.
- **Degeneracies:** coincident points and simultaneous crossings must produce a consistent final order; floating-point drift must be absorbed by the re-prediction rule of Section 34.7.

34.8.9 9. References

The kinetic sorted list is a central example of [BGH99, Gui98]; the order-maintenance labels are due to [DS87]. Its use inside larger kinetic structures is surveyed in [Gui98, MS05b].

34.9 Kinetic Convex Hulls

34.9.1 1. Overview

The kinetic convex hull problem maintains $\text{conv}(P(t))$ for points moving along known trajectories. Its certificates are orientation tests of hull edges and containment tests of interior points; its events are the moments when a point enters or leaves the hull; and its repair recomputes the affected tangent chains. The KDS of Basch, Guibas, and Hershberger [BGH99] is compact, local, responsive, and efficient.

34.9.2 2. Definitions

Definition 34.19 (Kinetic Convex Hull). A **kinetic convex hull** structure maintains $\text{conv}(P)$ of n moving points. Its certificates are of two kinds: for every hull edge uv , an orientation certificate asserting that all other points lie on the same side of the line through u and v ; and, for every interior point, a containment certificate asserting that it lies inside the hull. The **external events** are the times at which a point enters or leaves the hull.

34.9.3 3. Theory

The certificates are sufficient: while every hull edge is a supporting line and every interior point is contained, the stored hull is exact. A failure needs one of two repairs — a hull vertex moves so a neighbor leaves the boundary (find a new tangent pair), or an interior point crosses a hull edge (find the two tangents from the point to the hull). Both are local and use the tangent routines of Chapter 4 on the dynamic hull of Section 34.3.

Theorem 34.20 (Kinetic Hull Bounds). *The convex hull of n points moving with constant velocity along lines can change $\Theta(n^2)$ times in the worst case. The kinetic convex hull of Basch, Guibas, and Hershberger [BGH99] is compact ($O(n)$ certificates), local (each point in $O(\log n)$), responsive (each event in $O(\log^2 n)$), and efficient: the number of internal events is within a constant factor of the number of external hull events.*

Proof sketch. The lower bound adapts the crossing construction of Theorem 34.18 to the plane, making each of $\Theta(n^2)$ pairwise crossings change the hull boundary. For the upper bound, correctness holds until the first certificate failure; the repair touches only the vertices of the failed edge and its neighbors, and the new certificates' failure times are roots of constant-degree polynomials (Section 34.7). Each external event is triggered by a distinct failure, and internal events are bounded by locality. $\square$

34.9.4 4. Pseudocode

Algorithm 168 Kinetic Convex Hull

Input: n points with polynomial trajectories; a time horizon $[0, T]$.

Output: The convex hull as a function of time.

```

1: function KINETICHULL( $P$ )
2:   build  $\text{conv}(P(0))$ ; create edge and containment certificates
3:   schedule each certificate's failure time in a kinetic heap  $Q$ 
4:   while  $Q$  is not empty and  $Q.\text{min}() \leq T$  do
5:      $t \leftarrow Q.\text{extract-min}()$ ; verify the failure exactly
6:     if an interior point crossed a hull edge then
7:       recompute the two tangents from the point to the hull; insert it
8:     else if a hull vertex moved out then
9:       remove the vertex, connect its neighbors; recompute the local tangents
10:    end if
11:    replace the certificates of the changed edges; reschedule their failure times
12:  end while
13: end function

```

34.9.5 5. Parameter Selection

- **Interior maintenance:** interior points are tracked by containment certificates so no point can exit without an event; only the closest potential exit matters, so certificates fail in the right order.
- **Degenerate events:** simultaneous entry/exit and collinear triples are resolved by the ordering rules of Section 34.7.
- **Exact predicates:** orientation and containment tests use the adaptive exact methods of Section 3.4; an approximate test misorders the event sequence.

34.9.6 6. Complexity

- **Certificates:** $O(n)$; locality $O(\log n)$.
- **Responsiveness:** $O(\log^2 n)$ per event.
- **Events:** $\Theta(n^2)$ worst case for linear motion, so the total time over the horizon is $O(n^2 \log^2 n)$ and proportional to the number of hull changes plus polylogarithmic overhead.

34.9.7 7. Usages

- **Motion planning:** hulls of moving obstacles support the continuous collision checks of Chapter 26.

- **Bounding volumes:** the hull of a moving cluster is the tightest convex bounding volume and is maintained kinetically instead of resampled.
- **Peeling:** a kinetic hull yields kinetic hull layers of Section 4.5.

34.9.8 8. Testing Methods and Edge Cases

- **Static oracle:** at random times, recompute the hull from the positions and compare vertex sets.
- **Event oracle:** verify no point enters or leaves between consecutive events by re-checking all certificates at sample times.
- **Tangent correctness:** after each event, verify every point lies on one side of every hull edge.

34.9.9 9. References

The kinetic convex hull is developed in [BGH99] and surveyed in [Gui98]; the dynamic-hull subroutine is the structure of Section 34.3 ([OvL81, Ove83]). Textbook discussions appear in [BCKO08, MS05b].

34.10 Kinetic Closest Pairs

34.10.1 1. Overview

The kinetic closest-pair problem maintains the pair of points with minimum mutual distance among moving points. The key observation is that the closest pair is always a Delaunay edge (Chapter 13), so the problem reduces to maintaining the Delaunay triangulation kinetically and reporting its shortest edge. The kinetic Delaunay structure uses in-circle certificates and edge flips, as discussed in the Voronoi chapter (Section 11.4, Chapter 11).

34.10.2 2. Definitions

Definition 34.21 (Kinetic Closest Pair). A **kinetic closest-pair** structure maintains, for each time t , the pair of distinct points minimizing the Euclidean distance, together with the minimum value. Its certificates are those of an underlying kinetic Delaunay triangulation (in-circle tests) plus the tournament certificates of Section 34.6 used to select the shortest Delaunay edge.

34.10.3 3. Theory

The reduction is exact: the closest pair is a Delaunay edge, and its length is the minimum over all Delaunay edge lengths. The kinetic structure maintains the Delaunay triangulation — each edge carries an in-circle certificate asserting that the circumcircles of its adjacent triangles are empty — and a kinetic tournament over the edge lengths. When an in-circle certificate fails, the two affected triangles are flipped; when the tournament's minimum changes, the closest pair changes.

Theorem 34.22 (Kinetic Delaunay Complexity). *The Delaunay triangulation (equivalently the Voronoi diagram) of n points moving with constant velocity along lines can change $\Theta(n^2)$ times in the worst case [AGMR98]. For uniform linear motion, the number of events of a kinetic Delaunay triangulation is near-quadratic [Rub16]. Each event is an edge flip triggered by an in-circle certificate and repaired in $O(\log n)$ time.*

Theorem 34.23 (Kinetic Closest Pair Bounds). *The closest pair of n points moving with constant velocity can change $\Omega(n^2)$ times in the worst case [BGZ97]. The Delaunay-plus-tournament structure is compact ($O(n)$ certificates), local (each point in $O(\log n)$), and responsive ($O(\log^2 n)$ per event).*

Proof sketch. The lower bound of [BGZ97] adapts the crossing construction of Theorem 34.18 so that the closest pair shifts among $\Theta(n^2)$ distinct pairs. For the upper bound, the triangulation is exact while all in-circle certificates hold (the empty-circle property of Chapter 12); a failure flips one edge and reschedules the $O(1)$ affected certificates, and the tournament adds $O(\log n)$ per minimum change. Each certificate is a degree-at-most-four polynomial under linear motion (Section 34.7). $\square$

34.10.4 4. Pseudocode

Algorithm 169 Kinetic Closest Pair via Kinetic Delaunay

Input: n points with polynomial trajectories; time horizon $[0, T]$.

Output: The closest pair at all times in $[0, T]$.

```

1: function KINETICCLOSESTPAIR( $P$ )
2:   build the Delaunay triangulation  $DT$  of  $P(0)$ 
3:   for each edge  $e$  of  $DT$  do
4:     create an in-circle certificate for  $e$ ; schedule its failure time
5:   end for
6:   build a kinetic tournament  $M$  over the edge lengths of  $DT$ 
7:   while time  $\leq T$  and the event queue is nonempty do
8:     process the earliest event at time  $t$ 
9:     if an in-circle certificate fails on edge  $e$  then
10:      flip  $e$ ; remove the two old triangles, add the two new ones
11:      create certificates for the new edges; update  $M$  on the changed lengths
12:    end if
13:    report the minimum-length edge of  $M$  as the closest pair at  $t$ 
14:  end while
15: end function

```

34.10.5 5. Parameter Selection

- **In-circle polynomial:** under linear motion the certificate is a degree-four polynomial in t ; with approximate failure times, re-prediction (Section 34.7) keeps the structure exact.
- **All-nearest-neighbors:** the same Delaunay structure also supports kinetic nearest-neighbor queries per vertex; the tournament is replaced by a per-vertex report.
- **Higher dimensions:** in $\mathbb{R}^3$ the closest pair is not necessarily a Delaunay edge, so other certificate sets are used [BGZ97].

34.10.6 6. Complexity

- **Certificates:** $O(n)$ (one per Delaunay edge).
- **Responsiveness:** $O(\log^2 n)$ per event.
- **Events:** $\Theta(n^2)$ worst case for linear motion [AGMR98]; near-quadratic for uniform linear motion [Rub16].

34.10.7 7. Usages

- **Particle simulation:** the closest approaching pair, with its distance used as a time-step safety bound.
- **Tracking:** the pair of moving agents with minimal separation, reported as a continuous function of time.
- **Collision detection:** a threshold certificate on a pair's distance fails exactly when a collision becomes possible.

34.10.8 8. Testing Methods and Edge Cases

- **Brute-force oracle:** at dense sample times, compute the closest pair by all-pairs scanning (Section 13.1, Chapter 13).
- **Delaunay invariant:** after every flip, verify the empty-circle property by exact in-circle predicates.
- **Coincident points and ties:** two points meeting produce a zero-distance closest pair (no division by zero in the failure-time computation); equal minima are resolved by a fixed rule.

34.10.9 9. References

The kinetic closest-pair structure and its lower bound are due to [BGZ97]; the Delaunay complexity bounds are [AGMR98, Rub16]; the framework is [BGH99, Gui98]. Kinetic Voronoi diagrams are also treated in Section 11.4 of Chapter 11.

34.11 Applications to Tracking and Simulation

The dynamic and kinetic structures of this chapter are the algorithmic core of applications in which the world changes with time.

Tracking A moving target's position is sampled by sensors, and the system answers proximity queries about the current configuration. Dynamic range searching (Section 9.10, Chapter 9) and the dynamic nearest-neighbor structures of Section 34.4 answer the queries; when trajectories are known, kinetic versions *predict* the queries before they are asked.

Collision detection In robotics and animation, moving objects must be checked for collisions, and the checks should run only when the configuration changes. The kinetic free-space approach of Agarwal, Basch, Guibas, Hershberger, and Zhang [ABG⁺02] maintains a tiling of the free space whose cells' validity is certified; a certificate failure signals a potential contact and triggers a local repair. This is the continuous-motion companion of the static checks of Chapter 26.

Simulation Particle and agent-based models advance in discrete steps, but their combinatorial structure — neighbors, hulls, closest pairs — is best maintained kinetically so the time step is chosen adaptively from predicted events. The kinetic sorted list of Section 34.8 is the canonical scheduling device for such simulations.

Geographic information systems Moving objects (vehicles, animals) are queried over both space and time; kinetic range trees and the proximity queries of Section 34.10 extend the static GIS pipeline of Chapter 36.

Two lessons recur. First, the event-driven discipline of Section 34.7 is only as good as its exact predicates: a certificate that fails silently, or an event scheduled slightly too late, corrupts every later answer, so production systems couple the KDS with the adaptive exact predicates of Section 3.4 and with re-prediction. Second, the dynamic and kinetic views reinforce each other: a kinetic structure typically reduces to a dynamic structure (the hull of Section 34.3 inside the kinetic hull of Section 34.9), so progress on the dynamic problem directly improves the kinetic one. Open problems — robustness, event bounds for general motions, approximate kinetics — are surveyed in Chapter 41.

34.12 Exercises

1. Prove that the logarithmic method (Section 34.2) converts a static structure with query time $Q(n)$ and construction time $C(n)$ into a semidynamic structure with amortized update cost $O(C(n) \log n/n)$ and query cost $O(Q(n) \log n)$, and derive the bounds for 2D orthogonal range counting.
2. Argue that the Overmars–van Leeuwen dynamic hull (Section 34.3) answers a membership query in $O(\log n)$ time by binary search on the hull tree, and give an input on which a naive point-in-polygon test takes $O(n)$.
3. Show that a direction query in the dynamic hull is answered by a single root-to-leaf walk using stored bridges, and explain why a node's bridge must be recomputed after any update in its subtree.
4. Construct the two-velocity family of Theorem 34.18 explicitly for $n = 6$ and list the times at which the sorted order changes; verify that the kinetic sorted list processes exactly those events.
5. Prove the four quality measures (Definition 34.10) for the kinetic sorted list: compactness, locality $O(1)$, responsiveness $O(\log n)$ per event, and efficiency $O(1)$ relative to order changes.
6. Verify that the in-circle certificate under linear motion is a polynomial of degree at most four by writing the certificate determinant and counting degrees, and state the degree of an orientation certificate.
7. Design a kinetic structure maintaining the pair of points with maximum distance (the kinetic diameter) using a kinetic tournament, and discuss whether its event count can be quadratic in the worst case.
8. Modify the kinetic closest-pair algorithm (Section 34.10) to handle two points that coincide at a time, and state what the exact predicates and the failure-time computation must return.

Part XI

Applications

Chapter 35

Computational Geometry for Computer Graphics

- 35.1 Geometric Modeling
- 35.2 Spatial Acceleration Structures
- 35.3 Bounding Volume Hierarchies
- 35.4 Ray–Object Intersection
- 35.5 Hidden-Surface Problems
- 35.6 Clipping
- 35.7 Mesh Processing
- 35.8 Collision Detection
- 35.9 Real-Time Geometry
- 35.10 GPU-Based Geometric Computation
- 35.11 Case Study: Geometry Pipeline
- 35.12 Project

Chapter 36

Computational Geometry for GIS

- 36.1 Geographic Data
- 36.2 Spatial Indexing
- 36.3 Map Overlay
- 36.4 Point-in-Region Queries
- 36.5 Proximity Analysis
- 36.6 Voronoi-Based Service Regions
- 36.7 Terrain Representation
- 36.8 Triangulated Irregular Networks
- 36.9 Digital Elevation Models
- 36.10 Map Generalization
- 36.11 Large-Scale Spatial Data
- 36.12 Case Study: Geographic Facility Location
- 36.13 Project

Chapter 37

Computational Geometry for Robotics

- 37.1 Robot Geometry
- 37.2 Collision Detection
- 37.3 Configuration Spaces
- 37.4 Path Planning
- 37.5 Sensor Geometry
- 37.6 Localization
- 37.7 Mapping
- 37.8 SLAM and Geometric Structures
- 37.9 Manipulation Planning
- 37.10 Autonomous Navigation
- 37.11 Case Study: Mobile-Robot Planning
- 37.12 Project

Chapter 38

Computational Geometry for Scientific Computing

38.1 Geometry in Numerical Simulation

38.2 Mesh Generation

38.3 Finite-Element Geometry

38.4 Domain Discretization

38.5 Adaptive Refinement

38.6 Particle Methods

38.7 Nearest-Neighbor Computations

38.8 Spatial Decomposition

38.9 Parallel Geometric Algorithms

38.10 Large-Scale Scientific Geometry

38.11 Case Study: PDE Mesh Construction

38.12 Project

38.13 Stochastic and Meshless PDE Methods

38.14 Stochastic PDE Solvers

38.14.1 Overview

Stochastic PDE solvers use Monte Carlo methods to solve partial differential equations (like Laplace or Poisson) on geometric domains [SC20].

Unlike standard grid-based solvers (Finite Element/Difference), which require a high-quality global mesh, stochastic solvers like **Walk on Spheres (WoS)** and **Walk on Stars (WoSt)**

can solve for the value at a single point without constructing a global system of equations. This is particularly useful for complex or dirty geometries.

Definition 38.1 (Walk on Spheres). Walk on Spheres is a Monte Carlo method for solving the Laplace equation $\Delta u = 0$ in a domain Ω . Starting from an interior point x , a random walk iteratively jumps to a uniformly random point on the largest inscribed sphere centered at the current position, until the walk reaches the boundary $\partial\Omega$, where the Dirichlet data is evaluated.

Definition 38.2 (Walk on Stars). Walk on Stars generalizes WoS by allowing jumps to uniformly random points on the intersection of the largest inscribed sphere with a cone defined by the star-shaped region around the current point. This extension enables estimation of derivatives of the solution and supports Neumann boundary conditions [Y+25].

38.14.2 Implementation Details

The CompGeom implementation includes:

1. Walk on Spheres (WoS):

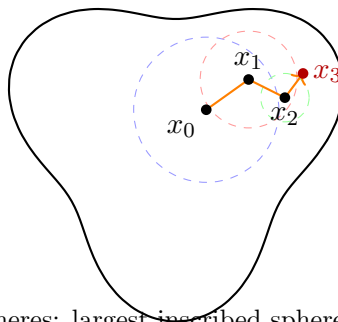
- **Laplace Solver:** Estimates $u(x)$ where $\Delta u = 0$.
- **Poisson Solver:** Estimates $u(x)$ where $\Delta u = f$ using the Walk on Balls variant with the Green's function representation.

2. Walk on Stars (WoSt):

- **Gradient Estimator:** Robustly estimates $\nabla u(x)$ using the Boundary Integral Equation (BIE).

Theorem 38.3 (WoS Convergence). *For a bounded domain Ω with Lipschitz boundary and continuous Dirichlet data g , the Walk on Spheres estimator converges almost surely to the unique harmonic solution $u(x)$ of $\Delta u = 0$ in Ω with $u|_{\partial\Omega} = g$, as the number of random walks $M \rightarrow \infty$.*

Proof. The proof relies on the mean value property of harmonic functions. At each step, the expected value of u at a random point on the sphere of radius r centered at x equals $u(x)$ by harmonicity. Therefore, the Walk on Spheres process is a martingale. By the martingale convergence theorem, the process converges as the number of steps grows. The boundary hitting point converges to $\partial\Omega$ because the domain is bounded, so the inscribed sphere radii decrease. The strong law of large numbers then gives almost sure convergence of the Monte Carlo average over M independent walks. $\square$



Walk on Spheres: largest inscribed sphere at each step

Figure 38.1: Walk on Spheres: from each interior point, jump to a random point on the largest inscribed sphere.

Algorithm 170 Stochastic PDE Solver Usage

```

1: from compgeom.mesh.algorithms.stochastic_pde import WalkOnSpheres,
   WalkOnStars
2: from compgeom import Point3D

3: # Initialize on a surface mesh
4: wos = WalkOnSpheres(mesh)
5: value = wos.solve_laplace(Point3D(0.5, 0.5, 0.5), boundary_func)

6: # Robust gradient estimation
7: wost = WalkOnStars(mesh)
8: grad = wost.solve_gradient(Point3D(0.5, 0.5, 0.5), boundary_func)

```

38.14.3 Usage

Example 38.4 (WoS on a Unit Disk). Consider the Laplace equation $\Delta u = 0$ on the unit disk $\Omega = \{(x, y) : x^2 + y^2 < 1\}$ with boundary data $g(\theta) = \cos(2\theta)$. The exact solution is $u(r, \theta) = r^2 \cos(2\theta)$. To estimate $u(0.5, 0)$ via WoS, we start at $(0.5, 0)$, which has inscribed sphere radius 0.5. A random walk jumps to a point on this sphere, then from the new position computes its inscribed sphere, and so on until hitting $\partial\Omega$. Repeating for $M = 10,000$ walks and averaging the boundary values yields an estimate close to $u(0.5, 0) = 0.25 \cdot 1 = 0.25$, demonstrating the convergence stated in Theorem 38.3.

Remark 38.5. The variance of WoS estimators decreases as $O(1/M)$ where M is the number of random walks. For Laplace’s equation, the walk is guaranteed to terminate almost surely for bounded domains. For Poisson’s equation, an additional correction term accumulates along each walk, increasing variance.

38.14.4 References

- Sawhney, R., & Crane, K. “Monte Carlo Geometry Processing: A Meshless Approach to Geometric PDE.” *ACM Transactions on Graphics (SIGGRAPH)*, 2020 [SC20].
- Yu, Y., et al. “Robust Derivative Estimation with Walk on Stars.” *ACM Transactions on Graphics (SIGGRAPH)*, 2025 [Y+25].

38.15 Wave Kernel Signature**38.15.1 Overview**

The Wave Kernel Signature (WKS) is a multi-scale geometric descriptor for vertices on a 3D mesh. Like the Heat Kernel Signature (HKS), it is based on the spectral properties of the Laplace–Beltrami operator. However, while HKS models heat diffusion (essentially a low-pass filter), WKS models the evolution of a quantum particle (wave function) with a specific energy distribution. This makes WKS better at capturing high-frequency geometric details and provides superior performance in non-rigid shape matching [ASC11].

38.15.2 Definitions

Definition 38.6 (Schrödinger Equation). The time-dependent Schrödinger equation on a Riemannian manifold (M, g) governs the evolution of a quantum state: $i\hbar \frac{\partial \psi}{\partial t} = -\Delta_M \psi$, where Δ_M is the Laplace–Beltrami operator Theorem 24.22 and $\hbar$ is the reduced Planck constant (set to 1 in the geometric setting).

Definition 38.7 (Wave Kernel Signature). The Wave Kernel Signature at vertex x and energy scale e is defined as

$$W(x, e) = \sum_{k=1}^{\infty} g_e(\lambda_k)^2 \phi_k(x)^2$$

where $\{\lambda_k\}$ and $\{\phi_k\}$ are the eigenvalues and orthonormal eigenfunctions of the Laplace–Beltrami operator, and g_e is an energy band-pass filter centered at energy e [ASC11].

Definition 38.8 (Energy Distribution). The energy distribution (or spectral filter) $g_e(\lambda)$ is typically a log-normal distribution:

$$g_e(\lambda) = \exp\left(-\frac{(\log e - \log \lambda)^2}{2\sigma^2}\right)$$

This ensures that the descriptor is sensitive to relative energy differences, providing a balanced representation of both coarse and fine geometric features [ASC11, SOG09].

38.15.3 Theory

The Wave Kernel Signature $W(x, e)$ at vertex x and energy scale e is defined as the average probability of a particle with energy distribution g_e being found at x :

$$W(x, e) = \sum_{k=1}^{\infty} g_e(\lambda_k)^2 \phi_k(x)^2$$

where λ_k and ϕ_k are the eigenvalues and eigenfunctions of the Laplace–Beltrami operator.

The filter g_e is usually a log-normal distribution Theorem 38.8:

$$g_e(\lambda) = \exp\left(-\frac{(\log e - \log \lambda)^2}{2\sigma^2}\right)$$

This ensures that the descriptor is sensitive to relative energy differences, providing a balanced representation of both coarse and fine geometric features.

The WKS can be interpreted as the diagonal of the spectral projection operator onto the energy band $[e - \sigma, e + \sigma]$. Unlike the HKS (which integrates over all time and thus acts as a low-pass filter), the WKS selects a specific frequency band, making it a “band-pass” descriptor that captures geometric information at a particular scale.

Theorem 38.9 (WKS Isometry Invariance). *The Wave Kernel Signature Theorem 38.7 is invariant under isometric deformations of the surface. That is, if (M, g) and (M, g') are isometric Riemannian manifolds with corresponding eigenpairs $\{(\lambda_k, \phi_k)\}$ and $\{(\lambda'_k, \phi'_k)\}$, then $W(x, e) = W'(f(x), e)$ for any isometry $f : M \rightarrow M'$.*

Proof. An isometry preserves the Laplace–Beltrami operator: $\Delta_{g'} = \Delta_g$ after the pullback by f . Hence the eigenvalues λ_k are identical ($\lambda'_k = \lambda_k$), and the eigenfunctions satisfy $\phi'_k(f(x)) = \phi_k(x)$. The WKS formula depends only on λ_k , $g_e(\lambda_k)$, and $\phi_k(x)^2$, all of which are invariant under the isometry. Therefore $W'(f(x), e) = \sum_k g_e(\lambda_k)^2 \phi'_k(f(x))^2 = \sum_k g_e(\lambda_k)^2 \phi_k(x)^2 = W(x, e)$. $\square$

Observation 38.10. While WKS Theorem 38.9 is isometry-invariant, it is *not* scale-invariant. Scaling the metric by a constant factor c scales eigenvalues as $\lambda_k \rightarrow c^{-2}\lambda_k$, which shifts the WKS along the energy axis. In contrast, the HKS with logarithmic time normalization achieves approximate scale invariance [BK10].

Algorithm 171 Wave Kernel Signature

```

1: function COMPUTEWKS(mesh, num_basis=100, num_scales=100)
2:    $L, M \leftarrow \text{BuildDiscreteLaplacian}(\text{mesh})$ 
3:    $\text{evals}, \text{Phis} \leftarrow \text{scipy.sparse.linalg.eigsh}(L, M, k=\text{num\_basis}, \sigma=-1e-6)$ 
4:    $e_{\min} \leftarrow \log(\text{evals}[1])$  ▷ Skip zero eigenvalue
5:    $e_{\max} \leftarrow \log(\text{evals}[-1])$ 
6:    $\text{scales} \leftarrow \text{np.linspace}(e_{\min}, e_{\max}, \text{num\_scales})$ 
7:    $\sigma \leftarrow (e_{\max} - e_{\min}) / \text{num\_scales} \times 7.0$  ▷ Heuristic width
8:    $\text{wks} \leftarrow \text{np.zeros}((\text{num\_vertices}, \text{num\_scales}))$ 
9:   for  $i, e \in \text{enumerate}(\text{scales})$  do
10:     $\text{weights} \leftarrow \exp(-(e - \log(\text{evals}))^2 / (2\sigma^2))$ 
11:     $\text{weights} \leftarrow \text{weights} / \sum(\text{weights})$ 
12:     $\text{wks}[:, i] \leftarrow (\text{Phis}^2) \cdot \text{weights}$ 
13:   end for
14:   return  $\text{wks}$ 
15: end function

```

38.15.4 Algorithm

Example 38.11 (WKS on a Sphere). On a unit sphere, the eigenvalues of the Laplace–Beltrami operator are $\lambda_k = k(k+1)$ for $k = 0, 1, 2, \dots$, with eigenfunctions being the spherical harmonics Y_k^m . Since the sphere is completely symmetric, the WKS at every vertex is identical: $W(x, e) = \sum_{k=0}^{\infty} g_e(k(k+1))^2 \sum_{m=-k}^k |Y_k^m(x)|^2$. By the addition theorem for spherical harmonics, $\sum_m |Y_k^m(x)|^2 = (2k+1)/(4\pi)$, which is constant in x . Therefore W is the same at all vertices—as expected for a shape with trivial isometry group orbits.

38.15.5 Parameter Selections

- **Number of Eigenfunctions (k):** Typically 100–300. More eigenfunctions are needed than for HKS to capture the “wave” behavior.
- **Filter Width (σ):** Controls the trade-off between spatial and spectral localization.
- **Scale Range:** Usually spans the entire computed spectrum in log-space.

38.15.6 Complexity

- **Eigen-decomposition:** $O(N^{1.5})$ for sparse matrices using Lanczos or LOBPCG solvers.
- **Signature Calculation:** $O(N \cdot k \cdot S)$ where S is the number of scales.
- **Space Complexity:** $O(N \cdot S)$ to store the descriptors.

38.15.7 Usages

- **Non-Rigid Shape Matching:** Finding corresponding points between different poses of the same object (e.g., a human walking) [ASC11].
- **Shape Retrieval:** Creating a global descriptor by pooling WKS values across the mesh [B⁺08].
- **Symmetry Detection:** Identifying vertices with similar geometric contexts.
- **Segment Classification:** Training machine learning models to identify parts of a mesh (e.g., “head,” “arm,” “leg”).

38.15.8 Testing Methods and Edge Cases

- **Isometry Invariance**Theorem 38.9: Verify that WKS is identical for a mesh in two different isometric poses.
- **Scale Invariance**: WKS is NOT naturally scale-invariant (unlike HKS with normalization). Test how it behaves when the mesh is scaled.
- **Locality**: Verify that WKS correctly identifies high-curvature regions (like fingertips or nose) as having distinct signatures.
- **Low Resolution**: Ensure the signature remains stable even on coarse triangulations.
- **Spectral Truncation**: Observe how the descriptor quality degrades as the number of basis functions k is reduced.

38.15.9 References

- Aubry, M., Schlickewei, U., & Cremers, D. (2011). “The wave kernel signature: A quantum mechanical approach to shape analysis.” *ICCV Workshops* [ASC11].
- Sun, J., Ovsjanikov, M., & Guibas, L. (2009). “A concise and provably informative multi-scale signature based on heat diffusion.” *Computer Graphics Forum* [SOG09].
- Bronstein, M. M., & Kokkinos, I. (2010). “Scale-invariant heat kernel signatures for non-rigid shape retrieval.” *CVPR* [BK10].
- Wikipedia: Spectral shape analysis.

Part XII

Modern and Research-Level Topics

Chapter 39

High-Dimensional Computational Geometry

39.1 Geometry beyond Three Dimensions

Almost every algorithm in this book was presented for the plane or for three dimensions, and its analysis relied on a geometric fact that quietly fails in high dimensions: a straight-line planar subdivision has only $O(n)$ edges, a triangulation of a point set in the plane has only $O(n)$ triangles, and two lines in the plane either meet or are parallel. When the dimension d is no longer a fixed constant but a quantity comparable to n , or even larger, the combinatorial explosion of the objects that computational geometry studies changes the rules of the game. This chapter studies what happens to the fundamental structures of the book — convex hulls, Delaunay triangulations, nearest-neighbor search, and range searching — as the dimension grows, and it develops the two families of ideas that make high-dimensional computation tractable in practice: geometric reduction, led by the Johnson–Lindenstrauss lemma, and data-dependent hashing, led by locality-sensitive hashing. Along the way we meet the curse of dimensionality, the concentration of measure that concentrates volume and distance where intuition does not expect them, and the reasons why approximate answers, not exact ones, are the right target in high dimension.

Definition 39.1 (Dimension regime). The dimension d is called *fixed* when it is a constant independent of the input size n , so that constants of the form $O(d^{O(d)})$ or $O(2^d)$ may hide inside the big- O of the running time. It is called *growing* or *high* when d is an input parameter that may be comparable to n ; the extreme case $d \approx n$ appears, for example, in the linear programming setting of Chapter 17 and in statistical applications.

The contrast with the rest of this book is sharpest when we count cells. In the plane the Voronoi diagram of n points has $O(n)$ vertices, and the combinatorial complexity of the convex hull is $O(n)$ (Chapters 13 and 4). In $\mathbb{R}^d$ the analogous counts grow like $n^{\lfloor d/2 \rfloor}$, which is linear only for $d \leq 3$. The complexity barrier is the subject of Section 39.2 and Section 39.3.

Theorem 39.2 (Complexity of convex hulls). *The number of facets of the convex hull of n points in $\mathbb{R}^d$ is at most $O(n^{\lfloor d/2 \rfloor})$, and this bound is tight: for every fixed d there are point sets whose hull has $\Theta(n^{\lfloor d/2 \rfloor})$ facets. The same bound holds for the number of d -simplices of the Delaunay triangulation and for the number of cells of the Voronoi diagram.*

Section 39.8 explains the counterintuitive geometry of high-dimensional balls, Gaussians, and convex bodies, and Section 39.9 converts that geometry into an algorithm analysis: why nearest neighbors lose their meaning, why range trees and kd -trees degrade to linear scans, and why worst-case approximation is hard. The two positive threads run through Sections 39.5, 39.6,

and 39.7 (random embeddings that compress the data while approximately preserving pairwise distances) and Sections 39.4 and 39.9 (hashing schemes whose query cost is a sublinear power of n). Section 39.10 ties the techniques to the machine learning pipeline: k -nearest-neighbor classifiers, feature maps, and the algorithmic geometry of learning.

Throughout the chapter the norm $\|\cdot\|$ is the Euclidean norm unless stated otherwise, and a point set is written $P \subset \mathbb{R}^d$ with $|P| = n$.

39.2 High-Dimensional Convex Hulls

39.2.1 1. Overview

The convex hull problem is the most basic one in computational geometry, and its high-dimensional version is the cleanest illustration of why dimension is a cost parameter. Chapter 4 showed that the planar hull of n points can be computed in $O(n \log n)$ time, matching the information theoretic lower bound, and that the hull has at most $2n$ vertices. In $\mathbb{R}^d$ the hull can have $\Theta(n^{\lfloor d/2 \rfloor})$ facets (Theorem 39.2); for $d = 4$ this is quadratic in n , and for d around $\log n$ it is $n^{\Omega(\log n / \log \log n)}$, larger than any polynomial. Any algorithm that writes the hull explicitly must therefore spend at least $n^{\lfloor d/2 \rfloor}$ time, so the only meaningful question is whether the combinatorial explosion of the output is the *only* cost. The answer is yes: randomized incremental construction achieves expected time proportional to the output size, plus an $O(n \log n)$ sorting term, for every fixed d .

Theorem 39.3 (Randomized incremental hulls). *For any fixed dimension d , the convex hull of n points in general position in $\mathbb{R}^d$ can be computed in expected $O(n \log n + n^{\lfloor d/2 \rfloor})$ time. For $d \leq 3$ the bound becomes $O(n \log n)$.*

The proof of Theorem 39.3 is a high-dimensional version of the randomized incremental argument of Chapter 31: insert the points one by one in random order, and amortize the cost of a point against the facets it destroys. The analysis depends on a critical fact called the Clarkson–Shor technique: the expected number of facets destroyed by a random insertion equals, up to constant factors, the expected number of facets of a random sample, and the latter can be bounded directly.

39.2.2 2. Definitions

Definition 39.4 (Facet, ridge, horizon). A *facet* of a d -dimensional convex polytope is a $(d - 1)$ -dimensional face. A *ridge* is a $(d - 2)$ -dimensional face. In a simplicial polytope every facet is a simplex spanned by d vertices and every ridge is spanned by $d - 1$ vertices. Given a point p outside a polytope H , a facet f is *visible from p* if p lies in the open halfspace outside the supporting hyperplane of f . The *horizon* of p on H is the set of ridges that belong to exactly one visible facet.

In the plane the horizon of an outside point is a pair of vertices and the visible facets form a contiguous chain along the polygon boundary. In $\mathbb{R}^d$ the visible facets still form a topological disk on the hull, bounded by the horizon ridges; this is the only structural fact the algorithm needs.

Definition 39.5 (Cyclic polytope). The *cyclic polytope* $C_d(n)$ is the convex hull of n points on the moment curve $\{(t, t^2, \dots, t^d) : t \in \mathbb{R}\}$. It is *neighborly*: every $\lfloor d/2 \rfloor$ vertices form a face. Its number of facets is $O(n^{\lfloor d/2 \rfloor})$, and it is the extremal polytope achieving the maximum number of facets among all d -polytopes with n vertices.

Cyclic polytopes show that the upper bound of Theorem 39.2 is attainable and that the answer to any algorithmic question about hulls in dimension $d \geq 4$ must be input-sensitive.

39.2.3 3. Theory

The high-dimensional hull inherits the polarity duality of Chapter 4: facets of the hull of P correspond to vertices of the polar polytope. The structural result underlying all exact algorithms is the upper bound theorem (Theorem 39.2), and the algorithmic result is that the hull can be found within the output bound plus $O(n \log n)$.

A word on lower bounds. Computing the hull of n points in $\mathbb{R}^d$ requires $\Omega(n \log n)$ time (sorting reduces to it) and $\Omega(n^{\lfloor d/2 \rfloor})$ time (the output may be that large). For fixed d the randomized incremental algorithm matches both bounds in expectation; the deterministic algorithm of Chazelle achieves the same asymptotics and establishes that the problem has optimal worst-case complexity in every fixed dimension.

39.2.4 4. Pseudocode

Algorithm 172 is the randomized incremental algorithm in its full generality. It reduces to the planar incremental algorithm of Chapter 4 for $d = 2$.

Algorithm 172 Randomized Incremental Convex Hull in $\mathbb{R}^d$

Input: A set P of n points in general position in $\mathbb{R}^d$.

Output: The facets of the convex hull of P .

```

1: function INCREMENTALHULL( $P$ )
2:   shuffle the points of  $P$  uniformly at random
3:   let  $H$  be the hull of the first  $d + 1$  points
4:   for each remaining point  $p$  do
5:     mark as visible every facet of  $H$  that has  $p$  on its outer side
6:     if no facet of  $H$  is visible from  $p$  then
7:       continue ▷  $p$  lies inside  $H$ 
8:     end if
9:     collect the horizon: ridges of  $H$  shared by one visible and one hidden facet
10:    for each horizon ridge  $r$  do
11:      add the new facet spanned by  $r$  and  $p$ 
12:    end for
13:    delete all visible facets from  $H$ 
14:  end for
15:  return  $H$ 
16: end function

```

The inner loop over facets is the whole algorithm in the plane; the only genuinely d -dimensional bookkeeping is locating the visible facets and walking their boundary to extract the horizon. Both are handled by a facet incidence graph maintained incrementally. Each inserted point either lies inside (and is discarded) or destroys the visible facets and creates the cone of new facets over the horizon.

39.2.5 5. Parameter Selection

The algorithm has no numerical parameters, but its analysis depends on the assumption of general position: no $d + 1$ points coplanar with a facet and no point lying on a supporting hyperplane. In practice one either perturbs the input symbolically or, better, works with an exact arithmetic layer that evaluates the orientation predicate of the facet with respect to p ; this is the d -dimensional determinant $\text{orient}(p_0, \dots, p_d)$, the natural generalization of the predicate used throughout this book. Exact computation is essential because a single misclassified visibility test invalidates the horizon walk and can destroy the combinatorial structure.

39.2.6 6. Complexity

Each point inserted destroys a set of facets and creates new ones; over the whole run the number of created facets is bounded by the expected output size. The Clarkson–Shor argument shows that the expected number of facets created by the i -th insertion is $O(i^{\lfloor d/2 \rfloor - 1})$, so the total expected number of created facets is $O(n^{\lfloor d/2 \rfloor})$, and each creation is charged a constant amount of work. The expected running time is therefore

$$O(n \log n + n^{\lfloor d/2 \rfloor}),$$

with the logarithmic term coming only from the initial shuffle and from accessing each point once. The space is proportional to the number of facets currently alive, $O(n^{\lfloor d/2 \rfloor})$ in the worst case.

39.2.7 7. Usages

The incremental hull is the engine behind fixed-dimensional Delaunay triangulations via lifting (Section 39.3), behind the computation of regular triangulations and secondary polytopes, and behind the fixed-dimensional linear programming algorithms studied in Chapter 17, whose randomized incremental structure is identical. It is also the standard route to the computation of the volume of convex polytopes in fixed dimension, where one triangulates the hull into d -simplices. For growing dimension the algorithm is unusable, since the output bound $n^{\lfloor d/2 \rfloor}$ is exponential in d ; the remedies are linear programming duality when only a single linear functional is sought (Section 39.8) and approximation, which is the subject of the rest of this chapter.

39.2.8 8. Testing Methods and Edge Cases

Edge cases to test: points all inside a small cloud (interior insertions must be rejected by the visibility test without modifying the hull); points lying on facet supporting hyperplanes (requires symbolic perturbation); duplicate and coplanar points (handled by the general position convention); and the degenerate starting configuration when the first $d + 1$ points are affinely dependent. A useful invariant-based test verifies, after every insertion, that every facet orientation is consistently outward and that the facet graph is manifold: every ridge bounds exactly two facets. For validation against brute force, enumerate all $d + 1$ -subsets and check that the facets computed are exactly the subsets whose supporting hyperplane leaves every other point on one side; this is exponential in n and only feasible for $n \leq 20$. The extremal cases are the cyclic polytopes of Definition 39.5, which exercise the upper-bound-size output and the horizon machinery maximally.

39.2.9 9. References

The incremental hull and its Clarkson–Shor analysis appear in [Mul94]; the deterministic optimal algorithm is due to [Cha93]. The upper bound theorem and cyclic polytopes are treated in [Mat02]. The standard planar treatment appears in [BCKO08], and the fixed-dimensional hull is surveyed in [dBCvKO08].

39.3 Voronoi and Delaunay Structures

39.3.1 1. Overview

Chapter 13 developed the planar Voronoi diagram and its dual, the Delaunay triangulation. Both objects remain well defined in $\mathbb{R}^d$: the Voronoi cell of a site is the set of points at least

as close to that site as to any other, and the Delaunay complex connects two sites when there is a sphere through them with no other site inside. The combinatorial complexity, however, explodes from $O(n)$ to $n^{\lceil d/2 \rceil}$, and — more dramatically — the Delaunay complex is no longer necessarily a triangulation of the space. Beyond the plane the Delaunay complex may have missing cells, and its topology is not in general a valid subdivision of $\mathbb{R}^d$. Understanding this failure is essential for meshing and for the geometry processing algorithms of Chapter 30.1 scaled to higher dimensions.

39.3.2 2. Definitions

Definition 39.6 (Voronoi cell, Delaunay triangulation). For a finite site set $P \subset \mathbb{R}^d$ and a site $p \in P$ the *Voronoi cell* of p is

$$\text{vor}(p) = \{x \in \mathbb{R}^d : \|x - p\| \leq \|x - q\| \text{ for all } q \in P\}.$$

The *Delaunay complex* of P is the collection of all subsets $S \subseteq P$ for which there exists a sphere passing through the points of S and containing no point of P in its interior. When every Delaunay cell is a d -simplex, the complex is a triangulation of $\text{conv}(P)$ and is called the *Delaunay triangulation*.

The k -face $(k + 1)$ -dimensional Voronoi cells are in bijection with the $(d - k)$ -dimensional Delaunay faces, exactly as in the plane, and the whole structure is captured by the lifting to the paraboloid.

Definition 39.7 (Lifting map). The *lifting map* sends $x \in \mathbb{R}^d$ to $\hat{x} = (x, \|x\|^2) \in \mathbb{R}^{d+1}$, the point on the paraboloid $y = \|x\|^2$. A sphere with center c and radius r in $\mathbb{R}^d$ lifts to a hyperplane in $\mathbb{R}^{d+1}$ that passes through the lifted images of all points of the sphere.

39.3.3 3. Theory

The lifting map is the master reduction. The Delaunay complex of P in $\mathbb{R}^d$ is the projection of the lower hull of the lifted points $\{\hat{p} : p \in P\}$ in $\mathbb{R}^{d+1}$: a k -simplex of the Delaunay complex corresponds to a $(k + 1)$ -facet of the lower hull. This is the statement proved in the plane in Chapter 13, and it extends verbatim to all dimensions. The consequence is an immediate transfer of every hull algorithm from Section 39.2:

Theorem 39.8 (Complexity of Delaunay and Voronoi). *For any fixed dimension d , the Delaunay complex of n points in $\mathbb{R}^d$ has at most $O(n^{\lceil d/2 \rceil})$ d -simplices and can be computed in expected $O(n \log n + n^{\lceil d/2 \rceil})$ time. The Voronoi diagram has the same complexity up to constant factors.*

Two structural warnings follow from the same theorem. First, for $d \geq 3$ the number of d -simplices can be superlinear, so the total work is dominated by the size of the output, which in turn can be made large by lifting cyclic polytopes. Second, the Delaunay complex need not be a triangulation of $\text{conv}(P)$: some d -cells may be missing, and the complex may fail to be a manifold. The Delaunay triangulation in $\mathbb{R}^3$ (for example) exists only for point sets that are inscribable in the right sense; this is a geometric obstruction with no planar analogue and is the reason meshing in higher dimension is qualitatively harder than the surface meshing of Chapter 22.

39.3.4 4. Pseudocode

Algorithm 173 reduces the Delaunay computation to the incremental hull of Algorithm 172 applied to lifted points. It is the same strategy used in the plane, now without the help of the planar structure.

Algorithm 173 Delaunay Complex in $\mathbb{R}^d$ by Lifting**Input:** A set P of n points in general position in $\mathbb{R}^d$.**Output:** The Delaunay complex of P .

```

1: function DELAUNAY( $P$ )
2:   compute  $\hat{P} = \{(p, \|p\|^2) : p \in P\} \subset \mathbb{R}^{d+1}$ 
3:   compute the convex hull  $H$  of  $\hat{P}$  using INCREMENTALHULL
4:   collect the facets of  $H$  whose outward normal has negative last coordinate  $\triangleright$  lower hull
5:   project those facets back to  $\mathbb{R}^d$  by dropping the last coordinate
6:   return the projected complex
7: end function

```

The orientation of the lower hull is decided by the sign of the last coordinate of the outward normal, which is computed from the facet vertices by the d -dimensional orientation predicate; this is the direct analogue of the empty-circle test used in the plane.

39.3.5 5. Parameter Selection

For exact construction one computes, for each candidate $(d+1)$ -simplex, the sign of the determinant of the lifted coordinates, an empty-sphere test that is the d -dimensional generalization of the incircle test. In floating point the signs are unstable when the radius is much larger than the local scale of the sites; symbolic perturbation of the coordinates resolves ties. The relevant scale parameter is the minimum feature size of the site set, which any robust implementation must estimate before choosing the working precision.

39.3.6 6. Complexity

Lifting adds one dimension, so the hull bound $O(n \log n + n^{\lfloor (d+1)/2 \rfloor})$ of Theorem 39.3 becomes $O(n \log n + n^{\lfloor d/2 \rfloor})$, matching Theorem 39.8. The Voronoi diagram is the dual cell complex and is computed in the same time. The constants hidden in the big- O grow super-exponentially with d (roughly $d^{d/2}$), so the algorithm is practical only for $d \leq 6$ or so; beyond that the output itself is astronomically large and approximation replaces construction.

39.3.7 7. Usages

Delaunay triangulations in $\mathbb{R}^3$ are a standard tool in mesh generation, and in dimension four they appear in the regular triangulations used to compute volumes and mixed volumes. The lifting reduction is the template for the more general concept of regular triangulations, where the vertical shift of each lifted point is allowed to vary, and it underlies the theory of secondary polytopes. In statistical geometry, Delaunay complexes of point clouds appear as the combinatorial part of persistent homology computations.

39.3.8 8. Testing Methods and Edge Cases

The primary correctness check is the empty-sphere predicate: every returned simplex must admit a sphere through its vertices with no site strictly inside. The dual check verifies that the projection of the lower hull has exactly the combinatorial type of the Delaunay complex, which is provably the union of the top faces of the so-called power diagram. Test configurations include regularly spaced grids (maximal number of simplices), cocircular sites in a hyperplane (flat lower hull), and the lifted cyclic polytope, which produces the maximal Delaunay complex and stresses the manifold test. A hidden trap is that the lower hull may project to a complex that is not a triangulation; the implementation must not assume that every d -cell is a simplex.

39.3.9 9. References

The lifting construction and its properties are treated in [BCKO08] for the plane and in [Mat02] for general dimension; the Delaunay computations via hull are surveyed in [dBCvKO08]. The randomized incremental analysis underlying Theorem 39.8 is in [Mul94].

39.4 Nearest-Neighbor Search

39.4.1 1. Overview

Chapter 13 solved planar nearest-neighbor queries with data structures of linear size and polylogarithmic query time, and Chapter 9 showed that range trees and kd -trees push the same paradigm into higher dimensions, at the cost of $O(n \log^{d-1} n)$ storage and $O(n^{1-1/d})$ query time. The degradation is structural: for d comparable to $\log n$ these bounds become polynomial and the structure degenerates into a linear scan. The solution for growing dimension is to give up exactness and to solve the approximate nearest-neighbor problem, for which sublinear query time is provably achievable by locality-sensitive hashing (LSH). This section develops the LSH framework of Indyk and Motwani, its instantiation on Hamming and Euclidean spaces, and the near-optimal schemes built on top of it.

39.4.2 2. Definitions

Definition 39.9 (Approximate near neighbor). Fix a metric space, a radius $r > 0$ and an approximation factor $c > 1$. The (r, c) -near-neighbor problem is: preprocess a set P of n points so that, given a query q , if there exists a point $p \in P$ with $\|p - q\| \leq r$, the structure returns a point $p' \in P$ with $\|p' - q\| \leq cr$. The answer is allowed to be any point within distance cr , and points within distance r may be missed. The *approximate nearest-neighbor* problem asks for a point within factor c of the true nearest neighbor.

Definition 39.10 (Sensitive hash family). A family $\mathcal{H}$ of hash functions on the point space is (r, cr, p_1, p_2) -sensitive for $p_1 > p_2$ if for uniformly random $h \in \mathcal{H}$ and all points p, q ,

$$\Pr[h(p) = h(q)] \geq p_1 \quad \text{whenever} \quad \|p - q\| \leq r,$$

$$\Pr[h(p) = h(q)] \leq p_2 \quad \text{whenever} \quad \|p - q\| \geq cr.$$

The two probabilities are the workhorse parameters: close points collide often, far points collide rarely.

39.4.3 3. Theory

The LSH construction amplifies a single weak hash family into a data structure. Concatenate k independent hash functions into $g(p) = (h_1(p), \dots, h_k(p))$ and build L independent hash tables under independent concatenations $g_1, \dots, g_L$. A query hashes itself with each g_i and inspects the points in the resulting buckets. The two design equations fix k and L :

$$k = \left\lceil \frac{\log n}{\log(1/p_2)} \right\rceil, \quad L = n^\rho, \quad \rho = \frac{\log(1/p_1)}{\log(1/p_2)}.$$

With these choices a fixed near neighbor p^* collides with q under a single g_i with probability $p_1^k \approx 1/L$, so it is found across the L tables with constant probability; meanwhile an arbitrary far point collides with probability at most $p_2^k = 1/n$, so the expected number of far candidates inspected in any one bucket is 1.

Theorem 39.11 (Indyk–Motwani bound). *Let $P \subset \{0, 1\}^D$ be a set of n points and let $\mathcal{H}$ be the family of coordinate projections (each h reads one coordinate). Then the (r, cr) -near-neighbor problem on the Hamming cube can be solved with a data structure of size $O(n^{1+\rho})$, query time $O(D \cdot n^\rho \cdot \log n)$ and constant success probability, where $\rho = \log(1/p_1)/\log(1/p_2)$ and $p_1 = 1 - r/D$, $p_2 = 1 - cr/D$. For small r/D the exponent is $\rho \approx 1/c$.*

The Euclidean analogue uses the p -stable families of Section 39.7, which give $\rho < 1/c$, and the asymptotically optimal schemes reach $\rho = 1/c^2 + o(1)$. An exact nearest-neighbor query is reduced to (r, c) -near-neighbor queries by a distance-doubling search: guess the distance to the nearest neighbor as powers of two and run the near-neighbor query for each guess.

39.4.4 4. Pseudocode

Algorithm 174 is the preprocessing and query template. The only free ingredients are the underlying family $\mathcal{H}$ and the parameter pair (k, L) .

Algorithm 174 Locality-Sensitive Hashing for (r, c) -Near Neighbor

Input: Points P , radius r , factor $c > 1$, and an (r, cr, p_1, p_2) -sensitive family $\mathcal{H}$.

Output: A structure returning a point within distance cr of q if one is within r .

```

1: function LSHPREPROCESS( $P$ )
2:    $k \leftarrow \lceil \log_{1/p_2} n \rceil$                                 ▷ concatenation length
3:    $\rho \leftarrow \log(1/p_1)/\log(1/p_2)$ 
4:    $L \leftarrow n^\rho$                                           ▷ number of tables
5:   for  $i = 1, \dots, L$  do
6:     draw  $h_{i,1}, \dots, h_{i,k}$  from  $\mathcal{H}$ 
7:     let  $g_i(p) = (h_{i,1}(p), \dots, h_{i,k}(p))$ 
8:     store every  $p \in P$  in table  $i$  under key  $g_i(p)$ 
9:   end for
10: end function
11: function LSHQUERY( $q$ )
12:    $seen \leftarrow 0$ 
13:   for  $i = 1, \dots, L$  do
14:     for each point  $p$  in bucket  $g_i(q)$  do
15:        $seen \leftarrow seen + 1$ 
16:       if  $\|p - q\| \leq cr$  then
17:         return  $p$ 
18:       end if
19:       if  $seen > 3L$  then
20:         return none
21:       end if
22:     end for
23:   end for
24:   return none
25: end function

```

39.4.5 5. Parameter Selection

The family $\mathcal{H}$ must be chosen for the metric at hand. On the Hamming cube $\{0, 1\}^D$ take coordinate projections; a point at distance u collides with the query with probability $1 - u/D$, yielding $p_1 = 1 - r/D$ and $p_2 = 1 - cr/D$. For Euclidean space the p -stable families of Section 39.7 hash by $\lfloor (a \cdot x + b)/W \rfloor$ with a p -stable random vector a ; the bucket width W trades the two

probabilities. Increasing k decreases both probabilities but sharpens the ratio; increasing L (equivalently increasing ρ) raises the success probability and the storage. In practice one chooses W by a small grid search over a holdout set and raises L until the empirical success probability is acceptable. The hash tables themselves are maintained as arrays indexed by g_i ; a secondary collision hash on the concatenated keys handles their variable size.

39.4.6 6. Complexity

Storage is $O(n \cdot (d + L)) = O(dn + n^{1+\rho})$. Each query evaluates L concatenations, each costing $O(dk)$ for the projections (or $O(k)$ per non-zero coordinate on sparse data), and inspects at most $O(L)$ candidate points, each verified in $O(d)$. The total query time is $O(d \cdot L \cdot k + dL) = O(dn^\rho \log n)$. The trade-off between the approximation factor and the exponent ρ is the central quantitative fact: $\rho \approx 1/c$ for the basic families, $\rho < 1/c$ for the p -stable families, and $\rho = 1/c^2 + o(1)$ for the near-optimal schemes. Lower bounds for hashing-based schemes state that $\rho \geq 1/c^2$ is unavoidable, so the near-optimal schemes are optimal within the LSH framework.

39.4.7 7. Usages

LSH is the workhorse of high-dimensional similarity search in practice: near duplicate detection in web corpora, image and video retrieval, de-duplication in databases, and genome sequence search. It is also the standard subroutine inside k -means-like clustering over high-dimensional data, where exact nearest-neighbor structures are useless. Section 39.10 returns to the machine learning applications; the connection to data structures for high-dimensional data is treated in the handbook [MS05a].

39.4.8 8. Testing Methods and Edge Cases

The canonical test compares the structure against a brute-force nearest neighbor on small random data, measuring both the success probability and the observed approximation factor over many queries. Edge cases: queries far from all data (the structure must terminate, bounded by the $3L$ cutoff); clusters of nearly coincident points (collision probability near 1 makes buckets large); and the parameter extremes $c \rightarrow 1^+$, where $\rho \rightarrow 1$, i.e. the structure degenerates to a linear scan. A robust implementation must also tolerate adversarial hash keys, where a secondary exact hash function collides, by comparing candidate points with the true distance rather than by trusting the bucket membership.

39.4.9 9. References

The framework and the Hamming analysis are due to [IM98] and [GIM99]. The Euclidean instantiation via p -stable distributions is [DIIM04], and the near-optimal schemes with $\rho = 1/c^2 + o(1)$ are [AI06]. The distance-concentration phenomenon that motivates approximation is analyzed in [BGRS99]. The modern practical HNSW graphs, which combine the LSH idea with greedy navigation, are presented in [MY20].

39.5 Dimensionality Reduction

39.5.1 1. Overview

The recurring obstacle in Sections 39.2 and 39.3 is that combinatorial objects in $\mathbb{R}^d$ have complexity exponential in d . The response of the field is *dimensionality reduction*: replace the point set by a small set of sketches in a low-dimensional space, chosen so that the quantities the application actually cares about (pairwise distances, nearest neighbors, dot products) are

approximately preserved. The guiding statement is the Johnson–Lindenstrauss lemma: n points can be embedded into $\mathbb{R}^k$ with $k = O(\log n)$ while preserving all pairwise distances within a factor $1 \pm \varepsilon$. This section states the lemma, discusses the information-theoretic intuition behind its logarithmic bound, and positions it relative to the exact theory of metric embeddings and to the linear algebraic methods used in practice.

39.5.2 2. Definitions

Definition 39.12 (Distortion of an embedding). A map $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ has *distortion* at most $1 + \varepsilon$ on a set P if for all $p, q \in P$,

$$(1 - \varepsilon) \|p - q\|^2 \leq \|f(p) - f(q)\|^2 \leq (1 + \varepsilon) \|p - q\|^2.$$

The squared normalization is a convention that removes square roots from the analysis; the equivalent statement for the un-squared distance holds with $(1 \pm \varepsilon)$ replaced by $(1 \pm 3\varepsilon)$.

A linear embedding is a matrix $A \in \mathbb{R}^{k \times d}$; it is *norm-preserving* for a unit vector x if $\|Ax\|^2$ is close to 1. Random constructions draw A entrywise from a zero-mean distribution scaled so each entry has variance $1/k$, in which case $\mathbb{E}[\|Ax\|^2] = \|x\|^2$ for every x : the random map is unbiased in expectation, and the question becomes how tightly it concentrates.

39.5.3 3. Theory

Why is the target dimension logarithmic in n and quadratic in $1/\varepsilon$? The lower bound is forced by two simple arguments. First, n equidistant points form a regular simplex in $\mathbb{R}^{n-1}$, and embedding it faithfully needs roughly $\log n$ dimensions once the distances may only be approximate. Second, the variance of the squared norm of a Gaussian projection of a unit vector is about $2/k$, so guaranteeing a $1 \pm \varepsilon$ multiplicative bound requires $k = \Omega(1/\varepsilon^2)$; concentration cannot do better. Both bounds are achieved, up to constants, by the Gaussian and Rademacher constructions of Section 39.7.

Theorem 39.13 (Johnson–Lindenstrauss). *Let P be a set of n points in $\mathbb{R}^d$ and let $\varepsilon \in (0, 1)$. For any $k \geq c\varepsilon^{-2} \log n$ with an absolute constant c , a random $k \times d$ matrix with independent Gaussian or Rademacher entries of variance $1/k$ satisfies the distortion bound of Definition 39.12 on P with probability at least $1 - 1/n$.*

The dimension reduction view extends from points to subspaces. A random matrix with $k = O(d/\varepsilon^2)$ rows is a subspace embedding: it preserves, up to factor $1 \pm \varepsilon$, the norm of every vector lying in a fixed d -dimensional subspace. This single statement packages the whole of the modern randomized linear algebra (approximate SVD, least-squares regression, low-rank approximation), where the subspace is the one spanned by the data columns and the random map compresses the problem before a deterministic solver runs on the sketch.

39.5.4 4. Pseudocode

The embedding itself is a single matrix multiplication (Algorithm 175); the data structure of this section is the selection of the matrix and the dimension, discussed next.

The reduction in dimension is dramatic when the data is intrinsically low-rank: computing pairwise distances in $\mathbb{R}^k$ with $k \approx \log n$ instead of $\mathbb{R}^d$ with d in the thousands is the difference between feasible and infeasible. The cost is a single $k \times d$ matrix multiplication per point, which on sparse data reduces to a sum over non-zero entries.

Algorithm 175 Johnson–Lindenstrauss Embedding**Input:** Points $p_1, \dots, p_n \in \mathbb{R}^d$, distortion $\varepsilon \in (0, 1)$.**Output:** Sketches $q_1, \dots, q_n \in \mathbb{R}^k$ with $k = O(\varepsilon^{-2} \log n)$.

```

1: function JLEMBED( $p_1, \dots, p_n, \varepsilon$ )
2:    $k \leftarrow \lceil c \ln n / \varepsilon^2 \rceil$  ▷ the constant from the theory
3:   draw a  $k \times d$  matrix  $A$  with i.i.d. entries of variance  $1/k$ 
4:   for  $i = 1, \dots, n$  do
5:      $q_i \leftarrow Ap_i$ 
6:   end for
7:   return  $q_1, \dots, q_n$ 
8: end function

```

39.5.5 5. Parameter Selection

The two parameters are the distortion ε and the constant c hidden in the bound. For most applications $\varepsilon \in (0.05, 0.2)$ is the practical range: smaller values push k above ≈ 100 even for moderate n , and larger values distort the metric enough to corrupt nearest neighbor or clustering results. The constant c is implementation dependent; choosing $k = 2\varepsilon^{-2} \ln n$ is the common heuristic and works in practice because the union bound over pairs is loose. When the downstream task is clustering or k -NN classification, the distortion can often be larger than the theory suggests, since only the local metric needs preservation.

39.5.6 6. Complexity

Embedding n points costs $O(ndk)$ time with a dense matrix and $O(\text{nnz} \cdot k)$ on sparse data, where nnz is the number of non-zeros of the data matrix. The resulting sketch space is $O(nk) = O(n\varepsilon^{-2} \log n)$ words. A random projection can never reduce the intrinsic rank of the data, but it does compress the ambient dimension to the information-theoretic minimum $k = \Theta(\log n)$ for distance preservation. When a deterministic guarantee is required, one may verify the distortion empirically on the pair set and rerun with fresh randomness until the bound holds.

39.5.7 7. Usages

Dimensionality reduction is the standard preprocessing in machine learning pipelines: before running a k -means clustering, a k -NN classifier, or a Gaussian kernel method, the data is embedded into $O(\log n)$ dimensions with negligible loss (Section 39.10). It is also the tool that makes high-dimensional range searching and nearest-neighbor structures (Sections 39.4 and 13.4) usable at scale, because the data structures of Chapters 9 and 13 depend exponentially on the ambient dimension.

39.5.8 8. Testing Methods and Edge Cases

Validate by computing the maximum pairwise distortion of the sketch against the original over held-out pairs; the empirical maximum should be within $1 \pm 2\varepsilon$. Edge cases: points already in low dimension (the map should leave them essentially unchanged); nearly collinear clouds (the sketched distances concentrate at the diameter, hiding the internal structure); and points with a huge spread in norms, where multiplicative error is acceptable but additive error is not. A hidden trap is numerical noise: entries of variance $1/k$ are small when k is large, and single precision can destroy the sketch; use double precision in the accumulation.

39.5.9 9. References

The foundational statement is [JL84]. The standard textbook treatment, including the constant and the lower bounds, is [Mat02]. Practical surveys and the connections to nearest neighbor and learning appear in [IM98] and [HP11].

39.6 Johnson–Lindenstrauss Lemma

39.6.1 1. Overview

This section proves Theorem 39.13. The proof is a model for the rest of high-dimensional analysis: it is a concentration argument (one-dimensional, then union-bounded over pairs) wrapped around a linear map. We give the two standard constructions — Gaussian entries and Rademacher entries — and the concentration inequality behind each, and we discuss tightness. The material of Section 39.8 supplies the probabilistic machinery, so this section may be read as its first application.

39.6.2 2. Definitions

Let A be a $k \times d$ matrix with i.i.d. entries A_{ij} of mean 0 and variance $1/k$. For a fixed unit vector x , the vector $y = Ax$ has independent coordinates

$$y_i = \sum_{j=1}^d A_{ij} x_j,$$

each of mean 0 and variance $\sum_j (1/k) x_j^2 = 1/k$. Its squared norm $\|y\|^2 = \sum_i y_i^2$ has expectation 1; the whole proof is the statement that $\|y\|^2$ is concentrated about 1. For Gaussian entries, $k\|y\|^2$ is a chi-squared random variable with k degrees of freedom. For Rademacher entries, a Hoeffding-type bound on y_i followed by a sub-exponential tail on y_i^2 gives the same result.

39.6.3 3. Theory

Theorem 39.14 (One-dimensional concentration). *Let $y = Ax$ for a fixed unit vector x and a $k \times d$ matrix with independent Gaussian or Rademacher entries of variance $1/k$. For any $t \in (0, 1)$,*

$$\Pr [\|y\|^2 \geq 1 + t] \leq e^{-ckt^2}, \quad \Pr [\|y\|^2 \leq 1 - t] \leq e^{-ckt^2},$$

for an absolute constant $c > 0$.

Proof of Theorem 39.14. For Gaussian entries, write $y_i = g_i/\sqrt{k}$ with g_i i.i.d. standard Gaussians, so $k\|y\|^2 = \sum_i g_i^2 \sim \chi_k^2$. The chi-squared tail bound

$$\Pr \left[\frac{1}{k} \sum_{i=1}^k g_i^2 - 1 \geq t \right] \leq \exp \left(-\frac{k}{2}(t - \ln(1+t)) \right),$$

together with $t - \ln(1+t) \geq t^2/4$ for $t \in (0, 1)$, gives the upper tail; the lower tail is symmetric by the same argument on $-y$. For Rademacher entries, y_i is a sum of independent bounded variables, so Hoeffding's inequality gives a Gaussian tail on y_i , and summing the truncated squares $y_i^2 \wedge M$ with a Bernstein bound yields the same exponential tail.

Theorem 39.15 (Johnson–Lindenstrauss, proved). *For $n \geq 2$, $\varepsilon \in (0, 1)$, and $k \geq 8\varepsilon^{-2} \ln n$, the map of Theorem 39.13 satisfies the distortion bound on all $\binom{n}{2}$ pairs simultaneously with probability at least $1 - 1/n$.*

Proof. For a fixed pair (p, q) , apply Theorem 39.14 to the unit vector $x = (p - q)/\|p - q\|$, noting that $A(p - q) = Ap - Aq$ and that the squared norm scales by $\|p - q\|^2$. The two tails each cost at most $e^{-ck\epsilon^2}$; union bound over the $\binom{n}{2}$ pairs gives failure probability at most $n^2 e^{-ck\epsilon^2} \leq 1/n$ for $k \geq (3/c)\epsilon^{-2} \ln n$. Absorbing the constants yields the claimed bound.

39.6.4 4. Pseudocode

The construction is Algorithm 175; its correctness follows from the two theorems. The Rademacher variant replaces each Gaussian by $\pm 1/\sqrt{k}$ with equal probability, which avoids random-number generation of continuous distributions and enables the hashed implementation of Section 39.7.

39.6.5 5. Parameter Selection

The constant 8 in the bound is an artifact of the union bound and the concentration constant; smaller values work in practice. Two practical refinements matter. First, centering: subtracting the mean point before embedding does not change distances but reduces the scale against which floating point error is measured. Second, when only k can be chosen (not the constant), setting $k = \lceil 2 \ln n / \epsilon^2 \rceil$ is the de facto standard; Section 39.7 discusses the consequences of the choice of the distribution.

39.6.6 6. Complexity

The embedding is a single dense multiplication, $O(ndk)$ time and $O(dk)$ space for the matrix; with $k = O(\log n)$ this is $O(nd \log n / \epsilon^2)$ for the whole data set. The Rademacher version has the same complexity but faster constants, and the sparse variants of Section 39.7 reduce the multiplication cost to the number of non-zeros.

39.6.7 7. Usages

The lemma is invoked wherever a high-dimensional metric is to be preserved at logarithmic cost: before any nearest-neighbor or clustering computation (Section 39.10), inside kernel methods via random features, and as the standard tool in data stream sketching, where only the projection of each incoming point is stored. It also underlies the theoretical reduction of many high-dimensional problems to their $d = O(\log n)$ instances.

39.6.8 8. Testing Methods and Edge Cases

Verify the embedding empirically on random Gaussian point sets with $n = 10^5$, $d = 10^3$, $\epsilon = 0.1$: the maximum distortion should stay below $1 + 0.1$ with high probability. Edge cases: nearly identical points (their difference is a near-unit vector, still concentrated); points at the origin (trivially preserved); adversarial points chosen after the matrix is fixed (the union bound fails, and one must either resample or verify deterministically). Numerical care is required: the entries of A scale as $1/\sqrt{k}$ and the products Ap accumulate round-off proportional to $\|p\|\sqrt{k}$.

39.6.9 9. References

The lemma is from [JL84]; the modern proof via concentration is standard and is presented in [Mat02]. The chi-squared tail bound used in the proof is derived in Section 39.8, and the Rademacher and sparse constructions are [Ach03]. The dimension bound is essentially optimal, as discussed in [Mat02].

39.7 Random Projections

39.7.1 1. Overview

A random projection is a concrete matrix distribution with the property that every norm is approximately preserved. Section 39.6 proved that Gaussian and Rademacher matrices work; this section is about the engineering: which distributions are used in practice, how to hash them so that a dense matrix is never materialized, how to use them for subspace embeddings and approximate linear algebra, and how the same idea instantiates the p -stable hash families of Section 39.4.

39.7.2 2. Definitions

Definition 39.16 (Random projection matrix). A *random projection* is a random $k \times d$ matrix A whose entries are independent, of mean 0 and variance $1/k$, drawn from one of the distributions below. Applying A to a point set is *random projection*; the image points are called *sketches*.

The three classical entry distributions are:

- *Gaussian*: $A_{ij} \sim \mathcal{N}(0, 1/k)$. Optimal constants, costs one normal draw per entry.
- *Rademacher*: $A_{ij} \in \{+1/\sqrt{k}, -1/\sqrt{k}\}$ each with probability $1/2$. No continuous sampling; supports hashing.
- *Sparse (Achlioptas)*: $A_{ij} \in \{0, \pm\sqrt{3/k}\}$ with $\Pr[\pm] = 1/6$ and $\Pr[0] = 2/3$. Each entry has variance $(3/k) \cdot (1/3) = 1/k$; only one third of the entries are non-zero.

A p -stable random vector $a \in \mathbb{R}^d$ is one such that $a \cdot x$ has the same distribution as $\|x\|_p X$ for a fixed p -stable random variable X ; Gaussians are 2-stable, Cauchy variables are 1-stable.

39.7.3 3. Theory

The subspace embedding theorem packages the norm-preservation guarantee uniformly over a subspace rather than over a finite set of points.

Theorem 39.17 (Subspace embedding). *Let $V \subseteq \mathbb{R}^d$ be a linear subspace of dimension D and let A be a $k \times d$ matrix with independent Gaussian entries of variance $1/k$ and $k = O(D/\varepsilon^2)$. Then with probability at least $1 - \exp(-\Omega(D))$,*

$$(1 - \varepsilon) \|x\|^2 \leq \|Ax\|^2 \leq (1 + \varepsilon) \|x\|^2 \quad \text{for all } x \in V.$$

The proof is an ε -net argument: a bounded net of $O(2^{O(D)})$ points in the unit sphere of V carries the norm of every vector up to an additive error of ε , and the union bound over the net absorbs the exponential dependence on D . The Gaussian distribution is rotation-invariant, which is what makes it the right choice for the uniform statement; the Rademacher version holds with slightly more rows.

The p -stable families are the bridge to LSH. Hash with

$$h_{a,b}(x) = \left\lfloor \frac{a \cdot x + b}{W} \right\rfloor, \quad a \sim p\text{-stable}, b \sim \text{Uniform}[0, W].$$

For two points at Euclidean distance u , the difference $a \cdot (x - y)$ is uX for a p -stable X , so the collision probability is

$$p(u) = \mathbb{E} \left[\left(1 - \frac{|uX|}{W} \right)_+ \right],$$

a decreasing function of u with $p(0) = 1$ and $p(u) \rightarrow 0$. Choosing $W \approx r$ makes $p_1 = p(r)$ and $p_2 = p(cr)$ small together, which yields $\rho < 1/c$ for the 2-stable (Gaussian) case.

39.7.4 4. Pseudocode

Algorithm 176 gives the generic application step; the distribution is a parameter.

Algorithm 176 Random Projection

Input: Points $p_1, \dots, p_n \in \mathbb{R}^d$, target dimension k .

Output: Sketches $q_1, \dots, q_n \in \mathbb{R}^k$.

```

1: function RANDOMPROJECT( $P, k$ )
2:   choose an entry distribution (Gaussian, Rademacher, or sparse Achlioptas)
3:   draw a  $k \times d$  matrix  $A$  from that distribution, scaled to variance  $1/k$ 
4:   for  $i = 1, \dots, n$  do
5:      $q_i \leftarrow Ap_i$ 
6:   end for
7:   return  $q_1, \dots, q_n$ 
8: end function

```

The hashed implementation avoids materializing A : instead of storing the matrix, store for each row a pseudo-random hash of the input coordinate into $\{+1, -1, 0\}$ (for the sparse family) together with the p -stable sample value (for the Gaussian family), and accumulate q_i coordinate by coordinate. This reduces the storage from $O(dk)$ to $O(k)$ words.

39.7.5 5. Parameter Selection

The target dimension k is set by the application: $\varepsilon^{-2} \log n$ for distance preservation, $O(D/\varepsilon^2)$ for a known subspace of dimension D , or a fixed budget in streaming settings. The bucket width W of the p -stable hash is chosen by grid search so that the ratio p_1/p_2 is maximized at the operating radius r ; in practice W a few times r balances the collision probability against the bucket noise. The sparse Achlioptas family is preferred over the dense Gaussian when d is large and the data is sparse, because the expected number of non-zero products per output coordinate is $d/3$ rather than d .

39.7.6 6. Complexity

Dense application costs $O(ndk)$; the sparse family costs $O(n \cdot \text{nnz})$ regardless of k when the hashed form is used, since each non-zero input coordinate touches at most one entry per output coordinate. The p -stable hash evaluates in $O(d)$ per query (one dot product per table), which is why the query time of Algorithm 174 carries the factor d . The subspace embedding compresses any $n \times d$ data matrix to $k \times d$ (or $n \times k$, whichever is smaller) before the expensive linear algebra runs, changing the dominant term of an SVD or least-squares solve from cubic in the ambient dimension to cubic in the sketch dimension.

39.7.7 7. Usages

Random projections are used for approximate SVD and principal component analysis, for least-squares regression on wide matrices, for k -means clustering at scale, for kernel methods via random features, and for similarity search on text and image data through the p -stable LSH families. The sparse matrix products that appear when sketching Gram matrices are sped up by the algorithms of [YZ04].

39.7.8 8. Testing Methods and Edge Cases

Test that $\mathbb{E}[\|Ax\|^2] = \|x\|^2$ on random unit vectors for each distribution (a statistical check of the scaling) and that the empirical distortion over the data set stays below $1 + \varepsilon$. Edge

cases: the zero vector (preserved exactly, do not divide by it in normalized checks); points with non-zero mean (center first, as in Section 39.6); and the pathological case where the data itself has an adversarial sparse structure aligned with the sampled coordinates, which the hashed implementation can under-save. Whenever the guarantee must be deterministic, run the map twice with independent randomness and keep the better sketch.

39.7.9 9. References

The entry distributions are analyzed in [Ach03]; the p -stable families are [DIIM04]; the subspace embedding view and its applications to randomized linear algebra are developed in [HP11]. The classical statements appear in [JL84] and [Mat02].

39.8 Concentration of Measure

The probabilistic machinery behind every random construction of this chapter is the concentration of measure: in high dimension, smooth functions of many independent (or independent-looking) variables are essentially constant. This section develops the phenomenon in its geometric forms — the Gaussian annulus, the volume of balls, Levy’s lemma on the sphere — and then shows how the same principle rescues algorithms whose worst-case analysis is catastrophic: the average-case and smoothed behavior of the simplex method, and the sampling-based computation of volumes of convex bodies.

Theorem 39.18 (Gaussian annulus). *Let $g \sim \mathcal{N}(0, I_d)$. Then with probability at least $1 - 2e^{-cd\epsilon^2}$,*

$$(1 - \epsilon)\sqrt{d} \leq \|g\| \leq (1 + \epsilon)\sqrt{d},$$

for an absolute constant $c > 0$.

The theorem says that a high-dimensional Gaussian vector is almost surely at distance $\sqrt{d}$ from the origin, within a multiplicative error that shrinks with d . The intuition is a variance computation: each coordinate contributes variance 1 to $\|g\|^2$, which has expectation d and standard deviation $\sqrt{2d}$, so the relative deviation $\sqrt{2d}/d = \sqrt{2/d}$ tends to zero. The proof is the chi-squared tail of Section 39.6: $\|g\|^2$ is a chi-squared variable with d degrees of freedom, and

$$\Pr[\|g\|^2 \geq (1 + t)d] \leq e^{-cdt^2}$$

follows from the exponential Markov inequality applied to $e^{\lambda g^2}$. This is exactly the one-dimensional statement (Theorem 39.14) that powered the Johnson–Lindenstrauss lemma, in its pure form.

The same phenomenon concentrates the volume of the unit ball. Writing

$$\text{vol}(B^d) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)},$$

the fraction of the volume lying within distance δ of the boundary of the unit ball is

$$1 - (1 - \delta)^d,$$

which tends to 1 exponentially fast. Equivalently, the ball of radius $1 - \delta$ contains an exponentially small fraction of the volume: high dimensions are empty in their interior and populated on their boundary shell. This is why a uniform grid with spacing ϵ in the unit cube needs ϵ^{-d} points — the volume-sampling version of the curse of dimensionality (Section 39.9).

Theorem 39.19 (Levy’s lemma). *Let $f : S^{d-1} \rightarrow \mathbb{R}$ be L -Lipschitz with respect to geodesic distance on the unit sphere. Then*

$$\Pr [|f - \mathbb{E}[f]| \geq t] \leq 2 \exp \left(-\frac{cdt^2}{L^2} \right)$$

for an absolute constant $c > 0$.

Levy’s lemma is the general concentration statement for the sphere: every Lipschitz function is nearly constant on most of the sphere. The proof passes through the isoperimetric inequality on the sphere, which says that among all sets of a given measure the caps minimize the boundary; integrating the isoperimetric profile along the level sets of f yields the exponential tail. The two corollaries most used in this chapter are the Gaussian annulus and the concentration of Lipschitz functions on the sphere, which together imply the one-dimensional concentration of Theorem 39.14 and hence the Johnson–Lindenstrauss lemma. A further corollary is that a random point on the sphere is within $O(1/\sqrt{d})$ of any fixed equator with overwhelming probability — high-dimensional spheres are concentrated near every great circle, a fact with no analogue in the plane.

Remark 39.20. Concentration is a double-edged sword. It makes random maps and random estimators reliable, which is what makes the algorithms of this chapter work, but it is also exactly the force that destroys nearest neighbors (Section 39.9): when all points of a Gaussian cloud are at distance $\sqrt{d}$ from one another up to a factor $1 + O(1/\sqrt{d})$, the distinction between “nearest” and “far” collapses.

39.8.1 Sampling convex bodies

The same concentration phenomenon, in its algorithmic form, is the engine of modern convex geometry. Consider the problem of approximating the volume of a convex body given only by a membership oracle — the generalization of the polyhedron volume computations of Chapter 16 to implicit bodies.

Theorem 39.21 (Approximate volume of convex bodies). *There is a randomized algorithm that, given a membership oracle for a convex body $K \subset \mathbb{R}^d$ of positive volume and an approximation factor $1 + \varepsilon$, outputs in time polynomial in d , $1/\varepsilon$, and the bit complexity of the input a value within a factor $1 \pm \varepsilon$ of $\text{vol}(K)$, with high probability. Computing the volume exactly is #P-hard.*

The algorithm is the paradigm of randomized geometric integration. It samples nearly uniform points from K using a random walk — the hit-and-run walk, which moves from the current point along a random direction to a random point inside K on that chord — and it combines the samples with a multiphase decomposition: K is intersected with a growing sequence of balls of radius $r_1 < r_2 < \dots < r_m = R$ such that each shell has bounded volume ratio, and the volume ratios are estimated by counting samples in the successive shells. The product of the ratios telescopes to $\text{vol}(K)$, and the concentration of the counts around their means is the only error source. The key analytic fact is that the hit-and-run walk mixes in $O(d^3)$ steps from a warm start — polynomial in the dimension, in stark contrast with the naive grid, which needs ε^{-d} points. This is concentration working in favor of the algorithm: it makes the sampled estimators reliable.

The same sampling machinery computes integrals over convex bodies and, via rejection, the probabilities of events in convex sets; it is the geometric core of the Monte Carlo methods in Bayesian statistics and of the counting problems reducible to volume computation.

39.8.2 Average case and smoothed analysis of the simplex method

The linear programming algorithms of Chapter 17 are polynomial for fixed dimension but the simplex method has exponential worst-case behavior on the Klee–Minty cubes. High-dimensional

geometry explains why the simplex method works anyway in practice: the average case is benign because the combinatorial objects that would make it slow — long chains of pivots — are exponentially unlikely under natural random models.

Theorem 39.22 (Borgwardt’s average-case bound). *Under a rotationally symmetric distribution on the coefficients, the simplex method performs an expected number of pivot steps that is polynomial in n and d .*

Borgwardt’s analysis shows that a random linear program has, with high probability, a shadow-vertex pivot path of polynomial length: projecting the feasible polyhedron onto the plane spanned by the objective and a fixed direction yields a polygon whose size is the number of pivots, and the spherical concentration of the coefficients keeps that projected polygon small. The geometry is pure concentration of measure on the sphere: the relevant pivots are those whose normal vectors lie in a thin band, and the probability that the band contains many of them is exponentially small.

Theorem 39.23 (Smoothed analysis of the simplex method). *Let a linear program be obtained by perturbing an arbitrary instance with independent Gaussian noise of standard deviation σ . The expected number of simplex pivots (under the shadow-vertex rule) is polynomial in n , d , and $1/\sigma$.*

Spielman–Teng’s smoothed analysis is the definitive resolution of the paradox: no instance is hard in the worst case once it is perturbed, and the perturbation need be only polynomially small. The proof analyzes the expected number of changes of the optimal basis under perturbation and again exploits that the expected number of vertices of a random projection of a polytope is polynomial. The simplex method and its smoothed behavior are treated in detail in Chapter 17; the point here is that concentration, which wrecks nearest-neighbor search, is the very mechanism that makes the simplex method fast. The worst-case exponential examples are artifacts of a negligible set of adversarial inputs.

Remark 39.24. The three theorems of this section — Gaussian concentration, polynomial volume computation, and polynomial smoothed pivoting — share a single methodological message: in high dimension, the space of inputs is dominated by typical inputs, and typical geometry is tractable. Algorithm design for high dimensions must therefore be *average-case* or *smoothed*, or it must be *approximate* with a randomized guarantee, as in Sections 39.4 and 39.6.

The references for this section are [Mat02] (Gaussian annulus, Levy’s lemma, volumes), [DFK91] (volume computation), [Bor87] (average case of the simplex method), and [ST04] (smoothed analysis).

39.9 Curse of Dimensionality

Definition 39.25 (Curse of dimensionality). The *curse of dimensionality* is the collection of phenomena, observed in Section 39.8, in which the running time or storage of an algorithm, or the sample complexity of a statistical estimator, grows like a quantity exponential in the dimension — $n^{\lfloor d/2 \rfloor}$, ϵ^{-d} , 2^d , or $\exp(cd)$ — even when the dependence on the number of points n is benign. It manifests in three distinct forms for this book: combinatorial (complexity of geometric structures), volumetric (the grid bound), and metric (the collapse of distance information).

39.9.1 The combinatorial curse

The combinatorial form is Theorem 39.2: hulls, Voronoi diagrams, and Delaunay complexes have $n^{\Theta(d/2)}$ cells. This is not an artifact of the specific structure; it is the number of cells of

any arrangement of n hyperplanes in $\mathbb{R}^d$, which is $\Theta(n^d)$, and the number of facets of the polar polytope, which is the same count. The complexity is intrinsic: even just listing the Delaunay complex of a cyclic-polytope sample requires $n^{\lceil d/2 \rceil}$ output, so no exact data structure in high dimension can beat a linear scan on the input of its own output. Section 39.3 and Section 39.2 pay the price in their complexity analyses; here we note that the bound is optimal and that approximation is the only escape.

The same combinatorial explosion governs the data structures of Chapter 9: a kd -tree answering orthogonal range queries in time $O(n^{1-1/d} + k)$ degrades to $O(n)$ as soon as d grows, and a range tree storing $n \log^{d-1} n$ points is pointless when $d \geq \log n$. Section 9.11 documented the transition, and Section 9.12 proved that the transition is unavoidable: any data structure with near-linear space must spend $\Omega(n^{1-1/d})$ time per simplex range query in the worst case. In high dimension, $n^{1-1/d}$ is n ; range searching in high dimensions is a linear scan.

39.9.2 The volumetric curse

The volumetric form is the grid bound derived in Section 39.8: covering the unit cube with boxes of side ε needs ε^{-d} boxes, and the fraction of the volume of the unit ball within distance δ of the boundary is $1 - (1 - \delta)^d$. Every algorithm that partitions space must pay this cost, because the volume itself is concentrated near the surface. This is the mechanism behind the ε^{-d} sample complexity of nonparametric statistics and behind the failure of uniform grid methods, and it is why Section 39.5 embeds before partitioning. The dimensionality reduction escape works because the *intrinsic* dimension of the data (the dimension of the manifold or subspace it occupies) is typically far smaller than the ambient dimension; the grid bound is a bound on the ambient dimension alone.

39.9.3 The metric curse

The metric form is the most damaging for machine learning, because it attacks the very meaning of similarity.

Theorem 39.26 (Distance concentration). *Let P be a set of n i.i.d. points from a distribution supported in $\mathbb{R}^d$, and let q be an independent point. Under mild moment and concentration conditions (satisfied by Gaussian data), for every fixed n the ratio*

$$\text{rel}(q) = \frac{\min_{p \in P} \|p - q\|}{\max_{p \in P} \|p - q\|}$$

converges to 1 in probability as $d \rightarrow \infty$.

The statement is the analysis of the phenomenon noticed in Section 39.8: all pairwise distances are within a relative window of width $O(1/\sqrt{d})$ around a common scale, so the nearest neighbor is only marginally closer than the farthest point. A nearest-neighbor classifier whose discrimination rests on the gap between nearest and second-nearest loses its meaning: there is no signal left. The practical consequence is not that high-dimensional nearest-neighbor search is impossible — Section 39.4 shows sublinear queries are achievable — but that the *queries* must be approximate and that the *data* must first be embedded so that the effective dimension of the distance is small. The results are due to [BGRS99], who also give the precise conditions under which the relative contrast degenerates, and the phenomenon is discussed further in Section 39.10.

39.9.4 Hardness of worst-case high-dimensional problems

Beyond the geometric and metric barriers sit the computational ones. The worst-case version of exact nearest-neighbor search is as hard as brute force: in a cell-probe model, any data

structure answering exact queries in the Hamming cube requires near-linear query time in the worst case, so the approximation guarantee of Section 39.4 is not a convenience but a necessity. More broadly, the PCP framework — the PCP machinery developed to prove hardness of approximation — established that many optimization problems over high-dimensional inputs are hard to approximate even approximately, in the worst case. The connection runs through the same high-dimensional geometry: a proof-checker verifies a witness by reading a few randomly chosen coordinates, and the soundness of the scheme rests on the concentration of the acceptance probability over the random coordinate choices. The foundational references are [Aro94] and the survey in [HP11]. The message for algorithm design is the same as in Section 39.8: the tractable regime in high dimensions is the typical case and the approximate case, not the worst exact case.

39.9.5 The escapes

Three complementary escapes from the curse run through this chapter:

- *Dimensionality reduction* (Sections 39.5, 39.6, 39.7): reduce the ambient dimension to the intrinsic one; the cost is a small relative distortion, which many downstream tasks tolerate.
- *Approximate and randomized queries* (Section 39.4): replace the exact answer by one that is correct up to a factor c with constant probability; the query cost drops from n to n^ρ with $\rho < 1$.
- *Data-dependent and worst-case-intrinsic algorithms*: exploit the intrinsic dimension or the smoothed complexity of the instance (Section 39.8), where the typical behavior is polynomial even though the worst case is exponential.

39.10 Applications to Machine Learning

The pipeline of modern machine learning — a high-dimensional feature vector, a similarity or kernel, and an optimization or search step — is, from the point of view of this chapter, a high-dimensional computational geometry problem with n points in $\mathbb{R}^d$, and nearly every technique of the preceding sections appears in it. This section walks through the pipeline and maps each stage to the geometric tool. The statistical and algorithmic foundations of the geometric view of learning are developed in Chapter 40 and in the approximation chapter’s treatment of high-dimensional algorithms (Section 32.12).

39.10.1 k -nearest-neighbor classifiers

The k -nearest-neighbor classifier labels a query by the majority label among its k nearest neighbors in the training set. Its accuracy depends completely on the geometry of the metric: if the nearest neighbor is not meaningfully closer than the rest, the classifier degenerates to the majority class. The distance-concentration analysis of Section 39.9 ([BGRS99]) makes the failure precise: on high-dimensional data with no intrinsic structure, the relative contrast between nearest and farthest points tends to 1, and the classifier has no signal. The remedies are the escapes of Section 39.9: embed the data first (Section 39.5), or use an approximate nearest-neighbor structure (Section 39.4). This is why, in practice, high-dimensional k -NN pipelines run a random projection (or a learned linear map) before the search: the projection is not a loss of information but a denoising of the metric. The state of the art in production systems combines these ideas in the graph-based HNSW structures of [MY20], whose greedy navigation over a proximity graph replaces the exact structures that the curse kills.

39.10.2 Dimensionality reduction as preprocessing

The same argument applies to clustering, outlier detection, and kernel methods. Before a Gaussian-kernel SVM on d -dimensional features, one can replace the exact kernel by a random-feature map: draw a random projection A and use the finite-dimensional feature map $\phi(x) = (\cos, \sin)$ -features derived from Ax , which by the Johnson–Lindenstrauss lemma approximates the kernel’s geometry (Section 39.7). This is a direct application of the lemmas of Sections 39.6 and 39.7 and is one of the few places where the concentration theorems pay for themselves twice: once for the map’s faithfulness and once for the kernel’s approximation. The data management side of these pipelines — indexing the sketches, hashing the features, and maintaining the tables — is surveyed in the handbook [MS05a].

39.10.3 Sparse data and fast linear algebra

Real feature vectors are sparse: each example has a handful of non-zero coordinates out of millions. Sparse random projections (Section 39.7) exploit this directly, and the matrix multiplications that build Gram matrices and covariance estimates from sparse inputs are accelerated by the sparse matrix product algorithms of [YZ04]. The interplay between sparsity and geometry is itself a high-dimensional phenomenon: the concentration results of Section 39.8 govern how many samples are needed to estimate inner products and distances from sketches, and the same machinery bounds the statistical error of the random-feature approximations.

39.10.4 The geometric picture of learning

The geometric reading of learning theory is the one this chapter makes natural. A data set is a point cloud; the tasks are geometric queries (nearest neighbors, kernels, clustering, projections); the sample complexity of a hypothesis class is governed by the combinatorial complexity of the geometry — the VC dimension of ranges in $\mathbb{R}^d$ is $O(d)$ and ε -nets of size $O(\frac{d}{\varepsilon} \log \frac{1}{\varepsilon})$ exist (Section 9.3 and the related material in Chapter 9), which is the polynomial-in- d side of the curse. The exponential side is the sample complexity of volume-based estimators (Section 39.8). Learning in high dimensions is possible exactly when the data has low intrinsic dimension, and dimensionality reduction is the algorithmic expression of that belief. The standard reference for the geometric viewpoint is Chapter 40; the approximation algorithms that make learning tractable in high dimensions are covered in Section 32.12 of the approximation chapter.

Remark 39.27. The most common mistake in high-dimensional practice is to assume that Euclidean distance on raw features is a meaningful measure of similarity. The analysis of Section 39.9 shows when it is not, and the techniques of this chapter show how to fix it: normalize, embed, and approximate. Each step is justified by a theorem in this chapter, and each step is cheap enough for production systems.

39.11 Exercises

1. Prove that computing the convex hull of n points in $\mathbb{R}^d$ requires $\Omega(n \log n)$ time even when the hull has $O(n)$ facets, by reducing sorting to a hull computation in a plane contained in $\mathbb{R}^d$.
2. Show that the expected number of facets created by the randomized incremental hull algorithm (Algorithm 172) on a random sample is $O(n^{\lfloor d/2 \rfloor})$, and derive the running time of Theorem 39.3 from the Clarkson–Shor charging argument.
3. Prove the lifting statement of Section 39.3 in full generality: a sphere in $\mathbb{R}^d$ with center c and radius r corresponds to a hyperplane through the lifted points, and a point lies inside

the sphere iff its lift lies below the hyperplane. Conclude the empty-sphere test from the orientation of the lifted simplex.

4. Let $\mathcal{H}$ be the family of coordinate projections on $\{0,1\}^D$. Compute p_1 , p_2 , and ρ for the (r, cr) -near-neighbor problem and verify the exponent $\rho \approx 1/c$ for small r/D in Theorem 39.11. Show that the query of Algorithm 174 inspects $O(L)$ points in expectation and terminates with constant success probability.
5. Prove the chi-squared tail bound used in Theorem 39.14 by applying the exponential Markov inequality to $\exp(\lambda \|g\|^2)$ and optimizing over λ . Derive the Gaussian annulus of Theorem 39.18 from it.
6. Give an ε -net proof of the subspace embedding theorem (Theorem 39.17): cover the unit sphere of a D -dimensional subspace by $O((3/\varepsilon)^D)$ points, apply the pointwise distortion bound, and take a union bound. Explain where the exponential dependence on D goes.
7. Derive the collision probability $p(u)$ of the p -stable hash family of Section 39.7, and show that it is decreasing in u . For the Cauchy (1-stable) case compute $p(u)$ explicitly and bound ρ for the (r, cr) -near-neighbor problem.
8. Argue from Theorem 39.26 that a k -nearest-neighbor classifier has error approaching the majority class on high-dimensional Gaussian data, and design a random-projection preprocessing that restores meaningful nearest neighbors on data lying on a k -dimensional affine subspace of $\mathbb{R}^d$. Justify the target dimension with Theorem 39.13 and analyze the resulting classification accuracy.

Chapter 40

Geometry of Data and Machine Learning

40.1 Geometric View of Data

Machine learning and data mining operate, at bottom, on **point sets**. A data record – a document, an image, a sensor reading, a spatial object – is represented as a vector of numeric features, and a data set is the finite collection of such vectors. The representation is chosen by the analyst, but once it is fixed the data becomes a geometric object: a cloud of n points in a feature space $\mathbb{R}^d$, and every learning task becomes a geometric question about that cloud. Classification asks where a new point lies relative to the labeled points; clustering asks how the cloud decomposes into natural groups; regression asks how one coordinate varies with the others; and outlier detection asks which points lie far from the mass of the data. The vocabulary of this book – distances, angles, convexity, hulls, proximity structures, decompositions – is exactly the vocabulary in which modern learning is described.

Definition 40.1 (Feature Space and Feature Vector). A **feature space** is a set of admissible data records together with a distance function. Given d numeric **features** $x_1, \dots, x_d$, a record is encoded as a **feature vector** $x = (x_1, \dots, x_d) \in \mathbb{R}^d$; the standard feature space is the Euclidean space $\mathbb{R}^d$ with the distance of choice (Section 40.2).

Definition 40.2 (Data Set as Point Cloud). A **data set** of size n is a multiset $P = \{p_1, \dots, p_n\} \subset \mathbb{R}^d$ of feature vectors, usually organized as an $n \times d$ **data matrix** whose rows are the points. We call P the **point cloud** of the data. In supervised problems each p_i carries an auxiliary **label** $y(p_i)$ drawn from a finite set (classification) or a real value (regression).

The geometric view has immediate consequences. Every algorithm of this book is a candidate subroutine once data is seen as points:

- **Proximity** (Chapter 13): nearest neighbors realize the intuition that nearby records behave alike, the hypothesis behind k -nearest-neighbor classification (Section 40.3) and clustering (Section 40.4).
- **Spatial structures** (Chapter 10): kd -trees, quadtrees, and R-trees accelerate the repeated distance queries that dominate learning pipelines (Section 40.11).
- **Hulls and halfspaces** (Chapter 4): linear classifiers are halfspaces, and their margins are governed by the distance between convex hulls (Section 40.5).
- **High-dimensional geometry** (Chapter 39): when d is large, concentration of measure and the curse of dimensionality (Section 39.9) shape everything from distance computation to generalization.

Remark 40.3 (The choice of representation is a geometric act). Normalization (rescaling each feature to unit range or unit variance), whitening (decorrelating the axes), and the choice of a distance are all geometric preprocessing. Standardizing axes is a linear map that replaces the Euclidean metric by a Mahalanobis metric (Definition 40.7); learning a good metric is itself a learning problem. The standard treatments of this pipeline are [Bis06, DHS01, HTF09]; the classical reference for the algorithmic geometry is [dBCvKO08, BCKO08].

Remark 40.4 (Labels and distances). In unsupervised settings there are no labels and the geometry is all there is: clustering and manifold learning (Section 40.7) must reconstruct structure from distances alone. The quality of the distance therefore limits the quality of everything downstream.

40.2 Metric Spaces

The first geometric primitive is the distance. A data set with a distance is a **metric space**, and virtually all geometry in this chapter assumes at least the triangle inequality, which is what makes pruning, approximation, and geometric data structures work (Sections 40.3 and 40.4).

Definition 40.5 (Metric). A function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ on a set X is a **metric** if for all $x, y, z \in X$: (i) $d(x, y) = 0$ iff $x = y$; (ii) $d(x, y) = d(y, x)$; (iii) $d(x, z) \leq d(x, y) + d(y, z)$ (the **triangle inequality**). If (i) is relaxed to allow $d(x, y) = 0$ for $x \neq y$, d is a **pseudo-metric**.

Definition 40.6 (ℓ_p Norms). For $p \geq 1$, the ℓ_p **norm** of $x \in \mathbb{R}^d$ is

$$\|x\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p},$$

and $d_p(x, y) = \|x - y\|_p$ is the induced metric. The cases $p = 1$ (Manhattan), $p = 2$ (Euclidean), and $p = \infty$ (Chebyshev, $\|x\|_\infty = \max_i |x_i|$) are the ones used in practice. The **cosine distance** $1 - \frac{x \cdot y}{\|x\| \|y\|}$ measures angular dissimilarity on the unit sphere.

Distance-based learning and retrieval are only as good as the metric, so a substantial literature **learns** the distance from data.

Definition 40.7 (Mahalanobis Metric and Metric Learning). Let M be a symmetric positive-semidefinite $d \times d$ matrix. The **Mahalanobis distance**

$$d_M(x, y) = \sqrt{(x - y)^\top M (x - y)}$$

is a pseudo-metric; if M is positive definite it is a metric. **Metric learning** is the problem of fitting M (subject to constraints such as $M \succeq 0$) so that prescribed pairs are close and non-similar pairs are far.

Theorem 40.8 (PSD Matrices Give Metrics). *For symmetric $M \succeq 0$, d_M satisfies the triangle inequality, and d_M is a metric exactly when M is positive definite. Writing $M = L^\top L$ with a $k \times d$ factor L gives $d_M(x, y) = \|Lx - Ly\|_2$, i.e., Mahalanobis distance is Euclidean distance after the linear embedding $x \mapsto Lx$.*

Proof. The identity $d_M(x, y) = \|Lx - Ly\|_2$ for any square root L of M (which exists for $M \succeq 0$, e.g., the Cholesky factor or the symmetric square root) is immediate from $x^\top M x = \|Lx\|_2^2$. The triangle inequality follows from the Euclidean triangle inequality; if $\ker M = \{0\}$ then L is injective and $d_M(x, y) = 0$ iff $x = y$. $\square$

Example 40.9 (Metric learning for kNN). The **large-margin nearest neighbor** (LMNN) method [WBS06] learns M so that each point's k target neighbors (same label) are pulled in while differently labeled points inside the neighborhood margin are pushed out. The result is a Mahalanobis metric under which k -nearest-neighbor classification of Section 40.3 is markedly more accurate.

Remark 40.10. Which metric to use is an algorithmic decision with algorithmic consequences. The Euclidean metric admits Voronoi diagrams, k d-trees, and the approximation machinery of Chapter 32; an ℓ_1 metric changes the shape of balls but preserves most low-dimensional structures; a learned Mahalanobis metric keeps all of them because it is Euclidean in a transformed space. Textbook treatments of distances and their role in classifiers are [Bis06, DHS01, HTF09].

40.3 Nearest-Neighbor Classification

40.3.1 1. Overview

Nearest-neighbor classification is the direct geometric translation of the heuristic “things that look alike are alike”. A query point is labeled by majority vote among the closest labeled points in the training set. The rule has no explicit training phase and no parameters beyond the metric and k , yet it is consistent: with enough data it approaches the best possible classifier. Its computational heart is the nearest-neighbor problem (Definition 13.7, Chapter 13), so the whole data-structure machinery of Chapters 10, 9, and 32 – and the high-dimensional analysis of Chapter 39 (Section 39.4) – becomes a machine-learning ingredient. The classical reference is Cover and Hart’s 1967 analysis [CH67]; the applications to vision and learning are collected in [SDI06].

40.3.2 2. Definitions

Definition 40.11 (k -Nearest-Neighbor Rule). Let P be a labeled training set and $k \geq 1$ an integer. The **k -nearest-neighbor rule** labels a query q by the majority class among the k points of P closest to q under the chosen metric d , breaking ties by distance and then arbitrarily.

Definition 40.12 (Bayes Risk). For a distribution μ over $X \times \{1, \dots, M\}$, the **Bayes risk** R^* is the misclassification probability of the optimal rule that labels x by the class maximizing the posterior probability $\mu(y | x)$. It is the theoretical minimum over all classifiers.

For the 1-nearest-neighbor rule the decision boundary is the Voronoi diagram of the labeled points (Chapter 11, Section 11.1): the query is labeled as the site of the Voronoi cell containing it (Definition 11.1). For $k > 1$ the boundary is a subset of the arrangement of bisectors of pairs of differently labeled points.

40.3.3 3. Theory

The learning-theoretic content of the rule is the classical theorem of Cover and Hart [CH67].

Theorem 40.13 (Cover–Hart). *For suitably smooth distributions in M -class problems, the asymptotic risk R of the 1-nearest-neighbor rule satisfies*

$$R^* \leq R \leq R^* \left(2 - \frac{M}{M-1} R^* \right),$$

and the bounds are tight. In particular for $M = 2$, $R^ \leq R \leq 2R^*(1 - R^*)$: the nearest-neighbor rule makes at most twice as many errors as the Bayes-optimal classifier, in the limit of infinite data.*

Proof sketch. For a fixed query x , condition on the nearest neighbor $p = p_n(x)$; under the smoothness assumptions the class of p converges in distribution to the class of x , so the asymptotic conditional error is $1 - \sum_y \mu(y | x)^2$. This quadratic in the posterior vector is bounded between $R^*(x)$ and $2R^*(x)(1 - R^*(x))$, where $R^*(x) = 1 - \max_y \mu(y | x)$; the general M -class bound follows by maximizing the gap over the simplex of posteriors. Tightness is exhibited by point-mass constructions attaining both extremes; the full argument is in [CH67]. $\square$

For $k > 1$ the risk decreases monotonically with k toward the Bayes risk: as $n \rightarrow \infty$ with $k = k(n) \rightarrow \infty$ and $k(n)/n \rightarrow 0$, the k -nearest-neighbor rule is **consistent**, i.e., its risk converges to R^* , under weak conditions on the distribution [DHS01, Bis06]. Two caveats temper the guarantee.

- **Curse of dimensionality:** in high dimension the nearest neighbors of a query are, with high probability, nearly as far as the farthest points of the data (Section 39.9), so the local information the rule exploits is diluted. This is the geometric reason that high-dimensional nearest-neighbor search is hard (Chapter 39, Section 39.4).
- **Nonparametric bias:** the rule is a histogram-density estimate in disguise; its error is governed by the bias–variance trade-off, and the metric choice (Section 40.2) is the main lever on the bias.

40.3.4 4. Pseudocode

Algorithm 177 k -Nearest-Neighbor Classification with a kd -Tree

Input: Labeled training set $P \subset \mathbb{R}^d$; query q ; integer $k \geq 1$; a kd -tree over P (Algorithm 14).

Output: The predicted label of q under the k -nearest-neighbor rule.

```

1: function KNNCLASSIFY( $q, k$ )
2:    $N \leftarrow$  the  $k$  nearest neighbors of  $q$  in the  $kd$ -tree ▷ Algorithm 14
3:   for each label  $\ell$  do
4:      $c_\ell \leftarrow |\{p \in N : y(p) = \ell\}|$ 
5:   end for
6:   return the label  $\ell$  maximizing  $c_\ell$  ▷ ties broken by smallest total distance
7: end function
```

40.3.5 5. Parameter Selection

- k : small k is sensitive to label noise, large k blurs the decision boundary; select k by cross-validation over a range of odd values. $k = 1$ coincides with the Voronoi-labeling rule.
- **Metric:** the Euclidean metric is the default; with features of different units, standardize, or learn a Mahalanobis metric (Section 40.2).
- **Weighting:** distance-weighted voting ($1/d(q, p)$ weights) smooths ties and improves robustness [DHS01].
- **Pruning:** the lazy-training view means every query does real work; for large n use the approximate structures of Section 32.5, which trade a factor c of distance for speed.

40.3.6 6. Complexity

- **Training:** $O(1)$ – the rule stores the data; building the kd -tree costs $O(dn \log n)$ (Theorem 10.5, Chapter 10).

- **Query:** $O(dn)$ by brute force; $O(d \log n)$ expected in low dimension via the kd -tree (Theorem 10.6, Section 10.1); $O(c^d \log n)$ via box-decomposition ANN (Theorem 32.15); and $O(dn^\rho)$ via LSH in high dimension (Theorem 32.17, Section 32.6, Definition 32.16).

40.3.7 7. Usages

- **Content-based retrieval:** the rule is similarity search with a label; image and document retrieval pipelines [SDI06].
- **Recommendation:** collaborative filtering labels items by the behavior of users with nearby feature vectors.
- **Geographic data:** spatial classification and map generalization in GIS (Chapter 36) are natural nearest-neighbor applications.
- **Baseline:** the rule is the standard sanity check that any learned classifier must beat; its accuracy upper-bounds simple geometry.

40.3.8 8. Testing Methods and Edge Cases

- Verify classification accuracy against brute-force nearest-neighbor computation for the same k and metric, on random and adversarial queries.
- Test ties on the decision boundary, duplicate points (they can fill several of the k slots), and the cases $k = 1$ and $k = n$.
- Test label noise: flipping labels of a fraction of the training set must degrade accuracy gracefully.
- Cross-validate k ; verify that accuracy as a function of k is unimodal, as the theory of Section 40.13 predicts asymptotically.

40.3.9 9. References

Cover and Hart’s consistency and error bounds [CH67]; the treatment in the pattern-classification and statistical-learning textbooks [DHS01, Bis06, HTF09]; the volume on nearest-neighbor methods in learning and vision [SDI06]; and the algorithmic core in Chapters 13, 10, and 32 with the high-dimensional view in Chapter 39.

40.4 Clustering Geometry

40.4.1 1. Overview

Clustering partitions a point cloud into groups of nearby points with no labels to guide it. Every concrete clustering algorithm is a geometric statement about how “group” should be defined: by a distance objective (k -means, k -median, k -center), by a hierarchy of nested partitions, or by a local density criterion (DBSCAN). The approximation-theoretic view of these objectives is Chapter 32, Section 32.9 (with the k -center/ k -median algorithms of Section 32.10 and the core-sets of Section 32.4); this section emphasizes the geometric content that the practitioner needs: objective functions, algorithm structure, parameter sensitivity, and the geometry each method implicitly assumes about the data.

40.4.2 2. Definitions

Definition 40.14 (*k*-Means). Let $P \subset \mathbb{R}^d$ and $k \geq 1$. The ***k*-means problem** asks for k centers $C = \{c_1, \dots, c_k\}$ minimizing

$$\Phi(C) = \sum_{p \in P} \min_{c \in C} \|p - c\|^2.$$

The optimum over all k -point center sets is Φ^* . Each point is assigned to its nearest center, inducing a Voronoi partition of P (a discrete Voronoi diagram in the sense of Chapter 11).

Definition 40.15 (*k*-Median and *k*-Center). ***k*-median** minimizes $\sum_{p \in P} \min_{c \in C} \|p - c\|$; ***k*-center** minimizes $\max_{p \in P} \min_{c \in C} \|p - c\|$ (Definition 32.22). The objectives differ in how they weight outliers: *k*-center is dominated by the single farthest point, *k*-means squares distances and emphasizes large clusters (Section 32.9).

Definition 40.16 (Single-Linkage Hierarchical Clustering). **Agglomerative hierarchical clustering** starts with each point as its own cluster and repeatedly merges the two clusters at minimum *linkage* distance. In **single linkage** the distance between clusters is $d(A, B) = \min_{a \in A, b \in B} \|a - b\|$, the nearest distance between the two sets; the result is a **dendrogram**, a tree of nested partitions that can be cut at any level.

Definition 40.17 (DBSCAN Clusters). Fix a radius $\varepsilon > 0$ and a threshold $MinPts \geq 1$. A point p is a **core point** if at least $MinPts$ points of P lie within distance ε of p . A point p is **density-reachable** from a core point o through a chain of core points, each within ε of the next. A **cluster** is a maximal set of points pairwise connected by density-reachability; every other point is **noise**.

40.4.3 3. Theory

All three objective-based problems are NP-hard even for points in the plane, and all three admit the approximation algorithms of Chapter 32: Gonzalez’s factor-2 *k*-center algorithm (Theorem 32.24), Lloyd-style iteration for *k*-means, and core-set reductions (Section 32.4) that yield $(1 + \varepsilon)$ -approximations in fixed dimension [HP11, Gon85, Llo82].

Theorem 40.18 (Lloyd Iteration is Monotone). *Lloyd’s algorithm (Algorithm 178), which alternates assigning points to nearest centers and moving each center to the centroid of its assigned points, never increases the *k*-means objective and converges to a fixed point. The fixed point need not be the global optimum [Llo82]; its quality depends on the seeding.*

Proof sketch. Each reassignment step moves every point to a center at least as close as before, weakly decreasing Φ . Each centroid step replaces, for every cluster P_i , the cost $\sum_{p \in P_i} \|p - c\|^2$ by its minimum over c , attained at the mean of P_i . Hence the two half-steps separately reduce Φ , and any convergent subsequence reaches a local fixed point of both maps. $\square$

Practical seeding runs the initialization several times and keeps the best run; the deterministic farthest-point seeding of Section 32.10 and randomized “*k*-means++”-style sampling are the standard choices.

DBSCAN’s cluster notion is well defined because density-reachability behaves like an equivalence relation among core points.

Theorem 40.19 (DBSCAN is Well-Defined). *For fixed ε and $MinPts$, the clusters of Definition 40.17 form a partition of the non-noise points, independently of the order in which points are scanned [EKSX96].*

Proof sketch. Two core points in the same connected component of the core-point graph are density-reachable from a common core point (e.g., along a path in the component), and density-reachability is transitive for core points, so “connected to a common core” is an equivalence relation. Border points attach to the unique cluster of their neighboring core point, and points reachable from no core point are noise. Determinism follows because the clusters are defined purely from P , ε , and $MinPts$. $\square$

40.4.4 4. Pseudocode

Algorithm 178 k -Means via Lloyd Iteration

Input: Point set $P \subset \mathbb{R}^d$; integer $k \geq 1$; initial centers $C = (c_1, \dots, c_k)$; tolerance δ .

Output: A locally optimal k -means clustering of P .

```

1: function KMEANS( $P, C$ )
2:   repeat
3:      $C' \leftarrow C$ 
4:     for each  $p \in P$  do assign  $p$  to the cluster  $P_i$  of its nearest center in  $C'$ 
5:   end for
6:   for  $i = 1$  to  $k$  do
7:      $c_i \leftarrow \frac{1}{|P_i|} \sum_{p \in P_i} p$  ▷ centroid of cluster  $P_i$ 
8:   end for
9:   until  $\max_i \|c_i - c'_i\| \leq \delta$ 
10:  return  $C$ 
11: end function

```

Algorithm 179 DBSCAN

Input: Point set P ; radius $\varepsilon > 0$; threshold $MinPts$; a range-searching structure over P (Chapter 9).

Output: A labeling of P into clusters and noise.

```

1: function DBSCAN( $P, \varepsilon, MinPts$ )
2:   mark all points of  $P$  as unvisited
3:   for each  $p \in P$  do
4:     if  $p$  is unvisited then
5:       mark  $p$  visited;  $N \leftarrow$  points within  $\varepsilon$  of  $p$ 
6:       if  $|N| < MinPts$  then
7:         mark  $p$  as noise
8:       else
9:         start a new cluster with seed  $p$ ; grow it over all density-reachable points using
           range queries
10:      end if
11:    end if
12:  end for
13:  return the cluster labels
14: end function

```

The range query is exactly the problem of Chapter 9; with a kd -tree (Section 9.3) or R-tree (Section 10.3) the neighborhood tests are logarithmic per point.

40.4.5 5. Parameter Selection

- k (**partitioning methods**): the objectives (Definition 40.14) are nonincreasing in k , so k is chosen by heuristics – the “elbow” of the cost curve, a silhouette score, or domain

knowledge; the approximation guarantees of Chapter 32 hold for every k .

- **Seeding:** Lloyd’s fixed point depends on the initial centers; rerun from several seeds (farthest-point or random-sampling seeds) and keep the run of smallest Φ . The Gonzalez centers of Algorithm 163 make robust seeds.
- ε and *MinPts* (**DBSCAN**): inspect the sorted k -nearest-neighbor distance plot (k -distance graph); the “knee” separates the density of clusters from that of noise. *MinPts* around $2d$ is the common default.
- **Linkage (hierarchical)**: single linkage finds elongated and non-convex clusters but is sensitive to noise (a single bridge merges clusters); complete and average linkage trade that sensitivity for convexity assumptions.

40.4.6 6. Complexity

- **k -means:** $O(nkt)$ for t iterations; t is not bounded polynomially in the worst case but is small in practice, and each iteration costs one nearest-center computation per point.
- **DBSCAN:** $O(n \log n)$ expected with a spatial index (range queries dominate), $O(n^2)$ with brute force; memory $O(n)$.
- **Hierarchical:** agglomerative single linkage is $O(n^2)$ time and space for the full dendrogram; variants on k -nearest neighbor graphs lower this at the cost of approximation.

40.4.7 7. Usages

- **Quantization and compression:** k -means centers are a vector quantizer (Lloyd–Max design) [Llo82]; the centers approximate the data distribution.
- **Image and video segmentation:** pixel features clustered into regions; geometric post-processing on the clusters.
- **Spatial data mining:** DBSCAN was designed for large spatial databases, discovering arbitrarily shaped clusters and labeling noise [EKSX96]; the spatial indexing needed for the range queries is the material of Chapters 10 and 9.
- **Anomaly detection:** points classified as noise by DBSCAN, or far from every k -means center, are natural outliers.

40.4.8 8. Testing Methods and Edge Cases

- Verify Lloyd’s monotonicity (Theorem 40.18): the objective must never increase between iterations; instrument and assert it.
- Test empty clusters (a center loses all points), $k \geq n$, and coincident points; define a deterministic convention (re-seed empty clusters).
- For DBSCAN, test varying-density data (uniform density is assumed), clusters connected by a thin bridge, and all-noise inputs; verify the output is independent of scan order (Theorem 40.19).
- Compare clusterings against ground-truth labels with adjusted Rand or mutual-information scores, not just the objective.

40.4.9 9. References

Lloyd’s quantization algorithm [Llo82], Gonzalez’s farthest-point method [Gon85], DBSCAN [EKSX96], the core-set theory of Har-Peled [HP11], and the textbook treatments [HTF09, Bis06]; the approximation algorithms behind the guarantees are Chapter 32.

40.5 Convex Geometry in Machine Learning

Linear classifiers – perceptrons, logistic regression, support vector machines – are halfspaces, and the theory of how well they separate point sets is convex geometry. This section develops the geometric backbone: when two point sets are linearly separable, what the margin is, and how the maximum-margin separator can be characterized purely in terms of convex hulls.

Definition 40.20 (Linear Separability and Margin). A labeled point set P with labels $y(p) \in \{-1, +1\}$ is **linearly separable** if there is a hyperplane $\{x : w \cdot x + b = 0\}$ with $y(p)(w \cdot p + b) > 0$ for all $p \in P$. For a unit vector w , the **geometric margin** of the separator is

$$\gamma = \min_{p \in P} \frac{y(p)(w \cdot p + b)}{\|w\|},$$

the minimum distance from a point to the hyperplane. The **maximum-margin** separator maximizes γ .

Theorem 40.21 (Separating Hyperplane). *Two finite point sets P^+, P^- are linearly separable if and only if their convex hulls are disjoint: $\text{conv}(P^+) \cap \text{conv}(P^-) = \emptyset$.*

Proof sketch. If a hyperplane $w \cdot x + b = 0$ strictly separates the points, then every convex combination of points of P^+ has $w \cdot x + b > 0$ and every convex combination from P^- has $w \cdot x + b < 0$, so the hulls cannot meet. Conversely, the classical separating hyperplane theorem for disjoint compact convex sets (see [CLRS09] for the algorithmic formulation) supplies the separator; for point clouds it can be obtained by computing the closest pair of points between the two hulls (Section 18.3). $\square$

The maximum-margin separator is a pure convex-geometric object.

Theorem 40.22 (Margin as Hull Distance). *Let P^+, P^- be finite point sets with disjoint convex hulls, and let $C^+ = \text{conv}(P^+)$, $C^- = \text{conv}(P^-)$. The maximum-margin separating hyperplane is perpendicular to the segment joining the closest pair of points between C^+ and C^- , and its margin equals half the distance $\text{dist}(C^+, C^-)$. In particular the hard-margin support vector machine reduces to a closest-point problem between two convex sets, the same reduction that underlies the Minkowski-difference machinery of Chapter 18 (Section 18.5, Theorem 18.5, Section 18.6).*

Proof sketch. The margin of a unit-normal separator equals $\gamma = \frac{1}{2}(u \cdot a - u \cdot b)$ for the two support points $a \in C^+$, $b \in C^-$ it touches, maximized over unit directions u . The supporting function of a convex set is maximized at its boundary, so the optimum pairs a, b are boundary points with u normal to both supporting hyperplanes; the value is maximized when a and b are closest points of the two hulls, giving $\gamma = \frac{1}{2}\|a - b\|$. Existence of the closest pair is by compactness; its computation is the Gilbert–Johnson–Keerthi style iteration of Section 18.6. $\square$

Example 40.23 (Hard-margin SVM). Two linearly separable clusters in the plane have their hulls separated by a slab. The maximum-margin hyperplane runs down the middle of the slab; the points that touch the slab boundaries are the **support vectors**. Only those points matter: the classifier is $y = \text{sign}(w \cdot x + b)$ with w a linear combination of the support vectors. This sparsity is exactly the statement that the optimum of a linear program is at a vertex, the geometric dual of the convex-geometry view [Bis06, MMR⁺01].

Example 40.24 (Perceptron convergence). The perceptron (Algorithm 180) is the simplest learning algorithm for a linearly separable set. Its analysis is a pure distance argument: if the separator w^* has margin γ , then after t mistakes the angle between the current weight w_t and w^* shrinks, and the process terminates after at most $(\|w^*\|^2/\gamma^2) \cdot \max_p \|p\|^2/(w^* \cdot p)$ mistakes.

Algorithm 180 Perceptron on a Linearly Separable Set

Input: Points $P \subset \mathbb{R}^d$ with labels $y(p) \in \{-1, +1\}$, linearly separable.

Output: A separator w, b with $y(p)(w \cdot p + b) > 0$ for all $p \in P$.

```

1: function PERCEPTRON( $P$ )
2:    $w \leftarrow 0; b \leftarrow 0$ 
3:   repeat
4:     find a point  $p$  with  $y(p)(w \cdot p + b) \leq 0$  ▷ a mistake
5:     if no point is a mistake then
6:       break
7:     end if
8:      $w \leftarrow w + y(p)p; b \leftarrow b + y(p)$  ▷ margin update
9:   until no mistakes
10:  return  $w, b$ 
11: end function

```

Real data is rarely exactly separable, and the convex geometry adapts in two ways. The **soft-margin** SVM allows a slack variable per point, penalizing violations of the margin; geometrically, the maximum-margin problem becomes one of pushing apart convex hulls after allowing a fraction of points to escape them. And **kernels** (Section 40.6) move the whole separation problem to a feature space where the hulls separate linearly; the margin theorem above holds verbatim there. The geometric connection to convex-hull and convex-distance computations makes the material of Chapters 4, 18, and 18.12 (Section 18.13 for the one-class minimum-enclosing-ball problem) directly relevant; the learning-theoretic framing is in [Bis06, DHS01, HTF09].

40.6 Kernel Geometry

The **kernel trick** lifts a linear method (Section 40.5) into a nonlinear one by moving the data through an implicit feature map and observing only inner products. Its geometric content is that the margin and separation theorems survive in a Hilbert space whose inner product is computed by a cheap function of the original coordinates. The standard introduction is [MMR⁺01]; the textbook treatments are [Bis06, HTF09].

Definition 40.25 (Kernel and Gram Matrix). A function $k : X \times X \rightarrow \mathbb{R}$ is a **positive-definite kernel** if for every finite set $\{x_1, \dots, x_m\}$ the **Gram matrix** G with $G_{ij} = k(x_i, x_j)$ is symmetric positive semidefinite. The **feature map** $\phi : X \rightarrow \mathcal{H}$ into a Hilbert space realizes k if $k(x, x') = \langle \phi(x), \phi(x') \rangle$.

Theorem 40.26 (Mercer's Theorem). *Let X be compact with a Borel measure μ and let k be a continuous positive-definite kernel. Then there is an orthonormal basis $\{\psi_j\}$ of $L^2(\mu)$ and nonnegative eigenvalues λ_j with*

$$k(x, x') = \sum_{j=1}^{\infty} \lambda_j \psi_j(x) \psi_j(x'),$$

the series converging absolutely, and the map $x \mapsto (\sqrt{\lambda_j} \psi_j(x))_j$ is a feature map into ℓ_2 . This is the kernel analogue of the separating-hyperplane theorem: it certifies that the implicit geometry is a genuine Hilbert geometry [MMR⁺01, Bis06].

Example 40.27 (Common kernels). The polynomial kernel $k(x, x') = (x \cdot x' + c)^m$ corresponds to the feature space of all monomials of degree at most m ; the Gaussian kernel $k(x, x') = \exp(-\frac{\|x - x'\|^2}{2\sigma^2})$ corresponds to an infinite-dimensional feature space and is universal – it can approximate any continuous function on a compact set uniformly. Both are used with the SVM of Section 40.5: the maximum-margin separator is computed in feature space, so the decision boundary is a curved surface in the original space.

Remark 40.28 (Geometry survives the trick). A kernel method is a linear method on the transformed points $\phi(p)$, and every theorem of Section 40.5 – separability, margin-as-hull-distance, support-vector sparsity – applies verbatim, with the hulls taken in $\mathcal{H}$. The computational geometry enters again at the data-structure level: exact nearest-neighbor search on kernel features is expensive, and the approximate structures of Chapter 32 (Sections 32.5 and 32.6) are the standard accelerants for large kernel pipelines. Kernel k -means runs Lloyd’s iteration (Algorithm 178) on the implicit features; kernel PCA diagonalizes the centered Gram matrix. Distance in feature space is $\|\phi(x) - \phi(x')\|^2 = k(x, x) + k(x', x') - 2k(x, x')$, so all of Section 40.2 applies to the induced pseudo-metric [MMR⁺01].

40.7 Manifold Learning

The **manifold hypothesis** asserts that high-dimensional data concentrates near a low-dimensional manifold embedded in the ambient space: a d -dimensional point cloud that “really” lives on a k -dimensional surface with $k \ll d$. The geometry of curved surfaces (Chapter 20) and of shape spaces (Chapter 30.12) then becomes a modeling tool, and **manifold learning** is the problem of recovering that intrinsic geometry from point samples.

Definition 40.29 (Intrinsic Dimension). A data set $P \subset \mathbb{R}^d$ has **intrinsic dimension** k if P lies approximately on a k -dimensional submanifold M of $\mathbb{R}^d$. A **manifold embedding** is a map $f : P \rightarrow \mathbb{R}^k$ that preserves the relevant geometry of M – at minimum the local distances, and ideally the geodesic distances of M .

Definition 40.30 (ISOMAP). **ISOMAP** embeds P into $\mathbb{R}^k$ by (i) building the k -nearest-neighbor graph on P , (ii) computing the shortest-path distances $d_G(p, q)$ over the graph as a proxy for the geodesic distance of M , and (iii) applying classical multidimensional scaling to the distance matrix, i.e., finding the points of $\mathbb{R}^k$ whose pairwise Euclidean distances best match d_G .

The geometric content is a sampling theorem: on an isometrically embedded manifold, graph distances converge to geodesic distances as the sampling density grows, so the embedding converges to the true isometry. The **Laplacian eigenmaps** and **LLE** family replaces the geodesic distances by the eigenvectors of a discrete Laplacian built on the proximity graph, which is precisely the cotangent Laplacian and Laplace–Beltrami operator of Chapter 30.12 (Definition 30.60); the two views – shortest paths and spectra – agree asymptotically because the graph Laplacian converges to the geodesic Laplacian as the mesh is refined.

Example 40.31 (Swiss roll). Sampling a rolled strip in $\mathbb{R}^3$ gives a point cloud whose Euclidean distance confuses nearby points across the roll with nearby points along it. ISOMAP restores the flat intrinsic coordinates by routing distances along the strip, undoing the fold; Laplacian eigenmaps and t-SNE achieve the same unrolling while emphasizing local neighborhoods.

Remark 40.32 (Data-structure costs). The pipeline is algorithmic geometry: the neighborhood graph is a range or nearest-neighbor construction (Chapters 13 and 10), the shortest paths are the algorithms of Chapter 24.11, and the spectral step is the machinery of Chapter 30.12. The connection to surface reconstruction (Chapter 28, Section 28.4) is direct: both fields infer a surface from a point sample, one to reconstruct shape, the other to reconstruct a coordinate chart. t-SNE and UMAP trade the global isometry guarantee for a local-neighborhood emphasis that is more robust on real data; see [Bis06, HTF09] for the statistical view.

40.8 Geometric Embeddings

Learning is easier when the ambient dimension is small, and a whole family of results shows that geometry can be compressed into few dimensions with controlled distortion. The master tool is the Johnson–Lindenstrauss lemma, developed in Chapter 39 (Section 39.6, with the random projections of Section 39.7 and the concentration machinery of Section 39.8).

Theorem 40.33 (Johnson–Lindenstrauss). *For any set of n points in $\mathbb{R}^d$ and any $\varepsilon \in (0, 1)$, a random projection to $m = O(\varepsilon^{-2} \log n)$ dimensions preserves all pairwise distances within a factor $1 + \varepsilon$, with high probability (Chapter 39, Section 39.6).*

Remark 40.34 (Why $O(\log n)$ dimensions). The bound is a concentration phenomenon: the projection of a single unit vector onto m standard directions has norm concentrated around $\sqrt{m/d}$, and the union bound over the n^2 pairwise differences forces $m = \Omega(\varepsilon^{-2} \log n)$. The number of points – not the ambient dimension – sets the target dimension, which is why random projections are a preprocessing standard in high-dimensional learning pipelines (Chapter 39, Section 39.5).

Embeddings of different flavors serve different learning goals:

- **Random projections:** distance-preserving maps used before clustering, k -means, and nearest-neighbor search; because they are linear and data-oblivious they compose with the metric structures of Sections 40.2 and 40.4.
- **Locality-sensitive hashing:** the random projections of Section 39.7 double as the hash functions of Section 32.6; LSH is a discrete, collision-based embedding of a metric into a Hamming space that drives approximate nearest-neighbor search and similarity retrieval [IM98, AI06].
- **Feature maps:** the kernel features of Section 40.6 are themselves embeddings; explicit finite-dimensional random feature maps approximate the infinite RKHS geometry of [MMR⁺01] at the cost of a bounded distortion.
- **Dimensionality reduction as learning:** principal component analysis embeds the data into the span of its top eigenvectors – an affine subspace best fit in the least-squares sense – and its geometry is the Eckart–Young optimality of the truncated data matrix; it is the linear baseline against which Section 40.7’s nonlinear embeddings are compared [Bis06, HTF09].

Remark 40.35. The geometric view of embeddings is one of guarantees: what distortion, for what distance, at what cost. This is the vocabulary in which dimension-reduction pipelines are analyzed, and it is exactly the vocabulary of Chapter 39; the practical tension between worst-case guarantees and empirical behavior on real data is the theme of Section 40.11.

40.9 Graph-Based Geometric Learning

A proximity graph turns a point cloud into a combinatorial object on which the learning problem becomes a graph problem. The standard constructions – the k -nearest-neighbor graph, the ε -ball graph, and the Delaunay graph (Chapter 12, Section 12.1) – each encode a different notion of neighborhood, and their spectral theory (Chapter 30.12) drives spectral clustering, label propagation, and graph neural networks.

Definition 40.36 (Proximity Graphs). For a point set $P \subset \mathbb{R}^d$: the **k -nearest-neighbor graph** connects p to q if q is among the k nearest neighbors of p ; the **ε -ball graph** connects p, q if $\|p - q\| \leq \varepsilon$; the **Delaunay graph** connects the pairs whose Voronoi cells share a boundary (Chapter 11). All are proximity queries of the kind built in Chapters 13 and 10.

Definition 40.37 (Graph Laplacian). For a graph $G = (V, E)$ with degree matrix D and adjacency matrix A , the **graph Laplacian** is $L = D - A$. For a point cloud with a proximity graph it is the discrete analogue of the Laplace–Beltrami operator of Chapter 30.12 (Definition 30.60).

Theorem 40.38 (Connectivity by Eigenvalue). *The graph Laplacian L is positive semidefinite, and its kernel has dimension equal to the number of connected components of G . In particular G is connected exactly when the second-smallest eigenvalue $\lambda_2 > 0$, and the associated eigenvector is a real-valued embedding whose sign already separates the two “most separated” parts of the graph.*

Proof sketch. $L = D - A$ satisfies $x^\top Lx = \sum_{(u,v) \in E} (x_u - x_v)^2 \geq 0$, so $L \succeq 0$. A vector in the kernel is constant on each connected component (all adjacent entries equal), giving one independent kernel vector per component. The Fiedler eigenvector of λ_2 is the relaxed solution of the balanced-cut problem; see [Bis06, HTF09] for the derivation. $\square$

Example 40.39 (Spectral clustering). Two interleaved rings in the plane are inseparable by any distance-based partition (Section 40.4) yet separated by the second eigenvector of the k -NN graph Laplacian: points on one ring have near-constant Fiedler coordinate, and the two rings take opposite signs. The pipeline – build the proximity graph, compute eigenvectors, cluster the embedded rows with k -means – is spectral clustering.

Remark 40.40. The proximity-graph choice is a geometric modeling decision: the Delaunay graph is the sparsest graph whose edges do not interfere (relative neighborhood graph), the k -NN graph is denser but captures anisotropic neighborhoods, and the ε -ball graph is controlled by a single scale parameter – the same scale that drives DBSCAN (Definition 40.17). The Delaunay graph’s proximity-preserving properties are the subject of Chapter 12 and the surveys [OBSC00, Aur91]. Diffusion and graph kernels on these graphs define the similarity measures used in label propagation and in graph-based semi-supervised learning [HTF09].

40.10 Optimal Transport: A Geometric Introduction

Optimal transport (OT) measures the distance between two probability measures by the minimal cost of moving one into the other. It is at once a generalization of the assignment and matching problems of Chapter 29 and a geometry on the space of distributions that machine learning uses for comparing histograms, images, and point clouds [PC19].

Definition 40.41 (Transport Plan and Cost). Given two probability measures μ, ν on metric spaces and a nonnegative cost $c(x, y)$, a **transport plan** is a coupling π of (μ, ν) (a joint measure with marginals μ and ν), and the **transport cost** of π is $\int c(x, y) d\pi(x, y)$. The **Wasserstein distance** of order $p \geq 1$ is

$$W_p(\mu, \nu) = \left(\inf_{\pi} \int \|x - y\|^p d\pi(x, y) \right)^{1/p},$$

the p -th root of the minimal transport cost; for $p = 1$ this is the **Earth mover’s distance**.

Theorem 40.42 (Wasserstein is a Metric). *For measures on $\mathbb{R}^d$ with finite moments of order p , the map $(\mu, \nu) \mapsto W_p(\mu, \nu)$ is a metric on probability measures: it is symmetric, $W_p(\mu, \nu) = 0$ iff $\mu = \nu$, and it satisfies the triangle inequality (via the gluing of two optimal plans through a common intermediate measure). Its geometry is the correct notion of distance for measures – unlike L^2 distances between densities, it is sensitive to the underlying geometry of the data space.*

Proof sketch. Symmetry is by transposing the plan. If $W_p(\mu, \nu) = 0$, a sequence of plans with cost tending to zero converges weakly to a plan supported on the diagonal $\{(x, x)\}$, forcing

$\mu = \nu$. The triangle inequality composes an optimal plan from μ to ν with an optimal plan from ν to ρ by disintegration through ν ; the resulting coupling has cost at most $W_p(\mu, \nu) + W_p(\nu, \rho)$, and Minkowski's inequality in L^p finishes the argument. $\square$

Theorem 40.43 (Kantorovich Duality). *For lower-semicontinuous costs, the OT optimum equals a dual supremum:*

$$W_1(\mu, \nu) = \sup_{\varphi} \left(\int \varphi d\mu - \int \varphi d\nu \right),$$

where φ ranges over 1-Lipschitz functions. This is a supporting-function duality: it identifies the Wasserstein distance with a difference of expectations, and it is the computational backbone of OT algorithms [PC19].

For discrete measures, optimal transport is a combinatorial problem that the computational-geometry reader already knows. With $\mu = \sum_i a_i \delta_{x_i}$ and $\nu = \sum_j b_j \delta_{y_j}$, a plan is a nonnegative matrix π_{ij} with row and column sums a and b , and the cost $\sum_{ij} \pi_{ij} c(x_i, y_j)$ is minimized over the transport polytope. When c is a metric on the points, the problem is the transportation problem of operations research; the case $n = m$, all weights 1, reduces to the assignment problem (Chapter 29, Section 29.5, Definition 29.10), solved by Hungarian-type algorithms, and the geometric variants on point sets are exactly the matching objectives of Chapter 29.

Remark 40.44 (Computational geometry meets OT). The Wasserstein distance between two point clouds, restricted to measures of equal mass, is a matching cost: the optimal plan pairs points and the value is a weighted assignment cost, so the algorithms and approximation ideas of Chapter 29 apply. Modern practice uses **entropic regularization**: adding $-\eta \sum_{ij} \pi_{ij} \log \pi_{ij}$ makes the problem strictly convex and smooth, and the Sinkhorn algorithm iterates row and column scalings of a kernel matrix – a sequence of matrix multiplications that scales to large data and approximates W_1 within a controlled entropy bias [PC19]. The resulting distances drive image registration, distributional robustness in learning, and generative-model objectives.

40.11 Computational Geometry for AI

40.11.1 1. Overview

This section inverts the relationship of the rest of the chapter. Where the earlier sections imported geometric structures into learning, here learning imports computation: geometric algorithms serve as subroutines inside learned pipelines, and – in the **learned data structures** program – the machine learns the data structure itself. The two directions share a methodology: understand the geometry of the data, then replace the worst-case-optimal structure by one tailored to the actual distribution.

40.11.2 2. Definitions

Definition 40.45 (Geometric Subroutine). A **geometric subroutine** is a fixed data structure or algorithm called inside a learning pipeline: a k d-tree serving nearest-neighbor queries for a k -NN classifier (Section 40.3, Algorithm 177), an R-tree serving the range queries of spatial data mining (Chapter 10, Section 10.3), or a Delaunay mesher feeding a geometric deep model.

Definition 40.46 (Learned Index). A **learned index** replaces a classical index by a model of the data's key distribution. Kraska, Beutel, Chi, Dean, and Polyzotis observed that an index is itself a function approximating key-to-position maps: a B-tree is a model predicting the rank of a key with bounded error, and a hash index predicts position in an unsorted array [KBC⁺18]. A learned index trains a model f on the cumulative distribution of the keys and answers lookups by evaluating f and correcting locally within a bounded error window.

40.11.3 3. Theory

The theory of learned data structures is the theory of **algorithms with predictions**: preprocess the data with a learned model whose predictions may be imperfect, and give guarantees that interpolate between “model perfect” and “model useless”.

Theorem 40.47 (Lookup with Bounded Prediction Error). *Let the sorted keys $x_1 < \dots < x_n$ have cumulative distribution F , and let $\hat{F}$ be a model with uniform error $\|\hat{F} - F\|_\infty \leq \varepsilon$. Then a lookup of key x reduces to a binary search on the predicted window $[\hat{F}(x) - \varepsilon, \hat{F}(x) + \varepsilon] \cdot n$, costing $O(\log(\varepsilon n))$ probes instead of $O(\log n)$. A model that is a perfect rank predictor gives $O(1)$ -time lookups; an adversarial model degrades gracefully back to binary search [KBC⁺18].*

Proof sketch. If the true rank satisfies $|\text{rank}(x) - \hat{F}(x) \cdot n| \leq \varepsilon n$, then the key x , if present, lies inside the predicted window, and binary search within a window of size $2\varepsilon n$ uses $O(\log(\varepsilon n))$ comparisons. The error bound is the model’s ℓ_∞ error against the empirical CDF; a hierarchical model (recursive model index) reduces the per-level error geometrically. $\square$

Remark 40.48 (Range search under predictions). Range queries use two lookups: predict the first key $\geq a$ and the last key $\leq b$, then report the interval between the corrected positions (Algorithm 181). This is the learned analogue of the range-reporting structures of Chapter 9: the classical kd-tree (Section 9.3) and R-tree report exactly with worst-case guarantees, while the learned index trades a bounded error for locality. The research program asks, systematically, which worst-case guarantees survive prediction – the theme of Section 40.12.

Learning geometric structures is the mirror image: instead of learning a data structure, the models learn the geometric object itself. Neural networks are trained to predict triangulations and meshes – a learned proxy for the Delaunay meshing and refinement algorithms of Chapters 12 and 22 – so that a trained model outputs a mesh in a forward pass rather than by running an incremental algorithm; surface reconstruction pipelines (Chapter 28) likewise learn signed-distance proxies for the restricted-Delaunay reconstruction of Section 28.4. Graph and point-cloud neural networks consume proximity graphs and point coordinates directly, making the geometric subroutine (neighborhood computation, range queries) the rate-limiting step that the fixed structures of Chapters 10 and 9 exist to serve.

40.11.4 4. Pseudocode

Algorithm 181 Learned Range Search

Input: Sorted keys $x_1 < \dots < x_n$ with records; learned rank model f with error bound ε ; query interval $[a, b]$.

Output: All records with keys in $[a, b]$.

```

1: function LEARNEDRANGE( $f, a, b$ )
2:    $\hat{i} \leftarrow \max(1, f(a) - \varepsilon n)$ ;  $\hat{j} \leftarrow \min(n, f(b) + \varepsilon n)$ 
3:    $i \leftarrow$  binary search for the first key  $\geq a$  in  $[\hat{i}, \hat{j}]$ 
4:    $j \leftarrow$  binary search for the last key  $\leq b$  in  $[\hat{i}, \hat{j}]$ 
5:   return records  $x_i, \dots, x_j$ 
6: end function

```

40.11.5 5. Parameter Selection

- **Model granularity:** a recursive model index layers models so that the residual error shrinks per level; the depth trades model-eval cost against the window correction cost [KBC⁺18].

- **Error bound ε :** overestimating enlarges windows and wastes probes; underestimating risks missing keys. Fit ε from held-out keys as the worst observed deviation.
- **Classical fallback:** combine the learned index with an authoritative small structure (e.g., an R-tree for bulk operations) so updates and distribution shifts do not silently corrupt lookups.
- **Updates:** learned indexes assume a static key distribution; handle inserts by buffering or by hybrid model/tree layouts.

40.11.6 6. Complexity

- **Learned lookup:** $O(\log(\varepsilon n))$ probes after evaluating the model; a perfect model gives $O(1)$.
- **Classical baseline:** B-trees answer range searches in $O(\log_B n + m)$ I/Os for m reported records; kd -trees and R-trees report in $O(\sqrt{n} + m)$ and $O(\log n + m)$ expected respectively (Chapters 9 and 10). Learned indexes win when the data distribution is learnable and the geometry of the queries matches it.
- **Geometric subroutines in pipelines:** k -NN queries dominate the cost of many learned models; the structures of Chapter 32 (Sections 32.5 and 32.6) reduce the constant at an approximation cost.

40.11.7 7. Usages

- **Database and storage systems:** learned indexes for key-value stores and columnar analytics [KBC⁺18].
- **Machine-learning pipelines:** kd -trees and R-trees accelerate the repeated proximity and range queries inside k -NN, clustering (Section 40.4), and DBSCAN (Algorithm 179).
- **Geometric deep learning:** point-cloud and mesh neural networks pair with the spatial structures and triangulation algorithms of Chapters 10 and 12; learned meshing predicts the mesh that refinement would build.
- **GIS:** range queries over spatial databases power map generalization and spatial joins (Chapter 36).

40.11.8 8. Testing Methods and Edge Cases

- Validate lookups against a ground-truth sorted array for keys on the window boundary, duplicates, and keys not present in the data.
- Stress-test distribution shift: train on one key distribution and query another; verify the error bound degrades gracefully rather than failing silently.
- Compare learned and classical structures on the same workload and measure both worst-case and average latency; the theoretical trade of Theorem 40.47 must be checked empirically.
- For geometric-subroutine usage, verify the subroutine's own edge cases: empty queries, all-points-in-range, and adversarial point layouts (Chapters 10 and 9).

40.11.9 9. References

Kraska, Beutel, Chi, Dean, and Polyzotis on learned index structures [KBC⁺18]; the classical structures in Chapters 9, 10, and 32; the textbooks [CLRS09, dBCvKO08] for the baseline bounds.

40.12 Research Directions

The frontier of this chapter is the feedback loop between geometry and learning, and several research programs illustrate it.

- **Algorithms with predictions.** The learned-index program of Section 40.11 (see [KBC⁺18]) generalizes to a methodology: every algorithm – caching, scheduling, routing, range search – may consume a prediction of the input distribution and must be analyzed for how its performance interpolates between the prediction-perfect and prediction-adversarial cases. The open question is which worst-case guarantees (Chapter 9) survive prediction and which must be abandoned.
- **Generalization through geometry.** Deep networks generalize despite having far more parameters than samples; one line of explanation is that trained nets admit succinct geometric reparametrizations, and compression-based generalization bounds of Arora, Ge, Neyshabur, and Zhang show that a compressible model has bounded sample complexity [AGNZ18]. This mirrors the core-set philosophy of Chapter 32: a small representative explains the behavior of the whole.
- **The geometry of attention.** Transformer architectures compute pairwise similarity weights – a geometric attention map – and their success in sequence and graph modeling has spawned geometric attention mechanisms on point clouds and meshes; the self-attention matrix is itself a proximity structure in the sense of Chapters 13 and 12 [VSP⁺17].
- **High-dimensional limits.** The curse of dimensionality (Chapter 39, Section 39.9) and the concentration phenomena of Section 39.8 set sharp limits on what any distance-based method can learn from data; the interaction of these limits with the approximation theory of Chapter 32 (Section 32.12) is an active area.
- **Geometric deep learning.** Architectures that respect the intrinsic geometry of their input – convolution on manifolds and meshes, equivariant networks on point clouds, graph neural networks on proximity graphs – fuse the material of Chapters 30.12, 30, and 22 with the learning pipeline of this chapter.

These directions are part of the wider research frontier surveyed in Chapter 41, in particular the machine-learning interfaces of Sections 41.6 and 41.9. The statistical foundations remain those of [Bis06, HTF09, DHS01]; the algorithmic geometry remains those of [dBCvKO08, HP11].

40.13 Project

Build a small geometric machine-learning system that exercises the pipeline of this chapter end to end, then replace one of its geometric components by a learned one and quantify the change.

1. **Data:** generate labeled random point sets (Chapter 27, Section 27.3) in $\mathbb{R}^2$ and $\mathbb{R}^d$ for $d = 2, 8, 32$ with $n = 10^3$ to 10^5 , using two or more well-separated Gaussians plus a noise class; load a real feature data set if available.

2. **Metric baseline:** implement brute-force k -nearest-neighbor classification (Section 40.3) as ground truth; measure accuracy and query time.
3. **Structures:** implement a kd -tree (Chapter 10, Section 10.1) and an R-tree (Section 10.3), and use them for the k -NN queries and for DBSCAN's range queries (Algorithm 179). Record query times and verify identical results to the brute-force baseline.
4. **Clustering:** run k -means (Algorithm 178) with several seeds and DBSCAN on the generated clouds; report objective values, cluster counts, and the sensitivity to k , ε , and $MinPts$ (Section 40.4).
5. **Convex view:** for a two-class subset, compute the maximum-margin separator of Section 40.5 by solving the closest-point problem between the two convex hulls (Section 18.3) and compare its accuracy and support vectors with a kernel version (Section 40.6).
6. **Learned component:** fit a learned index (Section 40.11, Algorithm 181) over the sorted keys of one feature, with an ℓ_∞ error bound fit on held-out keys; compare lookup latency and range-query cost against the classical kd -tree and R-tree, and test robustness to a shifted query distribution.
7. **Verification:** plot accuracy, query time, and window size against n and d ; explain the observed trade-offs using the bounds of Sections 40.3 and 40.11 and the curse of dimensionality (Section 39.9).

The project ties the proximity, spatial-structure, convexity, and learned data-structure material of this chapter to the approximation and high-dimensional machinery of Chapters 32 and 39, and it surfaces the gap between worst-case guarantees and empirical behavior that defines the field.

Chapter 41

Research Frontiers

This chapter surveys the active frontiers of computational geometry: the areas where the questions are open, the models are being renegotiated, and the textbook guarantees no longer apply directly. The common thread is that real data is massive, imprecise, moving, or generated by a learned model, while real hardware is hierarchical and real users demand certified answers.

41.1 Robust and Certified Geometry

Chapter 3 develops the machinery of exact predicates, adaptive arithmetic, and simulation of simplicity that turns geometric algorithms into reliable software. The frontier begins where that chapter stops: extending certification beyond single predicates, and coping with input that is itself uncertain.

A robust *predicate* certifies a sign; a robust *algorithm* certifies a combinatorial structure. Even with exact orientation and in-circle tests, an incremental algorithm can produce an inconsistent triangulation if its tie-breaking rules are not aligned across predicates, and a kinetic or dynamic algorithm can loop if a failure event is mishandled. Kettner, Mehlhorn, Pion, Schirra, and Yap collected small regression inputs—such as a convex-hull routine whose output topology changes under rotation—that now serve as a standard test suite for libraries such as CGAL [KMP⁺08].

Definition 41.1 (Exact Geometric Computation). The *exact geometric computation* (EGC) paradigm separates predicate evaluation from object construction: every predicate is computed exactly, and a constructed object such as an intersection point is represented by an exact expression evaluated to a certified precision on demand [Yap97, HMY04].

EGC converts robustness into a cost question: how many bits must an expression evaluation carry so that downstream predicates remain exact? Precision-driven evaluation answers this by evaluating each expression to successively higher precision until a requested error bound is met (Section 3.6).

A second line of work treats *data imprecision*: the input is not a set of points but a set of regions, each guaranteed to contain one unknown true point.

Definition 41.2 (Imprecise Point). An *imprecise point* is a connected region $\mathcal{R} \subset \mathbb{R}^d$ (for example an axis-aligned square, a disk, or a segment) known to contain the true point, whose exact location is unspecified. The set of all inputs obtained by choosing one point from each region is the *uncertainty set*.

Given imprecise points, one studies the *possible* values of geometric measures. For the diameter, width, closest pair, smallest enclosing circle, and bounding box, Löffler and van Kreveld give efficient algorithms for the largest and smallest possible values when the regions are squares or disks, and prove hardness for variants such as the largest possible width under segment

imprecision [LvK10b]. The analogous questions for convex hulls are polynomial for squares but NP-hard for some segment models [LvK10a]. The frontier combines exact predicates on uncertain regions, randomized robustness certificates, and near-float algorithms whose combinatorial output is certified with probability $1 - \delta$.

41.2 Massive Geometric Data Sets

Modern geometric inputs—lidar sweeps, sensor networks, simulation output, and transaction logs—routinely contain hundreds of millions of points. Three ideas make such data sets tractable: output-sensitive algorithms, parameterized complexity bounds, and sparse geometric structures.

An *output-sensitive* algorithm runs in time proportional to the size of its output. The canonical example is computing a convex hull in time $O(n \log h)$, where h is the number of hull vertices, achieved by Chan’s algorithm (Section 4.3); when the shape is simple, $h \ll n$ and the improvement over the worst-case $O(n \log n)$ is dramatic [BCKO08].

Theorem 41.3 (Output-Sensitive Hulls). *The convex hull of n points in the plane can be computed in $O(n \log h)$ time, where h is the number of hull vertices, and this is optimal in the comparison model; in $\mathbb{R}^3$ the hull can be computed in $O(n \log n)$ time, matching the worst-case output size [BCKO08].*

The same philosophy extends to *parameterized* bounds, where the running time is expressed as a function of both n and an independent parameter such as the number of reflex vertices, the treewidth of an underlying graph, or the number of outliers, so that near-linear behavior is provable on the structured instances that arise in practice.

A second cornerstone is the *geometric spanner*: a sparse graph that approximately preserves all pairwise distances.

Definition 41.4 (Geometric Spanner). A t -spanner of a point set P is a graph G with vertex set P such that for all $p, q \in P$ the graph distance satisfies

$$d_G(p, q) \leq t \cdot \|p - q\|.$$

The parameter $t \geq 1$ is the *stretch factor*.

Spanners with $O(n)$ edges and stretch $1 + \varepsilon$ can be built in $O(n \log n)$ time via the well-separated pair decomposition (WSPD) of Callahan and Kosaraju [CK95], and serve as compact stand-ins for the complete Euclidean graph in routing, motion planning, and proximity structures; the monograph of Narasimhan and Smid develops the full theory [NS07]. The frontier concerns spanners for *moving*, *streamed*, and high-dimensional point sets [NS07, Gui98].

Finally, when data is high-dimensional but intrinsically low-dimensional, one embeds it into a lower-dimensional space while preserving the geometry of interest. The Johnson–Lindenstrauss lemma (Section 39.6) preserves pairwise distances up to a factor $1 \pm \varepsilon$ in $O(\log n)$ dimensions; the frontier is *nonlinear* embedding, treated from the geometric point of view in Chapter 40, and sparse embedding schemes whose cost scales with the intrinsic rather than the ambient dimension.

41.3 Streaming Geometry

A *data stream* is a sequence of items that arrives once, in a fixed order, too quickly and too voluminously to store. Streaming geometry asks what can be computed about a point set that arrives as a stream, using memory polylogarithmic in n and a single (or few) passes over the data [Mut05]. The hallmark of the model is that exact answers are usually impossible: the algorithm must approximate, and the guarantee is a *summary* from which the desired measure can be recovered within error ε .

41.3.1 Definitions

Definition 41.5 (Streaming Summary). A *summary* of a stream of points is a data structure of size $s(n) \ll n$ from which one can answer the query of interest within a specified approximation factor. An algorithm is *single-pass* if it maintains the summary while reading each point exactly once.

Definition 41.6 (ε -Coreset). An ε -coreset of a point set $P \subset \mathbb{R}^d$ for a cost function f is a subset (or weighted subset) $Q \subseteq P$ such that for every candidate solution X ,

$$(1 - \varepsilon) f(P, X) \leq f(Q, X) \leq (1 + \varepsilon) f(P, X),$$

where $f(P, X)$ is the cost of X with respect to all of P . For k -means and k -median, X ranges over sets of k centers; for extent measures such as the diameter, X ranges over directions and the coreset is an ε -kernel.

41.3.2 Theory

Har-Peled and Mazumdar showed that a point set admits an ε -coreset for k -median and k -means clustering of size $O(k\varepsilon^{-d} \log n)$, giving the first streaming algorithm with provably sublinear memory for these problems [HPM04]. For extent measures, an ε -kernel of size $O(\varepsilon^{-(d-1)/2})$ can be maintained in a stream using $O(\log^d n)$ space and $O(1)$ amortized time per point [AGHV01].

Theorem 41.7 (Streaming k -Center). *There is a single-pass, $O(k)$ -space algorithm that, given the optimal k -center radius r^* , either outputs k centers of radius at most $2r^*$, or correctly reports that r^* was chosen too small. Guessing r^* from a geometric progression yields a 2-approximation in $O(k \log \Delta)$ space, where Δ is the spread of the input [Gon85, Mut05].*

41.3.3 Pseudocode

Algorithm 182 Streaming k -center (greedy, radius r)

Input: A stream S , an integer k , a radius candidate r .

Output: k centers covering the stream within radius $2r$, or FAIL.

```

1: function STREAMINGKCENTER( $S, k, r$ )
2:    $C \leftarrow \emptyset$ 
3:   for each point  $p$  in the stream do
4:     if  $d(p, C) > r$  then
5:        $C \leftarrow C \cup \{p\}$ 
6:       if  $|C| > k$  then
7:         return FAIL
8:       end if
9:     end if
10:  end for
11:  return  $C$ 
12: end function
```

The accepted centers form an r -separated set, so if the optimal radius is r , at most k centers are accepted and every stream point lies within $2r$ of some center [Gon85].

41.3.4 Parameters

- ε : the target approximation determines the coreset size; d -dependencies such as ε^{-d} make small fixed dimensions essential.

- k : if not known in advance, a range of guesses for r is maintained in parallel.
- **Pass count**: one pass forbids recovering from a bad partition, so randomized orderings and merge-reduce schedules bound the error.

41.3.5 Complexity

- **Space**: $O(k)$ for streaming k -center; $O(k\varepsilon^{-d} \log n)$ for k -means coresets; $O(\varepsilon^{-(d-1)/2} \log^d n)$ for extent kernels.
- **Lower bound**: any single-pass exact algorithm needs $\Omega(n)$ space, so the approximation is unavoidable.

41.3.6 Usages

- **Network monitoring**: summary statistics over IP-flow streams.
- **Large-scale lidar**: approximate density and extent summaries before full reconstruction (Chapter 28).

41.3.7 References

- Muthukrishnan [Mut05]: the standard survey of data-stream algorithms.
- Har-Peled and Mazumdar [HPM04]: coresets for k -means and k -median.
- Agarwal, Guibas, Hershberger, and Veach [AGHV01]: extent measures and ε -kernels for moving and streamed points.
- Chapter 32, Section 32.4 for the coreset framework in the batch setting.

41.4 External-Memory Geometry

When a data set exceeds main memory, the cost model changes: random-access memory is replaced by a two-level hierarchy in which a block transfer (an I/O) between fast internal memory and slow disk dominates all arithmetic. External-memory (EM) geometry designs algorithms whose I/O count is nearly optimal, exploiting the fact that block transfer is roughly as cheap as single-record transfer, so locality is the currency of the model [AV88, Vit01].

41.4.1 Definitions

Definition 41.8 (Parallel Disk Model). In the *parallel disk model* (PDM), N records reside on disk, at most M records fit in internal memory, and each I/O transfers a contiguous block of B records, where $B \leq M/2 < N$. The cost of an algorithm is measured by the number of I/O s, the disk space, and the internal CPU time [Vit01].

41.4.2 Theory

Theorem 41.9 (I/O Complexity of Sorting). *Sorting N records requires, up to a constant factor,*

$$\Theta\left(\frac{N}{B} \log_{M/B} \frac{N}{B}\right)$$

I/O s, and external merge and distribution sort achieve this bound. The bound also holds for permuting and for the Fourier transform [AV88].

The sorting bound is the transfer theorem of EM geometry: because most geometric primitives can be reduced to sorting, the same bound applies to constructing convex hulls, Voronoi diagrams, and Delaunay triangulations of massive point sets. Beyond sorting, the survey paradigms include the *distribution sweep* for spatial joins, *buffer trees* for batched sweep-line processing, persistent B-trees for batched point location, and *external fractional cascading* for red–blue segment intersection [Vit01]. Online, multidimensional indices such as the R-tree support range queries with linear space [Gut84, BKSS90].

41.4.3 Pseudocode

Algorithm 183 External merge sort (skeleton)

Input: Records on disk (N), memory M , block size B .

Output: The records in sorted order, on disk.

```

1: function EXTERNALMERGESORT( $N, M, B$ )
2:    $r \leftarrow 0$ 
3:   read  $\lfloor M/B \rfloor$  blocks, sort internally, write a run
4:   repeat
5:     merge  $\Theta(\lfloor M/B \rfloor)$  runs in one pass, outputting one block at a time
6:      $r \leftarrow \lceil r / \lfloor M/B \rfloor \rceil$ 
7:   until  $r = 1$ 
8:   return the sorted file
9: end function

```

41.4.4 Parameters

- **Block size B :** larger B amortizes transfer latency; the running time depends on B only through $\log_{M/B}$, so the exact value matters little once M/B is comfortable.
- **Memory M :** the fan-in $\lfloor M/B \rfloor$ determines the number of passes; memory must be shared between the buffer pool and the sort work area.
- **Locality of the geometry:** grid or space-filling-curve orderings (Chapter 10.10) make point sets block-local before the geometric algorithm begins.

41.4.5 Complexity

- **I/Os:** $\Theta(\frac{N}{B} \log_{M/B} \frac{N}{B})$ for sorting and most reduction problems; $O(\log_B N)$ I/Os per operation for B-tree-based dynamic dictionaries.
- **Space:** linear in blocks on disk; internal memory M as given.

41.4.6 Usages

- **Geographic information systems:** spatial joins over terrain and cadastral data (Chapter 36).
- **Spatial databases:** R-tree and B-tree based indexing of moving objects and map tiles [Gut84, BKSS90].

41.4.7 References

- Aggarwal and Vitter [AV88]: tight I/O bounds for sorting-related problems.
- Vitter [Vit01]: the standard survey of EM algorithms and data structures.
- Guttman [Gut84] and Beckmann et al. [BKSS90]: R-tree and R*-tree.
- Chapter 10 for the internal-memory structures that these indexes externalize.

41.5 Geometric Algorithms for Autonomous Systems

Autonomous systems—self-driving vehicles, warehouse robots, and drones—ask geometry to run on a moving platform with stale, noisy, and partial information. Two classical bodies of theory supply the foundations: motion planning (Chapter 26) and kinetic data structures (Chapter 34).

The configuration-space viewpoint reduces a robot and its obstacles to a point moving in a high-dimensional space (Section 26.1), and the classical exact algorithms of Lozano-Perez, Latombe, and successors solve the planar case [LP83, Lat91]. For articulated robots with many degrees of freedom, exact methods are intractable and the probabilistic roadmap (PRM) of Kavraki, Svestka, Latombe, and Overmars builds a random graph of feasible configurations (Section 26.4) [KSLO96]. The frontier is *closed-loop* planning: replanning as the world model is updated by perception, respecting both geometric constraints and the latency and uncertainty of the sensor loop.

When the environment moves—other vehicles, pedestrians, weather fronts—the geometric structures describing it must be maintained over time rather than rebuilt from scratch. The kinetic data structure (KDS) framework formalizes this.

Definition 41.10 (Kinetic Data Structure). A *kinetic data structure* (KDS) for a set of moving objects maintains a discrete structure S together with a set of *certificates*, each a geometric predicate whose truth at the current time guarantees the correctness of S . The motion is described by known trajectories (typically low-degree polynomials in time); a *failure event* is the first time a certificate becomes false, at which point the structure and the certificate set are updated [Gui98, BGH99].

Definition 41.11 (Certificate). A *certificate* for a structure S is a predicate $c(t)$ over the object positions at time t such that if all certificates hold, the structure is correct. The structure is *compact* if the number of certificates is near-linear, *local* if each object participates in few certificates, and *responsive* if a single event costs few certificate changes.

The classic KDSs for kinetic sorting, the convex hull, the closest pair, and the Delaunay triangulation all achieve near-linear storage [BGH99, Gui98]. Agarwal, Guibas, Hershberger, and Veatch proved that some extent measures are intrinsically costly to maintain exactly [AGHV01].

Theorem 41.12 (Kinetic Extent Bounds). *The diametral pair of n points in the plane, each moving with constant velocity, changes $\Theta(n^2)$ times in the worst case, so no KDS can maintain the diameter with fewer than $\Theta(n^2)$ events. This motivates approximate kinetic maintenance via ε -kernels that are valid for all time [AGHV01, Gui98].*

In collision detection, the kinetic tiling of free space by Agarwal, Basch, Guibas, Hershberger, and Zhang maintains a dynamic decomposition whose certificates certify that the moving objects remain separated [ABG⁺02]; kinetic Delaunay triangulations under uniform linear motion now achieve a near-quadratic event bound [Rub16].

The perception side of autonomy is also geometric: registering a live sensor scan to a stored map is the iterative closest point (ICP) problem (Section 29.8 and Chapter 29) [BM92], and

occupancy grids built from range data use the spatial structures of Chapter 10. The frontier combines these into pipelines whose primitives run at control rates with bounded latency, and whose failure modes—occlusion, sensor dropout, ambiguity—are certified rather than merely measured.

41.6 Geometry in Machine Learning

Chapter 40 develops the geometric view of data: points as samples, distances as similarity, and manifolds as structure. The research frontier runs in the opposite direction as well, using machine learning to improve geometric algorithms.

Classical analysis is worst-case, but real instances are rarely worst-case. *Algorithms with predictions* accept a machine-learned oracle and deliver guarantees that interpolate between ideal performance when the predictions are perfect and the classical worst case when they are useless [MV22].

Definition 41.13 (Learning-Augmented Algorithm). An algorithm *with predictions* receives, alongside its input, a set of predictions $\hat{y}_1, \dots, \hat{y}_m$ from an oracle. Its cost is a function of the input *and* of the prediction error η ; the design goal is that the cost is near-optimal for $\eta = 0$ and degrades gracefully to the classical worst case as η grows [MV22].

The cleanest geometric instantiation is the *learned index* of Kraska, Beutel, Chi, Dean, and Polyzotis, which observes that a B-tree is a model mapping keys to offsets and replaces it with a learned regressor followed by a small correction structure [KBC⁺18]. The same “model plus repair” pattern is being applied to spatial indexes, Bloom filters, and nearest-neighbor structures; the open problem is to give learned indexes worst-case guarantees matching classical ones [KBC⁺18, MV22].

The complementary frontier uses geometric data structures inside learned pipelines. Approximate nearest-neighbor search, the workhorse of embedding-based retrieval, is served by locality-sensitive hashing (Section 39.4) and by navigable small-world graphs such as HNSW [IM98, MY20]. For clustering, streaming, and range counting, the coresets and summaries of Sections 41.3–41.4 become the feature maps that learned models consume. Section 40.11 collects the applications of this chapter’s models to AI systems.

Remark 41.14 (Two Kinds of Learning in Geometry). It is important to separate *learning from geometry* (estimating a function on geometric data, as in geometric deep learning, Section 41.9) from *learning for geometry* (replacing a geometric data structure by a learned model). Both are active; the second has the sharper performance question, because a learned structure must still justify its worst-case behavior.

41.7 Topological Data Analysis

Topological data analysis (TDA) treats data as a sample of an unknown space and asks for its topological invariants, which are robust to noise precisely because they are coarse. The theoretical foundations appear in Chapter 30; the frontier is making persistent homology scale to massive and high-dimensional inputs.

Definition 41.15 (Filtration). A *filtration* of a simplicial complex is a nested sequence

$$K_1 \subseteq K_2 \subseteq \dots \subseteq K_m$$

of subcomplexes, typically induced by a function such as the radius of a ball around each data point (the Čech, Vietoris–Rips, and alpha complexes of Chapter 30).

Definition 41.16 (Persistent Homology). For a filtration as above, the p -th *persistence module* tracks each homology class: it is *born* at the first complex in which it appears and *dies* when it merges or vanishes. The *persistence diagram* of dimension p is the multiset of birth–death pairs (b, d) of these classes, including the diagonal.

Zomorodian and Carlsson showed that the persistence module of a filtered complex is a graded module over a polynomial ring, and that persistent homology over a field can therefore be computed from a boundary matrix in cubic time via a simple matrix-reduction algorithm [ZC05]; the pipeline is surveyed in Carlsson’s paper [Car09] and the monograph of Edelsbrunner and Harer [EH10].

Theorem 41.17 (Stability of Persistence Diagrams). *If two filtrations are induced by functions f and g on the same space that differ by at most $\|f - g\|_\infty$, then the corresponding persistence diagrams differ by at most that amount in bottleneck distance. Consequently, topological features whose birth–death intervals are long are robust to bounded noise [EH10, Car09].*

The frontier questions are computational: computing persistence in dimensions above three, streaming and distributed reductions for massive filtrations, approximate persistence with certified error, and the *Mapper* family of methods that summarize a data set at many scales. Applications span biology, materials science, and shape analysis, where alpha shapes and persistence barcodes bridge the geometry of Chapter 12 and the topology of Chapter 30 [EKS83, DSW10].

Remark 41.18 (Discrete Morse Theory). Discrete Morse theory gives a sparse, combinatorial description of the critical structure of a function on a complex, reducing the number of simplices needed to compute homology and persistence by orders of magnitude [For98, For02]. It is the main practical accelerator for persistence computations in three dimensions.

41.8 Shape and Surface Reconstruction

Chapter 28 develops reconstruction from first principles: the local feature size, the medial axis, and the certificate that a sample is dense enough for the crust or cocone algorithm to succeed [ABE98, AB99, Dey07]. The frontier is threefold.

First, *reconstruction without clean samples*: real scans contain noise, outliers, holes, and nonuniform density. Ball-pivoting reconstructs locally by rolling a ball of fixed radius over the samples [BMR⁺99]; Poisson reconstruction fits an implicit indicator function whose gradient matches the oriented sample normals and is robust to noise and nonuniformity [KBH06]. The frontier is making provable guarantees—sampling conditions, approximation bounds, and output quality—survive these practical relaxations [BO05].

Second, *reconstruction at scale*: the pipelines that win benchmarks—Voronoi filtering, Poisson, and their variants—all begin with a Delaunay or nearest-neighbor computation over millions of points, and the techniques of Sections 41.3–41.4 are being imported to push reconstruction into the out-of-core regime.

Third, *certified reconstruction*: recent directions certify the output surface’s Hausdorff closeness to the input and its topology, terminating with either a reconstruction or a statement about the sample that is provably insufficient. Section 28.3 lists the point-cloud models; Section 28.7 develops the implicit surface view that the frontier generalizes.

41.9 Geometric Deep Learning

Geometric deep learning (GDL) studies neural networks that respect the geometry of their input domain. The unifying blueprint of Bronstein, Bruna, Cohen, and Velickovic organizes the field by its *geometric priors*: symmetry and invariance under a domain’s group, stability under deformation, and scale separation [BBCV21]. On grids these priors give convolutional

networks; on sets they give permutation-invariant networks; on graphs and manifolds they give graph neural networks and intrinsic mesh CNNs [BBCV21, B⁺17].

Definition 41.19 (Geometric Prior). A *geometric prior* is a regularity of the input domain encoded into a network architecture, typically as invariance or equivariance to a symmetry group G : the network must satisfy $f(g \cdot x) = \rho(g) f(x)$ for a representation ρ , so that the model cannot waste capacity on transformations that leave the meaning of the data unchanged [BBCV21].

The connection to this book’s algorithms is direct. For point clouds, PointNet uses a symmetric aggregation function (a max over point features) to achieve permutation invariance, and then a spatial transformer that is a learned rigid alignment—the same registration problem of Chapter 29 [QSMG17]. For meshes, intrinsic networks operate on the Laplace–Beltrami spectrum of Chapter 30.12. The frontier is *expressivity*: which geometric structures can a network provably learn, and which classical algorithms (Delaunay tests, persistence diagrams, optimal transports) can be differentiated through so that a gradient-based optimizer can search over geometry itself. That last question is the subject of Section 41.10.

41.10 Differentiable Geometry

Modern 3D reconstruction is inverse graphics: given images or scans, estimate geometry, appearance, and motion so that a forward model reproduces the observations. This requires *differentiability*—the ability to compute gradients of an output image with respect to geometric parameters. The foundational issue is that discrete geometric events (which triangle is visible, which sample is the nearest neighbor) are not differentiable, so gradient information must be smoothed, relaxed, or randomized.

Definition 41.20 (Differentiable Renderer). A *differentiable renderer* is a forward model $R(\theta)$ of image synthesis parameterized by scene geometry θ such that the derivative $\partial R / \partial \theta$ is available, allowing loss gradients to be propagated back through the rendering process to the geometry [LB14].

OpenDR was the first general framework to expose this derivative, smoothing the discrete visibility function so that silhouette losses drive shape estimation [LB14]. The frontier now spans differentiable ray tracing with importance-weighted estimates, differentiable mesh simplification, and differentiable solvers for PDE-based geometry processing, where walk-on-spheres estimators provide derivatives with respect to the domain’s geometry [Y⁺25]. At the far end, neural field representations collapse the mesh into a learned simplicial or neural representation; Simplicitis shows that elastic simulation can be carried out on a learned reduced basis, agnostic to whether the input was a mesh, a point cloud, or a neural field [MSP⁺24]. The open problems are algorithmic: certificate guarantees for gradient estimates, and differentiable versions of combinatorial structures such as Delaunay triangulations and persistence diagrams.

41.11 Geometry Processing

Geometry processing is the discipline of operating on meshes and point clouds—remeshing, parameterization, deformation, simplification, and repair—with an emphasis on numerical quality and robustness rather than asymptotic complexity. Its foundations appear in Chapters 20.12, 20.1, and 30.1; the frontier is replacing fragile, hand-tuned pipelines with algorithms that work on *in-the-wild* geometry.

The classical guarantee of Chapter 22—a Delaunay tetrahedral mesh with bounded angle quality—is achieved by refinement for clean inputs, but real inputs arrive as self-intersecting, non-manifold, dirty meshes. The `fTetWild` and `TriWild` pipelines envelope the input geometry

with an ambient mesh and project back, producing robust tetrahedral and surface remeshes with angle bounds in the presence of degenerate input [H⁺18, H⁺21]. The frontier concerns output size and runtime, since the ambient space is typically much larger than the surface.

A second direction replaces the extrinsic (coordinate-based) view of a mesh with its intrinsic structure—geodesic distances and the Laplace–Beltrami operator—so that algorithms are invariant to how the shape is embedded; learned descriptors are trained directly on intrinsic quantities [S⁺25]. The frontier is the integration of these tools with the differentiable pipelines of Section 41.10.

Example 41.21 (Mesh Repair as Topological Repair). Repairing a triangle soup requires closing holes, resolving self-intersections, and fixing orientation. Volumetric repair rasterizes the soup into an occupancy grid and extracts an isosurface, which guarantees a watertight, manifold output at the price of a resolution parameter; this is the same march between a continuous surface and its discretization that Chapter 23 develops.

41.12 Open Problems in Computational Geometry

Every section of this chapter touches an open problem; this section collects the most famous and most structural ones. A fuller, living list appears in the open-problems chapter of the Handbook of Discrete and Computational Geometry [TOG17], and the long-standing conjectures trace back to Shamos’s thesis [Sha78].

1. **The $\Omega(n \log n)$ conjecture for triangulation.** Is there a true $O(n \log n)$ algorithm for computing the Delaunay triangulation of n points in the plane? The best algorithms run in $O(n \log n)$ randomized or deterministic expected time, and all rely on sorting; Shamos conjectured that $\Omega(n \log n)$ is inherent, but no conditional lower bound separates triangulation from the easier problems on which the bound is known [Sha78].
2. **Algebraic decision-tree lower bounds.** Ben-Or’s method proves $\Omega(n \log n)$ lower bounds for element distinctness and for convex hulls and closest pairs in the algebraic decision-tree model [BO83]. Extending such bounds to *randomized* and *quantum* models remains open.
3. **The complexity of robust computation.** Is there a quasi-linear-time algorithm for exact geometric predicates and constructed objects whose cost is bounded by a constant factor times the floating-point cost? Current adaptive predicates are near-float for signs, but certified *construction* of intersections and arrangements still carries large hidden constants [Yap97, HMY04].
4. **Kinetic structures with near-linear total cost.** The kinetic Delaunay triangulation under linear motion was recently shown to have $O(n^2)$ events, closing a long-standing gap [Rub16]. Whether a KDS exists whose *total* event processing over any polynomial motion is $o(n^2)$ for standard proximity structures remains open [Gui98, ABG⁺02].
5. **Streaming and external-memory trade-offs.** Tight space lower bounds for geometric streaming (whether $O(k \log n)$ bits suffice for approximate k -center in fixed dimension) and matching I/O lower bounds for geometric joins beyond the sorting barrier are unresolved [Mut05, Vit01].
6. **Spanners in low doubling dimension.** The spanner theory of Section 41.2 is Euclidean; whether $O(n)$ -edge spanners with stretch $1 + \varepsilon$ exist for arbitrary metric spaces of doubling dimension d , with running time $O(n \log n)f(d)$, is open in several regimes [NS07].

7. **The robustness of learned structures.** Whether learning-augmented data structures can simultaneously achieve worst-case guarantees matching classical indexes and average-case performance exceeding them is a formal question with only partial answers [KBC⁺18, MV22].
8. **Imprecision and the price of ignorance.** For the measures of Section 41.1, the complexity gap between computing worst-case and best-case values over an uncertainty set is far from understood; several natural variants such as the largest possible width remain open even for simple region models [LvK10b, LvK10a].

41.13 How to Read Computational Geometry Research

Computational geometry research is published in the venues that define the field: the Symposium on Computational Geometry (SoCG), the Symposium on Discrete Algorithms (SODA), and the Symposium on Theory of Computing (STOC), with the *Journal of the ACM* and *Discrete & Computational Geometry* carrying journal versions. A research paper is read in a specific order, and the order is not the order it is written:

1. **The title and abstract** tell you the problem, the model, and the headline bound. Ask whether the model matches the applications you care about.
2. **The statement of results** tells you what is claimed. Write it down with its parameters: what is held fixed, what is worst case, what is expected.
3. **The overview of techniques** (often Section 2) is the most valuable part for a reader who does not plan to rederive everything. It names the ideas: a duality, a randomized incremental construction, a charging argument.
4. **The proof** is read only as deeply as your goals require. For most readers the structure matters more than the algebra: where does the argument fail if the input is degenerate, if the dimension grows, or if the arithmetic is rounded?
5. **The applications and experiments**, when present, tell you how the theory maps to practice and where the hidden constants live.

The textbooks of this field remain the best gateway to the literature: de Berg, Cheong, van Kreveld, and Overmars for the classical algorithms [dBCvK008, BCK008], Matousek for discrepancy and its applications [Mat99], and the Handbook of Discrete and Computational Geometry for the breadth of the field [TOG17]. When reading a paper, maintain a *question log*: for each result, record what assumption is essential, what would change if it were relaxed, and whether the algorithm would survive contact with real data. Research often begins exactly at the point where a logged question has no answer in the literature.

41.14 Designing Computational Experiments

Computational geometry is unusual among algorithms disciplines in that worst-case analysis is both highly developed and frequently misleading: the notorious cases (cocircular points, collinear sets, degenerate coordinates) are exactly the ones that break floating-point implementations, and the worst-case instances of a randomized incremental algorithm may be vanishingly rare. Experimentation therefore has a specific shape.

1. **Choose a baseline.** Compare against a reference implementation, not just against a theoretical bound. CGAL, Boost Geometry, and the libraries of Appendix E provide the de facto baselines. Cite and version them.

2. **Generate structured instances.** Random uniform points are the least informative test set: they are in general position with probability one and are trivially well distributed. Use the generators of Chapter 27—Poisson-disk samples and Halton sequences (Section 27.1)—and add *adversarial families*: cocircular and collinear inputs, high aspect ratios, and clustered data.
3. **Measure what matters.** Report running time and peak memory, but also the number of predicate evaluations (the quantity the theory controls) and the number of degeneracies encountered. A timed run alone cannot distinguish an algorithm that is fast because it is clever from one that is fast because it is silently wrong.
4. **Test correctness, then robustness.** For every output structure, verify the certificates: is the reported triangulation a valid planar graph, and does every triangle satisfy the empty-circumcircle property? Then run the robustness suites of Section 41.1, including the regression inputs collected by Kettner et al. [KMP⁺08].
5. **Scale systematically.** Increase n geometrically and record the empirical order of growth, comparing it to the theoretical one. Report the crossover points between algorithms, which are where the hidden constants live.
6. **Reproduce and release.** Record random seeds, parameter values, and data generators so that every table is reproducible. The reproducible experiment is the unit of currency of the field's experimental literature.

A useful rule of thumb: if an experiment cannot be described in one sentence that states the question, the instances, the metric, and the baseline, it is not yet designed.

41.15 From Graduate Study to Research

This book ends where research begins. The transition from coursework to research in computational geometry is mostly a transition of attitude, and a few habits accelerate it.

First, *keep a research notebook* in which every problem you meet—in the exercises of this book, in a paper you read, in a colleague's seminar—is written down together with its parameters and its status. Open problems are not hidden in the literature; they accumulate in such notebooks. Second, *learn the theorems twice*: once for their statements, and once for the moment in the proof where an assumption is used. The second reading is what converts a theorem into a research tool, because most research questions are of the form "what happens if this assumption is relaxed?" Third, *implement what you study*; the distance between a bound that holds in the worst case and an algorithm that runs in the real world is measured in implemented experiments.

Fourth, *choose problems at the intersection of a model and a need*. The frontiers of Sections 41.3–41.11 all arose because a classical model stopped matching a real scale or a real source of noise. The most productive problems are typically ones where a classical algorithm is provably correct, empirically fragile, and structurally simple enough that its analysis can be repaired. Fifth, *publish by solving, not by polishing*: a short paper that resolves a clean question in a new model—streaming, kinetic, imprecise, learning-augmented—is the standard currency of the field.

Finally, the field rewards *transfer*: a technique developed for one problem (an ε -kernel for extent, a WSPD for spanners, an adaptive predicate for robustness) is frequently the key to an unrelated problem. The boundary between what is known and what is unknown is where the techniques of the previous parts of this book meet the questions of the current one. That boundary is, in the end, the natural starting point of a research career.

41.16 Computational Algebra and Similarity

41.17 Buchberger's Algorithm

41.17.1 Overview

Buchberger's algorithm is a fundamental method in computational algebraic geometry for computing a **Gröbner basis** of a polynomial ideal. Much like the Gaussian elimination algorithm for linear systems, Buchberger's algorithm transforms a set of multivariate polynomials into a “standard” form that makes it easy to solve systems of equations, perform polynomial division, and analyze the properties of the underlying algebraic variety. The algorithm was first described by Bruno Buchberger in his 1965 PhD thesis [Buc65] and has since become a cornerstone of computational algebra, extensively treated in standard references such as Cox, Little, and O'Shea [CLO07].

41.17.2 Definitions

Definition 41.22 (Polynomial Ideal). Let k be a field and let $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ be a set of polynomials. The **polynomial ideal** generated by these polynomials is the set

$$I = \langle f_1, \dots, f_m \rangle = \left\{ \sum_{i=1}^m h_i f_i \mid h_i \in k[x_1, \dots, x_n] \right\}.$$

The polynomials $f_1, \dots, f_m$ are called the *generators* of I .

Definition 41.23 (Monomial Order). A **monomial order** on $k[x_1, \dots, x_n]$ is a total ordering $\succ$ on the set of monomials $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ (where $\alpha \in \mathbb{N}^n$) that satisfies:

1. $x^\alpha \succ 1$ for all $\alpha \neq \mathbf{0}$.
2. If $x^\alpha \succ x^\beta$, then $x^{\alpha+\gamma} \succ x^{\beta+\gamma}$ for all $\gamma \in \mathbb{N}^n$.

Common choices include **lexicographic** (Lex) and **graded reverse lexicographic** (Degrevlex). For two monomials x^α and x^β , the Lex order compares first by x_1 -exponent, then x_2 , etc., while Degrevlex first compares by total degree and then breaks ties by the last differing exponent in reverse.

Definition 41.24 (Leading Term). Given a monomial order $\succ$ and a nonzero polynomial $f = \sum_{\alpha} c_{\alpha} x^{\alpha}$, the **leading term** $\text{LT}(f)$ is the term $c_{\alpha} x^{\alpha}$ with the greatest α under $\succ$. Similarly, $\text{LM}(f)$ denotes the leading monomial (coefficient removed) and $\text{LC}(f)$ the leading coefficient.

Definition 41.25 (Gröbner Basis). A finite set $G = \{g_1, \dots, g_t\} \subset I$ is a **Gröbner basis** for the ideal $I = \langle f_1, \dots, f_m \rangle$ with respect to a monomial order $\succ$ if

$$\langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle = \langle \text{LT}(I) \rangle,$$

where $\text{LT}(I) = \{\text{LT}(f) \mid f \in I \setminus \{0\}\}$ is the set of all leading terms of nonzero polynomials in I .

Definition 41.26 (S-Polynomial). Given two nonzero polynomials f, g and a monomial order $\succ$, the **S-polynomial** is defined as

$$S(f, g) = \frac{L}{\text{LT}(f)} f - \frac{L}{\text{LT}(g)} g,$$

where $L = \text{lcm}(\text{LM}(f), \text{LM}(g))$ is the least common multiple of the leading monomials of f and g . The S-polynomial is constructed precisely to cancel the leading terms of f and g .

41.17.3 Theory

The algorithm starts with an initial set of generators G . It systematically checks all pairs of polynomials (f, g) in G and calculates their S-polynomial:

$$S(f, g) = \frac{L}{LT(f)}f - \frac{L}{LT(g)}g$$

where $L = \text{lcm}(LM(f), LM(g))$.

The S-polynomial is then reduced by the current basis G using multivariate polynomial division. If the remainder is non-zero, it means G is not yet a Gröbner basis, and the remainder is added to G . This process continues until the S-polynomial of every pair in G reduces to zero.

Theorem 41.27 (Buchberger's Criterion). *A finite generating set G for an ideal I is a Gröbner basis with respect to a monomial order $\succ$ if and only if for every pair $f_i, f_j \in G$, the remainder of the S-polynomial $S(f_i, f_j)$ on division by G is zero; that is, $S(f_i, f_j) \xrightarrow{G} 0$ for all $i \neq j$.*

Proof. The forward direction is straightforward: if G is a Gröbner basis, then $LT(G)$ generates $LT(I)$, so every S-polynomial reduces to zero by the division algorithm. The converse relies on the fact that if $S(f_i, f_j) \rightarrow 0$ for all pairs, then the leading terms of G generate $LT(I)$, which is shown by induction on the monomial order applied to an arbitrary $h \in I$; see [CLO07, Chapter 2] for the complete proof. $\square$

Theorem 41.28 (Termination of Buchberger's Algorithm). *Buchberger's algorithm terminates after finitely many steps for any input set of polynomials F and any admissible monomial order.*

Proof. Each time a nonzero remainder r is produced, r is added to G , and $\langle LT(G) \rangle$ strictly increases in the monomial ideal lattice. Since every monomial ideal is finitely generated (by Dickson's lemma), this chain of strict inclusions must stabilize, and the algorithm terminates. $\square$

Generator	Leading	Remainder
f_1	x^2y	$S(\cancel{f_1}, f_2) \rightarrow r$
f_2	xy^2	

Figure 41.1: Buchberger's algorithm: S-polynomials are formed from pairs of generators and reduced.

41.17.4 Pseudocode

Example 41.29 (Computing an S-Polynomial). Let $f = x^2y - 1$ and $g = xy^2 - 1$ with Lex order ($x \succ y$). Then $LT(f) = x^2y$ and $LT(g) = xy^2$, so $L = \text{lcm}(x^2y, xy^2) = x^2y^2$. The S-polynomial is

$$S(f, g) = \frac{x^2y^2}{x^2y}f - \frac{x^2y^2}{xy^2}g = y \cdot f - x \cdot g = (x^2y^2 - y) - (x^2y^2 - x) = x - y.$$

Since $x - y$ is nonzero, it would be added to the basis G during Section 41.17.4.

41.17.5 Parameter Selection

- **Monomial Order: Lexicographic** (Lex) is used for solving systems of equations, while **Graded Reverse Lexicographical** (Degrevlex) is generally much faster for computing the basis.
- **Selection Heuristic:** Choosing pairs with the smallest LCM of leading terms can improve performance.

```
1: function BUCHBERGER( $F$ , monomial_order)
2:    $G \leftarrow F$ 
3:   pairs  $\leftarrow \{(f_i, f_j) \mid f_i, f_j \in G, i < j\}$ 
4:   while pairs  $\neq \emptyset$  do
5:      $(f, g) \leftarrow \text{pairs.pop}()$ 
6:      $S \leftarrow \text{S.Polynomial}(f, g, \text{monomial\_order})$ 
7:     remainder  $\leftarrow \text{MultivariateDivision}(S, G, \text{monomial\_order})$ 
8:     if remainder  $\neq 0$  then
9:       for each existing_g in  $G$  do
10:        pairs.add((existing_g, remainder))
11:      end for
12:       $G.\text{append}(\text{remainder})$ 
13:    end if
14:  end while
15:  return  $G$ 
16: end function
```

41.17.6 Complexity

- **Time Complexity:** In the worst case, the algorithm's complexity can be **doubly exponential** in the number of variables n . More precisely, for a zero-dimensional ideal with n variables and maximum degree d , the number of polynomials in a Gröbner basis can be $O(d^{2^n})$.
- **Space Complexity:** Can also grow extremely large as many intermediate polynomials are generated.

41.17.7 Usages

- **Solving Polynomial Systems:** Transforming a complex system into a Lex-ordered Gröbner basis (which looks like a “triangular” system) and solving by back-substitution.
- **Surface Implicitization:** Converting a parametric surface (e.g., Bézier or NURBS) into an implicit equation $f(x, y, z) = 0$.
- **Collision Detection:** Determining if two curved surfaces (defined by algebraic equations) intersect.
- **Robot Kinematics:** Solving the inverse kinematics problem for robot arms with multiple joints.
- **Motion Planning:** Checking for intersections between objects in configuration space.

41.17.8 Testing Methods and Edge Cases

- **Linear Systems:** On a system of linear equations, Buchberger should perform identically to Gaussian elimination.
- **Single Variable:** On polynomials in x , it should behave like the Euclidean algorithm for finding the GCD.
- **Inconsistent Systems:** If the system has no solution, the Gröbner basis should contain the constant 1.
- **Redundant Generators:** The algorithm should eliminate redundant polynomials.

- **Monomial Order Sensitivity:** Verify that the resulting basis is correct for different orders (Lex vs. Degrevlex).

41.17.9 References

- Buchberger, B. (1965). “An Algorithm for Finding the Basis Elements of the Residue Class Ring of a Zero Dimensional Polynomial Ideal”. PhD Thesis.
- Cox, D., Little, J., & O’Shea, D. (2007). “Ideals, Varieties, and Algorithms”. Springer.
- Wikipedia: [Buchberger’s algorithm](#)

41.18 Resultant-Based Elimination

41.18.1 Overview

Resultant-based elimination is a symbolic algebraic technique used to eliminate variables from a system of polynomial equations. The **resultant** of two polynomials is a scalar value (or another polynomial) that is zero if and only if the two original polynomials share a common root. This concept dates to the work of Sylvester [Syl40] and has been extensively developed in modern computational algebra [CLO07]. By calculating the resultant with respect to one variable, we “project” the intersection of the two polynomials onto a lower-dimensional space. It is a faster alternative to Gröbner bases for many problems in surface implicitization and collision detection [Man94].

41.18.2 Definitions

Definition 41.30 (Resultant). Let $f(x) = a_n x^n + \cdots + a_0$ and $g(x) = b_m x^m + \cdots + b_0$ be two polynomials in $k[x]$ with coefficients in a field k . The **resultant** $\text{Res}(f, g, x)$ is a polynomial in the coefficients a_i, b_j that vanishes if and only if f and g have a common root in some extension field of k , or equivalently, $\gcd(f, g)$ has positive degree.

Definition 41.31 (Sylvester Matrix). For $f(x) = a_n x^n + \cdots + a_0$ and $g(x) = b_m x^m + \cdots + b_0$, the **Sylvester matrix** is the $(m+n) \times (m+n)$ matrix formed by stacking m shifted copies of the coefficient vector of f followed by n shifted copies of the coefficient vector of g :

$$\mathbf{S}(f, g) = \begin{pmatrix} a_n & a_{n-1} & \cdots & a_0 & 0 & \cdots & 0 \\ 0 & a_n & a_{n-1} & \cdots & a_0 & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \\ 0 & \cdots & 0 & a_n & a_{n-1} & \cdots & a_0 \\ b_m & b_{m-1} & \cdots & b_0 & 0 & \cdots & 0 \\ 0 & b_m & b_{m-1} & \cdots & b_0 & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \\ 0 & \cdots & 0 & b_m & b_{m-1} & \cdots & b_0 \end{pmatrix}$$

The resultant is $\text{Res}(f, g, x) = \det(\mathbf{S}(f, g))$.

Definition 41.32 (Elimination). **Elimination** is the process of reducing a system of n polynomial equations in n variables to a system of fewer variables. In the context of resultants, computing $\text{Res}_x(f, g)$ eliminates the variable x , yielding a polynomial in the remaining variables whose roots encode the x -coordinates of intersection points.

41.18.3 Theory

The resultant of two polynomials $f(x) = \sum a_i x^i$ and $g(x) = \sum b_j x^j$ is the determinant of the Sylvester matrix.

For example, if $f(x, y)$ and $g(x, y)$ are two implicit curves, their resultant $\text{Res}_x(f, g)$ is a polynomial in y whose roots are the y -coordinates of all intersection points. This allows us to solve for y first and then back-substitute to find x .

For **Implicitization**, given parametric equations $x = X(t)$, $y = Y(t)$, $z = Z(t)$, we form the polynomials $f_1 = x - X(t)$, $f_2 = y - Y(t)$, $f_3 = z - Z(t)$ and calculate the resultant $\text{Res}_t(f_1, f_2, f_3)$ to eliminate the parameter t .

Theorem 41.33 (Resultant Characterization). *Let $f, g \in k[x]$ be nonzero polynomials of degrees n and m respectively. Then $\text{Res}(f, g, x) = 0$ if and only if f and g have a common root in the algebraic closure $\bar{k}$, or both $\text{LC}(f) = 0$ and $\text{LC}(g) = 0$.*

Proof. The result follows from the equivalence: f and g share a common root if and only if there exist nonzero polynomials u, v with $\deg(u) < m$ and $\deg(v) < n$ such that $uf + vg = 0$. This linear system in the coefficients of u and v has a nontrivial solution if and only if the $(m+n) \times (m+n)$ Sylvester matrix is singular, which occurs precisely when its determinant (the resultant) is zero. See [CLO07, Chapter 3]. $\square$

Example 41.34 (Resultant of Two Quadratics). Let $f(x) = x^2 + px + q$ and $g(x) = x^2 + rx + s$ (both degree $n = m = 2$). The Sylvester matrix is 4×4 :

$$\mathbf{S} = \begin{pmatrix} 1 & p & q & 0 \\ 0 & 1 & p & q \\ 1 & r & s & 0 \\ 0 & 1 & r & s \end{pmatrix}$$

Computing the determinant:

$$\text{Res}(f, g, x) = (q - s)^2 - (p - r)(ps - qr).$$

Setting this to zero yields the condition under which the two quadratics share a root.

41.18.4 Pseudocode

```

1: function RESULTANT( $f, g, x$ )
2:    $S \leftarrow \text{BuildSylvesterMatrix}(f, g, x)$ 
3:    $\text{res} \leftarrow \text{determinant}(S)$ 
4:   return res
5: end function
6: function BUILDSYLVESTERMATRIX( $f, g, x$ )
7:   Build matrix rows by shifting the coefficient vectors:
8:    $[a_n, a_{n-1}, \dots, a_0, 0, 0]$ 
9:    $[0, a_n, a_{n-1}, \dots, a_0, 0]$ 
10:   $\vdots$ 
11:   $[b_m, b_{m-1}, \dots, b_0, 0, 0]$ 
12:   $[0, b_m, b_{m-1}, \dots, b_0, 0]$ 
13:  return  $S$ 
14: end function

```

41.18.5 Parameter Selection

- **Matrix Type:** Sylvester matrices are large and sparse. Bezout or Dixon matrices are much smaller for higher-order polynomials but are more complex to construct.
- **Numerical Precision:** If coefficients are floating-point numbers, calculating the determinant can be numerically unstable. Using symbolic computation or arbitrary-precision arithmetic is recommended.

41.18.6 Complexity

- **Time Complexity:** $O(d^3)$ where $d = \deg(f) + \deg(g)$, based on the cost of the determinant calculation of the $(m+n) \times (m+n)$ Sylvester matrix.
- **Space Complexity:** $O(d^2)$ to store the Sylvester matrix.

41.18.7 Usages

- **Surface Implicitization:** Converting parametric Bézier/NURBS surfaces into implicit $f(x, y, z) = 0$ equations for faster inside/outside tests.
- **Curve/Surface Intersection:** Finding the exact points where a line (ray) intersects a curved surface.
- **Collision Detection:** Determining if two curved objects intersect by checking if their resultant has any real roots.
- **Robot Kinematics:** Solving inverse kinematics for complex linkages.
- **Computer Aided Design (CAD):** Boolean operations on curved surfaces.

41.18.8 Testing Methods and Edge Cases

- **Two Lines:** The resultant of two linear equations $a_1x + b_1$ and $a_2x + b_2$ should be $a_1b_2 - a_2b_1$.
- **Parallel Curves:** If curves never intersect, the resultant should be a non-zero constant.
- **Coincident Curves:** If two curves are the same, the resultant will be identically zero.
- **Leading Coefficient Zero:** Ensure the algorithm handles cases where the leading coefficient vanishes for certain values of the other variables.
- **Large Degrees:** Verify that the determinant calculation remains robust for polynomials of degree 10+.

41.18.9 References

- Sylvester, J. J. (1840). “A method of determining by inspection the form of the combined roots of two algebraic equations”. Philosophical Magazine.
- Cox, D., Little, J., & O’Shea, D. (2007). “Ideals, Varieties, and Algorithms”. Springer.
- Manocha, D. (1994). “Solving systems of polynomial equations using resultants”. SIG-GRAPH Course.
- Wikipedia: [Resultant](#)

Appendix A

Mathematical Reference

A.1 Linear Algebra

A.2 Vector Calculus

A.3 Determinants

A.4 Affine Geometry

A.5 Convexity

A.6 Probability

A.7 Graph Theory

A.8 Asymptotic Analysis

Appendix B

Algorithms and Data Structures

B.1 Sorting

B.2 Binary Search

B.3 Balanced Search Trees

B.4 Priority Queues

B.5 Hashing

B.6 Graph Algorithms

B.7 Union–Find

B.8 Randomized Algorithms

Appendix C

Geometry Formula Reference

C.1 Vector Operations

C.2 Orientation Tests

C.3 Distance Formulas

C.4 Intersection Formulas

C.5 Circle Predicates

C.6 Areas and Volumes

C.7 Coordinate Transformations

C.8 Barycentric Coordinates

Appendix D

Implementation Guide

- D.1 Designing a Geometry Kernel
- D.2 Choosing Numeric Types
- D.3 Floating-Point Error
- D.4 Exact Predicates
- D.5 Testing Geometric Software
- D.6 Generating Adversarial Test Cases
- D.7 Benchmarking
- D.8 Visualization and Debugging

Appendix E

Computational Geometry Software

E.1 CGAL

E.2 GEOSE

E.3 Boost.Geometry

E.4 Qhull

E.5 Triangle

E.6 TetGen

E.7 SciPy Spatial Algorithms

E.8 Python Geometry Ecosystem

E.9 Choosing a Library

Appendix F

Graduate Projects

- F.1 Convex Hull Laboratory
- F.2 Sweep-Line Intersection Engine
- F.3 Voronoi/Delaunay Package
- F.4 Spatial Database
- F.5 Mesh Generator
- F.6 Collision-Detection System
- F.7 Robot Path Planner
- F.8 Point-Cloud Reconstruction
- F.9 GIS Spatial Analysis System
- F.10 Persistent-Homology Explorer
- F.11 GPU Geometry Engine
- F.12 Research Replication Project

Appendix G

Selected Solutions and Hints

Bibliography

- [AB99] N. Amenta and M. Bern. Surface reconstruction by Voronoi filtering. *Discrete & Computational Geometry*, 22(4):481–504, 1999.
- [ABE98] N. Amenta, M. Bern, and D. Eppstein. The crust and the β -skeleton: Combinatorial curve reconstruction. *Graphical Models and Image Processing*, 60(2):125–135, 1998.
- [ABG⁺02] Pankaj K. Agarwal, Julien Basch, Leonidas J. Guibas, John Hershberger, and Li Zhang. Deformable free space tilings for kinetic collision detection. *International Journal of Computational Geometry & Applications*, 12(1–2):1–27, 2002.
- [ACDL00] N. Amenta, S. Choi, T. K. Dey, and N. Leekha. A simple algorithm for homeomorphic surface reconstruction. In *Proceedings of the 16th Annual ACM Symposium on Computational Geometry (SCG)*, pages 213–222, 2000.
- [ACH⁺91] E. M. Arkin, L. P. Chew, D. P. Huttenlocher, K. Kedem, and J. S. Mitchell. An efficiently computable metric for comparing polygonal shapes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1991.
- [Ach03] D. Achlioptas. Database-friendly random projections: Johnson-Lindenstrauss with binary coins. *Journal of Computer and System Sciences*, 66(4):671–687, 2003.
- [AF92] D. Avis and K. Fukuda. A pivoting algorithm for convex hulls and vertex enumeration of arrangements and polyhedra. *Discrete & Computational Geometry*, 8(3):295–313, 1992.
- [AG95] H. Alt and M. Godau. Computing the Fréchet distance between two polygonal curves. *International Journal of Computational Geometry & Applications*, 1995.
- [AG00] H. Alt and L. J. Guibas. Discrete geometric shapes: Matching, interpolation, and approximation. In J.-R. Sack and J. Urrutia, editors, *Handbook of Computational Geometry*, pages 121–153. Elsevier, 2000.
- [AGHV01] Pankaj K. Agarwal, Leonidas J. Guibas, John Hershberger, and Eric Veach. Maintaining the extent of a moving point set. *Discrete & Computational Geometry*, 26(3):353–374, 2001.
- [AGMR98] G. Albers, L. J. Guibas, J. S. Mitchell, and T. Roos. Voronoi diagrams of moving points. *International Journal of Computational Geometry & Applications*, 1998.
- [AGNZ18] Sanjeev Arora, Rong Ge, Behnam Neyshabur, and Yi Zhang. Stronger generalization bounds for deep nets via a compression approach. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 254–263, 2018.
- [AI06] A. Andoni and P. Indyk. Near-optimal hashing algorithms for approximate nearest neighbor in high dimensions. In *Proceedings of the 47th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 459–468, 2006.

- [AM01] T. Akenine-Möller. Fast 3D triangle-box overlap testing. *Journal of Graphics Tools*, 2001.
- [AMN⁺98] S. Arya, D. M. Mount, N. S. Netanyahu, R. Silverman, and A. Y. Wu. An optimal algorithm for approximate nearest neighbor searching in fixed dimensions. *Journal of the ACM*, 45(6):891–923, 1998.
- [And79] A. M. Andrew. Another efficient algorithm for convex hulls in two dimensions. *Information Processing Letters*, 1979.
- [Aro94] S. Arora. *Probabilistic Checking of Proofs and Hardness of Approximation Problems*. PhD thesis, University of California, Berkeley, 1994.
- [Aro98] S. Arora. Polynomial time approximation schemes for Euclidean traveling salesman and other geometric problems. *Journal of the ACM*, 45(5):753–782, 1998.
- [ASC11] M. Aubry, U. Schlickewei, and D. Cremers. The wave kernel signature: A quantum mechanical approach to shape analysis. In *ICCV Workshops*, 2011.
- [Ata85] M. J. Atallah. On symmetry detection. *IEEE Transactions on Computers*, 1985.
- [Aur91] F. Aurenhammer. Voronoi diagrams: A survey of a fundamental geometric data structure. *ACM Computing Surveys*, 1991.
- [AV88] Alok Aggarwal and Jeffrey S. Vitter. The input/output complexity of sorting and related problems. *Communications of the ACM*, 31(9):1116–1127, 1988.
- [B68] P. Bézier. How Renault uses numerical control for car body design and tooling. In *SAE Paper*, 1968.
- [B⁺08] A. M. Bronstein et al. *Numerical Geometry of Non-Rigid Shapes*. Springer, 2008.
- [B⁺17] M. M. Bronstein et al. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint*, 2017.
- [Bal95] I. J. Balaban. An optimal algorithm for finding segments intersections. In *Proceedings of the Eleventh Annual Symposium on Computational Geometry (SoCG)*, pages 211–219, 1995.
- [BBCV21] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Velickovic. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.
- [BBP01] H. Brönnimann, C. Burnikel, and S. Pion. Interval arithmetic yields efficient dynamic filters for computational geometry. In *Proceedings of the 17th Annual Symposium on Computational Geometry*, pages 165–174, 2001.
- [BC08] M. Badoiu and K. L. Clarkson. Optimal core-sets for balls. *Discrete & Computational Geometry*, 40(1):14–22, 2008. Preliminary version circulated as a manuscript in 2002.
- [BCKO08] M. de Berg, O. Cheong, M. van Kreveld, and M. Overmars. *Computational Geometry: Algorithms and Applications*. Springer-Verlag, 2008.
- [BDH96] C. B. Barber, D. P. Dobkin, and H. Huhdanpaa. The Quickhull algorithm for convex hulls. *ACM Transactions on Mathematical Software (TOMS)*, 1996.

-
- [Ben75] J. L. Bentley. Multidimensional binary search trees used for associative searching. *Communications of the ACM*, 1975.
 - [Ben77] J. L. Bentley. Algorithms for Klee’s measure problem. Technical report, Technical Report, 1977.
 - [Ben80] J. L. Bentley. Multidimensional divide-and-conquer. *Communications of the ACM*, 23(4):214–229, 1980.
 - [BGH99] J. Basch, L. J. Guibas, and J. Hershberger. Data structures for mobile data. *Journal of Algorithms*, 1999.
 - [BGRS99] K. S. Beyer, J. Goldstein, R. Ramakrishnan, and U. Shaft. When is “nearest neighbor” meaningful? In *Database Theory — ICDT ’99*, volume 1540 of *Lecture Notes in Computer Science*, pages 217–235. Springer, 1999.
 - [BGZ97] Julien Basch, Leonidas J. Guibas, and Li Zhang. Proximity problems on moving points. *Proceedings of the 13th Annual Symposium on Computational Geometry (SoCG 1997)*, pages 344–351, 1997.
 - [BHKW07] E. K. Burke, R. S. Hellier, G. Kendall, and G. Whitwell. Complete and robust no-fit polygon generation for the irregular cutting and packing problem. *European Journal of Operational Research*, 2007.
 - [Bis06] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.
 - [BJ02] Gerth Stølting Brodal and Riko Jacob. Dynamic planar convex hull. In *Proceedings of the 43rd Annual IEEE Symposium on Foundations of Computer Science (FOCS 2002)*, pages 617–626, 2002.
 - [BK10] M. M. Bronstein and I. Kokkinos. Scale-invariant heat kernel signatures for non-rigid shape retrieval. In *CVPR*, 2010.
 - [BKP⁺10] M. Botsch, L. Kobbelt, M. Pauly, P. Alliez, and B. Lévy. *Polygon Mesh Processing*. CRC Press, 2010.
 - [BKSS90] N. Beckmann, H. P. Kriegel, R. Schneider, and B. Seeger. The R^* -tree: An efficient and robust access method for points and rectangles. In *ACM SIGMOD*, 1990.
 - [BLP⁺13] D. Bommes, B. Lévy, N. Pietroni, E. Puppo, C. Silva, M. Tarini, and R. Zayer. State of the art in quad-meshing. *Eurographics STAR*, 2013.
 - [Blu67] H. Blum. A transformation for extracting new descriptors of shape. In *Models for the Perception of Speech and Visual Form*, 1967.
 - [BM92] P. J. Besl and N. D. McKay. A method for registration of 3-d shapes. In *IEEE Transactions on Pattern Analysis and Machine Intelligence*, volume 14, pages 239–256, 1992.
 - [BMR⁺99] V. Bernardini, J. Mittleman, H. Rushmeier, C. Silva, and G. Taubin. Ball-pivoting algorithm for surface reconstruction. In *IEEE Transactions on Visualization and Computer Graphics*, volume 5, pages 349–359, 1999.
 - [BO79] J. L. Bentley and T. A. Ottmann. Algorithms for reporting and counting geometric intersections. *IEEE Transactions on Computers*, 1979.

- [BO83] Michael Ben-Or. Lower bounds for algebraic computation trees. In *Proceedings of the Fifteenth Annual ACM Symposium on Theory of Computing (STOC)*, pages 80–86, 1983.
- [BO05] J.-D. Boissonnat and S. Oudot. Provably good sampling and meshing of surfaces. *Graphical Models*, 67(5):405–451, 2005.
- [BO08] J. A. Bennell and J. F. Oliveira. The geometry of nesting problems: A tutorial. *European Journal of Operational Research*, 2008.
- [Bor87] K.-H. Borgwardt. *The Simplex Method: A Probabilistic Analysis*, volume 1 of *Algorithms and Combinatorics*. Springer-Verlag, 1987.
- [Bow81] A. Bowyer. Computing Dirichlet tessellations. *The Computer Journal*, 1981.
- [Bri07] R. Bridson. Fast Poisson disk sampling in arbitrary dimensions. In *SIGGRAPH Sketches*, 2007.
- [Bri12] K. Bringmann. New upper bounds for Klee’s measure problem in small dimensions. *Computational Geometry*, 2012.
- [Bro79] K. Q. Brown. Voronoi diagrams from convex hulls. *Information Processing Letters*, 9(5):223–228, 1979.
- [BS76] J. L. Bentley and M. I. Shamos. Divide-and-conquer in multidimensional space. In *ACM Symposium on Theory of Computing (STOC)*, 1976.
- [BS07] A. I. Bobenko and B. A. Springborn. A discrete Laplace–Beltrami operator for simplicial surfaces. *Discrete & Computational Geometry*, 2007.
- [Buc65] B. Buchberger. *An Algorithm for Finding the Basis Elements of the Residue Class Ring of a Zero Dimensional Polynomial Ideal*. PhD thesis, University of Innsbruck, 1965.
- [Car09] Gunnar Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308, 2009.
- [Cat74] E. E. Catmull. *A Subdivision Algorithm for Computer Display of Curved Surfaces*. PhD thesis, University of Utah, 1974.
- [CB78] M. Cyrus and J. Beck. Generalized two- and three-dimensional clipping. *Computers and Graphics*, 3(1):23–28, 1978.
- [CC78] E. Catmull and J. Clark. Recursively generated B-spline surfaces on arbitrary topological meshes. *Computer-Aided Design*, 1978.
- [CF90] B. Chazelle and J. Friedman. A deterministic view of random sampling and its use in geometry. *Combinatorica*, 10(3):229–249, 1990.
- [CG86] B. Chazelle and L. J. Guibas. Fractional cascading: I. A data structuring technique. *Algorithmica*, 1(1):133–162, 1986.
- [CGL85] B. Chazelle, L. J. Guibas, and D. T. Lee. The power of geometric duality. *BIT Numerical Mathematics*, 1985.
- [CH67] Thomas M. Cover and Peter E. Hart. Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1):21–27, 1967.

-
- [Cha85] B. Chazelle. On the convex layers of a planar set. *IEEE Transactions on Information Theory*, 1985.
 - [Cha91] B. Chazelle. Triangulating a simple polygon in linear time. *Discrete & Computational Geometry*, 6(1):485–524, 1991.
 - [Cha92] B. Chazelle. Reporting and counting segment intersections. *Journal of Algorithms*, 13(1):38–56, 1992.
 - [Cha93] B. Chazelle. An optimal convex hull algorithm in any fixed dimension. *Discrete & Computational Geometry*, 10(1):377–409, 1993.
 - [Cha96] T. M. Chan. Optimal output-sensitive convex hull algorithms in two and three dimensions. *Discrete & Computational Geometry*, 1996.
 - [Cha06] Timothy M. Chan. A dynamic data structure for 3-d convex hulls and 2-d nearest neighbor queries. In *Proceedings of the 17th Annual ACM–SIAM Symposium on Discrete Algorithms (SODA 2006)*, pages 969–978, 2006.
 - [Che89] L. Paul Chew. Constrained Delaunay triangulations. *Algorithmica*, 1989.
 - [Chv75] V. Chvátal. A combinatorial theorem in plane geometry. *Journal of Combinatorial Theory, Series B*, 1975.
 - [CK70] D. R. Chand and S. S. Kapur. An algorithm for convex polytopes. *Journal of the ACM*, 17(1):78–86, 1970.
 - [CK95] P. B. Callahan and S. R. Kosaraju. A decomposition of multidimensional point sets with applications to k -nearest-neighbors and n -body potential fields. *Journal of the ACM*, 42(1):67–90, 1995.
 - [CL06] R. R. Coifman and S. Lafon. Diffusion maps. *Applied and Computational Harmonic Analysis*, 2006.
 - [Cla93] K. L. Clarkson. A randomized algorithm for closest-point queries. *SIAM Journal on Computing*, 22(4):719–760, 1993.
 - [CLO07] D. Cox, J. Little, and D. O’Shea. *Ideals, Varieties, and Algorithms*. Springer, 2007.
 - [CLRS09] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein. *Introduction to Algorithms*. MIT Press, 2009.
 - [CM69] E. Cuthill and J. McKee. Reducing the bandwidth of sparse symmetric matrices. In *ACM National Conference*, 1969.
 - [CM92] Y. Chen and G. Medioni. Object modeling by registration of multiple range images. In *IEEE International Conference on Robotics and Automation*, pages 2724–2729, 1992.
 - [Coo86] R. L. Cook. Stochastic sampling in computer graphics. *ACM Transactions on Graphics*, 1986.
 - [CS89] K. L. Clarkson and P. W. Shor. Applications of random sampling in computational geometry. *Discrete & Computational Geometry*, 4(5):387–421, 1989.
 - [CWW13a] K. Crane, C. Weischedel, and M. Wardetzky. Digital geometry processing with discrete exterior calculus. In *SIGGRAPH Course*, 2013.

- [CWW13b] K. Crane, C. Weischedel, and M. Wardetzky. Geodesics in heat: A new approach to computing distance based on heat flow. *ACM Transactions on Graphics*, 2013.
- [CWW17] K. Crane, C. Weischedel, and M. Wardetzky. Boundary first flattening. *ACM Transactions on Graphics*, 2017.
- [dBCvKO08] M. de Berg, O. Cheong, M. van Kreveld, and M. Overmars. *Computational Geometry: Algorithms and Applications*. Springer, 3rd edition, 2008.
- [Dev86] L. Devroye. *Non-Uniform Random Variate Generation*. Springer, 1986.
- [Dev02] O. Devillers. The delaunay hierarchy. *International Journal of Foundations of Computer Science*, 13(2):163–180, 2002.
- [Dey98] T. K. Dey. Improved bounds on planar k-sets and related problems. *Discrete & Computational Geometry*, 19(3):373–382, 1998.
- [Dey07] T. K. Dey. *Curve and Surface Reconstruction: Algorithms with Mathematical Analysis*. Cambridge University Press, 2007.
- [DFK91] M. E. Dyer, A. M. Frieze, and R. Kannan. A random polynomial-time algorithm for approximating the volume of convex bodies. *Journal of the ACM*, 38(1):1–17, 1991.
- [DHLM05] M. Desbrun, A. N. Hirani, M. Leok, and J. E. Marsden. Discrete exterior calculus. *arXiv preprint*, 2005.
- [DHS01] Richard O. Duda, Peter E. Hart, and David G. Stork. *Pattern Classification*. Wiley, 2nd edition, 2001.
- [Die17] R. Diestel. *Graph Theory*. Springer, 2017.
- [DIIM04] M. Datar, N. Immorlica, P. Indyk, and V. S. Mirrokni. Locality-sensitive hashing scheme based on p -stable distributions. In *Proceedings of the 20th Annual ACM Symposium on Computational Geometry (SoCG)*, pages 253–262, 2004.
- [Dij59] E. W. Dijkstra. A note on two problems in connexion with graphs. *Numerische Mathematik*, 1959.
- [DK83] D. P. Dobkin and D. G. Kirkpatrick. Fast detection of polyhedral intersection. *Theoretical Computer Science*, 27(3):241–253, 1983.
- [DK85] D. P. Dobkin and D. G. Kirkpatrick. A linear algorithm for determining the separation of convex polyhedra. *Journal of Algorithms*, 6(3):381–392, 1985.
- [DS65] H. Davenport and A. Schinzel. A combinatorial problem connected with differential equations. *American Journal of Mathematics*, 1965.
- [DS87] Paul F. Dietz and Daniel D. Sleator. Two algorithms for maintaining order in a list. In *Proceedings of the 19th Annual ACM Symposium on Theory of Computing (STOC 1987)*, pages 365–372, 1987.
- [DSW10] T. K. Dey, J. Sun, and Y. Wang. Approximating cycles in a shortest homology basis. *Discrete & Computational Geometry*, 2010.
- [Edd77] W. F. Eddy. A new convex hull algorithm for planar sets. *ACM Transactions on Mathematical Software (TOMS)*, 1977.

-
- [Ede80] H. Edelsbrunner. Dynamic data structures for orthogonal intersection queries. Technical report, Technical University of Graz, 1980.
 - [Ede85] H. Edelsbrunner. Computing the extreme distances between two convex polygons. *Journal of Algorithms*, 1985.
 - [Ede87] H. Edelsbrunner. *Algorithms in Combinatorial Geometry*. Springer-Verlag, 1987.
 - [EH10] Herbert Edelsbrunner and John L. Harer. *Computational Topology: An Introduction*. American Mathematical Society, 2010.
 - [EKS83] H. Edelsbrunner, D. G. Kirkpatrick, and R. Seidel. On the shape of a set of points in the plane. *IEEE Transactions on Information Theory*, 29(4):551–559, 1983.
 - [EK SX96] Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 226–231, 1996.
 - [EM90] H. Edelsbrunner and E. P. Mücke. Simulation of simplicity: A technique to cope with degenerate cases in geometric algorithms. *ACM Transactions on Graphics*, 9(1):66–104, 1990.
 - [EM94a] H. Edelsbrunner and E. P. Mücke. Three-dimensional alpha shapes. *ACM Transactions on Graphics*, 13(1):43–72, 1994.
 - [EM94b] T. Eiter and H. Mannila. Computing discrete Fréchet distance. Technical report, Technical University of Vienna, 1994.
 - [EOS86] H. Edelsbrunner, J. O’Rourke, and R. Seidel. Constructing arrangements of lines and hyperplanes with applications. *SIAM Journal on Computing*, 1986.
 - [Eri04] C. Ericson. *Real-Time Collision Detection*. Morgan Kaufmann, 2004.
 - [ES83] H. Edelsbrunner and R. Seidel. Voronoi diagrams and arrangements. *Discrete & Computational Geometry*, 1:25–44, 1983.
 - [Eul58] L. Euler. Elementa doctrinae solidorum. *Novi Commentarii Academiae Scientiarum Petropolitanae*, 1758.
 - [Far02] J. Farkas. Theorie der einfachen ungleichungen. *Journal für die Reine und Angewandte Mathematik*, 124:1–27, 1902.
 - [Far02] G. Farin. *Curves and Surfaces for CAGD: A Practical Guide*. Morgan Kaufmann, 2002.
 - [FB74] R. A. Finkel and J. L. Bentley. Quad trees: A data structure for retrieval on composite keys. *Acta Informatica*, 1974.
 - [FBF77] J. H. Friedman, J. L. Bentley, and R. A. Finkel. An algorithm for finding best matches in logarithmic expected time. *ACM Transactions on Mathematical Software*, 1977.
 - [FH12] P. F. Felzenszwalb and D. P. Huttenlocher. Distance transforms of sampled functions. *Theory of Computing*, 2012.
 - [Fis78] S. Fisk. A short proof of chvátal’s watchman theorem. *Journal of Combinatorial Theory, Series B*, 1978.

- [FKN80] H. Fuchs, Z. M. Kedem, and B. F. Naylor. On visible surface generation by a priori tree structures. In *Proceedings of the 7th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pages 124–133, 1980.
- [Flo53] P. J. Flory. *Principles of Polymer Chemistry*. Cornell University Press, 1953.
- [For87] S. Fortune. A sweepline algorithm for Voronoi diagrams. *Algorithmica*, 1987.
- [For98] R. Forman. Discrete Morse theory. *Advances in Mathematics*, 1998.
- [For02] R. Forman. A user’s guide to discrete Morse theory. *Sém. Lothar. Combin.*, 2002.
- [Fré06] M. Fréchet. Sur quelques points du calcul fonctionnel. *Rendiconti del Circolo Matematico di Palermo*, 1906.
- [FSBS07] M. Fisher, B. Springborn, A. Bobenko, and P. Schröder. An algorithm for the construction of intrinsic Delaunay triangulations with applications to digital geometry processing. *ACM Transactions on Graphics*, 2007.
- [FVW93] S. Fortune and C. J. Van Wyk. Efficient exact arithmetic for computational geometry. In *Proceedings of the 9th Annual Symposium on Computational Geometry*, pages 163–172, 1993.
- [FVW96] S. Fortune and C. J. Van Wyk. Static analysis yields efficient exact integer arithmetic for computational geometry. *ACM Transactions on Graphics*, 15(3):223–248, 1996.
- [GH86] M. Gage and R. S. Hamilton. The heat equation shrinking convex plane curves. *Journal of Differential Geometry*, 1986.
- [GH97] M. Garland and P. S. Heckbert. Surface simplification using quadric error metrics. *Proceedings of the 24th Annual Conference on Computer Graphics and Interactive Techniques*, pages 209–216, 1997.
- [GH98] G. Greiner and K. Hormann. Efficient clipping of arbitrary polygons. *ACM Transactions on Graphics*, 1998.
- [GHL⁺87] L. J. Guibas, J. Hershberger, D. Leven, M. Sharir, and R. E. Tarjan. Linear-time algorithms for visibility and shortest path problems inside triangulated simple polygons. *Algorithmica*, 2:209–233, 1987.
- [Gib98] S. F. F. Gibson. Constrained elastic surface nets: Generating smooth surfaces from binary segmented data. In *Proceedings of Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 888–898, 1998.
- [GIM99] A. Gionis, P. Indyk, and R. Motwani. Similarity search in high dimensions via hashing. In *Proceedings of the 25th International Conference on Very Large Data Bases (VLDB)*, pages 518–529. Morgan Kaufmann, 1999.
- [GJK88] E. G. Gilbert, D. W. Johnson, and S. S. Keerthi. A fast procedure for computing the distance between complex objects in three-dimensional space. *IEEE Journal on Robotics and Automation*, 1988.
- [GKS92] L. Guibas, D. E. Knuth, and M. Sharir. Randomized incremental construction of Delaunay and Voronoi diagrams. *Algorithmica*, 1992.
- [GM91] S. K. Ghosh and D. M. Mount. An output-sensitive algorithm for computing visibility graphs. *SIAM Journal on Computing*, 20(5):888–910, 1991.

-
- [GNPB08] A. Gyulassy, V. Natarajan, V. Pascucci, and P. T. Bremer. A practical approach to Morse-Smale complex computation. In *IEEE Transactions on Visualization and Computer Graphics*, 2008.
 - [Gol91] D. Goldberg. What every computer scientist should know about floating-point arithmetic. *ACM Computing Surveys*, 23(1):5–48, 1991.
 - [Gon85] T. F. Gonzalez. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306, 1985.
 - [GR09] C. Geuzaine and J.-F. Remacle. Gmsh: A 3-D finite element mesh generator with built-in pre- and post-processing facilities. *International Journal for Numerical Methods in Engineering*, 79(11):1309–1331, 2009.
 - [Gra72] R. L. Graham. An efficient algorithm for determining the convex hull of a finite planar set. *Information Processing Letters*, 1972.
 - [Gra87] M. A. Grayson. The heat equation shrinks embedded plane curves to round points. *Journal of Differential Geometry*, 1987.
 - [GS85] L. Guibas and J. Stolfi. Primitives for the manipulation of general subdivisions and the computation of Voronoi diagrams. *ACM Transactions on Graphics*, 1985.
 - [GS87] L. J. Guibas and R. Seidel. Computing convolutions by reciprocal search. *Discrete & Computational Geometry*, 1987.
 - [Gui98] L. J. Guibas. Kinetic data structures: A new recipe for dynamic geometric algorithms. *ACM SIGART Bulletin*, 1998.
 - [Gut84] A. Guttman. R-trees: A dynamic index structure for spatial searching. In *ACM SIGMOD*, 1984.
 - [H⁺18] Y. Hu et al. fTetWild: Robust tetrahedral meshing in the wild. In *ACM Transactions on Graphics (SIGGRAPH)*, 2018.
 - [H⁺21] Y. Hu et al. TriWild: Robust remeshing with angle bounds. In *ACM Transactions on Graphics (SIGGRAPH)*, 2021.
 - [HA01] K. Hormann and A. Agathos. The point in polygon problem for arbitrary polygons. *Computational Geometry*, 2001.
 - [Hal60] J. H. Halton. On the efficiency of certain quasi-random sequences of points in evaluating multi-dimensional integrals. *Numerische Mathematik*, 1960.
 - [Her89] J. Hershberger. Finding the upper envelope of n line segments in $O(n \log n)$ time. *Information Processing Letters*, 1989.
 - [Hig86] P. T. Highnam. Optimal algorithms for finding the symmetries of a planar point set. *Information Processing Letters*, 1986.
 - [Hil91] D. Hilbert. Über die stetige abbildung einer linie auf ein flächenstück. *Mathematische Annalen*, 1891.
 - [HMY04] D. Halperin, K. Mehlhorn, and C. K. Yap. Robust geometric computation. In J. E. Goodman and J. O’Rourke, editors, *Handbook of Computational Geometry*, pages 1119–1152. CRC Press, 2 edition, 2004.

- [HP11] S. Har-Peled. *Geometric Approximation Algorithms*, volume 173 of *Graduate Studies in Mathematics*. American Mathematical Society, 2011.
- [HPM04] S. Har-Peled and S. Mazumdar. On coresets for k -means and k -median clustering. In *Proceedings of the 36th Annual ACM Symposium on Theory of Computing (STOC)*, pages 291–300, 2004.
- [HTF09] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2nd edition, 2009.
- [HW87] D. Haussler and E. Welzl. ε -nets and simplex range queries. *Discrete & Computational Geometry*, 2(2):127–151, 1987.
- [IM98] P. Indyk and R. Motwani. Approximate nearest neighbors: Towards removing the curse of dimensionality. In *Proceedings of the 30th Annual ACM Symposium on Theory of Computing (STOC)*, pages 604–613, 1998.
- [JL84] W. B. Johnson and J. Lindenstrauss. Extensions of lipschitz mappings into a hilbert space. In *Conference in Modern Analysis and Probability (New Haven, Conn., 1982)*, volume 26 of *Contemporary Mathematics*, pages 189–206. American Mathematical Society, 1984.
- [Jyl10] J. Jylänki. A thousand ways to pack the bin - a practical approach to two-dimensional rectangle bin packing. Technical report, Technical Report, 2010.
- [Kac66] M. Kac. Can one hear the shape of a drum? *American Mathematical Monthly*, 1966.
- [KBC⁺18] Tim Kraska, Alex Beutel, Ed H. Chi, Jeffrey Dean, and Neoklis Polyzotis. The case for learned index structures. In *Proceedings of the 2018 ACM International Conference on Management of Data (SIGMOD)*, pages 489–504, 2018.
- [KBH06] M. Kazhdan, M. Bolitho, and H. Hoppe. Poisson surface reconstruction. *Proceedings of the Fourth Eurographics Symposium on Geometry Processing*, pages 61–70, 2006.
- [Kir83] D. G. Kirkpatrick. Optimal search in planar subdivisions. *SIAM Journal on Computing*, 12(1):28–35, 1983.
- [Kle77] V. Klee. Can the measure of $\cup[a_i, b_i]$ be computed in less than $O(n \log n)$ steps? *American Mathematical Monthly*, 1977.
- [KLN91] M. Karasick, D. Lieber, and L. R. Nackman. Efficient delaunay triangulation using rational arithmetic. *ACM Transactions on Graphics*, 10(1):71–91, 1991.
- [KMP⁺08] L. Kettner, K. Mehlhorn, S. Pion, S. Schirra, and C. K. Yap. Classroom examples of robustness problems in geometric computations. *Computational Geometry: Theory and Applications*, 40(1):61–90, 2008.
- [KMY03] P. Kumar, J. S. B. Mitchell, and E. A. Yildirim. Computing core-sets and approximate smallest enclosing hyperspheres in high dimensions. In *Proceedings of the Fifth Workshop on Algorithm Engineering and Experiments (ALENEX)*, pages 45–55. SIAM, 2003.
- [Koe36] P. Koebe. Kontaktprobleme der konformen abbildung. *Berichte Sächs. Akad. Wiss. Leipzig*, 1936.

-
- [KS98] R. Kimmel and J. A. Sethian. Computing geodesic paths on manifolds. *Proceedings of the National Academy of Sciences*, 1998.
- [KSLO96] Lydia E. Kavraki, Petr Svestka, Jean-Claude Latombe, and Mark H. Overmars. Probabilistic roadmaps for path planning in high-dimensional configuration spaces. *IEEE Transactions on Robotics and Automation*, 12(4):566–580, 1996.
- [Kuh55] H. W. Kuhn. The hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1-2):83–97, 1955.
- [Lam70] G. Laman. On graphs and rigidity of plane skeletal structures. *Journal of Engineering Mathematics*, 1970.
- [Las96] M. J. Laszlo. *Computational Geometry and Computer Graphics in C++*. Prentice Hall, 1996.
- [Lat91] Jean-Claude Latombe. *Robot Motion Planning*. Kluwer Academic Publishers, 1991.
- [Law77] C. L. Lawson. Software for C^1 surface interpolation. *Mathematical Software*, 1977.
- [LB14] Matthew M. Loper and Michael J. Black. Opendr: An approximate differentiable renderer. In *Proceedings of the European Conference on Computer Vision (ECCV)*, volume 8695 of *Lecture Notes in Computer Science*, pages 154–169. Springer, 2014.
- [LC87] W. E. Lorensen and H. E. Cline. Marching cubes: A high resolution 3d surface construction algorithm. In *ACM SIGGRAPH Computer Graphics*, volume 21, pages 163–169, 1987.
- [LD81] D. T. Lee and R. L. Drysdale. Generalization of Voronoi diagrams in the plane. *SIAM Journal on Computing*, 10(1):73–87, 1981.
- [LD08] A. Lagae and P. Dutré. A comparison of methods for generating Poisson disk distributions. *Computer Graphics Forum*, 2008.
- [Lee83] D. T. Lee. Visibility of a simple polygon. *Computer Vision, Graphics, and Image Processing*, 22(2):207–221, 1983.
- [Lev06] B. Levy. Laplace-Beltrami eigenfunctions towards an optimal tool for geology and geometry. In *SMA*, 2006.
- [Llo82] S. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 1982.
- [LMM02] A. Lodi, S. Martello, and M. Monaci. Two-dimensional packing problems: A survey. *European Journal of Operational Research*, 2002.
- [Loo87] C. Loop. Smooth subdivision surfaces based on triangle meshes. Master’s thesis, University of Utah, 1987.
- [LP79] D. T. Lee and F. P. Preparata. An optimal algorithm for finding the kernel of a polygon. *Journal of the ACM*, 26(4):702–712, 1979.
- [LP83] T. Lozano-Pérez. Spatial planning: A configuration space approach. *IEEE Transactions on Computers*, 1983.
- [Lue78] G. S. Lueker. A data structure for orthogonal range queries. In *IEEE Symposium on Foundations of Computer Science (FOCS)*, 1978.

- [LvK10a] Maarten Löffler and Marc van Kreveld. Largest and smallest convex hulls for imprecise points. *Algorithmica*, 56(2):235–269, 2010.
- [LvK10b] Maarten Löffler and Marc van Kreveld. Largest bounding box, smallest diameter, and related problems on imprecise points. *Computational Geometry: Theory and Applications*, 43(4):419–433, 2010.
- [Man94] D. Manocha. Solving systems of polynomial equations using resultants. In *SIGGRAPH Course*, 1994.
- [Mar72] G. Marsaglia. Choosing a point from the surface of a sphere. *The Annals of Mathematical Statistics*, 43(2):645–646, 1972.
- [Mat90] J. Matoušek. Construction of ε -nets. *Discrete & Computational Geometry*, 5(2):199–213, 1990.
- [Mat92] J. Matoušek. Reporting points in halfspaces. *Computational Geometry: Theory and Applications*, 2(3):169–186, 1992.
- [Mat95] J. Matoušek. Tight upper bounds for the discrepancy of half-spaces. *Discrete & Computational Geometry*, 13(3–4):593–601, 1995.
- [Mat99] J. Matoušek. *Geometric Discrepancy: An Illustrated Guide*. Springer, 1999.
- [Mat02] J. Matoušek. *Lectures on Discrete Geometry*, volume 212 of *Graduate Texts in Mathematics*. Springer, 2002.
- [McC85] E. M. McCreight. Priority search trees. *SIAM Journal on Computing*, 14(2):257–276, 1985.
- [Meg83] N. Megiddo. Linear-time algorithms for linear programming in R^3 and related problems. *SIAM Journal on Computing*, 1983.
- [Mei75] G. H. Meisters. Polygons have ears. *The American Mathematical Monthly*, 1975.
- [Mit99] J. S. B. Mitchell. Guillotine subdivisions approximate polygonal subdivisions: A simple polynomial-time approximation scheme for geometric TSP, k -MST, and related problems. *SIAM Journal on Computing*, 28(4):1298–1309, 1999.
- [MJFS01] B. Moon, H. V. Jagadish, C. Faloutsos, and J. H. Saltz. Analysis of the clustering properties of the Hilbert space-filling curve. *IEEE Transactions on Knowledge and Data Engineering*, 2001.
- [MMP87] J. S. B. Mitchell, D. M. Mount, and C. H. Papadimitriou. The discrete geodesic problem. *SIAM Journal on Computing*, 1987.
- [MMR⁺01] Klaus-Robert Müller, Sebastian Mika, Gunnar Rätsch, Koji Tsuda, and Bernhard Schölkopf. An introduction to kernel-based learning algorithms. *IEEE Transactions on Neural Networks*, 12(2):181–201, 2001.
- [Mor66] G. M. Morton. A computer-oriented geodetic data base and a new technique in file sequencing. Technical report, IBM Technical Report, 1966.
- [MR95] R. Motwani and P. Raghavan. *Randomized Algorithms*. Cambridge University Press, 1995.
- [MRH02] A. Meijster, J. B. Roerdink, and W. H. Hesselink. A general algorithm for computing distance transforms in linear time. In *Mathematical Morphology and its Applications to Image and Signal Processing*, 2002.

-
- [MRTT53] T. S. Motzkin, H. Raiffa, G. L. Thompson, and R. M. Thrall. The double description method. In *Contributions to the Theory of Games, Volume II*, volume 28 of *Annals of Mathematics Studies*, pages 51–73. Princeton University Press, 1953.
 - [MS88] H. G. Mairson and J. Stolfi. Reporting and counting intersections between two sets of line segments. In *Theoretical Foundations of Computer Graphics and CAD*, NATO ASI Series, pages 307–325. Springer, 1988.
 - [MS96] N. Madras and G. Slade. *The Self-Avoiding Walk*. Birkhäuser, 1996.
 - [MS05a] D. P. Mehta and S. Sahni, editors. *Handbook of Data Structures and Applications*. Chapman & Hall/CRC, 2005.
 - [MS05b] Dinesh P. Mehta and Sartaj Sahni. *Handbook of Data Structures and Applications*. Chapman & Hall/CRC, 2005.
 - [MSP⁺24] Vinicius C. Modi, Nicholas Sharp, Or Perel, Shinjiro Sueda, and David I. W. Levin. Simplicits: Mesh-free, geometry-agnostic, elastic simulation. *ACM Transactions on Graphics*, 43(4):117:1–117:15, 2024.
 - [MSZ99] E. P. Mücke, I. Saias, and B. Zhu. Fast randomized point location without pre-processing in two- and three-dimensional delaunay triangulations. *Computational Geometry: Theory and Applications*, 12(1–2):63–83, 1999.
 - [Mul90] K. Mulmuley. A fast planar partition algorithm, I. *Journal of Symbolic Computation*, 10:253–266, 1990.
 - [Mul94] K. Mulmuley. *Computational Geometry: An Introduction Through Randomized and Incremental Algorithms*. Prentice Hall, 1994.
 - [Mus13] K. Museth. VDB: High-resolution sparse volumes with dynamic topology. *ACM Transactions on Graphics*, 32(3):1–22, 2013.
 - [Mut05] S. Muthukrishnan. Data streams: Algorithms and applications. *Foundations and Trends in Theoretical Computer Science*, 1(2):117–236, 2005.
 - [MV22] Michael Mitzenmacher and Sergei Vassilvitskii. Algorithms with predictions. *Communications of the ACM*, 65(7):33–35, 2022.
 - [MY20] Y. A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(4):824–836, 2020.
 - [NAT90] B. Naylor, J. Amanatides, and W. Thibault. Merging bsp trees yields polyhedral set operations. In *Proceedings of the 17th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pages 115–124, 1990.
 - [Nie92] H. Niederreiter. *Random Number Generation and Quasi-Monte Carlo Methods*. SIAM, 1992.
 - [NNS72] M. E. Newell, R. G. Newell, and T. L. Sancha. A new approach to shaded picture problem. In *Proceedings of the ACM Annual Conference*, pages 443–450, 1972.
 - [NS07] Giri Narasimhan and Michiel Smid. *Geometric Spanner Networks*. Cambridge University Press, 2007.

- [NT03] F. S. Nooruddin and G. Turk. Simplification and repair of polygonal models using volumetric techniques. *IEEE Transactions on Visualization and Computer Graphics*, 2003.
- [OBCS⁺12] M. Ovsjanikov, M. Ben-Chen, J. Solomon, A. Butscher, and L. Guibas. Functional maps: A flexible representation of maps between shapes. *ACM Transactions on Graphics*, 2012.
- [OBSC00] A. Okabe, B. Boots, K. Sugihara, and S. N. Chiu. *Spatial Tessellations: Concepts and Applications of Voronoi Diagrams*. Wiley, 2nd edition, 2000.
- [OF03] S. Osher and R. Fedkiw. *Level Set Methods and Dynamic Implicit Surfaces*. Springer, 2003.
- [O’R87] J. O’Rourke. *Art Gallery Theorems and Algorithms*. Oxford University Press, 1987.
- [Ove83] M. H. Overmars. *The Design of Dynamic Data Structures*, volume 156 of *Lecture Notes in Computer Science*. Springer, 1983.
- [OvL81] M. H. Overmars and J. van Leeuwen. Maintenance of configurations in the plane. *Journal of Computer and System Sciences*, 1981.
- [OW92] M. H. Overmars and E. Welzl. New methods for computing visibility graphs. *Algorithmica*, 7:449–461, 1992.
- [OY91] M. H. Overmars and C. K. Yap. New upper bounds in Klee’s measure problem. *SIAM Journal on Computing*, 1991.
- [P²1] G. Pólya. Über eine aufgabe der Wahrscheinlichkeitsrechnung betreffend die Irrfahrt im Straßennetz. *Mathematische Annalen*, 1921.
- [P⁺14] D. Panozzo et al. Frame fields: An orientation-aware surface representation. In *ACM Transactions on Graphics (SIGGRAPH)*, 2014.
- [PC06] G. Peyré and L. D. Cohen. Geodesic remeshing using front propagation. *International Journal of Computer Vision*, 2006.
- [PC19] Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- [Pea90] G. Peano. Sur une courbe, qui remplit toute une aire plane. *Mathematische Annalen*, 1890.
- [Pea05] K. Pearson. The problem of the random walk. *Nature*, 1905.
- [PH77] F. P. Preparata and S. J. Hong. Convex hulls of finite sets of points in two and three dimensions. *Communications of the ACM*, 20(2):87–93, 1977.
- [PM92] W. B. Pennebaker and J. L. Mitchell. *JPEG: Still Image Data Compression Standard*. Van Nostrand Reinhold, 1992.
- [Poi95] H. Poincaré. Analysis situs. *Journal de l’École Polytechnique*, 1895.
- [PP93] U. Pinkall and K. Polthier. Computing discrete minimal surfaces and their conjugates. *Computing*, 1993.
- [PP03] K. Polthier and E. Preuss. Identifying vector field singularities using a discrete Hodge decomposition. In *Visualization and Mathematics III*, 2003.

-
- [Pri91] D. M. Priest. Algorithms for arbitrary precision floating-point arithmetic. In *Proceedings of the 10th IEEE Symposium on Computer Arithmetic*, pages 132–143, 1991.
 - [PS85] F. P. Preparata and M. I. Shamos. *Computational Geometry: An Introduction*. Springer-Verlag, 1985.
 - [PS04] P.-O. Persson and G. Strang. A simple mesh generator in MATLAB. *SIAM Review*, 46(2):329–345, 2004.
 - [PT97] L. Piegl and W. Tiller. *The NURBS Book*. Springer, 1997.
 - [QSMG17] Charles R. Qi, Hao Su, Kaichun Mo, and Leonidas J. Guibas. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 652–660, 2017.
 - [R⁺12] J. F. Remacle et al. A frontal Delaunay quad mesh generator. *International Journal for Numerical Methods in Engineering*, 2012.
 - [RP66] A. Rosenfeld and J. L. Pfaltz. Sequential operations in digital picture processing. *Journal of the ACM*, 1966.
 - [Rub16] Natan Rubin. On kinetic delaunay triangulations: A near-quadratic bound for uniform linear motion. *Discrete & Computational Geometry*, 55(4):918–943, 2016.
 - [Rup95] J. Ruppert. A delaunay refinement algorithm for quality 2-dimensional mesh generation. *Journal of Algorithms*, 18(3):548–585, 1995.
 - [RWP06] M. Reuter, F. E. Wolter, and N. Peinecke. Laplace-Beltrami spectra as Shape-DNA of surfaces and solids. *Computer-Aided Design*, 2006.
 - [S⁺25] N. Sharp et al. SpectralNet: Learning spectral descriptors for shape analysis. In *ACM Transactions on Graphics (SIGGRAPH)*, 2025.
 - [SA95] M. Sharir and P. K. Agarwal. *Davenport-Schinzel Sequences and Their Geometric Applications*. Cambridge University Press, 1995.
 - [Sag94] H. Sagan. *Space-Filling Curves*. Universitext, 1994.
 - [Sam84] H. Samet. The quadtree and related hierarchical data structures. *ACM Computing Surveys*, 1984.
 - [Sam06] H. Samet. *Foundations of Multidimensional and Metric Data Structures*. Elsevier, 2006.
 - [SC20] R. Sawhney and K. Crane. Monte carlo geometry processing: A meshless approach to geometric PDE. In *ACM Transactions on Graphics (SIGGRAPH)*, 2020.
 - [SDI06] Gregory Shakhnarovich, Trevor Darrell, and Piotr Indyk. *Nearest-Neighbor Methods in Learning and Vision: Theory and Practice*. MIT Press, 2006.
 - [Sei90] R. Seidel. Linear programming and convex hulls made easy. In *Proceedings of the Sixth Annual Symposium on Computational Geometry (SoCG)*, pages 86–92, 1990.
 - [Sei91] R. Seidel. A simple and fast incremental randomized algorithm for computing trapezoidal decompositions and for point location. *Computational Geometry: Theory and Applications*, 1(1):51–64, 1991.

- [Sei93] R. Seidel. Backwards analysis of randomized geometric algorithms. In J. Pach, editor, *New Trends in Discrete and Computational Geometry*, volume 10 of *Algorithms and Combinatorics*, pages 37–67. Springer-Verlag, 1993.
- [Set96] J. A. Sethian. A fast marching level set method for monotonically advancing fronts. *Proceedings of the National Academy of Sciences*, 1996.
- [SH74] I. E. Sutherland and G. W. Hodgman. Reentrant polygon clipping. *Communications of the ACM*, 17(1):32–42, 1974.
- [SH75] M. I. Shamos and D. Hoey. Closest-point problems. In *IEEE Symposium on Foundations of Computer Science (FOCS)*, 1975.
- [Sha78] M. I. Shamos. *Computational Geometry*. PhD thesis, Yale University, 1978.
- [She97] J. R. Shewchuk. Adaptive precision floating-point arithmetic and fast robust geometric predicates. *Discrete & Computational Geometry*, 1997.
- [She98] J. R. Shewchuk. *Delaunay Refinement Mesh Generation*. PhD thesis, Carnegie Mellon University, 1998.
- [She02] J. R. Shewchuk. Delaunay refinement algorithms for triangular mesh generation. *Computational Geometry: Theory and Applications*, 22(1–3):21–74, 2002.
- [Si15] H. Si. Tetgen, a delaunay-based quality tetrahedral mesh generator. *ACM Transactions on Mathematical Software*, 41(2):11:1–11:36, 2015.
- [SOG09] J. Sun, M. Ovsjanikov, and L. Guibas. A concise and provably informative multi-scale signature based on heat diffusion. *Computer Graphics Forum*, 2009.
- [Spi76] F. Spitzer. *Principles of Random Walk*. Springer, 1976.
- [ST04] D. A. Spielman and S.-H. Teng. Smoothed analysis of algorithms: Why the simplex algorithm usually takes polynomial time. *Journal of the ACM*, 51(3):385–463, 2004.
- [Ste05] K. Stephenson. *Introduction to Circle Packing: The Theory of Discrete Analytic Functions*. Cambridge University Press, 2005.
- [Sun01] D. Sunday. Inclusion of a point in a polygon. *Journal of Graphics Tools*, 2001.
- [SUW64] M. L. Stein, S. M. Ulam, and M. B. Wells. A visual display of some properties of the distribution of primes. *The American Mathematical Monthly*, 1964.
- [Syl40] J. J. Sylvester. A method of determining by inspection the form of the combined roots of two algebraic equations. *Philosophical Magazine*, 1840.
- [Thu85] W. Thurston. The finite Riemann mapping theorem. In *International Congress of Mathematicians*, 1985.
- [TLHD03] Y. Tong, S. Lombeyda, A. N. Hirani, and M. Desbrun. Discrete multiscale vector field decomposition. In *ACM Transactions on Graphics*, 2003.
- [TOG17] Csaba D. Tóth, Joseph O’Rourke, and Jacob E. Goodman, editors. *Handbook of Discrete and Computational Geometry*. CRC Press, third edition, 2017.
- [Tou83] G. T. Toussaint. Solving geometric problems with the rotating calipers. In *Proceedings of MELECON ’83*, 1983.

-
- [Tsi95] J. N. Tsitsiklis. Efficient algorithms for globally optimal trajectories. *IEEE Transactions on Automatic Control*, 1995.
 - [Tuk75] J. W. Tukey. Mathematics and the picturing of data. In *Proceedings of the International Congress of Mathematicians*, 1975.
 - [Tut63] W. T. Tutte. How to draw a graph. *Proceedings of the London Mathematical Society*, 1963.
 - [Vat92] B. R. Vatti. A generic solution to polygon clipping. *Communications of the ACM*, 1992.
 - [vdB04] G. van den Bergen. *Collision Detection in Interactive 3D Environments*. Morgan Kaufmann, 2004.
 - [VdC35] J. G. Van der Corput. Verteilungsfunktionen. *Proc. Akad. Wet. Amsterdam*, 1935.
 - [Vit01] Jeffrey S. Vitter. External memory algorithms and data structures: Dealing with massive data. *ACM Computing Surveys*, 33(2):209–271, 2001.
 - [Vor08] G. Voronoi. Nouvelles applications des paramètres continus à la théorie des formes quadratiques. *Journal für die reine und angewandte Mathematik*, 1908.
 - [VSP⁺17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pages 5998–6008, 2017.
 - [Wat81] D. F. Watson. Computing the n-dimensional Delaunay tessellation with application to Voronoi polytopes. *The Computer Journal*, 1981.
 - [WBS06] Kilian Q. Weinberger, John Blitzer, and Lawrence K. Saul. Distance metric learning for large margin nearest neighbor classification. In *Advances in Neural Information Processing Systems 18 (NIPS)*, pages 1473–1480, 2006.
 - [Wel91] E. Welzl. Smallest enclosing disks (balls and ellipsoids). In *Lecture Notes in Computer Science*, 1991.
 - [Wil85] D. E. Willard. New data structures for orthogonal range queries. *SIAM Journal on Computing*, 14(1):232–253, 1985.
 - [WLW22] X. Wei, J. Liu, and J. Wang. CoACD: Collision-oriented approximate convex decomposition. In *ACM Transactions on Graphics (SIGGRAPH)*, 2022.
 - [WP67] D. J. A. Welsh and M. B. Powell. An upper bound for the chromatic number of a graph and its application to timetabling problems. *The Computer Journal*, 1967.
 - [WR97] H. J. Wolfson and I. Rigoutsos. Geometric hashing: An overview. *IEEE Computational Science and Engineering*, 4(4):10–21, 1997.
 - [WWV85] J. D. Wolter, T. C. Woo, and R. A. Volz. Optimal algorithms for detecting relevant symmetries in polygons. *IEEE Transactions on Robotics and Automation*, 1985.
 - [Y⁺25] Y. Yu et al. Robust derivative estimation with walk on stars. In *ACM Transactions on Graphics (SIGGRAPH)*, 2025.
 - [Yap88] C. K. Yap. Symbolic treatment of geometric degeneracies. *Journal of Symbolic Computation*, 10(3-4):349–370, 1988.

- [Yap90] C. K. Yap. A geometric consistency theorem for a symbolic perturbation scheme. *Journal of Computer and System Sciences*, 40(1):2–18, 1990.
- [Yap97] C. K. Yap. Towards exact geometric computation. *Computational Geometry: Theory and Applications*, 7(1-2):3–23, 1997.
- [YZ04] R. Yuster and U. Zwick. Fast sparse matrix multiplication. In *Proceedings of the 12th Annual European Symposium on Algorithms (ESA)*, volume 3221 of *Lecture Notes in Computer Science*, pages 604–615. Springer, 2004.
- [Z⁺25] Y. Zhu et al. Designing 3D anisotropic frame fields with Odeco tensors. In *ACM Transactions on Graphics (SIGGRAPH)*, 2025.
- [ZC05] Afra Zomorodian and Gunnar Carlsson. Computing persistent homology. *Discrete & Computational Geometry*, 33(2):249–274, 2005.

Index

- $\lambda_s(n)$, 225
- k -d tree, 108
- x -node, 215
- y -node, 215
- 3D scanning, 346

- AABB, 197
- adaptive meshing, 377
- adaptive predicate, 8
- adjacency list, 478
- adjacency query, 357, 358
- advancing front, 376
- advancing-front, 376
- affine combination, 15
- affine dependence, 262
- affine heat flow, 333
- affine independence, 15
- affine space, 14
- affine transformation, 19
- algorithm families, 9
- algorithms with predictions, 635, 637, 645
- alpha shape, 186
- alpha-shape, 318, 456
- anchor point, 29
- Andrew’s algorithm, 32
- angle-bounded remeshing, 332
- anisotropy, 511
- applications, 9
- approximate convex decomposition, 307
- approximate nearest neighbor, 547, 549, 605
- approximation algorithm, 543
- approximation ratio, 543, 544
- arrangement, 241
- arrangements, 227
- art gallery problem, 396, 417
- art gallery theorem, 396
- aspect ratio, 371
- assignment problem, 468
- average-case analysis, 527
- AVL tree, 131

- B-spline, 450
- B-tree, 637
- backward analysis, 529, 530
- backwards analysis, 216
- ball pivoting, 456
- bandwidth, 496
- barycentric coordinates, 17, 77
- Bellman–Ford algorithm, 409
- Betti number, 517, 518, 522
- Bezier surface, 295
- bin packing, 562
- binary heap, 410
- binary space partition, 402
- binary space partitioning, 139
- bioinformatics, 416
- bipartite matching, 464
- bit-interleaving, 145
- blue noise, 429
- boolean operations, 367
- bottleneck matching, 464, 466, 467
- Bottom-Left-Fill, 566
- boundary, 355
- boundary alignment, 511
- boundary component, 356
- boundary cycle, 354
- boundary detection, 480
- boundary edge, 356, 480
- boundary integral equation, 592
- boundary loop, 480
- boundary operator, 515
- boundary penalty quadric, 502
- box decomposition, 549
- breadth-first search, 477
- bridge, 570
- broad phase, 281
- Brownian motion, 442
- Buchberger’s algorithm, 651

- can one hear the shape of a drum, 506
- Carathéodory’s theorem, 15, 261
- Catmull-Clark subdivision, 493

- certificate, 574
- certified computation, 639
- CGAL, 10
- Chamfer matching, 408
- character recognition, 70
- Chebyshev distance, 18
- Chernoff bound, 534
- chromatic number, 494
- circle contact graph, 559
- circle packing, 559
- circumcentric dual, 512
- circumradius, 318
- Clarkson–Shor technique, 532, 600
- clipping, 70
- closest points, 276
- clustering, 553, 629
- cochain, 512
- cocone, 456
- codifferential, 518
- coherence, 569, 573
- collinear, 239
- collinear points, 31
- collinearity, 251
- collision, 283
- collision detection, 63, 126, 136, 281, 288
- comb polygon, 396, 418
- complexity, 6, 48
- computational geometry, 3, 122
- computational topology, 344
- computer graphics, 122, 129, 408
- computer vision, 66, 70, 408, 416, 474, 475
- concentration of measure, 614
- concurrency, 251
- concurrent, 239
- configuration space, 63
- conflict graph, 530
- conformal parameterization, 322
- congruence, 468, 469
- conjugate gradient, 511
- connected component, 477, 509
- consistency, 624
- constrained Delaunay triangulation, 90, 180, 371, 372
- continuous collision detection, 281
- convex combination, 261
- convex decomposition, 72
- convex hull, 15, 29, 244, 318
- convex hull peeling, 37
- convex hull, separation, 630
- convex polytope, 257
- convex region, 247
- convex set, 15
- core point, 626
- core-set, 547
- coresets, 641
- correctness, 47
- cotangent weight, 513
- cotangent weights, 507
- counting query, 113
- coupling, 466
- Cox-de Boor recursion, 298
- cross-covariance matrix, 347
- cross-polytope, 258
- crust, 456
- CSG, 57
- curl, 515
- curl-free, 518
- curse of dimensionality, 114, 557, 599, 616
- cuspid vertex, 80
- cyclic polytope, 258, 259, 600
- DAG, 215
- data mining, 622
- data structures, 607
- database indexing, 132
- Davenport–Schinzel sequence, 225
- DBSCAN, 629
- DCEL, 58, 96
- degeneracy, 7
- degenerate intersection, 48
- Degrevlex, 652
- Delaunay property, 82
- Delaunay refinement, 184, 372
- Delaunay triangulation, 82, 165, 318, 371
- depth, 209
- depth-first search, 477
- derandomization, 534
- detection, 49
- determinant, 16
- differentiable geometry, 647
- differentiable rendering, 647
- differential geometry, 515
- diffusion, 442
- diffusion distance, 328
- diffusion equation, 443
- Dijkstra’s algorithm, 402, 408, 410
 - greedy, 409
- dimension reduction, 557
- dimensionality reduction, 607
- Dirichlet boundary condition, 507
- Dirichlet spectrum, 506
- discrepancy, 252, 433
- discrete exterior calculus, 512

- discrete Fréchet distance, 465, 466
- discrete Morse theory, 521
- distance, 623
- distance between convex sets, 276
- distance map, 406
- distance oracle, 553
- distance transform, 406
- divergence, 515
- divergence-free, 518
- divide and conquer, 35, 176, 193, 228
- dog-leash analogy, 414
- dot product, 14
- double wedge, 239
- doubly connected edge list, 96, 353, 354
- dual graph, 477, 483
- dual line, 238
- dual point, 238
- dual polytope, 263
- dual transform, 20, 237
- duality, 20, 60, 238, 244, 252
- dummy half-edge, 356
- duplicate points, 31
- dynamic closest pair, 572
- dynamic data structure, 114, 569
- dynamic nearest neighbor, 572
- dynamic point location, 104
- dynamic programming, 74, 414
- ear, 77
- ear clipping, 77
- Earth mover's distance, 634
- edge collapse, 501
- edge flip, 82, 171, 367, 377
- edge split, 367
- eigen-decomposition, 508, 595
- eigenfunction, 506, 594
- Eikonal equation, 402, 456
- elimination, 654
- embedding, 632
- empty circumcircle, 167
- envelope, 244, 251
- envelope complexity, 228
- epsilon-net, 535
- epsilon-sample, 315, 452
- Euclidean distance, 14, 193, 286
- Euclidean TSP, 544
- Euler characteristic, 20, 262, 356, 362, 486, 488, 491, 511, 522
- Euler formula, 488
- Euler operation, 365, 367
- Euler's formula, 262
- Euler–Poincaré formula, 262
- event, 43, 573
- event queue, 43, 44
- exact computation, 8
- exact geometric computation, 25
- exercises, 253, 475
- expanding polytope algorithm, 279
- expansion, 25
- expansion arithmetic, 24
- expected time, 527
- experiments, 649
- exterior calculus, 512
- exterior derivative, 515
- external memory, 642
- extraordinary vertex, 491
- extreme point, 258
- f-vector, 262
- face degree, 354
- face lattice, 257, 259
- face vector, 262
- face walk, 361
- face-edge incidence, 480
- facet, 259
- failure time, 573
- Farkas lemma, 260
- farthest-point Voronoi diagram, 162
- fast marching, 457
- Fast Marching Method, 402, 404
- feature, 621
- feature map, 631
- feature space, 621
- Fibonacci heap, 410
- Fiedler value, 507
- Fiedler vector, 509, 633
- finite element method, 378, 501
- fixed dimension, 599
- fixed-radius neighbor search, 137
- floating point, 7
- floating-point filter, 24
- Flory exponent, 446
- FMM
 - Accepted set, 403
 - Far set, 403
 - point classification, 403
 - Trial set, 403
- force-directed layout, 560
- Fréchet distance, 465
- Fréchet distance, 414, 415
- Fréchet mean, 411
- fractional cascading, 111
- frame field, 510
- frame field singularity, 511

- free space diagram, 414
- free-space diagram, 465
- functional maps, 326
- fundamental quadric, 502
- funnel algorithm, 401

- game development, 411
- Gauss–Bonnet theorem, 489
- general position, 7, 58
- generalized winding number, 302
- genus, 488
- geodesic, 401
- geodesic distance, 18, 328, 335, 405, 411
- geodesic path, 400
- geometric deep learning, 637, 646
- geometric hashing, 474
- geometry processing, 647
- GIS, 70, 122, 126, 129, 474, 475
- GIS clipping, 57
- GJK algorithm, 286
- GPS navigation, 410
- Gröbner basis, 651
- gradient, 515
- gradient estimation, 592
- graduate study, 650
- graph, 408
- graph coloring, 493
- graph Laplacian, 633
- graph partitioning, 507, 509
- graph search, 483
- gravity heuristic, 565
- greedy coloring, 494
- Green’s function, 592
- grid resolution, 403
- guard, 396
- guillotine cut, 563

- H-description, 257
- h-refinement, 377
- H-representation, 258
- half-edge, 95, 353, 354, 361, 478, 481, 492
- half-face, 361
- half-plane discrepancy, 231
- half-plane intersection, 247, 420
- half-space, 17
- Halton sequence, 432
- ham-sandwich, 252
- Handshaking Lemma, 362
- handwriting recognition, 416
- harmonic component, 518
- harmonic mapping, 323
- Hausdorff distance, 414, 464, 465

- heat kernel, 328, 507
- Heat Kernel Signature, 509, 593
- Heat Method, 412, 413
- Helly’s theorem, 261
- hex meshing, 511
- hexagonal packing, 559
- hexahedral remeshing, 511
- hidden surface removal, 140, 402
- hierarchical clustering, 626
- high-dimensional computational geometry, 600
- Hilbert curve, 142
- history, 4
- history DAG, 99, 101
- Hodge decomposition, 517, 518
- Hodge star, 512
- homogeneous coordinates, 19
- homology, 517, 521
- homology basis, 344
- hopper, 565
- hopper optimization, 565
- Hungarian algorithm, 468
- hyperplane, 14, 630
- hyperplane separation, 17

- I/O complexity, 642
- ICP, 471
- illegal edge, 87
- image compression, 126
- image processing, 406, 408
- image segmentation, 405
- implementation, 105
- implicit surface, 449
- implicitization, 655
- imprecise data, 639
- in-circle test, 82
- incidence, 239
- incident face, 355
- incremental construction, 172
- indexed face set, 353
- indicator function, 459
- inner product, 14
- inscribed polytope, 603
- intersection of convex sets, 261
- interval tree, 130
- intrinsic Delaunay edge, 188
- intrinsic Delaunay triangulation, 188
- intrinsic dimension, 618
- invariant, 474
- inverse Ackermann function, 225
- ISOMAP, 631
- isometry, 19, 469

-
- isometry invariance, 506, 507, 594
 - isospectral shapes, 507, 509
 - Iterative Closest Point, 346
 - iterative closest point, 471
 - Johnson–Lindenstrauss lemma, 608, 632
 - k-center, 553, 556
 - K-Geodesics, 412
 - k-level, 231, 241
 - k-means, 553, 626
 - K-Means clustering, 411
 - K-Means++, 413
 - k-median, 553, 556
 - k-nearest neighbor, 625
 - k-nearest neighbors, 196
 - k-nearest-neighbor graph, 632
 - k-set, 240
 - Kac, Mark, 506
 - KD-tree, 119, 196, 347, 350
 - Keil–Snoeyink algorithm, 74
 - kernel, 392, 630
 - kernel of a polygon, 420
 - kernel trick, 630
 - kinetic collision detection, 582
 - kinetic data structure, 227, 569, 573, 574, 644
 - kinetic event, 574, 575
 - kinetic heap, 574, 575
 - Klee’s measure problem, 233
 - Laplace equation, 591, 592
 - Laplace–Beltrami operator, 506, 593, 594
 - Laplace–Beltrami operator, 329, 513, 519
 - Laplacian eigenmaps, 631
 - lazy learning, 625
 - lazy propagation, 133
 - learned data structure, 637
 - learned index, 634
 - learning-augmented algorithm, 645
 - least-squares alignment, 471
 - Lee’s algorithm, 396
 - level, 252
 - level of detail, 501
 - level set, 457
 - level-set equation, 457
 - level-set problem, 252
 - library, 10
 - LiDAR, 313
 - lidar, 460
 - lifting, 177
 - lifting map, 603
 - line, 14
 - line arrangement, 58
 - linear programming, 251
 - linear separability, 629
 - link condition, 501
 - Lipschitz boundary, 592
 - Lloyd’s algorithm, 412, 556
 - local Delaunay criterion, 169
 - locality, 574
 - locality preservation, 142
 - locality-sensitive hashing, 549, 551, 605
 - Loop subdivision, 493
 - low-discrepancy sequence, 432
 - lower envelope, 225, 228, 240
 - machine learning, 122, 622
 - Mahalanobis distance, 622
 - Manhattan distance, 18
 - manifold, 20, 356, 485
 - manifold hypothesis, 631
 - manifold learning, 631
 - manifold mesh, 356
 - manifold verification, 485
 - marching cubes, 387, 457
 - margin, 629
 - martingale, 592
 - martingale convergence theorem, 592
 - mass matrix, 507
 - matching problem, 463
 - matching via the arrangement, 467
 - matrix construction, 508
 - MaxRects, 564
 - mean curvature flow, 329
 - mean value property, 592
 - medial axis, 315, 337, 408
 - medical imaging, 405, 408
 - Meijster’s algorithm, 407
 - Mercer kernel, 631
 - mesh, 369
 - mesh decimation, 501
 - mesh editing, 366, 367
 - mesh optimization, 377
 - mesh refinement, 303
 - mesh segmentation, 413, 509
 - mesh smoothing, 377
 - metric, 18
 - metric embedding, 640
 - metric learning, 622, 623
 - metric space, 622
 - min-Hausdorff matching, 464, 465
 - minimization diagram, 228
 - minimum bounding box, 206

- minimum enclosing ball, 557
- minimum-weight matching, 464
- Minkowski difference, 277, 286
- Minkowski sum, 18, 61
- Minkowski–Weyl theorem, 258
- model of computation, 6
- monomial order, 651
- monotone decomposition, 80
- Monte Carlo estimation, 233
- Monte Carlo method, 442, 591
- morphological operations, 63
- Morse theory, 521
- Morse–Smale complex, 521
- Morton curve, 145
- motion planning, 60
- multi-view reconstruction, 346
- multidimensional divide-and-conquer, 111
- multilevel data structure, 111
- multilevel structure, 111

- narrow phase, 281
- nearest neighbor, 196
- nearest neighbor search, 120
- nearest-neighbor classification, 623
- nesting, 565
- network routing, 411
- next operator, 355
- No-Fit Polygon, 565
- non-convex polygon, 61
- non-manifold edge, 365
- non-obtuse triangulation, 91
- non-rigid shape matching, 595
- non-zero winding rule, 202
- norm, 18
- normal cone, 260
- normal fan, 260
- normalization, 508
- NURBS, 298, 450

- octree, 123, 459
- Odeco frame fields, 510, 512
- Odeco tensor, 510
- onion peeling, 37
- OpenVDB, 385
- optimal packing, 559
- optimal transport, 634
- order maintenance problem, 570
- order reversal, 240
- orientability, 482
- orientation, 16
- orientation test, 32
- origin vertex, 355

- orthogonal matrix, 346
- orthogonal polygon, 418
- orthogonal range query, 199
- outlier detection, 39
- outliers, 466, 467
- output-sensitive, 6
- output-sensitive algorithm, 34, 640

- packing density, 559
- painter’s algorithm, 402
- parallel transport, 511
- parametric surface, 449
- partial differential equations, 512
- partial Hausdorff distance, 466
- partial matching, 466, 467
- partial rebuilding, 114, 569
- Peano curve, 148
- persistent homology, 645
- photogrammetry, 313
- physical security, 419
- physics engine, 283
- pinched vertex, 356
- plane, 14
- plane sweep, 80
- Poincaré–Hopf theorem, 511
- point at infinity, 19
- point cloud, 313, 346, 449, 510, 621
- point location, 95, 214
 - walking, 219
- point set, 621
- point-line duality, 237
- Poisson disk sampling, 429
- Poisson equation, 512, 591, 592
- Poisson reconstruction, 457
- polar, 263
- polar body, 251
- polar duality, 251
- polarity, 20, 257, 263
- poles, 456
- Polya’s recurrence theorem, 443
- polygon, 5
- polygon soup, 363, 365
- polygon triangulation, 417
- polynomial ideal, 651
- polynomial-time approximation scheme, 543
- predicate, 16
- prev operator, 355
- primal graph, 477
- primal mesh, 512
- primitive, 5
- priority queue, 403
- priority search tree, 112

-
- probabilistically checkable proofs, 618
 - projective space, 19
 - projective transformation, 19
 - PTAS, 544
 - quad meshing, 511
 - quad-edge, 355, 361
 - quad-edge data structure, 88
 - quadric, 501
 - quadric error metric, 501
 - quadtrees, 123, 544
 - quasi-Monte Carlo, 433
 - R-tree, 127, 197, 637
 - radial-edge, 361
 - radius-edge ratio, 371
 - Radon partition, 262
 - Radon's theorem, 262
 - random polygon generation, 440
 - random projection, 551, 632
 - random sampling, 534
 - random segment generation, 438
 - random walk, 442
 - randomization, 527
 - randomized algorithm, 6, 527
 - randomized incremental construction, 174, 215, 528
 - range minimum query, 134
 - range query, 107, 124
 - range search, 60, 199
 - range tree, 110, 200
 - rational arithmetic, 25
 - ray tracing, 132
 - real-RAM, 6
 - rectangle packing, 562
 - red-black tree, 131
 - reflectional symmetry, 65
 - reflex vertex, 73
 - registration, 467, 468, 474, 475
 - regular triangulation, 604
 - rejection sampling, 436
 - remeshing, 413
 - reporting, 49
 - reporting query, 113
 - research papers, 649
 - resource allocation, 132
 - restricted Delaunay, 452
 - resultant, 654
 - Reverse Cuthill–McKee, 496
 - reverse engineering, 460
 - ridge, 259
 - Riemannian geometry, 512
 - rigid transformation, 468, 469
 - RKHS, 631
 - roadmap, 398
 - robotics, 66, 346, 419
 - path planning, 405, 408, 411
 - robustness, 7
 - rotating calipers, 206, 287
 - rotational sweep, 396
 - rotational symmetry, 65
 - S-polynomial, 651
 - scale invariance, 594
 - Schrödinger equation, 593
 - search DAG, 100, 101
 - segment, 5
 - segment tree, 132, 209, 234
 - Seidel's lemma, 530
 - seismology, 405
 - self-avoiding walk, 445
 - semidynamic, 569
 - separable filter, 406
 - separating axis, 18, 286
 - separating axis theorem, 274
 - separating hyperplane, 260
 - separation theorem, 251, 260
 - shape, 449
 - shape alignment, 474, 475
 - shape analysis, 413
 - shape matching, 466, 467, 506
 - shape representation, 449
 - shape retrieval, 508, 595
 - shape spectrum, 506
 - Shape-DNA, 507, 508
 - shoelace formula, 16
 - shortest path, 399
 - signed area, 16
 - signed distance field, 383
 - signed distance function, 456
 - signed volume, 16
 - simple polytope, 259
 - simplicial complex, 20, 370
 - simplicial polytope, 259
 - Simulation of Simplicity, 26
 - singular value decomposition, 347
 - singularity, 491
 - size field, 377
 - skyline algorithm, 563
 - slab decomposition, 96
 - SLAM, 346
 - sliver, 371
 - sliver exudation, 377
 - smallest enclosing circle, 284, 557

- smoothed analysis, 618
- smoothness optimization, 511
- snake scan, 150
- social networks, 411
- software, 10
- space-filling curve, 142, 148
- space-partitioning, 119
- spanner, 553, 640
- sparse voxel octree, 384
- spatial database, 129
- spatial query, 123
- spectral clustering, 507, 633
- spectral projection, 594
- SpectralNet, 509, 510
- spherical Voronoi diagram, 413
- spiral walk, 153
- split tree, 553
- star polygon, 419
- star-shaped polygon, 392, 420
- status structure, 44
- Steiner point, 91
- stiffness matrix, 507
- stochastic PDE solver, 591
- straight skeleton, 340
- streaming algorithm, 547
- streaming algorithms, 640
- stretch factor, 553
- strip packing, 562
- subspace embedding, 608, 612
- support function, 260, 276
- support point, 276
- support points, 284
- support vector, 629
- support vector machine, 630
- supporting hyperplane, 260
- supporting line, 18
- surface implicitization, 653
- surface reconstruction, 313, 450
- sweep line, 211
- sweep status, 43
- sweep-line, 209, 211
- sweep-line algorithm, 234
- sweep-line paradigm, 41, 43
- Sylvester matrix, 654
- symbolic perturbation, 8
- symmetry, 468, 469, 508
- symmetry detection, 508, 595
- T-junction, 356
- t-SNE, 631
- target vertex, 355
- terrain, 460
- tetrahedral mesh, 370, 510
- texture atlas, 413
- time complexity
 - $O((n + k) \log n)$, 212
 - $O(n \log h)$, 34
 - $O(n \log n)$, 30, 84, 194
 - $O(n^2)$, 36, 39, 78, 84, 88
- TIN, 452
- tolerance, 348
- topological data analysis, 521, 645
- topological invariant, 362
- topology, 20
- tracking, 582
- trajectory analysis, 416
- transformation, 19
- trapezoid, 214
- trapezoidal map, 97, 214, 215
- traversal, 361
- triangle inequality, 543, 622
- triangle merging, 72
- triangle to quad conversion, 309
- triangle-box intersection, 300
- triangular mesh, 370
- Tukey depth, 37
- twin, 355
- two-pass algorithm, 406
- uncertainty, 639
- underlying space, 370
- uniform grid, 135
- uniform random sampling, 435
- union measure, 233
- union of rectangles, 51
- update time, 569
- upper bound theorem, 259
- upper envelope, 228
- upper hull, 240
- V-description, 257
- V-representation, 258
- valence, 358, 361, 362, 491
- van der Corput sequence, 432
- VC dimension, 534
- vector space, 14
- vertex star, 355, 358
- vertex walk, 361
- vertex welding, 365
- visibility, 391, 392
- visibility graph, 60, 396
- visibility polygon, 393
- visibility roadmap, 402
- VLSI design, 132

- Voronoi diagram, 84, 159
- Voronoi diagram, discrete, 629
- Voronoi surface, 465
- Voronoi vertex, 204
- voting, 474
- voxel geometry, 381
- voxel grid decimation, 311
- voxelization, 300, 381

- Walk on Spheres, 591, 592
- Walk on Stars, 591, 592
- Wasserstein distance, 634
- watertight, 450, 459
- wave front, 402
- Wave Kernel Signature, 509, 593, 595
- welded vertex, 363

- well-separated pair decomposition, 551, 553, 640
- Welzl's algorithm, 283
- Weyl's law, 507
- width, 251
- winding number, 202
- window, 393
- winged-edge, 361
- wireless networking, 419
- worst-case time, 527

- Yamabe equation, 322

- z-buffer, 402
- Z-order, 145
- zigzag curve, 150
- zone theorem, 241